Blueprint: Self-similar solutions for the gSQG patch equations

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November 5, 2026

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Chapter 1

Introduction and setup: self-similar reduction and Theorem 1.1 (paper §§1–2)

Throughout this chapter $\alpha \in (1,2)$ is fixed and we write $p = \alpha - 1 \in (0,1)$. Unless said otherwise, all functions are real valued. The Japanese bracket is $\langle y \rangle = (1+y^2)^{1/2}$. We set

$$c_\alpha = \frac{\Gamma(\alpha)\sin(\pi(\alpha-1)/2)}{\pi},$$

which is a strictly positive real number for $\alpha \in (1,2)$ (since $\Gamma(\alpha) > 0$ and $0 < \pi p/2 < \pi/2$). We write $W(y) = y_- = \max\{-y, 0\}$ for the model corner.

1.1 Fourier conventions and the fractional multiplier

Definition 1.1 (Model corner)**.** The model corner is $W : \mathbb{R} \to \mathbb{R}$, $W(y) = y_- := \max\{-y, 0\}$.

Definition 1.2 (Fourier transform)**.** For $f \in L^1(\mathbb{R};\mathbb{C})$ the Fourier transform is

$$\widehat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-ix\xi}\, dx, \qquad \xi \in \mathbb{R},$$

extended to tempered distributions $\mathcal{S}'(\mathbb{R})$ by duality, $\langle \widehat{u}, \varphi \rangle = \langle u, \widehat{\varphi} \rangle$. The inverse transform is $\check{g}(x) = \frac{1}{2\pi}\int_{\mathbb{R}} g(\xi) e^{ix\xi}\, d\xi$. With this convention $\widehat{\partial_x f}(\xi) = i\xi \widehat{f}(\xi)$. A Fourier multiplier with symbol $m \in L^\infty_{\mathrm{loc}}(\mathbb{R})$ of polynomial growth acts on $f \in \mathcal{S}(\mathbb{R})$ by $m(D)f = (m\widehat{f})^\vee$. (Mathlib's `Real.fourierIntegral` uses $e^{-2\pi i x\xi}$; the two conventions differ by the substitution $\xi \mapsto 2\pi\xi$.)

Definition 1.3 (Secant)**.** For $f : \mathbb{R} \to \mathbb{R}$ and $y \neq z$ put $S_f(y,z) = \dfrac{f(y) - f(z)}{y - z}$. If $f \in C^1$ we extend by $S_f(y,y) = f'(y)$; then $S_f(y,z) = \int_0^1 f'(z + s(y-z))\, ds$ for all y, z.

Definition 1.4 (Admissible profile class)**.** $\mathcal{A}$ denotes the set of $f \in C^1(\mathbb{R})$ such that $f' \in L^\infty(\mathbb{R})$ and f' is locally Lipschitz. Equivalently $f \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ with $f' \in L^\infty(\mathbb{R})$ (taking the C^1 representative). For $f \in \mathcal{A}$ and $R > 0$ write $\mathrm{Lip}_R(f')$ for the Lipschitz constant of f' on $[-R, R]$. Note that $\mathcal{A}$ contains functions with linear growth, e.g. any f with $f'(y) \to -1$ as $y \to -\infty$; no decay of f is required.

Definition 1.5 (The fractional multiplier $|D|^p$ and $|D|^p\partial_y$)**.** Let $0 < p < 1$, $\alpha = p + 1$.

1. For $g \in L^\infty(\mathbb{R})$ which is locally Lipschitz, define pointwise for every $y \in \mathbb{R}$

$$|D|^p g(y) := c_\alpha \int_{\mathbb{R}} \frac{g(y) - g(z)}{|y - z|^\alpha}\, dz.$$

   (The integral is absolutely convergent by Lemma 1.6; no principal value is needed because $\alpha < 2$.)

2. For $f \in \mathcal{A}$ define $|D|^p\partial_y f := |D|^p(f')$, i.e.

$$|D|^{\alpha-1}\partial_y f(y) = c_\alpha \int_{\mathbb{R}} \frac{f'(y) - f'(z)}{|y - z|^\alpha}\, dz.$$

   This is the meaning of $|D|^{\alpha-1}\partial_y f$ for non-decaying f (e.g. $f' \to -1$ at $-\infty$): only f' enters, and f' is bounded.

3. (Fourier side.) For $f \in \mathcal{S}(\mathbb{R})$, $|D|^p f = (|\xi|^p \widehat{f})^\vee$ and $|D|^p\partial_y f = (i\xi|\xi|^p \widehat{f})^\vee$; the two definitions agree on $\mathcal{S}(\mathbb{R})$ by Lemma 1.8.

Lemma 1.6 (Absolute convergence of the singular integral)**.** *Let $0 < p < 1$, $\alpha = p + 1$, and $g \in L^\infty(\mathbb{R})$ be Lipschitz on $[y - 1, y + 1]$ with constant Λ_y. Then $z \mapsto (g(y) - g(z))|y - z|^{-\alpha}$ is in $L^1(\mathbb{R})$ and*

$$\big||D|^p g(y)\big| \le c_\alpha\Big(\frac{2\Lambda_y}{2 - \alpha} + \frac{4\|g\|_{L^\infty}}{\alpha - 1}\Big).$$

Proof. Split $\mathbb{R} = \{|z - y| \le 1\} \cup \{|z - y| > 1\}$. On the first set $|g(y) - g(z)|\,|y - z|^{-\alpha} \le \Lambda_y |y - z|^{1-\alpha}$, and $\int_{|h|\le 1} |h|^{1-\alpha} dh = 2/(2 - \alpha)$ since $1 - \alpha > -1$. On the second set the integrand is bounded by $2\|g\|_\infty |y - z|^{-\alpha}$ and $\int_{|h|>1} |h|^{-\alpha} dh = 2/(\alpha - 1)$ since $\alpha > 1$. □

Lemma 1.7 (Affine invariance)**.** *If $f_1, f_2 \in \mathcal{A}$ and $f_1 - f_2$ is affine, then $|D|^p\partial_y f_1 = |D|^p\partial_y f_2$ pointwise. In particular $|D|^p\partial_y f = |D|^p\partial_y(f + y)$, so for f with $f'(y) \to -1$ as $y \to -\infty$ one may compute $|D|^p\partial_y f$ through the bounded function $f' + 1$.*

Proof. $f_1' - f_2'$ is constant, so $f_1'(y) - f_1'(z) = f_2'(y) - f_2'(z)$ for all y, z. □

Lemma 1.8 (Singular integral representation of $|\xi|^p$)**.** *Let $0 < p < 1$, $\alpha = p + 1$. For every $g \in \mathcal{S}(\mathbb{R})$ and $y \in \mathbb{R}$,*

$$(|\xi|^p \widehat{g})^\vee(y) = c_\alpha \int_{\mathbb{R}} \frac{g(y) - g(z)}{|y - z|^\alpha}\, dz, \qquad c_\alpha = \frac{\Gamma(1 + p)\sin(\pi p/2)}{\pi}.$$

Consequently, for $f \in \mathcal{S}(\mathbb{R})$, $|D|^p\partial_y f = (i\xi|\xi|^p \widehat{f})^\vee$, and the linearization at 0 of the gSQG nonlinearity is $|D|^{\alpha-1}\partial_x$ (see Lemma 1.15).

Proof. Write $z = y + h$ and symmetrize ($h \mapsto -h$, allowed since the integral converges absolutely by Lemma 1.6):

$$c_\alpha \int_{\mathbb{R}} \frac{g(y) - g(y+h)}{|h|^\alpha} dh = \frac{c_\alpha}{2} \int_{\mathbb{R}} \frac{2g(y) - g(y+h) - g(y-h)}{|h|^\alpha} dh.$$

Insert $g(x) = \frac{1}{2\pi} \int \widehat{g}(\xi) e^{ix\xi} d\xi$: $2g(y) - g(y+h) - g(y-h) = \frac{1}{2\pi} \int \widehat{g}(\xi) e^{iy\xi}\, 2(1 - \cos h\xi)\, d\xi$. Fubini is justified since $|\widehat{g}(\xi)|\,(1-\cos h\xi)|h|^{-\alpha} \le |\widehat{g}(\xi)| \min(h^2\xi^2, 2)|h|^{-\alpha}$, which is integrable on $\mathbb{R}^2$ ($\widehat{g}$ is Schwartz; $\int \min(h^2\xi^2, 2)|h|^{-\alpha} dh = C|\xi|^p$). Substituting $s = |h\xi|$,

$$\int_{\mathbb{R}} \frac{1 - \cos h\xi}{|h|^{1+p}} dh = 2|\xi|^p \int_0^\infty \frac{1 - \cos s}{s^{1+p}} ds = 2|\xi|^p \, \frac{\Gamma(1-p)\cos(\pi p/2)}{p}.$$

The last integral is classical (integrate by parts once to get $\frac{1}{p}\int_0^\infty s^{-p} \sin s\, ds = \frac{1}{p}\Gamma(1-p)\sin(\pi(1-p)/2)$). Hence the right side equals $c_\alpha \frac{2\Gamma(1-p)\cos(\pi p/2)}{p} (|\xi|^p \widehat{g})^\vee(y)$, and by $\Gamma(p)\Gamma(1-p) = \pi/\sin(\pi p)$ and $\sin(\pi p) = 2\sin(\pi p/2)\cos(\pi p/2)$,

$$\frac{p}{2\Gamma(1-p)\cos(\pi p/2)} = \frac{p\Gamma(p)\sin(\pi p)}{2\pi\cos(\pi p/2)} = \frac{\Gamma(1+p)\sin(\pi p/2)}{\pi} = c_\alpha.$$

For the second claim apply the first to $g = f'$ and use $\widehat{f'} = i\xi\widehat{f}$. □

Lemma 1.9 (Distributional meaning)**.** *Let $g \in L^\infty(\mathbb{R}) \cap W^{1,\infty}_{\mathrm{loc}}(\mathbb{R})$ with $|g'(y)| \le C\langle y\rangle^N$ a.e. for some $C, N \ge 0$. Then $|D|^p g$ (pointwise, Definition 1.5) is continuous, satisfies $\big||D|^p g(y)\big| \le C'\langle y\rangle^N$, and for every $\varphi \in \mathcal{S}(\mathbb{R})$*

$$\int_{\mathbb{R}} |D|^p g(y)\, \varphi(y)\, dy = \int_{\mathbb{R}} g(y)\, |D|^p \varphi(y)\, dy,$$

where $|D|^p\varphi = (|\xi|^p \widehat{\varphi})^\vee$ satisfies $|\,|D|^p\varphi(y)| \le C_\varphi \langle y\rangle^{-\alpha}$. In particular the pointwise $|D|^p g$ coincides with the distribution $\varphi \mapsto \langle g, |D|^p\varphi\rangle$, so a pointwise identity $\mathcal{E}_\epsilon(f) = 0$ on $\mathbb{R}$ (Definition 1.26) for $f \in \mathcal{A}$ with f'' of polynomial growth implies $\mathcal{E}_\epsilon(f) = 0$ in $\mathcal{D}'(\mathbb{R})$.

Proof. The pointwise bound follows from Lemma 1.6 with $\Lambda_y \le C\langle |y|+1\rangle^N \le C2^N\langle y\rangle^N$. Continuity: dominated convergence in the representation $c_\alpha \int (g(y) - g(y+h))|h|^{-\alpha} dh$, with dominating function $\Lambda|h|^{1-\alpha}\mathbf{1}_{|h|\le 1} + 2\|g\|_\infty |h|^{-\alpha}\mathbf{1}_{|h|>1}$ uniform for y in compacts. The decay of $|D|^p\varphi$: by Lemma 1.8, $|D|^p\varphi(y) = c_\alpha \int (\varphi(y) - \varphi(z))|y-z|^{-\alpha} dz$; for $|y| \ge 2$ split $|z-y| < |y|/2$ (use $|\varphi(y) - \varphi(z)| \le |y - z| \sup_{|w|\ge|y|/2} |\varphi'(w)|$, rapidly decaying) and $|z - y| \ge |y|/2$ (bounded by $\|\varphi\|_\infty \int_{|h|\ge|y|/2} |h|^{-\alpha} dh + (|y|/2)^{-\alpha}\|\varphi\|_{L^1}$). Duality: for $\delta > 0$ put $K_\delta(y,z) = c_\alpha |y-z|^{-\alpha}\mathbf{1}_{|y-z|>\delta}$. Then $\iint |g(y) - g(z)||\varphi(y)| K_\delta\, dz\, dy < \infty$ and likewise with $\varphi(z)$, so

$$\iint (g(y) - g(z))\varphi(y) K_\delta = \tfrac{1}{2} \iint (g(y) - g(z))(\varphi(y) - \varphi(z)) K_\delta = \iint g(y)(\varphi(y) - \varphi(z)) K_\delta$$

(swap $y \leftrightarrow z$, symmetry of K_δ). Let $\delta \to 0$: on the left the inner integral converges to $|D|^p g(y)$ and is dominated by $C'\langle y\rangle^N$, which is integrable against $|\varphi|$; on the right the inner integral converges to $|D|^p\varphi(y)$, dominated by $C_\varphi\langle y\rangle^{-\alpha}$, integrable against $|g| \le \|g\|_\infty$. For the last statement apply this with $g = f'$; the other terms of $\mathcal{E}_\epsilon(f)$ are continuous functions.

□

1.2 The gSQG front equation

Definition 1.10 (The coefficient A_ϵ)**.** For $\epsilon \geq 0$ and $u \in \mathbb{R}$ let $A_\epsilon(u) = (1 + \epsilon u^2)^{-\alpha/2}$ and $B_\epsilon(u) = A_\epsilon(u)^{-1} = (1+\epsilon u^2)^{\alpha/2}$. Then A_ϵ is smooth, $0 < A_\epsilon \leq 1$, $A_\epsilon(0) = 1$, $A_\epsilon'(0) = 0$, $A_0 \equiv 1$, and $|A_\epsilon'(u)| = \alpha\epsilon|u|(1+\epsilon u^2)^{-\alpha/2-1} \leq \alpha\epsilon|u|$.

Definition 1.11 (The nonlinearity $\mathcal{J}_\epsilon$)**.** For $\epsilon \geq 0$ and $f \in \mathcal{A}$ define, for every $y \in \mathbb{R}$,

$$\mathcal{J}_\epsilon(f)(y) = c_\alpha \int_{\mathbb{R}} \frac{f'(y) - f'(z)}{\big((y-z)^2 + \epsilon(f(y)-f(z))^2\big)^{\alpha/2}}\, dz = c_\alpha \int_{\mathbb{R}} \frac{f'(y)-f'(z)}{|y-z|^\alpha} A_\epsilon\big(S_f(y,z)\big)\, dz.$$

(The two integrands agree for $z \neq y$ since $(y-z)^2 + \epsilon(f(y)-f(z))^2 = |y-z|^2(1+\epsilon S_f(y,z)^2)$.) We write $\mathcal{J} := \mathcal{J}_1$. Note $\mathcal{J}_0(f) = |D|^p \partial_y f$.

Lemma 1.12 ($\mathcal{J}_\epsilon$ is well defined)**.** *For $\epsilon \geq 0$, $f \in \mathcal{A}$ and $y \in \mathbb{R}$, the integrand defining $\mathcal{J}_\epsilon(f)(y)$ is absolutely integrable and $|\mathcal{J}_\epsilon(f)(y)| \leq c_\alpha\big(\frac{2\Lambda_y}{2-\alpha} + \frac{4\|f'\|_\infty}{\alpha-1}\big)$, Λ_y the Lipschitz constant of f' on $[y-1, y+1]$.*

Proof. $0 < A_\epsilon \leq 1$, so the integrand is bounded in absolute value by $|f'(y) - f'(z)||y-z|^{-\alpha}$; apply Lemma 1.6 to $g = f'$. □

Definition 1.13 (Front equation; classical solution)**.** Let $I \subset \mathbb{R}$ be an open interval. A function $F : \mathbb{R} \times I \to \mathbb{R}$ is a *classical solution of the gSQG front equation*

$$\partial_t F(x,t) = c_\alpha \int_{\mathbb{R}} \frac{\partial_x F(x,t) - \partial_x F(x',t)}{\big((x-x')^2 + (F(x,t)-F(x',t))^2\big)^{\alpha/2}}\, dx' \tag{1.1}$$

on $\mathbb{R} \times I$ if

1. $F \in C^1(\mathbb{R} \times I)$, $F(\cdot,t) \in C^2(\mathbb{R})$ for every $t \in I$, and $\partial_x^2 F$ is continuous on $\mathbb{R} \times I$;
2. for every $t \in I$, $\partial_x F(\cdot,t) \in L^\infty(\mathbb{R})$ (hence $F(\cdot,t) \in \mathcal{A}$);
3. for every $(x,t) \in \mathbb{R} \times I$, $\partial_t F(x,t) = \mathcal{J}(F(\cdot,t))(x)$, the right side being the absolutely convergent integral of Lemma 1.12.

This is the graph (front) formulation of the gSQG patch problem with $\Omega(t) = \{x_2 < F(x_1,t)\}$; we take (1.1) as the definition and do not formalize its derivation from the gSQG system.

Lemma 1.14 (Agreement with the geometric front equation in the paper)**.** *Let $0 < p < 1$, $\epsilon \geq 0$, and $f \in \mathcal{A}$. For every $y \in \mathbb{R}$ the integral*

$$\mathcal{J}_\epsilon(f)(y) = c_\alpha \int_{\mathbb{R}} \frac{f'(y) - f'(z)}{((y-z)^2 + \epsilon(f(y)-f(z))^2)^{(p+1)/2}}\, dz$$

converges absolutely and equals the secant-factored operator used in Lean. Consequently every classical solution in Definition 1.13 satisfies the literal equation (1.1)*, with an absolutely convergent right-hand side.*

Proof. For $y \neq z$ factor the denominator as $|y-z|^2(1+\epsilon S_f(y,z)^2)$; positivity permits the real-power product rule. The two integrands also agree at $y=z$ under Lean's total division convention, since the numerator vanishes. Transfer absolute convergence from Lemma 1.12. For a classical solution, each spatial slice is C^2 with bounded first derivative, hence belongs to $\mathcal{A}$; apply the identity with $\epsilon = 1$. □

Lemma 1.15 (Linearization at the flat interface)**.** *Let $F \in \mathcal{A}$. Then $\mathcal{J}_0(F) = |D|^p \partial_x F$, and for every x, $\lim_{s\to 0} s^{-1}\mathcal{J}(sF)(x) = |D|^p \partial_x F(x)$. More generally $\mathcal{J}(sF) = s\,\mathcal{J}_{s^2}(F)$ for $s \in \mathbb{R}$.*

Proof. Directly from Definition 1.11: $(sF)' = sF'$ and $S_{sF} = sS_F$, and $A_1(sv) = A_{s^2}(v)$. The limit follows from dominated convergence, with the dominating function of Lemma 1.12, since $A_{s^2}(S_F(x,z)) \to 1$ pointwise. □

Lemma 1.16 (Symmetries of the front equation)**.** *Let F be a classical solution on $\mathbb{R}\times(a,b)$. Then the following are classical solutions:*

1. *$F(x-x_0, t-t_0) + h_0$ on $\mathbb{R}\times(a+t_0, b+t_0)$, for $x_0, t_0, h_0 \in \mathbb{R}$;*
2. *$F_c(x,t) = c^{-1}F(cx, c^\alpha t)$ on $\mathbb{R}\times(c^{-\alpha}a, c^{-\alpha}b)$, for $c>0$;*
3. *$\widetilde{F}(x,t) = F(-x,-t)$ on $\mathbb{R}\times(-b,-a)$.*

Proof. Regularity conditions (1)–(2) of Definition 1.13 are clearly preserved. (1): $\mathcal{J}$ only involves F' and differences of F, and is translation covariant (substitute $x' \mapsto x' - x_0$). (2): $\partial_t F_c = c^{\alpha-1}(\partial_t F)(cx, c^\alpha t)$. In $\mathcal{J}(F_c(\cdot,t))(x)$ we have $\partial_x F_c(x) = (\partial_x F)(cx)$, $(x-x')^2 + (F_c(x)-F_c(x'))^2 = c^{-2}\big((cx-cx')^2 + (F(cx)-F(cx'))^2\big)$; substituting $w = cx'$, $dx' = dw/c$ gives $\mathcal{J}(F_c(\cdot,t))(x) = c^{\alpha-1}\mathcal{J}(F(\cdot,c^\alpha t))(cx)$. (3): $\partial_t\widetilde{F}(x,t) = -(\partial_t F)(-x,-t)$ and $\partial_x\widetilde{F}(x,t) = -(\partial_x F)(-x,-t)$; substituting $x' \mapsto -x'$ gives $\mathcal{J}(\widetilde{F}(\cdot,t))(x) = -\mathcal{J}(F(\cdot,-t))(-x)$.
□

1.3 The self-similar ansatz and the profile equation

Definition 1.17 (Self-similar ansatz)**.** For $T \in \mathbb{R}$ let $L_T(t) = (\alpha(T-t))^{1/\alpha}$ for $t < T$. For $\epsilon > 0$ and $f : \mathbb{R} \to \mathbb{R}$ define $F = F_{\epsilon,T,f} : \mathbb{R}\times(-\infty,T) \to \mathbb{R}$ by

$$F(x,t) = \sqrt{\epsilon}\, L_T(t)\, f\Big(\frac{x}{L_T(t)}\Big). \tag{1.2}$$

We call $y = x/L_T(t)$ the self-similar variable and f the profile. The self-similar time is $\tau = -\log L_T(t)$.

Lemma 1.18 (Properties of L)**.** *$L = L_T \in C^\infty((-\infty,T))$, $L>0$, $L(t)\to 0$ as $t \uparrow T$, and $L'(t) = -L(t)^{1-\alpha}$.*

Proof. $L' = \frac{1}{\alpha}(\alpha(T-t))^{1/\alpha-1}\cdot(-\alpha) = -(\alpha(T-t))^{(1-\alpha)/\alpha} = -L^{1-\alpha}$. □

Lemma 1.19 (Time derivative of the ansatz)**.** *Let $\epsilon > 0$, $T \in \mathbb{R}$, $f \in C^1(\mathbb{R})$ and $F = F_{\epsilon,T,f}$. Then $F \in C^1(\mathbb{R} \times (-\infty, T))$ and, with $y = x/L(t)$,*

$$\partial_x F(x,t) = \sqrt{\epsilon} f'(y), \qquad \partial_t F(x,t) = -\sqrt{\epsilon}\, L(t)^{1-\alpha}\big(f(y) - y f'(y)\big).$$

If $f \in C^k(\mathbb{R})$ then $\partial_x^j F(x,t) = \sqrt{\epsilon} L(t)^{1-j} f^{(j)}(y)$ for $j \le k$, and $F \in C^k(\mathbb{R} \times (-\infty, T))$.

Proof. Chain rule: $\partial_t\big[Lf(x/L)\big] = L' f(y) + Lf'(y) \cdot (-xL'/L^2) = L'(f(y) - yf'(y))$; use Lemma 1.18. Joint continuity of the derivatives follows from continuity of $f^{(j)}$, L, and $(x,t) \mapsto x/L(t)$. □

Lemma 1.20 (Scaling of $\mathcal{J}$)**.** *Let $\epsilon > 0$, $\lambda > 0$, $f \in \mathcal{A}$ and $G(x) = \sqrt{\epsilon}\,\lambda f(x/\lambda)$. Then $G \in \mathcal{A}$ and for every $x \in \mathbb{R}$*

$$\mathcal{J}(G)(x) = \sqrt{\epsilon}\,\lambda^{1-\alpha}\, \mathcal{J}_\epsilon(f)(x/\lambda).$$

Proof. $G'(x) = \sqrt{\epsilon} f'(x/\lambda)$, bounded and locally Lipschitz. Put $y = x/\lambda$ and substitute $x' = \lambda z$, $dx' = \lambda dz$: $(x-x')^2 + (G(x) - G(x'))^2 = \lambda^2\big((y-z)^2 + \epsilon(f(y) - f(z))^2\big)$, so the integrand becomes $\sqrt{\epsilon}(f'(y) - f'(z))\lambda^{-\alpha}(\dots)^{-\alpha/2} \cdot \lambda$. The change of variables is legitimate for absolutely integrable functions (Lemma 1.12). □

Lemma 1.21 (Self-similar evolution)**.** *Let $\epsilon > 0$, $T \in \mathbb{R}$, and $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, $(y,\tau) \mapsto f(y,\tau)$, with $f, \partial_y f, \partial_y^2 f, \partial_\tau f$ continuous and $\partial_y f(\cdot,\tau)$ bounded for each τ. Let $F(x,t) = \sqrt{\epsilon} L(t) f(x/L(t), -\log L(t))$, $L = L_T$. Then F is a classical solution on $\mathbb{R} \times (-\infty, T)$ iff for all (y,τ)*

$$\partial_\tau f = f - y\partial_y f + \mathcal{J}_\epsilon(f(\cdot,\tau)). \tag{1.3}$$

Proof. With $\tau(t) = -\log L(t)$ we have $\tau'(t) = -L'/L = L^{-\alpha}$ (Lemma 1.18). The chain rule gives $\partial_t F = \sqrt{\epsilon}\big[L'(f - y\partial_y f) + L^{1-\alpha}\partial_\tau f\big] = \sqrt{\epsilon} L^{1-\alpha}\big[\partial_\tau f - f + y\partial_y f\big]$, and Lemma 1.20 with $\lambda = L(t)$ gives $\mathcal{J}(F(\cdot,t))(x) = \sqrt{\epsilon} L^{1-\alpha} \mathcal{J}_\epsilon(f(\cdot,\tau))(y)$. Divide by $\sqrt{\epsilon} L^{1-\alpha} > 0$. Since $t \mapsto (\tau(t))$ is a bijection $(-\infty, T) \to \mathbb{R}$ and $(x,t) \mapsto (x/L, \tau)$ is a C^∞ diffeomorphism of $\mathbb{R} \times (-\infty, T)$ onto $\mathbb{R}^2$, the identities are equivalent. Regularity: $\partial_x F = \sqrt{\epsilon}\partial_y f$ and $\partial_x^2 F = \sqrt{\epsilon} L^{-1} \partial_y^2 f$ are continuous, $\partial_x F(\cdot, t)$ bounded. □

Definition 1.22 (The lower-order nonlinearity $\widetilde{\mathcal{J}}_\epsilon$)**.** For $\epsilon \ge 0$, $f \in \mathcal{A}$ and $y \in \mathbb{R}$,

$$\widetilde{\mathcal{J}}_\epsilon(f)(y) = c_\alpha \int_{\mathbb{R}} \frac{f'(y) - f'(z)}{|y-z|^\alpha} \Big(A_\epsilon\big(S_f(y,z)\big) - A_\epsilon\big(f'(y)\big) \Big)\, dz.$$

Definition 1.23 (The cubic remainder $\mathcal{Q}_\epsilon$)**.** For $\epsilon \ge 0$, $f \in \mathcal{A}$ and $y \in \mathbb{R}$,

$$\mathcal{Q}_\epsilon(f)(y) = c_\alpha \int_{\mathbb{R}} \frac{f'(y) - f'(z)}{|y-z|^\alpha} \Big(A_\epsilon\big(S_f(y,z)\big) - 1 \Big)\, dz.$$

Lemma 1.24 (Decompositions of $\mathcal{J}_\epsilon$)**.** *For $\epsilon \ge 0$ and $f \in \mathcal{A}$ the integrals defining $\widetilde{\mathcal{J}}_\epsilon(f)(y)$ and $\mathcal{Q}_\epsilon(f)(y)$ converge absolutely for every y, and*

$$\mathcal{J}_\epsilon(f) = |D|^p \partial_y f + \mathcal{Q}_\epsilon(f) = A_\epsilon(f')\, |D|^p \partial_y f + \widetilde{\mathcal{J}}_\epsilon(f) \qquad \text{pointwise on } \mathbb{R}.$$

Proof. Since $|A_\epsilon(S)-1|\le 1$ and $|A_\epsilon(S)-A_\epsilon(f'(y))|\le 1$, both integrands are dominated by $|f'(y)-f'(z)||y-z|^{-\alpha}\in L^1(dz)$ (Lemma 1.6). Write $A_\epsilon(S_f)=1+(A_\epsilon(S_f)-1)=A_\epsilon(f'(y))+(A_\epsilon(S_f)-A_\epsilon(f'(y)))$ in the second form of Definition 1.11 and split the absolutely convergent integral by linearity; $A_\epsilon(f'(y))$ does not depend on z.

□

Lemma 1.25 ($\widetilde{\mathcal{J}}_\epsilon$ is of lower order)**.** *Let $\epsilon\ge 0$, $f\in\mathcal{A}$, $y\in\mathbb{R}$, and Λ the Lipschitz constant of f' on $[y-1,y+1]$. Then for $|z-y|\le 1$ the integrand of $\widetilde{\mathcal{J}}_\epsilon(f)(y)$ is bounded by $c_\alpha\,\alpha\epsilon\|f'\|_\infty\Lambda^2|y-z|^{2-\alpha}/2$.*

Proof. $|f'(y)-f'(z)|\le\Lambda|y-z|$. By the mean value theorem and Definition 1.10, $|A_\epsilon(S_f(y,z))-A_\epsilon(f'(y))|\le\alpha\epsilon\|f'\|_\infty|S_f(y,z)-f'(y)|$ (note $|S_f|\le\|f'\|_\infty$), and $|S_f(y,z)-f'(y)|=|\int_0^1(f'(z+s(y-z))-f'(y))ds|\le\Lambda|y-z|/2$.

□

Definition 1.26 (The semilinear residual $\mathcal{E}_\epsilon$)**.** For $\epsilon\ge 0$ and $f\in\mathcal{A}$ define, pointwise for $y\in\mathbb{R}$,

$$\mathcal{E}_\epsilon(f)(y)=|D|^{\alpha-1}\partial_y f(y)+\frac{1}{A_\epsilon(f'(y))}\Big(f(y)-yf'(y)+\widetilde{\mathcal{J}}_\epsilon(f)(y)\Big).$$

Definition 1.27 (Quasilinear residual)**.** For $\epsilon\ge 0$ and $f\in\mathcal{A}$ let $\widetilde{\mathcal{E}}_\epsilon(f)=|D|^{\alpha-1}\partial_y f+f-yf'+\mathcal{Q}_\epsilon(f)$, i.e. $\widetilde{\mathcal{E}}_\epsilon(f)=\mathcal{L}_0(f)+\mathcal{Q}_\epsilon(f)$ with $\mathcal{L}_0(f)=|D|^{\alpha-1}f'+f-yf'$.

Lemma 1.28 (Residual identities)**.** *For $\epsilon\ge 0$ and $f\in\mathcal{A}$, pointwise on $\mathbb{R}$:*

$$\widetilde{\mathcal{E}}_\epsilon(f)=f-yf'+\mathcal{J}_\epsilon(f)=A_\epsilon(f')\,\mathcal{E}_\epsilon(f).$$

In particular, since $A_\epsilon>0$, for each y: $\mathcal{E}_\epsilon(f)(y)=0\iff f(y)-yf'(y)+\mathcal{J}_\epsilon(f)(y)=0$.

Proof. First equality: Lemma 1.24, first form. Second: multiply Definition 1.26 by $A_\epsilon(f'(y))$ and use the second form of Lemma 1.24. □

Definition 1.29 (Profile equation)**.** Let $\epsilon\ge 0$. A function $f\in\mathcal{A}$ solves the self-similar profile equation if

$$|D|^{\alpha-1}\partial_y f+\frac{1}{A_\epsilon(\partial_y f)}\Big(f-y\partial_y f+\widetilde{\mathcal{J}}_\epsilon(f)\Big)=0\quad\text{i.e. }\ \mathcal{E}_\epsilon(f)(y)=0\ \forall y\in\mathbb{R},\ (1.4)$$

together with the boundary conditions $\lim_{y\to-\infty}f'(y)=-1$ and $\lim_{y\to+\infty}f(y)=0$. (The boundary conditions are not used in the reduction Lemma 1.30; they encode that F approaches the corner $\sqrt{\epsilon}\,x_-$.)

Lemma 1.30 (Self-similar reduction)**.** *Let $\epsilon>0$, $T\in\mathbb{R}$ and $f\in C^2(\mathbb{R})$ with $f'\in L^\infty(\mathbb{R})$ (so $f\in\mathcal{A}$). Then $F=F_{\epsilon,T,f}$ is a classical solution of (1.1) on $\mathbb{R}\times(-\infty,T)$ if and only if $\mathcal{E}_\epsilon(f)(y)=0$ for all $y\in\mathbb{R}$.*

Proof. Regularity: by Lemma 1.19 with $k=2$, $F\in C^2(\mathbb{R}\times(-\infty,T))$, in particular $F,\partial_x F\in C^1$, and $\|\partial_x F(\cdot,t)\|_\infty=\sqrt{\epsilon}\|f'\|_\infty<\infty$. Fix $t<T$, $x\in$

$\mathbb{R}$, $y = x/L(t)$. By Lemma 1.19, $\partial_t F(x,t) = -\sqrt{\epsilon}L^{1-\alpha}(f(y) - yf'(y))$; by Lemma 1.20 with $\lambda = L(t)$, $\mathcal{J}(F(\cdot,t))(x) = \sqrt{\epsilon}L^{1-\alpha}\mathcal{J}_\epsilon(f)(y)$. Hence

$$\partial_t F(x,t) - \mathcal{J}(F(\cdot,t))(x) = -\sqrt{\epsilon}L(t)^{1-\alpha}\big(f(y) - yf'(y) + \mathcal{J}_\epsilon(f)(y)\big) = -\sqrt{\epsilon}L(t)^{1-\alpha}A_\epsilon(f'(y))\,\mathcal{E}_\epsilon(f)(y)$$

by Lemma 1.28. The prefactor is nonzero. As (x,t) ranges over $\mathbb{R} \times (-\infty, T)$ (with any fixed t), $y = x/L(t)$ ranges over all of $\mathbb{R}$, so the equation holds for all (x,t) iff $\mathcal{E}_\epsilon(f) \equiv 0$. □

1.4 The main theorem

Theorem 1.31 (Self-similar corner formation)**.** *Let $\alpha \in (\frac{7}{4}, 2)$, $p = \alpha - 1$, $\beta = \frac{\alpha}{2(\alpha-1)}$. There exist $\epsilon_0 > 0$ and $C > 0$ (depending only on α) such that for every $0 < \epsilon < \epsilon_0$ there is a profile $f_\epsilon \in C^\infty(\mathbb{R})$ with the following properties.*

1. *(Profile bounds.) For $y \le 0$ and $0 \le j \le 2$: $|\partial^j(f_\epsilon(y) + y)| \le C\langle y\rangle^{-p-j}$. For $y \ge 0$: $|f_\epsilon(y)| \le C\langle y\rangle^{-p}$ and $|\partial^{j+1} f_\epsilon(y)| \le C\langle y\rangle^{-\beta+j/p}$ for $j \in \{0,1\}$.*
2. *(Solution.) For every $T > 0$, $F(x,t) = \sqrt{\epsilon}\,L(t)f_\epsilon(x/L(t))$, $L(t) = (\alpha(T-t))^{1/\alpha}$, is a classical solution of (1.1) on $\mathbb{R} \times (-\infty, T)$ (Definition 1.13), and $F \in C^\infty(\mathbb{R} \times (-\infty, T))$.*
3. *(Corner convergence.) For all $T > 0$ and $t < T$,*
$$\|F(\cdot,t) - \sqrt{\epsilon}\,x_-\|_{L^\infty(\mathbb{R})} \le C\sqrt{\epsilon}\,(T-t)^{1/\alpha}. \tag{1.5}$$
4. *(Curvature blow-up.) There is $c = c(\epsilon) > 0$ (independent of T and δ) such that for every $T > 0$ and every $\delta > 0$ there is $t_\delta < T$ with*
$$\sup_{x \in (-\delta,\delta)} |\partial_x^2 F(x,t)| \ge c\,(T-t)^{-1/\alpha}, \qquad t_\delta < t < T. \tag{1.6}$$

Proof. Let ϵ_0 be the minimum of the thresholds of Lemma 1.32 (inherited from Theorem 2.25) and 1, and for $0 < \epsilon < \epsilon_0$ let $f_\epsilon = \overline{f}_{\epsilon,\mu_\epsilon} + r_\epsilon$ be the profile given there; it lies in $\mathcal{A}$ and $\mathcal{E}_\epsilon(f_\epsilon) \equiv 0$. By Lemma 1.33, $f_\epsilon \in C^\infty(\mathbb{R})$. (1) is Lemma 1.34. (2): by (1) f_ϵ' is bounded, so Lemma 1.30 applies; smoothness of F follows from Lemma 1.19 for all k. (3) is Lemma 1.36 and (4) is Lemma 1.38, both applied with the bounds (1). □

Lemma 1.32 (The profile solves the profile equation)**.** *Fix $\alpha \in (\frac{7}{4}, 2)$ and the parameters $p, \beta, \gamma_1, \gamma_2$ of Definition 2.1. There exists $\epsilon_1 > 0$ such that for every $0 < \epsilon < \epsilon_1$, with μ_ϵ, $r_\epsilon \in X_{\gamma_2}$ given by Theorem 2.25 and $\overline{f}_{\epsilon,\mu_\epsilon}$ by Theorem 2.20, the function $f_\epsilon = \overline{f}_{\epsilon,\mu_\epsilon} + r_\epsilon$ belongs to $\mathcal{A}$, satisfies $\mathcal{E}_\epsilon(f_\epsilon)(y) = 0$ for every $y \in \mathbb{R}$, and $|f_\epsilon''(y)| \le C\langle y\rangle^{1/p}$ for a.e. y, with C independent of ϵ. Moreover $\lim_{y\to-\infty} f_\epsilon'(y) = -1$ and $\lim_{y\to+\infty} f_\epsilon(y) = 0$, so f_ϵ solves the profile equation (Definition 1.29).*

Proof. Take ϵ_1 from Theorem 2.25. Membership in $\mathcal{A}$: $\overline{f} = \overline{f}_{\epsilon,\mu_\epsilon}$ is C^∞ with $|\overline{f}'| \le C$ on $\mathbb{R}$ (item 2 of Theorem 2.20: $|\overline{f}'| \lesssim \langle y\rangle^{-\beta}$ for $y \ge 0$ and $|\overline{f}' + 1| \lesssim \langle y\rangle^{-p-1}$ for $y \le 0$), and $r = r_\epsilon \in X_{\gamma_2} \subset W^{2,\infty}_{\mathrm{loc}}$ with $|r'| \le \|r\|_{X_{\gamma_2}}\langle y\rangle^{-\gamma_2+1/p}$, $|r''| \le \|r\|_{X_{\gamma_2}}\langle y\rangle^{-\gamma_2+2/p}$ a.e.; since $\gamma_2 > \beta + 1/p > 1/p$ these are bounded by

$\|r\|_{X_{\gamma_2}}\langle y\rangle^{1/p}$, while $\overline{f}''$ has the stated polynomial growth and is bounded on every compact interval. So f' is bounded and locally Lipschitz. The equation: by Definition 2.18, $\mathcal{N}_{\epsilon,\overline{f}}(r) = \mathcal{E}_\epsilon(\overline{f}+r) - \mathcal{E}_\epsilon(\overline{f}) - \mathcal{L}_\epsilon(r)$ as functions on $\mathbb{R}$; inserting the correction equation $\mathcal{L}_\epsilon(r) = -\mathcal{E}_\epsilon(\overline{f}) - \mathcal{N}_{\epsilon,\overline{f}}(r)$ gives $\mathcal{E}_\epsilon(\overline{f}+r) = 0$ pointwise. Limits: $|r(y)| \le \|r\|\langle y\rangle^{-\gamma_2} \to 0$, $|r'(y)| \le \|r\|\langle y\rangle^{-\gamma_2-1} \to 0$ as $y \to -\infty$, combined with item 1–2 of Theorem 2.20.

□

Lemma 1.33 (Smoothness of the profile)**.** *The profile f_ϵ of Lemma 1.32 is in $C^\infty(\mathbb{R})$; hence also $r_\epsilon = f_\epsilon - \overline{f}_{\epsilon,\mu_\epsilon} \in C^\infty(\mathbb{R})$ and the a.e. bounds on r_ϵ'' coming from X_{γ_2} hold everywhere.*

Proof. $f_\epsilon \in W^{2,\infty}_{\mathrm{loc}}$, $f_\epsilon' \in L^\infty$ and $|f_\epsilon''| \le C\langle y\rangle^{1/p}$, so by Lemma 1.9 the pointwise identity $\mathcal{E}_\epsilon(f_\epsilon) \equiv 0$ implies $\mathcal{E}_\epsilon(f_\epsilon) = 0$ in $\mathcal{D}'(\mathbb{R})$. Apply Theorem 3.36. The bounds on r_ϵ'' hold a.e. and r_ϵ'' is continuous, hence they hold everywhere. □

Lemma 1.34 (Profile bounds)**.** *There exists $C > 0$, independent of $\epsilon \in (0, \epsilon_1)$, such that the profile f_ϵ of Lemma 1.32 satisfies the bounds of item (1) of Theorem 1.31.*

Proof. Let $\overline{f} = \overline{f}_{\epsilon,\mu_\epsilon}$, $r = r_\epsilon$, $\rho = \|r\|_{X_{\gamma_2}} \le C_0\epsilon \le C_0$ (Theorem 2.25). Recall $\beta + \frac{1}{p} < \gamma_2 < 2p+1$, so in particular $\gamma_2 > p$ (because $\beta + 1/p > 1/p > 1 > p$) and $\gamma_2 - \frac{1}{p} > \beta$. For $y \le 0$ and $j \le 2$: $|\partial^j(\overline{f}+y)| \le C_j\langle y\rangle^{-p-j}$ by Theorem 2.20, and $|\partial^j r| \le \rho\langle y\rangle^{-\gamma_2-j} \le \rho\langle y\rangle^{-p-j}$. For $y \ge 0$: $|\overline{f}| \le C\langle y\rangle^{-p}$ and $|r| \le \rho\langle y\rangle^{-\gamma_2} \le \rho\langle y\rangle^{-p}$; $|\overline{f}'| \le C\langle y\rangle^{-\beta}$ and $|r'| \le \rho\langle y\rangle^{-\gamma_2+1/p} \le \rho\langle y\rangle^{-\beta}$; $|\overline{f}''| \le C\langle y\rangle^{-\beta+1/p}$ and $|r''| \le \rho\langle y\rangle^{-\gamma_2+2/p} \le \rho\langle y\rangle^{-\beta+1/p}$. Add; all constants are uniform in ϵ and $\mu \in \overline{B}$ by Theorem 2.20. (Pointwise validity of the r'' bound: Lemma 1.33.) □

Lemma 1.35 (Bounded deviation from the corner)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ satisfy, for some $C > 0$ and $p > 0$, $|f(y) + y| \le C\langle y\rangle^{-p}$ for $y \le 0$ and $|f(y)| \le C\langle y\rangle^{-p}$ for $y \ge 0$. Then $\sup_{y\in\mathbb{R}} |f(y) - W(y)| \le C$, where $W(y) = \max\{-y, 0\}$.*

Proof. $W(y) = -y$ for $y \le 0$ and $W(y) = 0$ for $y \ge 0$; and $\langle y\rangle^{-p} \le 1$. □

Lemma 1.36 (Corner convergence)**.** *Let $\epsilon > 0$, $T \in \mathbb{R}$, and $f : \mathbb{R} \to \mathbb{R}$ with $M := \sup_y |f(y) - W(y)| < \infty$. Then for all $t < T$,*

$$\sup_{x\in\mathbb{R}} \big|F_{\epsilon,T,f}(x,t) - \sqrt{\epsilon}\, x_-\big| = \sqrt{\epsilon}\,\alpha^{1/\alpha} M\,(T-t)^{1/\alpha}.$$

In particular, under the bounds of Theorem 1.31(1), (1.5) *holds with constant $\alpha^{1/\alpha}C \le 2C$, and $F(\cdot,t) \to \sqrt{\epsilon}\,x_-$ uniformly as $t \uparrow T$.*

Proof. W is positively homogeneous of degree one: $\lambda W(y) = W(\lambda y)$ for $\lambda > 0$. Hence with $L = L_T(t)$ and $y = x/L$, $\sqrt{\epsilon}x_- = \sqrt{\epsilon}L\,W(y)$, so $F(x,t) - \sqrt{\epsilon}x_- = \sqrt{\epsilon}L\,(f(y) - W(y))$. Since $x \mapsto x/L$ is a bijection of $\mathbb{R}$, the supremum over x equals $\sqrt{\epsilon}LM$, and $L = \alpha^{1/\alpha}(T-t)^{1/\alpha}$. Finally $\alpha^{1/\alpha} \le 2$ for $\alpha \in (1,2)$, and use Lemma 1.35.

□

Lemma 1.37 (The profile is not affine)**.** *Let $f \in C^2(\mathbb{R})$ with $\lim_{y\to-\infty} f'(y) = -1$ and f bounded on $[0,\infty)$. Then there exists $y_0 \in \mathbb{R}$ with $f''(y_0) \ne 0$.*

Proof. If $f'' \equiv 0$ then f' is constant, and the limit forces $f' \equiv -1$, so $f(y) = b - y$ for some b, which is unbounded on $[0, \infty)$; contradiction. □

Lemma 1.38 (Curvature blow-up)**.** *Let $\epsilon > 0$ and $f \in C^2(\mathbb{R})$ with $f''(y_0) \neq 0$ for some $y_0 \in \mathbb{R}$. Put $c = \sqrt{\epsilon}\,\alpha^{-1/\alpha}|f''(y_0)| > 0$. Then for every $T \in \mathbb{R}$ and $\delta > 0$, with*

$$t_\delta = \begin{cases} T - \frac{1}{\alpha}\big(\delta/|y_0|\big)^\alpha, & y_0 \neq 0, \\ T - 1, & y_0 = 0, \end{cases}$$

we have $\sup_{|x|<\delta} |\partial_x^2 F_{\epsilon,T,f}(x,t)| \geq c\,(T-t)^{-1/\alpha}$ for all $t_\delta < t < T$. Moreover, if f' is bounded, the geometric curvature $\kappa(x,t) = \partial_x^2 F/(1+(\partial_x F)^2)^{3/2}$ satisfies $\sup_{|x|<\delta} |\kappa(x,t)| \geq c(1+\epsilon\|f'\|_\infty^2)^{-3/2}(T-t)^{-1/\alpha}$ for $t_\delta < t < T$.

Proof. By Lemma 1.19, $\partial_x^2 F(x,t) = \sqrt{\epsilon} L(t)^{-1} f''(x/L(t))$. For $t_\delta < t < T$ we have $L(t) < \delta/|y_0|$ (if $y_0 \neq 0$), i.e. $x_0 := L(t)y_0$ satisfies $|x_0| < \delta$; if $y_0 = 0$ take $x_0 = 0$. Then $|\partial_x^2 F(x_0,t)| = \sqrt{\epsilon} L(t)^{-1}|f''(y_0)| = c(T-t)^{-1/\alpha}$ since $L(t)^{-1} = \alpha^{-1/\alpha}(T-t)^{-1/\alpha}$. For κ use $|\partial_x F| = \sqrt{\epsilon}|f'(x/L)| \leq \sqrt{\epsilon}\|f'\|_\infty$. □

Corollary 1.39 (Desingularization of a corner)**.** *Let $\alpha \in (\frac{7}{4}, 2)$, $0 < \epsilon < \epsilon_0$, and F the solution of Theorem 1.31 with $T = 0$. Then $\widetilde{F}(x,t) = F(-x,-t)$ is a classical solution of (1.1) on $\mathbb{R} \times (0,\infty)$, $\widetilde{F} \in C^\infty$, and $\|\widetilde{F}(\cdot,t) - \sqrt{\epsilon}\,x_+\|_{L^\infty} \leq C\sqrt{\epsilon}\,t^{1/\alpha} \to 0$ as $t \downarrow 0$, where $x_+ = \max\{x, 0\}$.*

Proof. Lemma 1.16(3) with $(a,b) = (-\infty, 0)$, and $(-x)_- = x_+$ in (1.5). □

Remark 1.40 (On the restriction $\alpha > 7/4$). With $p = \alpha - 1$, the parameter interval $\beta + \frac{1}{p} < \gamma < 2p + 1$ of Definition 2.1, i.e. $\frac{\alpha+2}{2(\alpha-1)} < \gamma < 2\alpha - 1$, is nonempty iff $p + 3 < 4p^2 + 2p$, i.e. $(4p-3)(p+1) > 0$, i.e. $\alpha > \frac{7}{4}$.

Chapter 2

The fixed point argument: Theorems 3.1–3.5 (paper §3)

2.1 Parameters and function spaces

Definition 2.1 (Fixed parameters and cutoff)**.** Fix once and for all real numbers $p, \alpha, \beta, \gamma_1, \gamma_2$ with

$$\frac{3}{4} < p < 1, \qquad \alpha = p + 1, \qquad \beta = \frac{p+1}{2p}, \qquad \beta + \frac{1}{p} < \gamma_1 < \gamma_2 < 2p + 1.$$

(Thus $\alpha \in (\frac{7}{4}, 2)$.) We call γ *admissible* if $\beta + \frac{1}{p} < \gamma < 2p+1$; both γ_1 and γ_2 are admissible. Fix also a cutoff $\chi_+ \in C^\infty(\mathbb{R}; [0,1])$, nondecreasing, with $\chi_+(\xi) = 0$ for $\xi \le 1$ and $\chi_+(\xi) = 1$ for $\xi \ge 2$; in particular $\chi_+' \in C_c^\infty(\mathbb{R})$, $\operatorname{supp} \chi_+' \subset [1,2]$ and $\int_0^\infty \chi_+'(\xi)\, d\xi = 1$. (The same function χ_+ is used as a cutoff in the variable y elsewhere.) Finally let $\Phi_0(\xi) := \frac{1}{\alpha}\, \xi|\xi|^p$ for $\xi \in \mathbb{R}$.

Lemma 2.2 (The admissible range is nonempty)**.** *For every $p \in (\frac{3}{4}, 1)$ one has $\beta + \frac{1}{p} < 2p + 1$ with $\beta = \frac{p+1}{2p}$; hence there exist $\gamma_1 < \gamma_2$ as in Definition 2.1. Moreover $\gamma - \frac{1}{p} > \beta > 1$ for every admissible γ.*

Proof. $\beta + \frac{1}{p} = \frac{p+3}{2p}$, and $\frac{p+3}{2p} < 2p+1 \iff p+3 < 4p^2+2p \iff (4p-3)(p+1) > 0$, which holds since $p > \frac{3}{4}$. Also $\beta = \frac{1}{2} + \frac{1}{2p} > 1$ since $p < 1$. □

Definition 2.3 (Correction space X_γ)**.** Let $\gamma \in \mathbb{R}$. For $r \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R}; \mathbb{R})$ (equivalently: $r \in C^1(\mathbb{R})$ with r' locally Lipschitz; then r'' exists a.e. and is in L^∞_{loc}) set

$$\|r\|_{X_\gamma} := \operatorname*{ess\,sup}_{y \ge 0} \langle y\rangle^\gamma \big(|r(y)| + \langle y\rangle^{-1/p}|r'(y)| + \langle y\rangle^{-2/p}|r''(y)|\big) + \operatorname*{ess\,sup}_{y \le 0} \langle y\rangle^\gamma \big(|r(y)| + \langle y\rangle|r'(y)| + \langle y\rangle^2|r''(y)|\big) \in [0, \infty]$$

and $X_\gamma := \{r \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R}; \mathbb{R}) : \|r\|_{X_\gamma} < \infty\}$, a real normed space.

Definition 2.4 (Forcing space Y_γ)**.** Let $\gamma \in \mathbb{R}$. For $G \in W^{1,\infty}_{\mathrm{loc}}(\mathbb{R};\mathbb{R})$ (i.e. G locally Lipschitz, G' defined a.e.) set

$$\|G\|_{Y_\gamma} := \operatorname*{ess\,sup}_{y\geq 0}\langle y\rangle^{\gamma}\big(|G(y)|+\langle y\rangle^{-1/p}|G'(y)|\big)+\operatorname*{ess\,sup}_{y\leq 0}\langle y\rangle^{\gamma}\big(|G(y)|+\langle y\rangle|G'(y)|\big) \in [0,\infty],$$

and $Y_\gamma := \{G \in W^{1,\infty}_{\mathrm{loc}}(\mathbb{R};\mathbb{R}) : \|G\|_{Y_\gamma} < \infty\}$, a real normed space.

Lemma 2.5 (X_γ is a Banach space)**.** *For every $\gamma \in \mathbb{R}$, $(X_\gamma, \|\cdot\|_{X_\gamma})$ is a real Banach space.*

Proof. Norm axioms are immediate (definiteness: $\|r\|_{X_\gamma} = 0$ forces $r = 0$ a.e., hence everywhere by continuity). Put weights $w_0 = \langle y\rangle^{\gamma}$, and $w_1 = \langle y\rangle^{\gamma-1/p}$, $w_2 = \langle y\rangle^{\gamma-2/p}$ on $y \geq 0$, $w_1 = \langle y\rangle^{\gamma+1}$, $w_2 = \langle y\rangle^{\gamma+2}$ on $y < 0$. Then $\frac{1}{2}\max_k \|w_k r^{(k)}\|_{L^\infty} \leq \|r\|_{X_\gamma} \leq 6\max_k \|w_k r^{(k)}\|_{L^\infty}$, so it suffices to show completeness for the equivalent norm $\max_k \|w_k r^{(k)}\|_{L^\infty}$. Let (r_n) be Cauchy. Then $w_k r_n^{(k)} \to h_k$ in $L^\infty(\mathbb{R})$ for $k = 0, 1, 2$. On each compact $[a, b]$ the weights are bounded above and below by positive constants, so $r_n \to g_0 := h_0/w_0$ and $r_n' \to g_1 := h_1/w_1$ uniformly on $[a, b]$ (these are uniform limits of continuous functions, so g_0, g_1 are continuous after modification on a null set; in fact the convergence is uniform on compacts for every point since r_n, r_n' are continuous and Cauchy in sup on compacts), and $r_n'' \to g_2 := h_2/w_2$ in $L^\infty([a, b])$. Passing to the limit in $r_n(y) - r_n(a) = \int_a^y r_n'$ and $r_n'(y) - r_n'(a) = \int_a^y r_n''$ (FTC for C^1 functions, resp. for Lipschitz functions) gives $g_0 \in C^1$ with $g_0' = g_1$, and $g_1(y) - g_1(a) = \int_a^y g_2$; hence g_1 is locally Lipschitz with $g_1' = g_2$ a.e. Thus $r := g_0 \in W^{2,\infty}_{\mathrm{loc}}$, $w_k r^{(k)} = h_k$ a.e., so $r \in X_\gamma$ and $r_n \to r$ in X_γ. □

Lemma 2.6 (Y_γ is a Banach space)**.** *For every $\gamma \in \mathbb{R}$, $(Y_\gamma, \|\cdot\|_{Y_\gamma})$ is a real Banach space.*

Proof. Identical to Lemma 2.5 with only the two weights w_0, w_1 ($w_1 = \langle y\rangle^{\gamma-1/p}$ on $y \geq 0$, $\langle y\rangle^{\gamma+1}$ on $y < 0$), using $G_n(y) - G_n(a) = \int_a^y G_n'$ for locally Lipschitz G_n. □

Lemma 2.7 (Inclusion $X_{\gamma'} \subset X_\gamma$)**.** *If $\gamma \leq \gamma'$ then $X_{\gamma'} \subset X_\gamma$ and $\|r\|_{X_\gamma} \leq \|r\|_{X_{\gamma'}}$ for all $r \in X_{\gamma'}$. In particular the inclusion $X_{\gamma_2} \hookrightarrow X_{\gamma_1}$ is a bounded linear map of norm ≤ 1.*

Proof. $\langle y\rangle^{\gamma} \leq \langle y\rangle^{\gamma'}$ pointwise since $\langle y\rangle \geq 1$; all other factors are identical. □

Lemma 2.8 (Inclusion $Y_{\gamma'} \subset Y_\gamma$)**.** *If $\gamma \leq \gamma'$ then $Y_{\gamma'} \subset Y_\gamma$ and $\|G\|_{Y_\gamma} \leq \|G\|_{Y_{\gamma'}}$.*

Proof. As in Lemma 2.7. □

Lemma 2.9 (Schwartz functions lie in every Y_γ)**.** *If $G \in \mathcal{S}(\mathbb{R};\mathbb{R})$ then $G \in Y_\gamma$ for every $\gamma \in \mathbb{R}$, with $\|G\|_{Y_\gamma} \leq 2\sup_y \langle y\rangle^{|\gamma|+2}(|G(y)| + |G'(y)|) < \infty$.*

Proof. Since $p < 1$, $\langle y\rangle^{\gamma-1/p} \leq \langle y\rangle^{\gamma}$ and $\langle y\rangle^{\gamma+1} \leq \langle y\rangle^{|\gamma|+2}$; bound each supremum by $\sup\langle y\rangle^{|\gamma|+2}(|G| + |G'|)$, which is finite for Schwartz G. □

2.2 The functional ℓ, the functions $G_\pm$, G_μ and the projection Π

Definition 2.10 (The obstruction functional ℓ)**.** Let $\gamma > 1$ (so $Y_\gamma \subset L^1(\mathbb{R})$ and $\widehat{G}$ is a bounded continuous function, with the Fourier transform of Definition 1.2). For $G \in Y_\gamma$ define

$$\ell(G) := \lim_{R\to\infty} \int_0^R \xi\, e^{i\Phi_0(\xi)} \widehat{G}(\xi)\, d\xi \in \mathbb{C}$$

whenever the limit exists. ℓ is $\mathbb{R}$-linear (not $\mathbb{C}$-linear: Y_γ is a real space). For admissible γ the limit exists for all $G \in Y_\gamma$ and $\ell : Y_\gamma \to \mathbb{C}$ is continuous: this is Proposition 8.39 (where also the representation $\ell(G) = \int_{\mathbb{R}} G'(y)K(y)\,dy$, $K(y) = -i\int_0^\infty e^{i(\Phi_0(\xi)-y\xi)}d\xi$, is used; the normalising constants depend on Definition 1.2). The value $\ell(G)$ does not depend on γ: if $G \in Y_{\gamma'} \subset Y_\gamma$ the defining expression is the same. For admissible γ put $\widetilde{Y}_\gamma := \{G \in Y_\gamma : \ell(G) = 0\}$.

Definition 2.11 (The functions G_+, G_-)**.** Let $G_+, G_- : \mathbb{R} \to \mathbb{C}$ be the inverse Fourier transforms (Definition 1.2) of

$$\widehat{G_+}(\xi) = \frac{\chi_+'(\xi)}{\xi} e^{-i\Phi_0(\xi)}, \qquad \widehat{G_-}(\xi) = i\,\frac{\chi_+'(\xi)}{\xi} e^{-i\Phi_0(\xi)} \qquad (\xi \geq 0),$$

extended to $\xi < 0$ by $\widehat{G_\pm}(\xi) := \overline{\widehat{G_\pm}(-\xi)}$.

Lemma 2.12 (Properties of $G_\pm$)**.** *G_+ and G_- are real-valued Schwartz functions; $G_\pm \in Y_\gamma$ for every γ; and $\ell(G_+) = 1$, $\ell(G_-) = i$.*

Proof. $\widehat{G_\pm}$ are C^∞ with support in $\{1 \leq |\xi| \leq 2\}$ (the support of χ_+' and its reflection; division by ξ and the factor $e^{-i\Phi_0}$ are smooth there), so $G_\pm$ are Schwartz. The Hermitian symmetry $\widehat{G_\pm}(-\xi) = \overline{\widehat{G_\pm}(\xi)}$ is equivalent to $G_\pm$ being real-valued. By Lemma 2.9, $G_\pm \in Y_\gamma$. Finally, for $R \geq 2$, $\int_0^R \xi e^{i\Phi_0}\widehat{G_+}\,d\xi = \int_0^R \chi_+' = \chi_+(R) - \chi_+(0) = 1$, and similarly $\ell(G_-) = i$. □

Definition 2.13 (The functions G_μ)**.** Let $\mu_* \in \mathbb{C}\setminus\{0\}$ be as in Definition 4.15 (in the paper $\mu_* = -i/\sqrt{2\pi p}$). For $\mu \in \mathbb{C}$ let G_μ be the inverse Fourier transform of $\widehat{G_\mu}(\xi) = \big(1 - \frac{\mu}{\mu_*}\big)\frac{\chi_+'(\xi)}{\xi} e^{-i\Phi_0(\xi)}$ for $\xi \geq 0$, extended by $\widehat{G_\mu}(\xi) = \overline{\widehat{G_\mu}(-\xi)}$ for $\xi < 0$.

Lemma 2.14 (Decomposition of G_μ)**.** *Write $z(\mu) := 1 - \mu/\mu_*$. Then for all $\mu \in \mathbb{C}$:*

1. *$G_\mu = \mathrm{Re}(z(\mu))\, G_+ + \mathrm{Im}(z(\mu))\, G_-$; in particular G_μ is a real Schwartz function and $G_{\mu_*} = 0$;*
2. *$\ell(G_\mu) = z(\mu)$;*
3. *for every γ, $\mu \mapsto G_\mu$ is a continuous ($\mathbb{R}$-affine) map $\mathbb{C} \to Y_\gamma$, and $\sup_{|\mu-\mu_*|\leq\delta} \|G_\mu\|_{Y_\gamma} \leq \frac{\delta}{|\mu_*|}(\|G_+\|_{Y_\gamma} + \|G_-\|_{Y_\gamma})$ for any $\delta > 0$.*

Proof. (1) Write $z = a+ib$. For $\xi \geq 0$, $a\widehat{G_+}(\xi)+b\widehat{G_-}(\xi) = (a+ib)\frac{\chi_+'(\xi)}{\xi}e^{-i\Phi_0(\xi)} = \widehat{G_\mu}(\xi)$; for $\xi < 0$, since $a, b \in \mathbb{R}$, $a\widehat{G_+}(\xi) + b\widehat{G_-}(\xi) = \overline{a\widehat{G_+}(-\xi) + b\widehat{G_-}(-\xi)} = \overline{\widehat{G_\mu}(-\xi)} = \widehat{G_\mu}(\xi)$. Injectivity of the Fourier transform gives (1). (The factor i in G_- is exactly what makes the imaginary part of z pair with G_-.) $z(\mu_*) = 0$. (2) $\mathbb{R}$-linearity of ℓ and Lemma 2.12: $\ell(G_\mu) = a + ib$. (3) From (1), $\|G_\mu\|_{Y_\gamma} \leq |z(\mu)|(\|G_+\| + \|G_-\|)$ and $|z(\mu)| = |\mu - \mu_*|/|\mu_*|$; affine maps $\mathbb{C} \cong \mathbb{R}^2 \to Y_\gamma$ are continuous.

□

Definition 2.15 (The projection Π)**.** For admissible γ and $G \in Y_\gamma$ define

$$\Pi G := G - \operatorname{Re}(\ell(G))\, G_+ - \operatorname{Im}(\ell(G))\, G_-.$$

(Π is given by the same formula for every γ, so Π on Y_{γ_2} is the restriction of Π on Y_{γ_1}.)

Lemma 2.16 (Π is a bounded projection onto $\widetilde{Y}_\gamma$)**.** *Let γ be admissible and let $C_\ell(\gamma)$ be a constant with $|\ell(G)| \leq C_\ell(\gamma)\|G\|_{Y_\gamma}$ for $G \in Y_\gamma$ (which exists by Proposition 8.39). Then:*

1. *$\widetilde{Y}_\gamma$ is a closed $\mathbb{R}$-linear subspace of Y_γ (hence a Banach space);*
2. *$\Pi : Y_\gamma \to Y_\gamma$ is $\mathbb{R}$-linear with $\|\Pi G\|_{Y_\gamma} \leq \big(1+C_\ell(\gamma)(\|G_+\|_{Y_\gamma}+\|G_-\|_{Y_\gamma})\big)\|G\|_{Y_\gamma}$;*
3. *$\ell(\Pi G) = 0$, i.e. $\Pi(Y_\gamma) \subset \widetilde{Y}_\gamma$; $\Pi G = G$ for $G \in \widetilde{Y}_\gamma$; hence $\Pi^2 = \Pi$;*
4. *$\Pi G_\mu = 0$ for all $\mu \in \mathbb{C}$, and $\Pi G = G - G_\mu$ whenever $\ell(G) = 1 - \mu/\mu_*$.*

Proof. (1) $\widetilde{Y}_\gamma = \ker \ell$ with ℓ continuous and $\mathbb{R}$-linear; closed subspaces of Banach spaces are complete. (2) $G \mapsto \operatorname{Re}\ell(G), \operatorname{Im}\ell(G)$ are $\mathbb{R}$-linear with norm $\leq C_\ell(\gamma)$; triangle inequality. (3) $\ell(\Pi G) = \ell(G) - \operatorname{Re}\ell(G)\cdot 1 - \operatorname{Im}\ell(G)\cdot i = 0$ by Lemma 2.12; if $\ell(G) = 0$ the correction terms vanish. (4) If $\ell(G) = z(\mu)$ then $\Pi G = G - \operatorname{Re} z(\mu) G_+ - \operatorname{Im} z(\mu) G_- = G - G_\mu$ by Lemma 2.14(1); with $G = G_\mu$ (using (2) of that lemma) this gives $\Pi G_\mu = 0$.

□

2.3 The linear and nonlinear operators

Definition 2.17 (Linear operators $\mathcal{L}_0$, $\mathcal{L}_\epsilon$)**.** For $r \in X_\gamma$ (γ admissible) define the function $\mathcal{L}_0 r : \mathbb{R} \to \mathbb{R}$ by

$$\mathcal{L}_0 r := |D|^p \partial_y r + r - y\, \partial_y r,$$

where $|D|^p\partial_y$ is as in Definition 1.5. For admissible γ, r' is bounded, integrable and locally Lipschitz; its derivative need not be globally bounded. For $\epsilon \geq 0$ let $b_\epsilon(y) := c_\epsilon \chi_+(y) y^{-p}$ with c_ϵ defined in Definition 3.11 ($b_\epsilon := 0$ for $y \leq 1$), and define

$$\mathcal{L}_\epsilon r := \mathcal{L}_0 r + b_\epsilon\, \partial_y r + \mathcal{R}_\epsilon r,$$

where $\mathcal{R}_\epsilon$ is the remainder operator of Definition 8.9. In general $\mathcal{L}_\epsilon r \notin Y_\gamma$ (the term $y\, r'$ grows like $\langle y\rangle^{1+1/p-\gamma}$ relative to the Y_γ weight); $\mathcal{L}_\epsilon$ is only a linear map from X_γ into functions $\mathbb{R} \to \mathbb{R}$.

Definition 2.18 (Nonlinear operator $\mathcal{N}_{\epsilon,\overline{f}}$)**.** For $\epsilon \geq 0$, a profile $\overline{f}$ for which $\mathcal{E}_\epsilon(\overline{f})$ is defined (Definition 1.26), and $r \in X_\gamma$ such that $\mathcal{E}_\epsilon(\overline{f}+r)$ is defined, set

$$\mathcal{N}_{\epsilon,\overline{f}}(r) := \mathcal{E}_\epsilon(\overline{f}+r) - \mathcal{E}_\epsilon(\overline{f}) - \mathcal{L}_\epsilon(r),$$

so that $\mathcal{E}_\epsilon(\overline{f}+r) = \mathcal{E}_\epsilon(\overline{f}) + \mathcal{L}_\epsilon(r) + \mathcal{N}_{\epsilon,\overline{f}}(r)$ identically.

2.4 The three main theorems (statements)

Definition 2.19 (Approximation properties for fixed parameters)**.** For fixed P as in Definition 2.1 and real R, δ, ϵ_0, let $\mathsf{A}(P, R, \delta, \epsilon_0)$ denote the following conjunction. We use the explicit family $\overline{f}_{\epsilon,\mu}$ of Definition 4.38 at scale R and $\overline{B} = \{\mu \in \mathbb{C} : |\mu - \mu_*| \leq \delta\}$. First $\mu_* \neq 0$, and every $\overline{f}_{\epsilon,\mu}$ is real and C^∞ for $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$. The remaining properties are:

1. $\lim_{y\to-\infty}[\overline{f}_{\epsilon,\mu}(y) + y] = 0$ and $\lim_{y\to+\infty} \overline{f}_{\epsilon,\mu}(y) = 0$.
2. For every $j \geq 0$ there is $C_j > 0$ such that for all $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$: for $y \geq 0$, $|\overline{f}_{\epsilon,\mu}(y)| \leq C_j\langle y\rangle^{-p}$ and $|\partial_y^{j+1}\overline{f}_{\epsilon,\mu}(y)| \leq C_j\langle y\rangle^{-\beta+j/p}$; for $y \leq 0$, $|\partial_y^j(\overline{f}_{\epsilon,\mu}(y) + y)| \leq C_j\langle y\rangle^{-p-j}$.
3. For every admissible γ there is $C_\gamma > 0$ such that for all $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$: $\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) \in Y_\gamma$ and
$$\|\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu\|_{Y_\gamma} \leq C_\gamma\,\epsilon, \tag{2.1}$$
where G_μ is the real Schwartz function of Definition 2.13 (independent of ϵ, $G_{\mu_*} = 0$).
4. For every $\epsilon \in [0, \epsilon_0]$: the map $\mu \mapsto \overline{f}_{\epsilon,\mu}$ is continuous from $\overline{B}$ into C^2_{loc}, i.e. for every compact $K \subset \mathbb{R}$, $k \in \{0, 1, 2\}$ and $\mu_j \to \mu$ in $\overline{B}$, $\sup_K |\partial_y^k\overline{f}_{\epsilon,\mu_j} - \partial_y^k\overline{f}_{\epsilon,\mu}| \to 0$; and for every admissible γ the map $\mu \mapsto \mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu})$ is continuous from $\overline{B}$ into Y_γ (norm topology).

Theorem 2.20 (Existence of approximate solutions)**.** *For every parameter tuple P there exist $R_0, \delta_0 > 0$ such that for every $R \geq R_0$ and $0 < \delta \leq \delta_0$ there exists $\epsilon_0 > 0$ for which $\mathsf{A}(P, R, \delta, \epsilon_0)$ of Definition 2.19 holds. Thus the explicit family has all properties (1)–(4) there, with constants uniform in $\epsilon \in [0, \epsilon_0]$ and $\mu \in \overline{B}$.*

Proof. See Lemma 7.37. □

Definition 2.21 (Linear inverse properties for fixed parameters)**.** For fixed P as in Definition 2.1 and real δ, ϵ_0, let $\mathsf{L}(P, \delta, \epsilon_0)$ denote the following conjunction for the explicit operators of Definitions 8.7, 8.8 and 8.33 and $\overline{B} = \{\mu : |\mu - \mu_*| \leq \delta\}$. Every bound below has a positive constant, chosen before ϵ, μ and the input functions. Well-definedness of $\ell(G)$ means convergence of its defining truncated integrals. Membership of $\mathcal{T}_0 G$ includes the identity $\widehat{\mathcal{T}_0 G} = m_G$ in $\mathcal{S}'$. The properties are:

1. For every admissible γ, $\ell : Y_\gamma \to \mathbb{C}$ is well defined and continuous: there is $C_\ell(\gamma)$ with $|\ell(G)| \leq C_\ell(\gamma)\|G\|_{Y_\gamma}$.

2. There is an $\mathbb{R}$-linear map $\mathcal{T}_0$ (Definition 8.33), defined on $\bigcup_{\gamma \text{ adm.}} \widetilde{Y}_\gamma$ with values in functions $\mathbb{R} \to \mathbb{R}$ and given by a formula independent of γ, such that for every admissible γ: $\mathcal{T}_0(\widetilde{Y}_\gamma) \subset X_\gamma$, $\|\mathcal{T}_0 G\|_{X_\gamma} \le C_T(\gamma)\|G\|_{Y_\gamma}$ for $G \in \widetilde{Y}_\gamma$, and

$$\mathcal{L}_0(\mathcal{T}_0 G) = G \quad \text{pointwise on } \mathbb{R}, \qquad G \in \widetilde{Y}_\gamma. \tag{2.2}$$

3. For each $\epsilon \in [0, \epsilon_0]$ there are $\mathbb{R}$-linear bijections $\mathcal{U}_\epsilon, \mathcal{V}_\epsilon$ of the space of all functions $\mathbb{R} \to \mathbb{R}$ (Definitions 8.7 and 8.8; independent of γ; in the paper $\mathcal{U}_\epsilon r = r \circ \phi_\epsilon^{-1}$, $\mathcal{V}_\epsilon u = (\phi_\epsilon')^{p+1}\, u \circ \phi_\epsilon$) such that for every admissible γ: $\mathcal{U}_\epsilon$ maps X_γ bijectively onto X_γ, $\mathcal{V}_\epsilon$ maps Y_γ bijectively onto Y_γ, and there is $C_{UV}(\gamma)$ with

$$\|\mathcal{U}_\epsilon\|_{X_\gamma \to X_\gamma},\ \|\mathcal{U}_\epsilon^{-1}\|_{X_\gamma \to X_\gamma},\ \|\mathcal{V}_\epsilon\|_{Y_\gamma \to Y_\gamma},\ \|\mathcal{V}_\epsilon^{-1}\|_{Y_\gamma \to Y_\gamma} \le C_{UV}(\gamma) \qquad (0 \le \epsilon \le \epsilon_0);$$

moreover $\mathcal{L}_\epsilon r = \mathcal{V}_\epsilon\big(\mathcal{L}_0(\mathcal{U}_\epsilon r)\big)$ for all $r \in X_\gamma$.

4. For every admissible γ there is $C_G(\gamma)$ with

$$\sup_{\mu \in \overline{B}} \|(\mathcal{V}_\epsilon^{-1} - I) G_\mu\|_{Y_\gamma} \le C_G(\gamma)\, \epsilon, \qquad 0 \le \epsilon \le \epsilon_0. \tag{2.3}$$

Theorem 2.22 (Linear inverse)**.** *For every parameter tuple P there exists $\epsilon_0 > 0$, independent of δ, such that $\mathsf{L}(P, \delta, \epsilon_0)$ of Definition 2.21 holds for every $\delta > 0$. In the combined fixed-point argument one may take the minimum of this threshold and the approximation threshold. This gives all properties (1)–(4) of the definition.*

Proof. See Lemma 9.39. □

Definition 2.23 (Nonlinear properties for fixed parameters)**.** For fixed P as in Definition 2.1 and real R, δ, ϵ_0, let $\mathsf{N}(P, R, \delta, \epsilon_0)$ denote the following conjunction, using the explicit family of Definition 4.38 at scale R and $\overline{B} = \{\mu : |\mu - \mu_*| \le \delta\}$. The quantitative constants below are positive and uniform in ϵ, μ:

(1)–(2) For every admissible γ and every $N > 0$ there is $C_{\mathcal{N}}(\gamma, N)$ such that for all $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$ and $r, r_1, r_2 \in X_\gamma$ with $\|r\|_{X_\gamma}, \|r_1\|_{X_\gamma}, \|r_2\|_{X_\gamma} \le N$, writing $\overline{f} = \overline{f}_{\epsilon,\mu}$: $\mathcal{N}_{\epsilon,\overline{f}}(r)$ belongs to Y_γ, and

$$\|\mathcal{N}_{\epsilon,\overline{f}}(r)\|_{Y_\gamma} \le C_{\mathcal{N}}(\gamma, N)\, \epsilon\, \|r\|_{X_\gamma}, \tag{2.4}$$

$$\|\mathcal{N}_{\epsilon,\overline{f}}(r_1) - \mathcal{N}_{\epsilon,\overline{f}}(r_2)\|_{Y_\gamma} \le C_{\mathcal{N}}(\gamma, N)\, \epsilon\, \|r_1 - r_2\|_{X_\gamma}. \tag{2.5}$$

(3) For each $\epsilon \in [0, \epsilon_0]$ and each fixed $r \in X_{\gamma_2}$, the map $\mu \mapsto \mathcal{N}_{\epsilon,\overline{f}_{\epsilon,\mu}}(r)$ is continuous from $\overline{B}$ into Y_{γ_1}.

Theorem 2.24 (Nonlinear estimates)**.** *For every parameter tuple P there exist $R_0, \delta_0 > 0$ such that for every $R \ge R_0$ and $0 < \delta \le \delta_0$ there exists $\epsilon_0 > 0$ for which $\mathsf{N}(P, R, \delta, \epsilon_0)$ of Definition 2.23 holds. The threshold is chosen before the admissible exponent γ and the norm bound N; the constants in (1)–(2) may depend on $P, R, \delta, \epsilon_0, \gamma, N$.*

Proof. See Lemma 10.44. □

2.5 Existence of the correction

Theorem 2.25 (Existence of the correction)**.** *There exist $\epsilon_1 \in (0, \epsilon_0]$ and $C > 0$ such that for every $\epsilon \in [0, \epsilon_1]$ there exist $\mu_\epsilon \in \overline{B}$ and $r_\epsilon \in X_{\gamma_2}$ satisfying, as an identity of functions on $\mathbb{R}$,*

$$\mathcal{L}_\epsilon(r_\epsilon) = -\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu_\epsilon}) - \mathcal{N}_{\epsilon,\overline{f}_{\epsilon,\mu_\epsilon}}(r_\epsilon), \tag{2.6}$$

and

$$|\mu_\epsilon - \mu_*| + \|r_\epsilon\|_{X_{\gamma_2}} \le C\epsilon. \tag{2.7}$$

Proof. Take ϵ_1 and C_* from Lemma 2.40 (which also ensures the hypotheses of Lemma 2.34). Fix $\epsilon \in [0, \epsilon_1]$. By Lemmas 2.39 and 2.40, $\mathcal{K}_\epsilon : \overline{B} \to \overline{B}$ is continuous; $\overline{B}$ is a closed disc in $\mathbb{C}$, so by Lemma 2.41 there is $\mu_\epsilon \in \overline{B}$ with $\mathcal{K}_\epsilon(\mu_\epsilon) = \mu_\epsilon$, i.e. $\mu_*[1 - \ell(\mathcal{G}_{\epsilon,\mu_\epsilon}(w_{\epsilon,\mu_\epsilon}))] = \mu_\epsilon$, i.e. (dividing by $\mu_* \neq 0$) $\ell(\mathcal{G}_{\epsilon,\mu_\epsilon}(w_{\epsilon,\mu_\epsilon})) = 1 - \mu_\epsilon/\mu_*$. Together with $w_{\epsilon,\mu_\epsilon} = \mathcal{H}_{\epsilon,\mu_\epsilon}(w_{\epsilon,\mu_\epsilon})$ (Lemma 2.34) the pair $(w_{\epsilon,\mu_\epsilon}, \mu_\epsilon)$ solves the system of Lemma 2.28, so $r_\epsilon := \mathcal{U}_\epsilon^{-1} w_{\epsilon,\mu_\epsilon} \in X_{\gamma_2}$ solves (2.6). Bounds: $|\mu_\epsilon - \mu_*| = |\mathcal{K}_\epsilon(\mu_\epsilon) - \mu_*| \le C_*\epsilon$ by Lemma 2.40, and $\|r_\epsilon\|_{X_{\gamma_2}} \le C_{UV}(\gamma_2)\|w_{\epsilon,\mu_\epsilon}\|_{X_{\gamma_2}} \le C_{UV}(\gamma_2)C_0\epsilon$ by Theorem 2.22(3) and Lemma 2.34. Take $C := C_* + C_{UV}(\gamma_2)C_0$. □

Lemma 2.26 (The constructed profile solves the profile equation)**.** *For $\epsilon \in [0, \epsilon_1]$ and $(\mu_\epsilon, r_\epsilon)$ as in Theorem 2.25, $f_\epsilon := \overline{f}_{\epsilon,\mu_\epsilon} + r_\epsilon$ satisfies $\mathcal{E}_\epsilon(f_\epsilon) = 0$ on $\mathbb{R}$.*

Proof. By Definition 2.18, $\mathcal{E}_\epsilon(f_\epsilon) = \mathcal{E}_\epsilon(\overline{f}) + \mathcal{L}_\epsilon(r_\epsilon) + \mathcal{N}_{\epsilon,\overline{f}}(r_\epsilon)$, which vanishes by (2.6). □

2.5.1 Reduction to a fixed point system

Throughout this subsection $\epsilon_0, \delta, \overline{B}, \overline{f}_{\epsilon,\mu}$ are as in Theorem 2.20, and $\mathcal{T}_0, \mathcal{U}_\epsilon, \mathcal{V}_\epsilon$ as in Theorem 2.22.

Definition 2.27 (Forcing map and fixed point map)**.** For $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, an admissible γ and $w \in X_\gamma$ such that $\mathcal{N}_{\epsilon,\overline{f}_{\epsilon,\mu}}(\mathcal{U}_\epsilon^{-1} w)$ is defined and lies in Y_γ (e.g. any $w \in X_\gamma$ by Theorem 2.24), define

$$\mathcal{G}_{\epsilon,\mu}(w) := -\big[\mathcal{V}_\epsilon^{-1}\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu\big] - \mathcal{V}_\epsilon^{-1}\mathcal{N}_{\epsilon,\overline{f}_{\epsilon,\mu}}(\mathcal{U}_\epsilon^{-1} w) \ \in Y_\gamma, \tag{2.8}$$

and

$$\mathcal{H}_{\epsilon,\mu}(w) := \mathcal{T}_0 \Pi\, \mathcal{G}_{\epsilon,\mu}(w) \ \in X_\gamma. \tag{2.9}$$

Both are given by γ-independent formulas, so for $w \in X_{\gamma_2}$ the values computed in γ_2 and in γ_1 coincide.

Lemma 2.28 (The system implies the correction equation)**.** *Let $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, γ admissible, $w \in X_\gamma$, and suppose*

$$\begin{cases} w = \mathcal{T}_0 \Pi\, \mathcal{G}_{\epsilon,\mu}(w), \\ \ell(\mathcal{G}_{\epsilon,\mu}(w)) = 1 - \frac{\mu}{\mu_*}. \end{cases} \tag{2.10}$$

Then $r := \mathcal{U}_\epsilon^{-1} w \in X_\gamma$ satisfies $\mathcal{L}_\epsilon(r) = -\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) - \mathcal{N}_{\epsilon,\overline{f}_{\epsilon,\mu}}(r)$ on $\mathbb{R}$.

Proof. Since $\Pi\mathcal{G}_{\epsilon,\mu}(w) \in \widetilde{Y}_\gamma$ (Lemma 2.16(3)), (2.2) gives $\mathcal{L}_0 w = \mathcal{L}_0\mathcal{T}_0\Pi\mathcal{G}_{\epsilon,\mu}(w) = \Pi\mathcal{G}_{\epsilon,\mu}(w)$. By the second equation and Lemma 2.16(4), $\Pi\mathcal{G}_{\epsilon,\mu}(w) = \mathcal{G}_{\epsilon,\mu}(w) - G_\mu$, and by (2.8)

$$\mathcal{L}_0 w = -\mathcal{V}_\epsilon^{-1}\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) - \mathcal{V}_\epsilon^{-1}\mathcal{N}_{\epsilon,\overline{f}_{\epsilon,\mu}}(r) \qquad \text{(the } G_\mu \text{ terms cancel).}$$

Apply the linear map $\mathcal{V}_\epsilon$ (defined on all functions) and use $w = \mathcal{U}_\epsilon r$ and Theorem 2.22(3): $\mathcal{L}_\epsilon r = \mathcal{V}_\epsilon\mathcal{L}_0\mathcal{U}_\epsilon r = -\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) - \mathcal{N}_{\epsilon,\overline{f}_{\epsilon,\mu}}(r)$. Also $r \in X_\gamma$ since $\mathcal{U}_\epsilon^{-1}$ preserves X_γ.

□

2.5.2 Contraction for $\mathcal{H}$

Lemma 2.29 (Conjugated residual bound)**.** *For every admissible γ, with $C_R(\gamma) := C_{UV}(\gamma)C_\gamma + C_G(\gamma)$,*

$$\|\mathcal{V}_\epsilon^{-1}\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu\|_{Y_\gamma} \le C_R(\gamma)\,\epsilon \qquad (\epsilon \in [0,\epsilon_0],\ \mu \in \overline{B}). \tag{2.11}$$

Proof. Write $\mathcal{V}_\epsilon^{-1}\mathcal{E}_\epsilon(\overline{f}) - G_\mu = \mathcal{V}_\epsilon^{-1}(\mathcal{E}_\epsilon(\overline{f}) - G_\mu) + (\mathcal{V}_\epsilon^{-1} - I)G_\mu$ (both $\mathcal{E}_\epsilon(\overline{f})$, $G_\mu \in Y_\gamma$) and use (2.1), the bound on $\|\mathcal{V}_\epsilon^{-1}\|_{Y_\gamma\to Y_\gamma}$, and (2.3).

□

Lemma 2.30 (Bounds for the forcing map)**.** *Let γ be admissible and $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$. If $w, w_1, w_2 \in X_\gamma$ satisfy $C_{UV}(\gamma)\|w\|_{X_\gamma} \le 1$ and $C_{UV}(\gamma)\|w_i\|_{X_\gamma} \le 1$, then with $C_\mathcal{N} := C_\mathcal{N}(\gamma,1)$:*

$$\|\mathcal{G}_{\epsilon,\mu}(w)\|_{Y_\gamma} \le C_R(\gamma)\epsilon + C_{UV}(\gamma)^2 C_\mathcal{N}\,\epsilon\|w\|_{X_\gamma},$$
$$\|\mathcal{G}_{\epsilon,\mu}(w_1) - \mathcal{G}_{\epsilon,\mu}(w_2)\|_{Y_\gamma} \le C_{UV}(\gamma)^2 C_\mathcal{N}\,\epsilon\|w_1 - w_2\|_{X_\gamma},$$

and consequently, with $M(\gamma) := C_T(\gamma)\big(1 + C_\ell(\gamma)(\|G_+\|_{Y_\gamma} + \|G_-\|_{Y_\gamma})\big)$,

$$\|\mathcal{H}_{\epsilon,\mu}(w)\|_{X_\gamma} \le M(\gamma)C_R(\gamma)\epsilon + M(\gamma)C_{UV}(\gamma)^2 C_\mathcal{N}\epsilon\|w\|_{X_\gamma}, \tag{2.12}$$
$$\|\mathcal{H}_{\epsilon,\mu}(w_1) - \mathcal{H}_{\epsilon,\mu}(w_2)\|_{X_\gamma} \le M(\gamma)C_{UV}(\gamma)^2 C_\mathcal{N}\epsilon\|w_1 - w_2\|_{X_\gamma}. \tag{2.13}$$

Proof. $\|\mathcal{U}_\epsilon^{-1}w\|_{X_\gamma} \le C_{UV}(\gamma)\|w\|_{X_\gamma} \le 1$, so (2.4) with $N = 1$ gives $\|\mathcal{V}_\epsilon^{-1}\mathcal{N}(\mathcal{U}_\epsilon^{-1}w)\|_{Y_\gamma} \le C_{UV}C_\mathcal{N}\epsilon\, C_{UV}\|w\|$; add Lemma 2.29. The difference $\mathcal{G}(w_1) - \mathcal{G}(w_2) = -\mathcal{V}_\epsilon^{-1}[\mathcal{N}(\mathcal{U}_\epsilon^{-1}w_1) - \mathcal{N}(\mathcal{U}_\epsilon^{-1}w_2)]$ is bounded by (2.5) and linearity of $\mathcal{U}_\epsilon^{-1}$. Finally $\|\mathcal{T}_0\Pi G\|_{X_\gamma} \le M(\gamma)\|G\|_{Y_\gamma}$ by Theorem 2.22(2) and Lemma 2.16(2),(3), and $\mathcal{T}_0\Pi$ is linear.

□

Lemma 2.31 (Choice of constants: $\mathcal{H}$ is a contraction in both norms)**.** *Put $C_0 := 2M(\gamma_2)C_R(\gamma_2)$. There is $\epsilon_2 \in (0,\epsilon_0]$ such that for all $\epsilon \in [0,\epsilon_2]$ and $\mu \in \overline{B}$, writing $\mathcal{B}_\epsilon := \{w \in X_{\gamma_2} : \|w\|_{X_{\gamma_2}} \le C_0\epsilon\}$:*

1. *$\mathcal{H}_{\epsilon,\mu}(\mathcal{B}_\epsilon) \subset \mathcal{B}_\epsilon$;*
2. *$\|\mathcal{H}_{\epsilon,\mu}(w_1) - \mathcal{H}_{\epsilon,\mu}(w_2)\|_{X_{\gamma_2}} \le \frac{1}{2}\|w_1 - w_2\|_{X_{\gamma_2}}$ for $w_1, w_2 \in \mathcal{B}_\epsilon$;*
3. *$\|\mathcal{H}_{\epsilon,\mu}(w_1) - \mathcal{H}_{\epsilon,\mu}(w_2)\|_{X_{\gamma_1}} \le \frac{1}{2}\|w_1 - w_2\|_{X_{\gamma_1}}$ for $w_1, w_2 \in \mathcal{B}_\epsilon$;*
4. *$\|\mathcal{G}_{\epsilon,\mu}(w)\|_{Y_{\gamma_2}} \le 2C_R(\gamma_2)\epsilon$ for $w \in \mathcal{B}_\epsilon$.*

Proof. Choose ϵ_2 so small that, for $\gamma \in \{\gamma_1, \gamma_2\}$ and $C_{\mathcal{N}}(\gamma) := C_{\mathcal{N}}(\gamma, 1)$: (a) $C_{UV}(\gamma)C_0\epsilon_2 \le 1$; (b) $M(\gamma)C_{UV}(\gamma)^2 C_{\mathcal{N}}(\gamma)\epsilon_2 \le \frac{1}{2}$; (c) $C_{UV}(\gamma_2)^2 C_{\mathcal{N}}(\gamma_2)C_0\epsilon_2 \le C_R(\gamma_2)$. Note that $N = 1$ is fixed *before* choosing ϵ_2, so there is no circularity. For $w \in \mathcal{B}_\epsilon$: $\|w\|_{X_{\gamma_1}} \le \|w\|_{X_{\gamma_2}} \le C_0\epsilon$ (Lemma 2.7), so by (a) the hypotheses of Lemma 2.30 hold for both γ_1 and γ_2. (1) By (2.12) and (b): $\|\mathcal{H}(w)\|_{X_{\gamma_2}} \le \frac{C_0}{2}\epsilon + \frac{1}{2}C_0\epsilon = C_0\epsilon$. (2) and (3): (2.13) with $\gamma = \gamma_2$, resp. $\gamma = \gamma_1$, and (b). (4) By Lemma 2.30 and (c). □

Lemma 2.32 (Banach fixed point theorem)**.** *Let (E, d) be a complete metric space (e.g. a closed subset of a Banach space), $T : E \to E$ with $d(Tx, Ty) \le Kd(x, y)$ for some $K < 1$. Then T has a unique fixed point in E.*

Definition 2.33 (Selection of a fixed point when one exists)**.** For fixed parameters P, $R, \epsilon \in \mathbb{R}$, $\mu \in \mathbb{C}$ and radius $\rho \in \mathbb{R}$, select a function $w \in X_{\gamma_2}$ with $\|w\|_{X_{\gamma_2}} \le \rho$ and $\mathcal{H}_{\epsilon,\mu}(w) = w$ if such a function exists, and select $w = 0$ otherwise. This total choice is denoted $w^{\rho}_{\epsilon,\mu}$. Existence and uniqueness on the contraction ball are a separate assertion; subsequently $w_{\epsilon,\mu}$ abbreviates $w^{C_0\epsilon}_{\epsilon,\mu}$.

Lemma 2.34 (Solution of the first equation)**.** *For $\epsilon \in [0, \epsilon_2]$ and $\mu \in \overline{B}$ there is a unique $w_{\epsilon,\mu} \in \mathcal{B}_\epsilon$ with $w_{\epsilon,\mu} = \mathcal{H}_{\epsilon,\mu}(w_{\epsilon,\mu})$; in particular*

$$\|w_{\epsilon,\mu}\|_{X_{\gamma_2}} \le C_0\epsilon. \tag{2.14}$$

Proof. $\mathcal{B}_\epsilon$ is a closed ball in the Banach space X_{γ_2} (Lemma 2.5), hence complete; by Lemma 2.31(1),(2) $\mathcal{H}_{\epsilon,\mu}$ is a $\frac{1}{2}$-contraction of $\mathcal{B}_\epsilon$ into itself. Apply Lemma 2.32. (For $\epsilon = 0$, $\mathcal{B}_0 = \{0\}$ and $w_{0,\mu} = 0$.) □

2.5.3 Continuity in μ and the scalar equation

Lemma 2.35 (Continuity of $\mu \mapsto \mathcal{G}_{\epsilon,\mu}(w)$ for fixed w)**.** *Fix $\epsilon \in [0, \epsilon_0]$ and $w \in X_{\gamma_2}$. Then $\mu \mapsto \mathcal{G}_{\epsilon,\mu}(w)$ is continuous from $\overline{B}$ into Y_{γ_1}, and $\mu \mapsto \mathcal{H}_{\epsilon,\mu}(w)$ is continuous from $\overline{B}$ into X_{γ_1}.*

Proof. Three terms: (i) $\mu \mapsto \mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) \in Y_{\gamma_1}$ is continuous (Theorem 2.20(4) with $\gamma = \gamma_1$), and $\mathcal{V}_\epsilon^{-1}$ is a bounded linear operator on Y_{γ_1} independent of μ; (ii) $\mu \mapsto G_\mu \in Y_{\gamma_1}$ is continuous (Lemma 2.14(3)); (iii) $r := \mathcal{U}_\epsilon^{-1} w \in X_{\gamma_2}$ is independent of μ, so $\mu \mapsto \mathcal{N}_{\epsilon,\overline{f}_{\epsilon,\mu}}(r) \in Y_{\gamma_1}$ is continuous by Theorem 2.24(3), and then so is its image under $\mathcal{V}_\epsilon^{-1}$. Finally $\mathcal{T}_0\Pi : Y_{\gamma_1} \to X_{\gamma_1}$ is bounded linear (Theorem 2.22(2), Lemma 2.16).

□

Lemma 2.36 (Continuity of $\mu \mapsto w_{\epsilon,\mu}$ in X_{γ_1})**.** *For each $\epsilon \in [0, \epsilon_2]$ the map $\overline{B} \ni \mu \mapsto w_{\epsilon,\mu} \in X_{\gamma_1}$ is continuous.*

Proof. Let $\mu_j \to \mu$ in $\overline{B}$; write $w_j := w_{\epsilon,\mu_j}$, $w := w_{\epsilon,\mu}$, all in $\mathcal{B}_\epsilon$. Then

$$\|w_j - w\|_{X_{\gamma_1}} \le \|\mathcal{H}_{\epsilon,\mu_j}(w_j) - \mathcal{H}_{\epsilon,\mu_j}(w)\|_{X_{\gamma_1}} + \|\mathcal{H}_{\epsilon,\mu_j}(w) - \mathcal{H}_{\epsilon,\mu}(w)\|_{X_{\gamma_1}} \le \tfrac{1}{2}\|w_j - w\|_{X_{\gamma_1}} + \|\mathcal{H}_{\epsilon,\mu_j}(w) - \mathcal{H}_{\epsilon,\mu}(w)\|_{X_{\gamma_1}}$$

using Lemma 2.31(3) for the parameter μ_j (uniform in μ_j). All norms are finite (Lemma 2.7), hence $\|w_j - w\|_{X_{\gamma_1}} \le 2\|\mathcal{H}_{\epsilon,\mu_j}(w) - \mathcal{H}_{\epsilon,\mu}(w)\|_{X_{\gamma_1}} \to 0$ by Lemma 2.35 (applied with the fixed $w \in X_{\gamma_2}$). □

Definition 2.37 (Obstruction map $\mathcal{K}_\epsilon$)**.** For any $R, \epsilon, \rho \in \mathbb{R}$ and $\mu \in \mathbb{C}$, use the selected $w^\rho_{\epsilon,\mu}$ of Definition 2.33 and define

$$\mathcal{K}_\epsilon(\mu) := \mu_* \big[1 - \ell(\mathcal{G}_{\epsilon,\mu}(w^\rho_{\epsilon,\mu}))\big] \in \mathbb{C}. \tag{2.15}$$

The radius is subsequently $\rho = C_0\epsilon$.

Lemma 2.38 (Scalar fixed point identity)**.** *For the parameters of Definition 2.37, $\mathcal{K}_\epsilon(\mu) = \mu$ if and only if $\ell(\mathcal{G}_{\epsilon,\mu}(w^\rho_{\epsilon,\mu})) = 1 - \mu/\mu_*$.*

Proof. Since $p > 0$, $\mu_* = -i/\sqrt{2\pi p} \neq 0$. Divide the defining identity by μ_*. □

Lemma 2.39 (Continuity of $\mathcal{K}_\epsilon$)**.** *For each $\epsilon \in [0, \epsilon_2]$, $\mu \mapsto \mathcal{G}_{\epsilon,\mu}(w_{\epsilon,\mu})$ is continuous $\overline{B} \to Y_{\gamma_1}$, and $\mathcal{K}_\epsilon : \overline{B} \to \mathbb{C}$ is continuous.*

Proof. With notation as in Lemma 2.36,

$$\|\mathcal{G}_{\epsilon,\mu_j}(w_j) - \mathcal{G}_{\epsilon,\mu}(w)\|_{Y_{\gamma_1}} \le \|\mathcal{G}_{\epsilon,\mu_j}(w_j) - \mathcal{G}_{\epsilon,\mu_j}(w)\|_{Y_{\gamma_1}} + \|\mathcal{G}_{\epsilon,\mu_j}(w) - \mathcal{G}_{\epsilon,\mu}(w)\|_{Y_{\gamma_1}}.$$

The first term is $\le C_{UV}(\gamma_1)^2 C_{\mathcal{N}}(\gamma_1, 1)\epsilon \|w_j - w\|_{X_{\gamma_1}}$ by Lemma 2.30 with $\gamma = \gamma_1$ (hypotheses verified in the proof of Lemma 2.31; the constant is uniform in μ_j), and tends to 0 by Lemma 2.36. The second tends to 0 by Lemma 2.35. Finally ℓ is continuous on Y_{γ_1} (Theorem 2.22(1)) and its value does not depend on whether $\mathcal{G}_{\epsilon,\mu}(w_{\epsilon,\mu}) \in Y_{\gamma_2}$ is regarded in Y_{γ_1} or Y_{γ_2} (Definition 2.10).

□

Lemma 2.40 (Smallness of $\mathcal{K}_\epsilon - \mu_*$; $\mathcal{K}_\epsilon$ maps $\overline{B}$ into itself)**.** *Let $C_* := 2|\mu_*| C_\ell(\gamma_2) C_R(\gamma_2)$ and $\epsilon_1 := \min\{\epsilon_2, \delta/C_*\}$. For all $\epsilon \in [0, \epsilon_2]$ and $\mu \in \overline{B}$,*

$$|\mathcal{K}_\epsilon(\mu) - \mu_*| = |\mu_*|\,|\ell(\mathcal{G}_{\epsilon,\mu}(w_{\epsilon,\mu}))| \le C_*\epsilon, \tag{2.16}$$

and for $\epsilon \in [0, \epsilon_1]$, $\mathcal{K}_\epsilon(\overline{B}) \subset \overline{B}$.

Proof. $|\ell(G)| \le C_\ell(\gamma_2)\|G\|_{Y_{\gamma_2}}$ and $\|\mathcal{G}_{\epsilon,\mu}(w_{\epsilon,\mu})\|_{Y_{\gamma_2}} \le 2C_R(\gamma_2)\epsilon$ by Lemma 2.31(4) since $w_{\epsilon,\mu} \in \mathcal{B}_\epsilon$. If $C_*\epsilon \le \delta$ then $|\mathcal{K}_\epsilon(\mu) - \mu_*| \le \delta$, i.e. $\mathcal{K}_\epsilon(\mu) \in \overline{B}$.

□

Lemma 2.41 (Brouwer fixed point theorem for a closed disc in $\mathbb{C}$)**.** *Let $c \in \mathbb{C}$, $\delta > 0$, $\overline{B} = \{\mu : |\mu - c| \le \delta\}$ and $g : \overline{B} \to \overline{B}$ continuous. Then g has a fixed point.*

Proof. By the affine homeomorphism $\mu \mapsto (\mu - c)/\delta$ reduce to the closed unit disc $\overline{\mathbb{D}}$. If g had no fixed point, the map $\rho(\mu) :=$ the intersection of the ray from $g(\mu)$ through μ with $\partial\mathbb{D}$ (explicitly, $\rho(\mu) = \mu + t(\mu)(\mu - g(\mu))$ with $t(\mu) \ge 0$ the nonnegative root of a quadratic depending continuously on μ) would be a continuous retraction $\overline{\mathbb{D}} \to S^1$. No such retraction exists: composing with $\theta \mapsto e^{i\theta}$ and the homotopy $(s, \theta) \mapsto \rho(se^{i\theta})$ would show that the identity loop of S^1 (winding number 1) is homotopic to a constant loop (winding number 0), contradicting homotopy invariance of the winding number / $\pi_1(S^1) \cong \mathbb{Z}$. Reference: e.g. Milnor, *Topology from the differentiable viewpoint*, or any algebraic topology text. □

Remark 2.42 (Avoiding Brouwer). The only use of Lemma 2.41 is to solve the scalar (two real dimensional) equation $\mathcal{K}_\epsilon(\mu) = \mu$ on $\overline{B}$. If one had *Lipschitz* dependence on μ with small constant, namely

$$\|\mathcal{G}_{\epsilon,\mu}(w) - \mathcal{G}_{\epsilon,\nu}(w)\|_{Y_{\gamma_1}} \le C\epsilon|\mu - \nu| \qquad (\mu, \nu \in \overline{B},\ w \in \mathcal{B}_\epsilon)$$

(which would require Lipschitz-in-μ versions, into Y_{γ_1} with constant $O(\epsilon)$, of Theorem 2.20(4) for $\mu \mapsto \mathcal{V}_\epsilon^{-1}\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu$ and of Theorem 2.24(3); these are not established here), then the two-norm argument of Lemma 2.36 gives $\|w_{\epsilon,\mu} - w_{\epsilon,\nu}\|_{X_{\gamma_1}} \le C'\epsilon|\mu - \nu|$, hence $\mathcal{K}_\epsilon$ would be $C''\epsilon$-Lipschitz and Lemma 2.32 on $\overline{B}$ would replace Brouwer (and give uniqueness of μ_ϵ). Note that $\Pi G_\mu = 0$ (Lemma 2.16(4)), so the non-small μ-dependence of G_μ does not enter $\mathcal{H}_{\epsilon,\mu}$, and in $\mathcal{K}_\epsilon$ it enters only through the $O(\epsilon)$ difference $\mathcal{V}_\epsilon^{-1}\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu$. Alternatively, a formal proof of the 2D Brouwer theorem is a self-contained project.

Chapter 3

Analysis of the equation (paper §4) and the estimates of Appendices A–B

3.1 Notation specific to this chapter

Definition 3.1 (Local Hölder spaces)**.** Let $s > 0$ with $s \notin \mathbb{N}$, and write $s = k+\theta$ with $k \in \mathbb{N}_0$, $\theta \in (0,1)$. For a compact interval $I \subset \mathbb{R}$ and $g \in C^k(I;\mathbb{C})$ put

$$\|g\|_{C^s(I)} := \sum_{i=0}^{k} \sup_I |g^{(i)}| + \sup_{x \neq x' \in I} \frac{|g^{(k)}(x) - g^{(k)}(x')|}{|x-x'|^\theta},$$

and $C^s(I) = \{g \in C^k(I) : \|g\|_{C^s(I)} < \infty\}$. For an open interval U, $C^s_{\mathrm{loc}}(U)$ is the set of $g \in C^k(U)$ with $g|_I \in C^s(I)$ for every compact interval $I \subset U$. We write $C^s_{\mathrm{loc}} = C^s_{\mathrm{loc}}(\mathbb{R})$. For $0 < s_0 < 1$ every locally Lipschitz function lies in $C^{s_0}_{\mathrm{loc}}$.

Definition 3.2 (Secant)**.** For $f : \mathbb{R} \to \mathbb{R}$ and $y \neq z$ let $S_f(y,z) = \dfrac{f(y)-f(z)}{y-z}$. If $f \in C^1(\mathbb{R})$ then $S_f(y,z) = \int_0^1 f'(z+\tau(y-z))\,d\tau$, which extends continuously to $y = z$ by $S_f(y,y) = f'(y)$, and $|S_f(y,z)| \le \|f'\|_{L^\infty}$.

Definition 3.3 (The coefficient B_ϵ and the kernel function Γ_ϵ)**.** Let $\epsilon \ge 0$, $A_\epsilon(u) = (1+\epsilon u^2)^{-\alpha/2}$ (Definition 1.10) and $B_\epsilon(u) := A_\epsilon(u)^{-1} = (1+\epsilon u^2)^{\alpha/2}$. Let $c_\alpha = \Gamma(\alpha)\sin(\pi(\alpha-1)/2)/\pi$ be the constant of Definition 1.22. Define $P_\epsilon, \Gamma_\epsilon : \mathbb{R}^2 \to \mathbb{R}$ by

$$P_\epsilon(u,v) := \int_u^v \Big(1 - \frac{A_\epsilon(s)}{A_\epsilon(u)}\Big) ds,$$

$$\Gamma_\epsilon(u,v) := c_\alpha\Big[p\int_u^v \Big(1-\frac{A_\epsilon(s)}{A_\epsilon(u)}\Big)ds + (v-u)\Big(1-\frac{A_\epsilon(v)}{A_\epsilon(u)}\Big)\Big] = c_\alpha\Big[p\,P_\epsilon(u,v) + (v-u)\Big(1-\frac{A_\epsilon(v)}{A_\epsilon(u)}\Big)\Big], \tag{3.1}$$

and

$$\widetilde{\Gamma}_\epsilon(u,v) := -\frac{c_\alpha}{A_\epsilon(u)} \int_0^1 \big[1+p(1-t)\big] A'_\epsilon\big(u+t(v-u)\big)\, dt.$$

Since $A_\epsilon \in C^\infty(\mathbb{R})$ and $A_\epsilon > 0$, the functions $B_\epsilon, P_\epsilon, \Gamma_\epsilon, \widetilde{\Gamma}_\epsilon$ are C^∞ on their domains.

Definition 3.4 (The nonlocal operator $\mathcal{I}_\epsilon$)**.** For $f \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ with $f' \in L^\infty(\mathbb{R})$ and $y \in \mathbb{R}$ set

$$\mathcal{I}_\epsilon(f)(y) := \int_{\mathbb{R}} \frac{\Gamma_\epsilon\big(f'(y), S_f(y,z)\big)}{|y-z|^\alpha}\, dz = \int_{\mathbb{R}} \frac{\Gamma_\epsilon\big(f'(y), S_f(y,y-t)\big)}{|t|^\alpha}\, dt.$$

(Absolute convergence is part of Lemma 3.8.)

3.2 Analysis of the equation

3.2.1 Elementary properties of A_ϵ and Γ_ϵ

Lemma 3.5 (Bounds on A_ϵ)**.** *Let $1 < \alpha < 2$, $0 \le \epsilon \le 1$ and $R \ge 1$. Then $0 < A_\epsilon \le 1$, $A_\epsilon(0) = 1$, $A'_\epsilon(0) = 0$, $A'_\epsilon(s) = -\alpha\epsilon s(1+\epsilon s^2)^{-\alpha/2-1}$, so $|A'_\epsilon(s)| \le \alpha\epsilon|s|$ and $|1 - A_\epsilon(s)| \le \frac{\alpha}{2}\epsilon s^2$ for all $s \in \mathbb{R}$. Moreover for each $k \ge 0$ there is $C = C(k,R,\alpha)$ with $|\partial_s^k A'_\epsilon(s)| \le C\epsilon$, $|\partial_s^k(A_\epsilon(s)-1)| \le C\epsilon$ and $|\partial_s^k B_\epsilon(s)| \le C$ for all $|s| \le R$ and all $0 \le \epsilon \le 1$.*

Proof. Direct differentiation. $1-(1+x)^{-\alpha/2} \le \frac{\alpha}{2}x$ for $x \ge 0$ by concavity/mean value theorem. For the uniform bounds: $A'_\epsilon(s) = \epsilon \cdot g(\epsilon, s)$ with $g(\epsilon,s) = -\alpha s(1+\epsilon s^2)^{-\alpha/2-1}$ smooth on $[0,1]\times[-R,R]$, so every s-derivative of g is bounded on this compact set; similarly $A_\epsilon(s) - 1 = \int_0^s A'_\epsilon$ and $B_\epsilon(s) = (1+\epsilon s^2)^{\alpha/2}$ is jointly smooth in (ϵ, s). □

Lemma 3.6 (Quadratic factorisation of Γ_ϵ)**.** *For all $\epsilon \ge 0$ and $(u,v) \in \mathbb{R}^2$:*

$$\Gamma_\epsilon(u,v) = (u-v)^2 \widetilde{\Gamma}_\epsilon(u,v), \qquad \widetilde{\Gamma}_\epsilon(u,v) = -\frac{c_\alpha}{A_\epsilon(u)} \int_0^1 \big[1+p(1-t)\big] A'_\epsilon\big(u+t(v-u)\big)\, dt. \tag{3.2}$$

Moreover $|P_\epsilon(u,v)| \le \frac{1}{2}(v-u)^2 B_\epsilon(u) \sup_{[u,v]} |A'_\epsilon|$.

Proof. Put $g(s) = 1 - A_\epsilon(s)/A_\epsilon(u)$, so $g(u) = 0$, $g'(s) = -A'_\epsilon(s)/A_\epsilon(u)$. Substituting $s = u + t(v-u)$ and using $g(u+t(v-u)) = (v-u)\int_0^t g'(u+\tau(v-u))d\tau$ and Fubini,

$$P_\epsilon(u,v) = (v-u)\int_0^1 g(u+t(v-u))dt = (v-u)^2 \int_0^1 (1-\tau) g'(u+\tau(v-u))d\tau,$$

$$(v-u)g(v) = (v-u)^2 \int_0^1 g'(u+\tau(v-u))d\tau.$$

Hence $\Gamma_\epsilon/c_\alpha = (v-u)^2 \int_0^1 [1+p(1-\tau)] g'(u+\tau(v-u))d\tau$, which is the claim. The bound on P_ϵ follows from the first display. □

Lemma 3.7 (Kernel bounds; Remark "vanishing diagonal")**.** *Let $1 < \alpha < 2$ and $R \ge 1$. There are constants $C_j = C(j,R,\alpha)$ such that for all $0 \le \epsilon \le 1$ and all $|u|, |v| \le R$:*

1. $|\widetilde{\Gamma}_\epsilon(u,v)| \le C_0\epsilon(|u|+|v|)$ *and* $|\nabla^j\widetilde{\Gamma}_\epsilon(u,v)| \le C_j\epsilon$ *for all* $j \ge 0$*;*

2. *for every* $j \ge 0$*,*

$$|\nabla\Gamma_\epsilon(u,v)| \le C_1\epsilon|u-v|(|u|+|v|), \quad |\nabla^2\Gamma_\epsilon(u,v)| \le C_2\epsilon(|u|+|v|), \quad |\nabla^j\Gamma_\epsilon(u,v)| \le C_j\epsilon; \tag{3.3}$$

3. *for every* $j \ge 0$*,*

$$|(\partial_1+\partial_2)^j\Gamma_\epsilon(u,v)| \le C_j\epsilon|u-v|^2, \qquad |(\partial_1+\partial_2)\nabla\Gamma_\epsilon(u,v)| \le C_1\epsilon|u-v|. \tag{3.4}$$

Consequently, if $f \in W^{2,\infty}_{\mathrm{loc}}$*,* $f' \in L^\infty$*, then for* $|y| \le R$*,* $0 < |y-z| \le 2$*,* $|\Gamma_\epsilon(f'(y), S_f(y,z))|\,|y-z|^{-\alpha} \le C\epsilon|y-z|^{2-\alpha}$ *with* C *depending on* $R, \|f'\|_{L^\infty}, \|f''\|_{L^\infty(-R-2,R+2)}$*.*

Proof. (1) By Lemma 3.5, $|A'_\epsilon(w)| \le \alpha\epsilon|w| \le \alpha\epsilon(|u|+|v|)$ for w between u, v, and $B_\epsilon(u) \le (1+R^2)^{\alpha/2}$; this gives the first bound with $C_0 = c_\alpha(1+p)\alpha(1+R^2)^{\alpha/2}$. For derivatives, differentiate under the t-integral; every derivative falls on $B_\epsilon(u)$ or on A'_ϵ, and both are bounded with the factor ϵ coming from A'_ϵ (Lemma 3.5, applied on $[-R,R]$; note $u+t(v-u) \in [-R,R]$). (2) Write $\Gamma_\epsilon = (u-v)^2\widetilde{\Gamma}_\epsilon$ (Lemma 3.6). Then $\nabla\Gamma_\epsilon = 2(u-v)(1,-1)\widetilde{\Gamma}_\epsilon + (u-v)^2\nabla\widetilde{\Gamma}_\epsilon$; use (1) and $|u-v| \le |u|+|v|$. $\nabla^2\Gamma_\epsilon = 2(1,-1)^{\otimes 2}\widetilde{\Gamma}_\epsilon + 4(u-v)(1,-1)\otimes_s\nabla\widetilde{\Gamma}_\epsilon + (u-v)^2\nabla^2\widetilde{\Gamma}_\epsilon$; use (1) and $|u-v|^2 \le 2R(|u|+|v|)$. For $j \ge 3$ use Leibniz and (1). (3) $(\partial_1+\partial_2)(u-v)^2 = 0$, hence $(\partial_1+\partial_2)^j\Gamma_\epsilon = (u-v)^2(\partial_1+\partial_2)^j\widetilde{\Gamma}_\epsilon$, bounded by (1). Also $(\partial_1+\partial_2)\nabla\Gamma_\epsilon = \nabla[(u-v)^2(\partial_1+\partial_2)\widetilde{\Gamma}_\epsilon] = 2(u-v)(1,-1)(\partial_1+\partial_2)\widetilde{\Gamma}_\epsilon + (u-v)^2\nabla(\partial_1+\partial_2)\widetilde{\Gamma}_\epsilon$. Consequence: $|f'(y)-S_f(y,z)| = |\int_0^1 (f'(y)-f'(z+\tau(y-z)))d\tau| \le \frac{1}{2}\|f''\|_{L^\infty(-R-2,R+2)}|y-z|$, and $|f'(y)|, |S_f| \le \|f'\|_\infty$; apply (1) with R replaced by $\max(R, \|f'\|_\infty)$.

□

3.2.2 The semilinear form of the equation

Lemma 3.8 (Well-posedness and continuity of the terms of $\mathcal{E}_\epsilon$)**.** *Let* $\epsilon \ge 0$*,* $f \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ *with* $f' \in L^\infty(\mathbb{R})$*. Then:*

1. $|D|^p f'(y) = c_p\int_{\mathbb{R}} \frac{f'(y)-f'(y+z)}{|z|^{1+p}}dz$ *(Definition 1.5, Lemma 3.25) converges absolutely for every* y *and defines a continuous function;*

2. *the integral defining* $\widetilde{\mathcal{J}}_\epsilon(f)(y)$ *(Definition 1.22) and the integrals* $\int \frac{|f'(y)-f'(z)|}{|y-z|^\alpha}\left|\frac{A_\epsilon(S_f(y,z))}{A_\epsilon(f'(y))} - 1\right|dz$ *and* $\int \frac{|\Gamma_\epsilon(f'(y),S_f(y,z))|}{|y-z|^\alpha}dz$ *converge, locally uniformly in* y*, and* $\widetilde{\mathcal{J}}_\epsilon(f)$ *and* $\mathcal{I}_\epsilon(f)$ *are continuous;*

3. *consequently* $\mathcal{E}_\epsilon(f)$ *(Definition 1.26) is a continuous function on* $\mathbb{R}$*, and* $\mathcal{E}_\epsilon(f) = 0$ *in* $\mathcal{D}'(\mathbb{R})$ *if and only if* $\mathcal{E}_\epsilon(f)(y) = 0$ *for every* $y \in \mathbb{R}$*.*

Proof. Fix $R \ge 1$, $|y| \le R$; let $L = \|f''\|_{L^\infty(-R-2,R+2)}$, $K = \|f'\|_\infty$. (1) For $|z| \le 1$ the integrand is bounded by $L|z|^{-p}$, for $|z| \ge 1$ by $2K|z|^{-1-p}$; both integrable as $0 < p < 1$. Continuity in y follows from dominated convergence (the dominating function is uniform for $|y| \le R$). (2) For $|y-z| \le 2$: $|f'(y) - f'(z)| \le L|y-z|$ and, by the mean value theorem and $|A'_\epsilon| \le \alpha\epsilon\max(K,1)$ on $[-K,K]$, $|A_\epsilon(S_f) - A_\epsilon(f'(y))| \le C|S_f - f'(y)| \le CL|y-z|$; so the integrand

is $\le C|y-z|^{2-\alpha}$, integrable since $\alpha<3$. For $|y-z|\ge 2$ the integrands are $\le C|y-z|^{-\alpha}$, integrable as $\alpha>1$ ($A_\epsilon\le 1$, $B_\epsilon(f'(y))\le(1+\epsilon K^2)^{\alpha/2}$, Γ_ϵ bounded on $[-K,K]^2$). For Γ_ϵ near the diagonal use the consequence in Lemma 3.7 (the $\epsilon\le 1$ restriction there is only for uniformity; for fixed ϵ the same argument applies with ϵ-dependent constants). The integrands are continuous in y for $z\ne y$ ($f\in C^1$), so continuity follows by dominated convergence after the change of variables $z=y-t$. (3) $\mathcal{E}_\epsilon(f)=|D|^p f'+B_\epsilon(f')(f-yf'+\widetilde{\mathcal{J}}_\epsilon(f))$ is a sum of products of continuous functions. A continuous function vanishing as a distribution vanishes identically. □

Lemma 3.9 (Secant integration by parts identity)**.** *Let $f\in C^1(\mathbb{R})$, $y\in\mathbb{R}$, $u=f'(y)$. For $z\ne y$, $z\mapsto S_f(y,z)$ is C^1 with $\partial_z S_f(y,z)=\frac{S_f(y,z)-f'(z)}{y-z}$, hence*

$$(y-z)\partial_z P_\epsilon(u,S_f(y,z))=(S_f-u)\Big(1-\tfrac{A_\epsilon(S_f)}{A_\epsilon(u)}\Big)+(u-f'(z))\Big(1-\tfrac{A_\epsilon(S_f)}{A_\epsilon(u)}\Big),\quad(3.5)$$

with $S_f=S_f(y,z)$. Moreover $\partial_z\big((y-z)|y-z|^{-\alpha}\big)=p|y-z|^{-\alpha}$ for $z\ne y$.

Proof. Quotient rule; $\partial_v P_\epsilon(u,v)=1-A_\epsilon(v)/A_\epsilon(u)$ by the fundamental theorem of calculus; chain rule; and split $S_f-f'(z)=(S_f-u)+(u-f'(z))$. For the last identity write $w=y-z$: $\operatorname{sgn}(w)|w|^{1-\alpha}$ has w-derivative $(1-\alpha)|w|^{-\alpha}$ and $dw/dz=-1$. □

Proposition 3.10 (Semilinear kernel form)**.** *Let $1<\alpha<2$, $p=\alpha-1$, $\epsilon\ge 0$, $f\in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ with $f'\in L^\infty(\mathbb{R})$. Then for every $y\in\mathbb{R}$*

$$\frac{\widetilde{\mathcal{J}}_\epsilon(f)(y)}{A_\epsilon(f'(y))}=\mathcal{I}_\epsilon(f)(y),\qquad\text{hence}\qquad\mathcal{E}_\epsilon(f)=|D|^pf'+B_\epsilon(f')(f-yf')+\mathcal{I}_\epsilon(f).$$

Proof. Fix y, $u=f'(y)$, $S=S_f(y,z)$, $K=\|f'\|_\infty$, $L=\|f''\|_{L^\infty(y-1,y+1)}$. Step 1. Dividing the absolutely convergent integral (Lemma 3.8) by $A_\epsilon(u)>0$:

$$\frac{\widetilde{\mathcal{J}}_\epsilon(f)(y)}{A_\epsilon(u)}=c_\alpha\int_{\mathbb{R}}\frac{u-f'(z)}{|y-z|^\alpha}\Big(\frac{A_\epsilon(S)}{A_\epsilon(u)}-1\Big)dz.\qquad(3.6)$$

Step 2. By Lemma 3.9, for $z\ne y$ the integrand equals $c_\alpha|y-z|^{-\alpha}(S-u)(1-\frac{A_\epsilon(S)}{A_\epsilon(u)})-c_\alpha(y-z)|y-z|^{-\alpha}\partial_z P_\epsilon(u,S)$. The first term is absolutely integrable: $O(|y-z|^{2-\alpha})$ for $|y-z|\le 1$ (as $|S-u|\le L|y-z|/2$ and $|1-A_\epsilon(S)/A_\epsilon(u)|\le C|S-u|$) and $O(|y-z|^{-\alpha})$ at infinity. Hence so is the second. Step 3 (integration by parts on $\{\delta\le|y-z|\le M\}$). On each of $[y-M,y-\delta]$ and $[y+\delta,y+M]$, both factors are C^1; by Lemma 3.9

$$-\int_{\delta\le|y-z|\le M}\frac{y-z}{|y-z|^\alpha}\partial_z P_\epsilon(u,S)dz=p\int_{\delta\le|y-z|\le M}\frac{P_\epsilon(u,S)}{|y-z|^\alpha}dz-\Big[\frac{y-z}{|y-z|^\alpha}P_\epsilon(u,S)\Big]_{4\text{ endpoints}}.$$

Boundary terms: by Lemma 3.6, $|P_\epsilon(u,S)|\le C(S-u)^2\le CL^2|y-z|^2$, so the terms at $|y-z|=\delta$ are $O(\delta^{3-\alpha})\to 0$; $|P_\epsilon(u,S)|\le|S-u|(1+B_\epsilon(u))\le 2K(1+B_\epsilon(u))$, so the terms at $|y-z|=M$ are $O(M^{1-\alpha})=O(M^{-p})\to 0$. Step 4. Letting $\delta\to 0$, $M\to\infty$ (dominated convergence for the absolutely integrable pieces; $P_\epsilon(u,S)|y-z|^{-\alpha}$ is absolutely integrable by the same two bounds) gives $\widetilde{\mathcal{J}}_\epsilon(f)(y)/A_\epsilon(u)=\int|y-z|^{-\alpha}\Gamma_\epsilon(u,S)dz=\mathcal{I}_\epsilon(f)(y)$ by the definition of Γ_ϵ. Step 5. Substitute into $\mathcal{E}_\epsilon(f)=|D|^pf'+B_\epsilon(f')(f-yf'+\widetilde{\mathcal{J}}_\epsilon(f))$ (Definition 1.26). □

Definition 3.11 (Nonperturbative coefficient)**.** For real p, ϵ define

$$c_\epsilon = c_\alpha \int_0^1 \big[A_\epsilon(t) - 1\big](1-t)^{p-1}\, dt, \qquad \alpha = p + 1.$$

The integral is understood as the total Bochner integral in Lean; its role in the kernel and its quantitative bound for $0 < p < 1$, $0 \le \epsilon \le 1$ are proved separately below.

Lemma 3.12 (The nonperturbative coefficient)**.** *Let $1 < \alpha < 2$, $0 \le \epsilon \le 1$, $W(y) = \max\{-y, 0\}$ and let $\chi_+ \in C^\infty(\mathbb{R})$ with $\chi_+ = 1$ on $[2, \infty)$, $\chi_+ = 0$ on $(-\infty, 1]$. For $y > 1$ the integral $\int_{\mathbb{R}} \frac{\partial_1 \Gamma_\epsilon(W'(y), S_W(y,z))}{|y-z|^\alpha} dz$ converges absolutely; define $b_\epsilon(y)$ as $\chi_+(y)$ times this integral for $y > 1$, and $b_\epsilon(y) = 0$ for $y \le 1$. Then*

$$b_\epsilon(y) = c_\epsilon \chi_+(y) y^{-p} \ (y \in \mathbb{R}), \qquad c_\epsilon = c_\alpha \int_0^1 \big[(1+\epsilon t^2)^{-\alpha/2} - 1\big](1-t)^{p-1} dt,$$

and $|c_\epsilon| \le \frac{c_\alpha \alpha}{2p}\epsilon$. (In particular $c_0 = 0$.)

Proof. Step 1 (formula for $\partial_1 \Gamma_\epsilon$). Differentiating (3.1) in u:

$$\partial_1 \Gamma_\epsilon(u, v) = c_\alpha \Big(\frac{A_\epsilon(v)}{A_\epsilon(u)} - 1\Big) + c_\alpha \frac{A_\epsilon'(u)}{A_\epsilon(u)^2} \Big[p \int_u^v A_\epsilon(s) ds + (v-u) A_\epsilon(v)\Big]. \quad (3.7)$$

Since $A_\epsilon(0) = 1$, $A_\epsilon'(0) = 0$: $\partial_1 \Gamma_\epsilon(0, v) = c_\alpha(A_\epsilon(v) - 1)$; in particular $\partial_1 \Gamma_\epsilon(0, 0) = 0$. Step 2. Let $y > 1$; then $W'(y) = 0$. For $z \ge 0$, $W(z) = W(y) = 0$ so $S_W(y, z) = 0$ and the integrand vanishes. For $z < 0$, $S_W(y, z) = \frac{0-(-z)}{y-z} = \frac{z}{y-z} \in (-1, 0)$. The integrand is bounded by $c_\alpha \frac{\alpha}{2} \epsilon |y - z|^{-\alpha}$ (Lemma 3.5) and $|y - z| \ge y > 1$, so absolutely convergent. Step 3. Substitute $t = -z/(y-z) \in (0, 1)$, i.e. $z = -yt/(1-t)$, $y - z = y/(1-t)$, $dz = -y(1-t)^{-2} dt$; using that A_ϵ is even,

$$\int_{-\infty}^0 \frac{c_\alpha (A_\epsilon(\frac{z}{y-z}) - 1)}{|y-z|^\alpha} dz = c_\alpha \int_0^1 (A_\epsilon(t) - 1) \frac{(1-t)^\alpha}{y^\alpha} \frac{y}{(1-t)^2} dt = c_\epsilon y^{-p}.$$

Step 4. $|A_\epsilon(t) - 1| \le \frac{\alpha}{2} \epsilon t^2 \le \frac{\alpha}{2} \epsilon$ and $\int_0^1 (1-t)^{p-1} dt = 1/p$.

□

3.2.3 Smoothness of profiles: Hölder calculus

Lemma 3.13 (Product estimate at a fractional order below one)**.** *Let $0 < \theta < 1$, $I = [a, b]$ with $a \le b$, and $u, v \in C^\theta(I; \mathbb{C})$. Then $uv \in C^\theta(I)$ and*

$$\|uv\|_{C^\theta(I)} \le \|u\|_{L^\infty(I)} \|v\|_{C^\theta(I)} + \|v\|_{L^\infty(I)} \|u\|_{C^\theta(I)}.$$

Proof. Write $u(x)v(x) - u(x')v(x') = u(x)(v(x) - v(x')) + v(x')(u(x) - u(x'))$, divide by $|x - x'|^\theta$, and take suprema. The value norm is bounded by either of the two value-norm products on the right, so their sum also bounds it. □

Lemma 3.14 (Derivative recovery from a one-sided Taylor stencil)**.** *For every integer $k \ge 1$ and $0 < \theta < 1$, put $s = k + \theta$. There is $C = C(k, \theta) > 0$ such*

that for every interval $I = [a, b]$ of length at least 1, $g \in C^s(I; E)$ with values in a real normed space E that is complete (a Banach space, e.g. $\mathbb{C}$), $x \in I$, $0 < h \le 1/(2(k+1))$, and $0 \le i \le k$,

$$|g^{(i)}(x)| \le C\left(h^{-i}\|g\|_{L^\infty(I)} + h^{s-i}[g^{(k)}]_{C^\theta(I)}\right).$$

Endpoint derivatives are derivatives within I.

Proof. At least one of $[x, x+1/2]$ and $[x-1/2, x]$ lies in I. Choose its sign σ. Taylor's integral formula at x gives, for $j = 0, \dots, k$, $g(x+\sigma jh) = \sum_{i=0}^{k} (\sigma j)^i h^i g^{(i)}(x)/i! + E_j$ with $|E_j| \le C(k, \theta) h^s [g^{(k)}]_{C^\theta}$; for $j = 0$ the remainder is zero. Invert the fixed Vandermonde matrix $((\sigma j)^i/i!)_{0 \le i,j \le k}$. Its inverse norm depends only on k, and the two signs are conjugate by a diagonal sign matrix. This is the explicit finite-dimensional interpolation task required by the next lemma. □

Lemma 3.15 (Interpolation between the value norm and a noninteger Holder norm)**.** *For $s = k + \theta > 0$, $k \in \mathbb{N}$, $0 < \theta < 1$, there is $C = C(s) > 0$ such that for every compact interval I of length at least 1 and $g \in C^s(I; E)$, E a real normed space that is complete (e.g. $\mathbb{C}$), writing $U = \|g\|_\infty$ and $V = \|g\|_{C^s(I)}$, for $0 \le i \le k$,*

$$\|g^{(i)}\|_\infty \le C U^{1-i/s} V^{i/s}, \qquad [g^{(i)}]_{C^\theta(I)} \le C U^{1-(i+\theta)/s} V^{(i+\theta)/s}.$$

If $U = 0$, all the derivatives and seminorms in these inequalities vanish.

Proof. If $U > 0$, then $V \ge U$. Optimize the stencil bound at $h = \min\{1/(2(k+1)), (U/V)^{1/s}\}$; the cutoff case is absorbed using $V \ge U$. For $i < k$, split pairs of points at distance $(U/V)^{1/s}$: use the mean value theorem and the bound for $g^{(i+1)}$ at shorter distances, and $2\|g^{(i)}\|_\infty$ at longer distances. For $i = k$ use $[g^{(k)}]_{C^\theta} \le V$. The case $k = 0$ uses only this last observation.

□

Lemma 3.16 (Tame product estimate at every noninteger order)**.** *For every $s > 0$, $s \notin \mathbb{N}$, there is $C = C(s) > 0$ such that for every compact interval I of length at least 1 and $a, b \in C^s(I; \mathbb{C})$, $ab \in C^s(I)$ and*

$$\|ab\|_{C^s(I)} \le C\big(\|a\|_\infty \|b\|_{C^s(I)} + \|b\|_\infty \|a\|_{C^s(I)}\big).$$

Proof. Write $s = k + \theta$. Apply Leibniz to each derivative up to k and the fractional-order product estimate to each summand of $(ab)^{(k)}$. Interpolation bounds each top-order term by CX^tY^{1-t}, where $X = \|a\|_{C^s}\|b\|_\infty$, $Y = \|a\|_\infty \|b\|_{C^s}$, and $0 \le t \le 1$. Use $X^tY^{1-t} \le tX + (1-t)Y$; lower derivative terms are bounded the same way, using $\|a\|_\infty \le \|a\|_{C^s}$ and its analogue for b.

□

Lemma 3.17 (Composition at a fractional order below one)**.** *Let $0 < \theta < 1$, $K > 0$, $M \ge 0$, and let $G : \mathbb{R}^2 \to \mathbb{C}$ be C^1 with $|G(q)| + \|DG(q)\| \le M$ for $q \in [-K, K]^2$ (Euclidean operator norm). If $u, v \in C^\theta(I; \mathbb{R})$ and $|u|, |v| \le K$ on a compact interval I, then $G(u, v) \in C^\theta(I)$ and*

$$\|G(u,v)\|_{C^\theta(I)} \le M\big(1 + \|u\|_{C^\theta(I)} + \|v\|_{C^\theta(I)}\big).$$

Proof. Integrate DG along the segment between $(u(x), v(x))$ and $(u(x'), v(x'))$, which remains in the convex square. Divide the resulting difference bound by $|x - x'|^\theta$ and add the value bound. □

Lemma 3.18 (Composition by induction on the integer part of the Holder order)**.** *Fix $s = k + \theta$, $k \in \mathbb{N}$, $0 < \theta < 1$, $K > 0$, and $M \geq 0$. There is $C = C(s, K, M) > 0$, independent of the interval I of length at least 1, such that if $G \in C^\infty(\mathbb{R}^2; \mathbb{C})$, $\|D^j G\| \leq M$ on $[-K, K]^2$ for $0 \leq j \leq k + 2$, and $u, v \in C^s(I; \mathbb{R})$ with $|u|, |v| \leq K$, then*

$$G(u, v) \in C^s(I), \qquad \|G(u, v)\|_{C^s(I)} \leq C(1 + \|u\|_{C^s(I)} + \|v\|_{C^s(I)})^{k+1}.$$

Proof. Induct on k. For $k \geq 1$, use $(G(u, v))' = \partial_1 G(u, v)u' + \partial_2 G(u, v)v'$ and the induction hypothesis for each partial derivative of G at order $s - 1$. Apply the tame product bound at order $s - 1$, and $\|u'\|_{C^{s-1}} \leq \|u\|_{C^s}$, similarly for v. The value part of $G(u, v)$ is at most M; the derivative part has degree at most $k+1$ in $1 + \|u\|_{C^s} + \|v\|_{C^s}$. The norm identity $\|f\|_{C^s} = \|f\|_\infty + \|f'\|_{C^{s-1}}$ supplies the induction step. This route requires the interval derivative/norm API but no multivariate Faà di Bruno theorem.

□

Lemma 3.19 (Hölder product and composition estimates)**.** *Let $s > 0$, $s \notin \mathbb{N}$, and I a compact interval of length $|I| \geq 1$. There is $C = C(s)$ such that for $a, b \in C^s(I)$:*

1. $\|ab\|_{C^s(I)} \leq C(\|a\|_{L^\infty(I)}\|b\|_{C^s(I)} + \|a\|_{C^s(I)}\|b\|_{L^\infty(I)})$;
2. *if $G \in C^\infty(\mathbb{R}^2)$, $K > 0$, and $u, v \in C^s(I; \mathbb{R})$ with $|u|, |v| \leq K$ on I, then $\|G(u, v)\|_{C^s(I)} \leq C'(1+\|u\|_{C^s(I)}+\|v\|_{C^s(I)})^{\lceil s \rceil}$ with $C' = C'(s, K, \|G\|_{C^{\lceil s \rceil+1}([-K,K]^2)})$.*

Proof. (1) Write $s = k + \theta$. Interpolation (Gagliardo–Nirenberg on intervals of length ≥ 1): $\|g^{(i)}\|_{L^\infty(I)} \leq C\|g\|_{L^\infty}^{1-i/s}\|g\|_{C^s}^{i/s}$ for $0 \leq i \leq k$, and the analogous bound for the θ-seminorm of $g^{(i)}$ with exponent $(i + \theta)/s$. Expand $(ab)^{(k)}$ by Leibniz; each term $a^{(i)}b^{(k-i)}$ and its θ-Hölder quotient is bounded by products $\|a\|_\infty^{1-\sigma}\|a\|_{C^s}^{\sigma}\|b\|_\infty^{\sigma}\|b\|_{C^s}^{1-\sigma}$, $\sigma \in [0, 1]$, and $X^\sigma Y^{1-\sigma} \leq X + Y$ with $X = \|a\|_{C^s}\|b\|_\infty$, $Y = \|a\|_\infty\|b\|_{C^s}$. Lower derivatives are handled the same way. (2) Faà di Bruno for $\partial^i[G(u, v)]$, $i \leq k$, and the mean value theorem for the Hölder quotient of the top derivative; all derivatives of G are evaluated in $[-K, K]^2$. □

Lemma 3.20 (Integrals of Hölder families)**.** *Let $s = k + \theta \notin \mathbb{N}$, I a compact interval, (T, ν) a σ-finite measure space and $H : I \times T \to \mathbb{C}$ such that for ν-a.e. τ, $H(\cdot, \tau) \in C^s(I)$, the maps $(x, \tau) \mapsto \partial_x^i H(x, \tau)$, $i \leq k$, are measurable, and $\int_T \|H(\cdot, \tau)\|_{C^s(I)} d\nu(\tau) < \infty$. Then $F(x) = \int_T H(x, \tau)d\nu(\tau)$ lies in $C^s(I)$ with $\|F\|_{C^s(I)} \leq \int_T \|H(\cdot, \tau)\|_{C^s(I)} d\nu$.*

Proof. Differentiation under the integral sign k times (dominated by $\|H(\cdot, \tau)\|_{C^s}$, integrable), then the triangle inequality for the Hölder quotient of $F^{(k)}$. □

Lemma 3.21 (Hölder calculus on intervals of length at least ℓ)**.** *Fix $\ell \in (0, 1]$. The statements of Lemma 3.14, Lemma 3.15, Lemma 3.16 and Lemma 3.18 hold verbatim for all compact intervals $I = [a, b]$ of length $b - a \geq \ell$ (instead of ≥ 1), with constants depending additionally on ℓ (and not on I), where in the stencil*

lemma the step range is $0 < h \le \ell/(2(k+1))$ and in the composition lemma the exponent is $\lfloor s \rfloor + 1$ and $\|D^j G\| \le M$ on $[-K,K]^2$ for $j \le \lfloor s \rfloor + 2$. The four lemmas are the case $\ell = 1$ (in Lean they are derived from these variants).

Proof. Repeat the proofs of the four lemmas with 1 replaced by ℓ: in the stencil one of $[x, x+\ell/2]$, $[x-\ell/2, x]$ lies in I; the interpolation cutoff becomes $h = \min\{\ell/(2(k+1)), (U/V)^{1/s}\}$, which only changes the constants by powers of ℓ; the product and composition estimates are then derived exactly as before. (These variants are what the near/far regularity lemmas below need on $[-R,R]$ with $R < 1/2$.) □

Lemma 3.22 (Regularity of the near part of $\mathcal{I}_\epsilon$)**.** *Let $\epsilon \ge 0$, $f \in W^{2,\infty}_{\mathrm{loc}}$, $f' \in L^\infty$, $s \notin \mathbb{N}$, $s > 0$, and suppose $f' \in C^s_{\mathrm{loc}}$. Let $\chi \in C^\infty_c(\mathbb{R})$ with $\chi = 1$ on $[-1,1]$, $\operatorname{supp}\chi \subset [-2,2]$, $0 \le \chi \le 1$. Then*

$$\mathcal{I}^{\mathrm{near}}_\epsilon(f)(y) := \int_{\mathbb{R}} \frac{\chi(t)\Gamma_\epsilon(f'(y), S_f(y, y-t))}{|t|^\alpha}\, dt$$

lies in C^s_{loc}, with $\|\mathcal{I}^{\mathrm{near}}_\epsilon(f)\|_{C^s([-R,R])} \le C$ where C depends on $s, R, \epsilon, \alpha, \|f'\|_{L^\infty}$, $\|f''\|_{L^\infty(-R-2,R+2)}$ and $\|f'\|_{C^s([-R-2,R+2])}$.

Proof. Fix $R \ge 1$, $I = [-R,R]$, $J = [-R-2, R+2]$, $0 < |t| \le 2$. Put $v_t(y) = S_f(y, y-t) = \int_0^1 f'(y - \tau t)\, d\tau$ and $d_t = f' - v_t$. Step 1. $\|v_t\|_{C^s(I)} \le \|f'\|_{C^s(J)}$ (differentiate under the τ-integral; translation invariance of the Hölder quotient) and $|v_t| \le \|f'\|_\infty$. Step 2. $\|d_t\|_{L^\infty(I)} \le \frac{1}{2}\|f''\|_{L^\infty(J)}|t|$ and $\|d_t\|_{C^s(I)} \le 2\|f'\|_{C^s(J)}$. Step 3. By Lemma 3.6, $\Gamma_\epsilon(f', v_t) = d_t^2\, \widetilde{\Gamma}_\epsilon(f', v_t)$. By Lemma 3.19(2) with $K = \|f'\|_\infty$, $\|\widetilde{\Gamma}_\epsilon(f', v_t)\|_{C^s(I)} \le C$ uniformly in t; by (1) twice, $\|d_t^2\|_{C^s} \le 2C\|d_t\|_\infty \|d_t\|_{C^s} \le C|t|$ and $\|\Gamma_\epsilon(f', v_t)\|_{C^s(I)} \le C(\|d_t\|^2_\infty \|\widetilde{\Gamma}_\epsilon\|_{C^s} + \|d_t^2\|_{C^s}\|\widetilde{\Gamma}_\epsilon\|_\infty) \le C|t|$. Step 4. Apply Lemma 3.20 with $T = [-2,2] \setminus \{0\}$, $d\nu = \chi(t)|t|^{-\alpha} dt$: $\int_{|t|\le 2} |t|^{1-\alpha} dt = \int |t|^{-p} < \infty$. Measurability: $(y,t) \mapsto \partial_y^i H$ is continuous for $t \ne 0$.

□

Lemma 3.23 (Regularity of the far part of $\mathcal{I}_\epsilon$)**.** *Under the hypotheses of Lemma 3.22, $\mathcal{I}^{\mathrm{far}}_\epsilon(f) := \mathcal{I}_\epsilon(f) - \mathcal{I}^{\mathrm{near}}_\epsilon(f)$, i.e. $\mathcal{I}^{\mathrm{far}}_\epsilon(f)(y) = \int (1 - \chi(y - z))\Gamma_\epsilon(f'(y), S_f(y,z))|y-z|^{-\alpha} dz$, lies in C^s_{loc} with $\|\mathcal{I}^{\mathrm{far}}_\epsilon(f)\|_{C^s([-R,R])} \le C(s, R, \epsilon, \alpha, \|f'\|_{L^\infty}, \|f'\|_{C^s([-R,R])})$.*

Proof. Let $\widetilde{\chi} \in C^\infty_c$, $\widetilde{\chi} = 1$ on $[-\frac{1}{2}, \frac{1}{2}]$, $\operatorname{supp}\widetilde{\chi} \subset [-1,1]$, and $\psi(t) = (1 - \widetilde{\chi}(t))/t$, which is smooth with $|\psi^{(i)}(t)| \le C_i \langle t\rangle^{-1-i}$. Step 1. $V_z(y) := (f(y) - f(z))\psi(y-z)$ equals $S_f(y,z)$ when $|y - z| \ge 1$, satisfies $|V_z| \le \|f'\|_\infty$, and $\|V_z\|_{C^s(I)} \le C(\|f'\|_\infty + \|f'\|_{C^s(I)})$ uniformly in $z \in \mathbb{R}$: by Leibniz, $\partial_y^i V_z = \sum_{l<i} \binom{i}{l} f^{(i-l)}(y)\psi^{(l)}(y-z) + (f(y) - f(z))\psi^{(i)}(y-z)$ and $|f(y) - f(z)||\psi^{(i)}(y - z)| \le C\|f'\|_\infty$; the Hölder quotient is treated likewise with the mean value theorem. Step 2. $\omega(t) := (1 - \chi(t))|t|^{-\alpha}$ is smooth with $|\omega^{(i)}(t)| \le C_i \langle t\rangle^{-\alpha - i}$, and $\langle y - z\rangle^{-\alpha} \le 2^\alpha \langle R\rangle^\alpha \langle z\rangle^{-\alpha}$ for $|y| \le R$ (Peetre). Step 3. $H(y,z) = \Gamma_\epsilon(f'(y), V_z(y))\, \omega(y-z)$ equals the far integrand. By Lemma 3.19, $\|H(\cdot, z)\|_{C^s(I)} \le C\langle z\rangle^{-\alpha}$, integrable in z as $\alpha > 1$. Conclude with Lemma 3.20.

□

Lemma 3.24 (Regularity of the nonlinear term)**.** *Let $\epsilon \ge 0$, $f \in W^{2,\infty}_{\mathrm{loc}}$ with $f' \in L^\infty$, and $s > 0$, $s \notin \mathbb{N}$. If $f' \in C^s_{\mathrm{loc}}$ then $\mathcal{I}_\epsilon(f) \in C^s_{\mathrm{loc}}$ and $B_\epsilon(f')(f - yf') \in C^s_{\mathrm{loc}}$.*

Proof. $\mathcal{I}_\epsilon = \mathcal{I}_\epsilon^{\mathrm{near}} + \mathcal{I}_\epsilon^{\mathrm{far}}$ (Lemma 3.8 gives absolute convergence so the split is legitimate). For the local term: $f \in C^{s+1}_{\mathrm{loc}} \subset C^s_{\mathrm{loc}}$, $y \mapsto y$ smooth, B_ϵ smooth; use Lemma 3.19 on each $[-R, R]$.

□

3.2.4 Smoothness of profiles: the linear gain and the bootstrap

Lemma 3.25 (Singular-kernel formula for $|D|^p$)**.** *Let $0 < p < 1$ and $c_p := \big(\int_{\mathbb{R}} \frac{1-\cos z}{|z|^{1+p}} dz\big)^{-1} \in (0, \infty)$. Then $c_p \int_{\mathbb{R}} \frac{1-e^{i\lambda z}}{|z|^{1+p}} dz = |\lambda|^p$ for all $\lambda \in \mathbb{R}$ (absolutely convergent), and for every $g \in \mathcal{S}(\mathbb{R})$ (or $g \in C^1(\mathbb{R})$ with g, g' bounded and $\hat{g}$ defined as a tempered distribution, as in Definition 1.5)*

$$|D|^p g(y) = c_p \int_{\mathbb{R}} \frac{g(y) - g(y+z)}{|z|^{1+p}} dz.$$

Proof. The odd part $\sin(\lambda z)|z|^{-1-p}$ integrates to 0; scaling $z \mapsto z/|\lambda|$ gives $|\lambda|^p/c_p$ for the even part. For $g \in \mathcal{S}$: $g(y) - g(y+z) = \frac{1}{2\pi} \int \hat{g}(\xi) e^{iy\xi}(1 - e^{iz\xi}) d\xi$; Fubini is justified by $\int |1 - e^{iz\xi}||z|^{-1-p} dz = C|\xi|^p$ and $\hat{g} \in \mathcal{S}$. For bounded C^1 functions this is the definition/characterisation in Definition 1.5. □

Lemma 3.26 (Interior Schauder estimate for $|D|^p$)**.** *Let $0 < p < 1$, $\beta > 0$ with $\beta \notin \mathbb{N}$ and $\beta + p \notin \mathbb{N}$. There is $C = C(p, \beta)$ such that for every $w \in C^\infty(\mathbb{R})$ with $w \in C^\beta(\mathbb{R})$ (global Hölder norm finite),*

$$\|w\|_{C^{\beta+p}([-\frac{1}{2}, \frac{1}{2}])} \le C\big(\|w\|_{C^\beta(\mathbb{R})} + \||D|^p w\|_{C^\beta([-1,1])}\big).$$

Proof. Use the fixed cutoff and the construction of Chapter 11. The localized right-hand side is globally C^β with the required quantitative bound; apply the compact-input estimate and restrict to $[-1/2, 1/2]$. The target is Proposition 2.2 of the arXiv version of [ROS14], with $n = 1$, $s = p/2$. Corollary 2.4 in that version is the related weighted-tail estimate. □

Remark 3.27 (A direct proof). The proof in Chapter 11 is a direct one-dimensional argument and avoids the full boundary regularity theory. The following lemmas describe the use of this estimate in the smoothness bootstrap.

Lemma 3.28 (Localization of the fractional operator)**.** *Fix $0 < p < 1$, $s > 0$, $s \notin \mathbb{N}$, and $\eta \in C_c^\infty(\mathbb{R}; [0, 1])$, $\eta = 1$ on $[-2, 2]$, $\operatorname{supp} \eta \subset [-3, 3]$. Suppose g is bounded and locally Lipschitz, $g \in C^s_{\mathrm{loc}}(\mathbb{R})$, and $h = |D|^p g \in C^s_{\mathrm{loc}}(\mathbb{R})$, with the defining singular integral absolutely convergent. Then $v = \eta g$ is compactly supported and globally Lipschitz, $v \in C^s(\mathbb{R})$, and on $[-1, 1]$*

$$|D|^p v = h - T, \qquad T(y) = -c_p \int_{\mathbb{R}} (1 - \eta(z)) g(z) |z - y|^{-1-p} \, dz.$$

The displayed integral and all its y derivatives converge absolutely on $[-1, 1]$; T is smooth on $(-2, 2)$ and $\sup_{|y| \le 1} |T^{(j)}(y)| \le C(p, j, \eta)\|g\|_\infty$. In particular $|D|^p v \in C^s([-1, 1])$.

Proof. The cutoffs are constant on a neighbourhood of the evaluation interval. Split $g = \eta g + (1-\eta)g$ in the absolutely convergent kernel formula. For the tail, differentiate only $|z-y|^{-1-p}$; on $|y| \le 1$ its j-th derivative is bounded by $C(p,j)\mathbf{1}_{|z|\ge 2}|z|^{-1-p-j}$, an integrable majorant. Product estimates on $[-4,4]$, and zero extension across the region where $\eta = 0$, give the global Holder bound for v. □

Lemma 3.29 (Mollification commutes with the fractional operator)**.** *Fix $0 < p < 1$, $s > 0$, $s \notin \mathbb{N}$, and a nonnegative $\rho \in C_c^\infty((-1,1))$ with $\int \rho = 1$. Let $v : \mathbb{R} \to \mathbb{R}$ be compactly supported, globally Lipschitz, and in $C^s(\mathbb{R})$, with $|D|^p v \in C^s([-1,1])$. For $0 < \delta < 1/2$, set $\rho_\delta(x) = \delta^{-1}\rho(x/\delta)$ and $w_\delta = v * \rho_\delta$. Then $w_\delta \in C_c^\infty(\mathbb{R})$, $w_\delta \to v$ uniformly as $\delta \downarrow 0$, and*

$$|D|^p w_\delta = (|D|^p v) * \rho_\delta, \quad \|w_\delta\|_{C^s(\mathbb{R})} \le \|v\|_{C^s(\mathbb{R})}, \quad \||D|^p w_\delta\|_{C^s([-1/2,1/2])} \le \||D|^p v\|_{C^s([-1,1])}.$$

All convolutions and singular integrals in the identity converge absolutely.

Proof. In the double integral, dominate $|v(x-z)-v(x)|$ by $\min\{\mathrm{Lip}(v)|z|, 2\|v\|_\infty\}$ and integrate against $|z|^{-1-p}\rho_\delta$. Tonelli and Fubini justify commutation. Differentiation of the smooth compactly supported mollifier gives smoothness; averaging translates of v and of $|D|^p v$ gives the norm bounds. Uniform continuity of v gives uniform convergence, including outside its support. □

Lemma 3.30 (Closedness of a bounded Holder ball under uniform convergence)**.** *Let $s = k + \theta > 0$, $k \in \mathbb{N}$, $0 < \theta < 1$, $I = [a,b]$ with $a < b$, $M \ge 0$, and let E be a real normed space that is complete (e.g. $\mathbb{R}$ or $\mathbb{C}$). If $f_n \in C^s(I;E)$, $\|f_n\|_{C^s(I)} \le M$ for every n, and $f_n \to f$ uniformly on I, then $f \in C^s(I)$ and $\|f\|_{C^s(I)} \le M$.*

Proof. The top derivatives are equicontinuous by their common θ-Holder bound; lower derivatives are equicontinuous by the next derivative bound. Arzelà–Ascoli, applied finitely many times, gives a subsequence with uniform convergence of derivatives through order k. Pass to the limit in $f_n^{(i)}(x) - f_n^{(i)}(a) = \int_a^x f_n^{(i+1)}$ to identify the limits as the derivatives of f. For every finite collection of evaluation points and one Holder quotient, pass to the limit in their sum bounded by M; taking independent suprema gives the bound for the full norm, without multiplying M by the number of derivatives. □

Lemma 3.31 (Translation invariance of $|D|^p$)**.** *For every $p \in \mathbb{R}$, every $g : \mathbb{R} \to \mathbb{R}$ and all $x_0, y \in \mathbb{R}$, $|D|^p[g(\cdot + x_0)](y) = (|D|^p g)(y + x_0)$, where $|D|^p$ is the singular-integral expression $c_\alpha \int_{\mathbb{R}} \frac{g(y)-g(z)}{|y-z|^{1+p}}\,dz$ of Definition 1.5 (Lean `fracD`; both sides are the Bochner integral, so the identity holds also when the integral diverges).*

Proof. Substitute $z \mapsto z + x_0$ (translation invariance of Lebesgue measure) in the defining integral. □

Lemma 3.32 (Gluing local Hölder bounds)**.** *Let $s' > 0$, $s' \notin \mathbb{N}$, and $g : \mathbb{R} \to \mathbb{R}$. If $g \in C^{s'}([x_0 - \frac{1}{2}, x_0 + \frac{1}{2}])$ for every $x_0 \in \mathbb{R}$, then $g \in C^{s'}_{\mathrm{loc}}(\mathbb{R})$.*

Proof. g is $C^{\lfloor s' \rfloor}$ near every point, hence on $\mathbb{R}$, and the derivatives within the small intervals agree with the global ones. On a compact $[a,b]$ the values of the

derivatives are bounded by covering $[a,b]$ with finitely many intervals of length 1; for the Hölder quotient of $g^{(\lfloor s' \rfloor)}$ use the local Hölder constant of the interval centred at x when $|x-x'| \le \frac{1}{2}$, and the sup bound ($\le 8 \sup |g^{(\lfloor s' \rfloor)}|\,|x-x'|^{\theta}$) when $|x-x'| > \frac{1}{2}$. □

Lemma 3.33 (Localization of the fractional operator, wide version)**.** *Under the hypotheses of Lemma 3.28 (*$0 < p < 1$, $s > 0$, $s \notin \mathbb{N}$, $\eta \in C^\infty(\mathbb{R};[0,1])$, $\eta = 1$ *on* $[-2,2]$, $\eta = 0$ *for* $|x| > 3$*; g bounded, locally Lipschitz,* $g \in C^s_{\mathrm{loc}}(\mathbb{R})$, $z \mapsto \frac{g(y)-g(z)}{|y-z|^{1+p}}$ *integrable for every y, and* $|D|^p g \in C^s_{\mathrm{loc}}(\mathbb{R})$*), one has* $|D|^p(\eta g) \in C^s([-\frac{3}{2},\frac{3}{2}])$.

Proof. As for the localization lemma above (with a wider evaluation interval): on $[-\frac{3}{2},\frac{3}{2}]$, $|D|^p(\eta g) = |D|^p g - T$ with the tail $T(y) = -c_p \int (1-\eta(z))g(z)|z-y|^{-1-p}dz$ ($c_p = c_\alpha$ by Lemma 3.25). Since $|z-y| \ge \frac{1}{2}$ on the support of $1-\eta$, T is smooth on a neighbourhood of $[-\frac{3}{2},\frac{3}{2}]$, with integrable majorants $C\|g\|_\infty \mathbf{1}_{|z|\ge 2}\langle z\rangle^{-1-p-j}$ for its derivatives; its C^s norm is then bounded via Lemma 3.20.

□

Lemma 3.34 (Mollification, wide version)**.** *Under the hypotheses of Lemma 3.29, but with* $|D|^p v \in C^s([-\frac{3}{2},\frac{3}{2}])$ *in place of* $C^s([-1,1])$*, for every* $0 < \delta < \frac{1}{2}$ *the mollification* $w_\delta = v * \rho_\delta$ *satisfies* $w_\delta \in C^s(\mathbb{R})$, $|D|^p w_\delta \in C^s([-1,1])$ *and* $\||D|^p w_\delta\|_{C^s([-1,1])} \le \||D|^p v\|_{C^s([-\frac{3}{2},\frac{3}{2}])}$.

Proof. Lemma 3.29 (applied with the restriction of $|D|^p v$ to $[-1,1]$) gives smoothness and $|D|^p w_\delta = (|D|^p v) * \rho_\delta$. For $|x| \le 1$ and $\delta < \frac{1}{2}$ the convolution only samples $|D|^p v$ on $[-\frac{3}{2},\frac{3}{2}]$; averaging the translates (each with $C^s([-1,1])$ norm at most $\||D|^p v\|_{C^s([-\frac{3}{2},\frac{3}{2}])}$) against the probability density ρ_δ gives the bound. □

Lemma 3.35 (Regularity gain for the linear part)**.** *Let* $0 < p < 1$, $s > 0$ *with* $s \notin \mathbb{N}$, $s+p \notin \mathbb{N}$. *Let* $g \in L^\infty(\mathbb{R}) \cap W^{1,\infty}_{\mathrm{loc}}(\mathbb{R}) \cap C^s_{\mathrm{loc}}$ *and suppose the (pointwise, absolutely convergent) function* $h := |D|^p g$ *lies in* C^s_{loc}. *Then* $g \in C^{s+p}_{\mathrm{loc}}$.

Proof.

Lean route (no dilation): for each x_0 translate g (Lemma 3.31), localize with Lemma 3.33, mollify with Lemma 3.34, apply Lemma 3.26 with $\beta = s$ uniformly in $\delta_n = 1/(n+3)$, pass to the limit with Lemma 3.30 to get $g \in C^{s+p}([x_0 - \frac{1}{2}, x_0 + \frac{1}{2}])$, and conclude with Lemma 3.32. The argument with dilations is as follows. By translation and dilation invariance ($|D|^p[g(x_0 + r\,\cdot)] = r^p(|D|^p g)(x_0 + r\,\cdot)$) it suffices to show $g \in C^{s+p}([-\frac{1}{8},\frac{1}{8}])$ for g satisfying the hypotheses; covering a compact interval by finitely many such neighbourhoods then gives the claim. Step 1 (localisation). Let $\eta \in C^\infty_c$, $\eta = 1$ on $[-2,2]$, $\operatorname{supp}\eta \subset [-3,3]$. Then $\eta g \in C^s(\mathbb{R})$ (compact support) and, pointwise for $|y| < 1$, $|D|^p(\eta g)(y) = h(y) - |D|^p((1-\eta)g)(y)$ with $|D|^p((1-\eta)g)(y) = -c_p \int (1-\eta(w))g(w)|w-y|^{-1-p}dw$. The latter is C^∞ on $(-1,1)$, with k-th derivative bounded by $C_k\|g\|_{L^\infty}$ (as $|w-y| \ge 1$ on the support; differentiate under the integral). Hence $|D|^p(\eta g) \in C^s([-1,1])$. Step 2 (mollification). Let ρ_δ be a standard mollifier and $w_\delta = (\eta g) * \rho_\delta \in C^\infty_c$. Then $|D|^p w_\delta = (|D|^p(\eta g)) * \rho_\delta$, so $\||D|^p w_\delta\|_{C^s([-1/2,1/2])} \le \||D|^p(\eta g)\|_{C^s([-1,1])}$ for $\delta < 1/2$, and $\|w_\delta\|_{C^s(\mathbb{R})} \le \|\eta g\|_{C^s(\mathbb{R})}$. Step 3. Apply Lemma 3.26 (rescaled to the pair of intervals $[-\frac{1}{4},\frac{1}{4}] \subset [-\frac{1}{2},\frac{1}{2}]$): $\|w_\delta\|_{C^{s+p}([-1/4,1/4])} \le C$ uniformly in

δ. Step 4. $w_\delta \to \eta g = g$ uniformly on $[-\frac{1}{4}, \frac{1}{4}]$; by Arzelà–Ascoli the derivatives up to order $\lfloor s+p \rfloor$ converge uniformly on $[-\frac{1}{8}, \frac{1}{8}]$ and the Hölder bound passes to the limit. So $g \in C^{s+p}([-\frac{1}{8}, \frac{1}{8}])$.

□

Theorem 3.36 (Smoothness of profiles)**.** *Let $0 < p < 1$, $\alpha = p + 1$, $\epsilon \geq 0$. Suppose $f \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$, $f' \in L^\infty(\mathbb{R})$, and $\mathcal{E}_\epsilon(f) = 0$ in $\mathcal{D}'(\mathbb{R})$ (equivalently, by Lemma 3.8, pointwise everywhere). Then $f \in C^\infty(\mathbb{R})$. In particular, for ϵ small enough the profile $f = \overline{f}_{\epsilon,\mu_\epsilon} + r_\epsilon$ and the correction r_ϵ of Theorem 2.25 are C^∞.*

Proof. Step 0 (choice of exponents). The set $\{k - np : k, n \in \mathbb{N}_0\}$ is countable; pick $s_0 \in (0, 1)$ outside it, so $s_n := s_0 + np \notin \mathbb{N}$ for all $n \geq 0$. Step 1. By Lemma 3.8 and Proposition 3.10, for every y: $|D|^p f'(y) = -B_\epsilon(f'(y))(f(y) - yf'(y)) - \mathcal{I}_\epsilon(f)(y)$. Step 2 (base). f' is locally Lipschitz, so $f' \in C^{s_0}_{\mathrm{loc}}$. Step 3 (induction). Suppose $f' \in C^{s_n}_{\mathrm{loc}}$. By Lemma 3.24, the right side of Step 1 is in $C^{s_n}_{\mathrm{loc}}$. Apply Lemma 3.35 with $g = f'$ (bounded, $W^{1,\infty}_{\mathrm{loc}}$, $C^{s_n}_{\mathrm{loc}}$; $s_n, s_{n+1} \notin \mathbb{N}$) to get $f' \in C^{s_{n+1}}_{\mathrm{loc}}$. Step 4. Since $s_n \to \infty$, $f' \in C^k(\mathbb{R})$ for every k, so $f \in C^\infty$. Step 5 (application). By Theorem 2.20, $\overline{f} = \overline{f}_{\epsilon,\mu_\epsilon} \in C^\infty$ with $\overline{f}'$ bounded (bounds (2) there with $j = 0$). By Definition 2.3, $r_\epsilon \in W^{2,\infty}_{\mathrm{loc}}$ and for $y \geq 0$, $|r_\epsilon'| \leq \|r_\epsilon\|_{X_\gamma} \langle y \rangle^{1/p-\gamma} \leq \|r_\epsilon\|_{X_\gamma}$ since $\gamma = \gamma_2 > \beta + 1/p > 1/p$; for $y \leq 0$, $|r_\epsilon'| \leq \|r_\epsilon\|_{X_\gamma}$. So f satisfies the hypotheses. By definition of $\mathcal{N}_{\epsilon,\overline{f}}$ (Definition 2.18), $\mathcal{E}_\epsilon(\overline{f} + r) = \mathcal{E}_\epsilon(\overline{f}) + \mathcal{L}_\epsilon r + \mathcal{N}_{\epsilon,\overline{f}}(r)$, which vanishes by (2.6). Hence $f \in C^\infty$ and $r_\epsilon = f - \overline{f} \in C^\infty$. □

3.3 Appendix: elementary estimates

3.3.1 Fourier inverse of the non-smooth symbols $\chi\Phi_0^k$

Lemma 3.37 (Symbol bounds for $\xi^j \chi \Phi_0^k$)**.** *Let $0 < p < 1$, $\Phi_0(\xi) = \xi|\xi|^p/(p+1)$, and $\chi \in C^\infty_c(\mathbb{R})$ with $\operatorname{supp}\chi \subset [-2, 2]$, $\chi = 1$ on $[-1, 1]$. For $k \geq 1$, $j \in \{0, 1, 2\}$ let $a_{k,j}(\xi) = (i\xi)^j \chi(\xi)\Phi_0(\xi)^k$. Then $a_{k,j} \in C^{j+1}(\mathbb{R}) \cap C^\infty(\mathbb{R} \setminus \{0\})$, $\operatorname{supp} a_{k,j} \subset [-2, 2]$, $\partial^l a_{k,j}(0) = 0$ for $l \leq j + 1$, and there are $C_1 = C_1(p, \chi)$, $C = C(p)$ with*

$$|\partial^l a_{k,j}(\xi)| \leq C_1 C^k |\xi|^{p+1+j-l}, \qquad 0 < |\xi| \leq 2,\ 0 \leq l \leq j + 3.$$

Proof. For $\xi > 0$, $\xi^j \Phi_0^k = (p+1)^{-k} \xi^e$ with $e = j + (p+1)k$; for $\xi < 0$ it is $\pm(p+1)^{-k}|\xi|^e$. Hence for $\xi \neq 0$, $|\partial^l(\xi^j \Phi_0^k)| \leq e^l |\xi|^{e-l} \leq (5+2k)^5 |\xi|^{p+1+j-l} 2^{(p+1)(k-1)}$ using $|\xi| \leq 2$. Leibniz with χ: $|\xi|^{p+1+j-i} \leq 2^{l-i}|\xi|^{p+1+j-l}$. Absorb $(5 + 2k)^5 4^k$ into C^k. Regularity at 0: by induction on $l \leq j + 1$ (note $p + 1 + j - l \geq p > 0$), $\partial^{l-1} a_{k,j}(\xi)/\xi \to 0$, so $\partial^l a_{k,j}(0)$ exists and equals 0, and $\partial^l a_{k,j}(\xi) \to 0$ as $\xi \to 0$. □

Lemma 3.38 (Decay of the inverse Fourier transform of a symbol with a power singularity)**.** *Let $0 < p < 1$, $n \geq 1$, $M \geq 0$. Let $a : \mathbb{R} \to \mathbb{C}$ with $\operatorname{supp} a \subset [-2, 2]$, $a \in C^{n-1}(\mathbb{R}) \cap C^{n+1}(\mathbb{R} \setminus \{0\})$, $\partial^l a(0) = 0$ for $l \leq n - 1$, and $|\partial^l a(\xi)| \leq M|\xi|^{n-1+p-l}$ for $0 < |\xi| \leq 2$, $0 \leq l \leq n + 1$. Then for $|y| \geq 1$, $|\check{a}(y)| \leq C(n, p)M|y|^{-n-p}$, and $|\check{a}(y)| \leq C(n, p)M$ for all y.*

Proof. The second bound: $|\check{a}| \leq \frac{1}{2\pi} \int_{-2}^{2} M|\xi|^{n-1+p} d\xi$. Step 1. Integrate by parts n times: $\check{a}(y) = \frac{1}{2\pi}(-iy)^{-n} \int e^{iy\xi} \partial^n a(\xi) d\xi$; the first $n-1$ steps use $a \in C^{n-1}$ and

compact support; the n-th uses that $\partial^{n-1}a$ is absolutely continuous (it is C^1 off 0, continuous at 0, and $|\partial^n a| \le M|\xi|^{p-1} \in L^1$). Step 2. $\left|\int_{|\xi|\le|y|^{-1}} e^{iy\xi}\partial^n a\right| \le \int_{|\xi|\le|y|^{-1}} M|\xi|^{p-1} = \frac{2M}{p}|y|^{-p}$. Step 3. On $\{|y|^{-1} \le |\xi| \le 2\}$ integrate by parts once more: boundary terms at $\pm|y|^{-1}$ are $\le |y|^{-1}M|y|^{1-p}$ each (those at ± 2 vanish), and $|y|^{-1}\int_{|\xi|\ge|y|^{-1}} M|\xi|^{p-2} \le \frac{2M}{1-p}|y|^{-p}$. Combine: $|\check{a}(y)| \le CM|y|^{-n}|y|^{-p}$. □

Proposition 3.39 (Non-smooth Fourier inverse)**.** *Let $0 < p < 1$, χ as in Lemma 3.37, $\Phi_0(\xi) = \frac{\xi|\xi|^p}{p+1}$ and $m_k = \chi\Phi_0^k$, $k \ge 0$. Then $\check{m}_k \in C^\infty(\mathbb{R})$ and there are $C_0 = C_0(p,\chi)$, $C = C(p)$ such that*

$$|\partial^j \check{m}_k(y)| \le C_0 C^k \langle y\rangle^{-p-2-j}, \qquad y \in \mathbb{R},\ j \in \{0,1,2\},\ k \ge 0.$$

Proof. m_k is bounded and compactly supported, so $\check{m}_k$ is smooth and $\partial^j \check{m}_k = a_{k,j}\check{}$ (differentiation under the integral). $k = 0$: $m_0 = \chi \in C_c^\infty$, so $\check{\chi} \in \mathcal{S}$ and the bound holds with a constant depending on χ. $k \ge 1$: apply Lemma 3.38 to $a = a_{k,j}$ with $n = j+2$, $M = C_1 C^k$, using Lemma 3.37; for $|y| \ge 1$ use $|y|^{-1} \le \sqrt{2}\langle y\rangle^{-1}$, for $|y| \le 1$ the trivial bound.

□

3.3.2 Convergence in weighted spaces

Lemma 3.40 (Weighted tail lemma)**.** *Let $0 < \gamma' < \gamma$, $\omega : \mathbb{R} \to (0,\infty)$ locally bounded, and $h_k, h \in L^\infty_{\mathrm{loc}}(\mathbb{R})$ with $\sup_k \|\langle y\rangle^\gamma \omega h_k\|_{L^\infty} \le M$ and $h_k \to h$ in $L^\infty(-R,R)$ for every R. Then $\|\langle y\rangle^\gamma \omega h\|_{L^\infty} \le M$ and $\|\langle y\rangle^{\gamma'}\omega(h_k - h)\|_{L^\infty} \to 0$.*

Proof. The first claim: a.e. convergence on each $(-R,R)$. For $R > 0$: $\|\langle y\rangle^{\gamma'}\omega(h_k - h)\|_{L^\infty} \le \langle R\rangle^{\gamma'} \sup_{[-R,R]}\omega\, \|h_k - h\|_{L^\infty(-R,R)} + \langle R\rangle^{\gamma'-\gamma} 2M$. Take $\limsup_k$, then $R \to \infty$. □

Proposition 3.41 (Convergence in $X_{\gamma'}$, $Y_{\gamma'}$)**.** *Let $0 < \gamma' < \gamma < \infty$. If $(g_k) \subset X_\gamma$ with $\sup_k \|g_k\|_{X_\gamma} < \infty$ and $g_k \to g$ in $W^{2,\infty}_{\mathrm{loc}}$ (i.e. in $W^{2,\infty}(-R,R)$ for every R), then $g \in X_\gamma$ and $\|g_k - g\|_{X_{\gamma'}} \to 0$. If $(G_k) \subset Y_\gamma$ with $\sup_k \|G_k\|_{Y_\gamma} < \infty$ and $G_k \to G$ in $W^{1,\infty}_{\mathrm{loc}}$, then $G \in Y_\gamma$ and $\|G_k - G\|_{Y_{\gamma'}} \to 0$.*

Proof. Each of the (at most six) terms of the X_γ norm is $\sup_{y\in H}\langle y\rangle^\gamma \omega_i(y)|g^{(i)}(y)|$ with $H = [0,\infty)$ or $(-\infty,0]$, $i \in \{0,1,2\}$, $\omega_i = \langle y\rangle^{-i/p}$ resp. $\langle y\rangle^i$. Apply Lemma 3.40 to $h_k = \mathbb{1}_H g_k^{(i)}$. The norm of $X_{\gamma'}$ has the same weights with γ' in place of γ. Same for Y. □

3.4 Appendix: stationary and nonstationary phase

Throughout, phases Φ are real-valued and amplitudes are complex-valued. An integral $\int_a^\infty$ that is not absolutely convergent means $\lim_{R\to\infty}\int_a^R$ (improper Riemann integral), and its existence is part of each statement.

3.4.1 Van der Corput

Lemma 3.42 (Van der Corput, first derivative, on an interval)**.** *Let $a < b$, $\lambda > 0$, $\Phi \in C^2([a,b];\mathbb{R})$ with Φ' monotone and $|\Phi'| \geq \lambda$ on $[a,b]$. Then $\left|\int_a^b e^{i\Phi(\xi)} d\xi\right| \leq 2\max\{|\Phi'(a)|^{-1}, |\Phi'(b)|^{-1}\} \leq 2\lambda^{-1}$.*

Proof. Φ' is continuous and nonvanishing, so of constant sign, and $1/\Phi'$ is monotone. Write $e^{i\Phi} = \frac{1}{i\Phi'}\frac{d}{d\xi}e^{i\Phi}$ and integrate by parts: $\int_a^b e^{i\Phi} = \left[\frac{e^{i\Phi}}{i\Phi'}\right]_a^b - \int_a^b e^{i\Phi}\frac{d}{d\xi}\frac{1}{i\Phi'}$. The boundary term is $\leq |\Phi'(a)|^{-1} + |\Phi'(b)|^{-1}$; the integral is $\leq \int_a^b |(1/\Phi')'| = |1/\Phi'(b) - 1/\Phi'(a)|$ by monotonicity. With $x = |1/\Phi'(a)|$, $y = |1/\Phi'(b)|$ (same sign): $x + y + |x - y| = 2\max(x,y)$. □

Lemma 3.43 (Van der Corput, second derivative, on an interval)**.** *Let $a < b$, $\lambda > 0$, $\Phi \in C^2([a,b];\mathbb{R})$ with $|\Phi''| \geq \lambda$ on $[a,b]$. Then $\left|\int_a^b e^{i\Phi}\right| \leq 6\lambda^{-1/2}$.*

Proof. Replacing Φ by $-\Phi$ (complex conjugation) we may assume $\Phi'' \geq \lambda$, so Φ' is strictly increasing. Let $\delta = \lambda^{1/2}$ and $J = \{\xi \in [a,b] : |\Phi'(\xi)| < \delta\}$, an interval of length $\leq 2\delta/\lambda$ (mean value theorem). $[a,b]\backslash J$ consists of at most two intervals on which Φ' is monotone and $|\Phi'| \geq \delta$; by Lemma 3.42 each contributes $\leq 2/\delta$. Total $\leq 4/\delta + 2\delta/\lambda = 6\lambda^{-1/2}$. (If an endpoint of J is open, apply Lemma 3.42 on closed subintervals and pass to the limit.) □

Lemma 3.44 (Van der Corput with amplitude on an interval (sharp cutoff form))**.** *Let $a < b$, $\lambda > 0$, $\Phi \in C^2([a,b];\mathbb{R})$, $\psi \in C^1([a,b];\mathbb{C})$. Set $c_1 = 2$, $c_2 = 6$. If either ($k = 1$) Φ' is monotone and $|\Phi'| \geq \lambda$ on $[a,b]$, or ($k = 2$) $|\Phi''| \geq \lambda$ on $[a,b]$, then*

$$\left|\int_a^b \psi(\xi) e^{i\Phi(\xi)} d\xi\right| \leq c_k \lambda^{-1/k}\Big(|\psi(b)| + \int_a^b |\psi'(\xi)| d\xi\Big).$$

The same holds with $|\psi(b)|$ replaced by $|\psi(a)|$, and in particular with $\|\psi\|_{L^\infty(a,b)}$.

Proof. Let $F(x) = \int_a^x e^{i\Phi}$. The hypotheses hold on every $[a,x]$, so $|F(x)| \leq c_k\lambda^{-1/k}$. Integrate by parts: $\int_a^b \psi e^{i\Phi} = \psi(b)F(b) - \int_a^b F\psi'$. For the variant with $\psi(a)$ use $\widetilde{F}(x) = -\int_x^b e^{i\Phi}$. □

Lemma 3.45 (Van der Corput on a half-line)**.** *Let $a \in \mathbb{R}$, $\lambda > 0$, $\Phi \in C^2([a,\infty);\mathbb{R})$ with Φ' monotone, $|\Phi'| \geq \lambda$ on $[a,\infty)$, and $|\Phi'(\xi)| \to \infty$ as $\xi \to \infty$. Then $\lim_{R\to\infty}\int_a^R e^{i\Phi}$ exists and $\left|\int_a^\infty e^{i\Phi}\right| \leq 2|\Phi'(a)|^{-1} \leq 2\lambda^{-1}$.*

Proof. $|1/\Phi'|$ is nonincreasing (it is monotone and tends to 0). By Lemma 3.42, $|\int_R^{R'} e^{i\Phi}| \leq 2|\Phi'(R)|^{-1} \to 0$ (Cauchy criterion), and $|\int_a^R e^{i\Phi}| \leq 2|\Phi'(a)|^{-1}$ for all R. □

Lemma 3.46 (Van der Corput, paper form)**.** *Let $I \subset \mathbb{R}$ be a compact interval, $\lambda > 0$, $\Phi \in C^2(I;\mathbb{R})$, $\chi \in C^1(\mathbb{R};\mathbb{C})$ with $\operatorname{supp}\chi \subset I$.*

1. *If Φ' is monotone and $|\Phi'| \geq \lambda$ on I, then $\left|\int \chi e^{i\Phi}\right| \leq 2\lambda^{-1}(\|\chi\|_{L^\infty} + \|\chi'\|_{L^1})$.*
2. *If $|\Phi''| \geq \lambda$ on I, then $\left|\int \chi e^{i\Phi}\right| \leq 6\lambda^{-1/2}(\|\chi\|_{L^\infty} + \|\chi'\|_{L^1})$.*

The sharp-cutoff (interval) form, with $|\psi(b)| + \int_a^b |\psi'|$, is Lemma 3.44; the half-line form is Lemma 3.45.

Proof. Apply Lemma 3.44 on $I = [a, b]$ with $\psi = \chi|_I$, and $|\chi(b)| \le \|\chi\|_\infty$. Reference: [Ste93, Ch. VIII, Prop. 2 and Cor.]. □

3.4.2 The model integral $\int_\eta^\infty e^{i(\Phi_0 - y\xi)}$

Lemma 3.47 (Existence of the tail integrals)**.** *Let $0 < p < 1$, $\Phi_0(\xi) = \xi|\xi|^p/(p+1)$. For all $y \in \mathbb{R}$, $\eta > 0$, the improper integral $E_y(\eta) := \int_\eta^\infty e^{i(\Phi_0(\xi) - y\xi)} d\xi$ exists, and $\eta \mapsto E_y(\eta)$ is C^1 on $(0, \infty)$ with $\partial_\eta E_y(\eta) = -e^{i(\Phi_0(\eta) - y\eta)}$.*

Proof. On $\xi > 0$ the phase $\phi_y(\xi) = \Phi_0(\xi) - y\xi$ has $\phi_y' = \xi^p - y$ (increasing, $\to \infty$) and $\phi_y'' = p\xi^{p-1} > 0$. For $\xi_1 = \max(\eta, (|y| + 1)^{1/p})$, $\phi_y' \ge 1$ on $[\xi_1, \infty)$; apply Lemma 3.45 there and add the proper integral over $[\eta, \xi_1]$.

□

Lemma 3.48 (Tail integral, $y \le -1$)**.** *For $y \le -1$ and $\eta > 0$: $|E_y(\eta)| \le 2|y|^{-1}$.*

Proof. $\phi_y' = \xi^p + |y| \ge |y|$, increasing, $\to \infty$. Apply Lemma 3.45 on $[\eta, \infty)$.

□

Lemma 3.49 (Tail integral, $y \ge 1$)**.** *There is $C = C(p)$ such that for $y \ge 1$, $\eta > 0$: $|E_y(\eta)| \le C y^{\frac{1-p}{2p}}$.*

Proof. Let $\xi_0 = y^{1/p}$ (the stationary point) and split $[\eta, \infty)$ into its intersections with $I_1 = (0, \frac{1}{2}\xi_0]$, $I_2 = [\frac{1}{2}\xi_0, \frac{3}{2}\xi_0]$, $I_3 = [\frac{3}{2}\xi_0, \infty)$ (each intersection is an interval, possibly empty). On I_1: $\phi_y' = \xi^p - y \le -(1 - 2^{-p})y$, monotone; Lemma 3.42 gives $\le \frac{2}{(1-2^{-p})y}$. On I_2: $\phi_y'' = p\xi^{p-1} \ge p(3/2)^{p-1} y^{(p-1)/p}$; Lemma 3.43 gives $\le 6p^{-1/2}(3/2)^{(1-p)/2} y^{\frac{1-p}{2p}}$. On I_3: $\phi_y' \ge ((3/2)^p - 1)y$, increasing to ∞; Lemma 3.45 gives $\le \frac{2}{((3/2)^p - 1)y}$. Sum, using $y^{-1} \le 1 \le y^{\frac{1-p}{2p}}$.

□

Lemma 3.50 (Tail integral, $|y| \le 1$)**.** *For $|y| \le 1$, $\eta > 0$: $|E_y(\eta)| \le 2 + \frac{2}{2^p - 1}$.*

Proof. If $\eta \ge 2$: $\phi_y' \ge 2^p - 1 > 0$ on $[\eta, \infty)$, increasing to ∞; use Lemma 3.45. If $\eta < 2$: $E_y(\eta) = \int_\eta^2 e^{i\phi_y} + E_y(2)$ and $|\int_\eta^2| \le 2$.

□

Lemma 3.51 (Uniform bound for the tail integrals)**.** *Let $0 < p < 1$, $\Phi_0(\xi) = \frac{\xi|\xi|^p}{p+1}$. There is $C = C(p)$ such that for all $y \in \mathbb{R}$ the improper integrals $\int_\eta^\infty e^{i(\Phi_0(\xi) - y\xi)} d\xi$ exist for all $\eta > 0$ and*

$$\sup_{\eta > 0} \Big| \int_\eta^\infty e^{i(\Phi_0(\xi) - y\xi)} d\xi \Big| \le \begin{cases} C\langle y \rangle^{-1}, & y \le 0, \\ C\langle y \rangle^{\frac{1-p}{2p}}, & y \ge 0. \end{cases}$$

Proof. Combine the three regimes, using $|y|^{-1} \le \sqrt{2}\langle y \rangle^{-1}$ for $|y| \ge 1$ and $\langle y \rangle \le \sqrt{2}$ for $|y| \le 1$.

□

3.4.3 Shared finite-order symbol and integration-by-parts interfaces

Lemma 3.52 (Reciprocal derivative bounds by a finite recurrence)**.** *Fix $n \in \mathbb{N}$ and nonnegative constants $A_1, \dots, A_n$. There are constants $D_j = D_j(A_1, \dots, A_j) > 0$, $0 \le j \le n$, such that the following holds. Let $U \subset \mathbb{R}$ be open, $v \in C^n(U;\mathbb{C})$, $\lambda > 0$, and $w : U \to (0,\infty)$. If $|v(x)| \ge \lambda$ and $|v^{(j)}(x)| \le A_j \lambda w(x)^{-j}$ for $x \in U$, $1 \le j \le n$, then $v^{-1} \in C^n(U)$ and $|(v^{-1})^{(j)}(x)| \le D_j \lambda^{-1} w(x)^{-j}$ for $0 \le j \le n$. No regularity of the comparison weight w is required.*

Proof. The nonvanishing of v gives regularity. Differentiate $vv^{-1} = 1$: $(v^{-1})^{(j)} = -v^{-1} \sum_{\ell=1}^{j} \binom{j}{\ell} v^{(\ell)} (v^{-1})^{(j-\ell)}$. Starting with $D_0 = 1$, take $D_j = 1 + \sum_{\ell=1}^{j} \binom{j}{\ell} A_\ell D_{j-\ell}$. The powers of λ and $w(x)$ cancel pointwise. This recurrence is the implementation, not a citation of a general high-order chain rule. □

Lemma 3.53 (A finite sequence of integrations by parts)**.** *Let $U \subset \mathbb{R}$ be open, $n \in \mathbb{N}$, $\phi \in C^{n+1}(U;\mathbb{R})$ with $\phi' \ne 0$ on U, and $a \in C_c^n(U;\mathbb{C})$, extended by zero to $\mathbb{R}$. Set $L_\phi a = (a/(i\phi'))'$ on U and zero outside U, and iterate this definition. For $0 \le j \le n$, $L_\phi^j a$ is continuous, compactly supported in* $\operatorname{supp} a$*, and belongs to $C^{n-j}(\mathbb{R})$. All the following integrals converge absolutely:*

$$\int_{\mathbb{R}} a(x) e^{i\phi(x)}\, dx = (-1)^n \int_{\mathbb{R}} L_\phi^n a(x) e^{i\phi(x)}\, dx.$$

The values chosen for ϕ outside U do not affect either side.

Proof. Work on a finite union of bounded intervals compactly contained in U and containing the compact support of a. On each interval the quotient is C^n and vanishes near both endpoints. The ordinary interval integration-by-parts formula gives the case $n = 1$. Differentiation does not enlarge support, so zero extension is regular and induction gives the result. To obtain the finite union, cover the compact support by finitely many intervals whose closures lie in U, then merge overlapping intervals. Each endpoint lies outside the support after a small enlargement. Sum the interval identities; no partition of unity is needed. □

Lemma 3.54 (Symbol bounds after iterated integration by parts)**.** *Fix integers $n, j \ge 0$, $\sigma \in \mathbb{R}$, and nonnegative finite lists (A_ℓ) and (Q_ℓ) for $0 \le \ell \le n+j+1$. There is $C > 0$ depending only on these data such that: if $U \subset \mathbb{R} \setminus \{0\}$ is open, $a \in C_c^\infty(U;\mathbb{C})$, $\phi \in C^\infty(U;\mathbb{R})$, $B \ge 0$, $\Lambda > 0$, $\phi' \ne 0$ on U, and*

$$|a^{(\ell)}(z)| \le A_\ell B |z|^{\sigma-\ell}, \qquad |(1/\phi')^{(\ell)}(z)| \le Q_\ell \Lambda^{-1} |z|^{-\ell} \quad (z \in U,\ 0 \le \ell \le n+j+1),$$

then $|(L_\phi^n a)^{(j)}(z)| \le CB\Lambda^{-n} |z|^{\sigma-n-j}$ on U.

Proof. Use $(L_\phi a)^{(j)} = \sum_{\ell=0}^{j+1} \binom{j+1}{\ell} a^{(\ell)} (1/(i\phi'))^{(j+1-\ell)}$. Each summand has the same power $|z|^{\sigma-j-1}$ and one factor Λ^{-1}. Induct on n, carrying the finite list of constants for the derivative orders still required. No infinite-order estimate is needed. □

Lemma 3.55 (Zero extension of a supported oscillatory amplitude)**.** *Let $U \subset \mathbb{R}$ be open, $a \in C_c^\infty(U;\mathbb{C})$ extended by zero, and $\phi \in C^\infty(U;\mathbb{R})$ with arbitrary values outside U. Then $ae^{i\phi} \in C_c^\infty(\mathbb{R})$ and its support is contained in* $\operatorname{supp} a$*.*

Every derivative equals its ordinary product/chain-rule expression on U and is zero in a neighbourhood of each point outside $\operatorname{supp} a$.

Proof. Regularity is local: on U use the product and chain rules; outside the closed support of a the product is identically zero on a neighbourhood. These two open sets cover $\mathbb{R}$ because $\operatorname{supp} a \subset U$. Compact support also proves integrability of every derivative. □

3.4.4 Stationary phase with a homogeneous phase

Notation for this subsection: $0 < p < 1$, $h(s) := s - \frac{s^{p+1}}{p+1}$ for $s > 0$ (so $h'(s) = 1 - s^p$, $h(1) = \frac{p}{p+1}$, $h''(1) = -p$), and for $y > 0$

$$\phi_y(\xi) = y\xi - \frac{\xi^{p+1}}{p+1}, \quad \lambda_0(y) = y^{1/p}, \quad \mu(y) = y^{1+1/p} = \lambda_0^{p+1}, \quad \Psi_0(y) = \tfrac{p}{p+1} y^{1+1/p} = \mu h(1),$$

so that $\phi_y(\lambda_0 s) = \mu h(s)$. Fix $\chi_{\mathrm{sp}} \in C_c^\infty((\frac{1}{2}, 2))$ with $\chi_{\mathrm{sp}} = 1$ on $[\frac{3}{4}, \frac{3}{2}]$, $0 \le \chi_{\mathrm{sp}} \le 1$. For $c \in C^3(\mathbb{R})$ with $\operatorname{supp} c \subset (\frac{1}{2}, 2)$ and $\mu > 0$ define

$$\rho(\mu; c) := e^{i\pi/4} \Big(\frac{p\mu}{2\pi}\Big)^{1/2} e^{-i\mu h(1)} \int_0^\infty c(s) e^{i\mu h(s)} ds - c(1).$$

Definition 3.56 (Stationary-phase functions, amplitude class and Euler operator)**.** For $0 < p < 1$, the functions $h, \lambda_0, \Psi_0, \phi_y, \mu$ and the normalized remainder $\rho(\mu; c)$ have the formulas displayed at the start of this subsection. Put

$$P(y) = e^{i(\Psi_0(y) - \pi/4)} \sqrt{\frac{2\pi}{p \lambda_0(y)^{p-1}}}, \qquad b_y(s) = \lambda_0(y)^{-m} a(\lambda_0(y) s),$$

$$q(s) = \begin{cases} (h(s) - h(1))/(1 - s^p), & s \ne 1, \\ 0, & s = 1, \end{cases} \qquad Tc(s) = c(s)/2 - \partial_s(c(s) q(s)).$$

The symbol class $\mathcal{S}_+^m$ here consists of $a : (0, \infty) \to \mathbb{C}$ smooth, zero on $(0, 1]$, such that for every $k \in \mathbb{N}$ there is $C_k \ge 0$ with $|a^{(k)}(\xi)| \le C_k \xi^{m-k}$ for $\xi \ge 1$. These are data definitions only. Smoothness of the quotient at 1, the operator bounds, and the stationary-phase estimates are separate lemmas below.

Definition 3.57 (Near, far and full stationary integrals)**.** With the fixed cutoff χ_{sp}, set

$$I_{\mathrm{near}}(y) = \int_0^\infty a(\xi) \chi_{\mathrm{sp}}(\xi/\lambda_0(y)) e^{i\phi_y(\xi)} d\xi,$$

$$I_{\mathrm{far}}(y) = \lim_{R \to \infty} \int_0^R a(\xi)(1 - \chi_{\mathrm{sp}}(\xi/\lambda_0(y))) e^{i\phi_y(\xi)} d\xi, \quad I(y) = \lim_{R \to \infty} \int_0^R a(\xi) e^{i\phi_y(\xi)} d\xi, \quad r(y) = I(y)/P(y) - a(\lambda_0$$

The first formula is a total Bochner integral. The limits denote Lean's total limit operator; their existence is a separate assertion, not part of this definition.

Lemma 3.58 (Complex Gaussian integral)**.** *For $b \in \mathbb{C}$ with $\operatorname{Re} b > 0$: $\int_{\mathbb{R}} e^{-bx^2} dx = (\pi/b)^{1/2}$ (principal branch).*

Lemma 3.59 (Fresnel expansion with remainder)**.** *There is an absolute constant C such that for all $\nu > 0$, $L \geq 1$ and $e \in C^3(\mathbb{R};\mathbb{C})$ with $\operatorname{supp} e \subset [-L, L]$,*

$$\Big| \int_{\mathbb{R}} e(w) e^{-i\nu w^2/2} dw - \Big(\frac{2\pi}{\nu}\Big)^{1/2} e^{-i\pi/4} e(0) \Big| \leq C L \nu^{-3/2} \|e\|_{C^3(\mathbb{R})}.$$

Proof. Let $g(w) = e^{-w^2/2}$ and $d(w) = (e(w) - e(0)g(w))/w = \int_0^1 (e' - e(0)g')(tw)dt$ (smooth extension at 0). Step 1. By Lemma 3.58 with $b = (1 + i\nu)/2$: $\int g e^{-i\nu w^2/2} = (2\pi/(1+i\nu))^{1/2} = (2\pi/\nu)^{1/2} e^{-i\pi/4}(1 - i/\nu)^{-1/2}$, and $|(1 - i/\nu)^{-1/2} - 1| \leq 2/\nu$ (for $\nu \leq 1$ since $|1 - i/\nu| \geq 1$; for $\nu \geq 1$ by the mean value theorem). Step 2. $w e^{-i\nu w^2/2} = \frac{i}{\nu} \partial_w e^{-i\nu w^2/2}$; integrating by parts on $[-R, R]$ and letting $R \to \infty$ ($d(w) = O(|w|^{-1})$, $d' \in L^1$): $\int w d(w) e^{-i\nu w^2/2} dw = -\frac{i}{\nu} \int d'(w) e^{-i\nu w^2/2} dw$. Step 3. By Lemma 3.44 ($k = 2$, $\Phi'' = -\nu$) on $[-R, R]$ and $R \to \infty$: $|\int d' e^{-i\nu w^2/2}| \leq 6\nu^{-1/2}(\|d'\|_\infty + \|d''\|_{L^1})$. Step 4. $\|d'\|_\infty + \|d''\|_{L^\infty} \leq C\|e\|_{C^3}$ from the integral formula; for $|w| \geq 1$, $|d''(w)| \leq C(|e''| + |e'| + |e|)\mathbb{1}_{[-L,L]} + C|e(0)|(1 + |w|)^{-3}$, so $\|d''\|_{L^1} \leq CL\|e\|_{C^2}$. Combine: $e = e(0)g + wd$.

□

Lemma 3.60 (Morse coordinate for h near $s = 1$)**.** *Define on $(0, \infty)$ $w(s) := (s-1)\Big(\frac{2}{p}\int_0^1 (1-t)\, p\,(1 + t(s-1))^{p-1} dt\Big)^{1/2}$. Then $w \in C^\infty((0,\infty))$, $h(s) - h(1) = -\frac{p}{2} w(s)^2$, $w(1) = 0$, $w'(1) = 1$, and $w' > 0$ on $(0, \infty)$. Hence w is a C^∞ diffeomorphism of $[\frac{1}{2}, 2]$ onto $[w(\frac{1}{2}), w(2)]$; its inverse $s(w)$ has C^k norms bounded by constants depending only on p, k.*

Proof. Taylor: $h(s) - h(1) = (s-1)^2 \int_0^1 (1-t) h''(1 + t(s-1)) dt$ with $-h''(\sigma) = p\sigma^{p-1} > 0$; the radicand is positive and smooth, so w is smooth and $w^2 = \frac{2}{p}(h(1) - h(s))$. $w'(1) = (\frac{2}{p} \cdot \frac{p}{2})^{1/2} = 1$. For $s \neq 1$: $2ww' = -\frac{2}{p} h'(s)$ and $h'(s) = 1 - s^p$ has the sign of $1 - s$ while w has the sign of $s - 1$, so $w' > 0$. Inverse function theorem on the compact interval. □

Lemma 3.61 (Transport of a compact amplitude through the Morse coordinate)**.** *Fix $0 < p < 1$ and let w be the coordinate of Lemma 3.60. Write $J = (w(1/2), w(2))$ and let $s : J \to (1/2, 2)$ be its inverse. There are $L = L(p) \geq 1$ and $C = C(p) > 0$ such that for every $c \in C^3(\mathbb{R};\mathbb{C})$ with compact support in $(1/2, 2)$, the function $e(v) = c(s(v))s'(v)$ on J, extended by zero, lies in $C_c^3(\mathbb{R})$, has support in $[-L, L]$, satisfies $e(0) = c(1)$, and $\|e\|_{C^3(\mathbb{R})} \leq C\|c\|_{C^3(\mathbb{R})}$.*

Proof. The inverse coordinate has bounded derivatives through order four on the closed image of $[1/2, 2]$. Expand the first three derivatives of $c(s(v))s'(v)$ by product and chain rules. The extension is smooth at the endpoints because c vanishes in neighbourhoods of $1/2$ and 2; the bound uses only inverse-coordinate derivative bounds, not the size of those neighbourhoods. □

Lemma 3.62 (Morse change of variables in the stationary integral)**.** *With $0 < p < 1$, c and e as in Lemma 3.61, for every $\mu > 0$,*

$$\int_0^\infty c(s) e^{i\mu h(s)}\, ds = e^{i\mu h(1)} \int_{\mathbb{R}} e(v) e^{-ip\mu v^2/2}\, dv.$$

Both integrands are absolutely integrable.

Proof. Restrict to $(1/2, 2)$, apply the one-dimensional change-of-variables theorem for the increasing C^1 diffeomorphism w, and use $h(s) - h(1) = -pw(s)^2/2$. Extend the transformed integrand by zero and use compact support for absolute convergence. □

Lemma 3.63 (Stationary phase for compactly supported amplitudes)**.** *There is $C = C(p)$ such that for all $\mu > 0$ and $c \in C^3(\mathbb{R})$ with $\operatorname{supp} c \subset (\frac{1}{2}, 2)$: $|\rho(\mu; c)| \le C\mu^{-1}\|c\|_{C^3}$.*

Proof.

Change variables $s = s(w)$ (Lemma 3.60): $\int c(s)e^{i\mu h(s)}ds = e^{i\mu h(1)} \int e(w)e^{-ip\mu w^2/2}dw$ with $e(w) = c(s(w))s'(w)$ extended by 0, $e \in C^3$, $\operatorname{supp} e \subset [-L, L]$ with $L = L(p)$, $\|e\|_{C^3} \le C(p)\|c\|_{C^3}$, $e(0) = c(1)$. Apply Lemma 3.59 with $\nu = p\mu$; the main term times $e^{i\pi/4}(p\mu/2\pi)^{1/2}e^{-i\mu h(1)}$ equals $c(1)$. □

Lemma 3.64 (Removal of the apparent singularity in the Euler quotient)**.** *For $0 < p < 1$, define $N(s) = \int_0^1 h'(1+t(s-1))dt$ and $D(s) = \int_0^1 h''(1+t(s-1))dt$ for $s > 0$. Both are smooth, $D(s) < 0$, and $q = N/D$ agrees with $(h(s)-h(1))/h'(s)$ for $s \neq 1$. Moreover $q(1) = 0$, $q'(1) = 1/2$, and every derivative of q is bounded on $[1/2, 2]$ by a constant depending only on p and its order.*

Proof. Differentiate the parameter integrals on compact subintervals of $(0, \infty)$; their integrands and all derivatives have uniform integrable bounds on $t \in [0, 1]$. The fundamental theorem of calculus gives the quotient identities. At $s = 1$, $N = 0$, $N' = h''(1)/2$, and $D = h''(1) = -p$. Compactness gives the derivative bounds. □

Lemma 3.65 (Support, cancellation and norm bound for the Euler amplitude operator)**.** *For $0 < p < 1$ set $Tc = c/2 - (cq)'$, with q of Lemma 3.64. If $c \in C_c^\infty((1/2, 2); \mathbb{C})$, extended by zero, then $Tc \in C_c^\infty((1/2, 2))$ and $Tc(1) = 0$. For every integer $k \ge 0$ there is $C_k = C_k(p) > 0$, independent of c, with $\|Tc\|_{C^k(\mathbb{R})} \le C_k\|c\|_{C^{k+1}(\mathbb{R})}$.*

Proof. On the support interval use Leibniz and the bounds for $q^{(j)}$, $j \le k + 1$. Outside the support the expression is locally zero. The identity at 1 follows from $q(1) = 0$ and $q'(1) = 1/2$. □

Lemma 3.66 (Differentiation of the compact stationary integral)**.** *Fix $0 < p < 1$ and $c \in C_c^\infty((1/2, 2); \mathbb{C})$. Then $J(\mu; c) = \int_0^\infty c(s)e^{i\mu h(s)}ds$ is smooth for $\mu \in \mathbb{R}$ and*

$$\partial_\mu^j J(\mu; c) = \int_0^\infty (ih(s))^j c(s)e^{i\mu h(s)}ds \quad (j \in \mathbb{N}).$$

More generally, let $V \subset \mathbb{R}$ be open and $c : V \times \mathbb{R} \to \mathbb{C}$ smooth, with a common compact support in $(1/2, 2)$ for y in each compact subinterval of V. Then mixed derivatives in $y \in V$ and $\mu \in \mathbb{R}$ pass under the integral. The same assertions hold for the normalized remainder ρ when $\mu > 0$.

Proof. On the fixed support, each differentiated integrand is continuous and bounded uniformly for parameters in a compact set. Dominate by a constant times its indicator and iterate differentiation under the integral. The normalizing factor of ρ is smooth on $(0, \infty)$. □

Lemma 3.67 (Integration-by-parts identity for the normalized stationary remainder)**.** *For $0 < p < 1$, $c \in C_c^\infty((1/2,2);\mathbb{C})$, and $\mu > 0$, $\mu\partial_\mu\rho(\mu;c) = \rho(\mu;Tc)$.*

Proof. Differentiate the normalization of J. Its derivative contributes $J/2 - i\mu h(1)J$, while $\mu J'$ contributes $i\mu\int hce^{i\mu h}$. Use $qh' = h - h(1)$ and integrate the latter difference by parts in s; the boundary term vanishes because cq is compactly supported. The result is the normalized integral with amplitude Tc, with no extra value term since $Tc(1) = 0$. □

Lemma 3.68 (Euler identity for ρ)**.** *Let $q(s) := \frac{h(s)-h(1)}{h'(s)}$ for $s \in (0,\infty)\setminus\{1\}$, $q(1) := 0$. Then $q \in C^\infty((0,\infty))$, $q'(1) = \frac{1}{2}$. For $c \in C^\infty$ with $\operatorname{supp} c \subset (\frac{1}{2},2)$ set $Tc := \frac{1}{2}c - (cq)'$. Then $\operatorname{supp} Tc \subset (\frac{1}{2},2)$, $Tc(1) = 0$, $\|Tc\|_{C^k} \le C_k(p)\|c\|_{C^{k+1}}$, $\mu \mapsto \rho(\mu;c)$ is C^∞ on $(0,\infty)$, and $\mu\partial_\mu\rho(\mu;c) = \rho(\mu;Tc)$.*

Proof. $q = N/D$ with $N(s) = \int_0^1 h'(1+t(s-1))dt = \frac{h(s)-h(1)}{s-1}$ and $D(s) = \int_0^1 h''(1+t(s-1))dt = \frac{h'(s)}{s-1}$, both smooth, $D < 0$ on $(0,\infty)$ (as $h'' < 0$). $N(1) = 0$, $N'(1) = h''(1)/2$, $D(1) = h''(1)$, so $q'(1) = \frac{1}{2}$, and $Tc(1) = \frac{1}{2}c(1) - c(1)q'(1) - c'(1)q(1) = 0$. Differentiating under the (compactly supported) integral, with $J(\mu;c) = \int ce^{i\mu h}$: $\mu\partial_\mu J(\mu;c) = i\mu h(1)J(\mu;c) + \int c\, i\mu(h-h(1))e^{i\mu h}$, and $i\mu(h-h(1))e^{i\mu h} = q\,\partial_s e^{i\mu h}$, so the last integral is $-J(\mu;(cq)')$ (no boundary terms). Hence $\mu\partial_\mu[(p\mu/2\pi)^{1/2}e^{-i\mu h(1)}J(\mu;c)] = (p\mu/2\pi)^{1/2}e^{-i\mu h(1)}J(\mu;\frac{1}{2}c - (cq)')$; multiply by $e^{i\pi/4}$ and use $Tc(1) = 0$. □

Lemma 3.69 (Symbol bounds and Euler differentiation after rescaling)**.** *Fix $0 < p < 1$, $m \in \mathbb{R}$, and $a \in C^\infty((0,\infty);\mathbb{C})$ with $a = 0$ on $(0,1]$ and $|a^{(k)}(\xi)| \le C_k\xi^{m-k}$ for $\xi \ge 1$, where $C_k \ge 0$. For $y, s > 0$ put $b_y(s) = \lambda_0(y)^{-m}a(\lambda_0(y)s)$ and $\mathfrak{E} = p^{-1}(-m + s\partial_s)$. The map $(y,s) \mapsto b_y(s)$ is smooth and*

$$\partial_s^k b_y(s) = \lambda_0(y)^{k-m}a^{(k)}(\lambda_0(y)s), \quad |\partial_s^k b_y(s)| \le C_k s^{m-k}, \quad y\partial_y b_y = \mathfrak{E}b_y.$$

For each $j,k \in \mathbb{N}$ there is $D = D(p,m,j,k,C_0,\dots,C_{j+k})$, independent of y, with $\sup_{s\in[1/2,2]}|\partial_s^k\mathfrak{E}^j b_y(s)| \le D$.

Proof. The first formula follows by repeated differentiation of a linear rescaling. When $\lambda_0 s \le 1$, every derivative of a vanishes (including at 1 by smoothness). Otherwise apply the given symbol bound. The Euler identity is the chain rule and $y\lambda_0' = \lambda_0/p$. Expand the finite-order differential operator $\partial_s^k\mathfrak{E}^j$ inductively; it is a finite sum of bounded powers of s times derivatives of order at most $j+k$ on $[1/2,2]$. □

Lemma 3.70 (Exact scaling of the near stationary contribution)**.** *Under the hypotheses of Lemma 3.69, fix the cutoff χ_{sp} of this subsection. For every $y > 0$, $c_y = \chi_{\mathrm{sp}}b_y$ is smooth and compactly supported in $(1/2,2)$, the integral defining $I_{\mathrm{near}}(y)$ converges absolutely, $P(y) \ne 0$, and*

$$I_{\mathrm{near}}(y)/P(y) - a(\lambda_0(y)) = \lambda_0(y)^m\rho(\mu(y);c_y).$$

Both sides are smooth functions of $y > 0$.

Proof. Substitute $\xi = \lambda_0 s$ in the compact integral. Use $\phi_y(\lambda_0 s) = \mu h(s)$, $\mu = \lambda_0^{p+1}$ and $c_y(1) = \lambda_0^{-m} a(\lambda_0)$. Match the positive square-root factors in P and ρ. The family c_y has its support in the fixed support of χ_{sp}, so parameter differentiation proves smoothness without differentiating an improper integral. □

Lemma 3.71 (Euler recurrence for the parameter-dependent stationary remainder)**.** *With the same data, let $F(y) = \rho(\mu(y); \chi_{\mathrm{sp}} b_y)$ and $E = y\partial_y$. For every $l \in \mathbb{N}$ and $y > 0$,*

$$E^l F(y) = \sum_{i=0}^{l} \binom{l}{i} (1 + 1/p)^i \rho\big(\mu(y); T^i(\chi_{\mathrm{sp}} \mathfrak{E}^{l-i} b_y)\big).$$

For each l there is $C_l = C_l(p, m, \chi_{\mathrm{sp}}, C_0, \dots, C_{l+4}) > 0$ such that $|E^l F(y)| \le C_l \mu(y)^{-1}$ for all $y > 0$.

Proof. Differentiate both the parameter μ and the amplitude; $E\mu = (1+1/p)\mu$. The s-operator T commutes with y differentiation, while $E(\chi_{\mathrm{sp}} b_y) = \chi_{\mathrm{sp}} \mathfrak{E} b_y$. Induction and Pascal's identity give the displayed sum. Apply the compact stationary remainder bound, the T norm bound, and the rescaled-symbol estimate to each of its finitely many amplitudes.

□

Lemma 3.72 (Conversion of Euler bounds to ordinary derivatives)**.** *Let $F \in C^\infty((0,\infty); \mathbb{C})$ and $E = y\partial_y$. For $j \in \mathbb{N}$ define real coefficients $a_{j,l}$ by $\prod_{r=0}^{j-1}(X - r) = \sum_{l=0}^{j} a_{j,l} X^l$. Then $y^j F^{(j)}(y) = \sum_{l=0}^{j} a_{j,l} E^l F(y)$ for $y > 0$. Consequently, if $|E^l F(y)| \le B_l y^\sigma$ for $y \ge 1$, $0 \le l \le j$, $B_l \ge 0$, then $|F^{(j)}(y)| \le (\sum_{l=0}^{j} |a_{j,l}| B_l) y^{\sigma - j}$ for $y \ge 1$.*

Proof. Induct on j using $E(y^j F^{(j)}) = j y^j F^{(j)} + y^{j+1} F^{(j+1)}$. This is precisely multiplication of the coefficient polynomial by $X - j$. The estimate follows from the finite sum and division by the positive number y^j. □

Lemma 3.73 (Rescaling and the near part)**.** *Let $m \in \mathbb{R}$ and $a \in C^\infty((0,\infty))$ with $a = 0$ on $(0,1]$ and $|a^{(k)}(\xi)| \le C_k \xi^{m-k}$ for $\xi \ge 1$ (hence for all $\xi > 0$). For $y > 0$ let $b_y(s) = \lambda_0^{-m} a(\lambda_0 s)$, $c_y = \chi_{\mathrm{sp}} b_y$, and $I_{\mathrm{near}}(y) = \int_0^\infty a(\xi) \chi_{\mathrm{sp}}(\xi/\lambda_0) e^{i\phi_y(\xi)} d\xi$, $P(y) = e^{i(\Psi_0(y) - \pi/4)} \big(\frac{2\pi}{p\lambda_0^{p-1}}\big)^{1/2}$. Then*

1. *$|\partial_s^k b_y(s)| \le C_k s^{m-k}$ for $s > 0$, and $y\partial_y b_y = \mathfrak{E} b_y$ with $\mathfrak{E} b := \frac{1}{p}(-mb + sb')$;*
2. *$I_{\mathrm{near}}(y)/P(y) - a(\lambda_0(y)) = \lambda_0^m \rho(\mu(y); c_y)$, and $y \mapsto \lambda_0^m \rho(\mu(y); c_y)$ is C^∞ on $(0,\infty)$;*
3. *for each $j \ge 0$ there is C (depending on p, m, j and $C_0, \dots, C_{j+4}$) with $|\partial_y^j [\lambda_0^m \rho(\mu; c_y)]| \le C \lambda_0^{m-p-1} y^{-j}$ for $y \ge 1$.*

Proof.

(1) Chain rule; $y\partial_y \lambda_0 = \lambda_0/p$. (2) Substitute $\xi = \lambda_0 s$: $I_{\mathrm{near}} = \lambda_0^{1+m} \int c_y(s) e^{i\mu h(s)} ds$; and $\lambda_0^{1+m}(p\lambda_0^{p-1}/2\pi)^{1/2} = \lambda_0^m (p\mu/2\pi)^{1/2}$, $\Psi_0 = \mu h(1)$, $c_y(1) = \lambda_0^{-m} a(\lambda_0)$. (3) Let $F(y) = \rho(\mu(y); c_y)$. Since $y\partial_y \mu = (1 + \frac{1}{p})\mu$, ρ is linear in c, and $y\partial_y c_y = \chi_{\mathrm{sp}} \mathfrak{E} b_y$, Lemma 3.68 gives by induction $(y\partial_y)^l F = \sum_{i=0}^{l} \binom{l}{i} (1 +$

$\frac{1}{p})^i \rho(\mu; T^i(\chi_{\mathrm{sp}} \mathfrak{E}^{l-i} b_y))$. On $[\frac{1}{2}, 2]$, $\mathfrak{E}^{l-i} b_y$ has C^{3+i} norm bounded uniformly in $y > 0$ by (1); hence $\|T^i(\dots)\|_{C^3} \le C$ and by Lemma 3.63 $|(y\partial_y)^l F| \le C\mu^{-1}$. Next $(y\partial_y)^l(\lambda_0^m F) = \lambda_0^m(\frac{m}{p} + y\partial_y)^l F$, so $|(y\partial_y)^l(\lambda_0^m F)| \le C\lambda_0^m \mu^{-1} = C\lambda_0^{m-p-1}$. Finally $y^j \partial_y^j = \prod_{i=0}^{j-1}(y\partial_y - i)$. □

Lemma 3.74 (Nonstationarity and reciprocal phase bounds away from the critical point)**.** *For $0 < p < 1$ there is $c = c(p) > 0$ such that, for $y, \xi > 0$ with $\xi \le 3\lambda_0(y)/4$ or $\xi \ge 3\lambda_0(y)/2$, $|y - \xi^p| \ge c(y + \xi^p)$. For every $k, j \in \mathbb{N}$ there is $C_{k,j} = C_{k,j}(p) > 0$ with*

$$\left| \partial_y^j \partial_\xi^k \frac{1}{y - \xi^p} \right| \le C_{k,j} \xi^{-k} (y + \xi^p)^{-1-j}$$

on the same set. These bounds also hold with enlarged constants on the wider open set $\xi < 7\lambda_0(y)/8$ or $\xi > 5\lambda_0(y)/4$.

Proof. Divide by y and put $r = \xi/\lambda_0$ to obtain a positive lower bound on $|1 - r^p|/(1 + r^p)$ in each stated region. Differentiating in y gives a factorial times $(y - \xi^p)^{-1-j}$; apply the reciprocal recurrence in ξ, using $|\partial_\xi^k \xi^p| \le C(p,k)\xi^{p-k}$ and $\xi^p \le y + \xi^p$. □

Lemma 3.75 (Symbols produced by the far-region integration-by-parts operator)**.** *Let $0 < p < 1$, $m \in \mathbb{R}$, and a satisfy the symbol hypotheses of Lemma 3.69. Set $g_y(\xi) = a(\xi)(1 - \chi_{\mathrm{sp}}(\xi/\lambda_0(y)))$, $a_0 = g_y$, and $a_{n+1} = \partial_\xi(a_n/(i(y - \xi^p)))$, with value zero near $\xi = \lambda_0(y)$. The functions $a_n(y, \xi)$ are smooth for $y, \xi > 0$, vanish for $\xi \le 1$ and for $3\lambda_0(y)/4 \le \xi \le 3\lambda_0(y)/2$. For all $n, k, j \in \mathbb{N}$ there is $D = D(p, m, n, k, j, \chi_{\mathrm{sp}}, C_0, \dots, C_{n+k+j+2}) > 0$, independent of y, ξ, such that*

$$|\partial_y^j \partial_\xi^k a_n(y, \xi)| \le D\xi^{m-n-k}(y + \xi^p)^{-n} y^{-j} \quad (y > 0,\ \xi \ge 1).$$

Proof. Derivatives of the cutoff are supported where ξ and $\lambda_0(y)$ are comparable; every ξ derivative costs ξ^{-1} and every y derivative costs y^{-1}. The division is smooth after extension by zero because the numerator vanishes on a whole neighbourhood of the zero denominator. Induct on n, expanding the finite Leibniz sum; apply the reciprocal bounds and $(y + \xi^p)^{-1} \le y^{-1}$ for the extra y derivatives. Keep all mixed derivatives through the finite order needed in the induction. □

Lemma 3.76 (Boundary terms and the improper far integral)**.** *Under the preceding hypotheses with $m < p$, for every $y > 0$ and integer $n \ge 1$, the function $\xi \mapsto a_n(y, \xi)$ is integrable on $[1, \infty)$ and*

$$\lim_{R \to \infty} \int_1^R g_y(\xi) e^{i\phi_y(\xi)} d\xi = (-1)^n \int_1^\infty a_n(y, \xi) e^{i\phi_y(\xi)} d\xi.$$

Every boundary term $a_l(y, R)e^{i\phi_y(R)}/(i(y - R^p))$, $0 \le l < n$, tends to zero; the boundary terms at 1 vanish exactly.

Proof. Integrate by parts on $[1, R]$, interpreting the quotient as its smooth zero extension. The upper boundary term is $O_y(R^{m-l(p+1)-p})$, whose exponent is negative because $m < p$. The final amplitude is $O_y(\xi^{m-n(p+1)})$, integrable for $n \ge 1$. Flatness of a at 1 gives the lower boundary identities. Iterate the finite formula before taking limits. □

Lemma 3.77 (Integrable parameter-derivative majorants for the far representation)**.** *With a_n as in Lemma 3.75, for every $n, j \in \mathbb{N}$ there is $C = C(p, m, n, j, \chi_{\mathrm{sp}}, C_0, \dots, C_{n+j+2})$ such that for $y, \xi \geq 1$,*

$$|\partial_y^j (a_n(y,\xi) e^{i\phi_y(\xi)})| \leq C\xi^{m-n+j}(y+\xi^p)^{-n} \leq C y^{-n/2} \xi^{m-n+j-pn/2}.$$

On a fixed compact interval $J \subset (0,\infty)$ the same derivatives are bounded by $C_J \xi^{m-(p+1)n+j}$ for $\xi \geq 1$. Thus choosing n with $m-(p+1)n+j < -1$ gives an integrable majorant on J; choosing additionally $n/2 \geq M$ and $m-n+j-pn/2 < -1$ gives a bound $C_M y^{-M}$ for the integral of the derivative for $y \geq 1$. Constants are independent of the truncation radius.

Proof. Apply Leibniz; a derivative of $e^{i\phi_y}$ contributes $i\xi$, and the other factors are bounded by the preceding mixed-symbol estimate. On $y \geq 1$ discard factors y^{-1}; use $y+\xi^p \geq \sqrt{y\xi^p}$. On J, the factors y^{-1} are uniformly bounded. Integrate the resulting explicit powers on $[1,\infty)$. □

Lemma 3.78 (Finite-order differentiation of a parametric integral over $(1,\infty)$)**.** *Let $N \in \mathbb{N}$ and $F : \mathbb{R} \times \mathbb{R} \to \mathbb{C}$ with $(y,\xi) \mapsto F(y,\xi)$ of class C^∞ on $(0,\infty) \times (1,\infty)$. Assume that for every $j \leq N$ and all $0 < y_1$, $y_2 \in \mathbb{R}$ there is $g \in L^1((1,\infty))$ with $|\partial_y^j F(y,\xi)| \leq g(\xi)$ for all $y \in [y_1, y_2]$ and $\xi > 1$. Then $y \mapsto \int_{(1,\infty)} F(y,\xi)\, d\xi$ is C^N on $(0,\infty)$ and for every $j \leq N$ and $y > 0$,*

$$\partial_y^j \int_{(1,\infty)} F(y,\xi)\, d\xi = \int_{(1,\infty)} \partial_y^j F(y,\xi)\, d\xi.$$

Proof. Induction on $j < N$: the ξ-integrand $\partial_y^j F(y,\cdot)$ is continuous (hence measurable) and integrable by the majorant; its y-derivative $\partial_y^{j+1} F$ is dominated on $[y_1, y_2]$ by the majorant of order $j+1$, so the dominated differentiation theorem gives $\partial_y \int \partial_y^j F = \int \partial_y^{j+1} F$ on every interval $(y_1, y_2) \subset (0,\infty)$. Continuity of the order-N integral follows by dominated convergence, which gives C^N. □

Lemma 3.79 (Nonstationary (far) part)**.** *Let $m < p$ and a as in Lemma 3.73. For $y > 0$ the improper integral $I_{\mathrm{far}}(y) = \int_0^\infty a(\xi)(1-\chi_{\mathrm{sp}}(\xi/\lambda_0)) e^{i\phi_y(\xi)} d\xi$ exists, $I_{\mathrm{far}} \in C^\infty((0,\infty))$, and for all $j, M \geq 0$ there is C (depending on p, m, j, M and finitely many C_k) with $|\partial_y^j I_{\mathrm{far}}(y)| \leq C y^{-M}$ for $y \geq 1$.*

Proof.

Step 1 (nonstationarity). On the support ($\xi \geq 1$, $\xi \leq \frac{3}{4}\lambda_0$ or $\xi \geq \frac{3}{2}\lambda_0$): $|\phi_y'(\xi)| = |y - \xi^p| \geq c_p(y+\xi^p)$ with $c_p = \frac{1}{2}\min(1-(3/4)^p, 1-(2/3)^p)$. Step 2 (symbol class). Say G (smooth, supported in the set of Step 1) is of type (σ,τ) if $|\partial_\xi^k G| \leq C_k \xi^{\sigma-k}(y+\xi^p)^{-\tau}$ uniformly in $y \geq 1$. $g_y := a(1-\chi_{\mathrm{sp}}(\cdot/\lambda_0))$ is of type $(m,0)$ (derivatives of $\chi_{\mathrm{sp}}(\xi/\lambda_0)$ are supported where $\xi \approx \lambda_0$). Since $|\partial_\xi^k(1/\phi_y')| \leq C_k \xi^{-k}(y+\xi^p)^{-1}$ (Faà di Bruno with $|\partial^k \xi^p| \leq C\xi^{p-k}$), $L_y G := \partial_\xi(G/(i\phi_y'))$ maps type (σ,τ) to type $(\sigma-1, \tau+1)$. Step 3 (integration by parts). $\int_1^R g e^{i\phi_y} = [g e^{i\phi_y}/(i\phi_y')]_1^R - \int_1^R L_y g\, e^{i\phi_y}$; the boundary term at R is $O(R^{m-p}) \to 0$ since $m < p$, at 1 it vanishes. This proves existence and, iterating ($L_y^k g$ is absolutely integrable for $k \geq 1$), $I_{\mathrm{far}}(y) = (-1)^N \int_1^\infty (L_y^N g_y) e^{i\phi_y} d\xi$ for every $N \geq 1$. Step 4 (derivatives). ∂_y applied to the integrand produces: $i\xi$ from $e^{i\phi_y}$ (type $(\sigma+1,\tau)$), $\partial_y(1/\phi_y') = -1/\phi_y'^2$ (type raises τ), and $\partial_y[\chi_{\mathrm{sp}}(\xi/\lambda_0)] = -\frac{1}{py}\frac{\xi}{\lambda_0}\chi_{\mathrm{sp}}'(\xi/\lambda_0)$

(type $(0,0)$ times y^{-1}). So $\partial_y^j[(L_y^N g_y)e^{i\phi_y}]$ is a finite sum of terms bounded by $C\xi^{m-N+j}(y+\xi^p)^{-N} \le C\xi^{m-N+j-pN/2}y^{-N/2}$; for $N > 2(m+j+1)$ these are integrable uniformly for y in compacts of $[1,\infty)$ (similarly of $(0,\infty)$), which justifies differentiating under the integral, and give $|\partial_y^j I_{\mathrm{far}}| \le Cy^{-N/2}$. Take $N \ge 2M$. □

Lemma 3.80 (Homogeneous stationary phase)**.** *Let $0 < p < 1$, $m \le 0$, and $a \in C^\infty((0,\infty))$ with $a = 0$ on $(0,1]$ and $|\partial_\xi^k a(\xi)| \le C_k\xi^{m-k}$ for $\xi \ge 1$, $k \ge 0$. Let $\lambda_0(y) = y^{1/p}$, $\Psi_0(y) = \frac{p}{p+1}y^{1+1/p}$. For $y > 0$ the improper integral $I(y) = \int_0^\infty a(\xi)e^{i(y\xi-\xi^{p+1}/(p+1))}d\xi$ exists, and defining $r : (0,\infty) \to \mathbb{C}$ by*

$$I(y) = e^{i(\Psi_0(y)-\pi/4)}\Big(\frac{2\pi}{p\lambda_0(y)^{p-1}}\Big)^{1/2}\big[a(\lambda_0(y)) + r(y)\big], \tag{3.8}$$

r is C^∞ and for every $j \ge 0$ there is C depending only on p, m, j and finitely many C_k with

$$|\partial_y^j r(y)| \le C\lambda_0(y)^{m-(p+1)}y^{-j}, \qquad y \ge 1. \tag{3.9}$$

Proof. $I = I_{\mathrm{near}} + I_{\mathrm{far}}$ (the near part is a proper integral), so I exists and is smooth. Then $r = \lambda_0^m\rho(\mu; c_y) + I_{\mathrm{far}}/P$ by Lemma 3.73(2). The first term is bounded by Lemma 3.73(3). For the second, $1/P = e^{-i(\Psi_0-\pi/4)}(p\lambda_0^{p-1}/2\pi)^{1/2}$ with $\Psi_0' = \lambda_0$, so $|\partial_y^i(1/P)| \le Cy^{i/p}$ for $y \ge 1$; by Leibniz and Lemma 3.79 with M large, $|\partial_y^j(I_{\mathrm{far}}/P)| \le Cy^{j/p-M} \le C\lambda_0^{m-p-1}y^{-j}$. □

3.4.5 An oscillatory integral against a tail integral

Lemma 3.81 (Stationary phase against a tail integral)**.** *Let $x_0 \in \mathbb{R}$, $L > 0$, $I = (x_0 - L, x_0 + L)$, $\kappa > 0$, $A, M \ge 0$, $\omega, \omega_0 \in \mathbb{R}$, and constants $c_1, \dots, c_4 > 0$. Assume:*

1. *$\Phi \in C^3(\overline{I}; \mathbb{R})$, $\Phi'(x_0) = 0$, $c_1\kappa \le |\Phi''| \le c_2\kappa$ and $|\Phi'''| \le c_3\kappa L^{-1}$ on I;*
2. *$a \in C_c^2(I; \mathbb{C})$ with $|a^{(j)}| \le c_4AL^{-j}$, $j = 0, 1, 2$;*
3. *$\Psi \in C([x_0 - L, \infty); \mathbb{R})$ is such that $E(t) = \int_t^\infty e^{i\Psi(s)}ds$ exists (improper) for every $t \in I$, and $|E(t)| \le M$ on I;*
4. *$\Phi(t) + \Psi(t) = \omega t + \omega_0$ for $t \in I$.*

Then $\big|\int_I a(t)e^{i\Phi(t)}E(t)dt\big| \le C\,A(M\kappa^{-1/2} + \kappa^{-1})$ with $C = C(c_1, \dots, c_4)$.

Proof. WLOG $x_0 = 0$. $E \in C^1(I)$ with $E' = -e^{i\Psi}$. Φ'' is continuous and non-vanishing, hence of constant sign, so $|\Phi'(t)| \ge c_1\kappa|t|$. Case $L \le \kappa^{-1/2}$: bound absolutely, $\le 2Lc_4AM \le 2c_4AM\kappa^{-1/2}$. Case $L > \kappa^{-1/2}$. Step 1: $|\int_{|t|\le\kappa^{-1/2}}| \le 2c_4AM\kappa^{-1/2}$. Step 2 (bounds for $|t| \ge \kappa^{-1/2}$): $|a/\Phi'| \le CA/(\kappa|t|)$ and $|(a/\Phi')'| \le |a'|/|\Phi'| + |a||\Phi''|/\Phi'^2 \le CA/(\kappa L|t|) + CA/(\kappa t^2) \le CA/(\kappa t^2)$ (as $|t| < L$). Step 3 (integration by parts on $\kappa^{-1/2} < |t| < L$), writing $e^{i\Phi} = \frac{1}{i\Phi'}\partial_t e^{i\Phi}$ and using $aEe^{i\Phi} \to 0$ at $\pm L$ (compact support): the integral equals the boundary terms at $\pm\kappa^{-1/2}$, each $\le CAM/(\kappa\cdot\kappa^{-1/2})$, minus $\frac{1}{i}\int(a/\Phi')'Ee^{i\Phi}$, bounded by $M\int_{|t|>\kappa^{-1/2}} CA/(\kappa t^2) = 2CAM\kappa^{-1/2}$, plus $\frac{1}{i}\int_{\kappa^{-1/2}<|t|<L}\frac{a}{\Phi'}e^{i\Psi+i\Phi} = \frac{e^{i\omega_0}}{i}\int_{\kappa^{-1/2}<|t|<L}\frac{a(t)}{\Phi'(t)}e^{i\omega t}dt$. Step 4. Let $h(t) = ta(t)/\Phi'(t)$; since $\Phi'(t)/t = \int_0^1 \Phi''(\sigma t)d\sigma$ (modulus $\ge c_1\kappa$)

and $\partial_t(\Phi'(t)/t) = \int_0^1 \sigma\Phi'''(\sigma t)d\sigma$, h extends to a C^1 function on I with $h(0) = a(0)/\Phi''(0)$ and $|h'| \le CA/(\kappa L)$. Hence $|a/\Phi' - h(0)/t| \le CA/(\kappa L)$, whose integral over $|t| < L$ is $\le 2CA/\kappa$. Step 5. $\int_{\kappa^{-1/2}<|t|<L} \frac{e^{i\omega t}}{t}dt = 2i\int_{\kappa^{-1/2}}^{L} \frac{\sin \omega t}{t}dt$, bounded by $2\sup_{0\le u<v} |\int_u^v \frac{\sin x}{x}dx| \le 2\pi$ (finite absolute constant), and $|h(0)| \le c_4 A/(c_1\kappa)$. □

3.4.6 Oscillatory multiplier expansion for $|D|^p$

Notation: $0 < p < 1$, $m(\xi) = \xi^p$ for $\xi > 0$, c_p as in Lemma 3.25. For a measurable g with $g(z)|z|^{-1-p}e^{-\eta z^2} \in L^1$ for all $\eta > 0$ we write $\int^{\mathrm{osc}} g(z)|z|^{-1-p}dz := \lim_{\eta\downarrow 0} \int g(z)e^{-\eta z^2}|z|^{-1-p}dz$ when the limit exists.

Lemma 3.82 (Uniform Gaussian and cutoff symbol bounds)**.** *Fix $0 < p < 1$, $K > p$, an integer $n \ge 0$, and a smooth cutoff $\chi_1 : \mathbb{R} \to [0,1]$ equal to one on $[-1,1]$ and supported in $[-2,2]$. For every $j \le n$ there is $D_j > 0$ with $|\partial_z^j e^{-\eta z^2}| \le D_j|z|^{-j}$ for $\eta \ge 0$, $z \ne 0$. If $b \in C^\infty(\mathbb{R};\mathbb{C})$ and $|b^{(j)}(z)| \le C_j B|z|^{K-j}$ for $0 \le j \le n$, $z \ne 0$, $B \ge 0$, then for every $\Lambda > 0$, $\eta \ge 0$,*

$$g_{\eta,\Lambda}(z) = (1 - \chi_1(\Lambda z))e^{-\eta z^2}b(z)|z|^{-1-p}$$

extends smoothly through 0, vanishes on $|z| \le 1/\Lambda$, and $|g^{(j)}_{\eta,\Lambda}(z)| \le DB|z|^{K-1-p-j}$ for $z \ne 0$, $j \le n$, where D depends only on $p, K, n, \chi_1, C_0, \dots, C_n$.

Proof. For $\eta > 0$, induction expresses the Gaussian derivative as a polynomial in $\sqrt{\eta}z$ times $|z|^{-j}e^{-\eta z^2}$; each polynomial times the Gaussian is bounded on $\mathbb{R}$. At $\eta = 0$ the assertion is immediate. Nonzero derivatives of $\chi_1(\Lambda z)$ are supported where $1 \le \Lambda|z| \le 2$, so they cost $C_j|z|^{-j}$. Leibniz and the explicit derivatives of $|z|^{-1-p}$ give the finite-order bound. □

Lemma 3.83 (Absolute near-diagonal bound for the homogeneous kernel)**.** *Let $0 < p < 1$, $K > p$, $B, C_0 \ge 0$, $\Lambda > 0$, $\eta \ge 0$, and let b be measurable with $|b(z)| \le C_0 B|z|^K$ for $z \ne 0$. For a measurable real phase ϕ and any measurable cutoff $0 \le \chi_1 \le 1$ supported in $[-2,2]$, the near integrand is absolutely integrable and*

$$\left|\int_{\mathbb{R}} \chi_1(\Lambda z)e^{i\phi(z)-\eta z^2}b(z)|z|^{-1-p}dz\right| \le \frac{2C_0B}{K-p}(2/\Lambda)^{K-p}.$$

Proof. Bound the modulus by $C_0B\mathbf{1}_{|z|\le 2/\Lambda}|z|^{K-1-p}$. Integrate on the two half-lines; the exponent is greater than -1 because $K > p$. □

Lemma 3.84 (Far homogeneous-kernel bound after an explicit integer number of integrations)**.** *Let $0 < p < 1$, $K > p$, $B \ge 0$, $\Lambda > 0$, $c_0 > 0$, $U \subset \mathbb{R}$ open, $b \in C_c^\infty(U;\mathbb{C})$, and $\phi \in C^\infty(U;\mathbb{R})$. Assume on U, for $z \ne 0$, $|b^{(j)}(z)| \le C_jB|z|^{K-j}$ and $|(1/\phi')^{(j)}(z)| \le C_j\Lambda^{-1}|z|^{-j}$, with $c_0^{-1}\Lambda \le |\phi'| \le c_0\Lambda$. Fix χ_1 as in Lemma 3.82 and an integer $L > K - p$; the derivative hypotheses are required only for $0 \le j \le L+1$. There is $C = C(p, K, L, c_0, \chi_1, C_0, \dots, C_{L+1}) > 0$, independent of b, B, Λ, η, such that, with $g = g_{\eta,\Lambda}$,*

$$\int e^{i\phi}g = (-1)^L \int e^{i\phi}L_\phi^L g, \qquad |L_\phi^L g(z)| \le CB\Lambda^{-L}|z|^{K-1-p-L}\mathbf{1}_{|z|\ge 1/\Lambda}.$$

All integrals converge absolutely and their modulus is bounded by $2CB\Lambda^{p-K}/(L+p-K)$.

Proof. The cutoff removes the origin and b provides compact support, so the finite IBP identity has zero boundary terms. Apply the symbol iteration with $\sigma = K - 1 - p$ and integrate $|z|^{K-1-p-L}$ on $|z| \geq 1/\Lambda$. The denominator $L + p - K$ is strictly positive. □

Lemma 3.85 (Integration by parts against the homogeneous kernel)**.** *Let $0 < p < 1$, $\Lambda > 0$, $B \geq 0$, $K > p$, and $c_0, (C_s)_{s\geq 0}$ constants. Let $U \subset \mathbb{R}$ be open, $b \in C^\infty(\mathbb{R}; \mathbb{C})$ with $\operatorname{supp} b \subset U$ compact, $\phi \in C^\infty(U; \mathbb{R})$, such that on U: $|\partial_z^s b(z)| \leq C_s B|z|^{K-s}$, $c_0^{-1}\Lambda \leq |\phi'(z)| \leq c_0\Lambda$, $|\partial_z^s(1/\phi'(z))| \leq C_s\Lambda^{-1}|z|^{-s}$ ($s \geq 0$). Then for all $\eta \geq 0$*

$$\Big|\int_{\mathbb{R}} e^{i\phi(z)}e^{-\eta z^2}\frac{b(z)}{|z|^{1+p}}dz\Big| \leq CB\Lambda^{p-K},$$

with C depending only on p, K, c_0 and finitely many C_s; and the integral is continuous in $\eta \geq 0$.

Proof. Let $\chi_\Lambda(z) = \chi_1(\Lambda z)$ with $\chi_1 \in C_c^\infty([-2,2])$, $\chi_1 = 1$ on $[-1,1]$. Near part: $\leq \int_{|z|\leq 2/\Lambda} C_0 B|z|^{K-1-p} = CB\Lambda^{p-K}$ (as $K > p$). Far part: integrate by parts $L > K - p$ times with $e^{i\phi} = \frac{1}{i\phi'}\partial_z e^{i\phi}$ (no boundary terms: compact support, and the integrand vanishes near 0). The amplitude $g = (1-\chi_\Lambda)e^{-\eta z^2}b|z|^{-1-p}$ satisfies $|\partial^s g| \leq CB|z|^{K-1-p-s}$ on $|z| \geq 1/\Lambda$, uniformly in η (since $|\partial^s\chi_\Lambda| \leq C\Lambda^s \leq C|z|^{-s}$ on its support and $|\partial_z^s e^{-\eta z^2}| \leq C_s|z|^{-s}$ uniformly in $\eta \geq 0$). Each step multiplies by $\Lambda^{-1}|z|^{-1}$, giving $\leq CB\Lambda^{-L}\int_{|z|\geq 1/\Lambda}|z|^{K-1-p-L} = CB\Lambda^{p-K}$. Continuity in η: dominated convergence on the compact support. □

Lemma 3.86 (Moments of the kernel)**.** *For $\lambda > 0$ and integers $k \geq 1$, $\int^{\mathrm{osc}} e^{i\lambda z}z^k|z|^{-1-p}dz$ exists and*

$$-c_p\int^{\mathrm{osc}} e^{i\lambda z}\frac{z^k}{|z|^{1+p}}dz = (-i)^k m^{(k)}(\lambda), \tag{3.10}$$

locally uniformly in $\lambda \in (0,\infty)$. Also $c_p\int^{\mathrm{osc}}(1-e^{i\lambda z})|z|^{-1-p}dz = c_p\int(1-e^{i\lambda z})|z|^{-1-p}dz = \lambda^p$.

Proof. Let $g_\eta(\lambda) = c_p\int(1-e^{i\lambda z})e^{-\eta z^2}|z|^{-1-p}dz$, $\eta > 0$; it is C^∞ in λ with $g_\eta^{(k)}(\lambda) = -c_p\int(iz)^k e^{i\lambda z-\eta z^2}|z|^{-1-p}dz$ for $k \geq 1$, and $g_\eta \to \lambda^p$ pointwise by dominated convergence and Lemma 3.25. For $k \geq 1$, $g_\eta^{(k)}$ converges locally uniformly on $(0,\infty)$ as $\eta \downarrow 0$: split with $\chi_1(z)$; the part $|z| \leq 2$ converges by dominated convergence ($|z|^{k-1-p}$ integrable); on $|z| \geq 1$ integrate by parts $L > k - p$ times in z using $\partial_z e^{i\lambda z} = i\lambda e^{i\lambda z}$, obtaining absolutely convergent integrals, dominated uniformly in η and $\lambda \in [\lambda_1, \lambda_2]$. A sequence of functions converging pointwise whose derivatives converge locally uniformly has a differentiable limit with derivative the limit of the derivatives; iterate in k. Lean route: no integration by parts on $|z| \geq 1$. With $Q_m(\eta, l) = c_p\int P_m(l,z)e^{-\eta z^2}|z|^{-1-p}dz$ ($P_0 = 1 - e^{ilz}$, $P_m = -e^{ilz}z^m$): $\partial_l Q_m = iQ_{m+1}$ for $\eta > 0$; $|Q_m(\eta,l)| \leq C$ uniformly for $\eta > 0$, $l \geq a > 0$, $m \geq 1$ (from Lemma 3.85 applied to truncations $z^m\chi(z/R)$, $R \to \infty$); $k = 0$ by dominated convergence; induction on m using convergence of functions plus uniform second-derivative bounds; uniformity on $[\lambda_1, \lambda_2]$ from the scaling $Q_m(\eta, l) = l^{p-m}Q_m(\eta/l^2, 1)$. □

Definition 3.87 (Phase data for the oscillatory multiplier)**.** For $a : \mathbb{R} \to \mathbb{C}$, $\Psi : \mathbb{R} \to \mathbb{R}$, $Y \in \mathbb{R}$, put $\lambda = \Psi'$ and

$$\Theta_y(z) = \Psi(y+z) - \Psi(y) - \lambda(y)z, \qquad F_y(z) = \begin{cases} a(y+z)e^{i\Theta_y(z)}, & y+z \in (Y/4, 4Y), \\ 0, & \text{otherwise.} \end{cases}$$

For $k_0 \in \mathbb{N}$, $C_k \in \mathbb{R}$ and real Λ, A, c_0, the predicate $\mathsf{H}(Y, \Lambda, A, c_0, (C_k), k_0; a, \Psi)$ means that $a \in C^\infty(\mathbb{R})$, $\operatorname{supp} a \subset (Y/4, 4Y)$, $\Psi \in C^\infty((Y/4, 4Y))$, and for all $x \in (Y/4, 4Y)$ and $0 \le k \le k_0$,

$$c_0^{-1}\Lambda \le \lambda(x) \le c_0\Lambda, \quad |\partial^k \lambda(x)| \le C_k \Lambda Y^{-k}, \quad |\partial^k a(x)| \le C_k A Y^{-k}.$$

Positivity and size restrictions on Y, Λ, A, c_0 are hypotheses of the lemmas using this predicate.

Definition 3.88 (Multiplier coefficients and remainders)**.** Let $m(\xi) = \xi^p$. For $k \in \mathbb{N}$ and $N \ge 1$ define

$$B_k(y) = \frac{(-i)^k}{k!} m^{(k)}(\lambda(y)) F_y^{(k)}(0), \qquad R_{2N-1}(z) = F_y(z) - \sum_{k=0}^{2N-2} \frac{F_y^{(k)}(0)}{k!} z^k,$$

$$\mathcal{R}_{\mathrm{mult},N}(y) = e^{-i\Psi(y)} |D|^p (a e^{i\Psi})(y) - \sum_{k=0}^{2N-2} B_k(y), \qquad \widetilde{\mathcal{R}}_{\mathrm{mult},2}(y) = \mathcal{R}_{\mathrm{mult},2}(y) - \frac{p(p-1)}{2} \lambda(y)^{p-2} a''(y).$$

The finite index set consists of (r, q) with $r \in \{0, \dots, 2N-1\}$, $q = (q_1, \dots, q_{2N-1}) \in \{0, \dots, 2N-1\}^{2N-1}$, and $r + \sum_{j=1}^{2N-1} j q_j = 2N - 1$. Its coefficient is $d_{r,q} = i^{\sum_j q_j} / (r! \prod_j q_j! (j!)^{q_j})$. These are definitions; their integral representations and bounds require the lemmas below.

Definition 3.89 (Individual multiplier summands)**.** For an index (r, q) as above, let q_1 be its first component and $\ell = \sum_{j \ge 2} q_j$. Define

$$b_t(z) = z^{2N-1} a^{(r)}(y + tz) \prod_{j=1}^{2N-1} (\Theta_y^{(j)}(tz))^{q_j}$$

when $y + tz \in (Y/4, 4Y)$, and $b_t(z) = 0$ otherwise. Put

$$\phi_t(z) = \lambda(y)z + \Theta_y(tz), \qquad I_\eta(y, t) = \int_{\mathbb{R}} e^{i\phi_t(z) - \eta z^2} \frac{b_t(z)}{|z|^{1+p}} \, dz, \qquad B = A \Lambda^{q_1 + \ell} Y^{-2N+1+\ell} t^{q_1}.$$

Here I_η is the total Bochner integral; the hypotheses ensuring convergence and estimates are imposed in the following lemmas.

Lemma 3.90 (Recursive jets of an exponential)**.** *Let $U \subset \mathbb{R}$ be open and $\Theta \in C^\infty(U; \mathbb{R})$. Define $E_0 = 1$ and $E_{n+1} = E_n' + i\Theta' E_n$ on U. For every $n \in \mathbb{N}$, E_n is smooth and $\partial^n e^{i\Theta} = e^{i\Theta} E_n$ on U. In particular $E_1 = i\Theta'$ and $E_2 = i\Theta'' - (\Theta')^2$.*

Proof. Induct on n, applying the product rule and $(e^{i\Theta})' = i\Theta' e^{i\Theta}$. This fixes a recursive implementation of the jets before any closed combinatorial formula is used. □

Lemma 3.91 (Finite multiindex formula for exponential jets)**.** *For the jets of Lemma 3.90, and every $n \geq 1$,*

$$E_n = n! \sum_{\substack{q_1,\dots,q_n \geq 0 \\ \sum_{j=1}^n j q_j = n}} \frac{i^{\sum_j q_j}}{\prod_j q_j! (j!)^{q_j}} \prod_{j=1}^n (\Theta^{(j)})^{q_j}.$$

The index set is finite because each $q_j \leq n$.

Proof. Induct in the recursion. Multiplication by $i\Theta'$ increments q_1. Differentiating the j-th factor removes one $\Theta^{(j)}$ and adds one $\Theta^{(j+1)}$, contributing its multiplicity q_j. Reindex these finite sums; factorial cancellation gives the displayed coefficient at order $n+1$. The required finite-index reindexing and coefficient arithmetic are the concrete subtask here; no analytic estimates enter it. □

Lemma 3.92 (Absolutely convergent kernel identity for a localized oscillatory profile)**.** *Let $0 < p < 1$, $Y \geq 1$, $a \in C_c^\infty((Y/4, 4Y); \mathbb{C})$, and $\Psi \in C^\infty((Y/4, 4Y); \mathbb{R})$ with $\lambda = \Psi' > 0$. For every $y \in [Y, 2Y]$, $F_y \in C_c^\infty(\mathbb{R})$, $F_y(0) = a(y)$, and*

$$e^{-i\Psi(y)} |D|^p (a e^{i\Psi})(y) - \lambda(y)^p a(y) = -c_p \int e^{i\lambda(y)z} (F_y(z) - a(y)) |z|^{-1-p} dz.$$

The integral on the right converges absolutely, as does the defining kernel integral on the left.

Proof. Apply the fractional-kernel formula to the smooth supported product and subtract the plane-wave identity for λ^p. Near zero, $F_y(z) - F_y(0) = O(|z|)$; at infinity F_y vanishes and the remaining constant is integrable against $|z|^{-1-p}$. These two majorants justify subtraction and all changes of variables. □

Lemma 3.93 (Integral Taylor remainder with explicit amplitude and phase factors)**.** *With the data of Lemma 3.92, $N \geq 1$, and $y \in [Y, 2Y]$, put $n = 2N-1$. For every $z \in \mathbb{R}$,*

$$R_n(z) = n \sum_{r + \sum_{j=1}^n j q_j = n} d_{r,q} \int_0^1 (1-t)^{n-1} e^{i\Theta_y(tz)} b_t(z)\, dt.$$

Each integral is absolutely convergent. Here the amplitude b_t includes the factor z^n and is zero when $y + tz \notin (Y/4, 4Y)$, as in Definition 3.89.

Proof. Taylor's integral remainder for the globally smooth F_y gives $R_n(z) = z^n/(n-1)! \int_0^1 (1-t)^{n-1} F_y^{(n)}(tz) dt$. Expand by Leibniz and the exponential-jet formula. Outside the phase domain, every amplitude derivative vanishes on a neighbourhood, so the zero-extension convention is legitimate. Cancel $n!/(n-1)! = n$ after collecting the finite index set. □

Lemma 3.94 (Removal of Gaussian regularization from the Taylor remainder)**.** *Under the same hypotheses, for every $N \geq 1$, $y \in [Y, 2Y]$, and $\eta > 0$, $e^{i\lambda(y)z - \eta z^2} R_{2N-1}(z) |z|^{-1-p}$ is integrable, and*

$$\lim_{\eta \downarrow 0} -c_p \int e^{i\lambda(y)z - \eta z^2} R_{2N-1}(z) |z|^{-1-p} dz = \mathcal{R}_{\mathrm{mult,N}}(y).$$

Proof. The Taylor remainder vanishes to order $2N-1 \geq 1$ at zero and grows polynomially at infinity, giving integrability for $\eta > 0$. Expand it into $F_y - a(y)$ minus the finite polynomial with degrees 1 through $2N-2$. Dominated convergence applies to the first term, and the regularized moment lemma evaluates every polynomial term. This proves existence and value of the limit before any uniform remainder estimate is used. □

Lemma 3.95 (Kernel representation and Taylor expansion)**.** *Let $Y, \Lambda \geq 1$, $a \in C_c^\infty((Y/4, 4Y); \mathbb{C})$, $\Psi \in C^\infty((Y/4, 4Y); \mathbb{R})$, $\lambda = \Psi' > 0$ there. For $Y \leq y \leq 2Y$ set $\Theta_y(z) = \Psi(y+z) - \Psi(y) - \lambda(y)z$ and $F_y(z) = a(y+z)e^{i\Theta_y(z)}$ for $y + z \in (Y/4, 4Y)$, $F_y(z) = 0$ otherwise ($F_y \in C_c^\infty(\mathbb{R})$). Then*

$$e^{-i\Psi(y)}|D|^p(ae^{i\Psi})(y) - \lambda(y)^p a(y) = -c_p \int_{\mathbb{R}} e^{i\lambda(y)z} \frac{F_y(z) - a(y)}{|z|^{1+p}} dz \tag{3.11}$$

(absolutely convergent), and with B_k as in (3.20)*, $N \geq 1$, and $R_{2N-1}(z) := F_y(z) - \sum_{k=0}^{2N-2} \frac{F_y^{(k)}(0)}{k!} z^k$,*

$$\mathcal{R}_{\mathrm{mult},N}(y) := e^{-i\Psi(y)}|D|^p(ae^{i\Psi})(y) - \sum_{k=0}^{2N-2} B_k(y) = -c_p \int^{\mathrm{osc}} e^{i\lambda(y)z} \frac{R_{2N-1}(z)}{|z|^{1+p}} dz.$$

Moreover (3.12) *holds:*

$$R_{2N-1}(z) = (2N-1)z^{2N-1} \sum_{\substack{r, q_1, \dots, q_{2N-1} \geq 0 \\ r + \sum_j j q_j = 2N-1}} \frac{i^{\sum_j q_j}}{r! \prod_j q_j! (j!)^{q_j}} \int_0^1 (1-t)^{2N-2} e^{i\Theta_y(tz)} a^{(r)}(y+tz) \prod_{j=1}^{2N-1} \left(\Theta_y^{(j)}(tz)\right)^{q_j} dt, \tag{3.12}$$

where each summand is interpreted as 0 when $y + tz \notin (Y/4, 4Y)$.

Proof.

Step 1. $ae^{i\Psi} \in C_c^\infty(\mathbb{R})$; by Lemma 3.25, $e^{-i\Psi(y)}|D|^p(ae^{i\Psi})(y) = c_p \int (a(y) - e^{i\lambda z} F_y(z))|z|^{-1-p} dz$; subtract $\lambda^p a(y) = c_p a(y) \int (1 - e^{i\lambda z})|z|^{-1-p}$; both integrals absolutely convergent ($F_y(z) - a(y) = O(|z|)$ at 0). Step 2. Insert $e^{-\eta z^2}$ (dominated convergence) and expand $F_y - a(y) = \sum_{k=1}^{2N-2} \frac{F_y^{(k)}(0)}{k!} z^k + R_{2N-1}$; every piece is integrable against $e^{-\eta z^2}|z|^{-1-p}$ for $\eta > 0$. By Lemma 3.86, the k-th piece tends to $\frac{(-i)^k}{k!} m^{(k)}(\lambda) F_y^{(k)}(0) = B_k$ (and $B_0 = \lambda^p a$). So the limit for R_{2N-1} exists and gives the formula. Step 3. Taylor's formula with integral remainder, then Faà di Bruno/Leibniz for $F_y^{(2N-1)}(w) = \sum \frac{(2N-1)!}{r! \prod q_j! (j!)^{q_j}} a^{(r)}(y+w) e^{i\Theta_y(w)} \prod (i\Theta_y^{(j)}(w))^{q_j}$; and $(2N-1)!/(2N-2)! = 2N-1$. □

Lemma 3.96 (Support and mixed symbol bounds for a Taylor summand)**.** *Fix $N \geq 1$ and an index $r + \sum_{l=1}^{2N-1} l q_l = 2N-1$. For $Y, \Lambda \geq 1$, $A > 0$, $c_0 > 0$, assume the phase-data hypothesis H through order $2N + j + s + 2$, where $j, s \in \mathbb{N}$ are prescribed. Put $K = 2N - 1 + q_1$ and $B = A\Lambda^{q_1+\ell} Y^{-2N+1+\ell} t^{q_1}$. For $0 < t \leq 1$, $y \in [Y, 2Y]$, the amplitude b_t is smooth and compactly supported in $\{z : y + tz \in (Y/4, 4Y)\}$, and*

$$|\partial_y^j \partial_z^s b_t(z)| \leq CBY^{-j}|z|^{K-s} \qquad (z \neq 0).$$

Here C depends only on $N, j, s, c_0, C_0, \dots, C_{2N+j+s+2}$ and is uniform in Y, Λ, A, y, t. The same local smoothness and estimates, with enlarged constants, hold for $Y/2 < y < 3Y$.

Proof. Use $\Theta'_y(tz) = \lambda(y+tz) - \lambda(y)$ and the fundamental theorem of calculus along the segment inside the phase interval. Each mixed y derivative costs Y^{-1}; each z derivative costs $t/Y \le 4/|z|$ on the support. The factor $\Theta'_y(tz)$ contributes $\Lambda Y^{-1} t|z|$, and every higher phase derivative contributes ΛY^{1-l}. Leibniz and $r + \sum l q_l = 2N-1$ give the displayed powers. Smooth zero extension follows because every derivative of a is supported in the fixed compact support of a, strictly inside the phase interval. □

Lemma 3.97 (Mixed phase bounds and convex interpolation)**.** *Fix $j, s \in \mathbb{N}$, $Y, \Lambda \ge 1$, $c_0 > 0$, and assume the phase bounds in H through order $j+s+2$. For $0 < t \le 1$, $y \in [Y, 2Y]$, and $y + tz \in (Y/4, 4Y)$,*

$$\phi'_t(z) = (1-t)\lambda(y) + t\lambda(y+tz), \qquad c_0^{-1}\Lambda \le \phi'_t(z) \le c_0\Lambda,$$

$$|\partial_y^j \partial_z^s (1/\phi'_t)(z)| \le C\Lambda^{-1} Y^{-j} (t/Y)^s.$$

For $j \ge 1$, $z \ne 0$, also $|\partial_y^j \partial_z^s \phi_t(z)| \le C\Lambda Y^{-j}|z|^{1-s}$. These bounds hold uniformly for convex combinations of two such phases with the same constants, and extend to $Y/2 < y < 3Y$ with enlarged constants. Constants depend only on j, s, c_0 and the displayed finite list of symbol bounds.

Proof. The first identity is direct differentiation and positivity follows by convexity. Mixed derivatives of ϕ'_t cost Y^{-1} in y and t/Y in z. Apply the reciprocal recurrence, now inducting on the two derivative orders. For derivatives of ϕ_t itself use its vanishing at $z = 0$ and the fundamental theorem of calculus for the $s = 0$ bound; for $s \ge 1$ use $t|z| \le 4Y$ on the controlled domain. □

Lemma 3.98 (Pointwise parameter-derivative bound for one summand integral)**.** *Let $0 < p < 1$, $c_0 > 0$, $N \ge 1$ and an index (r, q), $q = (q_1, \dots, q_{2N-1})$, with $r + \sum_j j q_j = 2N-1$; put $K = 2N - 1 + q_1$. For every $j \in \mathbb{N}$ there is $k_0 \in \mathbb{N}$ such that for every sequence (C_k) there is $C > 0$ with the following property: for all $Y \ge 1$, $\Lambda \ge 1$, $A > 0$ and data (a, Ψ) satisfying $\mathsf{H}(Y, \Lambda, A, c_0, (C_k), k_0)$, all $0 < t \le 1$, $\eta \ge 0$ and $Y/2 < y < 3Y$,*

$$|\partial_y^j I_\eta(y,t)| \le C\, B\, Y^{-j} \Lambda^{p-K},$$

where $B = B(N, q, Y, \Lambda, A, t)$ is the size of Definition 3.89.

Proof. The differentiated amplitudes $E_{0,0} = b_t$, $E_{j+1,l} = \partial_y E_{j,l} + i(\partial_y \phi_t) E_{j,l-1}$ satisfy $\partial_y^j (e^{i\phi_t} b_t) = e^{i\phi_t} \sum_{l \le j} E_{j,l}$, and (using commutation of ∂_y and ∂_z together with Lemma 3.96 and Lemma 3.97) $E_{j,l}$ obeys symbol bounds of order $K + l$ and size $B\Lambda^l Y^{-j}$. Localize each $E_{j,l}$ to a C_c^∞ amplitude and apply Lemma 3.85 with order $K + l > p$; the factor Λ^l cancels the gain Λ^{-l}. The finitely many orders used fix k_0. □

Lemma 3.99 (Differentiated amplitudes and the summand integral bound)**.** *For $0 < p < 1$, fix the data and index of Lemma 3.96. For each $j \in \mathbb{N}$ there is a finite order $k_0 = k_0(p, N, j)$ such that bounds in H through order k_0 imply:*

$I_\eta(y,t)$ is smooth in $y \in (Y/2, 3Y)$ for each $\eta \geq 0$, $t > 0$, its derivatives are obtained by differentiating its absolutely integrable integrand, and

$$|\partial_y^j I_\eta(y,t)| \leq CBY^{-j}\Lambda^{p-K} \leq CAY^{-N-j}\Lambda^{p-N} \quad (y \in [Y, 2Y],\ 0 < t \leq 1,\ \eta \geq 0).$$

The constant depends only on $p, N, j, c_0, C_0, \dots, C_{k_0}$.

Proof. Define $b_{0,0} = b_t$ and recursively $b_{j+1,l} = \partial_y b_{j,l} + i(\partial_y \phi_t) b_{j,l-1}$, taking absent indices as zero. Then $\partial_y^j(e^{i\phi_t} b_t) = e^{i\phi_t} \sum_{l=0}^{j} b_{j,l}$. Mixed derivatives of $b_{j,l}$ have symbol order $K+l$, size $B\Lambda^l Y^{-j}$, by the two preceding lemmas. Choose an integer $L > K + j - p$ and use homogeneous IBP for every l; the factors Λ^l cancel the gain Λ^{-l}. For fixed $t > 0$ the support stays compact locally in y, and the integrable near-zero powers have order at least $K > p$, which justifies parameter differentiation even for $\eta = 0$. Finally $2\ell + q_1 \leq 2N-1$ gives $\ell \leq N-1$; since $t \leq 1$ and $Y\Lambda \geq 1$, $B\Lambda^{p-K} \leq AY^{-N}\Lambda^{p-N}$. The chosen L fixes a finite sufficient k_0. □

Lemma 3.100 (Bound for one remainder summand)**.** *In the setting of Lemma 3.95, assume moreover $c_0^{-1}\Lambda \leq \lambda \leq c_0\Lambda$, $|\lambda^{(k)}| \leq C_k \Lambda Y^{-k}$, $|a^{(k)}| \leq C_k A Y^{-k}$ on $(Y/4, 4Y)$, $k \geq 0$. Fix a summand (r, q) of (3.12), $\ell = \sum_{j \geq 2} q_j$, $K = 2N-1+q_1$, $B = A\Lambda^{q_1+\ell} Y^{-2N+1+\ell} t^{q_1}$, and for $0 < t \leq 1$ $\phi_t(z) = \lambda(y)z + \Theta_y(tz)$, $b_t(z) = z^{2N-1} a^{(r)}(y + tz) \prod_j (\Theta_y^{(j)}(tz))^{q_j}$. Then for $s \geq 0$, uniformly in $t \in (0, 1]$, $Y \leq y \leq 2Y$:*

$$|\partial_z^s b_t(z)| \leq CB|z|^{K-s}, \tag{3.13}$$

$$\phi_t'(z) = (1-t)\lambda(y) + t\lambda(y+tz), \quad c_0^{-1}\Lambda \leq \phi_t' \leq c_0\Lambda, \quad |\partial_z^s(1/\phi_t')| \leq C\Lambda^{-1}(t/Y)^s \leq C\Lambda^{-1}|z|^{-s} \tag{3.14}$$

on $\operatorname{supp} b_t$*, and, for $\eta \geq 0$ and $Y\Lambda \geq 1$,*

$$\Big| \int e^{i\phi_t(z) - \eta z^2} \frac{b_t(z)}{|z|^{1+p}} dz \Big| \leq CB\Lambda^{p-K} = CAt^{q_1}(Y\Lambda)^{-2N+1+\ell}\Lambda^p \leq CAY^{-N}\Lambda^{p-N}. \tag{3.15}$$

The same bounds hold for ∂_y^j of the integral ($j = 0, 1$, indeed any j) with an extra factor Y^{-j}. Constants depend on p, N, s, j, c_0 and finitely many C_k.

Proof.

On $\operatorname{supp} b_t$, $y + tz \in (Y/4, 4Y)$, so $t|z| \leq 4Y$. $\Theta_y'(w) = \lambda(y+w) - \lambda(y)$ gives $|\Theta_y'(tz)| \leq C\Lambda Y^{-1} t|z|$; $|\Theta_y^{(j)}| \leq C\Lambda Y^{1-j}$ for $j \geq 2$. Multiply: $|b_t| \leq CAY^{-r}(\Lambda Y^{-1} t|z|)^{q_1} \prod_{j \geq 2} (\Lambda Y^{1-j})^{q_j} |z|^{2N-1} = CB|z|^K$, using $r + \sum jq_j = 2N - 1$. Each ∂_z costs a factor $tY^{-1} \leq 4|z|^{-1}$ (on the a, $\Theta^{(j)}$, $j \geq 2$ factors) or $|z|^{-1}$ (on z^{2N-1}, and on $\Theta_y'(tz)$ which is $\leq Ct\Lambda Y^{-1}|z|$ with derivative $\leq Ct\Lambda Y^{-1}$). This is (3.13). (3.14): ϕ_t' is a convex combination of values of λ; derivatives of $1/\phi_t'$ by Faà di Bruno with $|\partial_z^s \phi_t'| \leq Ct^s \Lambda Y^{-s}$. (3.15): Lemma 3.85 ($K \geq 2N - 1 \geq 1 > p$). Last inequality: $\sum_j jq_j \leq 2N - 1$ forces $2\ell + q_1 \leq 2N - 1$, so $\ell \leq N - 1$; $t \leq 1$; $Y\Lambda \geq 1$ so $(Y\Lambda)^{-2N+1+\ell} \leq (Y\Lambda)^{-N}$. y-derivatives: $\partial_y \Theta_y(w) = \lambda(y+w) - \lambda(y) - \lambda'(y)w$, so $|\partial_y \phi_t| \leq C\Lambda Y^{-1}|z|$ with matching z-derivative bounds, and $|\partial_y \partial_z^s b_t| \leq CBY^{-1}|z|^{K-s}$. $\partial_y[e^{i\phi_t} b_t] = e^{i\phi_t}(\partial_y b_t + i\partial_y \phi_t\, b_t)$: amplitudes of type (BY^{-1}, K) and $(B\Lambda Y^{-1}, K+1)$; Lemma 3.85 gives $CBY^{-1}\Lambda^{p-K}$ for both. Differentiation under the integral is legitimate for $\eta > 0$ and $t > 0$ (compact support). Iterate for higher j. □

Lemma 3.101 (Difference symbols on the controlled phase domain)**.** *Let two data sets satisfy* H *with the same* Y, Λ, A, c_0, C_k*,* $Y, \Lambda \geq 1$*,* $A > 0$*, and assume* $\delta \geq 0$ *and* $|\partial^k(a_1 - a_0)| \leq C_k \delta A Y^{-k}$*,* $|\partial^k(\Psi_1 - \Psi_0)| \leq C_k \delta Y^{-k}$ *on* $(Y/4, 4Y)$*. For a fixed index* (r, q)*, each* $j, s \in \mathbb{N}$*, and symbol hypotheses through order* $2N + j + s + 3$*, there is* C *depending only on* N, j, s, c_0 *and those finitely many* C_k *such that*

$$|\partial_y^j \partial_z^s (b_{1,t} - b_{0,t})| \leq C \delta B Y^{-j} |z|^{K-s}$$

for all $y \in [Y, 2Y]$*,* $0 < t \leq 1$*,* $z \neq 0$*; and*

$$|\partial_y^j \partial_z^s (\phi_{1,t} - \phi_{0,t})| \leq C \delta Y^{-j-1} |z|^{1-s}$$

when additionally $y + tz \in (Y/4, 4Y)$*. All amplitudes are extended by zero as in their definitions.*

Proof. Telescope each finite product defining b_t, differentiating its factors with the finite Leibniz rule. A difference of amplitudes supplies δ, while a difference of phase derivatives supplies δ/Y in place of Λ; use $(Y\Lambda)^{-1} \leq 1$. For the second estimate write $\Delta\phi_t = (1-t)\Delta\Psi'(y)z + \Delta\Psi(y+tz) - \Delta\Psi(y)$ and integrate along the segment in the phase domain. Higher derivatives use the corresponding finite derivative bounds. No statement is made about the arbitrary extensions of the phases outside that domain. □

Lemma 3.102 (Absolutely integrable phase-interpolation identity)**.** *Let* $0 < p < 1$*,* $Y > 0$*,* $N \geq 1$*, and* $r + \sum l q_l = 2N - 1$*. Let* $a_0, a_1 \in C_c^\infty((Y/4, 4Y); \mathbb{C})$ *and* $\Psi_0, \Psi_1 \in C^\infty((Y/4, 4Y); \mathbb{R})$*. For* $y \in [Y, 2Y]$*,* $0 < t \leq 1$*, and* $\eta \geq 0$*, all integrals below converge absolutely, including the double integral with respect to* (σ, z)*, and*

$$I_{1,\eta} - I_{0,\eta} = \int e^{i\phi_{1,t} - \eta z^2} (b_{1,t} - b_{0,t}) |z|^{-1-p} dz + i \int_0^1 \int e^{i((1-\sigma)\phi_{0,t} + \sigma\phi_{1,t}) - \eta z^2} (\phi_{1,t} - \phi_{0,t}) b_{0,t} |z|^{-1-p} dz \, d\sigma.$$

The identity may be differentiated any fixed number of times in y *on* $(Y/2, 3Y)$*.*

Proof. Apply $e^{iu} - e^{iv} = i(u - v) \int_0^1 e^{i((1-\sigma)v + \sigma u)} d\sigma$ pointwise. For fixed $t > 0$, the union of the two amplitude supports is compact and varies within a common compact set on compact y intervals. At zero the amplitudes vanish to order at least $2N - 1 \geq 1 > p$, and $\Delta\phi_t = O(|z|)$. These bounds persist under any fixed number of y derivatives. They give integrable majorants uniform in $\sigma \in [0, 1]$, justifying Fubini and differentiation. □

Lemma 3.103 (Uniform estimate for the two interpolated difference contributions)**.** *Fix* $0 < p < 1$ *and* $\delta \geq 0$*. Under the hypotheses of Lemma 3.101, for every* $j \in \mathbb{N}$ *there is a finite* $k_0 = k_0(p, N, j)$ *such that the corresponding finite symbol and difference bounds give*

$$|\partial_y^j (I_{1,\eta} - I_{0,\eta})| \leq C \delta B Y^{-j} \Lambda^{p-K} + C \delta B Y^{-j-1} \Lambda^{p-K-1} \leq 2C \delta A Y^{-N-j} \Lambda^{p-N}$$

for $y \in [Y, 2Y]$*,* $0 < t \leq 1$*,* $\eta \geq 0$*. Here* $C = C(p, N, j, c_0, C_0, \ldots, C_{k_0}) > 0$ *is independent of* $\delta, Y, \Lambda, A, y, t, \eta$*.*

Proof. In the first interpolation term the amplitude has size δB and order K. In the second it has size $\delta B/Y$ and order $K+1$. The convexly interpolated phase has the same reciprocal derivative bounds, uniformly in σ. Repeat the explicit differentiated-amplitude recursion of the parameter-IBP lemma for each term, apply the homogeneous estimate, and integrate $\sigma \in [0,1]$. Each of the two final terms is at most $C\delta AY^{-N-j}\Lambda^{p-N}$; retain the factor 2. Lean route: instead of differentiating the σ-integral of the interpolation identity, interpolate both phase and amplitude, $\Phi_\sigma = (1-\sigma)\phi_0 + \sigma\phi_1$, $H_\sigma = (1-\sigma)b_0 + \sigma b_1$, write $\partial_y^j(I_1 - I_0) = \mathcal{R}(1) - \mathcal{R}(0)$ with $\mathcal{R}(\sigma) = \int e^{-\eta z^2}|z|^{-1-p}\partial_y^j(e^{i\Phi_\sigma}H_\sigma)\,dz$, and bound $\mathcal{R}'(\sigma)$ (differentiation under the z-integral) uniformly in $\sigma \in [0,1]$ by the two homogeneous-IBP estimates above; the interpolation identity of the preceding lemma is not used. □

Lemma 3.104 (Difference of remainders)**.** *Let (a_ν, Ψ_ν), $\nu = 0, 1$, satisfy the hypotheses of Lemma 3.100 with the same Y, Λ, A and constants, and suppose for some $\delta \in [0,1]$: $|\partial^k(a_1 - a_0)| \le C_k\delta AY^{-k}$, $|\partial^k(\Psi_1 - \Psi_0)| \le C_k\delta Y^{-k}$ on $(Y/4, 4Y)$, $k \ge 0$. Let $I_\nu(y,t)$ be the integral in (3.15) for (a_ν, Ψ_ν) (same summand). Then For $z \ne 0$, the amplitude bound below holds globally; the phase bound is asserted only when $y + tz \in (Y/4, 4Y)$.*

$$|\partial_y^j\partial_z^s(b_{1,t} - b_{0,t})| \le C\delta BY^{-j}|z|^{K-s}, \qquad |\partial_y^j\partial_z^s(\phi_{1,t} - \phi_{0,t})| \le C\delta Y^{-j-1}|z|^{1-s}, \tag{3.16}$$

$$I_1 - I_0 = \int e^{i\phi_{1,t}}\frac{b_{1,t} - b_{0,t}}{|z|^{1+p}}dz + i\int_0^1\int e^{i((1-\sigma)\phi_{0,t} + \sigma\phi_{1,t})}\frac{(\phi_{1,t} - \phi_{0,t})b_{0,t}}{|z|^{1+p}}dz\,d\sigma \tag{3.17}$$

(with $e^{-\eta z^2}$ inserted throughout), and

$$|\partial_y^j(I_1 - I_0)| \le C\delta BY^{-j}\Lambda^{p-K} + C\delta BY^{-j-1}\Lambda^{p-K-1} \le 2C\delta AY^{-N-j}\Lambda^{p-N}. \tag{3.18}$$

Proof.

Telescoping the products defining $b_{\nu,t}$: each difference of factors gains δ, since $|\Theta'_{1,y}(tz) - \Theta'_{0,y}(tz)| \le C\delta Y^{-2}t|z| \le C\delta\Lambda Y^{-1}t|z|$ (as $Y^{-2} \le \Lambda Y^{-1}$ by $Y\Lambda \ge 1$) and $|\Theta^{(k)}_{1,y} - \Theta^{(k)}_{0,y}| \le C\delta Y^{-k} \le C\delta\Lambda Y^{1-k}$. $\phi_{1,t} - \phi_{0,t} = (1-t)\Delta\Psi'(y)z + \Delta\Psi(y+tz) - \Delta\Psi(y)$ with $\Delta\Psi = \Psi_1 - \Psi_0$; its size is $\le C\delta Y^{-1}|z|$. (3.17): $e^{iu} - e^{iv} = i(u-v)\int_0^1 e^{i((1-\sigma)v+\sigma u)}d\sigma$ and Fubini. Each interpolated phase $(1-\sigma)\phi_{0,t} + \sigma\phi_{1,t}$ satisfies (3.14): its derivative is a convex combination of $\phi'_{0,t}, \phi'_{1,t} \in [c_0^{-1}\Lambda, c_0\Lambda]$, and the inverse-derivative bounds follow by Faà di Bruno as before. Apply Lemma 3.85 (and its y-differentiated version as in Lemma 3.100): the first integral with $(\delta B, K)$, the second with $(\delta BY^{-1}, K+1)$, giving (3.18); finally $Y^{-1}\Lambda^{-1} \le 1$ and the last step of (3.15). □

Lemma 3.105 (Parameter integral representation of the multiplier remainder)**.** *Fix $0 < p < 1$, $N \ge 1$, and $j \in \mathbb{N}$. There is an integer $k_0 = k_0(p, N, j)$ such that phase data satisfying H through order k_0, with $Y, \Lambda \ge 1$, $A > 0$, $c_0 > 0$, have the following properties. Put $n = 2N - 1$. For $y \in [Y, 2Y]$,*

$$\mathcal{R}_{\mathrm{mult,N}}(y) = -c_p n \sum_{r + \sum l q_l = n} d_{r,q}\int_0^1 (1-t)^{n-1} I_0(y,t)\,dt.$$

The t integrals converge absolutely; their value at the single point $t = 0$ is immaterial. The remainder is C^j on $(Y/2, 3Y)$ and its derivatives through order j are the same finite sums with the corresponding y derivatives of I_0 under the t integral. The sufficient finite order k_0 is chosen before the symbol constants, amplitude and phase. If the data satisfy such bounds at every finite order, this also gives smoothness.

Proof. First insert $\eta > 0$. Taylor's integral formula, the Gaussian factor and the near-zero vanishing give absolute double-integrability, so Fubini gives the parameter formula with I_η. The summand bound is uniform in $t \in (0, 1]$ and $\eta \geq 0$; for fixed $t > 0$, compact-support dominated convergence removes η. Apply dominated convergence in t and the regularized-limit identity. The all-order parameter bounds on a neighbourhood of $[Y, 2Y]$ justify repeated differentiation in y under the t integral and show smoothness. □

Lemma 3.106 (All-order remainder bounds and their difference estimate)**.** *Fix $0 < p < 1$, $N \geq 1$, $j \in \mathbb{N}$, $c_0 > 0$. There is an integer $k_0 = k_0(p, N, j)$ such that for every finite nonnegative list $C_0, \dots, C_{k_0}$ there is $C > 0$, independent of Y, Λ, A and of the data, with*

$$|\partial_y^j \mathcal{R}_{\mathrm{mult,N}}(y)| \leq CAY^{-N-j}\Lambda^{p-N}$$

whenever $Y, \Lambda \geq 1$, $A > 0$, H holds through order k_0, and $y \in [Y, 2Y]$. For two such data with difference bounds $|\partial^k(a_1 - a_0)| \leq C_k\delta AY^{-k}$ and $|\partial^k(\Psi_1 - \Psi_0)| \leq C_k\delta Y^{-k}$ on $(Y/4, 4Y)$, $k \leq k_0$, $\delta \geq 0$, the same conclusion holds for their remainder difference with right-hand side $C\min\{1, \delta\}AY^{-N-j}\Lambda^{p-N}$, after enlarging C.

Proof. Bound the finite sum and integrate $(1-t)^{2N-2}$, whose integral is $1/(2N-1)$. For $\delta \leq 1$ use the difference-integral bound with its factor 2. For $\delta \geq 1$ use the individual remainder estimates at the same derivative order j. This all-order individual statement is needed for that last argument even though the expansion theorem only requests individual bounds for $j = 0, 1$. □

Lemma 3.107 (First three coefficients and the two-term remainder)**.** *Under the multiplier phase hypotheses, for $y \in [Y, 2Y]$,*

$$B_0 = \lambda^p a, \quad B_1 = -ip\lambda^{p-1}a', \quad B_2 = -\frac{p(p-1)}{2}\lambda^{p-2}(a'' + i\lambda' a).$$

For $j = 0, 1$ there is $C = C(p, c_0, C_0, \dots, C_4) > 0$, independent of Y, Λ, A, such that

$$\left|\partial_y^j\left(\frac{p(p-1)}{2}\lambda^{p-2}a''\right)\right| \leq CAY^{-2-j}\Lambda^{p-2}.$$

Proof. At $z = 0$, $\Theta_y' = 0$ and $\Theta_y'' = \lambda'(y)$. Thus $F_y(0) = a$, $F_y'(0) = a'$, and $F_y''(0) = a'' + i\lambda' a$. Substitute in the coefficient definition. The last bound is the product rule, $\lambda \asymp \Lambda$, $|\lambda'| \leq C_1\Lambda/Y$, and the bounds for a'' and a'''. □

Lemma 3.108 (Oscillatory multiplier expansion)**.** *Let $0 < p < 1$, $A > 0$, $Y, \Lambda \geq 1$, $a \in C_c^\infty((Y/4, 4Y); \mathbb{C})$, $\Psi \in C^\infty((Y/4, 4Y); \mathbb{R})$, $\lambda = \Psi'$, and suppose that on $(Y/4, 4Y)$, for all $k \geq 0$,*

$$c_0^{-1}\Lambda \leq \lambda \leq c_0\Lambda, \qquad |\partial_y^k \lambda| \leq C_k\Lambda Y^{-k}, \qquad |\partial_y^k a| \leq C_k AY^{-k}.$$

Let $m(\xi)=\xi^p$, $\Theta_y(z)=\Psi(y+z)-\Psi(y)-\lambda(y)z$. For $N\ge 1$ and $Y\le y\le 2Y$ define $\mathcal{R}_{\mathrm{mult},N}$ by

$$|D|^p(ae^{i\Psi})(y)=e^{i\Psi(y)}\Big[\sum_{k=0}^{2N-2}B_k(y)+\mathcal{R}_{\mathrm{mult},N}(y)\Big], \tag{3.19}$$

$$B_k(y)=\frac{(-i)^k}{k!}m^{(k)}(\lambda(y))\,\partial_z^k\big[a(y+z)e^{i\Theta_y(z)}\big]\big|_{z=0}. \tag{3.20}$$

Then there is $C=C(p,N,j,c_0,(C_k)_{k\le k_0(N,j)})$ such that for $j=0,1$ and $Y\le y\le 2Y$:

$$|\partial_y^j\mathcal{R}_{\mathrm{mult},N}(y)|\le CAY^{-N-j}\Lambda^{p-N}. \tag{3.21}$$

In particular

$$|D|^p(ae^{i\Psi})=e^{i\Psi}\Big[\lambda^p a-ip\lambda^{p-1}a'-\frac{ip(p-1)}{2}\lambda^{p-2}\lambda' a+\widetilde{\mathcal{R}}_{\mathrm{mult},2}\Big], \tag{3.22}$$

with $\widetilde{\mathcal{R}}_{\mathrm{mult},2}=\mathcal{R}_{\mathrm{mult},2}-\frac{p(p-1)}{2}\lambda^{p-2}a''$ and $|\partial_y^j\widetilde{\mathcal{R}}_{\mathrm{mult},2}|\le CAY^{-2-j}\Lambda^{p-2}$, $j=0,1$. Moreover, if (a_ν,Ψ_ν), $\nu=0,1$, satisfy these hypotheses with the same Y,Λ,A,c_0,C_k and, for some $\delta\ge 0$, $|\partial^k(a_1-a_0)|\le C_k\delta AY^{-k}$, $|\partial^k(\Psi_1-\Psi_0)|\le C_k\delta Y^{-k}$ $(k\ge 0)$ on $(Y/4,4Y)$, then for every $j\ge 0$

$$|\partial_y^j(\mathcal{R}_{\mathrm{mult},N,1}-\mathcal{R}_{\mathrm{mult},N,0})(y)|\le C\min\{1,\delta\}AY^{-N-j}\Lambda^{p-N},\qquad Y\le y\le 2Y. \tag{3.23}$$

Proof.

Step 1. By Lemma 3.95, $\mathcal{R}_{\mathrm{mult},N}(y)=-c_p\lim_{\eta\downarrow 0}\int e^{i\lambda z-\eta z^2}R_{2N-1}(z)|z|^{-1-p}dz$. Insert (3.12); for fixed $\eta>0$ Fubini in (z,t) is justified since the t-integrand in the Taylor remainder is bounded by $\|F_y^{(2N-1)}\|_{L^\infty}|z|^{2N-1}$, and $|z|^{2N-2-p}e^{-\eta z^2}\in L^1$. Note $e^{i\lambda z}e^{i\Theta_y(tz)}=e^{i\phi_t(z)}$. So $\mathcal{R}_{\mathrm{mult},N}$ is a finite sum of $c\int_0^1(1-t)^{2N-2}I_\eta(y,t)dt$ with I_η as in Lemma 3.100; $|I_\eta(y,t)|\le CAY^{-N}\Lambda^{p-N}$ uniformly in η,t and $I_\eta(y,t)\to I_0(y,t)$, so by dominated convergence the bound passes to $\mathcal{R}_{\mathrm{mult},N}$. The y-derivative ($j=1$) is handled identically using the y-differentiated bounds of Lemma 3.100 (differentiate under the t-integral for $\eta>0$; the $\eta\to 0$ limit of the derivatives is locally uniform in y, so it equals the derivative of the limit). Step 2 (two-term form). With $N=2$, $\Theta_y'(0)=0$, $\Theta_y''(0)=\lambda'(y)$: $B_0=\lambda^p a$, $B_1=-ip\lambda^{p-1}a'$, $B_2=-\frac{p(p-1)}{2}\lambda^{p-2}(a''+i\lambda' a)$. And $|\partial_y^j(\lambda^{p-2}a'')|\le CAY^{-2-j}\Lambda^{p-2}$. Step 3 (differences). For $\delta\le 1$ sum Lemma 3.104 over summands and integrate in t (same limiting argument). For $\delta>1$ use the triangle inequality and Step 1 for each ν (for all j, by the any-j version of Lemma 3.100). □

Chapter 4

Approximate solution I: linear solution, family, $\mathcal{E}_1$–$\mathcal{E}_4$ (paper §§5.1–5.2, 5.4–5.7)

4.1 Standing conventions and auxiliary facts

Standing conventions. Throughout we fix the parameters of Definition 2.1: $\frac{3}{4} < p < 1$, $\alpha = p+1$, $\beta = \frac{p+1}{2p}$ (so $2\beta = 1 + \frac{1}{p}$, $1 < \beta < \frac{1}{p}$, $\beta \le p+1$), the cutoff χ_+ ($\chi_+ = 0$ on $(-\infty, 1]$, $\chi_+ = 1$ on $[2, \infty)$) and $\Phi_0(\xi) = \frac{\xi|\xi|^p}{p+1}$; note $\Phi_0'(\xi) = |\xi|^p$, Φ_0 is odd and C^1 on $\mathbb{R}$, C^∞ on $\mathbb{R}\setminus\{0\}$. $c_\alpha = \Gamma(1+p)\sin(\pi p/2)/\pi$ is the constant of Definition 1.5; $|D|^p g(y) = c_\alpha \int_{\mathbb{R}} (g(y) - g(z))|y-z|^{-\alpha}\, dz$ for bounded, locally Lipschitz g (complex-valued g: apply to real and imaginary parts). $A_\epsilon(u) = (1+\epsilon u^2)^{-\alpha/2}$ (Definition 1.10), $S_f(y,z)$ is the secant (Definition 1.3), $\mathcal{A}$ is the class of Definition 1.4, $\mathcal{Q}_\epsilon$ is Definition 1.23, and $\mathcal{L}_0 f = |D|^p f' + f - yf'$, $\widetilde{\mathcal{E}}_\epsilon(f) = \mathcal{L}_0 f + \mathcal{Q}_\epsilon(f)$ (Definition 1.27). Fourier transforms follow Definition 1.2: $\widehat{f}(\xi) = \int f(x) e^{-ix\xi} dx$, $\check{m}(y) = \frac{1}{2\pi}\int m(\xi) e^{iy\xi} d\xi$. $\langle y\rangle = (1+y^2)^{1/2}$. $W(y) := \max\{-y, 0\}$ (the corner), with $W'(y) := -\mathbf{1}_{(-\infty,0)}(y)$. For $\epsilon \ge 0$, $c_\epsilon = c_\alpha \int_0^1 [(1+\epsilon t^2)^{-\alpha/2} - 1](1-t)^{p-1} dt$ is the constant of Lemma 3.12; $|c_\epsilon| \le C\epsilon$ for $\epsilon \in [0,1]$, $c_0 = 0$.

An *admissible constant* is a positive number depending only on p, on the fixed cutoffs (χ_+, the truncation ρ of Definition 4.4, the cutoff χ of Definition 4.51), on δ, ϵ_0 of Definition 4.38 and on the derivative order j, but *not* on $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, $R \ge R_0$, Y, or the spatial variables.

Lemma 4.1 (Differentiating the fractional integral)**.** *Let $g \in C^2(\mathbb{R};\mathbb{C})$ with g, g' bounded and g'' locally bounded. Then $|D|^p g \in C^1(\mathbb{R})$ and $(|D|^p g)' = |D|^p(g')$ pointwise.*

Proof. Write $|D|^p g(y) = c_\alpha \int_{\mathbb{R}} (g(y) - g(y-h))|h|^{-\alpha} dh$. For $|y| \le Y_1$ the y-derivative of the integrand is $(g'(y) - g'(y-h))|h|^{-\alpha}$, bounded by $\sup_{|w| \le Y_1+1} |g''|\, |h|^{1-\alpha}\mathbf{1}_{|h|\le 1} + 2\|g'\|_\infty |h|^{-\alpha}\mathbf{1}_{|h|>1}$, which is integrable ($1-\alpha = -p > -1$, $\alpha > 1$) and independent of y. Dominated differentiation (Mathlib `hasDerivAt_integral_of_dominated_loc_of_deriv_le`) gives the derivative; continuity of $|D|^p g'$ follows from Lemma 1.6 and dominated convergence with the same majorant. □

Lemma 4.2 (Weak Fourier computation of $\mathcal{L}_0$ and of $|D|^p - y$)**.**

1. *Let $m \in C^1(\mathbb{R};\mathbb{C}) \cap L^1(\mathbb{R})$ vanish on a neighbourhood of 0, with $|m(\xi)| + |m'(\xi)| \le K\langle\xi\rangle^N$ for some K, N, and put $n := \xi m' + (2 + i\xi|\xi|^p)m$. Assume $n \in L^1(\mathbb{R})$ and that $g := \check{m}$ is C^2 with g, g' bounded and $|g''(y)| \le K\langle y\rangle^N$. Then $\mathcal{L}_0 g = \check{n}$ pointwise on $\mathbb{R}$.*

2. *Let $m \in C^1_c(\mathbb{R};\mathbb{C})$ and suppose $g := \check{m}$ is C^1 with g bounded and $|g'(y)| \le K\langle y\rangle^N$. Then $|D|^p g(y) - yg(y) = \check{n_b}(y)$ for all y, where $n_b := |\xi|^p m - im'$.*

Proof. Both sides are continuous in each case ($|D|^p g'$, $|D|^p g$ by Lemma 1.9; $\check{n}$, $\check{n_b}$ since $n, n_b \in L^1$), so it suffices to show $\int(\mathrm{LHS})\varphi = \int(\mathrm{RHS})\varphi$ for all $\varphi \in \mathcal{S}(\mathbb{R})$ (fundamental lemma; Mathlib `ae_eq_zero_of_integral_contDiff_smul_eq_zero` plus continuity). Write $\Phi(\xi) := \widehat{\varphi}(-\xi)$. For $\psi \in L^1$, Fubini ($m \in L^1$) gives $\int g\psi\, dy = \frac{1}{2\pi}\int m(\xi)\widehat{\psi}(-\xi)\, d\xi$.

(1) By Lemma 1.9 applied to g' (bounded, g'' polynomially bounded), $\int(|D|^p g')\varphi = \int g'|D|^p\varphi = -\int g\,(|D|^p\varphi)'$; the boundary terms vanish because g is bounded and $|D|^p\varphi$ decays like $\langle y\rangle^{-\alpha}$; moreover $(|D|^p\varphi)' = |D|^p\varphi'$ is $(i\xi|\xi|^p\widehat{\varphi})^\vee$ by Lemma 1.8, an L^1 function with L^1 Fourier transform. Also $\int yg'\varphi = -\int g(\varphi + y\varphi')$ (boundary terms vanish, g bounded, φ Schwartz). Hence $\int(\mathcal{L}_0 g)\varphi = \int g\psi$ with $\psi = -|D|^p\varphi' + 2\varphi + y\varphi'$, and $\widehat{\psi}(\eta) = (1 - i\eta|\eta|^p)\widehat{\varphi}(\eta) - \eta\widehat{\varphi}'(\eta)$ (using $\widehat{y\varphi'} = i\partial_\eta(i\eta\widehat{\varphi}) = -\widehat{\varphi} - \eta\widehat{\varphi}'$ and Fourier inversion for $|D|^p\varphi'$). At $\eta = -\xi$: $\widehat{\psi}(-\xi) = (1 + i\xi|\xi|^p)\Phi(\xi) - \xi\Phi'(\xi)$. Integrating by parts on $(-\infty, 0)$ and $(0, \infty)$ separately, $-\int m\xi\Phi' = \int(\xi m)'\Phi$: the boundary terms vanish at 0 ($m = 0$ near 0) and at $\pm\infty$ ($\xi m\Phi \to 0$ since m has polynomial growth and Φ is Schwartz). Thus $\int(\mathcal{L}_0 g)\varphi = \frac{1}{2\pi}\int n(\xi)\Phi(\xi)d\xi = \int \check{n}\,\varphi$ (Fubini, $n \in L^1$).

(2) By Lemma 1.9, $\int(|D|^p g)\varphi = \int g|D|^p\varphi = \frac{1}{2\pi}\int m|\xi|^p\Phi\, d\xi$ (since $\widehat{|D|^p\varphi} = |\eta|^p\widehat{\varphi}$, an even weight). Also $\widehat{y\varphi}(\eta) = i\widehat{\varphi}'(\eta)$, so $\widehat{y\varphi}(-\xi) = -i\Phi'(\xi)$ and $\int yg\varphi = \frac{1}{2\pi}\int m(-i\Phi') = \frac{1}{2\pi}\int im'\Phi$ (no boundary terms: m has compact support). Subtract.

□

Lemma 4.3 (A Dirichlet-type integral)**.** *For every $y \in \mathbb{R}$, $\int_0^\infty \frac{1-\cos(y\xi)}{\xi^2}d\xi = \frac{\pi}{2}|y|$ (absolutely convergent). Consequently $W(y) = -\frac{y}{2} + \frac{1}{\pi}\int_0^\infty \frac{1-\cos(y\xi)}{\xi^2}d\xi$.*

Proof. The integrand is $\le \min(y^2/2, 2\xi^{-2})$. For $y \neq 0$ substitute $s = |y|\xi$ to get $|y|\int_0^\infty(1 - \cos s)s^{-2}ds$; integrating by parts ($\frac{d}{ds}(-s^{-1})$, boundary terms vanish) gives $\int_0^\infty \frac{\sin s}{s}ds = \frac{\pi}{2}$ (Mathlib: `integral_sin_div` / Dirichlet integral, improper). Finally $-\frac{y}{2} + \frac{|y|}{2} = \max\{-y, 0\}$. □

Definition 4.4 (Oscillatory integrals with the dispersive phase)**.** For $m \in \mathbb{R}$ let S^m_+ be the set of $a \in C^\infty(\mathbb{R};\mathbb{C})$ with $a = 0$ on $(-\infty, 1]$ and $|\partial^l_\xi a(\xi)| \le C_l\xi^{m-l}$ for $\xi \ge 1$, $l \ge 0$. Fix $\rho \in C^\infty_c(\mathbb{R};[0,1])$ with $\rho = 1$ on $[-1,1]$ and $\operatorname{supp}\rho \subset [-2,2]$. For $a \in S^m_+$ and $y \in \mathbb{R}$ set

$$\mathrm{Osc}[a](y) := \lim_{M\to\infty}\int_0^\infty \rho(\xi/M)\, a(\xi)\, e^{i(y\xi - \Phi_0(\xi))}\, d\xi.$$

Lemma 4.5 (Symbol bounds under truncation and multiplication by frequency)**.** *Let $m \in \mathbb{R}$, $a \in S^m_+$, and let nonnegative numbers C_l satisfy $|a^{(l)}(\xi)| \le C_l\xi^{m-l}$ for $\xi \ge 1$ and $l \in \mathbb{N}$. For every $j \in \mathbb{N}$, $a_j(\xi) := (i\xi)^j a(\xi)$ belongs to S^{m+j}_+;*

for each l, its symbol constant depends only on $j, l, m, C_0, \dots, C_l$. Let ρ be any truncation of Definition 4.4. For $M \geq 1$ put $a_M(\xi) = \rho(\xi/M)a(\xi)$. Then $a_M \in C_c^\infty(\mathbb{R};\mathbb{C})$, $a_M = 0$ on $(-\infty, 1]$, and for each $l \in \mathbb{N}$ there is $B_l = B_l(m, \rho, C_0, \dots, C_l) > 0$, independent of M, such that

$$|a_M^{(l)}(\xi)| \leq B_l \xi^{m-l} \qquad (\xi \geq 1).$$

Also $a_M^{(l)}(\xi) = a^{(l)}(\xi)$ for $0 < \xi < M$. If $L \geq 1$ and $\theta \in C_c^\infty(\mathbb{R};[0,1])$ equals 1 on $[0, L]$ and vanishes for $\xi \geq 2L$, the same conclusions, with vanishing on $(-\infty, L]$, hold for $b_M = \rho(\cdot/M)(1-\theta)a$ and $b_\infty = (1-\theta)a$, except that compact support is asserted only for b_M. Their constants may additionally depend on L and finitely many derivatives of θ.

Proof. Expand every derivative by the finite Leibniz formula. On the support of a derivative of $\rho(\xi/M)$, $\xi \leq 2M$, hence $M^{-r} \leq 2^r \xi^{-r}$. Derivatives of θ are supported in a fixed compact interval bounded away from zero. For $(i\xi)^j$, use its explicit polynomial derivatives. Flatness at the endpoints of the constant regions gives smoothness and the asserted equality for $\xi < M$. Keep the finite sums of products of derivative bounds explicit; no constant may depend on M. □

Lemma 4.6 (Reciprocal derivatives of the nonstationary phase)**.** *Let $0 < p < 1$ and $\phi_y(\xi) = y\xi - \xi^{p+1}/(p+1)$ for $\xi > 0$. For every $l \in \mathbb{N}$ there is $D_l = D_l(p) > 0$ with the following properties. If $Y \geq 1$, $|y| \leq Y$, and $\xi \geq (2Y)^{1/p} + 2$, then*

$$|\phi_y'(\xi)| \geq \tfrac{1}{2}\xi^p, \qquad \left|\partial_\xi^l \frac{1}{y - \xi^p}\right| \leq D_l \xi^{-p-l}.$$

If $y \leq -1$ and $\xi \geq 1$, then

$$|\phi_y'(\xi)| = |y| + \xi^p, \qquad \left|\partial_\xi^l \frac{1}{y - \xi^p}\right| \leq D_l(|y| + \xi^p)^{-1}\xi^{-l} \leq D_l |y|^{-1}\xi^{-l}.$$

The reciprocal is smooth on each of these nonstationary regions.

Proof. Use $\partial_\xi^r \xi^p = p(p-1)\cdots(p-r+1)\xi^{p-r}$. Inductively differentiate $(y - \xi^p)q_y = 1$ to express $q_y^{(l)}$ through lower derivatives and one additional factor $(y-\xi^p)^{-1}$. In the first region use $|y-\xi^p| \geq \xi^p/2$; in the second use $\xi^p/(|y|+\xi^p) \leq 1$. This proves the bounds with D_l depending only on p, l. □

Lemma 4.7 (One integration-by-parts step: flat extension and symbol bound)**.** *Let $L \geq 1$, $m, \sigma \in \mathbb{R}$, $A, H > 0$, $b \in C^\infty(\mathbb{R};\mathbb{C})$ with $b = 0$ on $(-\infty, L]$, and $q \in C^\infty((L/2, \infty);\mathbb{R})$. Set $T_q b = -\partial_\xi(bq/i)$ on (L, ∞) and $T_q b = 0$ on $(-\infty, L]$. Then $T_q b \in C^\infty(\mathbb{R};\mathbb{C})$, every derivative of $T_q b$ vanishes on $(-\infty, L]$, and $\operatorname{supp} T_q b \subset \operatorname{supp} b$. For every $k \in \mathbb{N}$ and $\xi \geq L$,*

$$\partial_\xi^k T_q b(\xi) = i \sum_{r=0}^{k+1} \binom{k+1}{r} b^{(r)}(\xi) q^{(k+1-r)}(\xi).$$

If nonnegative numbers $B_0, \dots, B_{k+1}$ and $D_0, \dots, D_{k+1}$ satisfy

$$|b^{(r)}(\xi)| \leq B_r A \xi^{m-r}, \qquad |q^{(r)}(\xi)| \leq D_r H \xi^{-\sigma-r} \quad (\xi \geq L,\ 0 \leq r \leq k+1),$$

then

$$|\partial_\xi^k T_q b(\xi)| \le \left(\sum_{r=0}^{k+1} \binom{k+1}{r} B_r D_{k+1-r} \right) AH\xi^{m-1-\sigma-k} \qquad (\xi \ge L).$$

Thus this one-step constant involves only the displayed finite lists and k, and is independent of L, A, H, b, q once those bounds are fixed.

Proof. Glue bq on $(L/2, \infty)$ to zero on $(-\infty, L)$; the two definitions agree on their overlap because b vanishes there. The resulting global function is smooth. Since all derivatives of b vanish at L, its derivative multiplied by $i = -1/i$ equals the stated zero extension $T_q b$, including at L. Differentiation and multiplication do not enlarge the closed support of b. Apply the ordinary finite Leibniz formula at order $k+1$. Each product of bounds has exponent $(m-r)+(-\sigma-(k+1-r)) = m-1-\sigma-k$; summing gives the explicit constant. □

Lemma 4.8 (Iterated symbol bounds from a finite recurrence)**.** *Let L, m, σ, A, H, b, q satisfy the regularity and support hypotheses of Lemma 4.7, and let nonnegative sequences $(B_r), (D_r)$ satisfy its derivative bounds for every $r \in \mathbb{N}$. Define nonnegative coefficients by the finite recurrence*

$$E_{0,k} = B_k, \qquad E_{n+1,k} = \sum_{r=0}^{k+1} \binom{k+1}{r} E_{n,r} D_{k+1-r}.$$

For all $N, k \in \mathbb{N}$, $T_q^N b$ is smooth, vanishes on $(-\infty, L]$, has support contained in $\operatorname{supp} b$, and

$$|\partial_\xi^k T_q^N b(\xi)| \le E_{N,k} AH^N \xi^{m-N(1+\sigma)-k} \qquad (\xi \ge L).$$

The coefficient $E_{N,k}$ depends only on N, k and $B_0, \dots, B_{N+k}, D_0, \dots, D_{N+k}$. If $d := m - N(1+\sigma) < -1$, then $T_q^N b \in L^1((0,\infty))$ and

$$\int_0^\infty |T_q^N b(\xi)| d\xi \le E_{N,0} AH^N \frac{L^{d+1}}{-d-1}.$$

No compact-support hypothesis is imposed on b in this lemma.

Proof. Induct on N, simultaneously for all derivative orders k. At step $N+1$ apply the one-step lemma with symbol order $m-N(1+\sigma)$, amplitude scale AH^N, and derivative constants $E_{N,r}$. The resulting finite Leibniz sum is exactly the recurrence for $E_{N+1,k}$. The same induction proves smoothness, zero extension, support preservation, nonnegativity of the coefficients, and their dependence on the stated finite lists. For the last assertion, the function vanishes below L and is dominated above L by $E_{N,0} AH^N \xi^d$; integrate that power using $d < -1$. □

Lemma 4.9 (Repeated integration by parts with quantified symbol loss)**.** *Let $L \ge 1$, $m, \sigma \in \mathbb{R}$, and $A, H > 0$. Let $b \in C^\infty(\mathbb{R}; \mathbb{C})$ vanish on $(-\infty, L]$, and let $q \in C^\infty((L/2, \infty); \mathbb{R})$ satisfy, for every $r \in \mathbb{N}$ and $\xi \ge L$,*

$$|b^{(r)}(\xi)| \le B_r A \xi^{m-r}, \qquad |q^{(r)}(\xi)| \le D_r H \xi^{-\sigma-r},$$

where $B_r, D_r \ge 0$ are fixed. Define $T_q b = -\partial_\xi(bq/i)$ on (L, ∞), extended by zero to $(-\infty, L]$, and iterate this operator. Every $T_q^N b$ is smooth and vanishes

on $(-\infty, L]$. For all $N, k \in \mathbb{N}$ there is $C_{N,k} > 0$, depending only on N, k, m, σ and $B_0, \dots, B_{N+k}, D_0, \dots, D_{N+k}$, such that

$$|\partial_\xi^k T_q^N b(\xi)| \le C_{N,k} A H^N \xi^{m-N(1+\sigma)-k} \qquad (\xi \ge L).$$

Suppose in addition that b has compact support, $\phi \in C^\infty((0,\infty);\mathbb{R})$, and $q(\xi)\phi'(\xi) = 1$ for $\xi > L$. Then all the following integrals converge absolutely and

$$\int_0^\infty b(\xi) e^{i\phi(\xi)} d\xi = \int_0^\infty T_q^N b(\xi) e^{i\phi(\xi)} d\xi \qquad (N \in \mathbb{N}).$$

Compact support of b is not needed for the symbol bound; if $m - N(1+\sigma) < -1$, that bound makes $T_q^N b$ integrable on $(0,\infty)$.

Proof. Smoothness, support preservation, the symbol estimate, and the integrability assertion are Lemma 4.8; take $C_{N,k} = 1 + E_{N,k}$ to obtain a positive constant. For the compact-support integral identity, continuity extends $q\phi' = 1$ from (L,∞) to L, so $\phi'(L) \neq 0$. Consequently the compact support of b is contained in the nonstationary open set $U = (L/2,\infty) \cap \{\xi : \phi'(\xi) \neq 0\}$, even when that support contains L. Integrate by parts N times on U (compactly supported amplitudes, no boundary terms), with the operator $L_\phi a = (a/(i\phi'))'$, which has the opposite sign of ours. On (L,∞), $q = 1/\phi'$; below L, all derivatives of b and of its iterates vanish by Lemma 4.7. Thus $T_q^N b = (-1)^N L_\phi^N b$ by induction. The factor $(-1)^N$ in the generic integration-by-parts identity cancels this sign. Finally, all amplitudes vanish outside $(0,\infty)$, so its whole-line integrals equal the half-line integrals displayed here. Lean route: the identity is proved by direct interval integration by parts, without the generic compact-support iteration lemma of the analysis chapter. □

Lemma 4.10 (Locally uniform cutoff limit and independence of truncation)**.** *Let $a \in S_+^m$ and, for an admissible truncation ρ, write $F_M^\rho(a;y) = \int_0^\infty \rho(\xi/M) a(\xi) e^{i\phi_y(\xi)} d\xi$ for $M > 0$. These integrals converge absolutely. There is a continuous function $F : \mathbb{R} \to \mathbb{C}$ such that $F_M^\rho(a;\cdot) \to F$ locally uniformly as $M \to \infty$ for every admissible ρ, with the same F for all ρ. Thus $F = \mathrm{Osc}[a]$ for the fixed truncation of Definition 4.4. More precisely, fix $Y \ge 1$, $L \ge (2Y)^{1/p} + 2$, and a cutoff θ as in Lemma 4.5. Put $b_\infty = (1-\theta)a$ and $q_y(\xi) = (y - \xi^p)^{-1}$ on $(L/2,\infty)$; this interval is nonstationary because $p < 1$ and $L/2 > Y^{1/p}$. If $N \in \mathbb{N}$ and $m - N(1+p) < -1$, then for $|y| \le Y$,*

$$F(y) = \int_0^\infty \theta a e^{i\phi_y} d\xi + \int_0^\infty T_{q_y}^N b_\infty e^{i\phi_y} d\xi,$$

with both integrals absolutely convergent. For each fixed ρ there is $C = C(p, m, N, Y, L, \theta, \rho, C_0, \dots, C_N) > 0$ such that

$$\sup_{|y| \le Y} |F_M^\rho(a;y) - F(y)| \le C M^{m-N(1+p)+1} \qquad (M \ge 2L).$$

Proof. The compact part θa is unaffected by $\rho(\xi/M)$ once $M \ge 2L$. Apply the integration-by-parts lemma with $\sigma = p$, $A = H = 1$ to $b_M = \rho(\cdot/M) b_\infty$. Its N-fold transform equals that of b_∞ on $(0,M)$. Bound the difference on (M,∞) by $2C\xi^{m-N(1+p)}$ and integrate this explicit power. This proves the displayed error and a common limit independent of ρ. The same formula gives consistency on overlapping compact intervals. Each F_M^ρ is continuous by compact-support dominated convergence, hence the limit is continuous. □

Lemma 4.11 (Agreement with Lebesgue and improper integrals)**.** *Let $a \in S^m_+$. If $a \in L^1((0,\infty))$, then for every $y \in \mathbb{R}$ the function $\xi \mapsto a(\xi)e^{i\phi_y(\xi)}$ is integrable and*

$$\operatorname{Osc}[a](y) = \int_0^\infty a(\xi)e^{i\phi_y(\xi)}d\xi.$$

If $m < p$, then for every $y \in \mathbb{R}$,

$$\lim_{M\to\infty} \int_0^M a(\xi)e^{i\phi_y(\xi)}d\xi = \operatorname{Osc}[a](y).$$

The second assertion does not claim absolute convergence of the untruncated integral.

Proof. For the first assertion use $|\rho(\xi/M)ae^{i\phi_y}| \le |a|$ and dominated convergence. For the second choose a fixed large L beyond the stationary point and use the compact/tail splitting from Lemma 4.10 with one integration by parts. The boundary term at M is bounded by CM^{m-p} and tends to zero. The transformed tail has bound $C\xi^{m-p-1}$, which is integrable because $m < p$. The resulting limit is the same absolutely convergent representation as the cutoff limit. □

Lemma 4.12 (Differentiating the cutoff integrals and their limit)**.** *Let $a \in S^m_+$ and let $F^\rho_M(a;y)$ have the meaning of Lemma 4.10. For every $M > 0$, every $j \in \mathbb{N}$, and every $y \in \mathbb{R}$, its differentiated integrand is absolutely integrable, $F^\rho_M(a;\cdot)$ is smooth, and*

$$\partial_y^j F^\rho_M(a;y) = F^\rho_M((i\xi)^j a;y).$$

The function $\operatorname{Osc}[a]$ is smooth on $\mathbb{R}$, and for all $j \in \mathbb{N}$,

$$\partial_y^j \operatorname{Osc}[a] = \operatorname{Osc}[(i\xi)^j a].$$

In particular $i\xi a \in S^{m+1}_+$ and $\partial_y \operatorname{Osc}[a] = \operatorname{Osc}[i\xi a]$.

Proof. For fixed M, all ξ-moments of $\rho(\xi/M)a(\xi)$ have compact support, so dominated differentiation gives the displayed derivative at every order. The symbol-closure lemma permits applying locally uniform convergence to both $(i\xi)^j a$ and $(i\xi)^{j+1}a$. Use the theorem that locally uniform limits of differentiable functions and of their derivatives preserve the derivative, first for $j = 0$ and then inductively. Continuity of every limit supplied by Lemma 4.10 yields C^∞ regularity. □

Lemma 4.13 (Rapid decay of the oscillatory integral on the negative half-line)**.** *Let $a \in S^m_+$ with symbol constants C_l. For every real $N \ge 0$ there are $K \in \mathbb{N}$ and $C = C(p,m,N,\rho,C_0,\dots,C_K) > 0$ such that*

$$|F^\rho_M(a;y)| \le C|y|^{-N} \quad (M \ge 1,\ y \le -1), \qquad |\operatorname{Osc}[a](y)| \le C|y|^{-N} \quad (y \le -1).$$

One may choose any integer $K \ge N$ with $K > m+1$.

Proof. Here no stationary point or auxiliary tail cutoff is needed. Apply the transfer lemma with $L = 1$, $\sigma = 0$, $A = 1$, $H = |y|^{-1}$, and $b = \rho(\cdot/M)a$. The reciprocal-phase lemma supplies derivative constants independent of $y \le -1$. After K integrations by parts the absolute value of the integrand is bounded by $C|y|^{-K}\xi^{m-K}$ on $[1,\infty)$, uniformly in M. Integrate this power, use $|y|^{-K} \le |y|^{-N}$, and pass to the cutoff limit. □

Lemma 4.14 (Properties of Osc)**.** *Let $a \in S^m_+$.*

1. *The limit defining* $\operatorname{Osc}[a](y)$ *exists, locally uniformly in $y \in \mathbb{R}$, and does not depend on ρ. If $a \in L^1$ it equals the Lebesgue integral $\int_0^\infty a e^{i(y\xi-\Phi_0)}d\xi$; if $m < p$ it equals the improper Riemann integral $\lim_{M\to\infty}\int_0^M a e^{i(y\xi-\Phi_0)}d\xi$.*
2. $\operatorname{Osc}[a] \in C^\infty(\mathbb{R})$ *and* $\partial_y \operatorname{Osc}[a] = \operatorname{Osc}[i\xi a]$ *(note $i\xi a \in S^{m+1}_+$).*
3. *For all $N \geq 0$ there is C (depending on N, m, p and finitely many C_l) with* $|\operatorname{Osc}[a](y)| \leq C|y|^{-N}$ *for $y \leq -1$.*

Proof. The locally uniform limit and independence of the truncation are exactly Lemma 4.10. Its compatibility with the two ordinary integration conventions is Lemma 4.11. The frequency-shift and differentiation statements follow from Lemma 4.12. The negative-half-line estimate is Lemma 4.13. Their proofs isolate the cutoff, reciprocal-phase, and repeated-integration-by-parts interfaces needed for these conclusions. □

4.2 The linear solution

Definition 4.15 (The constant μ_*)**.** $\mu_* := -\dfrac{i}{\sqrt{2\pi p}} \in \mathbb{C}\setminus\{0\}$.

Definition 4.16 (Linear seed and its pieces)**.** For $y \in \mathbb{R}$ define

$$f_0(y) = -\frac{y}{2}+\frac{1}{\pi}\int_0^\infty \frac{1-\cos(y\xi-\Phi_0(\xi))}{\xi^2}\,d\xi, \qquad f_{0,+}(y) = -\frac{1}{2\pi}\int_0^\infty \frac{\chi_+(\xi)}{\xi^2}e^{i(y\xi-\Phi_0(\xi))}\,d\xi, \tag{4.1}$$

$f_{0,\mathrm{high}} := 2\operatorname{Re} f_{0,+}$, $f_{0,\mathrm{low}} := f_0 - f_{0,\mathrm{high}}$, and $\Gamma_+(y) := \frac{1}{2\pi}\int_0^\infty \frac{\chi_+'(\xi)}{\xi}e^{i(y\xi-\Phi_0(\xi))}d\xi$ (a Schwartz function, the integrand being smooth with compact support in $[1,2]$). For $y > 0$ let

$$\lambda_0(y) = y^{1/p}, \qquad \Psi_0(y) = \frac{p}{p+1}y^{1+1/p}, \qquad a_*(y) = \frac{e^{-i(\Psi_0(y)-\pi/4)}}{\mu_*\lambda_0(y)^{-(p+1)/2}}\,f_{0,+}'(y),$$

so that, identically for $y > 0$, $f_{0,+}'(y) = \mu_*\lambda_0(y)^{-(p+1)/2}a_*(y)e^{i(\Psi_0(y)-\pi/4)}$ (note $\lambda_0^{-(p+1)/2} = y^{-\beta}$, $\Psi_0' = \lambda_0$). Finally for $\mu \in \mathbb{C}$ put

$$f_{0,\mu} = f_0 + 2\operatorname{Re}\Big[\Big(\frac{\mu}{\mu_*}-1\Big)f_{0,+}\Big]. \tag{4.2}$$

Lemma 4.17 (Absolute convergence and the high-frequency seed symbol)**.** *Let $\frac{3}{4} < p < 1$. For every $y \in \mathbb{R}$, both integrands*

$$\xi \longmapsto \frac{1-\cos(y\xi-\Phi_0(\xi))}{\xi^2}, \qquad \xi \longmapsto \frac{\chi_+(\xi)}{\xi^2}e^{i(y\xi-\Phi_0(\xi))}$$

belong to $L^1((0,\infty))$. The function $a_{-2}(\xi) = \chi_+(\xi)/\xi^2$, extended by zero for $\xi \leq 0$, belongs to $L^1(\mathbb{R}) \cap S^{-2}_+$. For each $l \in \mathbb{N}$ there is $C_l = C_l(l,\chi_+) > 0$ with $|a_{-2}^{(l)}(\xi)| \leq C_l\xi^{-2-l}$ for $\xi \geq 1$. For every $Y \geq 1$, the first integrand is dominated uniformly in $|y| \leq Y$ by

$$\tfrac{1}{2}(Y+1)^2\mathbf{1}_{(0,1]}(\xi) + 2\xi^{-2}\mathbf{1}_{(1,\infty)}(\xi),$$

and the absolute value of the second is dominated by $\xi^{-2}\mathbf{1}_{(1,\infty)}(\xi)$.

Proof. Use $0 \le 1-\cos u \le u^2/2$ and $|y\xi - \Phi_0(\xi)| \le (Y+1)\xi$ on $(0,1]$; outside that interval use $|1-\cos u| \le 2$. The high-frequency cutoff removes the origin and the exponential has modulus one. On $[2,\infty)$ the derivatives of a_{-2} are explicit derivatives of ξ^{-2}. On $[1,2]$, compactness bounds each derivative. Near $\xi = 1$ and on $(-\infty, 1]$, the cutoff is flat, which justifies the zero extension. □

Lemma 4.18 (Oscillatory representation and regularity of the positive seed)**.** *For $\frac{3}{4} < p < 1$, write $a_{-2}(\xi) = \chi_+(\xi)/\xi^2$, extended by zero for $\xi \le 0$. Then $f_{0,+} \in C^\infty(\mathbb{R};\mathbb{C})$, and for all $j \in \mathbb{N}$ and $y \in \mathbb{R}$,*

$$f_{0,+}(y) = -\frac{1}{2\pi}\operatorname{Osc}[a_{-2}](y), \qquad \partial_y^j f_{0,+}(y) = -\frac{1}{2\pi}\operatorname{Osc}[(i\xi)^j a_{-2}](y).$$

Furthermore $f_{0,+}(y) \to 0$ as $|y| \to \infty$ (equivalently, along the cocompact filter on $\mathbb{R}$).

Proof. The first equality uses the absolutely convergent case of the oscillatory-integral compatibility lemma. Differentiate it using the all-order derivative formula. For decay, the function $\mathbf{1}_{\xi>0} a_{-2}(\xi) e^{-i\Phi_0(\xi)}$ is in $L^1(\mathbb{R})$; apply the Riemann–Lebesgue lemma to its inverse Fourier transform with the fixed $1/(2\pi)$ normalization. No differentiation of an untruncated, nonabsolute integral is needed. □

Lemma 4.19 (An absolutely convergent representation of the low-frequency seed)**.** *Let $\frac{3}{4} < p < 1$ and*

$$H_0(y,\xi) = \frac{\operatorname{Re}[1-(1-\chi_+(\xi))e^{i(y\xi-\Phi_0(\xi))}]}{\xi^2} \qquad (\xi > 0).$$

For every $y \in \mathbb{R}$, $H_0(y,\cdot) \in L^1((0,\infty))$, and

$$f_{0,\mathrm{low}}(y) = -\frac{y}{2} + \frac{1}{\pi}\int_0^\infty H_0(y,\xi)\,d\xi, \qquad f_0(y) = f_{0,\mathrm{low}}(y) + 2\operatorname{Re} f_{0,+}(y).$$

For every $Y \ge 1$ there is an integrable function $B_Y : (0,\infty) \to [0,\infty)$, depending only on p, Y, χ_+, with $|H_0(y,\xi)| \le B_Y(\xi)$ for $|y| \le Y$.

Proof. Write the numerator as $1-\cos(y\xi-\Phi_0(\xi)) + \chi_+(\xi)\cos(y\xi-\Phi_0(\xi))$. The previous lemma bounds both resulting integrands and their sum uniformly on compact y-intervals. Split the absolutely convergent integral and use $f_{0,\mathrm{low}} = f_0 - 2\operatorname{Re} f_{0,+}$. □

Lemma 4.20 (Derivative formulas for the low-frequency seed)**.** *For $\frac{3}{4} < p < 1$, $f_{0,\mathrm{low}} \in C^\infty(\mathbb{R})$ and*

$$f'_{0,\mathrm{low}}(y) = -\frac{1}{2} + \frac{1}{\pi}\int_0^\infty (1-\chi_+(\xi))\frac{\sin(y\xi-\Phi_0(\xi))}{\xi}\,d\xi.$$

For all integers $k \ge 2$,

$$f^{(k)}_{0,\mathrm{low}}(y) = -\frac{1}{\pi}\int_0^\infty (1-\chi_+(\xi))\operatorname{Re}[(i\xi)^k e^{i(y\xi-\Phi_0(\xi))}]\xi^{-2}\,d\xi.$$

Every displayed integrand belongs to $L^1((0,\infty))$ for every y. The first is dominated uniformly in $|y| \le Y$ by $(Y+\xi^p)\mathbf{1}_{(0,2]}(\xi)$; the kth one, $k \ge 2$, by $\xi^{k-2}\mathbf{1}_{(0,2]}(\xi)$. In particular, for every $k \ge 2$ there is $C_k = C_k(k,\chi_+) > 0$ with $|f^{(k)}_{0,\mathrm{low}}(y)| \le C_k$ for all $y \in \mathbb{R}$.

Proof. Differentiate H_0 first on a fixed compact y-interval. Since $|\sin(y\xi - \Phi_0)|/\xi \le |y| + \xi^p$, its first derivative has the stated majorant. Every subsequent derivative is a compactly supported power moment; its majorant is integrable even at $\xi = 0$ because $k - 2 \ge 0$. Apply dominated differentiation successively, retaining continuity at each stage. Integrating the majorant for $k \ge 2$ gives a bound independent of y. □

Lemma 4.21 (Fourier symbol of the second low-frequency derivative)**.** *For $\frac{3}{4} < p < 1$, define $m_{\mathrm{low}}(\xi) = (1-\chi_+(|\xi|))e^{-i\Phi_0(\xi)}$ on $\mathbb{R}$. Then $m_{\mathrm{low}} \in C_c^1(\mathbb{R};\mathbb{C})$, its support lies in $[-2,2]$, $m_{\mathrm{low}}(-\xi) = \overline{m_{\mathrm{low}}(\xi)}$, and*

$$f''_{0,\mathrm{low}}(y) = m\check{}_{\mathrm{low}}(y) \qquad (y \in \mathbb{R}).$$

The inverse transform is absolutely convergent for every y, and is real-valued.

Proof. The even cutoff is constant near zero and smooth away from zero; the phase is odd and C^1 with derivative $|\xi|^p$. Thus the symbol is C^1 and compactly supported. Use the $k = 2$ formula from the preceding lemma. Split the whole-line inverse transform into its positive and negative halves, substitute $\xi \mapsto -\xi$ in the latter, and use Hermitian symmetry to obtain π^{-1} times the real positive-half integral. The factor $(i\xi)^2 = -\xi^2$ cancels the minus sign in that derivative formula. □

Lemma 4.22 (Corner comparison as two absolutely convergent integrals)**.** *Let $\frac{3}{4} < p < 1$ and, on $(0,\infty)$, put*

$$b(\xi) = (1-\chi_+(\xi))\frac{1-e^{-i\Phi_0(\xi)}}{\xi^2}, \qquad a_{-2}(\xi) = \frac{\chi_+(\xi)}{\xi^2}.$$

Both b and a_{-2} are integrable on $(0,\infty)$, and $|b(\xi)| \le \xi^{p-1}\mathbf{1}_{(0,2]}(\xi)$. Define the absolutely convergent integrals

$$\mathcal{I}_1(y) = \frac{1}{\pi}\operatorname{Re}\int_0^\infty e^{iy\xi}b(\xi)d\xi, \qquad \mathcal{I}_2(y) = \frac{1}{\pi}\operatorname{Re}\int_0^\infty e^{iy\xi}a_{-2}(\xi)d\xi.$$

Then $f_{0,\mathrm{low}}(y) - W(y) = \mathcal{I}_1(y) + \mathcal{I}_2(y)$ for every $y \in \mathbb{R}$.

Proof. The bound $|1-e^{-it}| \le |t|$ gives the claimed bound for b because $\Phi_0(\xi) = \xi^{p+1}/(p+1)$ on $\xi > 0$ and $p > 0$. Subtract the Dirichlet integral formula for W from the low-frequency representation. All integrals involved are absolutely convergent. Use $e^{iy\xi} - (1-\chi_+)e^{iy\xi - i\Phi_0} = e^{iy\xi}[(1-\chi_+)(1-e^{-i\Phi_0}) + \chi_+]$ and distribute the integral and real part.

□

Lemma 4.23 (Well-posedness and smoothness of the seed)**.**

1. *The integrals defining f_0 and $f_{0,+}$ converge absolutely for every y; $f_{0,+} = -\frac{1}{2\pi}\operatorname{Osc}[\chi_+\xi^{-2}]$ is C^∞ with $\partial_y^j f_{0,+} = -\frac{1}{2\pi}\operatorname{Osc}[(i\xi)^j\chi_+\xi^{-2}]$, and $f_{0,+}(y) \to 0$ as $|y| \to \infty$.*
2. *$f_{0,\mathrm{low}}(y) = -\frac{y}{2} + \frac{1}{\pi}\int_0^\infty \operatorname{Re}\big[1-(1-\chi_+(\xi))e^{i(y\xi-\Phi_0(\xi))}\big]\xi^{-2}d\xi$; $f_{0,\mathrm{low}} \in C^\infty(\mathbb{R})$ with $f'_{0,\mathrm{low}}(y) = -\frac{1}{2} + \frac{1}{\pi}\int_0^\infty(1-\chi_+)\frac{\sin(y\xi-\Phi_0)}{\xi}d\xi$ and $f^{(k)}_{0,\mathrm{low}}(y) = -\frac{1}{\pi}\int_0^\infty(1-\chi_+)\operatorname{Re}[(i\xi)^k e^{i(y\xi-\Phi_0)}]\xi^{-2}d\xi$ for $k \ge 2$, all absolutely convergent. In particular $f''_{0,\mathrm{low}} = m\check{}_{\mathrm{low}}$ with $m_{\mathrm{low}}(\xi) := (1-\chi_+(|\xi|))e^{-i\Phi_0(\xi)} \in C_c^1(\mathbb{R})$, and $f^{(k)}_{0,\mathrm{low}}$ is bounded for $k \ge 2$.*

3. *For all* $y \in \mathbb{R}$: $f_{0,\mathrm{low}}(y) - W(y) = \mathcal{I}_1(y) + \mathcal{I}_2(y)$, *where* $\mathcal{I}_1(y) = \frac{1}{\pi}\operatorname{Re}\int_0^\infty e^{iy\xi} b(\xi)\,d\xi$, $b(\xi) = (1-\chi_+(\xi))\frac{1-e^{-i\Phi_0(\xi)}}{\xi^2}$, *and* $\mathcal{I}_2(y) = \frac{1}{\pi}\operatorname{Re}\int_0^\infty e^{iy\xi}\chi_+(\xi)\xi^{-2}\,d\xi$, *both absolutely convergent.*

4. $f_0 = f_{0,\mathrm{low}} + 2\operatorname{Re} f_{0,+} \in C^\infty(\mathbb{R};\mathbb{R})$.

Proof. Item (1) combines Lemma 4.17 and Lemma 4.18. Item (2) combines the low-frequency representation, derivative, and Fourier-symbol lemmas Lemmas 4.19 to 4.21. Item (3) is Lemma 4.22. Finally the defining identity $f_0 = f_{0,\mathrm{low}} + 2\operatorname{Re} f_{0,+}$ and the smoothness of both summands prove item (4). □

Lemma 4.24 (Low-frequency remainder $\mathcal{I}_1$)**.** $\mathcal{I}_1 \in C^\infty(\mathbb{R})$ *and for every* $j \ge 0$ *there is* $C_j = C_j(p,\chi_+)$ *with* $|\partial_y^j \mathcal{I}_1(y)| \le C_j |y|^{-p-j}$ *for* $|y| \ge 1$.

Proof. First, $|\partial_\xi^k b(\xi)| \le C_k \xi^{p-1-k}$ on $(0,2]$ and $b = 0$ on $[2,\infty)$: $1 - e^{-i\Phi_0} = F(\Phi_0)$ with $F(s) = 1 - e^{-is}$, $F(0) = 0$, and by Faà di Bruno $\partial^k[F(\Phi_0)]$ is a sum of terms $F^{(r)}(\Phi_0)\prod_{i\le r}\Phi_0^{(l_i)}$, $r \ge 1$, $\sum l_i = k$, each bounded by $C\xi^{r(p+1)-k} \le C\xi^{p+1-k}$ for $\xi \le 2$; then use Leibniz with $(1-\chi_+)\xi^{-2}$. Hence $|\partial^k(\xi^j b)| \le C\xi^{p-1+j-k}$. Since $\xi^j b \in L^1$ with compact support, $\partial_y^j \mathcal{I}_1 = \frac{1}{\pi}\operatorname{Re}\int_0^\infty (i\xi)^j e^{iy\xi} b\,d\xi$. Let $|y| \ge 1$, $\rho_y = |y|^{-1} \le 1$. Then $|\int_0^{\rho_y}| \le C\int_0^{\rho_y}\xi^{p+j-1} = C|y|^{-p-j}$. On $[\rho_y, 2]$ integrate by parts $j+1$ times with $e^{iy\xi} = (iy)^{-1}\partial_\xi e^{iy\xi}$: the boundary terms at 2 vanish (all derivatives of b vanish there), those at ρ_y are $\le C|y|^{-k-1}\rho_y^{p+j-1-k} = C|y|^{-p-j}$ $(0 \le k \le j)$, and the final integral is $\le C|y|^{-j-1}\int_{\rho_y}^2 \xi^{p-2}d\xi \le \frac{C}{1-p}|y|^{-j-1}\rho_y^{p-1} = C'|y|^{-p-j}$ (using $p < 1$). □

Lemma 4.25 (High-frequency remainder $\mathcal{I}_2$)**.** $\mathcal{I}_2 \in C^\infty(\mathbb{R}\setminus\{0\})$ *and for all* $j, N \ge 0$ *there is* $C_{j,N}$ *with* $|\partial_y^j \mathcal{I}_2(y)| \le C_{j,N}|y|^{-N}$ *for* $|y| \ge 1$.

Proof. Let $c = \chi_+\xi^{-2}$; $|c^{(l)}(\xi)| \le C_l\xi^{-2-l}$, $c = 0$ on $(-\infty, 1]$. For $y \ne 0$ and $L \ge 0$, integrating by parts L times (boundary terms vanish: $c^{(k)} = 0$ near 1 and $c^{(k)}(\xi) \to 0$ as $\xi \to \infty$) gives $\int_0^\infty e^{iy\xi}c\,d\xi = (-iy)^{-L}\int_0^\infty e^{iy\xi}c^{(L)}d\xi =: (-iy)^{-L}G_L(y)$. Since $\xi^l c^{(L)} \in L^1$ for $l \le L$, $G_L \in C^L$ with $|G_L^{(l)}| \le \int \xi^l |c^{(L)}| < \infty$. Take $L \ge \max(j, N)$ and apply Leibniz to $y^{-L}G_L$ for $|y| \ge 1$. □

Lemma 4.26 (Rapid decay of $f_{0,+}$ on the left)**.** *For all* $j, N \ge 0$ *there is* $C_{j,N}$ *with* $|\partial_y^j f_{0,+}(y)| \le C_{j,N}|y|^{-N}$ *for* $y \le -1$.

Proof. $\partial_y^j f_{0,+} = -\frac{1}{2\pi}\operatorname{Osc}[(i\xi)^j\chi_+\xi^{-2}]$ with $(i\xi)^j\chi_+\xi^{-2} \in S_+^{j-2}$; apply Lemma 4.14(3). □

Lemma 4.27 (The slowly varying amplitude a_*)**.** *There are* $y_0 \ge 2$ *(depending only on* p, χ_+*) and constants* C_j *such that for* $y \ge y_0$ *and* $j \ge 0$*:* $|\partial_y^j(a_* - 1)(y)| \le C_j y^{-2\beta-j}$*. In particular* $|a_*| \le 2$ *and* $|\partial_y^j a_*| \le C_j y^{-j}$ *on* $[y_0,\infty)$*, and* $|\partial_y^j f_{0,+}'(y)| \le C_j y^{-\beta+j/p}$ *for* $y \ge y_0$.

Proof. By Lemma 4.23(1), $f_{0,+}' = -\frac{i}{2\pi}\operatorname{Osc}[a]$ with $a = \chi_+/\xi \in S_+^{-1}$; since $-1 < p$ this is the improper integral $\int_0^\infty a(\xi)e^{i(y\xi - \xi^{p+1}/(p+1))}d\xi$ of Lemma 3.80

(with $m=-1$; a vanishes for $\xi\le 1$, $|\partial^k a|\le C_k\xi^{-1-k}$). That lemma gives y_1 with, for $y\ge y_1$,

$$f'_{0,+}(y)=-\frac{i}{2\pi}\Big(\frac{2\pi}{p}\Big)^{1/2}\lambda_0^{(1-p)/2}e^{i(\Psi_0-\pi/4)}\big(a(\lambda_0)+r\big)=\mu_*\lambda_0^{(1-p)/2}e^{i(\Psi_0-\pi/4)}\big(a(\lambda_0)+r\big),$$

$|\partial^j r|\le C_j\lambda_0^{-1-(p+1)}y^{-j}$. Hence $a_*=\lambda_0^{(1-p)/2+(p+1)/2}(a(\lambda_0)+r)=\lambda_0(a(\lambda_0)+r)$. For $y\ge 2^p$, $\lambda_0\ge 2$ so $a(\lambda_0)=\lambda_0^{-1}$ and $a_*-1=\lambda_0 r$. Leibniz with $|\partial^i\lambda_0|\le C_i y^{1/p-i}$: $|\partial^j(\lambda_0 r)|\le C\sum_i y^{1/p-i}y^{-(p+2)/p-(j-i)}=Cy^{-(p+1)/p-j}=Cy^{-2\beta-j}$. Take $y_0\ge\max(y_1,2^p,2)$ so large that $C_0y_0^{-2\beta}\le 1$. For the last claim, $f'_{0,+}=\mu_* a_* y^{-\beta}e^{i(\Psi_0-\pi/4)}$ and $|\partial^k e^{i\Psi_0}|\le C_k y^{k/p}$ (Faà di Bruno, $|\partial^l\Psi_0|\le Cy^{1+1/p-l}$, and $1/p-(l-1)\le l/p$); Leibniz. □

Lemma 4.28 (Global bounds for the high-frequency part)**.** *There is C with $|f_{0,+}(y)|+|f'_{0,+}(y)|\le C$ and $|f''_{0,+}(y)|+|f'''_{0,+}(y)|\le C\langle y\rangle^{2/p}$ for all $y\in\mathbb{R}$.*

Proof. By Lemma 4.23, $f_{0,+}$ is continuous and tends to zero as $|y|\to\infty$, hence is globally bounded. For derivatives, combine the positive-tail estimates of Lemma 4.27 with the negative-tail estimates of Lemma 4.26 and continuity on the intervening compact interval. The first derivative has exponent $-\beta<0$; the second and third derivatives have exponents $-\beta+1/p$ and $-\beta+2/p$, both at most $2/p$. Since $\langle y\rangle\ge 1$, enlarge the constant to combine the bounds.

□

Lemma 4.29 ($\mathcal{L}_0$ applied to $f_{0,+}$)**.** $\mathcal{L}_0 f_{0,+}=-\Gamma_+$ *pointwise on* $\mathbb{R}$.

Proof. Apply Lemma 4.2(1) with $m(\xi)=-\chi_+(\xi)\xi^{-2}e^{-i\Phi_0(\xi)}$ ($\xi>0$), $m=0$ ($\xi\le 0$): $m\in C^\infty\cap L^1$, $m=0$ on $(-\infty,1]$, $|m|+|m'|\le C$, $f_{0,+}=\check{m}$, and the regularity of $g=f_{0,+}$ is Lemma 4.28. With $E=e^{-i\Phi_0}$, $E'=-i\xi^p E$ for $\xi>0$: $\xi m'=-\chi'_+\xi^{-1}E+2\chi_+\xi^{-2}E+i\chi_+\xi^{p-1}E$ and $(2+i\xi^{1+p})m=-2\chi_+\xi^{-2}E-i\chi_+\xi^{p-1}E$, so $n=-\chi'_+\xi^{-1}E\,\mathbf{1}_{\xi>0}\in C_c^\infty$ and $\check{n}=-\Gamma_+$.

□

Lemma 4.30 ($\mathcal{L}_0 f_0$ is constant)**.** $f_0\in\mathcal{A}$, $\mathcal{L}_0 f_0\in C^1(\mathbb{R})$ *and* $(\mathcal{L}_0 f_0)'\equiv 0$.

Proof. $f'_{0,\mathrm{low}}$ is bounded: for $|y|\ge 1$ it equals $W'+\mathcal{I}'_1+\mathcal{I}'_2$, bounded by Lemmas 4.24 and 4.25, and it is continuous on $[-1,1]$; $f^{(k)}_{0,\mathrm{low}}$, $k\ge 2$, are bounded by Lemma 4.23(2). With Lemma 4.28, f'_0 is bounded and C^1, so $f_0\in\mathcal{A}$. By $\mathbb{R}$-linearity and since $|D|^p$ commutes with Re, $\mathcal{L}_0 f_0=\mathcal{L}_0 f_{0,\mathrm{low}}+2\operatorname{Re}\mathcal{L}_0 f_{0,+}=\mathcal{L}_0 f_{0,\mathrm{low}}-2\operatorname{Re}\Gamma_+$ (Lemma 4.29). By Lemma 4.1 with $g=f'_{0,\mathrm{low}}$, $(\mathcal{L}_0 f_{0,\mathrm{low}})'=|D|^p f''_{0,\mathrm{low}}+f'_{0,\mathrm{low}}-f'_{0,\mathrm{low}}-yf''_{0,\mathrm{low}}=(|D|^p-y)f''_{0,\mathrm{low}}$. Apply Lemma 4.2(2) with $m=m_{\mathrm{low}}$: for $\xi>0$, $m'_{\mathrm{low}}=-\chi'_+E-i\xi^p(1-\chi_+)E$, so $n_b=|\xi|^p m_{\mathrm{low}}-im'_{\mathrm{low}}=i\chi'_+(\xi)E$; for $\xi<0$, $n_b(\xi)=-i\chi'_+(-\xi)e^{-i\Phi_0(\xi)}=\overline{n_b(-\xi)}$. Hence $(|D|^p-y)f''_{0,\mathrm{low}}=\frac{1}{\pi}\operatorname{Re}\int_0^\infty i\chi'_+(\xi)e^{i(y\xi-\Phi_0)}d\xi=2\operatorname{Re}\Gamma'_+$ (differentiate Γ_+ under the integral). Thus $(\mathcal{L}_0 f_0)'=2\operatorname{Re}\Gamma'_+-2\operatorname{Re}\Gamma'_+=0$.

□

Lemma 4.31 (Tails of the seed)**.** *For every $j\ge 0$ there is C_j such that $|\partial_y^j(f_{0,\mathrm{low}}-W)(y)|\le C_j|y|^{-p-j}$ for $|y|\ge 1$, and $|\partial_y^j(f_0(y)+y)|\le C_j|y|^{-p-j}$ for $y\le -1$.*

Proof. The first claim is Lemma 4.23(3) with Lemmas 4.24 and 4.25 ($N = p+j$). For $y \le -1$, $W(y) = -y$, so $f_0 + y = (f_{0,\mathrm{low}} - W) + 2\,\mathrm{Re}\, f_{0,+}$; use Lemma 4.26 with $N = p + j$.

□

Lemma 4.32 (f_0 solves the linearized equation)**.** *$\mathcal{L}_0 f_0 \equiv 0$ on $\mathbb{R}$.*

Proof. By Lemma 4.30, $\mathcal{L}_0 f_0 \equiv \kappa$; we show $\mathcal{L}_0 f_0(y) \to 0$ as $y \to -\infty$. Write $\mathcal{L}_0 f_0 = |D|^p f_0' + (f_0 + y) - y(f_0' + 1)$. For $y \le -2$, Lemma 4.31 gives $|f_0 + y| \le C|y|^{-p}$, $|y(f_0'+1)| \le C|y|^{-p}$. In $|D|^p f_0'(y) = c_\alpha \int (f_0'(y) - f_0'(y-h))|h|^{-\alpha} dh$: for $|h| \le |y|/2$ both $y, y-h \le y/2 \le -1$, so $|f_0'(y) - f_0'(y-h)| \le |h| \sup_{w \le y/2} |f_0''(w)| \le C|h||y|^{-p-2}$, contributing $\le C|y|^{-p-2}|y|^{1-p}$; for $|h| > |y|/2$ the integrand is $\le 2\|f_0'\|_\infty |h|^{-\alpha}$, contributing $\le C|y|^{-p}$. All terms tend to 0, so $\kappa = 0$.

□

Lemma 4.33 (The linear solution)**.** *$f_0 \in C^\infty(\mathbb{R};\mathbb{R}) \cap \mathcal{A}$, $\mathcal{L}_0(f_0) = 0$, $\lim_{y\to-\infty} f_0'(y) = -1$, $\lim_{y\to\infty} f_0'(y) = 0$, and, with y_0 from Lemma 4.27, for every $j \ge 0$ there is C_j (depending only on p, χ_+, j) with:*

1. *$|\partial_y^j (a_* - 1)(y)| \le C_j y^{-2\beta - j}$ for $y \ge y_0$;*
2. *$|\partial_y^j (f_{0,\mathrm{low}} - W)(y)| \le C_j |y|^{-p-j}$ for $|y| \ge 1$, and $|\partial_y^j (f_0(y) + y)| \le C_j |y|^{-p-j}$ for $y \le -1$;*
3. *for every $\mu \in \mathbb{C}$, $f_{0,\mu} \in C^\infty(\mathbb{R};\mathbb{R}) \cap \mathcal{A}$ and $\mathcal{L}_0(f_{0,\mu}) = G_\mu$.*

Proof. Smoothness: Lemma 4.23(4); $\mathcal{L}_0 f_0 = 0$: Lemma 4.32; (1): Lemma 4.27; (2): Lemma 4.31. Limits of f_0': at $-\infty$ from (2); at $+\infty$, $f_0' = f_{0,\mathrm{low}}' + 2\,\mathrm{Re}\, f_{0,+}'$ with $|f_{0,\mathrm{low}}'| \le C y^{-p-1}$ ((2), $W' = 0$) and $|f_{0,+}'| \le C y^{-\beta}$ (Lemma 4.27). (3) $f_{0,\mu}$ is real, smooth, in $\mathcal{A}$ by Lemma 4.28. By linearity and Lemma 4.29, $\mathcal{L}_0 f_{0,\mu} = 0 + 2\,\mathrm{Re}[(\mu/\mu_* - 1)(-\Gamma_+)] = 2\,\mathrm{Re}[(1 - \mu/\mu_*)\Gamma_+]$. By Definition 2.13 and the conjugate symmetry of $\widehat{G_\mu}$, $G_\mu(y) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{G_\mu} e^{iy\xi} d\xi = \frac{1}{\pi}\,\mathrm{Re}\int_0^\infty (1 - \frac{\mu}{\mu_*})\frac{\chi_+'}{\xi} e^{i(y\xi - \Phi_0)} d\xi = 2\,\mathrm{Re}[(1 - \frac{\mu}{\mu_*})\Gamma_+(y)]$.

□

4.3 The approximate solution family

Definition 4.34 (Cubic part of the nonlinearity)**.** For $\epsilon \ge 0$ and $v \in \mathcal{A}$ (real or complex valued) let

$$\mathcal{Q}_{\epsilon,3}(v)(y) = -\frac{(p+1)\epsilon c_\alpha}{2} \int_{\mathbb{R}} \frac{v'(y) - v'(z)}{|y - z|^\alpha}\, S_v(y,z)^2\, dz, \tag{4.3}$$

and more generally the trilinear form $T(u,v,w)(y) = \int_{\mathbb{R}} \frac{u'(y) - u'(z)}{|y-z|^\alpha} S_v(y,z) S_w(y,z)\, dz$, so that $\mathcal{Q}_{\epsilon,3}(v) = \kappa_\epsilon T(v,v,v)$ with $\kappa_\epsilon := -\frac{(p+1)\epsilon c_\alpha}{2}$. The integrals converge absolutely since $|S_v| \le \|v'\|_\infty$ and by Lemma 1.6.

Lemma 4.35 (Cubic response to a sine)**.** *The coefficients $c_1(\epsilon), c_2(\epsilon)$ defined by the integral formulas in Definition 4.36 are the unique real numbers with $\mathcal{Q}_{\epsilon,3}(\sin)(y) = c_1 \cos y + c_2 \cos(3y)$ for all $y \in \mathbb{R}$. There is $C_3 = C_3(p)$ such that, for every $\epsilon \ge 0$, $c_i(\epsilon) = \epsilon c_i(1)$ and $|c_1|, |c_2| \le C_3 \epsilon$.*

Proof. Let $e_\sigma(y) = e^{i\sigma y}$, $\sigma \in \{\pm 1\}$. Substituting $z = y - h$: $e'_\sigma(y) - e'_\sigma(y-h) = i\sigma e^{i\sigma y}(1 - e^{-i\sigma h})$, $S_{e_\sigma}(y, y-h) = e^{i\sigma y}(1-e^{-i\sigma h})/h$. Hence $T(e_{\sigma_1}, e_{\sigma_2}, e_{\sigma_3})(y) = e^{i(\sigma_1+\sigma_2+\sigma_3)y}\tau(\sigma)$ with $\tau(\sigma) = i\sigma_1 \int_{\mathbb{R}} \prod_{k=1}^{3}(1-e^{-i\sigma_k h})\,|h|^{-\alpha}h^{-2}dh$, and $|\tau(\sigma)| \le C_\tau := \int_{\mathbb{R}} \min(2,|h|)^3|h|^{-\alpha-2}dh < \infty$ ($3-\alpha-2 = -p > -1$ near 0, $-\alpha-2 < -1$ at ∞). Since $\sin = \frac{1}{2i}\sum_\sigma \sigma e_\sigma$ and T is $\mathbb{C}$-trilinear on bounded C^2 functions with bounded derivatives (absolute convergence), $\mathcal{Q}_{\epsilon,3}(\sin)(y) = \sum_{k\in\{\pm1,\pm3\}} d_k e^{iky}$ with $|d_k| \le |\kappa_\epsilon| C_\tau$. The function $F = \mathcal{Q}_{\epsilon,3}(\sin)$ is real (sin real) and even: for odd v, v' is even and $S_v(-y,-z) = S_v(y,z)$, so substituting $z \mapsto -z$ shows $\mathcal{Q}_{\epsilon,3}(v)(-y) = \mathcal{Q}_{\epsilon,3}(v)(y)$. The odd part $i\sum d_k \sin(ky)$ of F vanishes, so $F = (d_1+d_{-1})\cos y + (d_3+d_{-3})\cos 3y$; taking real parts, $c_1 = \mathrm{Re}(d_1+d_{-1})$, $c_2 = \mathrm{Re}(d_3+d_{-3})$, $|c_i| \le 2|\kappa_\epsilon|C_\tau$. Uniqueness: $\cos y, \cos 3y$ are linearly independent. Linearity in ϵ: $\mathcal{Q}_{\epsilon,3} = \epsilon \mathcal{Q}_{1,3}$. □

Definition 4.36 (Correction constants)**.** For $\epsilon \ge 0$ define the coefficients directly by

$$c_1(\epsilon) := \frac{1}{\pi}\int_0^{2\pi} \mathcal{Q}_{\epsilon,3}(\sin)(y)\cos y\,dy, \qquad c_2(\epsilon) := \frac{1}{\pi}\int_0^{2\pi} \mathcal{Q}_{\epsilon,3}(\sin)(y)\cos(3y)\,dy.$$

Set $n_\epsilon := 4c_1(\epsilon)$ and $c_{\mathrm{cubic}} := -4c_2(\epsilon)/(3^p-1)$ ($3^p > 1$). The sine-response lemma then identifies these explicit coefficients with the paper's coefficients; its normalization reads

$$\mathcal{Q}_{\epsilon,3}(2\sin(\cdot))(y) = 2n_\epsilon \cos y - 2(3^p-1)c_{\mathrm{cubic}}\cos(3y) \tag{4.4}$$

(since $\mathcal{Q}_{\epsilon,3}$ is 3-homogeneous). For $\mu \in \mathbb{C}$ set

$$c_{\log}(\mu) = -\frac{n_\epsilon|\mu|^2}{p}, \quad c_{\mathrm{power}} = -\frac{c_\epsilon}{1-p^2}, \quad c_+ = -\frac{c_\epsilon}{p+1}, \quad c_- = \frac{c_\alpha}{p+1}\int_0^1 [A_\epsilon(t)-1]t^{p-1}dt, \tag{4.5}$$

with c_ϵ from Lemma 3.12, and $\widetilde{\lambda}_{\epsilon,\mu}(y) := -\frac{c_\epsilon}{p}y^{-p-1} + c_{\log}(\mu)y^{-1-1/p}$ for $y > 0$.

Lemma 4.37 (Size of the constants)**.** *Let $K_\mu > 0$. There is C (depending on p, K_μ) such that for $\epsilon \in [0,1]$, $|\mu| \le K_\mu$: $|c_\epsilon|, |n_\epsilon|, |c_{\mathrm{cubic}}|, |c_{\log}(\mu)|, |c_{\mathrm{power}}|, |c_\pm| \le C\epsilon$; all vanish at $\epsilon = 0$; $c_{\log}$ is continuous in μ. Moreover for $y \ge 1$, $k \ge 0$: $|\partial_y^k \widetilde{\lambda}_{\epsilon,\mu}(y)| \le C_k \epsilon y^{-p-1-k}$.*

Proof. $|c_\epsilon| \le C\epsilon$, $c_0 = 0$: Lemma 3.12 ($A_0 \equiv 1$). $n_\epsilon, c_{\mathrm{cubic}}$: Lemma 4.35. $|A_\epsilon(t) - 1| \le \frac{\alpha}{2}\epsilon t^2$ gives $|c_-| \le C\epsilon$. For $\widetilde{\lambda}$: $1 + 1/p > p+1$, so $y^{-1-1/p-k} \le y^{-p-1-k}$ for $y \ge 1$. □

Definition 4.38 (Approximate solution family)**.** Fix $\delta \in (0, |\mu_*|/2]$, $\overline{B} = \overline{B}(\mu_*, \delta) = \{\mu \in \mathbb{C} : |\mu - \mu_*| \le \delta\}$ (so $|\mu| \le 2|\mu_*|$). Let C_λ be the constant of Lemma 4.37 for $k = 0$, $K_\mu = 2|\mu_*|$, and fix $\epsilon_0 := \min\{1, (2C_\lambda)^{-1}\}$ (so $|\widetilde{\lambda}_{\epsilon,\mu}(y)| \le \frac{1}{2}$ for $y \ge 1$; later chapters may decrease ϵ_0). Let $R_0 := \max\{y_0, 2\}$ with y_0 from Lemma 4.27, and fix $R \ge R_0$ (in particular $R \ge 2$; later chapters may enlarge R_0). Let $\chi_R(y) := \chi_+(y/R)$ (so $\chi_R = 0$ on $(-\infty, R]$, $\chi_R = 1$ on $[2R, \infty)$, $|\partial^j \chi_R| \le \|\chi_+^{(j)}\|_\infty R^{-j}$). For $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$ and $y > R/2$ define

$$\lambda_{\epsilon,\mu}(y) := \Psi'_{\epsilon,\mu}(y) = y^{1/p} + c_{\mathrm{power}}\Big(\frac{1-p^2}{p}\Big)y^{1/p-p-1} + \frac{c_{\log}(\mu)}{y}, \qquad \Psi_{\epsilon,\mu}(R) = \Psi_0(R), \tag{4.6}$$

$$Z_{\epsilon,\mu}(y) = \mu\, a_*(y)\, \lambda_{\epsilon,\mu}(y)^{-(p+1)/2} e^{i(\Psi_{\epsilon,\mu}(y)-\pi/4)}. \tag{4.7}$$

We abbreviate $f = \overline{f}_{\epsilon,\mu}$, $Z = Z_{\epsilon,\mu}$, $\lambda = \lambda_{\epsilon,\mu}$, $\Psi = \Psi_{\epsilon,\mu}$, and write $Z_0 := Z_{0,\mu_*}$ (also λ_0, Ψ_0 agree with $\lambda_{0,\mu}, \Psi_{0,\mu}$, see Lemma 4.39). Define

$$f_{\mathrm{slow}}(y) = c_+ y^{-p}\chi_R(y) + c_-|y|^{-p}\chi_R(-y), \tag{4.8}$$

$$f_{\mathrm{osc}}(y) = -2\chi_R(y)\,\mathrm{Re}\int_y^\infty \Big(Z_{\epsilon,\mu}(t) + c_{\mathrm{cubic}} Z_{\epsilon,\mu}(t)^3 - \frac{\mu}{\mu_*} Z_0(t)\Big)dt \quad (y > R), \qquad f_{\mathrm{osc}}(y) = 0 \quad (y \le R), \tag{4.9}$$

(absolutely convergent, see Lemma 4.40) and

$$\overline{f}_{\epsilon,\mu} = f_{0,\mu} + f_{\mathrm{slow}} + f_{\mathrm{osc}}. \tag{4.10}$$

By Lemma 4.41, for $y \ge 2R$

$$\overline{f}_{\epsilon,\mu}(y) = f_{0,\mathrm{low}}(y) + c_+ y^{-p} - 2\,\mathrm{Re}\int_y^\infty \big(Z(t) + c_{\mathrm{cubic}} Z(t)^3\big)dt, \tag{4.11}$$

and for $y \le -2R$

$$\overline{f}_{\epsilon,\mu}(y) = f_0(y) + 2\,\mathrm{Re}\Big[\Big(\frac{\mu}{\mu_*} - 1\Big) f_{0,+}(y)\Big] + c_-|y|^{-p}. \tag{4.12}$$

Lemma 4.39 (Basic properties of the modified phase)**.** *For $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, $y > R/2$ (so $y \ge 1$):*

1. *$\lambda = \lambda_0(1 + \widetilde{\lambda}_{\epsilon,\mu})$, hence $\frac{1}{2}y^{1/p} \le \lambda(y) \le \frac{3}{2}y^{1/p}$, and $|\partial^k \lambda| \le C_k y^{1/p-k}$, $|\partial^k(\lambda^{-q})| \le C_{k,q} y^{-q/p-k}$ for every $q > 0$;*
2. *$\Psi_{\epsilon,\mu}(y) = \Psi_0(y) + c_{\mathrm{power}}(y^{1/p-p} - R^{1/p-p}) + c_{\log}(\mu)\log(y/R)$, and $|\partial^k e^{i\Psi}| \le C_k y^{k/p}$;*
3. *at $\epsilon = 0$: $\lambda_{0,\mu} = \lambda_0$, $\Psi_{0,\mu} = \Psi_0$, $Z_{0,\mu} = \frac{\mu}{\mu_*} Z_0$, $c_{\mathrm{cubic}} = c_\pm = 0$; and $Z_0 = f'_{0,+}$ on $(R/2, \infty)$.*

Constants are admissible.

Proof. (1) $c_{\mathrm{power}}\frac{1-p^2}{p} = -\frac{c_\epsilon}{p}$ and $\frac{c_{\log}}{y} = \lambda_0 c_{\log} y^{-1-1/p}$; $|\widetilde{\lambda}| \le \frac{1}{2}$ by the choice of ϵ_0. Derivatives by Leibniz/Faà di Bruno using Lemma 4.37 and $|\partial^k \lambda_0| \le C_k y^{1/p-k}$, $1 + \widetilde{\lambda} \in [\frac{1}{2}, \frac{3}{2}]$. (2) Differentiate the right side: $c_{\mathrm{power}}(\frac{1}{p} - p)y^{1/p-p-1} + \frac{c_{\log}}{y}$ matches (4.6), and the value at R is $\Psi_0(R)$. For $e^{i\Psi}$: Faà di Bruno, each factor $\partial^l \Psi = \partial^{l-1}\lambda$ is $\le C y^{1/p-l+1} \le C y^{l/p}$ $(1/p \ge 1)$. (3) $c_0 = 0$ and $n_0 = 4c_1(0) = 0$; the identity $Z_0 = f'_{0,+}$ is the definition of a_*. □

Lemma 4.40 (Bounds for Z)**.** *For $j \ge 0$ there are admissible C_j with, for $y \ge R$, $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$: $|\partial^j Z(y)| \le C_j y^{-\beta+j/p}$, $|\partial^j(Z^3)(y)| \le C_j y^{-3\beta+j/p}$, $|\partial^j(Z/\lambda)(y)| \le C_j y^{-\beta-1/p+j/p}$, and $|\partial_y^j(a_*\lambda^{-q})(y)| \le C_{j,q} y^{-q/p-j}$ $(q > 0)$. In particular $\int_y^\infty |Z| \le C y^{1-\beta}$ $(\beta > 1)$.*

Proof. Leibniz with $|\partial^k a_*| \le C_k y^{-k}$ (Lemma 4.27, $R \ge y_0$), Lemma 4.39(1),(2), $|\mu| \le 2|\mu_*|$ and $\lambda^{-(p+1)/2} \approx y^{-\beta}$; for Z^3 use $Z^3 = \mu^3 a_*^3 \lambda^{-3(p+1)/2} e^{3i(\Psi-\pi/4)}$ and $|\partial^k e^{3i\Psi}| \le C_k y^{k/p}$; $y^{-k} \le y^{k/p}$.

□

Lemma 4.41 (Regional forms and smoothness of the family)**.** *For $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$: $\overline{f}_{\epsilon,\mu} \in C^\infty(\mathbb{R};\mathbb{R}) \cap \mathcal{A}$; $f_{\mathrm{slow}} = 0$ on $[-R,R]$, $f_{\mathrm{osc}} = 0$ on $(-\infty,R]$; (4.11) holds for $y \geq 2R$ and (4.12) for $y \leq -2R$; $\overline{f}_{0,\mu_*} = f_0$. For $y \geq 2R$, $\overline{f}' = f'_{0,\mathrm{low}} - pc_+y^{-p-1} + 2\,\mathrm{Re}(Z + c_{\mathrm{cubic}}Z^3)$.*

Proof. The integrand of f_{osc} is smooth on $(R/2,\infty)$ and $O(t^{-\beta})$, so $F(y) := \int_y^\infty(\cdots)$ is smooth on $(R/2,\infty)$ with $F' = -(\cdots)$; since χ_R vanishes on $(-\infty,R]$, $\chi_R F$ extended by 0 is smooth on $\mathbb{R}$. f_{slow} is smooth since $\chi_R(\pm y)$ vanish near 0. Bounded derivative: Lemma 4.40, Lemma 4.28, $f_0 \in \mathcal{A}$. For $y \geq 2R$: $\chi_R(y) = 1$, $\chi_R(-y) = 0$, and $f_{0,\mu} = f_0 - 2\,\mathrm{Re}\,f_{0,+} + 2\,\mathrm{Re}[\frac{\mu}{\mu_*}f_{0,+}]$ with $f_{0,+}(y) = -\int_y^\infty f'_{0,+} = -\int_y^\infty Z_0$ (Lemma 4.23(1): $f_{0,+} \to 0$; Lemma 4.39(3)); the $\frac{\mu}{\mu_*}Z_0$ terms cancel and $f_0 - 2\,\mathrm{Re}\,f_{0,+} = f_{0,\mathrm{low}}$. For $y \leq -2R$: $f_{\mathrm{osc}} = 0$, $f_{\mathrm{slow}} = c_-|y|^{-p}$. At $(0,\mu_*)$: $c_\pm = c_{\mathrm{cubic}} = 0$ and $Z_{0,\mu_*} = Z_0$. The derivative formula follows from the fundamental theorem of calculus.

□

Lemma 4.42 (Frequency and phase differences)**.** *For $j \geq 0$ there are admissible C_j with, for $y \geq R$, $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$,*

$$|\partial_y^j(\lambda_{\epsilon,\mu} - \lambda_0)(y)| \leq C_j\epsilon y^{\frac{1}{p}-p-1-j}, \tag{4.13}$$

$$|\partial_y^j(\Psi_{\epsilon,\mu} - \Psi_0)(y)| \leq C_j\epsilon y^{\frac{1}{p}-p-j}. \tag{4.14}$$

Proof. $\lambda - \lambda_0 = -\frac{c_\epsilon}{p}y^{1/p-p-1} + c_{\log}y^{-1}$ and $-1 < \frac{1}{p} - p - 1$, so for $y \geq 1$ both terms are $\leq C\epsilon y^{1/p-p-1-j}$ after j derivatives. For $j \geq 1$, $\partial^j(\Psi - \Psi_0) = \partial^{j-1}(\lambda - \lambda_0)$. For $j = 0$, by Lemma 4.39(2), $|\Psi - \Psi_0| \leq |c_{\mathrm{power}}|y^{1/p-p} + |c_{\log}|\log y \leq C\epsilon y^{1/p-p}$ (as $\log y \leq \frac{1}{1/p-p}y^{1/p-p}$ for $y \geq 1$). □

Lemma 4.43 (Reciprocal-frequency amplitudes and their differences)**.** *Fix $\frac{3}{4} < p < 1$ and $0 < \delta \leq |\mu_*|/2$. There are $\epsilon_0 \in (0,1]$ and $R_0 \geq 2$, depending only on p,δ and the fixed cutoffs, with the following property. For every $q > 0$ and $j \in \mathbb{N}$ there is $C_{j,q} > 0$, depending only on those fixed data and j,q, such that for all $R \geq R_0$, $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$, and $y \geq R$,*

$$F_q = a_*\lambda^{-q}, \qquad F_{q,0} = a_*\lambda_0^{-q}$$

are smooth on $(R/2,\infty)$ and satisfy

$$|\partial_y^j F_q(y)| + |\partial_y^j F_{q,0}(y)| \leq C_{j,q}y^{-q/p-j}, \qquad |\partial_y^j(F_q - F_{q,0})(y)| \leq C_{j,q}\epsilon y^{-q/p-p-1-j}.$$

The thresholds ϵ_0, R_0 are chosen before j,q; the constants are independent of R,ϵ,μ,y.

Proof. Use $\lambda = \lambda_0(1+\widetilde{\lambda})$ with $|\widetilde{\lambda}| \leq 1/2$ and $|\partial^k\widetilde{\lambda}| \leq C_k\epsilon y^{-p-1-k}$. The smooth functions $(1+t)^{-q}$ and $[(1+t)^{-q}-1]/t$, the latter extended at zero, have bounded derivatives of every fixed order on $[-1/2,1/2]$. Apply the finite chain-rule expansion to obtain $|\partial^k[(1+\widetilde{\lambda})^{-q} - 1]| \leq C_{k,q}\epsilon y^{-p-1-k}$: every term contains at least one derivative of $\widetilde{\lambda}$, and additional factors ϵy^{-p-1} are bounded by one after a fixed threshold choice. Multiply by $a_*y^{-q/p}$ and use the Leibniz formula and the symbol bounds on a_*. Smoothness follows from the positive frequency and the smooth positive seed. This is the place to reuse an all-order product/composition API for symbol bounds.

□

Lemma 4.44 (Interpolated phases and exponential differences)**.** *Fix $\frac{3}{4} < p < 1$ and $0 < \delta \le |\mu_*|/2$. There are $\epsilon_0 \in (0,1]$ and $R_0 \ge 2$, depending only on p, δ and the fixed cutoffs, such that for every $j \in \mathbb{N}$ there is $C_j > 0$ with the following bounds, uniformly for $R \ge R_0$, $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, and $y \ge R$. Put $d = 1/p - p > 0$, $\Delta\Psi = \Psi - \Psi_0$, and $\Psi_s = \Psi_0 + s\Delta\Psi$ for $s \in [0,1]$. Then*

$$\sup_{0\le s\le 1} |\partial_y^j e^{i\Psi_s(y)}| \le C_j y^{j/p}, \qquad |\partial_y^j (e^{i\Psi(y)} - e^{i\Psi_0(y)})| \le C_j \epsilon y^{d+j/p}.$$

These functions are smooth in $y \in (R/2, \infty)$, and on that interval

$$e^{i\Psi(y)} - e^{i\Psi_0(y)} = i \int_0^1 \Delta\Psi(y) e^{i\Psi_s(y)} ds.$$

Every y-derivative can be passed through this integral on $[0,1]$; on compact y-intervals its derivative integrands have a common finite majorant independent of s. All constants in the displayed bounds are independent of s, R, ϵ, μ, y.

Proof. The derivative $\Psi_s' = \lambda_0 + s(\lambda - \lambda_0)$ has uniform symbol bounds $|\partial_y^k \Psi_s'| \le C_k y^{1/p-k}$. Induct on j in $\partial_y e^{i\Psi_s} = i\Psi_s' e^{i\Psi_s}$ to prove the first bound, retaining the finite polynomial in phase derivatives. The identity follows from the fundamental theorem of calculus in s. Joint smoothness on a neighbourhood of each compact subset of $(R/2,\infty) \times [0,1]$ permits differentiation under that proper integral. Use Leibniz and $|\partial_y^k \Delta\Psi| \le C_k \epsilon y^{d-k}$: the term with k derivatives on $\Delta\Psi$ has exponent $d - k + (j-k)/p \le d + j/p$ because $y \ge R \ge 2$. □

Lemma 4.45 (The two product terms in the amplitude-phase difference)**.** *Fix $\frac{3}{4} < p < 1$ and $0 < \delta \le |\mu_*|/2$. There are thresholds $\epsilon_0 \in (0,1]$, $R_0 \ge 2$, chosen before derivative orders and powers, such that for every $q > 0$, $j \in \mathbb{N}$ there is $C_{j,q} > 0$ satisfying the following for every $R \ge R_0$, $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, and $y \ge R$. With $F_q = a_* \lambda^{-q}$, $F_{q,0} = a_* \lambda_0^{-q}$, put*

$$B_q = \mu(F_q - F_{q,0}) e^{i\Psi}, \qquad P_q = \mu F_{q,0}(e^{i\Psi} - e^{i\Psi_0}).$$

Both terms are smooth on $(R/2,\infty)$ and

$$|\partial_y^j B_q(y)| \le C_{j,q} \epsilon y^{-q/p-p-1+j/p}, \qquad |\partial_y^j P_q(y)| \le C_{j,q} \epsilon y^{1/p-p-q/p+j/p}.$$

These constants depend only on p, δ, j, q and the fixed cutoffs. Defining $D_q = \mu a_* [\lambda^{-q} e^{i\Psi} - \lambda_0^{-q} e^{i\Psi_0}]$, one has $D_q = B_q + P_q$ and, on $(R/2,\infty)$,

$$Z_{\epsilon,\mu} - \frac{\mu}{\mu_*} Z_0 = e^{-i\pi/4} D_{(p+1)/2}.$$

Proof. Take the minimum of the two epsilon thresholds and the maximum of the two radius thresholds from the preceding lemmas. Bound $|\mu|$ by $|\mu_*| + \delta$. In the Leibniz sum for B_q, the term with k derivatives on its amplitude has exponent $-q/p - p - 1 - k + (j-k)/p \le -q/p - p - 1 + j/p$. For P_q, the corresponding exponent is $-q/p - k + (1/p - p) + (j-k)/p \le 1/p - p - q/p + j/p$. Sum the finite binomial coefficients. The first identity is algebraic; the second uses $\lambda_{0,\mu_*} = \lambda_0$, $\Psi_{0,\mu_*} = \Psi_0$, and $\mu_* \ne 0$. □

Lemma 4.46 (Amplitude-phase differences)**.** *For $q > 0$ let $D_q := \mu a_* \big[\lambda^{-q} e^{i\Psi} - \lambda_0^{-q} e^{i\Psi_0}\big]$. For $j \ge 0$ there are admissible $C_{j,q}$ with $|\partial_y^j D_q(y)| \le C_{j,q}\epsilon y^{\frac{1}{p}-p-\frac{q}{p}+\frac{j}{p}}$ for $y \ge R$. Moreover $|\partial_y^j \partial_y[a_*\lambda^{-q} - a_*\lambda_0^{-q}]| \le C\epsilon y^{-q/p-p-2-j}$. In particular ($q = \frac{p+1}{2}$, $q/p = \beta$)*

$$\Big|\partial_y^j\Big(Z_{\epsilon,\mu} - \frac{\mu}{\mu_*}Z_0\Big)(y)\Big| \le C_j\epsilon y^{\frac{1}{p}-p-\beta+\frac{j}{p}}. \tag{4.15}$$

Proof. Use common thresholds obtained by a finite minimum in ϵ_0 and maximum in R_0. By Lemma 4.45, $D_q = B_q + P_q$ with both summands smooth. Differentiate the sum and add their bounds. The exponent of B_q is smaller than the stated exponent by $1 + 1/p$, so $y \ge R \ge 2$ gives the first estimate after enlarging the constant. The second estimate is the reciprocal-amplitude difference bound of Lemma 4.43 at order $j + 1$. For the last assertion take $q = (p + 1)/2$ in the exact identity for Z from Lemma 4.45; multiplication by $e^{-i\pi/4}$ preserves norms, and $q/p = \beta$. □

Lemma 4.47 (Primitive of Z by integration by parts)**.** *For $y \ge R$, with $q_1 = \frac{p+3}{2}$ (so $q_1/p = \beta + \frac{1}{p}$),*

$$\int_y^\infty Z(t)dt = \frac{iZ(y)}{\lambda(y)} + i\mu e^{-i\pi/4}\int_y^\infty \partial_t\big(a_*\lambda^{-q_1}\big)(t)e^{i\Psi(t)}dt,$$

the last integral converging absolutely. Hence, with admissible C: (a) $|\int_y^\infty Z| \le Cy^{-\beta-1/p}$; (b) $|\int_y^\infty (Z - \frac{\mu}{\mu_*}Z_0)| \le C\epsilon y^{-p-\beta}$; (c) $|c_{\mathrm{cubic}}\int_y^\infty Z^3| \le C\epsilon y^{1-3\beta} \le C\epsilon \min\{y^{-p-\beta}, y^{-\beta-1/p}\}$.

Proof. On $[y, M]$ write $Z = \mu e^{-i\pi/4}(a_*\lambda^{-(p+1)/2})\frac{\partial_t e^{i\Psi}}{i\lambda}$ and integrate by parts; the boundary term at M is $O(M^{-\beta-1/p}) \to 0$ and $|\partial_t(a_*\lambda^{-q_1})| \le Ct^{-\beta-1/p-1}$ (Lemma 4.40), integrable. Since $-1/i = i$, the formula follows. (a): $|Z/\lambda| \le Cy^{-\beta-1/p}$ and $\int_y^\infty t^{-\beta-1/p-1}dt \le Cy^{-\beta-1/p}$. (b): apply the formula to Z and to $\frac{\mu}{\mu_*}Z_0$ (the latter is the formula at $\epsilon = 0$ multiplied by μ/μ_*) and subtract: the boundary terms give $e^{-i\pi/4}iD_{q_1}(y)$, bounded by $C\epsilon y^{1/p-p-\beta-1/p} = C\epsilon y^{-p-\beta}$ (Lemma 4.46); the integrands differ by $\partial_t(a_*\lambda^{-q_1})(e^{i\Psi} - e^{i\Psi_0}) + \partial_t(a_*\lambda^{-q_1} - a_*\lambda_0^{-q_1})e^{i\Psi_0}$, bounded by $Ct^{-\beta-1/p-1}\cdot\epsilon t^{1/p-p} + C\epsilon t^{-\beta-1/p-p-2} \le C\epsilon t^{-p-\beta-1}$; integrate. (c): $|c_{\mathrm{cubic}}| \le C\epsilon$, $|Z|^3 \le Ct^{-3\beta}$, $3\beta > 1$; and $1 - 3\beta \le -p - \beta \iff p + 1 \le 2\beta$ (true, $2\beta = 1 + 1/p$), $1 - 3\beta = -\beta - 1/p$.

□

Lemma 4.48 (Approximate phase differences)**.** *For every $j \ge 0$ there is an admissible C_j such that for $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, $R \ge R_0$ and $y \ge R$,* (4.14), (4.13), (4.15) *hold, and*

$$|\partial_y^j f_{\mathrm{osc}}(y)| \le C_j \min\big\{\epsilon\langle y\rangle^{-p-\beta+\frac{j}{p}},\ \langle y\rangle^{-\beta+\frac{j-1}{p}}\big\}. \tag{4.16}$$

($f_{\mathrm{osc}} = 0$ on $(-\infty, R]$.) Consequently

$$\|f_{\mathrm{osc}}\|_{L^1(\mathbb{R})} \le C\epsilon R^{1-p-\beta} \le C\epsilon. \tag{4.17}$$

Proof. The first three are Lemmas 4.42 and 4.46. For $y \ge R \ge 2$, $y \le \langle y\rangle \le \sqrt{2}y$, so it suffices to prove the bounds with y. $j=0$: $|f_{\rm osc}| \le 2|\int_y^\infty (Z - \frac{\mu}{\mu_*}Z_0)| + 2|c_{\rm cubic}\int_y^\infty Z^3| \le C\epsilon y^{-p-\beta}$ by Lemma 4.47(b),(c); and $|f_{\rm osc}| \le 2|\int Z| + 2|\frac{\mu}{\mu_*}||\int Z_0| + 2|c_{\rm cubic}\int Z^3| \le Cy^{-\beta-1/p}$ by (a) (also for Z_0), (c). $j \ge 1$: by Leibniz, $\partial^j f_{\rm osc} = -2\sum_k \binom{j}{k}\partial^{j-k}\chi_R\,\partial^k \operatorname{Re} F$ with $F(y) = \int_y^\infty (Z + c_{\rm cubic}Z^3 - \frac{\mu}{\mu_*}Z_0)$ and $\partial^k F = -\partial^{k-1}(Z + c_{\rm cubic}Z^3 - \frac{\mu}{\mu_*}Z_0)$ for $k \ge 1$. By (4.15) and Lemma 4.40, $|\partial^{k-1}(Z - \frac{\mu}{\mu_*}Z_0)| \le C\epsilon y^{-p-\beta+k/p}$, $|c_{\rm cubic}\partial^{k-1}Z^3| \le C\epsilon y^{-3\beta+(k-1)/p} \le C\epsilon y^{-p-\beta+k/p}$ (as $2\beta+1/p \ge p$), and also $|\partial^{k-1}(\cdots)| \le Cy^{-\beta+(k-1)/p}$. The factors $\partial^{j-k}\chi_R$, $j-k \ge 1$, are supported in $[R,2R]$ where $R^{-(j-k)} \le 2^{j-k}y^{-(j-k)} \le 2^{j-k}y^{(j-k)/p}$; combine with the k-th term bounds (the $k=0$ term uses the $j=0$ bounds). Finally $\int_R^\infty C\epsilon y^{-p-\beta}dy = \frac{C\epsilon}{p+\beta-1}R^{1-p-\beta}$, and $p+\beta > 1$, $R \ge 1$. □

Proposition 4.49 (Profile bounds)**.** *For every $j \ge 0$ there is an admissible C_j such that for all $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$, $R \ge R_0$:*

$$|\overline{f}_{\epsilon,\mu}(y)| \le C_0\langle y\rangle^{-p}, \qquad |\partial^{j+1}\overline{f}_{\epsilon,\mu}(y)| \le C_j\langle y\rangle^{-\beta+\frac{j}{p}} \qquad (y \ge 0),$$

$$|\partial^j(\overline{f}_{\epsilon,\mu}(y) + y)| \le C_j\langle y\rangle^{-p-j} \qquad (y \le 0).$$

Moreover, for each ϵ and $k \ge 0$, $\mu \mapsto \overline{f}_{\epsilon,\mu}$ is continuous from $\overline{B}$ into $C^k(K)$ for every compact $K \subset \mathbb{R}$ (in particular into $C^2_{\rm loc}$). $\overline{f}_{\epsilon,\mu}$ is real-valued and C^∞.

Proof. Smoothness/reality: Lemma 4.41. Bounds for $f_{\rm slow}$ (all y): since $\chi_R(\pm y)$ is supported in $|y| \ge R \ge 2$, and $|\partial^l\chi_R| \le CR^{-l} \le C2^l y^{-l}$ on its support (where $|y| \le 2R$ when $l \ge 1$), $|\partial^j f_{\rm slow}| \le C_j\epsilon\langle y\rangle^{-p-j}$. *Case $y \ge 0$.* Using $f_0 = f_{0,\rm low} + 2\operatorname{Re} f_{0,+}$: $\overline{f} = f_{0,\rm low} + 2\operatorname{Re}[\frac{\mu}{\mu_*}f_{0,+}] + f_{\rm slow} + f_{\rm osc}$. Here $W = 0$ so $|\partial^j f_{0,\rm low}| \le C\langle y\rangle^{-p-j}$ (Lemma 4.33(2) for $y \ge 1$, continuity on $[0,1]$), and $\langle y\rangle^{-p-1-j} \le \langle y\rangle^{-\beta+j/p}$ ($\beta \le p+1$). For $y \ge R_0$: $f_{0,+}(y) = -\int_y^\infty Z_0$, so $|f_{0,+}| \le Cy^{-\beta-1/p} \le Cy^{-p}$ (Lemma 4.47(a) at $\epsilon = 0$ with $R = R_0$; Z_0 does not depend on R), and $|\partial^{j+1}f_{0,+}| \le Cy^{-\beta+j/p}$ (Lemma 4.27); on $[0,R_0]$ use continuity. $|\mu/\mu_*| \le 2$. $f_{\rm osc}$: (4.16), second entry ($j=0$: $\langle y\rangle^{-\beta-1/p} \le \langle y\rangle^{-p}$). *Case $y \le 0$.* $f_{\rm osc} = 0$ and $\overline{f} + y = (f_0 + y) + 2\operatorname{Re}[(\frac{\mu}{\mu_*} - 1)f_{0,+}] + f_{\rm slow}$; use Lemma 4.33(2) for $y \le -1$ and smoothness on $[-1,0]$, Lemma 4.26 (with $N = p+j$) and continuity on $[-1,0]$, and the $f_{\rm slow}$ bound. Constants: none depends on ϵ, μ, R. *Continuity in μ.* $f_{0,\mu}$ is $\mathbb{R}$-affine in μ with fixed smooth coefficients; $f_{\rm slow}$ is independent of μ. In $f_{\rm osc}$, μ enters through μ, μ/μ_* and $c_{\log}(\mu)$, so $(\mu,t) \mapsto Z_{\epsilon,\mu}(t)$ and all its t-derivatives are jointly continuous on $\overline{B} \times (R/2,\infty)$. On a compact K, $\partial^k f_{\rm osc}$ is a finite combination of $\partial^l\chi_R$ times either $\int_y^\infty(\cdots)$ (continuous in μ uniformly in $y \in K$ by dominated convergence, majorant $Ct^{-\beta}$, Lemma 4.40) or t-derivatives of the integrand (uniformly continuous on $\overline{B} \times K$). □

4.4 Decomposition of the residual for $y \ge 8R$

Lemma 4.50 ($\mathcal{Q}_\epsilon$ at the corner)**.** *For $\epsilon \ge 0$ and $y > 0$, the integral $\mathcal{Q}_\epsilon(W)(y) := c_\alpha\int_{\mathbb{R}} \frac{W'(y)-W'(z)}{|y-z|^\alpha}\big(A_\epsilon(S_W(y,z)) - 1\big)dz$ converges absolutely and equals $c_\epsilon y^{-p}$.*

Proof. For $z \ge 0$, $W'(y) = W'(z) = 0$. For $z < 0$, $W'(y) - W'(z) = 1$ and $S_W(y,z) = \frac{0-(-z)}{y-z} = \frac{z}{y-z} \in (-1,0)$, so the integrand is bounded by $(y-z)^{-\alpha}$, integrable on $(-\infty, 0)$. Substituting $t = -z/(y-z)$ ($y - z = y/(1-t)$, $dz = -y(1-t)^{-2}dt$) gives $c_\alpha y^{-p} \int_0^1 [A_\epsilon(-t) - 1](1-t)^{p-1}dt = c_\epsilon y^{-p}$, as in the proof of Lemma 3.12.

□

Definition 4.51 (Residual decomposition)**.** Fix $\chi \in C_c^\infty((\frac{1}{4}, 4); [0,1])$ with $\chi = 1$ on $[\frac{1}{2}, 3]$. For $Y \ge 8R$ let $\chi_Y(z) := \chi(z/Y)$ (so $|\partial^j \chi_Y| \le \|\chi^{(j)}\|_\infty Y^{-j}$) and

$$v_Y(z) = -2\chi_Y(z) \operatorname{Re} \int_z^\infty \big(Z(t) + c_{\text{cubic}} Z(t)^3\big) dt \quad (z > R), \qquad v_Y(z) = 0 \ (z \le R). \tag{4.18}$$

$\mathcal{Q}_\epsilon$ is Definition 1.23 and $\mathcal{Q}_{\epsilon,3}$ is (4.3). For $h \in \mathcal{A}$ and $y > 0$ define (the Gateaux derivative of $\mathcal{Q}_\epsilon$ at the corner W, written with $W'(y) - W'(z) = \mathbf{1}_{z<0}$, $A_\epsilon(S_W) - 1 = 0$ for $z \ge 0$)

$$D\mathcal{Q}_\epsilon(W)[h](y) := c_\alpha \int_{-\infty}^0 \Big[(h'(y) - h'(z)) \Big(A_\epsilon\Big(\frac{z}{y-z}\Big) - 1 \Big) + A_\epsilon'\Big(\frac{z}{y-z}\Big) S_h(y,z) \Big] \frac{dz}{(y-z)^\alpha},$$

absolutely convergent ($|h'|, |S_h| \le \|h'\|_\infty$, $(y-z)^{-\alpha} \in L^1(-\infty, 0)$). With $f = \overline{f}_{\epsilon,\mu}$, for $y > R$ set

$$\mathcal{E}_1 = \mathcal{L}_0(f_{\text{slow}}) + \mathcal{Q}_\epsilon(W), \tag{4.19}$$

$$\mathcal{E}_2 = 2\Big(\lambda^p - y - p\lambda_0^{p-1} \frac{c_{\log}(\mu)}{y} + c_\epsilon y^{-p} \Big) \operatorname{Re} Z, \tag{4.20}$$

$$\mathcal{E}_3 = 2\Big(p\lambda_0^{p-1} \frac{c_{\log}(\mu)}{y} + n_\epsilon \lambda^p |Z|^2 \Big) \operatorname{Re} Z, \tag{4.21}$$

$$\mathcal{E}_4 = 2c_{\text{cubic}} \big(\lambda^p - y + c_\epsilon y^{-p} \big) \operatorname{Re}(Z^3), \tag{4.22}$$

$$\mathcal{E}_5^Y = \mathcal{L}_0(f_{\text{osc}}) + \mathcal{Q}_{\epsilon,3}(v_Y) - 2\big(\lambda^p - y + n_\epsilon \lambda^p |Z|^2 \big) \operatorname{Re} Z - 2c_{\text{cubic}} (\lambda^p - y) \operatorname{Re}(Z^3), \tag{4.23}$$

$$\mathcal{E}_6 = D\mathcal{Q}_\epsilon(W)[f - W] - 2c_\epsilon y^{-p} \operatorname{Re}\big(Z + c_{\text{cubic}} Z^3 \big), \tag{4.24}$$

$$\mathcal{E}_7^Y = \mathcal{Q}_{\epsilon,3}(\chi_Y f) - \mathcal{Q}_{\epsilon,3}(v_Y), \tag{4.25}$$

$$\mathcal{E}_8^Y = \mathcal{Q}_\epsilon(\chi_Y f) - \mathcal{Q}_{\epsilon,3}(\chi_Y f), \tag{4.26}$$

$$\mathcal{E}_9^Y = \mathcal{Q}_\epsilon(f) - \mathcal{Q}_\epsilon(W) - D\mathcal{Q}_\epsilon(W)[f - W] - \mathcal{Q}_\epsilon(\chi_Y f). \tag{4.27}$$

(Here $f - W$ is used only through $(f-W)'$ and S_{f-W}, which are bounded since f' is bounded; $f - W \notin C^1$ at 0 but $D\mathcal{Q}_\epsilon(W)[f-W](y)$ only involves $z < 0 < y$, where the formula makes sense with $(f-W)'(y) = f'(y)$, $(f-W)'(z) = f'(z) + 1$.) We write $\mathcal{E}_k$ for $\mathcal{E}_k^Y$ when Y is fixed.

Lemma 4.52 (Convergence needed for the residual decomposition)**.** *Fix $\frac{3}{4} < p < 1$ and $0 < \delta \le |\mu_*|/2$. There are $\epsilon_0 \in (0,1]$ and $R_0 \ge 2$ such that, for every $R \ge R_0$, $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}(\mu_*, \delta)$, $Y \ge 8R$, and $y > R$, f_{slow} and f_{osc} are differentiable on $\mathbb{R}$ and the following kernels are in $L^1(dz)$ (write $f = \overline{f}_{\epsilon,\mu}$):*

1. $\dfrac{u'(y) - u'(z)}{|y-z|^{p+1}}$ *for* $u \in \{f_{0,\mu}, f_{\text{slow}}, f_{\text{osc}}\}$*;*

2. $\dfrac{u'(y)-u'(z)}{|y-z|^{p+1}}[A_\epsilon(S_u(y,z))-1]$ *for* $u \in \{f, \chi_Y f, W\}$;

3. $\dfrac{v'(y)-v'(z)}{|y-z|^{p+1}} S_v(y,z)^2$ *for* $v \in \{v_Y, \chi_Y f\}$;

4. *on* $(-\infty, 0)$, *with* $h = f - W$,

$$\frac{(h'(y)-h'(z))[A_\epsilon(z/(y-z))-1] + A'_\epsilon(z/(y-z))S_h(y,z)}{(y-z)^{p+1}}.$$

The choices of ϵ_0, R_0 *are made before* R, ϵ, μ, Y, y.

Proof. The seed and family have bounded first derivatives and locally Lipschitz first derivatives; the same is true for the slow and oscillatory pieces. The functions $\chi_Y f, v_Y$ are smooth with compact support. For each such profile the difference of first derivatives is $O(|y-z|)$ locally, and the secant is bounded. Thus the first three kernel types are dominated near $z = y$ by $C|y-z|^{-p}$ and at infinity by $C|y-z|^{-p-1}$. For W, use Lemma 4.50. The last kernel only involves $z < 0 < y$: h' and S_h are bounded, and $(y-z)^{-p-1}$ is integrable on $(-\infty, 0)$. The family lemmas supply common thresholds after taking a minimum in ϵ_0 and a maximum in R_0. This node records the analytic convergence obligations explicitly.

□

Lemma 4.53 (Linear reduction of the residual)**.** *Let* $p, R, \epsilon, y \in \mathbb{R}$ *and* $\mu \in \mathbb{C}$. *Suppose that* $f_{0,\mu}, f_{\mathrm{slow}}, f_{\mathrm{osc}}$ *are differentiable on* $\mathbb{R}$, *that* $z \mapsto (u'(y)-u'(z))|y-z|^{-p-1}$ *is integrable for each of these three functions, and that* $\mathcal{L}_0 f_{0,\mu}(y) = G_\mu(y)$. *Then, with* $f = f_{0,\mu} + f_{\mathrm{slow}} + f_{\mathrm{osc}}$,

$$\widetilde{\mathcal{E}}_\epsilon(f)(y) - G_\mu(y) = \mathcal{L}_0 f_{\mathrm{slow}}(y) + \mathcal{L}_0 f_{\mathrm{osc}}(y) + \mathcal{Q}_\epsilon(f)(y).$$

Proof. Differentiate the sum. Absolute convergence permits addition of the three fractional derivative integrals. Expand $\mathcal{L}_0$ and $\widetilde{\mathcal{E}}_\epsilon(f) = \mathcal{L}_0 f + \mathcal{Q}_\epsilon(f)$, then cancel $\mathcal{L}_0 f_{0,\mu} = G_\mu$. □

Lemma 4.54 (Algebraic cancellation of the nine residual pieces)**.** *For all* $p, R, \epsilon, Y, y \in \mathbb{R}$ *and* $\mu \in \mathbb{C}$, *the expressions in Definition 4.51 satisfy the algebraic identity*

$$\mathcal{L}_0 f_{\mathrm{slow}}(y) + \mathcal{L}_0 f_{\mathrm{osc}}(y) + \mathcal{Q}_\epsilon(f)(y) = \sum_{k=1}^{9} \mathcal{E}_k^Y(y).$$

Here the expressions are the total Lean definitions; convergence in the analytic parameter range is supplied separately by Lemma 4.52.

Proof. Expand the nine definitions and use $\mathrm{Re}(Z + c_{\mathrm{cubic}} Z^3) = \mathrm{Re}\, Z + c_{\mathrm{cubic}}\, \mathrm{Re}(Z^3)$. The λ^p, y, $c_{\log}$, n_ϵ, and c_ϵ terms cancel. The remaining terms $\mathcal{Q}_\epsilon(W)$, $D\mathcal{Q}_\epsilon(W)[f-W]$, $\mathcal{Q}_{\epsilon,3}(v_Y)$, and $\mathcal{Q}_{\epsilon,3}(\chi_Y f)$ cancel pairwise. This is a polynomial identity in the values of the named expressions. □

Lemma 4.55 (Residual identity)**.** *For* $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, $Y \geq 8R$ *and every* $y > R$, *all terms in Definition 4.51 are finite and*

$$\widetilde{\mathcal{E}}_\epsilon(f)(y) - G_\mu(y) = \mathcal{L}_0(f_{\mathrm{slow}})(y) + \mathcal{L}_0(f_{\mathrm{osc}})(y) + \mathcal{Q}_\epsilon(f)(y) = \sum_{k=1}^{9} \mathcal{E}_k^Y(y). \quad (4.28)$$

Proof. Choose the common thresholds from Lemma 4.52; all integral expressions then converge absolutely. The seed is smooth and $\mathcal{L}_0 f_{0,\mu} = G_\mu$ by Lemma 4.33(3), so the first equality is Lemma 4.53. The second equality is exactly Lemma 4.54. No additional parameter threshold is needed.

□

Lemma 4.56 (C^1 regularity of the residual terms near the window $Y \le y \le 2Y$)**.** *Let $\frac{3}{4} < p < 1$. For every $\delta \in (0, |\mu_*|/2]$ there are $\epsilon_0 \in (0,1]$ and $R_0 \ge 2$ such that for all $R \ge R_0$, $\epsilon \in [0, \epsilon_0]$ and $\mu \in \overline{B}$:*

1. *$\mathcal{E}_2$, $\mathcal{E}_3$, $\mathcal{E}_4$ are C^1 on (R, ∞);*
2. *for every $Y \ge 8R$: $\mathcal{E}_5^Y$ is C^1 at every $y \in [Y, 2Y]$, and $\mathcal{E}_7^Y$, $\mathcal{E}_8^Y$, $\mathcal{E}_9^Y$ are C^1 on $J_Y = (3Y/4, 5Y/2)$;*
3. *consequently, for every $Y \ge 8R$, $\mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_4, \mathcal{E}_7^Y, \mathcal{E}_8^Y, \mathcal{E}_9^Y$ are C^1 on J_Y and $\mathcal{E}_5^Y$ is C^1 at every point of $[Y, 2Y]$.*

Proof. (1) $\mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_4$ are explicit expressions in y, λ, λ_0, $|Z|^2$ and Z, all C^∞ on (R, ∞) with $\lambda > 0$ there (Lemma 4.39). (2) Take ϵ_0 below the thresholds of Lemma 7.30 and $\min\{1, \epsilon_0^*(\delta)\}$ of the chapter on $\mathcal{E}_6$–$\mathcal{E}_9$, and R_0 above all thresholds. $\mathcal{Q}_{\epsilon,3}(v_Y)$ is C^1 on $\mathbb{R}$ (Lemma 6.25), so Lemma 7.30 applies at each $y \in [Y, 2Y]$; $\mathcal{E}_7^Y$, $\mathcal{E}_8^Y$ are C^1 on $\mathbb{R}$ (Lemma 6.21, Lemma 6.24) and $\mathcal{E}_9^Y$ on J_Y (Lemma 6.33). (3) $J_Y \subset (R, \infty)$ since $3Y/4 \ge 6R$.

□

Proposition 4.57 (Residual bound for $y \ge 8R$)**.** *There is an admissible constant C (it may also depend on R through the constants of Propositions 5.19, 5.30 and 6.15 and Lemmas 6.20, 6.23 and 6.31) such that for all $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, the function $\widetilde{\mathcal{E}}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu$ is C^1 on $(8R, \infty)$ and*

$$|\partial^j(\widetilde{\mathcal{E}}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu)(y)| \le C\epsilon\langle y\rangle^{-2p-1+\frac{j}{p}}, \qquad y \ge 8R,\ j = 0, 1.$$

In particular, for every $\gamma \le 2p+1$ (hence every admissible γ), $|\partial^j(\widetilde{\mathcal{E}}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu)(y)| \le C\epsilon\langle y\rangle^{-\gamma+j/p}$ for $y \ge 8R$, $j = 0, 1$.

Proof.

Fix $y_1 \ge 8R$ and take $Y = y_1$. By Lemma 4.55 the identity (4.28) holds on (R, ∞) for this Y; each $\mathcal{E}_k^Y$ is C^1 near $[Y, 2Y]$ (Lemma 4.56 for $k = 2, 3, 4, 5, 7, 8, 9$; Proposition 4.61 and Proposition 6.15 for $k = 1, 6$) and satisfies at $y = y_1 \in [Y, 2Y]$ (with $\epsilon^2 \le \epsilon$): $\mathcal{E}_1$: $C\epsilon y^{-2p-1-j}$ (Proposition 4.61); $\mathcal{E}_2$: $C\epsilon y^{-2p-1-\beta+j/p}$ (Proposition 4.63); $\mathcal{E}_3$: $C\epsilon(y^{-p-3\beta+j/p} + y^{1-5\beta+j/p})$ (Proposition 4.65); $\mathcal{E}_4$: $C\epsilon y^{1-5\beta+j/p}$ (Proposition 4.66); $\mathcal{E}_5$: $C\epsilon y^{-2p-1+j/p}$ (Proposition 5.30, which combines Proposition 5.19); $\mathcal{E}_6$: $C\epsilon y^{-2p-1+j/p}$ (Proposition 6.15); $\mathcal{E}_7$: $C\epsilon y^{-2p-1+j/p}$ (Lemma 6.20); $\mathcal{E}_8$: $C\epsilon y^{-2p-1+j/p}$ (Lemma 6.23); $\mathcal{E}_9$: $C\epsilon y^{-p-2+j/p}$ (Lemma 6.31). Each exponent is $\le -2p-1+j/p$: $-2p-1-j \le -2p-1+j/p$; $p+3\beta-(2p+1) = (p+1)(\frac{3}{2p}-1) > 0$; $5\beta-1-(2p+1) = \frac{5+p-4p^2}{2p} > 0$ for $p < 1$; $p+2 > 2p+1$. Since $y \ge 8R \ge 16$, $y^{-a} \le 2^{a/2}\langle y\rangle^{-a}$ for $a \ge 0$ and all exponents $-2p-1+j/p$ are negative ($1/p < 2p+1$). Sum the nine bounds. The last claim: $\langle y\rangle^{-2p-1} \le \langle y\rangle^{-\gamma}$ for $\gamma \le 2p+1$.

□

4.5 Control of $\mathcal{E}_1$

Lemma 4.58 ($\mathcal{E}_1$ is a fractional derivative of f_{slow})**.** *For $y \geq 2R$: $\mathcal{E}_1(y) = |D|^p f'_{\mathrm{slow}}(y)$, and the same holds for the first derivatives (both sides are C^1 on (R,∞) and their derivatives agree on $[2R,\infty)$). Moreover $|\partial^k f_{\mathrm{slow}}(y)| \leq C_k \epsilon \langle y\rangle^{-p-k}$ for all $y \in \mathbb{R}$, $k \geq 0$.*

Proof. By Lemma 4.50, $\mathcal{Q}_\epsilon(W) = c_\epsilon y^{-p}$. On $[2R,\infty)$, $f_{\mathrm{slow}} = c_+ y^{-p}$ together with all derivatives ($\chi_R - 1$ vanishes on $[2R,\infty)$), so $f_{\mathrm{slow}} - y f'_{\mathrm{slow}} = (1+p)c_+ y^{-p} = -c_\epsilon y^{-p}$ there, also after one differentiation. C^1: Lemma 4.1 with $g = f'_{\mathrm{slow}}$. The bound on f_{slow} is as in the proof of Proposition 4.49. □

Lemma 4.59 (Near-diagonal part of $|D|^p$ on decaying functions)**.** *Fix $\chi^\flat \in C_c^\infty((-\frac12,\frac12);[0,1])$ with $\chi^\flat = 1$ on $[-\frac14,\frac14]$. Let $k \in \{0,1\}$, $K > 0$, $h \in C^3(\mathbb{R})$ with $|\partial^l h(z)| \leq K\langle z\rangle^{-p-l}$ for $0 \leq l \leq 3$, $z \in \mathbb{R}$. Then for $y \geq 2$,*

$$\Big| c_\alpha \int_{\mathbb{R}} \chi^\flat\Big(\frac{y-z}{y}\Big) \frac{h^{(k+1)}(y) - h^{(k+1)}(z)}{|y-z|^{1+p}} dz \Big| \leq C(p) K y^{-2p-1-k}.$$

Proof. On the support, $|y-z| \leq y/2$, so $z \in [y/2, 3y/2]$ and $|h^{(k+1)}(y) - h^{(k+1)}(z)| \leq |y-z| \sup_{[y/2,3y/2]} |h^{(k+2)}| \leq CK|y-z| y^{-p-2-k}$. Then $\int_{|y-z|\leq y/2} |y-z|^{-p} dz = \frac{2(y/2)^{1-p}}{1-p}$. □

Lemma 4.60 (Far part of $|D|^p$ on decaying functions)**.** *With $\chi^\flat, k, K, h$ as in Lemma 4.59 and $y \geq 2$,*

$$\Big| c_\alpha \int_{\mathbb{R}} \Big(1 - \chi^\flat\Big(\frac{y-z}{y}\Big)\Big) \frac{h^{(k+1)}(y) - h^{(k+1)}(z)}{|y-z|^{1+p}} dz \Big| \leq C(p,\chi^\flat) K y^{-2p-1-k}.$$

Proof. Let $\kappa_y(z) := (1 - \chi^\flat(\frac{y-z}{y}))|y-z|^{-1-p}$, smooth (it vanishes for $|y-z| \leq y/4$). The $h^{(k+1)}(y)$ term: $|h^{(k+1)}(y)| \int_{|y-z|\geq y/4} |y-z|^{-1-p} \leq CK y^{-p-1-k} y^{-p}$. The $h^{(k+1)}(z)$ term: integrate by parts $k+1$ times in z (boundary terms at $\pm\infty$ vanish since $h^{(l)}$ is bounded and $\kappa_y^{(l)}$ decays); $|\partial_z^l \kappa_y(z)| \leq C_l |y-z|^{-1-p-l} \mathbf{1}_{|y-z|\geq y/4}$, because the derivatives falling on $\chi^\flat$ produce factors $y^{-1} \leq \frac12 |y-z|^{-1}$ on $\mathrm{supp}(\chi^\flat)'(\frac{y-\cdot}{y})$. Thus the term is $\leq CK \int_{|y-z|\geq y/4} \langle z\rangle^{-p} |y-z|^{-2-p-k} dz$. Split: $|z| \leq 2y$ gives $\leq (y/4)^{-2-p-k} \int_{|z|\leq 2y} \langle z\rangle^{-p} \leq C y^{-2-p-k} y^{1-p}$; $|z| > 2y$ gives $|y-z| \geq |z|/2$ and $\leq C \int_{|z|>2y} |z|^{-2-2p-k} \leq C y^{-1-2p-k}$. □

Proposition 4.61 (Control of $\mathcal{E}_1$)**.** *There is an admissible C with, for $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$, $y \geq 2R$, $j = 0,1$:*

$$|\partial^j \mathcal{E}_1(y)| \leq C\epsilon \langle y\rangle^{-2p-1-j}, \tag{4.29}$$

hence $|\partial^j \mathcal{E}_1(y)| \leq C\epsilon\langle y\rangle^{-\gamma+j/p}$ for every $\gamma \leq 2p+1$.

Proof. By Lemma 4.58, $\partial^j \mathcal{E}_1 = \partial^j |D|^p f'_{\mathrm{slow}} = |D|^p f^{(j+1)}_{\mathrm{slow}}$ on $[2R,\infty)$ (Lemma 4.1), and $h = f_{\mathrm{slow}}$ satisfies the hypotheses of Lemma 4.59 with $K = C\epsilon$. Split $|D|^p h^{(j+1)}$ with $\chi^\flat + (1-\chi^\flat)$ and apply Lemmas 4.59 and 4.60 with $k = j$; $y \geq 2R \geq 4$, $y \approx \langle y\rangle$. Finally $-2p-1-j \leq -\gamma + j/p$. □

4.6 Control of $\mathcal{E}_2$, $\mathcal{E}_3$, $\mathcal{E}_4$

Lemma 4.62 (Second-order Taylor remainder of λ^p)**.** *Let $g(x) = (1+x)^p - 1 - px$. For $y \geq R$: $\lambda^p - y - p\lambda_0^{p-1}\frac{c_{\log}(\mu)}{y} + c_\epsilon y^{-p} = y\, g(\widetilde{\lambda})$, and $\lambda^p - y + c_\epsilon y^{-p} = pc_{\log}(\mu)y^{-1/p} + y\, g(\widetilde{\lambda})$, with $|y\, g(\widetilde{\lambda})| \leq C\epsilon^2 y^{-2p-1}$, $|\partial_y(y\, g(\widetilde{\lambda}))| \leq C\epsilon^2 y^{-2p-2}$, and $|\partial_y^k(\lambda^p - y + c_\epsilon y^{-p})| \leq C\epsilon y^{-1/p-k}$ ($k = 0, 1$).*

Proof. $\lambda^p = \lambda_0^p(1+\widetilde{\lambda})^p = y(1+\widetilde{\lambda})^p$ and $py\widetilde{\lambda} = -c_\epsilon y^{-p} + pc_{\log}y^{-1/p}$, while $p\lambda_0^{p-1}c_{\log}/y = pc_{\log}y^{(p-1)/p-1} = pc_{\log}y^{-1/p}$. Taylor: for $|x| \leq \frac{1}{2}$, $g(x) = p(p-1)x^2\int_0^1(1-t)(1+tx)^{p-2}dt$, so $|g(x)| \leq Cx^2$, $|g'(x)| = p|(1+x)^{p-1} - 1| \leq C|x|$. With $|\widetilde{\lambda}| \leq C\epsilon y^{-p-1}$, $|\widetilde{\lambda}'| \leq C\epsilon y^{-p-2}$: $|yg| \leq C\epsilon^2 y^{-2p-1}$, $|(yg)'| \leq |g| + y|g'||\widetilde{\lambda}'| \leq C\epsilon^2 y^{-2p-2}$. The last claim uses $|c_{\log}| \leq C\epsilon$ and $2p+1 > 1/p$ (as $p > \frac{1}{2}$), $\epsilon \leq 1$. □

Proposition 4.63 (Control of $\mathcal{E}_2$)**.** *There is an admissible C with $|\partial^j\mathcal{E}_2(y)| \leq C\epsilon^2\langle y\rangle^{-2p-1-\beta+j/p}$ for $y \geq 2R$, $j = 0, 1$.*

Proof. $\mathcal{E}_2 = 2yg(\widetilde{\lambda})\operatorname{Re} Z$ (Lemma 4.62); Leibniz with $|\partial^j Z| \leq Cy^{-\beta+j/p}$ (Lemma 4.40): $j = 1$ gives $C\epsilon^2(y^{-2p-2-\beta} + y^{-2p-1-\beta+1/p})$.

□

Lemma 4.64 (Cancellation in $\mathcal{E}_3$)**.** *For $y \geq R$,*

$$\mathcal{E}_3 = 2n_\epsilon|\mu|^2\Big(\frac{|a_*|^2}{\lambda} - \frac{1}{\lambda_0}\Big)\operatorname{Re} Z, \tag{4.30}$$

and for $k = 0, 1$

$$\Big|\partial_y^k\Big(\frac{|a_*|^2}{\lambda} - \frac{1}{\lambda_0}\Big)\Big| \leq Cy^{-1/p-k}\big(y^{-2\beta} + \epsilon y^{-p-1}\big). \tag{4.31}$$

Proof. $p\lambda_0^{p-1}c_{\log}/y = -n_\epsilon|\mu|^2\lambda_0^{p-1}y^{-1} = -n_\epsilon|\mu|^2/\lambda_0$ (as $\lambda_0^p = y$), and $\lambda^p|Z|^2 = |\mu|^2|a_*|^2\lambda^{-1}$ from (4.7). Next $\frac{|a_*|^2}{\lambda} - \frac{1}{\lambda_0} = \frac{1}{\lambda_0}\cdot\frac{|a_*|^2-1-\widetilde{\lambda}}{1+\widetilde{\lambda}}$, with $|\partial^k(|a_*|^2-1)| \leq Cy^{-2\beta-k}$ (Lemma 4.27, $|a_*|^2 - 1 = 2\operatorname{Re}(a_*-1) + |a_*-1|^2$), $|\partial^k\widetilde{\lambda}| \leq C\epsilon y^{-p-1-k}$, $1+\widetilde{\lambda} \geq \frac{1}{2}$, $|\partial^k\lambda_0^{-1}| \leq Cy^{-1/p-k}$. Leibniz.

□

Proposition 4.65 (Control of $\mathcal{E}_3$)**.** *There is an admissible C with, for $y \geq 2R$, $j = 0, 1$,*

$$|\partial^j\mathcal{E}_3(y)| \leq C\epsilon^2\langle y\rangle^{-p-3\beta+\frac{j}{p}} + C\epsilon\langle y\rangle^{1-5\beta+\frac{j}{p}}. \tag{4.32}$$

Proof. By (4.30), (4.31), $|n_\epsilon| \leq C\epsilon$, $|\mu| \leq 2|\mu_*|$ and $|\partial^j Z| \leq Cy^{-\beta+j/p}$: $|\partial^j\mathcal{E}_3| \leq C\epsilon y^{-3\beta-1/p+j/p} + C\epsilon^2 y^{-\beta-1/p-p-1+j/p}$ (the derivative of the coefficient costs $y^{-1} \leq y^{1/p}$). Now $3\beta + \frac{1}{p} = 5\beta - 1$ and $\beta + \frac{1}{p} + p + 1 = p + 3\beta$ (as $\frac{1}{p} = 2\beta - 1$).

□

Proposition 4.66 (Control of $\mathcal{E}_4$)**.** *There is an admissible C with, for $y \geq 2R$, $j = 0, 1$,*

$$|\partial^j\mathcal{E}_4(y)| \leq C\epsilon^2\langle y\rangle^{1-5\beta+\frac{j}{p}}. \tag{4.33}$$

Proof. $|c_{\text{cubic}}| \leq C\epsilon$, $|\partial^k(\lambda^p - y + c_\epsilon y^{-p})| \leq C\epsilon y^{-1/p-k}$ (Lemma 4.62), $|\partial^j(Z^3)| \leq Cy^{-3\beta+j/p}$; Leibniz gives $C\epsilon^2 y^{-1/p-3\beta+j/p}$ and $\frac{1}{p} + 3\beta = 5\beta - 1$.

□

Chapter 5

Approximate solution II: control of $\mathcal{E}_5$ (paper §5.8)

5.1 Setting and the decomposition of $\mathcal{E}_5$

The term $\mathcal{E}_5$ of the residual decomposition (Definition 4.51) is the error made when the linear operator $\mathcal{L}_0$ and the cubic operator $\mathcal{Q}_{\epsilon,3}$ are applied to the oscillatory correction and replaced by their "frozen-frequency" symbols. We split $\mathcal{E}_5 = \mathcal{R}_{\mathrm{lin}} + \mathcal{R}_{\mathrm{cub}}$ and estimate the two pieces separately (Propositions 5.19 and 5.30).

Definition 5.1 (Standing hypotheses and notation for this chapter)**.** Fix $p \in (3/4, 1)$, $\alpha = 1 + p$, $\beta = \frac{p+1}{2p}$ (so $2\beta = 1 + \frac{1}{p}$). Let $c_\alpha > 0$ be the constant with $|D|^p h(y) = c_\alpha \int_{\mathbb{R}} \frac{h(y)-h(z)}{|y-z|^{1+p}}\, dz$ for bounded C^1 functions h (Definition 1.5); for complex-valued h the same formula is used, so that $|D|^p(\mathrm{Re}\, h) = \mathrm{Re}\, |D|^p h$. For g with g' bounded and C^1 put $\mathcal{L}_0 g := |D|^p g' + g - y g'$ (this is $\mathcal{L}_\epsilon$ of Definition 2.17 at $\epsilon = 0$).

Fix $\delta \in (0, |\mu_*|/2]$, $\mu_* = -i/\sqrt{2\pi p}$ (Definition 4.15), and $\epsilon_0 \in (0, \epsilon_0^*(\delta)]$, where $\epsilon_0^*(\delta) \in (0, 1]$ is the minimum of the smallness thresholds for ϵ provided (for this δ) by Lemma 4.42, Lemma 4.40, Lemma 4.45, Lemma 4.46, Lemma 4.62, Lemma 4.47, Lemma 4.44 and Lemma 4.48 (Lean: `c05_eps0_star`); put $\overline{B} = \overline{B}(\mu_*, \delta)$, the cutoff χ_+ and the profile of χ_R as in Definition 4.38. For every $Y > 0$ let $\chi_Y \in C^\infty(\mathbb{R}; [0,1])$ be the cutoff of Definition 4.51: $\operatorname{supp} \chi_Y \subset (Y/4, 4Y)$, $\chi_Y \equiv 1$ on $[Y/2, 3Y]$. We assume constants K_j ($j \geq 0$), independent of R, Y, with $|\partial^j \chi_R| \leq K_j R^{-j}$ and $|\partial^j \chi_Y| \leq K_j Y^{-j}$.

A *parameter tuple* is (ϵ, μ, R, Y) with $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, $R \geq R_0$, $Y \geq 8R$, where $R_0 \geq 1$ depends only on $(p, \epsilon_0, \delta, \chi_+)$ and is so large that for all $y \geq R_0$: (R1) the bounds of Lemma 4.33(1) hold and $|a_*(y) - 1| \leq \frac{1}{2}$; (R2) writing $\lambda = \lambda_{\epsilon,\mu} = y^{1/p}(1 + \widetilde{\lambda})$ with $\widetilde{\lambda}(y) = -\frac{c_\epsilon}{p} y^{-p-1} + c_{\log}(\mu) y^{-1-1/p}$ one has $|\widetilde{\lambda}(y)| \leq \frac{1}{2}$; (R3) the estimates of Lemma 4.48 hold on $[R, \infty)$.

Notation (all for $y \geq R$): $Z = Z_{\epsilon,\mu}$, $\lambda = \lambda_{\epsilon,\mu}$, $\Psi = \Psi_{\epsilon,\mu}$, $\Phi := \Psi - \pi/4$, $A := \mu a_* \lambda^{-(p+1)/2}$, so $Z = A e^{i\Phi}$; $\lambda_0 = y^{1/p}$, Ψ_0, $\Phi_0 := \Psi_0 - \pi/4$, $A_0 := \mu a_* \lambda_0^{-(p+1)/2}$, so $\frac{\mu}{\mu_*} Z_0 = A_0 e^{i\Phi_0}$. The constants $c_\epsilon, n_\epsilon, c_{\mathrm{cubic}}, c_{\log}(\mu)$, the cubic operator $\mathcal{Q}_{\epsilon,3}$, the function f_{osc}, the local front v_Y and $\mathcal{E}_5$ are those of Definitions 4.38 and 4.51;

in particular $\mathcal{Q}_{\epsilon,3}(2\sin)(x) = 2n_\epsilon \cos x - 2(3^p-1)c_{\mathrm{cubic}}\cos(3x)$ and $v_Y(z) = -2\chi_Y(z)\,\mathrm{Re}\int_z^\infty (Z + c_{\mathrm{cubic}}Z^3)\,dt$.

An *admissible constant* is a number $C \in (0,\infty)$ depending only on $p, c_\alpha, \epsilon_0, \delta, \chi_+, (K_j)_j$ (and, where indicated, on an integer k), but not on the tuple (ϵ, μ, R, Y) nor on the points y, s, t, z.

Lemma 5.2 (Smallness of the correction constants)**.** *There is an admissible C_0 such that for all $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$: $|c_\epsilon| + |n_\epsilon| + |c_{\mathrm{cubic}}| + |c_{\log}(\mu)| \le C_0\epsilon$.*

Proof. $|c_\epsilon| \lesssim \epsilon$ is Lemma 3.12. By the lemma following the definition of $\mathcal{Q}_{\epsilon,3}$ in Chapter 4, $\mathcal{Q}_{\epsilon,3}(\sin) = c_1\cos + c_2\cos(3\cdot)$ with $|c_1|, |c_2| \lesssim \epsilon$. Cubic homogeneity gives $\mathcal{Q}_{\epsilon,3}(2\sin) = 8\mathcal{Q}_{\epsilon,3}(\sin)$, so $n_\epsilon = 4c_1$ and $c_{\mathrm{cubic}} = -4c_2/(3^p-1)$ (note $3^p - 1 > 0$). Finally $|c_{\log}(\mu)| = |n_\epsilon||\mu|^2/p \le |n_\epsilon|(|\mu_*| + \delta)^2/p$. □

Definition 5.3 (Components of $\mathcal{E}_5$)**.** For a parameter tuple define the fundamental and third-harmonic parts of f_{osc},

$$f_{\mathrm{osc},1}(y) = -2\chi_R(y)\,\mathrm{Re}\int_y^\infty \Big(Z - \tfrac{\mu}{\mu_*}Z_0\Big)\,dt, \qquad f_{\mathrm{osc},3}(y) = -2c_{\mathrm{cubic}}\chi_R(y)\,\mathrm{Re}\int_y^\infty Z^3\,dt \tag{5.1}$$

(both $:= 0$ for $y \le R$; the improper integrals converge by Lemma 5.8). For $y > 2R$ define

$$\mathcal{R}_{\mathrm{lin}} := \mathcal{L}_0(f_{\mathrm{osc}}) - 2(\lambda^p - y)\,\mathrm{Re}\,Z - 2c_{\mathrm{cubic}}(3^p\lambda^p - y)\,\mathrm{Re}(Z^3), \tag{5.2}$$

$$\mathcal{R}_{\mathrm{cub}} := \mathcal{Q}_{\epsilon,3}(v_Y) - 2n_\epsilon\lambda^p|Z|^2\,\mathrm{Re}\,Z + 2c_{\mathrm{cubic}}(3^p-1)\lambda^p\,\mathrm{Re}(Z^3), \tag{5.3}$$

and

$$\begin{aligned}\mathcal{R}_{\mathrm{lin},1} &:= |D|^p f'_{\mathrm{osc},1} + f_{\mathrm{osc},1} - 2\lambda^p\,\mathrm{Re}\,Z + 2y\,\mathrm{Re}\Big(\tfrac{\mu}{\mu_*}Z_0\Big),\\ \mathcal{R}_{\mathrm{lin},3} &:= |D|^p f'_{\mathrm{osc},3} + f_{\mathrm{osc},3} - 2c_{\mathrm{cubic}}3^p\lambda^p\,\mathrm{Re}(Z^3).\end{aligned} \tag{5.4}$$

($\mathcal{R}_{\mathrm{cub}}$ depends on Y through v_Y; $\mathcal{R}_{\mathrm{lin}}$ does not.)

Definition 5.4 (Localization, multiplier and primitive remainders)**.** For a parameter tuple, a function $g : \mathbb{R} \to \mathbb{R}$ and $y \in (Y/2, 3Y)$ put

$$\mathcal{R}_{\mathrm{far},Y}[g](y) := c_\alpha \int_{\mathbb{R}} g(z)\,\partial_z\Big[\frac{1-\chi_Y(z)}{|y-z|^{1+p}}\Big]dz. \tag{5.5}$$

Regard $\chi_Y Z$, $\chi_Y\frac{\mu}{\mu_*}Z_0$, $\chi_Y Z^3$ as functions in $C_c^\infty((Y/4, 4Y))$ (extended by 0). For $y \in (Y/2, 3Y)$ define

$$\begin{aligned}\mathcal{E}_{\mathrm{mult}} &:= |D|^p(\chi_Y Z) - \Big[\lambda^p Z + i\frac{(\lambda^p)'}{\lambda}Z - ip\mu a'_*\lambda^{(p-3)/2}e^{i\Phi}\Big],\\ \mathcal{E}_{\mathrm{mult},0} &:= |D|^p\Big(\chi_Y\tfrac{\mu}{\mu_*}Z_0\Big) - \Big[y\tfrac{\mu}{\mu_*}Z_0 + \frac{i}{\lambda_0}\tfrac{\mu}{\mu_*}Z_0 - ip\mu a'_*\lambda_0^{(p-3)/2}e^{i\Phi_0}\Big],\end{aligned}$$

and, for $y \ge R$,

$$\mathcal{R}_{\mathrm{prim}}(y) := 2\,\mathrm{Re}\int_y^\infty \Big[e^{i\Phi(s)}\partial_s\Big(\frac{A(s)}{i\lambda(s)}\Big) - e^{i\Phi_0(s)}\partial_s\Big(\frac{A_0(s)}{i\lambda_0(s)}\Big)\Big]ds. \tag{5.6}$$

(Note $A/(i\lambda) = \mu a_*/(i\lambda^{(p+3)/2})$, matching the paper.)

Definition 5.5 (Frozen plane wave and cubic integrands)**.** For a parameter tuple and $y \geq R$ write $\lambda = \lambda(y)$ and define $P_y(z) := 2\,\mathrm{Re}\big[\frac{Z(y)}{i\lambda} e^{i\lambda(z-y)}\big]$, $z \in \mathbb{R}$. For a function h set $\Delta h(y,s) := h(y) - h(y-s)$; for P_y the increment is always taken at the same base point y that freezes P_y. Put $K_\epsilon := -\frac{(p+1)\epsilon c_\alpha}{2}$ and

$$N_0(y,s) := \Delta v_Y'(\Delta v_Y)^2 - \Delta P_y'(\Delta P_y)^2,$$
$$N_1(y,s) := \big[\Delta v_Y''(\Delta v_Y)^2 - \Delta P_y''(\Delta P_y)^2\big] + 2\big[(\Delta v_Y')^2 \Delta v_Y - (\Delta P_y')^2 \Delta P_y\big].$$

Finally $G(y) := 2n_\epsilon \lambda^p |Z|^2\,\mathrm{Re}\,Z - 2c_{\mathrm{cubic}}(3^p-1)\lambda^p\,\mathrm{Re}(Z^3)$ and $T_P(y) := \partial_z\big[\mathcal{Q}_{\epsilon,3}(P_y)(z)\big]_{z=y}$ (y held fixed inside P_y).

Lemma 5.6 (Algebraic splitting of $\mathcal{E}_5$)**.** *For every parameter tuple and $y > 2R$: $\mathcal{E}_5(y) = \mathcal{R}_{\mathrm{lin}}(y) + \mathcal{R}_{\mathrm{cub}}(y)$ and $\mathcal{R}_{\mathrm{lin}}(y) = \mathcal{R}_{\mathrm{lin},1}(y) + \mathcal{R}_{\mathrm{lin},3}(y)$ (the latter is the splitting (eqn:approx_E5_linear_split) of the paper).*

Proof. Adding (5.2) and (5.3), the $\mathrm{Re}(Z^3)$ coefficients combine to $-2c_{\mathrm{cubic}}(3^p\lambda^p - y) + 2c_{\mathrm{cubic}}(3^p-1)\lambda^p = -2c_{\mathrm{cubic}}(\lambda^p - y)$, and the $\mathrm{Re}\,Z$ coefficients to $-2(\lambda^p - y + n_\epsilon\lambda^p|Z|^2)$, which is the definition of $\mathcal{E}_5$ (eqn:approx_E5). For the second identity: $f_{\mathrm{osc}} = f_{\mathrm{osc},1} + f_{\mathrm{osc},3}$ by Definition 4.38; for $y > 2R$, $\chi_R \equiv 1$ near y, so $f_{\mathrm{osc}}'(y) = 2\,\mathrm{Re}(Z + c_{\mathrm{cubic}}Z^3 - \frac{\mu}{\mu_*}Z_0)$ (fundamental theorem of calculus). Hence $-yf_{\mathrm{osc}}' + 2y\,\mathrm{Re}\,Z + 2c_{\mathrm{cubic}}y\,\mathrm{Re}\,Z^3 = 2y\,\mathrm{Re}(\frac{\mu}{\mu_*}Z_0)$, and $\mathcal{R}_{\mathrm{lin}} = |D|^p f_{\mathrm{osc}}' + f_{\mathrm{osc}} - 2\lambda^p\,\mathrm{Re}\,Z - 2c_{\mathrm{cubic}}3^p\lambda^p\,\mathrm{Re}\,Z^3 + 2y\,\mathrm{Re}(\frac{\mu}{\mu_*}Z_0)$; linearity of $|D|^p$ gives the split.

□

5.2 Symbol bounds and primitives

Lemma 5.7 (Symbol bounds for Z, λ, A)**.** *For every integer $k \geq 0$ there is an admissible C_k such that for every parameter tuple and every $y \geq R$:*

1. $\frac{1}{2}y^{1/p} \leq \lambda(y) \leq 2y^{1/p}$, $|\partial^k\lambda(y)| \leq C_k y^{1/p-k}$, $|\partial^k(\lambda-\lambda_0)(y)| \leq C_k\epsilon y^{1/p-p-1-k}$, $|\partial^k(\Psi-\Psi_0)(y)| \leq C_k\epsilon y^{1/p-p-k}$;
2. $|\partial^k A(y)| + |\partial^k A_0(y)| \leq C_k y^{-\beta-k}$, $|\partial^k(A-A_0)(y)| \leq C_k\epsilon y^{-\beta-p-1-k}$, $|\partial^k a_*'(y)| \leq C_k y^{-2\beta-1-k}$;
3. $|\partial^k Z(y)| + |\partial^k(\frac{\mu}{\mu_*}Z_0)(y)| \leq C_k y^{-\beta+k/p}$, $|\partial^k(Z^3)(y)| \leq C_k y^{-3\beta+k/p}$; *moreover* $Z' - i\lambda Z = A'e^{i\Phi}$, *hence* $|Z'(y) - i\lambda(y)Z(y)| \leq C_0 y^{-\beta-1}$;
4. *for* $k \in \{0,1\}$: $|\partial^k((\lambda^p)'-1)(y)| \leq C\epsilon y^{-p-1-k}$;
5. $|e^{i(\Psi-\Psi_0)(y)} - 1| \leq C\epsilon y^{1/p-p}$;
6. *if* $t \geq R$ *and* $|t-y| \leq y/2$, *then* $\frac{1}{4}\lambda(y) \leq \lambda(t) \leq 4\lambda(y)$ *and* $|\lambda(t)-\lambda(y)| \leq C\lambda(y)|t-y|/y$.

Proof. (1) From (eqn:approx_modified_frequency) and $c_{\mathrm{power}}\frac{1-p^2}{p} = -\frac{c_\epsilon}{p}$ we get $\lambda = y^{1/p}(1+\widetilde{\lambda})$ with $\widetilde{\lambda}$ as in (R2). Since $1 + \frac{1}{p} > p+1$ and by Lemma 5.2, $|\partial^k\widetilde{\lambda}| \leq C_k\epsilon y^{-p-1-k}$, and $|\widetilde{\lambda}| \leq \frac{1}{2}$ by (R2); the bounds on λ and $\lambda - \lambda_0 = y^{1/p}\widetilde{\lambda}$ follow by Leibniz; the Ψ bound is (eqn:approx_phase_difference). (2)

a_*: Lemma 4.33(1) with $|a_*| \le \frac{3}{2}$ (R1). $\lambda^{-(p+1)/2} = y^{-\beta}(1+\widetilde{\lambda})^{-(p+1)/2}$ and $(1+x)^{-(p+1)/2}$ is smooth on $|x| \le \frac{1}{2}$ (Faà di Bruno); $|\mu| \le |\mu_*| + \delta$. For the difference, $A - A_0 = \mu a_* y^{-\beta}[(1+\widetilde{\lambda})^{-(p+1)/2} - 1]$ and $|\partial^k[(1+\widetilde{\lambda})^{-(p+1)/2} - 1]| \le C_k \epsilon y^{-p-1-k}$. (3) $Z = Ae^{i\Phi}$, $\Phi' = \lambda$; $\partial^k e^{i\Phi}$ is $e^{i\Phi}$ times a polynomial in $\lambda, \dots, \lambda^{(k-1)}$ of weighted degree k, bounded by $C_k y^{k/p}$ (as $y^{1/p-1} \le y^{1/p}$); Leibniz. $Z^3 = A^3 e^{3i\Phi}$ likewise. $Z' = (A' + i\lambda A)e^{i\Phi}$. (4) $\lambda^p = y(1+\widetilde{\lambda})^p$, so $(\lambda^p)' - 1 = [(1+\widetilde{\lambda})^p - 1] + py(1+\widetilde{\lambda})^{p-1}\widetilde{\lambda}'$; use (1). (5) $|e^{ix} - 1| \le |x|$ and (1). (6) $\lambda(t)/\lambda(y) = (t/y)^{1/p}(1+\widetilde{\lambda}(t))/(1+\widetilde{\lambda}(y))$ with $t/y \in [\frac{1}{2}, \frac{3}{2}]$, giving the two-sided bound; the mean value theorem with $|\lambda'| \le C_1 \tau^{1/p-1}$, $\tau \in [y/2, 3y/2]$, gives the last bound.

□

Lemma 5.8 (Primitives of Z and Z^3)**.** *For every parameter tuple and $z \ge R$ the improper integrals $\int_z^\infty Z$, $\int_z^\infty Z^3$ converge,*

$$\int_z^\infty Z(t)\,dt = -\frac{Z(z)}{i\lambda(z)} - \int_z^\infty e^{i\Phi(t)}\partial_t\Big(\frac{A(t)}{i\lambda(t)}\Big)dt, \qquad \int_z^\infty Z(t)^3 dt = -\frac{Z(z)^3}{3i\lambda(z)} - \int_z^\infty e^{3i\Phi}\partial_t\Big(\frac{A^3}{3i\lambda}\Big)dt,$$

and there is an admissible C with $|\int_z^\infty Z| \le Cz^{-\beta-1/p}$, $|\int_z^\infty Z^3| \le Cz^{-3\beta-1/p}$. The same holds with (Z, A, Φ, λ) replaced by $(\frac{\mu}{\mu_*}Z_0, A_0, \Phi_0, \lambda_0)$.

Proof. On $[z, M]$ write $Z = \frac{A}{i\lambda}\partial_t e^{i\Phi}$ and integrate by parts: $\int_z^M Z = [\frac{A}{i\lambda}e^{i\Phi}]_z^M - \int_z^M e^{i\Phi}\partial_t(\frac{A}{i\lambda})$. By Lemma 5.7, $|A/\lambda| \le Ct^{-\beta-1/p} \to 0$ and $|\partial_t(A/\lambda)| \le Ct^{-\beta-1/p-1}$, which is integrable; let $M \to \infty$. The bound follows: $Cz^{-\beta-1/p} + C\int_z^\infty t^{-\beta-1/p-1}dt$. For $Z^3 = \frac{A^3}{3i\lambda}\partial_t e^{3i\Phi}$ the same argument with $|A^3/\lambda| \le Ct^{-3\beta-1/p}$ and $|\partial_t(A^3/\lambda)| \le Ct^{-3\beta-1/p-1}$. Identical for the 0-quantities.

□

Lemma 5.9 (Bounds on the two parts of f_{osc})**.** *For every parameter tuple: $f_{\mathrm{osc},1}, f_{\mathrm{osc},3} \in C^\infty(\mathbb{R})$ vanish on $(-\infty, R]$, are bounded with bounded first derivative, $f_{\mathrm{osc}} = f_{\mathrm{osc},1} + f_{\mathrm{osc},3}$, and for $y \ge 2R$, $f'_{\mathrm{osc},1}(y) = 2\,\mathrm{Re}(Z - \frac{\mu}{\mu_*}Z_0)(y)$, $f'_{\mathrm{osc},3}(y) = 2c_{\mathrm{cubic}}\,\mathrm{Re}(Z^3)(y)$. There is an admissible C such that for all $y \ge R$*

$$|f_{\mathrm{osc},3}(y)| \le C\epsilon y^{-3\beta-1/p}, \qquad |f_{\mathrm{osc},1}(y)| \le C\epsilon y^{-p-\beta}, \qquad \|f_{\mathrm{osc},1}\|_{L^1(\mathbb{R})} + \|f_{\mathrm{osc},3}\|_{L^1(\mathbb{R})} \le C\epsilon.$$

Proof. Smoothness: $\int_y^\infty Z = \int_y^{y_1} Z + \int_{y_1}^\infty Z$ is smooth in y; χ_R vanishes on $(-\infty, R]$. The sum identity is (4.9). Derivatives: product rule, with $\chi'_R = 0$ on $[2R, \infty)$. Boundedness of f': $|\chi'_R| \le K_1 R^{-1}$ times the primitive bounds, plus $|Z| + |Z_0| + |Z|^3$. The bound on $f_{\mathrm{osc},3}$ is Lemma 5.8 with $|c_{\mathrm{cubic}}| \le C_0\epsilon$. Then $|f_{\mathrm{osc},1}| \le |f_{\mathrm{osc}}| + |f_{\mathrm{osc},3}| \le C\epsilon\langle y\rangle^{-p-\beta} + C\epsilon y^{-3\beta-1/p}$ by (eqn:approx_fosc_bounds) with $j = 0$; and $3\beta + \frac{1}{p} \ge p + \beta$ because $2\beta + \frac{1}{p} = 1 + \frac{2}{p} > p$. The L^1 bounds follow by integrating over (R, ∞), since $p + \beta > 1$ and $3\beta + \frac{1}{p} > 1$.

□

Lemma 5.10 (Localization of $|D|^p$ away from $[Y/2, 3Y]$)**.** *Let (ϵ, μ, R, Y) be a parameter tuple and $g \in C^\infty(\mathbb{R}; \mathbb{R})$ with g, g' bounded, $g \in L^1(\mathbb{R})$, $g = 0$ on $(-\infty, R]$. Then for every $y \in (Y/2, 3Y)$,*

$$|D|^p g'(y) = |D|^p(\chi_Y g')(y) + \mathcal{R}_{\mathrm{far},Y}[g](y), \tag{5.7}$$

both terms are C^1 on $(Y/2, 3Y)$, and there is an admissible C with $|\partial_y^j \mathcal{R}_{\mathrm{far},Y}[g](y)| \le CY^{-p-2-j}\|g\|_{L^1}$ for $y \in [Y, 2Y]$, $j = 0, 1$.

Proof. By linearity $|D|^p g' = |D|^p(\chi_Y g') + |D|^p((1-\chi_Y)g')$. Fix $y \in (Y/2, 3Y)$; $(1-\chi_Y)g'$ vanishes on a neighbourhood of y, so $|D|^p((1-\chi_Y)g')(y) = -c_\alpha \int (1-\chi_Y(z))g'(z)|y-z|^{-1-p}dz$, absolutely convergent (g' bounded, the integrand vanishes near $z = y$). Integrate by parts on $[-M, M]$: the kernel $k_y(z) = (1-\chi_Y(z))|y-z|^{-1-p}$ is smooth, the boundary terms $g(\pm M)k_y(\pm M) \to 0$ (g bounded, $k_y(z) = O(|z|^{-1-p})$), giving (5.5). For $y \in [Y, 2Y]$ and $z \in \mathrm{supp}(1-\chi_Y)$ one has $|y-z| \ge Y/2$, so $|\partial_z k_y| \le K_1 Y^{-1}(Y/2)^{-1-p} + (1+p)(Y/2)^{-2-p} \le CY^{-p-2}$, and $|\partial_y \partial_z k_y| \le CY^{-p-3}$; differentiation under the integral is justified by these uniform bounds and $g \in L^1$. Regularity of $|D|^p(\chi_Y g')$: $\chi_Y g' \in C_c^\infty$, so $|D|^p(\chi_Y g')$ is C^1 with derivative $|D|^p((\chi_Y g')')$; it equals $|D|^p g' - \mathcal{R}_{\mathrm{far},Y}[g]$ on $(Y/2, 3Y)$. □

5.3 The linear remainder $\mathcal{R}_{\mathrm{lin}}$

Lemma 5.11 (Hypotheses of the oscillatory multiplier lemma)**.** *Let (ϵ, μ, R, Y) be a parameter tuple, $I_Y = (Y/4, 4Y)$, $\Lambda = Y^{1/p}$, $\delta_Y = \epsilon Y^{1/p-p}$. Put $a_1 = \chi_Y A$, $a_0 = \chi_Y A_0$, $a_3 = \chi_Y A^3$ (in $C_c^\infty(I_Y)$). Then, with constants c_*, C_k admissible:*

1. *(a_1, Φ) and (a_0, Φ_0) satisfy the hypotheses of Lemma 3.108 on I_Y with amplitude size $Y^{-\beta}$ and frequency size Λ: $c_*\Lambda \le \Phi' \le c_*^{-1}\Lambda$, $|\partial^k \Phi'| \le C_k \Lambda Y^{-k}$, $|\partial^k a_\nu| \le C_k Y^{-\beta-k}$ (same for Φ_0);*
2. *the difference hypotheses hold with $\delta = \delta_Y$: $|\partial^k(a_1 - a_0)| \le C_k \delta_Y Y^{-\beta} Y^{-k}$, $|\partial^k(\Phi - \Phi_0)| \le C_k \delta_Y Y^{-k}$ on I_Y;*
3. *$(a_3, 3\Phi)$ satisfies the hypotheses with amplitude size $Y^{-3\beta}$ and frequency size Λ.*

Proof. On I_Y one has $y \ge Y/4 \ge 2R \ge R$ and $y \in (Y/4, 4Y)$, so $y^\theta \le C_\theta Y^\theta$ for every real θ (with $C_\theta = \max(4^\theta, 4^{-\theta})$). Apply Lemma 5.7(1)–(2) and Leibniz with $|\partial^k \chi_Y| \le K_k Y^{-k}$. For (2): $\epsilon Y^{-\beta-p-1-k} \le \delta_Y Y^{-\beta-k}$ since $-1 \le \frac{1}{p}$. For (3): $|\partial^k A^3| \le C_k Y^{-3\beta-k}$, $(3\Phi)' = 3\lambda$.

□

Lemma 5.12 (Multiplier expansion for the fundamental harmonic)**.** *For every parameter tuple and $y \in (Y/2, 3Y)$ (with the notation of Definition 5.4),*

$$\mathcal{E}_{\mathrm{mult}} = e^{i\Phi}\widetilde{\mathcal{R}}_{\mathrm{mult},2}[a_1, \Phi], \qquad \mathcal{E}_{\mathrm{mult},0} = e^{i\Phi_0}\widetilde{\mathcal{R}}_{\mathrm{mult},2}[a_0, \Phi_0], \tag{5.8}$$

where $\widetilde{\mathcal{R}}_{\mathrm{mult},2}[a, \Phi]$ is the remainder of (eqn:appendix_oscillatory_multiplier_two_term_expansion). Moreover there is an admissible C such that for $y \in [Y, 2Y]$ and $j = 0, 1$

$$|\partial_y^j(\mathcal{E}_{\mathrm{mult}} - \mathcal{E}_{\mathrm{mult},0})(y)| \le C\epsilon Y^{-p-3\beta+j/p}. \tag{5.9}$$

Proof. Identity. By Lemma 5.11(1) the two-term expansion applies: $|D|^p(a_1 e^{i\Phi}) = e^{i\Phi}[\lambda^p a_1 - ip\lambda^{p-1}a_1' - \frac{ip(p-1)}{2}\lambda^{p-2}\lambda' a_1 + \widetilde{\mathcal{R}}]$. For $y \in (Y/2, 3Y)$, $\chi_Y \equiv 1$ near y, so $a_1 = A$ near y, and $A' = \mu a_*' \lambda^{-(p+1)/2} - \frac{p+1}{2}A\lambda'/\lambda$. Hence $-ip\lambda^{p-1}A' - \frac{ip(p-1)}{2}\lambda^{p-2}\lambda' A = -ip\mu a_*' \lambda^{(p-3)/2} + ip\lambda^{p-2}\lambda' A$ and $ip\lambda^{p-2}\lambda' = i(\lambda^p)'/\lambda$. Multiplying by $e^{i\Phi}$ gives the claimed form. For $\nu = 0$ use $\lambda_0^p = y$, $(\lambda_0^p)' = 1$.

Difference. Write $\widetilde{\mathcal{R}}_\nu = \mathcal{R}_{\mathrm{mult},2,\nu} - \frac{p(p-1)}{2}\lambda_\nu^{p-2}a_\nu''$ ($\lambda_1 := \lambda$) and

$$\mathcal{E}_{\mathrm{mult}} - \mathcal{E}_{\mathrm{mult},0} = e^{i\Phi}(\widetilde{\mathcal{R}}_1 - \widetilde{\mathcal{R}}_0) + (e^{i\Phi} - e^{i\Phi_0})\widetilde{\mathcal{R}}_0.$$

By (eqn:appendix_oscillatory_multiplier_amplitude_difference) with $N = 2$ and Lemma 5.11(2), $|\partial^j(\mathcal{R}_{\mathrm{mult},2,1} - \mathcal{R}_{\mathrm{mult},2,0})| \le C\min\{1,\delta_Y\}Y^{-\beta-2-j}\Lambda^{p-2}$. By Lemma 5.7, $|\partial^j(\lambda^{p-2}a_1'' - \lambda_0^{p-2}a_0'')| \le C\delta_Y Y^{-\beta-2-j}\Lambda^{p-2}$. Also $|\partial^j\widetilde{\mathcal{R}}_0| \le CY^{-\beta-2-j}\Lambda^{p-2}$ (appendix lemma), $|e^{i\Phi} - e^{i\Phi_0}| \le |\Psi - \Psi_0| \le C\delta_Y$ and $|\partial_y(e^{i\Phi} - e^{i\Phi_0})| \le \lambda|e^{i\Phi} - e^{i\Phi_0}| + |\lambda - \lambda_0| \le C\Lambda\delta_Y$, $|\partial_y e^{i\Phi}| \le C\Lambda$. Leibniz (with $Y^{-1} \le \Lambda$) gives $|\partial^j(\cdots)| \le C\delta_Y Y^{-\beta-2}\Lambda^{p-2+j}$, and $\delta_Y Y^{-\beta-2}\Lambda^{p-2+j} = \epsilon Y^{\frac{1}{p}-p-\beta-2+1-\frac{2}{p}+\frac{j}{p}} = \epsilon Y^{-p-3\beta+j/p}$ using $3\beta = \beta + 1 + \frac{1}{p}$.

□

Lemma 5.13 (Integration by parts in $f_{\mathrm{osc},1}$)**.** *For every parameter tuple and $y \ge 2R$,*

$$f_{\mathrm{osc},1}(y) = -2\,\mathrm{Re}\Big[i\Big(\frac{Z}{\lambda} - \frac{\mu}{\mu_*}\frac{Z_0}{\lambda_0}\Big)\Big](y) + \mathcal{R}_{\mathrm{prim}}(y). \tag{5.10}$$

$\mathcal{R}_{\mathrm{prim}} \in C^1([R,\infty))$ with $\mathcal{R}'_{\mathrm{prim}} = -2\,\mathrm{Re}[e^{i\Phi}\partial(\frac{A}{i\lambda}) - e^{i\Phi_0}\partial(\frac{A_0}{i\lambda_0})]$, and there is an admissible C with, for $y \ge R$, $|\mathcal{R}_{\mathrm{prim}}(y)| \le C\epsilon y^{-p-3\beta}$ and $|\mathcal{R}'_{\mathrm{prim}}(y)| \le C\epsilon y^{-p-3\beta+1/p}$.

Proof. Identity. For $y \ge 2R$, $\chi_R(y) = 1$; insert Lemma 5.8 for Z and $\frac{\mu}{\mu_*}Z_0$ into (5.1), using $-\frac{Z}{i\lambda} = \frac{iZ}{\lambda}$.

Derivative. The integrand $F(s) = e^{i\Phi}g - e^{i\Phi_0}g_0$, $g := \partial_s(A/(i\lambda))$, $g_0 := \partial_s(A_0/(i\lambda_0))$, is continuous and absolutely integrable; fundamental theorem of calculus. By Lemma 5.7, $|g| + |g_0| \le Cs^{-\beta-1/p-1}$, $|g - g_0| \le C\epsilon s^{-\beta-1/p-p-2}$ (from $|\partial^k(A - A_0)|$ and $|\partial^k(\lambda^{-1} - \lambda_0^{-1})| \le C\epsilon s^{-1/p-p-1-k}$). Hence $|F(y)| \le |g - g_0| + |g_0||e^{i(\Psi-\Psi_0)} - 1| \le C\epsilon y^{-\beta-1/p-p-2} + C\epsilon y^{-\beta-1-p} \le C\epsilon y^{-p-\beta-1}$, and $-p-\beta-1 = -p-3\beta+\frac{1}{p}$.

Value. Set $b = g/(i\lambda)$, $b_0 = g_0/(i\lambda_0)$, so $e^{i\Phi}g = b\,\partial_s e^{i\Phi}$. Then $|\partial^k b| + |\partial^k b_0| \le Cs^{-\beta-1-2/p-k}$ and $|\partial^k(b - b_0)| \le C\epsilon s^{-p-\beta-2-2/p-k}$ ($k = 0, 1$). Integrating by parts on $[y, M]$ and letting $M \to \infty$ ($b(M) \to 0$): $\int_y^\infty F = -(be^{i\Phi} - b_0e^{i\Phi_0})(y) - \int_y^\infty(b'e^{i\Phi} - b_0'e^{i\Phi_0})$. As in the derivative step, $|be^{i\Phi} - b_0e^{i\Phi_0}| \le |b - b_0| + |b_0||e^{i(\Psi-\Psi_0)} - 1| \le C\epsilon y^{-p-\beta-1-1/p} = C\epsilon y^{-p-3\beta}$, and $|b'e^{i\Phi} - b_0'e^{i\Phi_0}| \le C\epsilon s^{-p-3\beta-1}$, whose integral over $[y,\infty)$ is $\le C\epsilon y^{-p-3\beta}$.

□

Lemma 5.14 (Cancellation identity for $\mathcal{R}_{\mathrm{lin},1}$)**.** *For every parameter tuple and $y \in (Y/2, 3Y)$,*

$$\begin{aligned}\mathcal{R}_{\mathrm{lin},1} = {} & 2\,\mathrm{Re}\Big[i\big((\lambda^p)' - 1\big)\frac{Z}{\lambda} - ip\mu a_*'\big(\lambda^{(p-3)/2}e^{i\Phi} - \lambda_0^{(p-3)/2}e^{i\Phi_0}\big)\Big] \\ & + 2\,\mathrm{Re}(\mathcal{E}_{\mathrm{mult}} - \mathcal{E}_{\mathrm{mult},0}) + \mathcal{R}_{\mathrm{prim}} + \mathcal{R}_{\mathrm{far},Y}[f_{\mathrm{osc},1}].\end{aligned} \tag{5.11}$$

Proof. By Lemma 5.9 $g = f_{\mathrm{osc},1}$ satisfies the hypotheses of Lemma 5.10, and on $\mathrm{supp}\,\chi_Y \subset (Y/4, 4Y) \subset (2R,\infty)$, $\chi_Y f'_{\mathrm{osc},1} = 2\,\mathrm{Re}[\chi_Y Z - \chi_Y\frac{\mu}{\mu_*}Z_0]$. Thus $|D|^p f'_{\mathrm{osc},1} = 2\,\mathrm{Re}[|D|^p(\chi_Y Z) - |D|^p(\chi_Y\frac{\mu}{\mu_*}Z_0)] + \mathcal{R}_{\mathrm{far},Y}[f_{\mathrm{osc},1}]$. Substitute the definitions of $\mathcal{E}_{\mathrm{mult}}, \mathcal{E}_{\mathrm{mult},0}$ and (5.10) into (5.4): the terms $\pm 2\lambda^p\,\mathrm{Re}\,Z$ and

$\pm 2y\,\mathrm{Re}(\frac{\mu}{\mu_*}Z_0)$ cancel, $2\,\mathrm{Re}[i\frac{(\lambda^p)'}{\lambda}Z]-2\,\mathrm{Re}[i\frac{Z}{\lambda}] = 2\,\mathrm{Re}[i((\lambda^p)'-1)\frac{Z}{\lambda}]$, and $-2\,\mathrm{Re}[\frac{i}{\lambda_0}\frac{\mu}{\mu_*}Z_0]+2\,\mathrm{Re}[i\frac{\mu}{\mu_*}\frac{Z_0}{\lambda_0}] = 0$.

□

Lemma 5.15 (Explicit terms in $\mathcal{R}_{\mathrm{lin},1}$)**.** *Let $T_1 = i((\lambda^p)'-1)Z/\lambda$ and $T_2 = -ip\mu a_*'(\lambda^{(p-3)/2}e^{i\Phi} - \lambda_0^{(p-3)/2}e^{i\Phi_0})$. There is an admissible C such that for every parameter tuple, $y \geq R$, $j = 0, 1$: $|\partial^j T_1(y)| + |\partial^j T_2(y)| \leq C\epsilon y^{-p-3\beta+j/p}$.*

Proof. T_1: $|(\lambda^p)'-1| \leq C\epsilon y^{-p-1}$, $|\partial((\lambda^p)'-1)| \leq C\epsilon y^{-p-2}$, $|Z/\lambda| \leq Cy^{-\beta-1/p}$, $|\partial(Z/\lambda)| \leq Cy^{-\beta}$; and $p+1+\beta+\frac{1}{p} = p+3\beta$. T_2: split $\lambda^{(p-3)/2}e^{i\Phi} - \lambda_0^{(p-3)/2}e^{i\Phi_0} = (\lambda^{(p-3)/2} - \lambda_0^{(p-3)/2})e^{i\Phi} + \lambda_0^{(p-3)/2}(e^{i\Phi} - e^{i\Phi_0})$. The first is $\leq C\epsilon y^{(p-3)/(2p)-p-1}$, the second $\leq C\epsilon y^{(p-3)/(2p)+1/p-p}$; with $|a_*'| \leq Cy^{-2\beta-1}$ and $-2\beta-1+\frac{p-3}{2p}+\frac{1}{p}-p = -p-3\beta$ the claim for $j=0$ follows. For $j=1$ each derivative costs at most $Cy^{1/p}$: $\partial_y(e^{i\Phi}-e^{i\Phi_0}) = i\lambda(e^{i\Phi}-e^{i\Phi_0}) + i(\lambda-\lambda_0)e^{i\Phi_0}$, and derivatives of the amplitudes cost $Cy^{-1} \leq Cy^{1/p}$.

□

Lemma 5.16 (Bound on $\mathcal{R}_{\mathrm{lin},1}$)**.** *There is an admissible C such that for every parameter tuple, $y \in [Y, 2Y]$, $j = 0, 1$,*

$$|\partial_y^j \mathcal{R}_{\mathrm{lin},1}(y)| \leq C\epsilon\big(Y^{-p-3\beta+j/p} + Y^{-p-2-j}\big) \leq C\epsilon Y^{-2p-1+j/p}. \tag{5.12}$$

Proof. Differentiate (5.11) term by term (all terms are C^1 on $(Y/2, 3Y)$): Lemma 5.15, (5.9), Lemma 5.13 (with $y \approx Y$), and Lemma 5.10 with $\|f_{\mathrm{osc},1}\|_{L^1} \leq C\epsilon$. The last inequality: $p+3\beta \geq 2p+1$ iff $3\beta \geq p+1$ iff $\frac{3}{2p} \geq 1$; and $p+2+j \geq 2p+1-\frac{j}{p}$ as $p < 1$; and $Y \geq 1$.

□

Lemma 5.17 (Third-harmonic multiplier bound)**.** *There is an admissible C such that for every parameter tuple, $y \in [Y, 2Y]$, $j = 0, 1$, $\big|\partial_y^j\big[|D|^p(\chi_Y Z^3) - 3^p\lambda^p Z^3\big](y)\big| \leq CY^{1-5\beta+j/p}$.*

Proof. Apply (eqn:appendix_oscillatory_multiplier_two_term_expansion) to $(a_3, 3\Phi)$ (Lemma 5.11(3)), frequency 3λ. On $[Y, 2Y]$, $(3\lambda)^p a_3 e^{3i\Phi} = 3^p\lambda^p Z^3$; the difference equals $e^{3i\Phi}[B+\widetilde{\mathcal{R}}]$ where $B = -ip(3\lambda)^{p-1}a_3' - \frac{ip(p-1)}{2}(3\lambda)^{p-2}3\lambda' a_3$ satisfies $|\partial^j B| \leq CY^{-3\beta-1-j}\Lambda^{p-1}$ and $|\partial^j\widetilde{\mathcal{R}}| \leq CY^{-3\beta-2-j}\Lambda^{p-2}$. Since $|\partial_y e^{3i\Phi}| = 3\lambda \leq C\Lambda$, $|\partial^j(\cdot)| \leq CY^{-3\beta-1}\Lambda^{p-1+j}$, and $-3\beta-1+\frac{p-1}{p} = -3\beta-\frac{1}{p} = 1-5\beta$. □

Lemma 5.18 (Bound on $\mathcal{R}_{\mathrm{lin},3}$)**.** *There is an admissible C such that for every parameter tuple, $y \in [Y, 2Y]$, $j = 0, 1$,*

$$|\partial_y^j \mathcal{R}_{\mathrm{lin},3}(y)| \leq C\epsilon\big(Y^{1-5\beta+j/p} + Y^{-p-2-j}\big) \leq C\epsilon Y^{-2p-1+j/p}. \tag{5.13}$$

Proof. By Lemma 5.10 with $g = f_{\mathrm{osc},3}$ and $\chi_Y f_{\mathrm{osc},3}' = 2c_{\mathrm{cubic}}\,\mathrm{Re}(\chi_Y Z^3)$,

$$\mathcal{R}_{\mathrm{lin},3} = 2c_{\mathrm{cubic}}\,\mathrm{Re}\big[|D|^p(\chi_Y Z^3) - 3^p\lambda^p Z^3\big] + f_{\mathrm{osc},3} + \mathcal{R}_{\mathrm{far},Y}[f_{\mathrm{osc},3}]$$

on $[Y, 2Y]$. First term: $|c_{\mathrm{cubic}}| \leq C_0\epsilon$ and Lemma 5.17. Second: $|f_{\mathrm{osc},3}| \leq C\epsilon Y^{-3\beta-1/p} = C\epsilon Y^{1-5\beta}$ and $|f_{\mathrm{osc},3}'| = 2|c_{\mathrm{cubic}}||\,\mathrm{Re}\,Z^3| \leq C\epsilon Y^{-3\beta} = C\epsilon Y^{1-5\beta+1/p}$. Third: $CY^{-p-2-j}\|f_{\mathrm{osc},3}\|_{L^1} \leq C\epsilon Y^{-p-2-j}$. Finally $5\beta-1 \geq 2p+1$ iff $\frac{5}{2p} \geq 2$, true for $p < 1$.

□

Proposition 5.19 (Linear remainder of $\mathcal{E}_5$)**.** *There is an admissible constant C such that for every parameter tuple (ϵ, μ, R, Y) (so $Y \geq 8R$), every $y \in [Y, 2Y]$ and $j \in \{0, 1\}$, $\mathcal{R}_{\mathrm{lin}}$ is C^1 near y and*

$$|\partial_y^j \mathcal{R}_{\mathrm{lin}}(y)| \leq C\epsilon \langle y \rangle^{-2p-1+\frac{j}{p}}. \tag{5.14}$$

Proof. $\mathcal{R}_{\mathrm{lin}} = \mathcal{R}_{\mathrm{lin},1} + \mathcal{R}_{\mathrm{lin},3}$ (Lemma 5.6); add (5.12) and (5.13). Since $Y \geq 8$ and $y \in [Y, 2Y]$, $Y \leq \langle y \rangle \leq 3Y$, so $Y^\theta \leq 3^{|\theta|}\langle y \rangle^\theta$ for $\theta = -2p - 1 + j/p$. □

5.4 The cubic remainder $\mathcal{R}_{\mathbf{cub}}$

Lemma 5.20 (Basic properties of $\mathcal{Q}_{\epsilon,3}$)**.** *Let $v \in C^2(\mathbb{R}; \mathbb{R})$ with v, v', v'' bounded. Then for every y the integral defining $\mathcal{Q}_{\epsilon,3}(v)(y)$ converges absolutely and*

$$\mathcal{Q}_{\epsilon,3}(v)(y) = K_\epsilon \int_{\mathbb{R}} |s|^{-p-3}\, \Delta v'(y,s) \big(\Delta v(y,s)\big)^2 ds. \tag{5.15}$$

If moreover $v \in C^3$ with v''' bounded, then $\mathcal{Q}_{\epsilon,3}(v) \in C^1(\mathbb{R})$ and $\partial_y \mathcal{Q}_{\epsilon,3}(v)(y) = K_\epsilon \int |s|^{-p-3}[\Delta v''(\Delta v)^2 + 2(\Delta v')^2 \Delta v](y,s)\, ds$. Finally, for $c \in \mathbb{R}$, $\kappa > 0$, $x_0 \in \mathbb{R}$ and $g \in C_b^2$, the function $v(z) = c\, g(\kappa z + x_0)$ satisfies $\mathcal{Q}_{\epsilon,3}(v)(z) = c^3 \kappa^{p+3} \mathcal{Q}_{\epsilon,3}(g)(\kappa z + x_0)$.

Proof. Substitute $z = y - s$ in (eqn:approx_cubic_operator); $|y-z|^{-\alpha}(y-z)^{-2} = |s|^{-p-3}$. Domination: $|\Delta v'| \leq \min\{\|v''\|_\infty |s|, 2\|v'\|_\infty\}$, $|\Delta v| \leq \min\{\|v'\|_\infty |s|, 2\|v\|_\infty\}$, so the integrand is $\leq \|v''\|\|v'\|^2 |s|^{-p}$ for $|s| \leq 1$ and $\leq 8\|v'\|\|v\|^2 |s|^{-p-3}$ for $|s| \geq 1$, both integrable since $p < 1$. The y-derivative of the integrand is $[\Delta v''(\Delta v)^2 + 2(\Delta v')^2 \Delta v]|s|^{-p-3}$, dominated (uniformly in y) by the same type of bound with $\|v'''\|$; differentiate under the integral (dominated convergence) and continuity follows likewise. Scaling: substitute $w = \kappa z + x_0$ in the original integral: $v(y) - v(z) = c(g(x) - g(w))$, $v' = c\kappa g'$, $y - z = (x - w)/\kappa$, $dz = dw/\kappa$, giving the factor $c^3 \kappa^{1+\alpha+2-1} = c^3 \kappa^{p+3}$. □

Lemma 5.21 (Response to the frozen plane wave)**.** *For every parameter tuple, $y \geq R$, $\lambda = \lambda(y)$ and all $z \in \mathbb{R}$:*

$$\mathcal{Q}_{\epsilon,3}(P_y)(z) = 2n_\epsilon \lambda^p |Z(y)|^2 \operatorname{Re}\big(Z(y) e^{i\lambda(z-y)}\big) - 2(3^p - 1) c_{\mathrm{cubic}} \lambda^p \operatorname{Re}\big(Z(y)^3 e^{3i\lambda(z-y)}\big). \tag{5.16}$$

In particular

$$\mathcal{Q}_{\epsilon,3}(P_y)(y) = G(y), \qquad T_P(y) = -2n_\epsilon \lambda^{p+1} |Z|^2 \operatorname{Im} Z + 6(3^p - 1) c_{\mathrm{cubic}} \lambda^{p+1} \operatorname{Im}(Z^3). \tag{5.17}$$

Proof. If $Z(y) = 0$ then $P_y = 0$ and both sides vanish. Otherwise write $Z(y) = |Z| e^{i\theta}$; then $P_y(z) = \frac{|Z|}{\lambda} 2\sin(\lambda z + \theta - \lambda y)$. By the scaling law of Lemma 5.20 with $c = |Z|/\lambda$, $\kappa = \lambda$, $g = 2\sin$ and the normalization (eqn:approx_cubic_normalization), $\mathcal{Q}_{\epsilon,3}(P_y)(z) = |Z|^3 \lambda^p [2n_\epsilon \cos\vartheta - 2(3^p - 1) c_{\mathrm{cubic}} \cos 3\vartheta]$ with $\vartheta = \lambda(z - y) + \theta$. Use $|Z|^3 \cos\vartheta = |Z|^2 \operatorname{Re}(Z e^{i\lambda(z-y)})$ and $|Z|^3 \cos 3\vartheta = \operatorname{Re}(Z^3 e^{3i\lambda(z-y)})$. For T_P differentiate in z at $z = y$: $\operatorname{Re}(i\lambda Z) = -\lambda \operatorname{Im} Z$, $\operatorname{Re}(3i\lambda Z^3) = -3\lambda \operatorname{Im} Z^3$. □

Lemma 5.22 (Dependence of the frozen response on the base point)**.** *$G \in C^1([R,\infty))$ and there is an admissible C such that for every parameter tuple and $y \in [Y,2Y]$, $|T_P(y) - G'(y)| \le C\epsilon Y^{-3\beta} = C\epsilon Y^{1-5\beta+1/p}$.*

Proof. Writing $T_P = 2n_\epsilon \lambda^p |Z|^2 \operatorname{Re}(i\lambda Z) - 2(3^p-1)c_{\mathrm{cubic}}\lambda^p \operatorname{Re}(3i\lambda Z^3)$ and differentiating G,

$$T_P - G' = -2n_\epsilon(\lambda^p|Z|^2)' \operatorname{Re} Z - 2n_\epsilon \lambda^p |Z|^2 \operatorname{Re}(Z' - i\lambda Z) + 2(3^p-1)c_{\mathrm{cubic}}\big[(\lambda^p)' \operatorname{Re} Z^3 + 3\lambda^p \operatorname{Re}(Z^2(Z' - i\lambda Z))\big].$$

Now $\lambda^p|Z|^2 = |\mu|^2|a_*|^2/\lambda$, so $|(\lambda^p|Z|^2)'| \le Cy^{-1/p-1}$; $|Z| \le Cy^{-\beta}$; $|Z' - i\lambda Z| \le Cy^{-\beta-1}$; $|(\lambda^p)'| \le C$; $\lambda^p \le Cy$. Each term is $\le C\epsilon y^{-3\beta}$ (using $\beta + 1 + \frac{1}{p} = 3\beta$), and $-3\beta = 1 - 5\beta + \frac{1}{p}$ because $2\beta = 1 + \frac{1}{p}$.

□

Lemma 5.23 (Integral representation of $\mathcal{R}_{\mathrm{cub}}$ and of its derivative)**.** *For every parameter tuple, $\mathcal{R}_{\mathrm{cub}} \in C^1((R,\infty))$ and for $y \in [Y,2Y]$*

$$\mathcal{R}_{\mathrm{cub}}(y) = \mathcal{Q}_{\epsilon,3}(v_Y)(y) - \mathcal{Q}_{\epsilon,3}(P_y)(y) = K_\epsilon \int_{\mathbb{R}} |s|^{-p-3} N_0(y,s)\, ds, \tag{5.18}$$

$$\mathcal{R}'_{\mathrm{cub}}(y) = K_\epsilon \int_{\mathbb{R}} |s|^{-p-3} N_1(y,s)\, ds + \big(T_P(y) - G'(y)\big).$$

Proof. $v_Y \in C_c^\infty$ and P_y is a trigonometric function, so Lemma 5.20 applies to both. By definition and (5.17), $\mathcal{R}_{\mathrm{cub}} = \mathcal{Q}_{\epsilon,3}(v_Y) - G$; apply (5.15) to v_Y at y and to P_y at $z = y$ and subtract. Derivative: $\mathcal{R}'_{\mathrm{cub}} = \partial_y \mathcal{Q}_{\epsilon,3}(v_Y) - T_P + (T_P - G')$, with both $\partial_y \mathcal{Q}_{\epsilon,3}(v_Y)(y)$ and $T_P(y)$ given by the derivative formula of Lemma 5.20.

□

Lemma 5.24 (Size of v_Y, P_y and their increments)**.** *There is an admissible C such that for every parameter tuple, $y \in [Y,2Y]$, $\lambda = \lambda(y)$:*

1. *on $[Y/2,3Y]$: $v_Y' = 2\operatorname{Re}(Z + c_{\mathrm{cubic}}Z^3)$ and $v_Y'' = 2\operatorname{Re}(Z' + 3c_{\mathrm{cubic}}Z^2Z')$;*
2. *$\|v_Y\|_\infty \le CY^{-\beta}\lambda^{-1}$, $\|v_Y'\|_\infty \le CY^{-\beta}$, $\|v_Y''\|_\infty \le CY^{-\beta}\lambda$; and $\|P_y^{(k)}\|_\infty \le 2|Z(y)|\lambda^{k-1} \le CY^{-\beta}\lambda^{k-1}$, $k = 0,1,2$;*
3. *for $h \in \{v_Y, P_y\}$ and all $s \in \mathbb{R}$: $|\Delta h(y,s)| \le CY^{-\beta}\min\{|s|, \lambda^{-1}\}$, $|\Delta h'(y,s)| \le CY^{-\beta}\min\{1, \lambda|s|\}$, $|\Delta h''(y,s)| \le CY^{-\beta}\lambda\min\{1,\lambda|s|\}$.*

Proof. (1) $\chi_Y \equiv 1$ there. (2) On $\operatorname{supp}\chi_Y$, $z \in (Y/4,4Y)$, $z \ge R$, and by Lemma 5.8 and $|c_{\mathrm{cubic}}| \le C_0\epsilon \le C_0$, $|\int_z^\infty (Z + c_{\mathrm{cubic}}Z^3)| \le CY^{-\beta-1/p}$; $\lambda \le 2(2Y)^{1/p}$ gives $Y^{-1/p} \le C\lambda^{-1}$. Then $v_Y' = -2\chi_Y' \operatorname{Re}\int + 2\chi_Y \operatorname{Re}(Z + c_{\mathrm{cubic}}Z^3)$ and $v_Y'' = -2\chi_Y'' \operatorname{Re}\int + 4\chi_Y' \operatorname{Re}(Z + c_{\mathrm{cubic}}Z^3) + 2\chi_Y \operatorname{Re}(Z' + 3c_{\mathrm{cubic}}Z^2Z')$; use $|\partial^k \chi_Y| \le K_k Y^{-k}$, Lemma 5.7(3) and $Y^{-1} \le Y^{1/p} \le C\lambda$. P_y: direct. (3) $|\Delta h| \le \min\{|s|\|h'\|_\infty, 2\|h\|_\infty\}$ and similarly for h', h''; note $\min\{1,\lambda|s|\} = \lambda\min\{|s|,\lambda^{-1}\}$.

□

Lemma 5.25 (Frozen-frequency comparison for Z)**.** *Let a parameter tuple, $y \in [Y,2Y]$, $\lambda = \lambda(y)$ and $w(t) := Z'(t) - i\lambda Z(t)$ $(t \ge R)$ be given. There is an admissible C such that for $|s| \le Y/4$ and t between $y - s$ and y:*

1. $Z(y)e^{-i\lambda s} - Z(y-s) = \int_{y-s}^{y} e^{-i\lambda(t-y+s)} w(t)\,dt$;

2. $|w(t)| \le CY^{-\beta-1}(1+\lambda|t-y|)$ *and* $|w'(t)| \le CY^{-\beta-1}\lambda(1+\lambda|t-y|)$;

3. $|Z(y)e^{-i\lambda s} - Z(y-s)| \le CY^{-\beta-1}(|s|+\lambda s^2)$ *and* $|w(y)-w(y-s)| \le CY^{-\beta-1}\lambda(|s|+\lambda s^2)$;

4. $|Z(y)^3 - Z(y-s)^3| \le CY^{-3\beta}\min\{1,\lambda|s|\}$ *and* $|(Z^2Z')(y) - (Z^2Z')(y-s)| \le CY^{-3\beta}\lambda\min\{1,\lambda|s|\}$.

Proof. $t \in [3Y/4, 9Y/4]$, so $t \ge R$ and $|t-y| \le Y/4 \le y/2$. (1) $\partial_t[Z(t)e^{-i\lambda(t-y+s)}] = e^{-i\lambda(t-y+s)}w(t)$; integrate over $[y-s, y]$. (2) $w = A'e^{i\Phi} + i(\lambda(t)-\lambda)Z(t)$; use Lemma 5.7(2),(3),(6). Also $w' = [A'' + i(2\lambda(t)-\lambda)A' + i\lambda'(t)A + \lambda(t)(\lambda - \lambda(t))A]e^{i\Phi}$, each term $\le CY^{-\beta-1}\lambda(1+\lambda|t-y|)$ (using $Y^{-1} \le C\lambda$, $|\lambda'| \le C\lambda/Y$). (3) Integrate (2): $\int_0^{|s|}(1+\lambda\tau)d\tau \le |s| + \lambda s^2$; for w use the mean value inequality. (4) Mean value inequality with $|(Z^3)'| \le CY^{-3\beta}\lambda$, $|(Z^2Z')'| \le CY^{-3\beta}\lambda^2$, combined with the sup bounds $CY^{-3\beta}$, $CY^{-3\beta}\lambda$. □

Lemma 5.26 (Comparison of v_Y with the frozen plane wave)**.** *There is an admissible C such that for every parameter tuple, $y \in [Y, 2Y]$, $\lambda = \lambda(y)$ and $|s| \le Y/4$, with $m_0 = \min\{|s|, \lambda^{-1}\}$, $m_1 = \min\{1, \lambda|s|\}$:*

$$\begin{aligned} |\Delta v_Y - \Delta P_y| &\le C\big[Y^{-\beta-1}s^2 + \epsilon Y^{-3\beta}m_0\big], \\ |\Delta v_Y' - \Delta P_y'| &\le C\big[Y^{-\beta-1}(|s|+\lambda s^2) + \epsilon Y^{-3\beta}m_1\big], \\ |\Delta v_Y'' - \Delta P_y''| &\le C\lambda\big[Y^{-\beta-1}(|s|+\lambda s^2) + \epsilon Y^{-3\beta}m_1\big]. \end{aligned} \tag{5.19}$$

Proof. The segment between $y-s$ and y lies in $[Y/2, 3Y]$, where Lemma 5.24(1) holds; also $P_y'(z) = 2\,\mathrm{Re}(Z(y)e^{i\lambda(z-y)})$, $P_y''(z) = 2\,\mathrm{Re}(i\lambda Z(y)e^{i\lambda(z-y)})$. Hence the identities

$$\Delta v_Y' - \Delta P_y' = 2\,\mathrm{Re}\big[Z(y)e^{-i\lambda s} - Z(y-s) + c_{\mathrm{cubic}}(Z(y)^3 - Z(y-s)^3)\big], \tag{5.20}$$

$$\Delta v_Y - \Delta P_y = 2\,\mathrm{Re}\int_{y-s}^{y}\big[Z(t) - Z(y)e^{i\lambda(t-y)} + c_{\mathrm{cubic}}Z(t)^3\big]dt, \tag{5.21}$$

$$\Delta v_Y'' - \Delta P_y'' = 2\,\mathrm{Re}\big[w(y) - w(y-s) + i\lambda(Z(y)e^{-i\lambda s} - Z(y-s)) + 3c_{\mathrm{cubic}}((Z^2Z')(y) - (Z^2Z')(y-s))\big].$$

The second and third bounds follow from Lemma 5.25(3),(4) and $|c_{\mathrm{cubic}}| \le C_0\epsilon$. For the first: since $\int_{y-s}^{y} Z = \frac{1}{i\lambda}[Z(y) - Z(y-s) - \int_{y-s}^{y} w]$ and $\int_{y-s}^{y} Z(y)e^{i\lambda(t-y)} = \frac{Z(y)(1-e^{-i\lambda s})}{i\lambda}$, Lemma 5.25(1) gives $\int_{y-s}^{y}[Z(t) - Z(y)e^{i\lambda(t-y)}]dt = \frac{1}{i\lambda}\int_{y-s}^{y}[e^{-i\lambda(t-y+s)} - 1]w(t)dt$. With $|e^{-i\lambda(t-y+s)} - 1|/\lambda \le 2m_0$ and Lemma 5.25(2) this is $\le CY^{-\beta-1}m_0(|s| + \lambda s^2) \le CY^{-\beta-1}s^2$ (cases $\lambda|s| \le 1$, $\lambda|s| \ge 1$). The cubic part: $|\int_{y-s}^{y} Z^3| \le CY^{-3\beta}|s|$ and, by Lemma 5.8, $\le |\int_{y-s}^{\infty} Z^3| + |\int_{y}^{\infty} Z^3| \le CY^{-3\beta-1/p} \le CY^{-3\beta}\lambda^{-1}$. □

Lemma 5.27 (Elementary kernel integrals)**.** *For $p \in (0,1)$, $\lambda > 0$, $L > 0$, with m_0, m_1 as above: $\int_{\mathbb{R}}(|s|+\lambda s^2)m_0^2|s|^{-p-3}ds \le (\frac{4}{1-p} + \frac{4}{p})\lambda^{p-1}$, $\int_{\mathbb{R}} m_1 m_0^2|s|^{-p-3}ds \le (\frac{2}{1-p} + \frac{2}{p+2})\lambda^p$, and $\int_{|s|\ge L}|s|^{-p-3}ds = \frac{2}{p+2}L^{-p-2}$.*

Proof. Split at $|s| = \lambda^{-1}$. First: for $|s| \le \lambda^{-1}$ the integrand is $\le 2|s|^{-p}$; for $|s| \ge \lambda^{-1}$ it is $\le 2\lambda^{-1}|s|^{-p-1}$. Second: $\lambda|s|^{-p}$ resp. $\lambda^{-2}|s|^{-p-3}$. Integrate explicitly. □

Lemma 5.28 (Near region)**.** *There is an admissible C such that for every parameter tuple, $y \in [Y, 2Y]$, $j \in \{0, 1\}$,*

$$|K_\epsilon| \int_{|s| \le Y/4} |s|^{-p-3} |N_j(y, s)|\, ds \le C\epsilon Y^{1-5\beta+j/p}. \tag{5.22}$$

Proof. Write $X_0 = Y^{-\beta-1}s^2 + \epsilon Y^{-3\beta} m_0$, $X_1 = Y^{-\beta-1}(|s| + \lambda s^2) + \epsilon Y^{-3\beta} m_1$ and recall $m_1 = \lambda m_0$. Using $a_1 b_1^2 - a_2 b_2^2 = (a_1 - a_2)b_1^2 + a_2(b_1 - b_2)(b_1 + b_2)$,

$$N_0 = (\Delta v' - \Delta P')(\Delta v)^2 + \Delta P'(\Delta v - \Delta P)(\Delta v + \Delta P),$$
$$N_1 = (\Delta v'' - \Delta P'')(\Delta v)^2 + \Delta P''(\Delta v - \Delta P)(\Delta v + \Delta P) + 2(\Delta v' - \Delta P')(\Delta v' + \Delta P')\Delta v + 2(\Delta P')^2(\Delta v - \Delta$$

($v = v_Y$, $P = P_y$). By Lemma 5.24(3) and Lemma 5.26, every term is bounded by $C\lambda^j Y^{-2\beta} m_0^2 (X_1 + \lambda X_0)$, and $Y^{-2\beta} m_0^2 (X_1 + \lambda X_0) \le C[Y^{-3\beta-1}(|s| + \lambda s^2) m_0^2 + \epsilon Y^{-5\beta} m_1 m_0^2]$ (as $\lambda s^2 \le |s| + \lambda s^2$ and $\lambda m_0 = m_1$). By Lemma 5.27 and $|K_\epsilon| \le C\epsilon$, the left side is $\le C\epsilon\lambda^j (Y^{-3\beta-1}\lambda^{p-1} + \epsilon Y^{-5\beta}\lambda^p)$. With $\frac{1}{2} Y^{1/p} \le \lambda \le 2^{1+1/p} Y^{1/p}$: $Y^{-3\beta-1}\lambda^{p-1} \le C Y^{-3\beta-1/p} = C Y^{1-5\beta}$, $Y^{-5\beta}\lambda^p \le C Y^{1-5\beta}$, $\lambda^j \le C Y^{j/p}$, $\epsilon \le 1$.

□

Lemma 5.29 (Far region)**.** *There is an admissible C such that for every parameter tuple, $y \in [Y, 2Y]$, $j \in \{0, 1\}$, $|K_\epsilon| \int_{|s| \ge Y/4} |s|^{-p-3} |N_j(y, s)|\, ds \le C\epsilon Y^{1-5\beta+j/p}$.*

Proof. Triangle inequality and the sup bounds of Lemma 5.24(2): $|\Delta h| \le C Y^{-\beta}\lambda^{-1}$, $|\Delta h'| \le C Y^{-\beta}$, $|\Delta h''| \le C Y^{-\beta}\lambda$, so $|N_j| \le C Y^{-3\beta}\lambda^{j-2}$. With Lemma 5.27 ($L = Y/4$) the left side is $\le C\epsilon Y^{-3\beta-p-2} Y^{(j-2)/p}$, and $-3\beta - p - 2 + \frac{j-2}{p} \le -3\beta - \frac{1}{p} + \frac{j}{p} = 1 - 5\beta + \frac{j}{p}$ since $p + 2 + \frac{1}{p} \ge 0$.

□

Proposition 5.30 (Cubic remainder of $\mathcal{E}_5$; bound on $\mathcal{E}_5$)**.** *There is an admissible constant C such that for every parameter tuple (ϵ, μ, R, Y), every $y \in [Y, 2Y]$ and $j \in \{0, 1\}$,*

$$|\partial_y^j \mathcal{R}_{\mathrm{cub}}(y)| \le C\epsilon \langle y \rangle^{1-5\beta+\frac{j}{p}}, \tag{5.23}$$

and consequently

$$|\partial_y^j \mathcal{E}_5(y)| \le C\epsilon \langle y \rangle^{-2p-1+\frac{j}{p}}. \tag{5.24}$$

Proof. By Lemma 5.23, $|\partial^j \mathcal{R}_{\mathrm{cub}}(y)| \le |K_\epsilon| \int_{\mathbb{R}} |s|^{-p-3} |N_j|\, ds + j|T_P - G'|(y)$; split the integral at $|s| = Y/4$ and use Lemmas 5.22, 5.28 and 5.29; convert Y to $\langle y \rangle$ as in Proposition 5.19. For (5.24): $y \ge Y \ge 8R > 2R$, so $\mathcal{E}_5 = \mathcal{R}_{\mathrm{lin}} + \mathcal{R}_{\mathrm{cub}}$ (Lemma 5.6); add (5.14), using $1 - 5\beta \le -2p - 1$ ($\Leftrightarrow \frac{5}{2p} \ge 2$) and $\langle y \rangle \ge 1$. □

Chapter 6

Approximate solution III: control of $\mathcal{E}_6$–$\mathcal{E}_9$ (paper §§5.9–5.12)

6.1 Standing conventions and elementary tools

Standing assumptions. Throughout this chapter we fix the parameters of Definition 2.1: $\frac{3}{4} < p < 1$, $\alpha = 1 + p$, $\beta = \frac{p+1}{2p}$ (so $2\beta = 1 + \frac{1}{p}$, $1 < \beta < \frac{1}{p}$ and $\beta < 1 + p$). $c_\alpha > 0$ is the constant of Definition 1.22 (it depends only on p) and $A_\epsilon(u) = (1 + \epsilon u^2)^{-\alpha/2}$ (Definition 1.10). $\delta \in (0, |\mu_*|/2]$, μ_* (Definition 4.15) and $\overline{B} = \overline{B}(\mu_*, \delta)$ are as in Definition 4.38, and $\epsilon_0 \in (0, \epsilon_0^*(\delta)]$, where $\epsilon_0^*(\delta) \in (0, 1]$ is the smallness threshold for ϵ provided by Proposition 4.49 and the other results of the construction chapter used here (the minimum of the thresholds of Proposition 4.49, Lemma 4.41, Lemma 4.40, Lemma 4.47; Lean: `c06_eps0_star`); every statement of this chapter about the family $\overline{f}_{\epsilon,\mu}$ assumes $\epsilon_0 \le \epsilon_0^*(\delta)$; $R \ge R_0$ is the large parameter of that definition, where $R_0 \ge 1$ is so large that the estimates of Lemma 4.33 for $f_{0,\mathrm{low}} - W$ hold for $|y| \ge R_0$ and those of Lemma 4.48, Proposition 4.49 hold. For $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$ we write $f = \overline{f}_{\epsilon,\mu}$, $Z = Z_{\epsilon,\mu}$, and c_{cubic}, c_+, c_ϵ, $f_{0,\mathrm{low}}$ are as in Definition 4.38. $W(y) = \max\{-y, 0\}$. For $Y \ge 8R$, the cutoff χ_Y, the localized front v_Y, the operators $\mathcal{Q}_\epsilon, \mathcal{Q}_{\epsilon,3}$ and the residuals $\mathcal{E}_6, \dots, \mathcal{E}_9$ are those of Definition 4.51. We assume (as we may) that $\chi_Y(z) = \chi(z/Y)$ for one fixed $\chi \in C_c^\infty((1/4, 4))$ with $\chi = 1$ on $[1/2, 3]$, $0 \le \chi \le 1$; hence $|\partial^j \chi_Y| \le C_j Y^{-j}$ with C_j independent of Y. For $w : \mathbb{R} \to \mathbb{R}$ the secant is $S_w(y, z) = \frac{w(y) - w(z)}{y - z}$ $(y \ne z)$; for $w \in C^1$ we also use $S_w(y, y - s) = \int_0^1 w'(y - \theta s)\, d\theta$, which extends it to $s = 0$.

An *admissible constant* is a positive constant depending only on p, ϵ_0, δ, R and the fixed cutoffs χ_+, χ_R, χ, and never on $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, Y, or the spatial variables y, z, s, t.

Lemma 6.1 (Bounds for the coefficient A_ϵ)**.** *Let $0 < p < 1$, $\alpha = 1 + p$, $0 \le \epsilon \le 1$, and define the quintic remainder*

$$\widetilde{A}_\epsilon(u) := c_\alpha \Big[A_\epsilon(u) - 1 + \tfrac{\alpha}{2} \epsilon u^2 \Big].$$

There is a constant C_A depending only on p such that for all $u \in \mathbb{R}$:

1. $0 \le 1 - A_\epsilon(u) \le \frac{\alpha}{2}\epsilon u^2$;
2. $|A'_\epsilon(u)| \le \alpha\epsilon|u|$;
3. $|A''_\epsilon(u)| + |A'''_\epsilon(u)| \le C_A\epsilon$;
4. $|\widetilde{A}_\epsilon(u)| \le C_A\epsilon^2u^4$ *and* $|\widetilde{A}'_\epsilon(u)| \le C_A\epsilon^2|u|^3$.

In particular $A_\epsilon(0) = 1$, $A'_\epsilon(0) = 0$.

Proof. Write $A_\epsilon(u) = g(\epsilon u^2)$ with $g(\sigma) = (1+\sigma)^{-\alpha/2}$, $\sigma \ge 0$; then $g^{(k)}(\sigma) = (-1)^k\frac{\alpha}{2}(\frac{\alpha}{2}+1)\cdots(\frac{\alpha}{2}+k-1)(1+\sigma)^{-\alpha/2-k}$, so $|g^{(k)}(\sigma)| \le c_k(1+\sigma)^{-\alpha/2-k}$ with $c_1 = \alpha/2$. (1) is $|g(\sigma)-g(0)| \le \sup|g'|\,\sigma$ and g decreasing. (2): $A'_\epsilon(u) = 2\epsilon u\, g'(\epsilon u^2)$ and $|g'| \le \alpha/2$. (3): $A''_\epsilon = 2\epsilon g' + 4\epsilon^2u^2g''$ and $A'''_\epsilon = 12\epsilon^2u\,g'' + 8\epsilon^3u^3g'''$, all evaluated at ϵu^2. Put $x = \sqrt{\epsilon}\,u$; then $\epsilon^2u^2|g''| = \epsilon\,x^2(1+x^2)^{-\alpha/2-2}c_2 \le c_2\epsilon$, $\epsilon^2|u||g''| = \epsilon^{3/2}|x|(1+x^2)^{-\alpha/2-2}c_2 \le c_2\epsilon$ and $\epsilon^3|u|^3|g'''| \le c_3\epsilon^{3/2} \le c_3\epsilon$ (using $\epsilon \le 1$). (4): $\widetilde{A}_\epsilon(u) = c_\alpha[g(\sigma) - g(0) - g'(0)\sigma]$ with $\sigma = \epsilon u^2$ (note $g'(0) = -\alpha/2$), so Taylor's formula gives $|\widetilde{A}_\epsilon(u)| \le \frac{c_\alpha c_2}{2}\epsilon^2u^4$. Also $\widetilde{A}'_\epsilon(u) = 2c_\alpha\epsilon u\,[g'(\epsilon u^2) - g'(0)]$ and $|g'(\sigma) - g'(0)| \le c_2\sigma$. □

Lemma 6.2 (Secant calculus)**.** *Let $w \in C^1(\mathbb{R})$ with $\sup|w'| < \infty$. Then:*

1. $|S_w(y,z)| \le \sup_{\mathbb{R}}|w'|$ *for all $y \ne z$; more precisely $|S_w(y,z)| \le \sup_I|w'|$ where I is the segment between y and z;*
2. *if moreover $w \in C^2(\mathbb{R})$ with $\sup|w''| < \infty$, then for every $s \in \mathbb{R}$ the map $y \mapsto S_w(y, y-s)$ is C^1 with $\partial_y S_w(y,y-s) = S_{w'}(y,y-s)$, and $|S_{w'}(y,y-s)| \le \sup|w''|$;*
3. *for fixed z and $y \ne z$, $\partial_y S_w(y,z) = \dfrac{w'(y) - S_w(y,z)}{y-z}$;*
4. $S_{w_1+w_2} = S_{w_1} + S_{w_2}$ *and* $S_{cw} = cS_w$ *for $c \in \mathbb{R}$.*

Proof. (1) mean value theorem. (2) differentiate $S_w(y,y-s) = \int_0^1 w'(y-\theta s)\,d\theta$ under the integral (the integrand's y-derivative $w''(y-\theta s)$ is bounded), obtaining $\int_0^1 w''(y-\theta s)\,d\theta = S_{w'}(y,y-s)$. (3) quotient rule. (4) linearity. □

Lemma 6.3 (Fractional difference integral)**.** *Let $0 < p < 1$, $A, B > 0$, and let $w : \mathbb{R} \to \mathbb{R}$ satisfy $|w| \le A$ and $|w(a) - w(b)| \le B|a-b|$ for all a, b. Then for every $y \in \mathbb{R}$*

$$\int_{\mathbb{R}} \frac{|w(y) - w(y-s)|}{|s|^{1+p}}\,ds \le C_p A^{1-p}B^p, \qquad C_p = 2^{2-p}\Big(\frac{1}{1-p} + \frac{1}{p}\Big).$$

Proof. $|w(y)-w(y-s)| \le \min\{2A, B|s|\}$. Put $r = 2A/B$. Then $\int_{|s|<r} B|s|^{-p}\,ds = \frac{2Br^{1-p}}{1-p} = \frac{2^{2-p}}{1-p}A^{1-p}B^p$ and $\int_{|s|\ge r} 2A|s|^{-1-p}\,ds = \frac{4Ar^{-p}}{p} = \frac{2^{2-p}}{p}A^{1-p}B^p$. □

Lemma 6.4 (Differentiating translated fractional integrals)**.** *Let $0 < p < 1$ and $w \in C^3(\mathbb{R})$ with $\sup|w^{(k)}| \le B_k < \infty$, $k = 1,2,3$ ($B_k > 0$). Let $\Phi : \mathbb{R}\times\mathbb{R} \to \mathbb{R}$*

be measurable, such that for every s the map $y \mapsto \Phi(y,s)$ is differentiable, $y \mapsto \partial_y\Phi(y,s)$ is measurable in (y,s), and $|\Phi| \le M_0$, $|\partial_y\Phi| \le M_1$ on $\mathbb{R}\times\mathbb{R}$. Set

$$I(y) = \int_{\mathbb{R}} \frac{w'(y) - w'(y-s)}{|s|^{1+p}} \Phi(y,s)\, ds.$$

Then the integral converges absolutely for every y, I is differentiable on $\mathbb{R}$,

$$I'(y) = \int_{\mathbb{R}} \frac{(w''(y) - w''(y-s))\Phi(y,s) + (w'(y) - w'(y-s))\partial_y\Phi(y,s)}{|s|^{1+p}}\, ds,$$

and, with C_p as in Lemma 6.3,

$$|I(y)| \le C_p M_0 B_1^{1-p} B_2^p, \qquad |I'(y)| \le C_p\big(M_0 B_2^{1-p} B_3^p + M_1 B_1^{1-p} B_2^p\big).$$

Proof. Absolute convergence and the bound for I: apply Lemma 6.3 to w' ($A = B_1$, $B = B_2$). For differentiability use the dominated differentiation theorem (Mathlib: `hasDerivAt_integral_of_dominated_loc_of_deriv_le`): the y-derivative of the integrand is bounded, uniformly in y, by $\big(M_0 \min\{2B_2, B_3|s|\} + M_1 \min\{2B_1, B_2|s|\}\big)|s|^{-1-p}$, which is integrable in s by the computation in Lemma 6.3; the integrand at a base point is integrable likewise. The bound for I' follows by applying Lemma 6.3 to w'' ($A = B_2$, $B = B_3$) and to w'. □

Lemma 6.5 (Continuity of the derivative of translated fractional integrals)**.** *Under the hypotheses of Lemma 6.4 ($0 < p < 1$, $w \in C^3(\mathbb{R})$ with $\sup|w^{(k)}| \le B_k$, $k = 1,2,3$; Φ, $\partial_y\Phi$ jointly measurable and bounded by M_0, M_1, $y \mapsto \Phi(y,s)$ differentiable), assume in addition that $y \mapsto \partial_y\Phi(y,s)$ is continuous for every $s \in \mathbb{R}$. Then $I(y) = \int_{\mathbb{R}} \frac{w'(y)-w'(y-s)}{|s|^{1+p}}\Phi(y,s)\, ds$ is C^1 on $\mathbb{R}$.*

Proof. By Lemma 6.4, I is differentiable with $I'(y)$ given by the integral of $\big[(w''(y) - w''(y-s))\Phi(y,s) + (w'(y) - w'(y-s))\partial_y\Phi(y,s)\big]|s|^{-1-p}$. For each $s \neq 0$ this integrand is continuous in y (w', w'' continuous, $\Phi(\cdot,s)$ differentiable, $\partial_y\Phi(\cdot,s)$ continuous) and it is dominated, uniformly in y, by $\big(M_0 \min\{2B_2, B_3|s|\} + M_1 \min\{2B_1, B_2|s|\}\big)|s|^{-1-p} \in L^1$ (Lemma 6.3). Dominated convergence gives continuity of I'. □

Lemma 6.6 (Elementary tail integrals)**.** *Let $0 < p < 1$.*

1. *For $a > 1$ and $y > 0$: $\int_{-\infty}^{0} (y-z)^{-a}\, dz = \frac{y^{1-a}}{a-1}$.*
2. *For $a > 1$, $r > 0$ and $y \in \mathbb{R}$: $\int_{|y-z|\ge r} |y-z|^{-a}\, dz = \frac{2r^{1-a}}{a-1}$.*
3. *For $a > 0$ and $y > 0$: $\int_{-\infty}^{0} \langle z\rangle^{-1-p}(y-z)^{-a}\, dz \le (1 + \frac{1}{p})\, y^{-a}$.*
4. *For $a > 1-p$ and $y > 0$: $\int_0^{\infty} \langle t\rangle^{-p}(y+t)^{-a}\, dt \le \big(\frac{1}{1-p} + \frac{1}{a+p-1}\big) y^{1-p-a}$.*

Proof. (1),(2) direct integration. (3) $(y-z)^{-a} \le y^{-a}$ for $z \le 0$ and $\int_{-\infty}^{0}\langle z\rangle^{-1-p}\, dz \le 1 + \int_1^{\infty} z^{-1-p}\, dz = 1 + \frac{1}{p}$. (4) On $[0,y]$ use $(y+t)^{-a} \le y^{-a}$ and $\langle t\rangle^{-p} \le t^{-p}$, giving $y^{-a}\frac{y^{1-p}}{1-p}$; on $[y,\infty)$ use $\langle t\rangle^{-p}(y+t)^{-a} \le t^{-p-a}$, giving $\frac{y^{1-p-a}}{a+p-1}$. □

Lemma 6.7 (Consequences of the profile bounds)**.** *Let $\epsilon_0 \le \epsilon_0^*(\delta)$ (standing assumptions). There is an admissible constant $C_0 \ge 1$ such that for all $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, $f = \overline{f}_{\epsilon,\mu}$ is a real C^∞ function and*

1. $|f'(z)| \le C_0$ *and* $|f(z) - W(z)| \le C_0\langle z\rangle^{-p}$ *for all* $z \in \mathbb{R}$;

2. $|f(z)| \le C_0\langle z\rangle^{-p}$ *and* $|f'(z)| \le C_0\langle z\rangle^{-\beta}$ *for* $z \ge 0$, *and* $|f'(z)+1| \le C_0\langle z\rangle^{-p-1}$ *for* $z \le 0$;

3. *for* $z \ge 1$ *and* $k = 0, 1, 2$: $|f(z)| \le C_0 z^{-p}$ *and* $|f^{(k+1)}(z)| \le C_0 z^{-\beta+k/p}$.

Proof. Smoothness and the bounds for $y \ge 0$ (with $j \le 2$) and $y \le 0$ (with $j \le 1$) are Proposition 4.49; its constants are uniform in ϵ, μ. For $z \le 0$, $W(z) = -z$, so $f - W = f + z$ and $|f'| \le 1 + |f'+1|$; for $z \ge 0$, $W = 0$ and $\langle z\rangle^{-\beta} \le 1$. For (3): for $z \ge 1$, $z \le \langle z\rangle \le \sqrt{2}z$, so $\langle z\rangle^b \le 2^{|b|/2} z^b$ for every real b. □

Lemma 6.8 (Representation of f on the positive side)**.** *Let* $P(y) = -2\,\mathrm{Re}\int_y^\infty \big(Z(t) + c_{\mathrm{cubic}} Z(t)^3\big)\,dt$ *for* $y \ge R$ *(absolutely convergent). There is an admissible constant* C_1 *such that for all* $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$:

1. $|c_\epsilon| + |c_+| \le C_1\epsilon$;

2. *for* $y \ge 2R$: $f(y) = f_{0,\mathrm{low}}(y) + c_+ y^{-p} + P(y)$ *and*

$$f'(y) - 2\,\mathrm{Re}\big(Z(y) + c_{\mathrm{cubic}} Z(y)^3\big) = f'_{0,\mathrm{low}}(y) - p c_+ y^{-p-1};$$

3. *for* $y \ge 2R$ *and* $0 \le k \le 3$: $|\partial_y^k (f_{0,\mathrm{low}} + c_+ y^{-p})(y)| \le C_1 y^{-p-k}$;

4. *for* $y \ge 2R$ *and* $j = 0, 1$: $\big|\partial_y^j\big[f' - 2\,\mathrm{Re}(Z + c_{\mathrm{cubic}} Z^3)\big](y)\big| \le C_1 y^{-p-1-j}$.

Proof. (1) $|c_\epsilon| \lesssim \epsilon$ is Lemma 3.12 and $c_+ = -c_\epsilon/(p+1)$ (eqn:approx_correction_constants). Absolute convergence of P: $|Z(t)| \le Ct^{-\beta}$ for $t \ge R$ (from Lemma 4.48 with $j = 0$ and $|Z_0| = |f'_{0,+}| \lesssim t^{-\beta}$, μ/μ_* bounded on $\overline{B}$) and $\beta > 1$. (2) is eqn:approx_family_positive of Definition 4.38; differentiating (fundamental theorem of calculus for the absolutely convergent tail integral) gives the second identity. (3) For $y \ge 2R \ge R_0$, $W(y) = 0$ and Lemma 4.33(2) gives $|\partial^k f_{0,\mathrm{low}}(y)| \lesssim y^{-p-k}$; also $|\partial^k y^{-p}| \le C_k y^{-p-k}$ and (1). (4) follows from (2), (3) with $k = j + 1$. □

6.2 Control of $\mathcal{E}_6$

Recall $\mathcal{E}_6 = D\mathcal{Q}_\epsilon(W)[f - W] - 2c_\epsilon y^{-p}\,\mathrm{Re}(Z + c_{\mathrm{cubic}} Z^3)$.

Definition 6.9 (Wedge kernel)**.** For $\epsilon \ge 0$, $y > 0$, $t \ge 0$ set

$$K_\epsilon(y, t) = \frac{A_\epsilon\big(-t/(y+t)\big) - 1}{(y+t)^{1+p}}.$$

Lemma 6.10 (Wedge kernel bounds and identities)**.** *There is a constant* C_K *depending only on* p *such that for* $0 \le \epsilon \le 1$, $y > 0$, $t \ge 0$:

1. $K_\epsilon(y, 0) = 0$ *and* K_ϵ *is* C^∞ *on* $(0, \infty) \times [0, \infty)$;

2. $|\partial_y^k K_\epsilon(y, t)| \le C_K \epsilon (y + t)^{-1-p-k}$ *for* $k = 0, 1, 2$;

3. $\partial_y K_\epsilon(y,t) - \partial_t K_\epsilon(y,t) = \dfrac{A'_\epsilon(-t/(y+t))}{(y+t)^{p+2}}$.

Proof. Write $\sigma = t/(y+t) \in [0,1)$, $G(\sigma) = A_\epsilon(-\sigma) - 1$, so $K = G(\sigma)(y+t)^{-1-p}$. By Lemma 6.1 (with $|\sigma| \le 1$): $|G| \le \frac{\alpha}{2}\epsilon$, $|G'| \le \alpha\epsilon$, $|G''| \le C_A\epsilon$ on $[0,1]$. (1): $G(0) = 0$. (2): $\partial_y\sigma = -\sigma/(y+t)$, so $\partial_y K = -(y+t)^{-2-p}\big[\sigma G'(\sigma) + (1+p)G(\sigma)\big]$ and $\partial_y^2 K = (y+t)^{-3-p}\big[\sigma^2 G'' + (2p+4)\sigma G' + (1+p)(2+p)G\big]$ (using $\partial_y(\sigma G'(\sigma)) = -(\sigma G' + \sigma^2 G'')/(y+t)$); bound each bracket using $0 \le \sigma \le 1$. (3): $\partial_y\sigma - \partial_t\sigma = -\frac{t}{(y+t)^2} - \frac{y}{(y+t)^2} = -\frac{1}{y+t}$ and the factor $(y+t)^{-1-p}$ has equal y- and t-derivatives, so $\partial_y K - \partial_t K = -G'(\sigma)(y+t)^{-2-p} = A'_\epsilon(-\sigma)(y+t)^{-2-p}$. □

Lemma 6.11 (Mass of the wedge kernel)**.** *For $\epsilon \ge 0$ and $y > 0$: $c_\alpha \int_0^\infty K_\epsilon(y,t)\,dt = c_\epsilon y^{-p}$, where $c_\epsilon = c_\alpha \int_0^1 [(1+\epsilon\sigma^2)^{-\alpha/2} - 1](1-\sigma)^{p-1}\,d\sigma$ is the constant of Lemma 3.12.*

Proof. Substitute $t = y\sigma/(1-\sigma)$, $\sigma \in [0,1)$: $y + t = y/(1-\sigma)$, $dt = y(1-\sigma)^{-2}d\sigma$, so $\int_0^\infty K\,dt = y^{-p}\int_0^1 [A_\epsilon(-\sigma) - 1](1-\sigma)^{p-1}\,d\sigma$, and A_ϵ is even. This is the computation in the proof of Lemma 3.12. The integral converges since $p - 1 > -1$.

□

Lemma 6.12 (First variation of $\mathcal{Q}_\epsilon$ at the wedge)**.** *Let $0 \le \epsilon \le 1$ and let $h : \mathbb{R} \to \mathbb{R}$ be bounded and Lipschitz, with $h \in C^2((0,\infty))$. Fix $y > 0$. Then $s \mapsto \mathcal{Q}_\epsilon(W + sh)(y)$ is differentiable at $s = 0$, and its derivative $D\mathcal{Q}_\epsilon(W)[h](y)$ equals*

$$c_\alpha\Big(\int_0^\infty K_\epsilon(y,t)\big(h'(y) - h'(-t)\big)\,dt\Big) + c_\alpha\Big(\int_0^\infty \frac{A'_\epsilon(-t/(y+t))}{(y+t)^{p+2}}\big(h(y) - h(-t)\big)\,dt\Big) \tag{6.1}$$

(here $h'(-t)$ is the a.e. derivative), equivalently, in the z variable, $c_\alpha \int_{-\infty}^0 \frac{h'(y)-h'(z)}{(y-z)^{1+p}}[A_\epsilon(S_W(y,z)) - 1]\,dz + c_\alpha \int_{-\infty}^0 \frac{A'_\epsilon(S_W(y,z))S_h(y,z)}{(y-z)^{1+p}}\,dz$, with $S_W(y,z) = z/(y-z)$. Moreover the function $y \mapsto \int_0^\infty K_\epsilon(y,t)(h(y) - h(-t))\,dt$ is differentiable on $(0,\infty)$ and

$$D\mathcal{Q}_\epsilon(W)[h](y) = c_\alpha \int_0^\infty \big[K_\epsilon(y,t)h'(y) + \partial_y K_\epsilon(y,t)(h(y) - h(-t))\big]\,dt = c_\alpha\,\partial_y \int_0^\infty K_\epsilon(y,t)\big(h(y) - h(-t)\big)\,dt. \tag{6.2}$$

Proof. Step 1 (Gâteaux derivative). For $y > 0$: $W'(y) = 0$; for $z > 0$, $S_W(y,z) = 0$; for $z < 0$, $W'(z) = -1$ and $S_W(y,z) = z/(y-z) \in (-1,0)$. Let L be the Lipschitz constant of h. The $z > 0$ part of $\mathcal{Q}_\epsilon(W + sh)(y)$ is $c_\alpha s \int_0^\infty \frac{h'(y)-h'(z)}{|y-z|^{1+p}}[A_\epsilon(sS_h(y,z)) - 1]\,dz$, whose modulus is $\le c_\alpha \frac{\alpha}{2}\epsilon L^2|s|^3 \int_0^\infty |h'(y) - h'(z)||y-z|^{-1-p}dz = O(|s|^3)$ (the last integral is finite: near $z = y$ use $h \in C^2$ near y, elsewhere $|h'| \le L$); hence it contributes 0 to the derivative. The $z < 0$ part is $c_\alpha \int_{-\infty}^0 \big(1 + s(h'(y) - h'(z))\big)[A_\epsilon(S_W + sS_h) - 1](y-z)^{-1-p}\,dz$; for $|s| \le 1$ the s-derivative of the integrand is bounded by $C(L)\epsilon(y-z)^{-1-p}$ (using $|A_\epsilon - 1| \le \frac{\alpha}{2}\epsilon u^2$, $|A'_\epsilon(u)| \le \alpha\epsilon|u|$ with $|u| \le 1 + L$, $|S_h| \le L$), which is integrable on $(-\infty, 0)$ since $y - z \ge y > 0$. Differentiating under the integral at $s = 0$ and substituting $z = -t$ gives (6.1).

Step 2 (integration by parts). By Lemma 6.10(3) the second integral in (6.1) is $c_\alpha \int_0^\infty (\partial_y K - \partial_t K)(h(y) - h(-t))\,dt$, each part absolutely convergent

($|\partial_y K|, |\partial_t K| \lesssim \epsilon(y+t)^{-2-p}$, h bounded; for $\partial_t K$ use (3) and Lemma 6.1(2)). On $[0,T]$, $t \mapsto h(y)-h(-t)$ is Lipschitz with a.e. derivative $h'(-t)$, so $-\int_0^T \partial_t K\,(h(y)-h(-t))\,dt = -K(y,T)(h(y)-h(-T)) + K(y,0)(h(y)-h(0)) + \int_0^T K\,h'(-t)\,dt$. The boundary terms vanish: $K(y,0)=0$ and $|K(y,T)|\,2\sup|h| \le 2C_K\epsilon \sup|h|\,(y+T)^{-1-p} \to 0$. Adding to the first integral of (6.1) the terms $\pm Kh'(-t)$ cancel, giving the first equality in (6.2) (all integrals absolutely convergent since $\int_0^\infty |K|\,dt < \infty$).

Step 3 (divergence form). For y in a compact $J \subset (0,\infty)$ with $\min J = y_1$, $|\partial_y[K(y,t)(h(y)-h(-t))]| \le C_K\epsilon\big[(y_1+t)^{-2-p}2\sup|h| + (y_1+t)^{-1-p}L\big]$, integrable in t; dominated differentiation gives the second equality.

□

Lemma 6.13 (Exact decomposition of $\mathcal{E}_6$)**.** *For $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$ and $y \ge 2R$,*

$$\mathcal{E}_6(y) = c_\epsilon y^{-p}\big[f'_{0,\mathrm{low}}(y) - pc_+ y^{-p-1}\big] - pc_\epsilon y^{-p-1} f(y) - c_\alpha \int_0^\infty \partial_y K_\epsilon(y,t)\big(f(-t)-t\big)\,dt,$$

and the last integral converges absolutely.

Proof. $h = f - W$ is bounded and Lipschitz (Lemma 6.7(1); W is 1-Lipschitz) and $h = f$ is smooth on $(0,\infty)$. For $y > 0$, $h(y) = f(y)$ and $h(-t) = f(-t) - t$. By (6.2) and Lemma 6.11,

$$D\mathcal{Q}_\epsilon(W)[f-W](y) = \partial_y\big(c_\epsilon y^{-p} f(y)\big) - c_\alpha\,\partial_y \int_0^\infty K_\epsilon(y,t)(f(-t)-t)\,dt,$$

and the last derivative may be taken under the integral (dominated by $C_K\epsilon(y_1+t)^{-2-p}C_0$ on compacts, using $|f(-t)-t| \le C_0$). Hence $D\mathcal{Q}_\epsilon(W)[f-W] = c_\epsilon y^{-p} f' - pc_\epsilon y^{-p-1} f - c_\alpha \int_0^\infty \partial_y K(f(-t)-t)\,dt$. Subtract $2c_\epsilon y^{-p}\,\mathrm{Re}(Z + c_{\mathrm{cubic}} Z^3)$ and use Lemma 6.8(2).

□

Lemma 6.14 (Remote wedge interaction)**.** *Let $\mathcal{R}_6(y) = c_\alpha \int_0^\infty \partial_y K_\epsilon(y,t)(f(-t)-t)\,dt$ for $y > 0$. Then $\mathcal{R}_6 \in C^1((0,\infty))$, $\mathcal{R}_6'(y) = c_\alpha \int_0^\infty \partial_y^2 K_\epsilon(y,t)(f(-t)-t)\,dt$, and there is an admissible C with*

$$|\partial_y^j \mathcal{R}_6(y)| \le C\epsilon\, y^{-2p-1-j}, \qquad y > 0,\ j = 0,1,$$

for all $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$.

Proof. By Lemma 6.7(1) with $z = -t$: $|f(-t)-t| = |f(-t)-W(-t)| \le C_0\langle t\rangle^{-p}$. By Lemma 6.10(2), $|\partial_y^{1+j} K| \le C_K\epsilon(y+t)^{-2-p-j}$; on compact subsets of $(0,\infty)$ this gives an integrable dominating function, justifying differentiation under the integral. Then $|\partial_y^j \mathcal{R}_6(y)| \le c_\alpha C_K C_0\,\epsilon \int_0^\infty \langle t\rangle^{-p}(y+t)^{-p-2-j}dt$ and Lemma 6.6(4) with $a = p+2+j$ gives the bound $\lesssim \epsilon y^{1-p-(p+2+j)} = \epsilon y^{-2p-1-j}$.

□

Proposition 6.15 (Control of $\mathcal{E}_6$)**.** *There is an admissible constant C such that for all $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$, $\mathcal{E}_6$ is C^1 on $(2R,\infty)$ and*

$$|\partial_y^j \mathcal{E}_6(y)| \le C\epsilon\langle y\rangle^{-2p-1+\frac{j}{p}}, \qquad y \ge 8R, \quad j = 0,1.$$

In fact $|\mathcal{E}_6(y)| \le C\epsilon y^{-2p-1}$ and $|\partial_y \mathcal{E}_6(y)| \le C\epsilon(y^{-2p-2} + y^{-p-1-\beta})$.

Proof. Use Lemma 6.13: $\mathcal{E}_6 = T_1 - T_2 - \mathcal{R}_6$ with $T_1 = c_\epsilon y^{-p} g$, $g = f'_{0,\mathrm{low}} - pc_+ y^{-p-1}$, and $T_2 = pc_\epsilon y^{-p-1} f$. By Lemma 6.8(1),(4): $|c_\epsilon| \le C_1\epsilon$ and $|g| \le C_1 y^{-p-1}$, $|g'| \le C_1 y^{-p-2}$; so $|T_1| \le C\epsilon y^{-2p-1}$, $|T_1'| \le C\epsilon(py^{-2p-2} + y^{-2p-2})$. By Lemma 6.7(3) ($y \ge 2R \ge 1$): $|f| \le C_0 y^{-p}$, $|f'| \le C_0 y^{-\beta}$, so $|T_2| \le C\epsilon y^{-2p-1}$ and $|T_2'| \le C\epsilon(y^{-2p-2} + y^{-p-1-\beta})$. Lemma 6.14 bounds $\mathcal{R}_6$. This proves the "in fact" bounds. Conversion: $y^{-2p-2} \le y^{-2p-1+1/p}$ and $y^{-p-1-\beta} \le y^{-2p-1+1/p}$, the latter because $p - \beta < 1/p$ (as $p < 1 < 1/p$). Finally, for $y \ge 1$ and $b > 0$, $y^{-b} \le 2^{b/2}\langle y\rangle^{-b}$, and $b = 2p+1-j/p > 0$ for $j = 0, 1$ (since $1/p < 4/3 < 2p+1$). □

6.3 Control of $\mathcal{E}_7$

Recall $\mathcal{E}_7 = \mathcal{Q}_{\epsilon,3}(\chi_Y f) - \mathcal{Q}_{\epsilon,3}(v_Y)$ with $\mathcal{Q}_{\epsilon,3}(w)(y) = -\frac{(p+1)\epsilon c_\alpha}{2} \int_{\mathbb{R}} \frac{w'(y)-w'(z)}{|y-z|^{1+p}} S_w(y,z)^2\, dz$.

Lemma 6.16 (Bounds for the localized pieces)**.** *Let $Y \ge 8R$, $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, and $h_Y := \chi_Y f - v_Y$. Then:*

1. *$\chi_Y f$, v_Y, h_Y are C^∞ with support in $(Y/4, 4Y) \subset (2R, \infty)$, and $h_Y = \chi_Y\,(f_{0,\mathrm{low}} + c_+ z^{-p})$;*
2. *there is an admissible constant C_2 such that for all $z \in \mathbb{R}$:*

$$|\chi_Y f(z)| \le C_2 Y^{-p}, \qquad \begin{aligned} |\partial^j h_Y(z)| &\le C_2 Y^{-p-j} \quad (0 \le j \le 3),\\ |\partial^{j+1}(\chi_Y f)(z)| &\le C_2 Y^{-\beta+j/p} \quad (0 \le j \le 2),\\ |\partial^{j+1} v_Y(z)| &\le C_2 Y^{-\beta+j/p} \quad (0 \le j \le 2). \end{aligned}$$

Proof. (1) $\operatorname{supp}\chi_Y \subset (Y/4, 4Y)$ and $Y/4 \ge 2R$. By definition $v_Y = \chi_Y P$ with P as in Lemma 6.8, so Lemma 6.8(2) gives $h_Y = \chi_Y(f - P) = \chi_Y(f_{0,\mathrm{low}} + c_+ z^{-p})$. Smoothness of P on (R, ∞) follows from $P' = 2\operatorname{Re}(Z + c_{\mathrm{cubic}} Z^3)$ and smoothness of Z. (2) Leibniz rule with $|\partial^k \chi_Y| \le C_k Y^{-k}$ and, on $[Y/4, 4Y]$ (where $z \simeq Y$ up to factor 4): for h_Y, Lemma 6.8(3) gives $|\partial^k(f_{0,\mathrm{low}} + c_+ z^{-p})| \le C_1 4^{p+k} Y^{-p-k}$; each Leibniz term is $\lesssim Y^{-m} Y^{-p-(j-m)} = Y^{-p-j}$. For $\chi_Y f$, Lemma 6.7(3): the term with m derivatives on χ_Y is $\lesssim Y^{-m} Y^{-\beta+(j-m)/p} \le Y^{-\beta+j/p}$ for $m \le j$, and for $m = j+1$ it is $\lesssim Y^{-j-1} Y^{-p} \le Y^{-\beta+j/p}$ because $\beta < 1 + p$ and $Y \ge 1$. For $v_Y = \chi_Y f - h_Y$ combine the two, using $Y^{-p-1-j} \le Y^{-\beta+j/p}$.

□

Lemma 6.17 (Algebraic form of $\mathcal{E}_7$)**.** *With $h = h_Y$, $v = v_Y$ and $\kappa := -\frac{(p+1)\epsilon c_\alpha}{2}$, for every $y \in \mathbb{R}$*

$$\mathcal{E}_7(y) = \kappa\,(T_{7,1}(y) + T_{7,2}(y)),$$

$$T_{7,1}(y) = \int_{\mathbb{R}} \frac{h'(y) - h'(y-s)}{|s|^{1+p}} S_{v+h}(y, y-s)^2\, ds, \quad T_{7,2}(y) = \int_{\mathbb{R}} \frac{v'(y) - v'(y-s)}{|s|^{1+p}} S_h(y, y-s)\, S_{2v+h}(y, y-s)\, ds,$$

all integrals (including those defining $\mathcal{Q}_{\epsilon,3}(\chi_Y f)$ and $\mathcal{Q}_{\epsilon,3}(v_Y)$) converging absolutely.

Proof. Absolute convergence: for $w \in \{v, \chi_Y f\}$, $|S_w| \le \sup|w'|$ and Lemma 6.3 applies to w'. Substitute $z = y - s$ (so $|y - z|^\alpha = |s|^{1+p}$). With $a_1 = (v + h)'(y) - (v + h)'(z)$, $b_1 = S_{v+h}$, $a_2 = v'(y) - v'(z)$, $b_2 = S_v$ use $a_1 b_1^2 - a_2 b_2^2 = (a_1 - a_2) b_1^2 + a_2(b_1 - b_2)(b_1 + b_2)$ together with $v + h = \chi_Y f$, $a_1 - a_2 = h'(y) - h'(z)$, $b_1 - b_2 = S_h$, $b_1 + b_2 = S_{2v+h}$ (Lemma 6.2(4)). □

Lemma 6.18 (Size of $\mathcal{E}_7$)**.** *There is an admissible C such that for $Y \geq 8R$, all $y \in \mathbb{R}$, $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$: $|T_{7,1}(y)| \leq CY^{-2p-1-2\beta}$, $|T_{7,2}(y)| \leq CY^{-p-2\beta}$, hence $|\mathcal{E}_7(y)| \leq C\epsilon Y^{-p-2\beta}$.*

Proof. By Lemma 6.16 and Lemma 6.2(1): $|S_{v+h}| = |S_{\chi_Y f}| \leq C_2 Y^{-\beta}$, $|S_h| \leq C_2 Y^{-p-1}$, $|S_{2v+h}| \leq 3C_2 Y^{-\beta}$. Lemma 6.3 with $w = h'$ ($A = C_2 Y^{-p-1}$, $B = C_2 Y^{-p-2}$) gives $\int |h'(y) - h'(y-s)||s|^{-1-p} ds \leq C_p C_2 Y^{-(p+1)(1-p)-(p+2)p} = C_p C_2 Y^{-2p-1}$; with $w = v'$ ($A = C_2 Y^{-\beta}$, $B = C_2 Y^{-\beta+1/p}$) it gives $C_p C_2 Y^{1-\beta}$. Hence $|T_{7,1}| \lesssim Y^{-2\beta} Y^{-2p-1}$ and $|T_{7,2}| \lesssim Y^{-p-1} Y^{-\beta} Y^{1-\beta}$. Since $2p + 1 + 2\beta \geq p + 2\beta$ and $Y \geq 1$, and $|\kappa| \leq C\epsilon$, the claim follows. □

Lemma 6.19 (Derivative of $\mathcal{E}_7$)**.** *For $Y \geq 8R$, $\mathcal{E}_7$ is differentiable on $\mathbb{R}$ and there is an admissible C with $|\partial_y \mathcal{E}_7(y)| \leq C\epsilon Y^{-p-2\beta+1/p}$ for all $y \in \mathbb{R}$, $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$.*

Proof. Apply Lemma 6.4 to $T_{7,1}$ with $w = h$, $\Phi(y, s) = S_{\chi_Y f}(y, y-s)^2$: by Lemma 6.2(2), $\partial_y \Phi = 2S_{\chi_Y f} S_{(\chi_Y f)'}$, so $M_0 = C_2^2 Y^{-2\beta}$, $M_1 = 2C_2^2 Y^{-2\beta+1/p}$, and $B_k = C_2 Y^{-p-k}$ ($k = 1, 2, 3$). Thus $|T'_{7,1}| \lesssim Y^{-2\beta} Y^{-(p+2)(1-p)-(p+3)p} + Y^{-2\beta+1/p} Y^{-2p-1} = Y^{-2\beta-2p-2} + Y^{-2\beta-2p-1+1/p}$. Apply it to $T_{7,2}$ with $w = v$, $\Phi = S_h S_{2v+h}$, $\partial_y \Phi = S_{h'} S_{2v+h} + S_h S_{2v'+h'}$: $M_0 = 3C_2^2 Y^{-p-1-\beta}$, $M_1 \lesssim Y^{-p-2-\beta} + Y^{-p-1-\beta+1/p}$, and $B_k = C_2 Y^{-\beta+(k-1)/p}$. Since $B_2^{1-p} B_3^p \lesssim Y^{1-\beta+1/p}$ and $B_1^{1-p} B_2^p \lesssim Y^{1-\beta}$: $|T'_{7,2}| \lesssim Y^{-p-2\beta+1/p} + Y^{-p-1-2\beta}$. All four exponents are $\leq -p - 2\beta + 1/p$ (using $p \leq 1/p$, $Y \geq 1$). Multiply by $|\kappa| \leq C\epsilon$. □

Lemma 6.20 (Control of $\mathcal{E}_7$)**.** *There is an admissible C such that for all $Y \geq 8R$, $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$,*

$$|\partial_y^j \mathcal{E}_7(y)| \leq C\epsilon \langle y \rangle^{-p-2\beta+\frac{j}{p}} \leq C\epsilon \langle y \rangle^{-2p-1+\frac{j}{p}}, \qquad Y \leq y \leq 2Y,\ j = 0, 1.$$

Proof. By Lemma 6.18, Lemma 6.19, $|\partial^j \mathcal{E}_7| \leq C\epsilon Y^{-p-2\beta+j/p}$. The exponent $-p-2\beta+j/p = -p-1-(1-j)/p$ is negative, and for $1 \leq Y \leq y \leq 2Y$ we have $\langle y \rangle \leq \sqrt{2} y \leq 2\sqrt{2} Y$, so $Y^{-b} \leq (2\sqrt{2})^b \langle y \rangle^{-b}$ for $b \geq 0$. The second inequality is $p + 2\beta \geq 2p + 1$, i.e. $1/p \geq p$. □

Lemma 6.21 ($\mathcal{E}_7$ is C^1)**.** *Under the standing assumptions of this chapter (in particular $\delta \in (0, |\mu_*|/2]$ and $\epsilon_0 \in (0, \epsilon_0^*(\delta)]$) there is $R_0 \geq 1$ such that for all $R \geq R_0$, $Y \geq 8R$, $\epsilon \in [0, \epsilon_0]$ and $\mu \in \overline{B}$, with $f = \overline{f}_{\epsilon,\mu}$: $\mathcal{E}_7^Y$ is C^1 on $\mathbb{R}$.*

Proof. By Lemma 6.17, $\mathcal{E}_7^Y = \kappa(T_{7,1} + T_{7,2})$. Each $T_{7,i}$ has the form of Lemma 6.5 with $w = h_Y$, $\Phi = S_{v+h}(y, y-s)^2$, resp. $w = v_Y$, $\Phi = S_h(y, y-s) S_{2v+h}(y, y-s)$; the bounds on $w^{(k)}$, Φ, $\partial_y \Phi$ come from Lemma 6.16 and Lemma 6.2, and $\partial_y \Phi$ is continuous in y because the secants of the C^∞ functions v_Y, h_Y have continuous y-derivatives $\partial_y S_w(y, y-s) = \int_0^1 w''(y - \theta s)\, d\theta$. □

6.4 Control of $\mathcal{E}_8$

Recall $\mathcal{E}_8 = \mathcal{Q}_\epsilon(\chi_Y f) - \mathcal{Q}_{\epsilon,3}(\chi_Y f)$.

Lemma 6.22 (Quintic remainder representation)**.** *Let $0 \leq \epsilon \leq 1$ and $w \in C^2(\mathbb{R})$ with w', w'' bounded. Then for every $y \in \mathbb{R}$*

$$\mathcal{Q}_\epsilon(w)(y) - \mathcal{Q}_{\epsilon,3}(w)(y) = \int_{\mathbb{R}} \frac{w'(y) - w'(y-s)}{|s|^{1+p}}\, \widetilde{A}_\epsilon\big(S_w(y, y-s)\big)\, ds,$$

with $\widetilde{A}_\epsilon$ as in Lemma 6.1, all three integrals converging absolutely.

Proof. The integrands of $\mathcal{Q}_\epsilon(w)$ and $\mathcal{Q}_{\epsilon,3}(w)$ are $c_\alpha \frac{w'(y)-w'(z)}{|y-z|^{1+p}}[A_\epsilon(S_w)-1]$ and $c_\alpha \frac{w'(y)-w'(z)}{|y-z|^{1+p}}[-\frac{\alpha}{2}\epsilon S_w^2]$ (note $\frac{p+1}{2} = \frac{\alpha}{2}$); each is bounded by a constant times $|w'(y)-w'(z)||y-z|^{-1-p}$ since S_w is bounded, which is integrable by Lemma 6.3. Subtract and substitute $z = y - s$. □

Lemma 6.23 (Control of $\mathcal{E}_8$)**.** *There is an admissible C such that for all $Y \geq 8R$, $\epsilon \in [0,\epsilon_0]$, $\mu \in \overline{B}$, $\mathcal{E}_8$ is differentiable, $|\mathcal{E}_8(y)| \leq C\epsilon^2 Y^{1-5\beta}$, $|\partial_y \mathcal{E}_8(y)| \leq C\epsilon^2 Y^{1-5\beta+1/p}$ for all $y \in \mathbb{R}$, and*

$$|\partial_y^j \mathcal{E}_8(y)| \leq C\epsilon^2 \langle y\rangle^{-2p-1+\frac{j}{p}}, \qquad Y \leq y \leq 2Y,\ j = 0, 1.$$

Proof. Apply Lemma 6.22 with $w = \chi_Y f$ and then Lemma 6.4 with $\Phi(y,s) = \widetilde{A}_\epsilon(S_w(y, y-s))$, $\partial_y\Phi = \widetilde{A}'_\epsilon(S_w)S_{w'}$ (Lemma 6.2(2)). By Lemma 6.16, $|S_w| \leq C_2 Y^{-\beta}$, $|S_{w'}| \leq C_2 Y^{-\beta+1/p}$, and $B_k = C_2 Y^{-\beta+(k-1)/p}$; by Lemma 6.1(4), $M_0 = C_A C_2^4 \epsilon^2 Y^{-4\beta}$, $M_1 = C_A C_2^4 \epsilon^2 Y^{-4\beta+1/p}$. Hence $|\mathcal{E}_8| \lesssim \epsilon^2 Y^{-4\beta} Y^{1-\beta}$ and $|\mathcal{E}_8'| \lesssim \epsilon^2(Y^{-4\beta}Y^{1-\beta+1/p} + Y^{-4\beta+1/p}Y^{1-\beta})$. Finally $5\beta > 2p+2$ (equivalently $5/p + 1 > 4p$, true as $p < 1$), so $Y^{1-5\beta+j/p} \leq Y^{-2p-1+j/p}$, and the conversion to $\langle y\rangle$ is as in the proof of Lemma 6.20 ($2p + 1 - j/p > 0$). □

Lemma 6.24 ($\mathcal{E}_8$ is C^1)**.** *Under the standing assumptions of this chapter (in particular $\delta \in (0, |\mu_*|/2]$ and $\epsilon_0 \in (0, \epsilon_0^*(\delta)]$) there is $R_0 \geq 1$ such that for all $R \geq R_0$, $Y \geq 8R$, $\epsilon \in [0,\epsilon_0]$ and $\mu \in \overline{B}$, with $f = \overline{f}_{\epsilon,\mu}$: $\mathcal{E}_8^Y$ is C^1 on $\mathbb{R}$.*

Proof. By Lemma 6.22 with $w = \chi_Y f$, $\mathcal{E}_8^Y$ is the integral of Lemma 6.5 with $\Phi(y,s) = \widetilde{A}_\epsilon(S_w(y, y-s))$, whose y-derivative $\widetilde{A}'_\epsilon(S_w)\partial_y S_w$ is continuous ($\widetilde{A}_\epsilon$ smooth, $w \in C^\infty$) and bounded (Lemma 6.1, Lemma 6.16), with the constants computed in the proof of the bound for $\mathcal{E}_8$. □

Lemma 6.25 (The cubic term of the local front is C^1)**.** *Under the standing assumptions of this chapter (in particular $\delta \in (0, |\mu_*|/2]$ and $\epsilon_0 \in (0, \epsilon_0^*(\delta)]$) there is $R_0 \geq 1$ such that for all $R \geq R_0$, $Y \geq 8R$, $\epsilon \in [0,\epsilon_0]$ and $\mu \in \overline{B}$, with $f = \overline{f}_{\epsilon,\mu}$: $\mathcal{Q}_{\epsilon,3}(v_Y)$ is C^1 on $\mathbb{R}$.*

Proof. After the substitution $z = y-s$, $\mathcal{Q}_{\epsilon,3}(v_Y)(y) = \kappa \int_{\mathbb{R}} \frac{v_Y'(y)-v_Y'(y-s)}{|s|^{1+p}} S_{v_Y}(y, y-s)^2\, ds$ with $\kappa = -\frac{(p+1)\epsilon c_\alpha}{2}$; apply Lemma 6.5 with $w = v_Y$ and $\Phi = S_{v_Y}^2$ (bounds from Lemma 6.16 and Lemma 6.2); $\partial_y\Phi = 2S_{v_Y}\partial_y S_{v_Y}$ is continuous in y since $\partial_y S_{v_Y}(y, y-s) = \int_0^1 v_Y''(y-\theta s)\, d\theta$. □

6.5 Control of $\mathcal{E}_9$

Recall $\mathcal{E}_9 = \mathcal{Q}_\epsilon(f) - \mathcal{Q}_\epsilon(W) - D\mathcal{Q}_\epsilon(W)[f-W] - \mathcal{Q}_\epsilon(\chi_Y f)$. Throughout this section $Y \geq 8R$ and $J_Y := (3Y/4, 5Y/2)$; note $\chi_Y = 1$ on a neighbourhood of $\overline{J_Y} \subset (Y/2, 3Y)$, and for $y \in J_Y$, $z \in (0,\infty) \setminus [Y/2, 3Y]$ we have $|y-z| \geq Y/4$; for $z < 0$, $y - z \geq y \geq 3Y/4$.

Lemma 6.26 (Splitting of $\mathcal{E}_9$)**.** *For $y \in J_Y$ (write $S_\bullet = S_\bullet(y,z)$), $\mathcal{E}_9(y) = \mathcal{E}_{9,+}(y) + \mathcal{I}_1(y) + \mathcal{I}_2(y) + \mathcal{I}_3(y)$, where*

$$\mathcal{E}_{9,+}(y) = c_\alpha \int_{(0,\infty)\setminus[Y/2,3Y]} \frac{(f'(y)-f'(z))[A_\epsilon(S_f)-1] - (f'(y)-(\chi_Y f)'(z))[A_\epsilon(S_{\chi_Y f})-1]}{|y-z|^{1+p}}\,dz,$$

$$\mathcal{I}_1(y) = c_\alpha \int_{-\infty}^{0} \frac{A_\epsilon(S_f) - A_\epsilon(S_W) - A'_\epsilon(S_W)S_{f-W}}{(y-z)^{1+p}}\,dz,$$

$$\mathcal{I}_2(y) = c_\alpha \int_{-\infty}^{0} \frac{(f'(y)-f'(z)-1)[A_\epsilon(S_f)-A_\epsilon(S_W)]}{(y-z)^{1+p}}\,dz,$$

$$\mathcal{I}_3(y) = -c_\alpha \int_{-\infty}^{0} \frac{f'(y)}{(y-z)^{1+p}}\Big[A_\epsilon\Big(\frac{f(y)}{y-z}\Big) - 1\Big]dz,$$

with all integrals absolutely convergent.

Proof. Split each of $\mathcal{Q}_\epsilon(f)(y)$, $\mathcal{Q}_\epsilon(\chi_Y f)(y)$ into $z>0$ and $z<0$. For $y>0$: $\mathcal{Q}_\epsilon(W)(y) = c_\alpha\int_{-\infty}^0 [A_\epsilon(S_W)-1](y-z)^{-1-p}dz$ (since $W'(y)-W'(z)=1$ for $z<0$ and $S_W=0$ for $z>0$), and by Lemma 6.12 (z-form, $h=f-W$, $h'(y)-h'(z) = f'(y)-f'(z)-1$ for $z<0$) $D\mathcal{Q}_\epsilon(W)[f-W](y) = c_\alpha\int_{-\infty}^0 \frac{(f'(y)-f'(z)-1)[A_\epsilon(S_W)-1]+A'_\epsilon(S_W)S_{f-W}}{(y-z)^{1+p}}dz$. Writing $f'(y)-f'(z) = 1 + (f'(y)-f'(z)-1)$ in the $z<0$ part of $\mathcal{Q}_\epsilon(f)$ and using $S_f - S_W = S_{f-W}$, the $z<0$ parts of $\mathcal{Q}_\epsilon(f) - \mathcal{Q}_\epsilon(W) - D\mathcal{Q}_\epsilon(W)[f-W]$ combine to $\mathcal{I}_1+\mathcal{I}_2$. For $z<0$, $\chi_Y(z)=0$ and $\chi_Y=1$ near y, so the $z<0$ part of $-\mathcal{Q}_\epsilon(\chi_Y f)(y)$ is $\mathcal{I}_3$. For $z\in[Y/2,3Y]$ and $y\in J_Y$, $(\chi_Y f)(z) = f(z)$, $(\chi_Y f)'(z) = f'(z)$, $(\chi_Y f)(y) = f(y)$, so the two $z>0$ integrands cancel there, leaving $\mathcal{E}_{9,+}$. Absolute convergence: the $z<0$ integrands are $O(\epsilon(y-z)^{-2-p})$ by Lemma 6.7(1) and Lemma 6.1(1),(2) (all secants are bounded, $|S_{f-W}| \le 2C_0/(y-z)$); the integrand of $\mathcal{E}_{9,+}$ is $O(\epsilon|y-z|^{-3-p})$ on $|y-z|\ge Y/4$; and $\mathcal{Q}_\epsilon(f)(y)$, $\mathcal{Q}_\epsilon(\chi_Y f)(y)$ converge near $z=y$ since $|f'(y)-f'(z)|\le C_y|y-z|$ there and $|A_\epsilon - 1|\le 1$, and at infinity since f' is bounded and $|A_\epsilon-1|\le 1$.

□

Lemma 6.27 (Far positive contribution)**.** *$\mathcal{E}_{9,+}$ is differentiable on J_Y (derivative under the integral sign) and there is an admissible C with $|\partial_y^j \mathcal{E}_{9,+}(y)| \le C\epsilon Y^{-p-2+j/p}$ for $y\in[Y,2Y]$, $j=0,1$, all $\epsilon\in[0,\epsilon_0]$, $\mu\in\overline{B}$.*

Proof. Let $g\in\{f,\chi_Y f\}$. On $[0,\infty)$: $|g|\le C_0$ and $|g'|\le C_0 + \sup|\chi'_Y|C_0 \le C$ (Lemma 6.7(2)); at $y\in J_Y$, $g(y)=f(y)$, $g'(y)=f'(y)$, $g''(y)=f''(y)$ with $|f'(y)|\le CY^{-\beta}$, $|f''(y)|\le CY^{-\beta+1/p}$ (Lemma 6.7(3)). For $z>0$ with $|y-z|\ge Y/4\ge 1$: $|S_g(y,z)|\le 2C_0|y-z|^{-1}$ and, by Lemma 6.2(3), $|\partial_y S_g(y,z)|\le (|f'(y)|+|S_g|)|y-z|^{-1}\le C|y-z|^{-1}$; by Lemma 6.1(1),(2), $|A_\epsilon(S_g)-1|\le C\epsilon|y-z|^{-2}$ and $|A'_\epsilon(S_g)|\le C\epsilon|y-z|^{-1}$. With $F_g(y,z) = (f'(y)-g'(z))[A_\epsilon(S_g)-1]|y-z|^{-1-p}$ we get $|F_g|\le C\epsilon|y-z|^{-3-p}$ and

$$|\partial_y F_g| \le |f''(y)||A_\epsilon(S_g)-1||y-z|^{-1-p} + C|A'_\epsilon(S_g)||\partial_y S_g||y-z|^{-1-p} + (1+p)C|A_\epsilon(S_g)-1||y-z|^{-2-p}$$

$$\le C\epsilon(Y^{-\beta+1/p}+1)|y-z|^{-3-p},$$ uniformly for $y\in J_Y$ (this also justifies differentiating under the integral). By Lemma 6.6(2) with $r=Y/4$, $a=3+p$: $|\partial^j\mathcal{E}_{9,+}| \le C\epsilon(1+Y^{-\beta+1/p})^j Y^{-2-p}$, and $1+Y^{-\beta+1/p}\le 2Y^{1/p}$.

□

Lemma 6.28 (Second-order Taylor term $\mathcal{I}_1$)**.** *$\mathcal{I}_1$ is differentiable on J_Y and there is an admissible C with $|\mathcal{I}_1(y)| \le C\epsilon Y^{-p-2}$, $|\partial_y \mathcal{I}_1(y)| \le C\epsilon Y^{-p-3}$ for $y \in [Y, 2Y]$.*

Proof. For $y \in J_Y$, $z < 0$: $S_W = z/(y-z) \in (-1, 0)$, $\partial_y S_W = -S_W/(y-z)$ so $|\partial_y S_W| \le (y-z)^{-1}$; $|S_{f-W}| \le 2C_0(y-z)^{-1}$ (Lemma 6.7(1)); and since $W'(y) = 0$, $\partial_y S_{f-W} = (f'(y) - S_{f-W})/(y-z)$, so $|\partial_y S_{f-W}| \le CY^{-\beta}(y-z)^{-1} + C(y-z)^{-2}$. Let $T = A_\epsilon(S_f) - A_\epsilon(S_W) - A'_\epsilon(S_W)S_{f-W}$. Taylor (eqn:approx_E9_I1_taylor): $T = S_{f-W}^2 \int_0^1 (1-\theta) A''_\epsilon(S_W + \theta S_{f-W}) d\theta$, so $|T| \le \frac{C_A}{2}\epsilon S_{f-W}^2 \le C\epsilon(y-z)^{-2}$. Using $S_f = S_W + S_{f-W}$, $\partial_y T = [A'_\epsilon(S_f) - A'_\epsilon(S_W) - A''_\epsilon(S_W)S_{f-W}]\partial_y S_W + [A'_\epsilon(S_f) - A'_\epsilon(S_W)]\partial_y S_{f-W}$, and by Taylor with $|A''_\epsilon| + |A'''_\epsilon| \le C_A\epsilon$: $|\partial_y T| \le C\epsilon[(y-z)^{-3} + Y^{-\beta}(y-z)^{-2}]$. Hence $|\partial_y(T(y-z)^{-1-p})| \le C\epsilon[(y-z)^{-4-p} + Y^{-\beta}(y-z)^{-3-p}]$ (uniformly on J_Y, justifying differentiation), and Lemma 6.6(1) gives $|\mathcal{I}_1| \le C\epsilon y^{-2-p}$, $|\mathcal{I}'_1| \le C\epsilon(y^{-3-p} + Y^{-\beta}y^{-2-p}) \le C\epsilon Y^{-p-3}$ ($\beta > 1$, $y \ge Y$).

□

Lemma 6.29 (Mixed term $\mathcal{I}_2$)**.** *$\mathcal{I}_2$ is differentiable on J_Y and there is an admissible C with $|\partial_y^j \mathcal{I}_2(y)| \le C\epsilon Y^{-p-2+j/p}$ for $y \in [Y, 2Y]$, $j = 0, 1$.*

Proof. With the secant bounds from the proof of Lemma 6.28: all arguments $S_W + \theta S_{f-W}$ lie in $[-1-2C_0, 2C_0]$, so $|A'_\epsilon| \le \alpha(1+2C_0)\epsilon$ there and (eqn:approx_E9_I2_taylor) $|A_\epsilon(S_f) - A_\epsilon(S_W)| \le C\epsilon(y-z)^{-1}$. Also $|f'(y) - f'(z) - 1| \le |f'(y)| + |f'(z) + 1| \le C(Y^{-\beta} + \langle z\rangle^{-p-1})$ (Lemma 6.7(2),(3)). $j = 0$: Lemma 6.6(1),(3) with $a = 2 + p$ give $|\mathcal{I}_2| \le C\epsilon(Y^{-\beta-1-p} + Y^{-2-p}) \le C\epsilon Y^{-p-2}$. $j = 1$: $\partial_y[A_\epsilon(S_f) - A_\epsilon(S_W)] = [A'_\epsilon(S_f) - A'_\epsilon(S_W)]\partial_y S_W + A'_\epsilon(S_f)\partial_y S_{f-W}$ (eqn:approx_E9_I2_coefficient_derivative), bounded by $C\epsilon[(y-z)^{-2} + Y^{-\beta}(y-z)^{-1}]$. Differentiating the integrand (three terms: $f''(y)$; the coefficient; the kernel, giving $(1+p)(y-z)^{-2-p}$) and using $|f''(y)| \le CY^{-\beta+1/p}$, Lemma 6.6(1),(3) give $|\mathcal{I}'_2| \le C\epsilon(Y^{-p-1-\beta+1/p} + Y^{-p-2-\beta} + Y^{-p-3} + Y^{-p-1-2\beta}) \le C\epsilon Y^{-p-2+1/p}$, using $\beta > 1$. The same pointwise bounds, uniform on J_Y, justify differentiation under the integral.

□

Lemma 6.30 (Truncation term $\mathcal{I}_3$)**.** *$\mathcal{I}_3$ is differentiable on J_Y and there is an admissible C with $|\partial_y^j \mathcal{I}_3(y)| \le C\epsilon Y^{-p-2+j/p}$ for $y \in [Y, 2Y]$, $j = 0, 1$.*

Proof. By Lemma 6.1(1),(2) (eqn:approx_E9_I3_coefficient_bounds) with $u = f(y)/(y-z)$ and $|f(y)| \le CY^{-p}$, $|f'(y)| \le CY^{-\beta}$, $|f''(y)| \le CY^{-\beta+1/p}$: $|\mathcal{I}_3| \le C\epsilon|f'(y)||f(y)|^2 \int_{-\infty}^0 (y-z)^{-3-p} dz \le C\epsilon Y^{-\beta-2p} Y^{-2-p}$. Using $\partial_y \frac{f(y)}{y-z} = \frac{f'(y)}{y-z} - \frac{f(y)}{(y-z)^2}$, $|\mathcal{I}'_3| \le C\epsilon\big[(|f''||f|^2 + |f'|^2|f|)\int_{-\infty}^0 (y-z)^{-3-p} dz + |f'||f|^2 \int_{-\infty}^0 (y-z)^{-4-p} dz\big] \le C\epsilon[(Y^{-\beta-2p+1/p} + Y^{-2\beta-p})Y^{-p-2} + Y^{-\beta-2p}Y^{-p-3}] \le C\epsilon Y^{-p-2+1/p}$.

□

Lemma 6.31 (Control of $\mathcal{E}_9$)**.** *There is an admissible C such that for all $Y \ge 8R$, $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, $\mathcal{E}_9$ is differentiable on $[Y, 2Y]$ and*

$$|\partial_y^j \mathcal{E}_9(y)| \le C\epsilon\langle y\rangle^{-p-2+\frac{j}{p}}, \qquad Y \le y \le 2Y,\ j = 0, 1.$$

Proof. Sum the four bounds (the identity of Lemma 6.26 holds on the open set $J_Y \supset [Y, 2Y]$, so derivatives add); $Y^{-p-3} \le Y^{-p-2+1/p}$. Convert to $\langle y\rangle$ as in Lemma 6.20, since $p + 2 - j/p > 0$.

□

Lemma 6.32 (C^1 regularity of the pieces of $\mathcal{E}_9$)**.** *Under the standing assumptions of this chapter (in particular $\delta \in (0, |\mu_*|/2]$ and $\epsilon_0 \in (0, \epsilon_0^*(\delta)]$) there is $R_0 \geq 1$ such that for all $R \geq R_0$, $Y \geq 8R$, $\epsilon \in [0, \epsilon_0]$ and $\mu \in \overline{B}$, with $f = \overline{f}_{\epsilon,\mu}$: each of the four functions $\mathcal{E}_{9,+}$, $\mathcal{I}_1$, $\mathcal{I}_2$, $\mathcal{I}_3$ of Lemma 6.26 is C^1 on $J_Y = (3Y/4, 5Y/2)$.*

Proof. The proofs of Lemma 6.27, Lemma 6.28, Lemma 6.29 and Lemma 6.30 differentiate under the integral sign with majorants uniform for $y \in J_Y$. The differentiated integrands are continuous in $y \in J_Y$ for each fixed z ($f \in C^\infty$, A_ϵ smooth, secants $S_g(y, z)$ with z off the diagonal), and are dominated by the same integrable majorants; dominated convergence gives continuity of the derivatives on J_Y.

□

Lemma 6.33 ($\mathcal{E}_9$ is C^1 on J_Y)**.** *Under the standing assumptions of this chapter (in particular $\delta \in (0, |\mu_*|/2]$ and $\epsilon_0 \in (0, \epsilon_0^*(\delta)]$) there is $R_0 \geq 1$ such that for all $R \geq R_0$, $Y \geq 8R$, $\epsilon \in [0, \epsilon_0]$ and $\mu \in \overline{B}$, with $f = \overline{f}_{\epsilon,\mu}$: $\mathcal{E}_9^Y$ is C^1 on $J_Y = (3Y/4, 5Y/2) \supset [Y, 2Y]$.*

Proof. On the open set J_Y, $\mathcal{E}_9^Y = \mathcal{E}_{9,+} + \mathcal{I}_1 + \mathcal{I}_2 + \mathcal{I}_3$ (Lemma 6.26); each summand is C^1 on J_Y by Lemma 6.32.

□

Chapter 7

Approximate solution IV: residual bounds and Theorem 3.1 (paper §§5.3, 5.13–5.15)

7.1 Standing conventions and auxiliary objects

Throughout this chapter we use the following standing assumptions.

- p, β are as in Definition 2.1: $\frac{3}{4} < p < 1$, $\alpha = p + 1$, $\beta = \frac{p+1}{2p}$. Note $\beta > 1$ (since $p < 1$). We write $\theta := \frac{1}{p} - \beta = \frac{1-p}{2p}$; since $p > \frac{1}{2}$ we have $0 < \theta < p$.
- $c_\alpha = \Gamma(\alpha)\sin(\pi(\alpha-1)/2)/\pi > 0$ is the constant of Definition 1.5, so that for g bounded and locally Lipschitz $|D|^p g(y) = c_\alpha \int_{\mathbb{R}} \frac{g(y)-g(z)}{|y-z|^{1+p}}\, dz$ (absolutely convergent).
- $A_\epsilon(u) = (1+\epsilon u^2)^{-\alpha/2}$ (Definition 1.10) and $B_\epsilon(u) = A_\epsilon(u)^{-1} = (1+\epsilon u^2)^{\alpha/2}$.
- $\mu_* = -i/\sqrt{2\pi p}$ (Definition 4.15); the parameters $R \geq 1$, $\epsilon_0 \in (0,1]$, $\delta > 0$, the cutoffs χ_+, χ_R, the closed ball $\overline{B} = \{\mu \in \mathbb{C} : |\mu - \mu_*| \leq \delta\}$, and the family $\overline{f}_{\epsilon,\mu}$ with its pieces $f_{0,\mu}, f_{0,+}, f_{\mathrm{slow}}, f_{\mathrm{osc}}, Z_{\epsilon,\mu}, \lambda_{\epsilon,\mu}, \Psi_{\epsilon,\mu}$ and constants $c_\pm, c_{\mathrm{cubic}}, n_\epsilon, c_{\log}(\mu)$ are those of Definition 4.38. Here R is large and ϵ_0 small, chosen in Chapter 4 so that Lemma 4.48 and Propositions 4.49 and 4.57 hold; we additionally use that ϵ_0 is small enough that $\lambda_{\epsilon,\mu}(t) \geq \frac{1}{2}t^{1/p}$ for $t \geq R$ (this follows from (4.13) with $j = 0$).
- $\chi \in C_c^\infty((-\frac{1}{2}, \frac{1}{2}))$ is a fixed even cutoff with $0 \leq \chi \leq 1$ and $\chi = 1$ on $[-\frac{1}{4}, \frac{1}{4}]$.
- A constant is called *admissible* if it depends only on p, R, δ and the fixed cutoffs χ_+, χ_R, χ (and, when indicated, on γ, j, M, K); it never depends on $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, the point y, or a sequence index n.

- For $M>0$, $k\ge 0$, $\|u\|_{C^k([-M,M])}:=\sum_{i\le k}\sup_{|y|\le M}|u^{(i)}(y)|$. A sequence $u_n\to u$ *in* C^k_{loc} means $\|u_n-u\|_{C^k([-M,M])}\to 0$ for every $M>0$; a map $\mu\mapsto u_\mu$ from $\overline{B}$ is continuous into C^k_{loc} if $\mu_n\to\mu$ in $\overline{B}$ implies $u_{\mu_n}\to u_\mu$ in C^k_{loc} ($\overline{B}$ is a metric space, so sequential continuity suffices).

Definition 7.1 (Free operator, cubic remainder, divided residual)**.** Let $f\in C^2(\mathbb{R};\mathbb{R})$ with $f'\in L^\infty(\mathbb{R})$ and $0\le\epsilon\le 1$.

1. The secant is $S_f(y,z):=\frac{f(y)-f(z)}{y-z}$ for $y\neq z$.
2. The free operator is $\mathcal{L}_0(f):=|D|^p f'+f-yf'$ (this is $\mathcal{L}_\epsilon$ of Definition 2.17 at $\epsilon=0$).
3. The nonlinear remainder is $\mathcal{Q}_\epsilon(f)(y):=c_\alpha\int_{\mathbb{R}}\frac{f'(y)-f'(z)}{|y-z|^\alpha}\big[A_\epsilon(S_f(y,z))-1\big]\,dz$.
4. The divided residual is $\widetilde{\mathcal{E}}_\epsilon(f):=A_\epsilon(f')\,\mathcal{E}_\epsilon(f)$, where $\mathcal{E}_\epsilon$ is the residual of Definition 1.26.
5. The corner profile is $W(y):=\max\{-y,0\}$.

Definition 7.2 (Secant kernel operator)**.** For $f\in C^2(\mathbb{R};\mathbb{R})$ with f' bounded and $\Gamma\in C^1(\mathbb{R};\mathbb{R})$, set

$$\mathcal{T}_\Gamma(f)(y):=c_\alpha\int_{\mathbb{R}}\frac{f'(y)-f'(y-s)}{|s|^\alpha}\,\Gamma\big(S_f(y,y-s)\big)\,ds,\qquad S_f(y,y-s)=\frac{f(y)-f(y-s)}{s}.$$

Thus $\mathcal{T}_1(f)=|D|^pf'$, $\mathcal{T}_{A_\epsilon-1}(f)=\mathcal{Q}_\epsilon(f)$ and, by linearity in Γ, $\mathcal{T}_{A_\epsilon}(f)=|D|^pf'+\mathcal{Q}_\epsilon(f)$ (whenever the integrals converge absolutely, see Lemma 7.7). For $K>0$ write $\|\Gamma\|_K:=\sup_{|u|\le K}(|\Gamma(u)|+|\Gamma'(u)|)$.

Definition 7.3 (Admissible profile class)**.** For $K\ge 1$, $\mathcal{P}(K)$ is the set of $f\in C^3(\mathbb{R};\mathbb{R})$ such that for all $y\in\mathbb{R}$:

1. $|f'(y)|\le K$;
2. $|f''(y)|\le K\langle y\rangle^\theta$ if $y\ge 0$, and $|f''(y)|\le K\langle y\rangle^{-1}$ if $y\le 0$ (recall $\theta=\frac{1}{p}-\beta$);
3. $|f'''(y)|\le K\langle y\rangle^{2/p}$.

Lemma 7.4 (Elementary bounds on A_ϵ, B_ϵ)**.** *There is $C=C(p)$ such that for all $0\le\epsilon\le 1$ and $u\in\mathbb{R}$:*

$$|A_\epsilon(u)-1|\le\tfrac{\alpha}{2}\epsilon u^2,\quad |A_\epsilon'(u)|\le\alpha\epsilon|u|,\quad |A_\epsilon''(u)|\le C\epsilon,\quad 0\le B_\epsilon(u)-1\le\tfrac{\alpha}{2}\epsilon u^2,\quad |B_\epsilon'(u)|\le\alpha\epsilon|u|.$$

In particular, for $K\ge 1$, $\|A_\epsilon-1\|_K+\|A_\epsilon'\|_K\le CK^2\epsilon$ and $\|A_\epsilon\|_K\le 1+CK^2$.

Proof. Put $x=\epsilon u^2\ge 0$. Since $x\mapsto(1+x)^{-\alpha/2}$ is convex with derivative $-\alpha/2$ at 0, $0\le 1-(1+x)^{-\alpha/2}\le\frac{\alpha}{2}x$. Since $\alpha/2<1$, $x\mapsto(1+x)^{\alpha/2}$ is concave, so $0\le(1+x)^{\alpha/2}-1\le\frac{\alpha}{2}x$. Directly, $A_\epsilon'(u)=-\alpha\epsilon u(1+\epsilon u^2)^{-\alpha/2-1}$, $B_\epsilon'(u)=\alpha\epsilon u(1+\epsilon u^2)^{\alpha/2-1}$, and both factors $(1+x)^{-\alpha/2-1},(1+x)^{\alpha/2-1}$ are ≤ 1. Finally $A_\epsilon''(u)=-\alpha\epsilon(1+x)^{-\alpha/2-1}+\alpha(\alpha+2)\epsilon\frac{x}{1+x}(1+x)^{-\alpha/2-1}$, so $|A_\epsilon''|\le\alpha(\alpha+3)\epsilon$. □

Lemma 7.5 (Secant calculus)**.** *Let $f \in C^1(\mathbb{R})$ with f' bounded. For $y \neq z$: $|S_f(y,z)| \le \|f'\|_{L^\infty}$, and*

$$\partial_y S_f = \frac{f'(y) - S_f}{y - z}, \qquad \partial_z S_f = \frac{S_f - f'(z)}{y - z}, \qquad \partial_y \partial_z S_f = \frac{f'(y) + f'(z) - 2S_f}{(y-z)^2}.$$

Moreover $S_f(y, y-s) = \int_0^1 f'(y - ts)\, dt$, $\partial_y[S_f(y, y-s)] = \frac{f'(y) - f'(y-s)}{s}$, and $S_f - S_g = S_{f-g}$ (linearity). If f is only continuous and piecewise C^1 with bounded derivative (e.g. $f = W$) the same holds wherever the derivatives exist.

Proof. Mean value theorem / fundamental theorem of calculus for the first claims, quotient rule for the derivatives: e.g. $\partial_z S_f = \frac{-f'(z)(y-z) + (f(y) - f(z))}{(y-z)^2}$; then $\partial_y \partial_z S_f = \frac{\partial_y S_f}{y-z} - \frac{S_f - f'(z)}{(y-z)^2}$ and insert $\partial_y S_f$. □

Lemma 7.6 (Divided residual identity)**.** *Let $0 \le \epsilon \le 1$ and $f \in C^2(\mathbb{R}; \mathbb{R})$ with f' bounded. Then for every $y \in \mathbb{R}$ the integrals defining $|D|^p f'(y)$, $\mathcal{Q}_\epsilon(f)(y)$, $\widetilde{\mathcal{J}}_\epsilon(f)(y)$ and $\mathcal{T}_{A_\epsilon}(f)(y)$ converge absolutely, and*

$$\widetilde{\mathcal{E}}_\epsilon(f) = \mathcal{L}_0(f) + \mathcal{Q}_\epsilon(f) = f - yf' + \mathcal{T}_{A_\epsilon}(f), \qquad \mathcal{E}_\epsilon(f) = B_\epsilon(f')\, \widetilde{\mathcal{E}}_\epsilon(f).$$

Proof. Fix y. For $|y-z| \le 1$, $|f'(y) - f'(z)| \le |y-z| \sup_{[y-1, y+1]} |f''|$, so each integrand is $O(|y-z|^{1-\alpha}) = O(|y-z|^{-p})$, integrable; for $|y-z| > 1$ each integrand is $\le 2\|f'\|_\infty (1 + \sup |A_\epsilon|) |y-z|^{-\alpha}$, integrable since $\alpha > 1$ (and $0 < A_\epsilon \le 1$). By Definition 1.22, $\widetilde{\mathcal{J}}_\epsilon(f)(y) = c_\alpha \int \frac{f'(y) - f'(z)}{|y-z|^\alpha} A_\epsilon(S_f)\, dz - A_\epsilon(f'(y)) |D|^p f'(y)$, splitting an absolutely convergent integral. Hence by Definition 1.26 $A_\epsilon(f') \mathcal{E}_\epsilon(f) = A_\epsilon(f') |D|^p f' + f - yf' + \widetilde{\mathcal{J}}_\epsilon(f) = f - yf' + \mathcal{T}_{A_\epsilon}(f)$. Writing $A_\epsilon(S) = 1 + (A_\epsilon(S) - 1)$ gives $\mathcal{T}_{A_\epsilon}(f) = |D|^p f' + \mathcal{Q}_\epsilon(f)$, hence $\widetilde{\mathcal{E}}_\epsilon(f) = \mathcal{L}_0(f) + \mathcal{Q}_\epsilon(f)$. Dividing by $A_\epsilon(f') > 0$ gives the last identity. □

Lemma 7.7 (C^1 regularity and local bounds for $\mathcal{T}_\Gamma$)**.** *Let $K \ge 1$, $f \in \mathcal{P}(K)$ and $\Gamma \in C^1(\mathbb{R}; \mathbb{R})$. Then for every y the integral defining $\mathcal{T}_\Gamma(f)(y)$ converges absolutely, $\mathcal{T}_\Gamma(f) \in C^1(\mathbb{R})$, and*

$$\partial_y \mathcal{T}_\Gamma(f)(y) = c_\alpha \int_{\mathbb{R}} \Big[\frac{f''(y) - f''(y-s)}{|s|^\alpha} \Gamma(S) + \frac{(f'(y) - f'(y-s))^2}{s|s|^\alpha} \Gamma'(S) \Big] ds, \qquad S = S_f(y, y-s),$$

again absolutely convergent. Moreover, for every $M \ge 1$ there is $C = C(p, K, M)$ with $\|\mathcal{T}_\Gamma(f)\|_{C^1([-M,M])} \le C \|\Gamma\|_K$.

Proof. Write $k(y,s)$ for the integrand of $\mathcal{T}_\Gamma(f)$ and $k_1(y,s)$ for the integrand above. By Lemma 7.5, $|S| \le K$ and $\partial_y k = k_1$ for $s \neq 0$ (chain rule, using $\partial_y S_f(y, y-s) = (f'(y) - f'(y-s))/s$). Fix $M \ge 1$ and let $|y| \le M + 1$. Set $K_2 := K\langle M+2 \rangle^{1/p} \ge \sup_{|x| \le M+2} |f''(x)|$ and $K_3 := K\langle M+2 \rangle^{2/p} \ge \sup_{|x| \le M+2} |f'''(x)|$. *Near region $|s| \le 1$:* by the mean value theorem $|f'(y) - f'(y-s)| \le K_2 |s|$ and $|f''(y) - f''(y-s)| \le K_3 |s|$, hence $|k| \le c_\alpha K_2 \|\Gamma\|_K |s|^{-p}$ and $|k_1| \le c_\alpha (K_3 + K_2^2) \|\Gamma\|_K |s|^{-p}$. *Far region $|s| > 1$:* $|f'(y) - f'(y-s)| \le 2K$, so $|k| \le 2c_\alpha K \|\Gamma\|_K |s|^{-1-p}$; and by Definition 7.3(2), $|f''(x)| \le K\langle x \rangle^\theta$ for all x, and $\langle y - s \rangle \le \sqrt{2} \langle y \rangle \langle s \rangle$, so $|f''(y)| + |f''(y-s)| \le 4K \langle M+1 \rangle^\theta |s|^\theta$; hence $|k_1| \le c_\alpha (4K \langle M+1 \rangle^\theta |s|^\theta + 4K^2 |s|^{-1}) \|\Gamma\|_K |s|^{-1-p}$. Let $D_M(s)$ be the resulting majorant (independent of $y \in [-M-1, M+1]$). Since $p < 1$ and $\theta < p$

(here $p > \frac{1}{2}$ is used), $D_M \in L^1(\mathbb{R})$ with $\|D_M\|_{L^1} \le C(p,K,M)\|\Gamma\|_K$. Absolute convergence and the bound $|\mathcal{T}_\Gamma(f)(y)| \le \|D_M\|_{L^1}$ follow. For each $s \neq 0$, $y \mapsto k(y,s)$ is C^1 with derivative k_1 and $|k_1(y,s)| \le D_M(s)$ for $|y| < M+1$; the theorem on differentiation under the integral sign (dominated derivative; Mathlib `hasDerivAt_integral_of_dominated_loc_of_deriv_le`) gives differentiability with the stated formula. Continuity of $y \mapsto \int k_1(y,s)ds$ follows from continuity of $k_1(\cdot,s)$ for $s \neq 0$ and dominated convergence with D_M. The $C^1([-M,M])$ bound is $2\|D_M\|_{L^1}$. □

Lemma 7.8 (Continuity of $\mathcal{T}_\Gamma$ under C^2_{loc} convergence)**.** *Let $K \ge 1$, $\Gamma \in C^1(\mathbb{R};\mathbb{R})$, and $f_n, f \in \mathcal{P}(K)$ with $f_n \to f$ in C^2_{loc}. Then $\mathcal{T}_\Gamma(f_n) \to \mathcal{T}_\Gamma(f)$ in C^1_{loc}.*

Proof. Fix $M \ge 1$; let $k_n, k_{1,n}$ (resp. k, k_1) be the integrands of Lemma 7.7 for f_n (resp. f). Since all functions lie in $\mathcal{P}(K)$, the majorant D_M of that proof serves for all n simultaneously. Define $e_n(s) := \sup_{|y|\le M}\big(|k_n(y,s) - k(y,s)| + |k_{1,n}(y,s) - k_1(y,s)|\big)$, which is continuous in $s \neq 0$ (supremum over a compact set of a jointly continuous function), hence measurable, and $e_n \le 2D_M$. *Claim: $e_n(s) \to 0$ for each fixed $s \neq 0$.* Indeed on the compact interval $I_s = [-M-|s|, M+|s|]$ we have $f_n^{(i)} \to f^{(i)}$ uniformly for $i \le 2$; hence $S_{f_n}(y, y-s) \to S_f(y, y-s)$ uniformly in $|y| \le M$ (it is $(f_n(y) - f_n(y-s))/s$ with s fixed), all secants lie in $[-K,K]$, and Γ, Γ' are uniformly continuous on $[-K,K]$; the integrands are finite sums of products of these uniformly convergent bounded quantities and the fixed factor $|s|^{-\alpha}$ or $s^{-1}|s|^{-\alpha}$. By dominated convergence $\int e_n \to 0$, and $\|\mathcal{T}_\Gamma(f_n) - \mathcal{T}_\Gamma(f)\|_{C^1([-M,M])} \le 2c_\alpha \int e_n(s)\,ds \to 0$, using the derivative formula of Lemma 7.7. Lean route: dominated convergence along the filter $n \to \infty$ jointly with $y \to x$ within $[-M,M]$, plus the compact local-uniform convergence criterion; this avoids the measurability of the supremum e_n. □

Lemma 7.9 (Elementary weighted integrals)**.** *Let $0 < p < 1$ and $q > 1$. There is $C = C(p,q)$ such that for all $|y| \ge 1$:*

1. $\int_{|y-z|\ge|y|/4} \langle z\rangle^{-p}|y-z|^{-q}\,dz \le C|y|^{1-p-q}$;
2. $\int_{|y-z|\ge|y|/4} |y-z|^{-q}\,dz \le C|y|^{1-q}$ *and* $\int_{|s|\le|y|/2} |s|^{-p}\,ds \le C|y|^{1-p}$;
3. *if $y < 0$:* $\int_0^\infty \big(|y|^{-p} + \langle z\rangle^{-p}\big)(|y|+z)^{-q}\,dz \le C|y|^{1-p-q}$.

Proof. (2) is explicit integration. (1): on $\{|z| \le 2|y|\}$ bound $|y-z|^{-q} \le 4^q|y|^{-q}$ and $\int_{|z|\le 2|y|}\langle z\rangle^{-p}dz \le \int_{|z|\le 2|y|}|z|^{-p}dz = \frac{2(2|y|)^{1-p}}{1-p}$; on $\{|z| > 2|y|\}$, $|y-z| \ge |z|/2$ and $\langle z\rangle^{-p} \le |z|^{-p}$, so the integral is $\le 2^q\int_{|z|>2|y|}|z|^{-p-q}dz = C|y|^{1-p-q}$ since $p+q > 1$. (3): the first part is $|y|^{-p}\frac{|y|^{1-q}}{q-1}$; for the second split at $z = |y|$ as in (1). □

Lemma 7.10 (Decay of $|D|^p h'$ for decaying h)**.** *Let $0 < p < 1$, $K \ge 0$, and $h \in C^3(\mathbb{R};\mathbb{R})$ with $|h^{(i)}(y)| \le K\langle y\rangle^{-p-i}$ for $i = 0,1,2,3$ and all $y \in \mathbb{R}$. Then $|D|^p h' \in C^1(\mathbb{R})$, $\partial_y|D|^p h' = |D|^p h''$, and there is $C = C(p)$ such that for all $|y| \ge 1$ and $j \in \{0,1\}$,*

$$|\partial_y^j |D|^p h'(y)| \le CK\langle y\rangle^{-2p-1-j}.$$

Proof. Differentiation under the integral in $|D|^p h'(y) = c_\alpha \int (h'(y) - h'(y-s))|s|^{-1-p} ds$ is justified by the majorant $\|h'''\|_\infty |s|^{-p}\mathbf{1}_{|s|\le 1} + 2\|h''\|_\infty |s|^{-1-p}\mathbf{1}_{|s|>1}$, giving $\partial_y |D|^p h' = |D|^p h''$, continuous by dominated convergence. Fix $|y| \ge 1$ (so $\langle y\rangle \le \sqrt{2}|y|$), let $g \in \{h', h''\}$, i.e. $g = h^{(1+j)}$, and split with $\chi_y(z) := \chi((y-z)/|y|)$:

$$|D|^p g(y) = \underbrace{c_\alpha \int \chi_y \frac{g(y)-g(z)}{|y-z|^{1+p}} dz}_{I_{\mathrm{near}}} + \underbrace{c_\alpha g(y) \int \frac{1-\chi_y}{|y-z|^{1+p}} dz}_{I_{\mathrm{far},1}} \underbrace{- c_\alpha \int \frac{(1-\chi_y)\, g(z)}{|y-z|^{1+p}} dz}_{I_{\mathrm{far},2}}.$$

Near: on $\operatorname{supp}\chi_y$, $|y-z| \le |y|/2$, so $|z| \ge |y|/2$ and the segment $[y,z]$ stays in $\{|x| \ge |y|/2\}$; thus $|g(y)-g(z)| \le CK\langle y\rangle^{-p-2-j}|y-z|$ and, by Lemma 7.9(2), $|I_{\mathrm{near}}| \le CK\langle y\rangle^{-p-2-j}|y|^{1-p}$. *Far 1:* $1-\chi_y$ is supported in $|y-z| \ge |y|/4$, so $|I_{\mathrm{far},1}| \le CK\langle y\rangle^{-p-1-j}|y|^{-p}$. *Far 2:* integrate by parts $1+j$ times in z to move all derivatives from $g = h^{(1+j)}$ onto $\kappa(z) := (1-\chi_y(z))|y-z|^{-1-p}$; boundary terms vanish since κ vanishes near $z = y$ and $h^{(i)}(z)\kappa^{(l)}(z) \to 0$ as $|z| \to \infty$. Since $\partial_z \chi_y = -\chi'(\cdot)/|y|$ and $|y|^{-1} \le (2|y-z|)^{-1}$ on $\operatorname{supp}\chi_y'$, we get $|\partial_z^{1+j}\kappa(z)| \le C\mathbf{1}_{|y-z|\ge|y|/4}|y-z|^{-2-p-j}$. Then $|I_{\mathrm{far},2}| \le CK\int_{|y-z|\ge|y|/4}\langle z\rangle^{-p}|y-z|^{-2-p-j}dz \le CK|y|^{-1-2p-j}$ by Lemma 7.9(1) with $q = 2+p+j$. Summing, $|D|^p g(y) = O(K\langle y\rangle^{-2p-1-j})$. □

Lemma 7.11 (Local estimate for $|D|^p u'$ and $\mathcal{L}_0 u$)**.** *Let $0 < p < 1$, $M \ge 1$, and $u \in C^3(\mathbb{R};\mathbb{R})$ with u, u' bounded on $\mathbb{R}$. Then $|D|^p u' \in C^1(\mathbb{R})$ and there is $C = C(p)$ with*

$$\big\||D|^p u'\big\|_{C^1([-M,M])} \le C\big(\|u\|_{C^3([-M-1,M+1])} + \|u\|_{L^\infty(\mathbb{R})}\big), \qquad \|\mathcal{L}_0 u\|_{C^1([-M,M])} \le C(1+M)\big(\|u\|_{C^3([-M-1,M+$$

(The finiteness of $\|u'\|_{L^\infty(\mathbb{R})}$ is only used to make the integrals absolutely convergent; it does not enter the constant.)

Proof. For $|y| \le M$ write $|D|^p u'(y) = c_\alpha \int_{|s|\le 1} \frac{u'(y)-u'(y-s)}{|s|^{1+p}} ds + \frac{2c_\alpha}{p} u'(y) - c_\alpha \int_{|s|>1} \frac{u'(y-s)}{|s|^{1+p}} ds$. The first term is $\le C\|u''\|_{L^\infty[-M-1,M+1]}$. In the last, integrate by parts on $s > 1$ and $s < -1$ separately ($\partial_s[u(y-s)] = -u'(y-s)$): $\int_{|s|>1} \frac{u'(y-s)}{|s|^{1+p}} ds = u(y-1) - u(y+1) - (1+p)\int_{|s|>1} \frac{\operatorname{sgn}(s)\, u(y-s)}{|s|^{2+p}} ds$, bounded by $C(\|u\|_{L^\infty[-M-1,M+1]} + \|u\|_{L^\infty(\mathbb{R})})$. For the derivative (differentiation under the integral is justified by the bounded u', u'' on compacts and $u' \in L^\infty$), differentiate this representation in y: the near term becomes $\int_{|s|\le 1} \frac{u''(y)-u''(y-s)}{|s|^{1+p}} ds$, bounded by $C\|u'''\|_{L^\infty[-M-1,M+1]}$; the local terms give $u''(y), u'(y \pm 1)$; and $\partial_y \int_{|s|>1} \operatorname{sgn}(s) u(y-s)|s|^{-2-p} ds = \int_{|s|>1} \operatorname{sgn}(s) u'(y-s)|s|^{-2-p} ds$, which after one more integration by parts is bounded by $C(\|u\|_{L^\infty[-M-1,M+1]} + \|u\|_{L^\infty(\mathbb{R})})$. Finally $u - yu'$ and $(u-yu')' = -yu''$ are bounded on $[-M,M]$ by $(1+2M)\|u\|_{C^2([-M,M])}$. □

7.2 Uniform properties of the approximate family

Lemma 7.12 (Smoothness of $f_{0,+}$)**.** *The function $f_{0,+}(y) = -\frac{1}{2\pi}\int_0^\infty \frac{\chi_+(\xi)}{\xi^2} e^{i(y\xi - \Phi_0(\xi))} d\xi$ (absolutely convergent) belongs to $C^\infty(\mathbb{R};\mathbb{C})$.*

Proof. Fix $M \geq 1$, $k \geq 0$, and $\Xi := (2M)^{1/p} + 2$; let $\eta \in C_c^\infty([0, 2\Xi))$ with $\eta = 1$ on $[0, \Xi]$. The part with amplitude $\eta\chi_+\xi^{-2}$ is an integral over a compact ξ-interval of a smooth function of (y, ξ), hence smooth in y. For the part with amplitude $a(\xi) = (1-\eta)\chi_+\xi^{-2}$ (supported in $[\Xi, \infty)$) and phase $\varphi = y\xi - \Phi_0(\xi)$, for $|y| < M$ we have $|\partial_\xi\varphi| = |y - \xi^p| \geq \xi^p/2$ on $\operatorname{supp} a$. Integrating by parts N times with $e^{i\varphi} = \frac{\partial_\xi e^{i\varphi}}{i(y-\xi^p)}$ (no boundary terms: a vanishes near Ξ, decay at ∞) gives $\int a e^{i\varphi} = \int (\mathcal{D}^t)^N a\, e^{i\varphi}$ with $\mathcal{D}^t g = -\partial_\xi\big(g/(i(y-\xi^p))\big)$, and $|\partial_y^l (\mathcal{D}^t)^N a| \leq C_{N,l,M}\xi^{-2-Np}$. Therefore the y-derivatives up to order k of $(\mathcal{D}^t)^N a\, e^{i\varphi}$ are bounded by $C\xi^{k-2-Np}$, integrable on $[\Xi, \infty)$ once $Np > k$; differentiation under the integral sign gives C^k on $(-M, M)$. As M, k are arbitrary, $f_{0,+} \in C^\infty$. □

Lemma 7.13 (The family is real and smooth)**.** *For all $0 \leq \epsilon \leq \epsilon_0$ and $\mu \in \overline{B}$, $\overline{f}_{\epsilon,\mu} \in C^\infty(\mathbb{R}; \mathbb{R})$, and the pieces $f_{0,\mu}$, f_{slow}, f_{osc} of (4.10) are real-valued and C^∞. Moreover $f_{\mathrm{osc}} = 0$ on $(-\infty, R]$ and $f_{\mathrm{slow}} = 0$ on $[-R, R]$.*

Proof. f_0 is real and smooth by Lemma 4.33 (its formula is real); $f_{0,+}$ is smooth by Lemma 7.12, so $f_{0,\mu} = f_0 + 2\operatorname{Re}[(\mu/\mu_* - 1)f_{0,+}]$ is real and smooth. χ_R is smooth with closed support in (R, ∞), so it vanishes on a neighbourhood of $(-\infty, R]$; hence $y^{-p}\chi_R(y)$ and $|y|^{-p}\chi_R(-y)$ are smooth on $\mathbb{R}$ and vanish for $|y| \leq R$. For $t \geq R$, $\lambda_{\epsilon,\mu}(t) \geq \frac{1}{2}t^{1/p} > 0$ (standing assumption), so $Z_{\epsilon,\mu}$ in (4.7) is smooth on $[R, \infty)$ and $|Z_{\epsilon,\mu}(t)| \leq Ct^{-\beta}$ (using that a_* is bounded on $[R, \infty)$ by Lemma 4.33(1) and continuity). As $\beta > 1$, $H_\mu := Z_{\epsilon,\mu} + c_{\mathrm{cubic}} Z_{\epsilon,\mu}^3 - \frac{\mu}{\mu_*} Z_0 \in L^1([R, \infty))$, so $F_\mu(y) := \int_y^\infty H_\mu$ is C^∞ on $[R, \infty)$ with $F_\mu' = -H_\mu$, and $f_{\mathrm{osc}} = -2\chi_R \operatorname{Re} F_\mu$ (extended by 0) is real and C^∞ on $\mathbb{R}$.

□

Lemma 7.14 (Uniform bounds for the family)**.** *There is an admissible $K_0 \geq 1$ such that for all $0 \leq \epsilon \leq \epsilon_0$ and $\mu \in \overline{B}$, with $f = \overline{f}_{\epsilon,\mu}$ and $g := f - W$:*

1. *$f \in \mathcal{P}(K_0)$;*
2. *$|g(y)| \leq K_0\langle y\rangle^{-p}$ for all $y \in \mathbb{R}$;*
3. *$|g^{(i)}(y)| \leq K_0\langle y\rangle^{-p-i}$ for $y \leq 0$ and $i = 0, 1, 2, 3$ (note $g = f + y$ on $(-\infty, 0]$).*

Proof. By Lemma 7.13, $f \in C^\infty$. By Proposition 4.49 (whose constants are uniform in ϵ, μ) with $j = 0, 1, 2$: for $y \geq 0$, $|f| \lesssim \langle y\rangle^{-p}$, $|f'| \lesssim \langle y\rangle^{-\beta} \leq 1$, $|f''| \lesssim \langle y\rangle^{-\beta+1/p} = \langle y\rangle^{\theta}$, $|f'''| \lesssim \langle y\rangle^{-\beta+2/p} \leq \langle y\rangle^{2/p}$; for $y \leq 0$, $|\partial^i(f + y)| \lesssim \langle y\rangle^{-p-i}$, so $|f'| \leq 1 + C$, $|f''| \lesssim \langle y\rangle^{-p-2} \leq \langle y\rangle^{-1}$, $|f'''| \lesssim \langle y\rangle^{-p-3}$. Since $W = 0$ on $[0, \infty)$ and $W = -y$ on $(-\infty, 0]$, (2)–(3) follow. Take K_0 the maximum of the constants. □

Lemma 7.15 (Continuity of the family in μ)**.** *Fix $0 \leq \epsilon \leq \epsilon_0$ and $k \geq 0$. The map $\mu \mapsto \overline{f}_{\epsilon,\mu}$ is continuous from $\overline{B}$ into C^k_{loc}. In particular (case $k = 2$) it is continuous into C^2_{loc}.*

Proof. Let $\mu_n \to \mu$ in $\overline{B}$. (i) $f_{0,\mu} = f_0 + 2\operatorname{Re}[(\mu/\mu_* - 1)f_{0,+}]$ is affine in μ with smooth coefficient $f_{0,+}$, so $\|f_{0,\mu_n} - f_{0,\mu}\|_{C^k([-M,M])} \leq \frac{2|\mu_n - \mu|}{|\mu_*|}\|f_{0,+}\|_{C^k([-M,M])} \to 0$. (ii) f_{slow} does not depend on μ ($c_\pm$ depend only on ϵ, by (4.5)). (iii) For f_{osc}: by (4.5), $c_{\log}(\mu) = -n_\epsilon|\mu|^2/p$ is continuous in μ; by (4.6), $\lambda_{\epsilon,\mu}(t)$ and $\Psi_{\epsilon,\mu}(t) = \Psi_0(R) + \int_R^t \lambda_{\epsilon,\mu}$ (explicitly a combination of powers of t and

$c_{\log}(\mu)\log(t/R)$) and all their t-derivatives are jointly continuous in $(\mu,t)\in\overline{B}\times[R,\infty)$; since $\lambda_{\epsilon,\mu}\geq\frac{1}{2}t^{1/p}>0$, the same holds for $Z_{\epsilon,\mu}(t)$ in (4.7) and for $H_\mu=Z_{\epsilon,\mu}+c_{\mathrm{cubic}}Z_{\epsilon,\mu}^3-\frac{\mu}{\mu_*}Z_0$. By uniform continuity on compacts $\overline{B}\times[R,T]$, $\partial_t^l H_{\mu_n}\to\partial_t^l H_\mu$ uniformly on $[R,T]$ for each l,T. Moreover $|H_\nu(t)|\leq Ct^{-\beta}$ uniformly in $\nu\in\overline{B}$ (as in Lemma 7.13), and $\beta>1$; by dominated convergence, $\sup_{y\geq R}|F_{\mu_n}(y)-F_\mu(y)|\leq\int_R^\infty|H_{\mu_n}-H_\mu|\,dt\to0$. Since $\partial_y^l F_\nu=-\partial_y^{l-1}H_\nu$ for $l\geq1$, the Leibniz rule applied to $f_{\mathrm{osc}}=-2\chi_R\operatorname{Re}F_\nu$ gives $f_{\mathrm{osc}}^{(\mu_n)}\to f_{\mathrm{osc}}^{(\mu)}$ in C^k_{loc}. Summing (i)–(iii) via (4.10) proves the claim.

□

Lemma 7.16 (Bounds on the slow correction)**.** *There is an admissible C such that for all $0\leq\epsilon\leq\epsilon_0$: $|c_+|+|c_-|\leq C\epsilon$, and $|f_{\mathrm{slow}}^{(i)}(y)|\leq C\epsilon\langle y\rangle^{-p-i}$ for all $y\in\mathbb{R}$ and $i=0,1,2,3$. Moreover $f_{\mathrm{slow}}(y)=c_-|y|^{-p}$ for $y\leq-2R$.*

Proof. $c_+=-c_\epsilon/(p+1)$ and $|c_\epsilon|\lesssim\epsilon$ by Lemma 3.12 (directly: $|(1+\epsilon t^2)^{-\alpha/2}-1|\leq\frac{\alpha}{2}\epsilon t^2$ and $\int_0^1 t^2(1-t)^{p-1}dt<\infty$). $|c_-|\leq\frac{c_\alpha}{p+1}\int_0^1\frac{\alpha}{2}\epsilon t^2t^{p-1}dt\leq C\epsilon$ by Lemma 7.4. For $y^{-p}\chi_R(y)$: by Leibniz, $\partial^i(y^{-p}\chi_R)=\sum_{l\leq i}\binom{i}{l}\partial^{i-l}(y^{-p})\chi_R^{(l)}$; $\chi_R^{(l)}$ for $l\geq1$ is supported in $[R,2R]$ and bounded by $C_lR^{-l}\leq C_l2^ly^{-l}$ there, so $|\partial^i(y^{-p}\chi_R)|\leq Cy^{-p-i}\leq C\langle y\rangle^{-p-i}$ on $\operatorname{supp}\chi_R\subset[R,\infty)$, $R\geq1$. The same for $|y|^{-p}\chi_R(-y)$. Finally $\chi_R(-y)=1$, $\chi_R(y)=0$ for $y\leq-2R$.

□

Lemma 7.17 (Residual splitting)**.** *For all $0\leq\epsilon\leq\epsilon_0$, $\mu\in\overline{B}$, with $f=\overline{f}_{\epsilon,\mu}$: the functions f, $f_{0,\mu}$, f_{slow}, f_{osc} are C^∞ with bounded first derivative, and for every $y\in\mathbb{R}$*

$$\widetilde{\mathcal{E}}_\epsilon(f)(y)-G_\mu(y)=\mathcal{L}_0(f_{\mathrm{slow}})(y)+\mathcal{L}_0(f_{\mathrm{osc}})(y)+\mathcal{Q}_\epsilon(f)(y).$$

Proof. Boundedness of f': Lemma 7.14; of f'_{slow}: Lemma 7.16; of f'_{osc}: (4.16) with $j=1$ gives $|f'_{\mathrm{osc}}|\lesssim\epsilon\langle y\rangle^{-p-\beta+1/p}$ and $-p-\beta+\frac{1}{p}=-p+\theta<0$; hence also $f'_{0,\mu}=f'-f'_{\mathrm{slow}}-f'_{\mathrm{osc}}$ is bounded. By Lemma 7.6, $\widetilde{\mathcal{E}}_\epsilon(f)=\mathcal{L}_0(f)+\mathcal{Q}_\epsilon(f)$; $\mathcal{L}_0$ is linear on C^2 functions with bounded derivative (each singular integral converges absolutely), so $\mathcal{L}_0(f)=\mathcal{L}_0(f_{0,\mu})+\mathcal{L}_0(f_{\mathrm{slow}})+\mathcal{L}_0(f_{\mathrm{osc}})$, and $\mathcal{L}_0(f_{0,\mu})=G_\mu$ by Lemma 4.33(3). This is the first equality of (4.28). □

Lemma 7.18 (Properties of G_μ)**.** *For $\mu\in\mathbb{C}$ let G_μ be as in Definition 2.13: $\widehat{G_\mu}(\xi)=(1-\frac{\mu}{\mu_*})\frac{\chi_+'(\xi)}{\xi}e^{-i\Phi_0(\xi)}$ for $\xi>0$, $\widehat{G_\mu}(\xi)=\overline{\widehat{G_\mu}(-\xi)}$ for $\xi<0$, $\widehat{G_\mu}(0)=0$, with $\Phi_0(\xi)=\xi|\xi|^p/\alpha$. Then:*

1. *$\mu_*\neq0$; $\widehat{G_\mu}\in C_c^\infty(\mathbb{R})$ with support in $[-2,-1]\cup[1,2]$; G_μ is a real-valued Schwartz function, independent of ϵ;*
2. *$G_\mu=\operatorname{Re}(1-\frac{\mu}{\mu_*})G_+ + \operatorname{Im}(1-\frac{\mu}{\mu_*})G_-$ with $G_\pm$ of Definition 2.11; in particular $G_{\mu_*}=0$;*
3. *for every $\gamma>0$ and $N\geq0$ there is $C=C(p,\gamma,N)$ with $\sup_y\langle y\rangle^N(|G_\mu(y)|+|G'_\mu(y)|)+\|G_\mu\|_{Y_\gamma}\leq C|1-\mu/\mu_*|$ and $\|G_\mu-G_\nu\|_{Y_\gamma}+\|G'_\mu-G'_\nu\|_{C^1(\mathbb{R})}\leq C|\mu-\nu|/|\mu_*|$; in particular $\sup_{\mu\in\overline{B}}\|G_\mu\|_{Y_\gamma}<\infty$ and $\mu\mapsto G_\mu$ is continuous into Y_γ and into C^1_{loc}.*

Proof. (1) $\mu_* = -i/\sqrt{2\pi p} \neq 0$. χ_+' is smooth and supported in $[1,2]$ (as $\chi_+ = 0$ on $(-\infty,1]$, $= 1$ on $[2,\infty)$), and $\xi \mapsto \xi^{-1}e^{-i\Phi_0(\xi)}$ is smooth on $(0,\infty)$; the conjugate-symmetric extension is smooth and compactly supported away from 0. The inverse Fourier transform (Definition 1.2) of a C_c^∞ function is Schwartz, and conjugate symmetry $\widehat{G}(-\xi) = \overline{\widehat{G}(\xi)}$ is equivalent to G real. No ϵ appears. (2) Write $1-\mu/\mu_* = a+ib$, $a,b \in \mathbb{R}$. On $\xi > 0$, $(a+ib)\frac{\chi_+'}{\xi}e^{-i\Phi_0} = a\widehat{G_+} + b\widehat{G_-}$ by Definition 2.11; both sides are conjugate-symmetric (real coefficients), so they agree on $\mathbb{R}$. For $\mu = \mu_*$, $a = b = 0$. (3) $G_\pm$ are fixed Schwartz functions, so (2) gives the bounds with $C = \sup_y \langle y\rangle^N(|G_+|+|G_+'|+|G_-|+|G_-'|)$ etc.; the Y_γ norm (Definition 2.4) is bounded by $\sup\langle y\rangle^{\gamma+1}(|G|+|G'|)$ since $\langle y\rangle^{-1/p} \leq 1 \leq \langle y\rangle$. Linearity in μ gives the difference bounds. □

7.3 Residual bounds for $y \leq -8R$

In this section $f = \overline{f}_{\epsilon,\mu}$, $g = f - W$, and K_0 is the constant of Lemma 7.14.

Lemma 7.19 ($\mathcal{Q}_\epsilon(W)$ on the negative axis)**.** *For $0 \leq \epsilon \leq 1$ and $y < 0$, the integral defining $\mathcal{Q}_\epsilon(W)(y)$ converges absolutely and*

$$\mathcal{Q}_\epsilon(W)(y) = -c_\alpha|y|^{-p}\int_0^1 [A_\epsilon(t)-1]t^{p-1}\,dt = -(p+1)c_-|y|^{-p}.$$

Proof. For $y < 0$: $W'(y) = -1$, and $W'(z) = -1$ for $z < 0$, $W'(z) = 0$ for $z > 0$; so the integrand vanishes for $z < 0$. For $z > 0$: $W'(y) - W'(z) = -1$, $|y-z| = |y| + z$, $S_W(y,z) = \frac{-y-0}{y-z} = -\frac{|y|}{|y|+z}$, and A_ϵ is even, so $\mathcal{Q}_\epsilon(W)(y) = -c_\alpha\int_0^\infty \frac{A_\epsilon(|y|/(|y|+z))-1}{(|y|+z)^{1+p}}dz$, absolutely convergent since $|A_\epsilon - 1| \leq \epsilon$ on $[0,1]$. Substituting $t = |y|/(|y|+z)$ ($dz = -|y|t^{-2}dt$, $(|y|+z)^{-1-p} = t^{1+p}|y|^{-1-p}$) gives $|y|^{-p}\int_0^1(A_\epsilon(t)-1)t^{p-1}dt$. The last equality is the definition of c_- in (4.5). □

Lemma 7.20 (Slow part, negative side)**.** *There is an admissible C such that for all ϵ,μ, $y \leq -8R$ and $j \in \{0,1\}$: $\mathcal{L}_0(f_{\mathrm{slow}})(y) + \mathcal{Q}_\epsilon(W)(y) = |D|^p f_{\mathrm{slow}}'(y)$ and*

$$\left|\partial_y^j\big(\mathcal{L}_0(f_{\mathrm{slow}}) + \mathcal{Q}_\epsilon(W)\big)(y)\right| \leq C\epsilon\langle y\rangle^{-2p-1-j}.$$

Proof. On $(-\infty,-2R]$, $f_{\mathrm{slow}} = c_-(-y)^{-p}$ (Lemma 7.16), so $f_{\mathrm{slow}}' = pc_-(-y)^{-p-1}$ and $f_{\mathrm{slow}} - yf_{\mathrm{slow}}' = (1+p)c_-|y|^{-p}$, which cancels $\mathcal{Q}_\epsilon(W)$ by Lemma 7.19. The identity holds on the open set $y < -2R$, so derivatives may be taken. Apply Lemma 7.10 to $h = f_{\mathrm{slow}}$ with $K = C\epsilon$ (Lemma 7.16); note $|y| \geq 8R \geq 1$. □

Lemma 7.21 (Oscillatory part, negative side)**.** *There is an admissible C such that for all ϵ,μ, $y \leq -8R$ and $j \in \{0,1\}$:*

$$\mathcal{L}_0(f_{\mathrm{osc}})(y) = -c_\alpha\alpha\int_R^\infty \frac{f_{\mathrm{osc}}(z)}{(z-y)^{\alpha+1}}dz, \qquad |\partial_y^j\mathcal{L}_0(f_{\mathrm{osc}})(y)| \leq C\epsilon\langle y\rangle^{-p-2-j} \leq C\epsilon\langle y\rangle^{-2p-1-j}.$$

Proof. By Lemma 7.13, $f_{\mathrm{osc}} = f_{\mathrm{osc}}' = 0$ on $(-\infty,R]$, so for $y \leq -8R$, $\mathcal{L}_0(f_{\mathrm{osc}})(y) = |D|^p f_{\mathrm{osc}}'(y) = -c_\alpha\int_R^\infty f_{\mathrm{osc}}'(z)(z-y)^{-\alpha}dz$ (non-singular, $z - y \geq |y|$). Integrate by parts on $[R,T]$ and let $T \to \infty$: $f_{\mathrm{osc}}(R) = 0$ and $f_{\mathrm{osc}}(T) \to 0$ by (4.16), giving

the formula. By (4.17), $\|f_{\rm osc}\|_{L^1} \le C\epsilon$. Since $\partial_y (z-y)^{-\alpha-1} = (\alpha+1)(z-y)^{-\alpha-2}$, differentiation under the integral is justified (majorant $|f_{\rm osc}(z)|(8R)^{-\alpha-2}$), and $|\partial_y^j \mathcal{L}_0(f_{\rm osc})(y)| \le C\|f_{\rm osc}\|_{L^1}|y|^{-p-2-j}$. Finally $p+2 > 2p+1$ as $p<1$, and $|y| \ge 1$. □

Lemma 7.22 (Splitting of $\mathcal{Q}_\epsilon(f) - \mathcal{Q}_\epsilon(W)$)**.** *For $y<0$ let $\chi_y(z) := \chi((y-z)/|y|)$. Then*

$$\mathcal{Q}_\epsilon(f)(y) - \mathcal{Q}_\epsilon(W)(y) = \mathcal{I}_1(y) + \mathcal{I}_2(y) + \mathcal{I}_3(y), \tag{7.1}$$

where

$$\begin{aligned}
\mathcal{I}_1 &= c_\alpha \int_{\mathbb{R}} \chi_y(z) \frac{f'(y)-f'(z)}{|y-z|^\alpha} [A_\epsilon(S_f(y,z)) - 1] dz, \\
\mathcal{I}_2 &= c_\alpha \int_{\mathbb{R}} (1-\chi_y(z)) \frac{g'(y)-g'(z)}{|y-z|^\alpha} [A_\epsilon(S_f(y,z)) - 1] dz, \\
\mathcal{I}_3 &= -c_\alpha \int_0^\infty \frac{A_\epsilon(S_f(y,z)) - A_\epsilon(S_W(y,z))}{(z-y)^\alpha} dz,
\end{aligned}$$

all absolutely convergent ($g'(z)$ is defined for $z \ne 0$).

Proof. Absolute convergence: near $z=y$ only $\mathcal{I}_1$ has a singular kernel, with integrand $O(|y-z|^{-p})$; elsewhere integrands are $O(\epsilon|y-z|^{-\alpha})$ with bounded numerators. On $\operatorname{supp}\chi_y$, $|y-z| \le |y|/2$, so $z \le y/2 < 0$ and $W'(y) - W'(z) = 0$; hence $\mathcal{Q}_\epsilon(W)(y) = c_\alpha \int (1-\chi_y) \frac{W'(y)-W'(z)}{|y-z|^\alpha} [A_\epsilon(S_W) - 1] dz$. Split $\mathcal{Q}_\epsilon(f)$ with $\chi_y + (1-\chi_y)$, write $f' = W' + g'$ and $A_\epsilon(S_f) - 1 - (A_\epsilon(S_W) - 1) = A_\epsilon(S_f) - A_\epsilon(S_W)$: the far part of $\mathcal{Q}_\epsilon(f) - \mathcal{Q}_\epsilon(W)$ equals $\mathcal{I}_2 + c_\alpha \int (1-\chi_y) \frac{W'(y)-W'(z)}{|y-z|^\alpha} [A_\epsilon(S_f) - A_\epsilon(S_W)] dz$. In the last integral $W'(y) - W'(z)$ is 0 for $z<0$ and -1 for $z>0$, and for $z>0$, $|y-z| = |y| + z > |y|/2$ so $1-\chi_y = 1$; this is $\mathcal{I}_3$.

□

Lemma 7.23 (Near term $\mathcal{I}_1$)**.** *There is an admissible C with $|\partial_y^j \mathcal{I}_1(y)| \le C\epsilon|y|^{-2p-1-j}$ for all ϵ, μ, $y \le -8R$, $j \in \{0,1\}$. Moreover $\mathcal{I}_1$ is differentiable at every $y \le -8R$.*

Proof. Substitute $z = y - s$: $\mathcal{I}_1 = c_\alpha \int \chi(s/|y|) \frac{f'(y)-f'(y-s)}{|s|^\alpha} [A_\epsilon(S_f(y, y-s)) - 1] ds$. On the support, $|s| \le |y|/2$, so $y - s \in [\frac{3}{2}y, \frac{1}{2}y] \subset (-\infty, 0)$ and $|y-s| \ge |y|/2$. There $f' = -1 + g'$, so by Lemma 7.14(3) and the mean value theorem, $|\partial_y^j (f'(y) - f'(y-s))| = |g^{(1+j)}(y) - g^{(1+j)}(y-s)| \le CK_0|y|^{-p-2-j}|s|$ for $j = 0, 1$. By Lemma 7.4 (with $|S_f| \le K_0$), $|A_\epsilon(S_f) - 1| + |A_\epsilon'(S_f)| \le C\epsilon$, and $|\partial_y [S_f(y, y-s)]| = |f'(y) - f'(y-s)|/|s| \le C|y|^{-p-2}$. The y-derivative of $\chi(s/|y|)$ is bounded by $C|y|^{-1}$ (as $|s| \le |y|$). Differentiating under the integral (justified by these bounds, uniform for y in compact subsets of $(-\infty, -8R]$), every term in ∂_y^j of the integrand is bounded by $C\epsilon|y|^{-p-2-j}|s|^{1-\alpha} = C\epsilon|y|^{-p-2-j}|s|^{-p}$ on $|s| \le |y|/2$. Integrating with Lemma 7.9(2) gives $C\epsilon|y|^{-p-2-j}|y|^{1-p}$.

□

Lemma 7.24 (Far term $\mathcal{I}_2$)**.** *For $y \le -8R$ let $\kappa(y,z) := (1-\chi_y(z)) [A_\epsilon(S_f(y,z)) - 1] |y-z|^{-\alpha}$. Then*

1. $$\mathcal{I}_2(y) = c_\alpha g'(y) \int_{\mathbb{R}} \kappa(y,z)\, dz + c_\alpha \int_{\mathbb{R}} g(z)\, \partial_z \kappa(y,z)\, dz; \tag{7.2}$$

2. *for* $j,l\in\{0,1\}$*:* $|\partial_y^j\partial_z^l\kappa(y,z)|\le C\epsilon\,\mathbf{1}_{\{|y-z|\ge|y|/4\}}|y-z|^{-p-1-j-l}$*;*

3. $|\partial_y^j\mathcal{I}_2(y)|\le C\epsilon|y|^{-2p-1-j}$ *for* $j\in\{0,1\}$*,*

with C *admissible, uniformly in* ϵ,μ*. Moreover* $\mathcal{I}_2$ *is differentiable at every* $y\le -8R$*.*

Proof. (1) Split $\mathcal{I}_2 = c_\alpha g'(y)\int\kappa\,dz - c_\alpha\int g'(z)\kappa\,dz$. In the second, integrate by parts separately on $(-\infty,0)$ and $(0,\infty)$ (g is smooth on each closed half-line, continuous at 0). Since $|y|\ge 8R$, $z=0$ lies in $\{1-\chi_y=1\}$ and $\kappa(y,\cdot)$ is smooth near 0, so the boundary terms $\pm g(0)\kappa(y,0)$ cancel; $\kappa(y,\cdot)$ vanishes near $z=y$; and $|g(z)\kappa(y,z)|\le CK_0\epsilon\langle z\rangle^{-p}|y-z|^{-\alpha}\to 0$ as $|z|\to\infty$. (2) Use Lemma 7.4 ($|A_\epsilon-1|,|A_\epsilon'|,|A_\epsilon''|\le C\epsilon$ on $[-K_0,K_0]$), Lemma 7.5 ($|\partial_y S_f|,|\partial_z S_f|\le 2K_0|y-z|^{-1}$, $|\partial_y\partial_z S_f|\le 4K_0|y-z|^{-2}$), and the cutoff: with $w=(y-z)/|y| = -1+z/y$ (as $y<0$), $\partial_z w=-1/|y|$, $\partial_y w=-z/y^2$, and on $\operatorname{supp}\chi'(w)$ one has $|y|/4\le|y-z|\le|y|/2$, $|z|\le\frac{3}{2}|y|$, so every derivative of χ_y of total order $m\le 2$ is bounded by $C|y|^{-m}\le C|y-z|^{-m}$; finally $|\partial_y^j\partial_z^l|y-z|^{-\alpha}|\le C|y-z|^{-\alpha-j-l}$. Leibniz gives (2). (3) Differentiating (1) under the integral (justified by (2)): $\partial_y\mathcal{I}_2 = c_\alpha g''(y)\int\kappa + c_\alpha g'(y)\int\partial_y\kappa + c_\alpha\int g\,\partial_y\partial_z\kappa$. By Lemma 7.14(3), $|g^{(1+i)}(y)|\le K_0\langle y\rangle^{-p-1-i}$; by (2) and Lemma 7.9(2), $\int|\partial_y^i\kappa|dz\le C\epsilon|y|^{-p-i}$, so the local terms are $\le C\epsilon|y|^{-2p-1-j}$. By Lemma 7.14(2) and Lemma 7.9(1) with $q=p+2+j$: $\int|g(z)||\partial_y^j\partial_z\kappa|dz\le C\epsilon\int_{|y-z|\ge|y|/4}\langle z\rangle^{-p}|y-z|^{-p-2-j}dz\le C\epsilon|y|^{-2p-1-j}$.

□

Lemma 7.25 (Corner term $\mathcal{I}_3$)**.** *There is an admissible* C *with* $|\partial_y^j\mathcal{I}_3(y)|\le C\epsilon|y|^{-2p-1-j}$ *for all* ϵ,μ*,* $y\le -8R$*,* $j\in\{0,1\}$*. Moreover* $\mathcal{I}_3$ *is differentiable at every* $y\le -8R$*.*

Proof. Let $y\le -8R$, $z>0$, so $z-y=|y|+z$. By Lemma 7.5, $S_f - S_W = S_g$ and $|S_g(y,z)|\le\frac{|g(y)|+|g(z)|}{|y|+z}\le K_0\frac{|y|^{-p}+\langle z\rangle^{-p}}{|y|+z}$. All secants lie in $[-K_0,K_0]$ ($K_0\ge 1$), so by Lemma 7.4 and the mean value theorem $|A_\epsilon(S_f)-A_\epsilon(S_W)|\le C\epsilon\frac{|y|^{-p}+\langle z\rangle^{-p}}{|y|+z}$. Next $\partial_y(A_\epsilon(S_f)-A_\epsilon(S_W)) = A_\epsilon'(S_f)\partial_y S_g + [A_\epsilon'(S_f)-A_\epsilon'(S_W)]\partial_y S_W$, with $\partial_y S_g=\frac{g'(y)-S_g}{y-z}$, $|g'(y)|\le K_0|y|^{-p-1}$, $|A_\epsilon'(S_f)-A_\epsilon'(S_W)|\le C\epsilon|S_g|$ and $|\partial_y S_W|\le\frac{2}{|y|+z}$; hence $|\partial_y(A_\epsilon(S_f)-A_\epsilon(S_W))|\le C\epsilon\big[\frac{|y|^{-p-1}}{|y|+z}+\frac{|y|^{-p}+\langle z\rangle^{-p}}{(|y|+z)^2}\big]$. Also $\partial_y(z-y)^{-\alpha}=\alpha(z-y)^{-\alpha-1}$. Differentiating under the integral (majorants as below, locally uniform in y) and using Lemma 7.9(3) with $q=p+2$ and $q=p+3$: $|\mathcal{I}_3|\le C\epsilon\int_0^\infty\frac{|y|^{-p}+\langle z\rangle^{-p}}{(|y|+z)^{p+2}}dz\le C\epsilon|y|^{-2p-1}$ and $|\partial_y\mathcal{I}_3|\le C\epsilon\int_0^\infty\big[\frac{|y|^{-p-1}}{(|y|+z)^{p+2}}+\frac{|y|^{-p}+\langle z\rangle^{-p}}{(|y|+z)^{p+3}}\big]dz\le C\epsilon|y|^{-2p-2}$.

□

Proposition 7.26 (Residual bound for $y\le -8R$)**.** *There is an admissible constant* C *such that for all* $0\le\epsilon\le\epsilon_0$*,* $\mu\in\overline{B}$*,* $y\le -8R$ *and* $j\in\{0,1\}$*, with* $f=\overline{f}_{\epsilon,\mu}$*,*

$$|\partial_y^j(\widetilde{\mathcal{E}}_\epsilon(f)-G_\mu)(y)|\le C\epsilon\langle y\rangle^{-2p-1-j}.$$

In particular, for every $\gamma\le 2p+1$ *(so for every* γ *with* $\beta+\frac{1}{p}<\gamma<2p+1$*),* $|\partial_y^j(\widetilde{\mathcal{E}}_\epsilon(f)-G_\mu)(y)|\le C\epsilon\langle y\rangle^{-\gamma-j}$*.*

Proof. By Lemma 7.17 and Lemma 7.22, $\widetilde{\mathcal{E}}_\epsilon(f) - G_\mu = [\mathcal{L}_0(f_{\text{slow}}) + \mathcal{Q}_\epsilon(W)] + \mathcal{L}_0(f_{\text{osc}}) + \mathcal{I}_1 + \mathcal{I}_2 + \mathcal{I}_3$ pointwise on $(-\infty, -8R]$ (each summand is C^1 there, by the lemmas cited). Sum the bounds of Lemmas 7.20, 7.21 and 7.23 to 7.25, using $|y|^{-1} \le \sqrt{2}\langle y\rangle^{-1}$ for $|y| \ge 1$. The last statement follows from $\langle y\rangle^{-2p-1} \le \langle y\rangle^{-\gamma}$. □

7.4 Residual bounds for $|y| \le 8R$

Lemma 7.27 (Linear corrections on the compact region)**.** *There is an admissible C such that for all ϵ, μ:*

$$\|f_{\text{osc}}\|_{L^\infty(\mathbb{R})} + \|f_{\text{osc}}\|_{C^3([-8R-1,8R+1])} \le C\epsilon, \tag{7.3}$$

and $\|\mathcal{L}_0(f_{\text{slow}})\|_{C^1([-8R,8R])} + \|\mathcal{L}_0(f_{\text{osc}})\|_{C^1([-8R,8R])} \le C\epsilon$.

Proof. Apply Lemma 7.11 with $M = 8R$. For $u = f_{\text{slow}}$: by Lemma 7.16, u, u' are bounded and $\|u\|_{C^3(\mathbb{R})} + \|u\|_{L^\infty} \le C\epsilon$. For $u = f_{\text{osc}}$: it is C^∞, vanishes on $(-\infty, R]$ (Lemma 7.13), and by (4.16), $|f_{\text{osc}}^{(i)}(y)| \le C\epsilon\langle y\rangle^{-p-\beta+i/p}$ for $y \ge R$; hence $\|u\|_{L^\infty} \le C\epsilon$, u' bounded (as $-p-\beta+\frac{1}{p} < 0$), and $\|u\|_{C^3([-8R-1,8R+1])} \le C\epsilon\langle 8R+1\rangle^{3/p}$, which is (7.3). □

Lemma 7.28 (Nonlinear remainder on the compact region)**.** *For every $M \ge 1$ there is an admissible $C = C(M)$ such that for all ϵ, μ: $\|\mathcal{Q}_\epsilon(\overline{f}_{\epsilon,\mu})\|_{C^1([-M,M])} \le C\epsilon$.*

Proof. $\mathcal{Q}_\epsilon = \mathcal{T}_{A_\epsilon - 1}$ (Definition 7.2), $\overline{f}_{\epsilon,\mu} \in \mathcal{P}(K_0)$, and $\|A_\epsilon - 1\|_{K_0} \le CK_0^2\epsilon$ by Lemma 7.4; apply Lemma 7.7. □

Proposition 7.29 (Residual bound for $|y| \le 8R$)**.** *There is an admissible C (depending on R) such that for all $0 \le \epsilon \le \epsilon_0$ and $\mu \in \overline{B}$, $\widetilde{\mathcal{E}}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu \in C^1(\mathbb{R})$ and*

$$\|\widetilde{\mathcal{E}}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu\|_{C^1([-8R,8R])} \le C\epsilon.$$

Proof. C^1 regularity: $\widetilde{\mathcal{E}}_\epsilon(f) = f - yf' + \mathcal{T}_{A_\epsilon}(f)$ (Lemma 7.6) with $\mathcal{T}_{A_\epsilon}(f) \in C^1$ (Lemma 7.7, $f \in \mathcal{P}(K_0)$), and G_μ is Schwartz. The bound: by Lemma 7.17, on $[-8R, 8R]$ the difference is $\mathcal{L}_0(f_{\text{slow}}) + \mathcal{L}_0(f_{\text{osc}}) + \mathcal{Q}_\epsilon(f)$; apply Lemma 7.27 and Lemma 7.28 with $M = 8R$. □

7.5 C^1 regularity of $\mathcal{E}_5^Y$ for $y \ge 8R$

Lemma 7.30 ($\mathcal{E}_5^Y$ is C^1 where the cubic term is)**.** *Let $\frac{3}{4} < p < 1$. For every $\delta \in (0, |\mu_*|/2]$ there are $\epsilon_0 \in (0,1]$ and $R_0 \ge 2$ such that for all $R \ge R_0$, $\epsilon \in [0, \epsilon_0]$, $\mu \in \overline{B}$, $Y \ge 8R$ and $y \in [Y, 2Y]$: if $\mathcal{Q}_{\epsilon,3}(v_Y)$ is C^1 at y, then $\mathcal{E}_5^Y$ is C^1 at y.*

Proof. Take $\epsilon_0 = \min\{1, \epsilon_0^*(\delta)\}$ with $\epsilon_0^*(\delta)$ of Definition 5.1, so that the standing assumptions of the chapter on $\mathcal{E}_5$ hold. By Lemma 5.6, near $y > 2R$, $\mathcal{E}_5^Y = \mathcal{R}_{\text{lin}} + \mathcal{R}_{\text{cub}}$. $\mathcal{R}_{\text{lin}}$ is C^1 near y by Proposition 5.19, and $\mathcal{R}_{\text{cub}}$ is $\mathcal{Q}_{\epsilon,3}(v_Y)$ minus explicit expressions in λ and Z, which are C^∞ on (R, ∞) (Lemma 4.39).

□

7.6 Assembly: proof of Theorem 2.20

Lemma 7.31 (Patching three regions into Y_γ)**.** *Let $\gamma > 0$, $\rho \geq 0$, $a, b \geq 0$, and $H \in C^1(\mathbb{R};\mathbb{R})$ with: $|H^{(j)}(y)| \leq a\langle y\rangle^{-\gamma+j/p}$ for $y \geq \rho$, $|H^{(j)}(y)| \leq a\langle y\rangle^{-\gamma-j}$ for $y \leq -\rho$ ($j = 0, 1$), and $\|H\|_{C^1([-\rho,\rho])} \leq b$. Then $H \in Y_\gamma$ and $\|H\|_{Y_\gamma} \leq 4a + 4\langle\rho\rangle^{\gamma+1}b$.*

Proof. $H \in C^1 \subset W^{1,\infty}_{\mathrm{loc}}$. In the norm of Definition 2.4, for $y \geq \rho$: $\langle y\rangle^\gamma(|H| + \langle y\rangle^{-1/p}|H'|) \leq 2a$; for $0 \leq y \leq \rho$: $\leq \langle\rho\rangle^\gamma \cdot 2b$; for $y \leq -\rho$: $\langle y\rangle^\gamma(|H| + \langle y\rangle|H'|) \leq 2a$; for $-\rho \leq y \leq 0$: $\leq \langle\rho\rangle^\gamma(1 + \langle\rho\rangle)b$. Each of the two suprema is $\leq 2a + 2\langle\rho\rangle^{\gamma+1}b$. □

Lemma 7.32 (Divided residual in Y_γ)**.** *For every γ with $\beta + \frac{1}{p} < \gamma < 2p+1$ there is an admissible $C = C(\gamma)$ such that for all $0 \leq \epsilon \leq \epsilon_0$, $\mu \in \overline{B}$: $\widetilde{\mathcal{E}}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu \in Y_\gamma$ and $\|\widetilde{\mathcal{E}}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu\|_{Y_\gamma} \leq C\epsilon$.*

Proof. $H := \widetilde{\mathcal{E}}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu \in C^1$ by Proposition 7.29. Apply Lemma 7.31 with $\rho = 8R$, $a = C\epsilon$ from Proposition 4.57 (region $y \geq 8R$, weight $\langle y\rangle^{-\gamma+j/p}$) and Proposition 7.26 (region $y \leq -8R$, weight $\langle y\rangle^{-\gamma-j}$), and $b = C\epsilon$ from Proposition 7.29. □

Lemma 7.33 (Multiplication by $B_\epsilon(f')$ on Y_γ)**.** *Let $\gamma > 0$, $K \geq 1$, $0 \leq \epsilon \leq 1$, $f \in \mathcal{P}(K)$. There is $C = C(p, K)$ such that:*

1. *if $H \in Y_\gamma \cap C^1$ then $B_\epsilon(f')H \in Y_\gamma \cap C^1$ and $\|B_\epsilon(f')H\|_{Y_\gamma} \leq C\|H\|_{Y_\gamma}$;*
2. *if $G \in C^1$ with $N_\gamma(G) := \sup_y \langle y\rangle^{\gamma+1}(|G(y)|+|G'(y)|) < \infty$, then $\|(B_\epsilon(f')-1)G\|_{Y_\gamma} \leq C\epsilon N_\gamma(G)$.*

Proof. By Lemma 7.4 and $|f'| \leq K$: $|B_\epsilon(f')| \leq 1 + K^2$, $|B_\epsilon(f') - 1| \leq K^2\epsilon$, $|\partial_y B_\epsilon(f')| = |B'_\epsilon(f')f''| \leq \alpha\epsilon K|f''|$, and by Definition 7.3(2) $|f''| \leq K\langle y\rangle^\theta \leq K\langle y\rangle^{1/p}$ for $y \geq 0$, $|f''| \leq K\langle y\rangle^{-1}$ for $y \leq 0$. (1) $(BH)' = B'(f')f''H + BH'$. For $y \geq 0$: $\langle y\rangle^{\gamma-1/p}|B'(f')f''H| \leq \alpha\epsilon K^2\langle y\rangle^\gamma|H|$; for $y \leq 0$: $\langle y\rangle^{\gamma+1}|B'(f')f''H| \leq \alpha\epsilon K^2\langle y\rangle^\gamma|H|$. The terms BH, BH' are bounded by $(1 + K^2)$ times the corresponding terms of H. (2) Same computation with H replaced by G and B by $B-1$, all terms carrying a factor ϵ: for $y \geq 0$, $\langle y\rangle^\gamma|B-1||G| + \langle y\rangle^{\gamma-1/p}(|B'f''||G| + |B-1||G'|) \leq C\epsilon N_\gamma(G)$, and for $y \leq 0$ the weight $\langle y\rangle^{\gamma+1}$ is absorbed by $N_\gamma(G)$. □

Lemma 7.34 (Residual in Y_γ)**.** *For every γ with $\beta + \frac{1}{p} < \gamma < 2p+1$ there is an admissible $C = C(\gamma)$ such that for all $0 \leq \epsilon \leq \epsilon_0$, $\mu \in \overline{B}$: $\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) \in Y_\gamma \cap C^1(\mathbb{R})$,*

$$\|\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu\|_{Y_\gamma} \leq C\epsilon, \qquad \|\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu})\|_{Y_\gamma} \leq C.$$

Proof. By Lemma 7.6, $\mathcal{E}_\epsilon(f) - G_\mu = B_\epsilon(f')[\widetilde{\mathcal{E}}_\epsilon(f) - G_\mu] + [B_\epsilon(f') - 1]G_\mu$. Apply Lemma 7.33(1) with $H = \widetilde{\mathcal{E}}_\epsilon(f) - G_\mu$ (Lemma 7.32) and Lemma 7.33(2) with $G = G_\mu$, $N_\gamma(G_\mu) \leq C$ uniformly in $\mu \in \overline{B}$ (Lemma 7.18(3)), $K = K_0$. Then $\|\mathcal{E}_\epsilon(f)\|_{Y_\gamma} \leq C\epsilon + \sup_{\overline{B}}\|G_\mu\|_{Y_\gamma}$. C^1: $B_\epsilon(f') \in C^1$ as $f \in C^2$, and $\widetilde{\mathcal{E}}_\epsilon(f) \in C^1$ (Proposition 7.29). □

Lemma 7.35 (Continuity of the residuals in C^1_{loc})**.** *Fix $0 \leq \epsilon \leq \epsilon_0$. The maps $\mu \mapsto \widetilde{\mathcal{E}}_\epsilon(\overline{f}_{\epsilon,\mu})$ and $\mu \mapsto \mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu})$ are continuous from $\overline{B}$ into C^1_{loc}.*

Proof. Let $\mu_n \to \mu$, $f_n = \overline{f}_{\epsilon,\mu_n}$, $f = \overline{f}_{\epsilon,\mu}$. By Lemma 7.15, $f_n \to f$ in C^2_{loc}, hence $f_n - y f_n' \to f - yf'$ in C^1_{loc} (its derivative is $-yf_n''$). All $f_n, f \in \mathcal{P}(K_0)$ (Lemma 7.14), so Lemma 7.8 with $\Gamma = A_\epsilon$ gives $\mathcal{T}_{A_\epsilon}(f_n) \to \mathcal{T}_{A_\epsilon}(f)$ in C^1_{loc}. By Lemma 7.6 $\widetilde{\mathcal{E}}_\epsilon(f_n) \to \widetilde{\mathcal{E}}_\epsilon(f)$ in C^1_{loc}. Next $B_\epsilon(f_n') \to B_\epsilon(f')$ in C^1_{loc} (as $f_n' \to f'$ in C^1_{loc}, B_ϵ is smooth and B_ϵ, B_ϵ' are uniformly continuous on $[-K_0, K_0]$), and products of C^1_{loc}-convergent sequences converge in C^1_{loc}; so $\mathcal{E}_\epsilon(f_n) = B_\epsilon(f_n')\widetilde{\mathcal{E}}_\epsilon(f_n) \to \mathcal{E}_\epsilon(f)$. □

Lemma 7.36 (Continuity of the residual in Y_γ)**.** *Fix $0 \le \epsilon \le \epsilon_0$ and γ with $\beta + \frac{1}{p} < \gamma < 2p+1$. The map $\mu \mapsto \mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu})$ is continuous from $\overline{B}$ into Y_γ.*

Proof. Let $\gamma'' := \frac{1}{2}(\gamma + 2p + 1) \in (\gamma, 2p+1)$; this uses that the admissible range of γ is open. Let $\mu_n \to \mu$. By Lemma 7.34 at γ'', $\sup_n \|\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu_n})\|_{Y_{\gamma''}} \le C(\gamma'')$ and the limit lies in $Y_{\gamma''}$. By Lemma 7.35, the sequence converges in C^1_{loc}, hence in $W^{1,\infty}_{\mathrm{loc}}$. Proposition 3.41 (second statement, with exponents $\gamma < \gamma''$) gives convergence in Y_γ. □

Lemma 7.37 (Existence of approximate solutions: assembly)**.** *Let $\mu_* = -\frac{i}{\sqrt{2\pi p}}$ (Definition 4.15), and let $\epsilon_0, \delta > 0$, $\overline{B} = \{\mu \in \mathbb{C} : |\mu - \mu_*| \le \delta\}$ and the functions $\overline{f}_{\epsilon,\mu}$ ($0 \le \epsilon \le \epsilon_0$, $\mu \in \overline{B}$) be as in Definition 4.38. For $\mu \in \overline{B}$ let G_μ be as in Definition 2.13. Then $\mu_* \in \mathbb{C} \setminus \{0\}$, every $\overline{f}_{\epsilon,\mu}$ is a real-valued C^∞ function on $\mathbb{R}$, and for every γ with $\beta + \frac{1}{p} < \gamma < 2p+1$:*

1. *$\lim_{y\to-\infty}[\overline{f}_{\epsilon,\mu}(y) + y] = 0$ and $\lim_{y\to\infty} \overline{f}_{\epsilon,\mu}(y) = 0$;*
2. *for every $j \ge 0$ there is C_j, independent of $\epsilon \in [0, \epsilon_0]$ and $\mu \in \overline{B}$, such that for $y \ge 0$: $|\overline{f}_{\epsilon,\mu}(y)| \le C_0\langle y\rangle^{-p}$ and $|\partial^j \overline{f}'_{\epsilon,\mu}(y)| \le C_j\langle y\rangle^{-\beta+j/p}$, and for $y \le 0$: $|\partial^j(\overline{f}_{\epsilon,\mu}(y) + y)| \le C_j\langle y\rangle^{-p-j}$;*
3. *$\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) \in Y_\gamma$ for all ϵ, μ; each G_μ is a real-valued Schwartz function independent of ϵ, with $\widehat{G_\mu}(\xi) = (1 - \frac{\mu}{\mu_*})\frac{\chi_+'(\xi)}{\xi}e^{-i\Phi_0(\xi)}$, $\Phi_0(\xi) = \frac{1}{\alpha}\xi|\xi|^p$, for $\xi > 0$ and $\widehat{G_\mu}(\xi) = \overline{\widehat{G_\mu}(-\xi)}$ for $\xi < 0$; $G_{\mu_*} = 0$; and there is C_γ independent of ϵ, μ with (2.1): $\sup_{\mu\in\overline{B}} \|\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu}) - G_\mu\|_{Y_\gamma} \le C_\gamma\epsilon$ for all $0 \le \epsilon \le \epsilon_0$;*
4. *for each fixed $\epsilon \in [0, \epsilon_0]$, the map $\mu \mapsto \overline{f}_{\epsilon,\mu}$ is continuous from $\overline{B}$ into C^2_{loc}, and $\mu \mapsto \mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu})$ is continuous from $\overline{B}$ into C^1_{loc} and into Y_γ.*

This is exactly the conclusion of Theorem 2.20.

Proof. Smoothness and reality: Lemma 7.13; $\mu_* \ne 0$: Lemma 7.18(1). (2) is Proposition 4.49 (with $\partial^j\overline{f}' = \partial^{j+1}\overline{f}$), its constants being uniform in ϵ, μ. (1) follows from (2) with $j = 0$: $|\overline{f}(y)| \le C_0\langle y\rangle^{-p} \to 0$ as $y \to \infty$ and $|\overline{f}(y) + y| \le C_0\langle y\rangle^{-p} \to 0$ as $y \to -\infty$ (as $p > 0$). (3): membership and the bound (2.1) are Lemma 7.34; the properties of G_μ (Schwartz, real, independent of ϵ, explicit form, $G_{\mu_*} = 0$) are Lemma 7.18(1)–(2). (4): continuity into C^2_{loc} is Lemma 7.15 with $k = 2$; continuity of the residual into C^1_{loc} is Lemma 7.35, and into Y_γ is Lemma 7.36. □

Chapter 8

Linear inverse I: conjugation, ℓ, low frequency (paper §§6.1–6.3)

Standing notation. Throughout this chapter $p, \alpha = p+1, \beta = \frac{p+1}{2p}$, the cutoff χ_+ and the phase $\Phi_0(\xi) = \frac{1}{\alpha}\xi|\xi|^p$ are those of Definition 2.1; note $\Phi_0'(\xi) = |\xi|^p$ and $\Phi_0(t\xi) = t^\alpha \Phi_0(\xi)$ for $t > 0$. We write $c_\alpha = \Gamma(1+p)\sin(\pi p/2)/\pi$ for the constant of Definition 1.5, and $S_f(y,z) = \frac{f(y)-f(z)}{y-z}$ $(y \neq z)$ for the secant of a function f. The norms $\|\cdot\|_{X_\gamma}$, $\|\cdot\|_{Y_\gamma}$ (Definitions 2.3 and 2.4) are used with the same formulas for complex-valued functions; we then speak of $X_\gamma^{\mathbb{C}}$, $Y_\gamma^{\mathbb{C}}$. By definition of the norms, for $r \in X_\gamma$, $G \in Y_\gamma$ and (a.e.) $y \in \mathbb{R}$:

$$|r(y)| \le \|r\|_{X_\gamma}\langle y\rangle^{-\gamma}, \quad |r'(y)| \le \|r\|_{X_\gamma} w_1(y), \quad |r''(y)| \le \|r\|_{X_\gamma} w_2(y), \quad |G(y)| \le \|G\|_{Y_\gamma}\langle y\rangle^{-\gamma}, \quad |G'(y)| \le \|G\|_{Y_\gamma} w_1(y),$$

where $w_j(y) = \langle y\rangle^{-\gamma+j/p}$ for $y \ge 0$ and $w_j(y) = \langle y\rangle^{-\gamma-j}$ for $y \le 0$. Note $w_j(y) \le \langle y\rangle^{-\gamma+j/p}$ for all y.

Lemma 8.1 (Integrability of elements of X_γ, Y_γ)**.** *Let $\gamma > 1 + \frac{1}{p}$ (in particular every admissible γ). There is $C = C(p,\gamma)$ such that for $r \in X_\gamma$ and $G \in Y_\gamma$: $\|r\|_{L^1} + \|r'\|_{L^1} + \|r\|_{L^\infty} + \|r'\|_{L^\infty} \le C\|r\|_{X_\gamma}$ and $\|G\|_{L^1} + \|G'\|_{L^1} + \|G\|_{L^\infty} \le C\|G\|_{Y_\gamma}$. Moreover $G(y) \to 0$ as $|y| \to \infty$, and $\widehat{G'}(\xi) = i\xi\widehat{G}(\xi)$ for all ξ; similarly $\widehat{r'} = i\xi\widehat{r}$.*

Proof. The weights $\langle y\rangle^{-\gamma}$ and w_1 are integrable since $\gamma > 1$ and $\gamma - \frac{1}{p} > 1$, and bounded since $\gamma - \frac{1}{p} > 0$. For the Fourier identity integrate by parts on $[-R,R]$: $\int_{-R}^{R} G' e^{-iy\xi}dy = [Ge^{-iy\xi}]_{-R}^{R} + i\xi\int_{-R}^{R} Ge^{-iy\xi}dy$ (G is locally absolutely continuous), and let $R \to \infty$ using $G \to 0$ and $G, G' \in L^1$. □

8.1 Conjugation of the linear operator

The perturbed operator is $\mathcal{L}_\epsilon = \mathcal{L}_0 + b_\epsilon\partial_y + \mathcal{R}_\epsilon$ (Definition 2.17), where $b_\epsilon = c_\epsilon\chi_+ y^{-p}$ with c_ϵ from Lemma 3.12, and $\mathcal{R}_\epsilon$ is defined in Definition 8.9 below. The idea: find an increasing diffeomorphism ϕ such that $(\phi')^{p+1}\,[\mathcal{L}_0(r\circ\phi^{-1})]\circ$

$\phi = \mathcal{L}_0 r + b_\epsilon r' + (\text{small})$. Everything is first done for a real parameter c in place of c_ϵ; then one substitutes $c = c_\epsilon$, using $|c_\epsilon| \le C_0\epsilon$ (Lemma 3.12).

8.1.1 The diffeomorphism ϕ_c

Definition 8.2 (The conjugating diffeomorphism)**.** Let $c \in \mathbb{R}$ with $|c| \le \frac{1}{2}$. Define $b_c : \mathbb{R} \to \mathbb{R}$ by $b_c(y) = c\,\chi_+(y)y^{-p}$ for $y > 1$ and $b_c(y) = 0$ for $y \le 1$ (so $b_c \in C^\infty(\mathbb{R})$ since $\chi_+ = 0$ on $(-\infty, 1]$, and $|b_c(y)| \le |c|$). Since $t - b_c(t) \ge t - \frac{1}{2} > 0$ for $t \ge 1$, we may define

$$\phi_c(y) := y \quad (y \le 1), \qquad \phi_c(y) := \Big[1 + \frac{p+1}{p}\int_1^y \big(t - b_c(t)\big)^{1/p} dt\Big]^{\frac{p}{p+1}} \quad (y > 1).$$

Put $\psi_c := \phi_c^{-1}$ (a bijection of $\mathbb{R}$ by Lemma 8.4). For $\epsilon \ge 0$ with $|c_\epsilon| \le \frac{1}{2}$ we write $\phi_\epsilon := \phi_{c_\epsilon}$, $\psi_\epsilon := \psi_{c_\epsilon}$, and $b_\epsilon = b_{c_\epsilon}$ (which is the coefficient of Definition 2.17).

Lemma 8.3 (The integrand)**.** *For $|c| \le \frac{1}{2}$ let $g_c(t) := (t - b_c(t))^{1/p} - t^{1/p}$ for $t \ge \frac{1}{2}$. Then $g_c \in C^\infty((\frac{1}{2}, \infty))$, $g_c = 0$ on $(\frac{1}{2}, 1]$, and for every $j \ge 0$ there is $C_j = C_j(p, \chi_+)$ with*

$$|g_c^{(j)}(t)| \le C_j|c|\, t^{\frac{1}{p}-p-1-j}, \qquad t \ge 1.$$

Proof. On $(\frac{1}{2}, 1]$, $b_c = 0$ so $g_c = 0$. For $t > 1$ write $g_c(t) = t^{1/p}h(c\,\varrho(t))$ with $\varrho(t) = \chi_+(t)t^{-1-p}$ and $h(s) = (1-s)^{1/p} - 1$. Since $|c\varrho| \le \frac{1}{2}$ and h is smooth on $[-\frac{1}{2}, \frac{1}{2}]$ with $h(0) = 0$, write $h(s) = s\tilde{h}(s)$ with $\tilde{h}$ smooth. Each derivative of ϱ obeys $|\varrho^{(i)}(t)| \le C_i t^{-1-p-i}$ for $t \ge 1$ (Leibniz; $\chi_+^{(i)}$ bounded). By Faà di Bruno, $|\partial_t^i[h(c\varrho)]| \le C_i|c|t^{-1-p-i}$; multiply by $t^{1/p}$ (with $|\partial^i t^{1/p}| \le Ct^{1/p-i}$) and use Leibniz. Smoothness across $t = 1$: g_c is the composition of smooth maps on $(\frac{1}{2}, \infty)$ (the expression $(t-b_c(t))^{1/p} - t^{1/p}$ is smooth on $(\frac{1}{2}, \infty)$ since $t - b_c(t) > 0$ there). □

Lemma 8.4 (ϕ_c is a smooth increasing diffeomorphism solving the ODE)**.** *Let $|c| \le \frac{1}{2}$. Then $\phi_c \in C^\infty(\mathbb{R})$, $\phi_c' > 0$ on $\mathbb{R}$, $\phi_c(y) = y$ for $y \le 1$, ϕ_c maps $(1, \infty)$ onto $(1, \infty)$, $\phi_c : \mathbb{R} \to \mathbb{R}$ is a bijection, $\psi_c \in C^\infty(\mathbb{R})$, $\psi_c(y) = y$ for $y \le 1$, and*

$$\phi_c(y)\,\phi_c'(y)^p = y - b_c(y) \qquad \text{for all } y \in \mathbb{R}. \tag{8.1}$$

Proof. On $(\frac{1}{2}, \infty)$ let $F(y) := y^{\frac{p+1}{p}} + \frac{p+1}{p}\int_1^y g_c(t)\,dt$. Since $\int_1^y t^{1/p}dt = \frac{p}{p+1}(y^{\frac{p+1}{p}} - 1)$, $F(y)$ equals the bracket in Definition 8.2 for $y > 1$, and $F(y) = y^{(p+1)/p}$ for $y \in (\frac{1}{2}, 1]$ ($g_c = 0$ there). F is smooth on $(\frac{1}{2}, \infty)$ by Lemma 8.3, and $F \ge 1$ on $[1, \infty)$ (integrand $(t - b_c)^{1/p} > 0$), $F > 0$ on $(\frac{1}{2}, 1]$. Hence $\phi_c = F^{p/(p+1)}$ on $(\frac{1}{2}, \infty)$, which is smooth and coincides with y on $(\frac{1}{2}, 1]$; so $\phi_c \in C^\infty(\mathbb{R})$. For $y > 1$, $F' = \frac{p+1}{p}(y - b_c)^{1/p}$, and $F' = \frac{p+1}{p}\phi_c^{1/p}\phi_c'$, so $\phi_c^{1/p}\phi_c' = (y - b_c)^{1/p} > 0$; raising to the power p (both sides positive, $\phi_c \ge 1$) gives (8.1) on $y > 1$; on $y \le 1$ it is $y \cdot 1 = y$. Thus $\phi_c' > 0$ everywhere, $\phi_c(1) = 1$, and $\phi_c(y) \ge \big[1 + \frac{p+1}{p}\int_1^y (t/2)^{1/p}dt\big]^{p/(p+1)} \to \infty$, so ϕ_c is an increasing bijection with $\phi_c((1, \infty)) = (1, \infty)$. By the inverse function theorem ψ_c is smooth, and $\psi_c = \mathrm{id}$ on $(-\infty, 1]$. □

Lemma 8.5 (Bounds for ϕ_c)**.** *There are $c_0 \in (0, \frac{1}{2}]$ and constants $C_j = C_j(p, \chi_+)$, $j \geq 0$, such that for all $|c| \leq c_0$ and $y \in \mathbb{R}$:*

$$|\partial_y^j(\phi_c(y) - y)| \leq C_j|c|\langle y\rangle^{-p-j} \qquad (j \geq 0), \tag{8.2}$$

and moreover $|\phi_c' - 1| \leq \frac{1}{2}$, $|\phi_c(y) - y| \leq \frac{1}{2}$, $\frac{1}{2}\langle y\rangle \leq \langle \phi_c(y)\rangle \leq 2\langle y\rangle$, and

$$|(\phi_c')^{p+1} - 1| + \langle y\rangle\,|\partial_y(\phi_c')^{p+1}| \leq C|c|\langle y\rangle^{-p-1}.$$

All these quantities vanish identically on $y \leq 1$.

Proof. On $y \leq 1$ everything is trivial ($\phi_c = \mathrm{id}$). For $y > 1$ (where $\langle y\rangle \approx y$), let $I(y) = \int_1^y g_c$ and $E(y) = \frac{p+1}{p} y^{-\frac{p+1}{p}} I(y)$, so $\phi_c(y) = y(1 + E(y))^{p/(p+1)}$ (from $F = y^{(p+1)/p}(1 + E)$). Since $\frac{1}{p} - p > 0$, Lemma 8.3 gives $|I(y)| \leq C|c| \int_1^y t^{1/p-p-1}dt \leq C|c|y^{1/p-p}$ and $|I^{(j)}| = |g_c^{(j-1)}| \leq C|c|y^{1/p-p-j}$ for $j \geq 1$. By Leibniz, $|E^{(j)}(y)| \leq C_j|c|y^{-1-p-j}$. Choose c_0 so that $|E| \leq \frac{1}{2}$. Writing $(1 + s)^{p/(p+1)} - 1 = s\,k(s)$ with k smooth on $[-\frac{1}{2}, \frac{1}{2}]$, Faà di Bruno and Leibniz give $|\partial^j[yE\,k(E)]| \leq C_j|c|y^{-p-j}$, which is (8.2). Shrinking c_0 so that $C_0c_0, C_1c_0 \leq \frac{1}{2}$ gives the remaining claims ($\langle\cdot\rangle$ is 1-Lipschitz and $\langle y\rangle \geq 1$). The last estimate follows from $\phi_c' - 1 = O(|c|\langle y\rangle^{-p-1})$, $\phi_c'' = O(|c|\langle y\rangle^{-p-2})$ and smoothness of $s \mapsto s^{p+1}$ near 1. □

Lemma 8.6 (Bounds for $\psi_c = \phi_c^{-1}$)**.** *For $|c| \leq c_0$ and all y: $\psi_c = \mathrm{id}$ on $(-\infty, 1]$, $\psi_c((1, \infty)) = (1, \infty)$,*

$$\tfrac{1}{2}\langle y\rangle \leq \langle \psi_c(y)\rangle \leq 2\langle y\rangle, \quad \psi_c' = \frac{1}{\phi_c' \circ \psi_c} \in [\tfrac{2}{3}, 2], \quad \psi_c'' = -\frac{\phi_c'' \circ \psi_c}{(\phi_c' \circ \psi_c)^3}, \tag{8.3}$$

and $|\psi_c(y) - y| \leq C|c|\langle y\rangle^{-p}$, $|\psi_c'(y) - 1| \leq C|c|\langle y\rangle^{-p-1}$, $|\psi_c''(y)| \leq C|c|\langle y\rangle^{-p-2}$, $|(\psi_c')^{p+1} - 1| + |(\psi_c')^{p+2} - 1| \leq C|c|\langle y\rangle^{-p-1}$.

Proof. Differentiate $\phi_c(\psi_c(y)) = y$ twice. $\psi_c(y) - y = -(\phi_c - \mathrm{id})(\psi_c(y))$, and $\langle \psi_c(y)\rangle \approx \langle y\rangle$ because $|\psi_c(y) - y| \leq \frac{1}{2}$. Then insert Lemma 8.5 at the point $\psi_c(y)$. □

8.1.2 The operators $\mathcal{U}_\epsilon, \mathcal{V}_\epsilon, \mathcal{R}_\epsilon$

Let $\epsilon_c > 0$ be such that $|c_\epsilon| \leq c_0$ for $0 \leq \epsilon \leq \epsilon_c$ (possible since $|c_\epsilon| \leq C_0\epsilon$ by Lemma 3.12); in this section always $0 \leq \epsilon \leq \epsilon_c$. Let $\mathcal{F}(\mathbb{R})$ denote the real vector space of all functions $\mathbb{R} \to \mathbb{R}$.

Definition 8.7 ($\mathcal{U}_\epsilon$)**.** For $u \in \mathcal{F}(\mathbb{R})$ let $\mathcal{U}_\epsilon u := u \circ \psi_\epsilon$. This is an $\mathbb{R}$-linear bijection of $\mathcal{F}(\mathbb{R})$ with inverse $\mathcal{U}_\epsilon^{-1}u = u \circ \phi_\epsilon$; it does not depend on γ. (In the paper's notation $\mathcal{U} = \mathcal{T}_\phi^{-1}$, $\mathcal{T}_\phi u = u \circ \phi$; this is (8.4).)

Definition 8.8 ($\mathcal{V}_\epsilon$)**.** For $u \in \mathcal{F}(\mathbb{R})$ let $(\mathcal{V}_\epsilon u)(y) := \phi_\epsilon'(y)^{p+1}\,u(\phi_\epsilon(y))$. This is an $\mathbb{R}$-linear bijection of $\mathcal{F}(\mathbb{R})$ with inverse $(\mathcal{V}_\epsilon^{-1}u)(y) = \psi_\epsilon'(y)^{p+1}u(\psi_\epsilon(y))$ (check: $\phi_\epsilon'(y)\psi_\epsilon'(\phi_\epsilon(y)) = 1$). It does not depend on γ.

$$\mathcal{V}_\epsilon = (\phi_\epsilon')^{p+1}\mathcal{T}_{\phi_\epsilon}, \qquad \mathcal{U}_\epsilon = \mathcal{T}_{\phi_\epsilon}^{-1}. \tag{8.4}$$

Definition 8.9 (Remainder $\mathcal{R}_\epsilon$)**.** For $y \neq z$ let

$$K_\epsilon(y,z) := c_\alpha \Big[1 - \Big(\frac{\phi_\epsilon'(y)}{S_{\phi_\epsilon}(y,z)}\Big)^{p+1}\Big] \frac{1}{|y-z|^{1+p}}.$$

For $r \in C^1(\mathbb{R})$ with r' bounded define

$$\mathcal{R}_{1,\epsilon} r := \big((\phi_\epsilon')^{p+1} - 1\big) r, \tag{8.5}$$

$$\mathcal{R}_{2,\epsilon} r(y) := \int_{\mathbb{R}} K_\epsilon(y,z)\, r'(z)\, dz = c_\alpha \int_{\mathbb{R}} \Big[1 - \Big(\frac{\phi_\epsilon'(y)}{S_{\phi_\epsilon}(y,z)}\Big)^{p+1}\Big] \frac{r'(z)}{|y-z|^{1+p}}\, dz, \tag{8.6}$$

(absolutely convergent by Lemmas 8.19 and 8.20) and $\mathcal{R}_\epsilon := \mathcal{R}_{1,\epsilon} + \mathcal{R}_{2,\epsilon}$. For $\epsilon = 0$ ($c_0 = 0$, $\phi_0 = \mathrm{id}$) one has $\mathcal{R}_0 = 0$.

8.1.3 The conjugation identity

Lemma 8.10 (Local part of the conjugation)**.** *Let $r \in C^1(\mathbb{R})$ and $v := \mathcal{U}_\epsilon r$. Then for all $y \in \mathbb{R}$, with $x = \phi_\epsilon(y)$,*

$$\phi_\epsilon'(y)^{p+1}\big(v(x) - x\, v'(x)\big) = r(y) - y\, r'(y) + b_\epsilon(y) r'(y) + \mathcal{R}_{1,\epsilon} r(y).$$

Proof. $v(\phi_\epsilon(y)) = r(y)$ and $v'(\phi_\epsilon(y)) = r'(y)/\phi_\epsilon'(y)$. Hence the left side equals $(\phi_\epsilon')^{p+1} r - \phi_\epsilon(\phi_\epsilon')^p r' = r + ((\phi_\epsilon')^{p+1} - 1) r - (y - b_\epsilon) r'$ by (8.1). □

Lemma 8.11 (A vanishing integral)**.** *Let $|c| \le c_0$, $\phi = \phi_c$, $y \in \mathbb{R}$. The function $z \mapsto \frac{\phi'(y)^p \phi'(z)}{|\phi(y) - \phi(z)|^{1+p}} - \frac{1}{|y-z|^{1+p}}$ is in $L^1(\mathbb{R})$ and its integral is 0.*

Proof. Write it as $|y-z|^{-1-p} B(z)$, $B(z) = \phi'(y)^p \phi'(z) S_\phi(y,z)^{-1-p} - 1$. Since $S_\phi(y,z) = \int_0^1 \phi'(z + s(y-z)) ds \in [\frac{1}{2}, \frac{3}{2}]$ is smooth in z (including $z = y$, value $\phi'(y)$), B is C^1 with $B(y) = 0$ and B' bounded (bounds of Lemma 8.5), so $|B(z)| \le C|y-z|$; also $|B| \le C$. Hence the integrand is $O(|y-z|^{-p})$ near y and $O(|y-z|^{-1-p})$ at infinity: L^1. For $\delta \in (0,1)$, by the substitution $x' = \phi(z)$ (ϕ increasing, $\phi(\pm\infty) = \pm\infty$),

$$\int_{|z-y|>\delta} \frac{\phi'(z)\, dz}{|\phi(y) - \phi(z)|^{1+p}} = \frac{1}{p}\big[(\phi(y) - \phi(y-\delta))^{-p} + (\phi(y+\delta) - \phi(y))^{-p}\big], \qquad \int_{|z-y|>\delta} \frac{dz}{|y-z|^{1+p}} = \frac{2}{p\delta^p}.$$

So the truncated integral equals $\frac{\delta^{-p}}{p}\big[a_-^p + a_+^p - 2\big]$ with $a_\pm = \phi'(y)/S_\phi(y, y \pm \delta)$. Since $|S_\phi(y, y \pm \delta) - \phi'(y)| \le \frac{1}{2}\|\phi''\|_\infty \delta$ and $S_\phi \ge \frac{1}{2}$, $|a_\pm - 1| \le C\delta$, so the truncated integral is $O(\delta^{1-p}) \to 0$. By dominated convergence the truncated integrals converge to the full integral, which is therefore 0.

□

Lemma 8.12 (Nonlocal part of the conjugation)**.** *Let $r \in C^1(\mathbb{R})$ with r' bounded and locally Lipschitz, $v = \mathcal{U}_\epsilon r$ and $w := v' = (r' \circ \psi_\epsilon)\psi_\epsilon'$. Then w is bounded and locally Lipschitz, and for every $y \in \mathbb{R}$*

$$\phi_\epsilon'(y)^{p+1}\, (|D|^p w)(\phi_\epsilon(y)) = (|D|^p r')(y) + \mathcal{R}_{2,\epsilon} r(y),$$

with $|D|^p$ the absolutely convergent singular integral of Definition 1.5.

Proof. Write $\phi = \phi_\epsilon$, $S = S_\phi(y,z)$. Substitute $x' = \phi(z)$ in $|D|^p w(\phi(y)) = c_\alpha \int \frac{w(\phi(y))-w(x')}{|\phi(y)-x'|^{1+p}}dx'$, using $w(\phi(z)) = r'(z)/\phi'(z)$, $dx' = \phi'(z)dz$, $|\phi(y)-\phi(z)| = S|y-z|$:

$$\phi'(y)^{p+1}|D|^p w(\phi(y)) = c_\alpha \int_{\mathbb{R}} \frac{r'(y)\phi'(y)^p\phi'(z)S^{-1-p} - \phi'(y)^{p+1}S^{-1-p}r'(z)}{|y-z|^{1+p}}dz.$$

Decompose the numerator as $r'(y)\big[\phi'(y)^p\phi'(z)S^{-1-p}-1\big] + \big[r'(y)-r'(z)\big] + \big[1-(\phi'(y)/S)^{p+1}\big]r'(z)$. Each of the three resulting integrals converges absolutely (first: Lemma 8.11; second: Lemma 1.6; third: $|1-(\phi'(y)/S)^{p+1}| \le C\min(1,|y-z|)$, r' bounded), so splitting is legitimate. The first is 0 by Lemma 8.11, the second is $|D|^p r'(y)$, the third is $\mathcal{R}_{2,\epsilon}r(y)$. □

Lemma 8.13 (Conjugation identity)**.** *Let $\gamma > 1+\frac{1}{p}$, $0 \le \epsilon \le \epsilon_c$ and $r \in X_\gamma$. Then $\mathcal{U}_\epsilon r \in X_\gamma$ (Lemma 8.14) and, pointwise on $\mathbb{R}$,*

$$\mathcal{V}_\epsilon\big(\mathcal{L}_0(\mathcal{U}_\epsilon r)\big) = \mathcal{L}_0 r + b_\epsilon r' + \mathcal{R}_\epsilon r = \mathcal{L}_\epsilon r. \tag{8.7}$$

Proof. By Lemma 8.1 r' is bounded, and it is locally Lipschitz as $r \in W^{2,\infty}_{\mathrm{loc}}$. With $v = \mathcal{U}_\epsilon r$, $\mathcal{L}_0 v = |D|^p v' + v - xv'$; evaluate at $x = \phi_\epsilon(y)$, multiply by $\phi_\epsilon'(y)^{p+1}$, and add Lemma 8.10 and Lemma 8.12. □

8.1.4 Boundedness of $\mathcal{U}_\epsilon$, $\mathcal{V}_\epsilon$ and the transport of G_μ

Lemma 8.14 ($\mathcal{U}_\epsilon$ is a uniform isomorphism of X_γ)**.** *Let $\gamma \in \mathbb{R}$. There is $C = C(p,\gamma,\chi_+)$ such that for $0 \le \epsilon \le \epsilon_c$ and all $r \in X_\gamma$: $\mathcal{U}_\epsilon r$, $\mathcal{U}_\epsilon^{-1} r \in X_\gamma$ and $\|\mathcal{U}_\epsilon r\|_{X_\gamma} + \|\mathcal{U}_\epsilon^{-1} r\|_{X_\gamma} \le C\|r\|_{X_\gamma}$. Hence $\mathcal{U}_\epsilon$ restricts to a linear bijection $X_\gamma \to X_\gamma$ with $\|\mathcal{U}_\epsilon\|, \|\mathcal{U}_\epsilon^{-1}\| \le C$.*

Proof. $\mathcal{U}_\epsilon r = r$ on $(-\infty,1]$, which covers $y \le 0$ and $0 \le y \le 1$. For $y > 1$, $\psi = \psi_\epsilon$ maps into $(1,\infty)$, $\mathcal{U}_\epsilon r \in W^{2,\infty}_{\mathrm{loc}}$, and $(\mathcal{U}_\epsilon r)' = (r'\circ\psi)\psi'$, $(\mathcal{U}_\epsilon r)'' = (r''\circ\psi)(\psi')^2 + (r'\circ\psi)\psi''$ a.e. By (8.3), $|\partial^j(\mathcal{U}_\epsilon r)(y)| \le C\|r\|_{X_\gamma}\langle y\rangle^{-\gamma+j/p}$, $j = 0,1,2$ (the constant absorbs $2^{|\gamma|+2/p}$ from $\langle\psi(y)\rangle \approx \langle y\rangle$). Same argument for $\mathcal{U}_\epsilon^{-1} r = r\circ\phi_\epsilon$ with Lemma 8.5. □

Lemma 8.15 ($\mathcal{V}_\epsilon$ is a uniform isomorphism of Y_γ)**.** *Let $\gamma \in \mathbb{R}$. There is $C = C(p,\gamma,\chi_+)$ such that for $0 \le \epsilon \le \epsilon_c$ and $G \in Y_\gamma$: $\mathcal{V}_\epsilon G, \mathcal{V}_\epsilon^{-1} G \in Y_\gamma$ with $\|\mathcal{V}_\epsilon G\|_{Y_\gamma} + \|\mathcal{V}_\epsilon^{-1} G\|_{Y_\gamma} \le C\|G\|_{Y_\gamma}$. Hence $\mathcal{V}_\epsilon$ restricts to a linear bijection of Y_γ with $\|\mathcal{V}_\epsilon\|, \|\mathcal{V}_\epsilon^{-1}\| \le C$.*

Proof. $\mathcal{V}_\epsilon G = G$ on $(-\infty,1]$. For $y > 1$: $(\mathcal{V}_\epsilon G)' = (p+1)(\phi_\epsilon')^p\phi_\epsilon''(G\circ\phi_\epsilon) + (\phi_\epsilon')^{p+2}(G'\circ\phi_\epsilon)$ a.e.; with $\phi_\epsilon' \in [\frac{1}{2},\frac{3}{2}]$, $|\phi_\epsilon''| \le C\langle y\rangle^{-p-2}$, $\langle\phi_\epsilon(y)\rangle \approx \langle y\rangle$ we get $|\partial^j\mathcal{V}_\epsilon G(y)| \le C\|G\|_{Y_\gamma}\langle y\rangle^{-\gamma+j/p}$, $j = 0,1$. Same for $\mathcal{V}_\epsilon^{-1}G = (\psi_\epsilon')^{p+1}G\circ\psi_\epsilon$ with (8.3). □

Lemma 8.16 ($\mathcal{V}_\epsilon^{-1} - I$ is small on smooth forcings)**.** *Let $\gamma \in \mathbb{R}$. There is $C = C(p,\gamma,\chi_+,C_0)$ such that for $0 \le \epsilon \le \epsilon_c$ and every $G \in Y_\gamma$ with $G \in C^2((1,\infty))$ and $M(G) := \sup_{y>1}\langle y\rangle^\gamma(|G(y)| + |G'(y)| + |G''(y)|) < \infty$:*

$$\|(\mathcal{V}_\epsilon^{-1} - I)G\|_{Y_\gamma} \le C\,\epsilon\,M(G).$$

Proof. On $y \le 1$ the function vanishes. For $y > 1$, with $\psi = \psi_\epsilon$ (so $y, \psi(y) \in (1,\infty)$, and the segment between them lies in $(1,\infty)$, where $\langle\cdot\rangle \approx \langle y\rangle$): $(\mathcal{V}_\epsilon^{-1} - I)G = [(\psi')^{p+1} - 1]G\circ\psi + (G\circ\psi - G)$, so by the mean value theorem and Lemma 8.6, $|\cdot| \le C|c_\epsilon| M(G)(\langle y\rangle^{-p-1-\gamma} + \langle y\rangle^{-p-\gamma})$. Its derivative is $(p+1)(\psi')^p\psi'' \, G\circ\psi + [(\psi')^{p+2}-1]G'\circ\psi + (G'\circ\psi - G')$, bounded by $C|c_\epsilon|M(G)(\langle y\rangle^{-\gamma-p-2} + \langle y\rangle^{-\gamma-p-1} + \langle y\rangle^{-\gamma-p}) \le C|c_\epsilon|M(G)\langle y\rangle^{-\gamma+1/p}$. Conclude with $|c_\epsilon| \le C_0\epsilon$. □

Lemma 8.17 (Transport of G_μ)**.** *Let $\gamma \in \mathbb{R}$. There is $C = C(p,\gamma,\chi_+,C_0)$ such that for $0 \le \epsilon \le \epsilon_c$ and all $\mu \in \mathbb{C}$,*

$$\|(\mathcal{V}_\epsilon^{-1} - I)G_\mu\|_{Y_\gamma} \le C\,\epsilon\left|1 - \frac{\mu}{\mu_*}\right|.$$

In particular, for any closed disc $\overline{B} = \overline{B}(\mu_*,\delta)$, $\sup_{\mu\in\overline{B}} \|(\mathcal{V}_\epsilon^{-1} - I)G_\mu\|_{Y_\gamma} \le C\,\epsilon\,\delta/|\mu_*|$.

Proof. $\widehat{G_\pm}$ are smooth and compactly supported (in $1 \le |\xi| \le 2$; χ_+' is supported in $[1,2]$), so $G_\pm$ are Schwartz; they are real-valued by the Hermitian extension. With $z = 1 - \mu/\mu_*$, comparing Fourier transforms on $\xi > 0$ and using Hermitian symmetry, $G_\mu = (\operatorname{Re} z)G_+ + (\operatorname{Im} z)G_-$. Apply Lemma 8.16 to $G_\pm$ ($M(G_\pm) < \infty$, $G_\pm \in Y_\gamma$) and linearity.

□

8.1.5 Bounds for the remainder $\mathcal{R}_\epsilon$

Lemma 8.18 ($\mathcal{R}_{1,\epsilon}$)**.** *Let $\gamma \in \mathbb{R}$. There is C such that for $0 \le \epsilon \le \epsilon_c$, $r \in X_\gamma$:*

$$\|\mathcal{R}_{1,\epsilon}r\|_{Y_\gamma} \le C\epsilon\|r\|_{X_\gamma}. \tag{8.8}$$

Proof. $\mathcal{R}_{1,\epsilon}r = 0$ on $y \le 1$. For $y > 1$, by Lemma 8.5 and the product rule, $|\mathcal{R}_{1,\epsilon}r| \le C|c_\epsilon|\langle y\rangle^{-p-1-\gamma}\|r\|$ and $|\partial_y\mathcal{R}_{1,\epsilon}r| \le C|c_\epsilon|(\langle y\rangle^{-p-2}|r| + \langle y\rangle^{-p-1}|r'|) \le C|c_\epsilon|\|r\|\langle y\rangle^{-\gamma+1/p}$. □

Lemma 8.19 (Kernel bounds, near region)**.** *There is $C = C(p,\chi_+)$ such that for $0 \le \epsilon \le \epsilon_c$, $y \in \mathbb{R}$ and $0 < |y-z| \le \frac{1}{4}\langle y\rangle$:*

1. *$K_\epsilon(y,z) = 0$ and $(\partial_y + \partial_z)K_\epsilon(y,z) = 0$ whenever $\max(y,z) < 1$; in particular for all $y \le 0$ (then $z \le y + \frac{1}{4}\langle y\rangle \le \frac{1}{4}$).*
2. *$|K_\epsilon(y,z)| \le C|c_\epsilon|\langle y\rangle^{-p-2}|y-z|^{-p}$.*
3. *$(\partial_y + \partial_z)K_\epsilon(y,z) = -c_\alpha(p+1)|y-z|^{-1-p}\big(\frac{\phi'(y)}{S_\phi}\big)^{p+1}\big(\frac{\phi''(y)}{\phi'(y)} - \frac{S_{\phi'}(y,z)}{S_\phi(y,z)}\big)$ ($\phi = \phi_\epsilon$), and $|(\partial_y+\partial_z)K_\epsilon(y,z)| \le C|c_\epsilon|\langle y\rangle^{-p-3}|y-z|^{-p}$.*

Moreover, for all $y \ne z$: $|K_\epsilon(y,z)| + |(\partial_y+\partial_z)K_\epsilon(y,z)| \le C\min(|y-z|^{-p}, |y-z|^{-1-p})$.

Proof. (1) $\phi = \mathrm{id}$ on $(-\infty,1)$. For $y \le 0$: $\langle y\rangle \le 1-y$ gives $y + \frac{1}{4}\langle y\rangle \le \frac{3y+1}{4} \le \frac{1}{4}$. For (2),(3): every ζ between y and z satisfies $\frac{3}{4}\langle y\rangle \le \langle\zeta\rangle \le \frac{5}{4}\langle y\rangle$ ($\langle\cdot\rangle$ is 1-Lipschitz). Since $S_\phi(y,z) = \int_0^1 \phi'(z + s(y-z))ds$ and $S_{\phi'}(y,z) = \int_0^1 \phi''(z + s(y-z))ds$, (8.2) gives $|\phi'(y) - S_\phi| \le C|c_\epsilon|\langle y\rangle^{-p-2}|y-z|$ and $|\phi''(y) - S_{\phi'}| \le C|c_\epsilon|\langle y\rangle^{-p-3}|y-z|$. As $a := \phi'(y)/S_\phi \in [\frac{1}{3},3]$, $|1-a^{p+1}| \le C|1-a| \le C|\phi'(y) - S_\phi|$, giving (2). The formula in (3) follows from $(\partial_y+\partial_z)|y-z|^{-1-p} = 0$,

$(\partial_y+\partial_z)S_\phi(y,z)=S_{\phi'}(y,z)$, and $\frac{\phi''}{\phi'}-\frac{S_{\phi'}}{S_\phi}=\frac{\phi''(S_\phi-\phi')+\phi'(\phi''-S_{\phi'})}{\phi' S_\phi}$, whose modulus is $\le C|c_\epsilon|\langle y\rangle^{-p-3}|y-z|$ (using also $|\phi''|\le C\langle y\rangle^{-p-2}$). The global bound: ϕ'' is bounded, so $|S_\phi-\phi'(y)|\le C\min(1,|y-z|)$ and $|S_{\phi'}|\le C$.

□

Lemma 8.20 (Kernel bounds, far region)**.** *There is $C=C(p,\chi_+)$ such that for $0\le\epsilon\le\epsilon_c$ and $|y-z|\ge\frac18\langle y\rangle$:*

$$|K_\epsilon(y,z)|\le C|c_\epsilon|\Big(\frac{\langle y\rangle^{-p-1}}{|y-z|^{p+1}}+\frac{\langle z\rangle^{-p}}{|y-z|^{p+2}}\Big),\qquad |\partial_y K_\epsilon(y,z)|\le C|c_\epsilon|\Big(\frac{\langle y\rangle^{-p-2}}{|y-z|^{p+1}}+\frac{\langle z\rangle^{-p}}{|y-z|^{p+3}}\Big).$$

Proof. $\phi'(y)-S_\phi(y,z)=(\phi'(y)-1)-\frac{(\phi-\mathrm{id})(y)-(\phi-\mathrm{id})(z)}{y-z}$, so by (8.2) $|\phi'(y)-S_\phi|\le C|c_\epsilon|(\langle y\rangle^{-p-1}+\frac{\langle y\rangle^{-p}+\langle z\rangle^{-p}}{|y-z|})\le C|c_\epsilon|(\langle y\rangle^{-p-1}+\frac{\langle z\rangle^{-p}}{|y-z|})$ using $|y-z|\ge\langle y\rangle/8$. This and $|1-a^{p+1}|\le C|1-a|$ give the first bound. For fixed z, $\partial_y S_\phi(y,z)=\frac{\phi'(y)-S_\phi}{y-z}$, hence $|\partial_y K_\epsilon|\le C\big(|\phi''(y)||y-z|^{-1-p}+|\phi'(y)-S_\phi||y-z|^{-2-p}\big)$

$$\le C|c_\epsilon|\Big(\frac{\langle y\rangle^{-p-2}}{|y-z|^{p+1}}+\frac{\langle y\rangle^{-p-1}}{|y-z|^{p+2}}+\frac{\langle z\rangle^{-p}}{|y-z|^{p+3}}\Big),\tag{8.9}$$

and the middle term is $\le 8\langle y\rangle^{-p-2}|y-z|^{-p-1}$.

□

Lemma 8.21 (Derivative of $\mathcal{R}_{2,\epsilon}r$)**.** *Let γ be admissible, $0\le\epsilon\le\epsilon_c$, $r\in X_\gamma$. Then $\mathcal{R}_{2,\epsilon}r$ is locally Lipschitz and for a.e. y*

$$\partial_y(\mathcal{R}_{2,\epsilon}r)(y)=\int_{\mathbb{R}}\big[(\partial_y+\partial_z)K_\epsilon(y,z)\,r'(z)+K_\epsilon(y,z)\,r''(z)\big]dz,$$

the integral being absolutely convergent, locally uniformly in y.

Proof. Write $\mathcal{R}_{2,\epsilon}r(y)=\int k(y,u)r'(y-u)du$ with $k(y,u)=K_\epsilon(y,y-u)$; then $\partial_y k(y,u)=[(\partial_y+\partial_z)K_\epsilon](y,y-u)$. By the global bound of Lemma 8.19, $|k|+|\partial_y k|\le C\min(|u|^{-p},|u|^{-1-p})\in L^1(du)$. Also $|r''(y-u)|\le\|r\|w_2(y-u)\le\|r\|\langle y-u\rangle^s$ with $s=\max(0,\frac2p-\gamma)<\frac1p-\beta=\frac{1-p}{2p}<p$ (since $\gamma>\beta+\frac1p$ and $2p^2+p-1>0$), so $\int\min(|u|^{-p},|u|^{-1-p})\langle y-u\rangle^s du<\infty$ locally uniformly in y. For $y_0<y_1$, $u\mapsto$ fixed, $y\mapsto k(y,u)r'(y-u)$ is absolutely continuous with derivative $\partial_y k\,r'+k\,r''$ a.e.; integrate in $y\in[y_0,y_1]$ and use Fubini to get $\mathcal{R}_{2,\epsilon}r(y_1)-\mathcal{R}_{2,\epsilon}r(y_0)=\int_{y_0}^{y_1}(\mathrm{RHS})\,dy$. Change back $z=y-u$. Lean route (Lipschitz domination, no Fubini over $[y_0,y_1]\times\mathbb{R}$): at every x_0 apply differentiation under the integral in the variable $u=y-z$ with a Lipschitz majorant: for a.e. u, $x\mapsto K_\epsilon(x,x-u)r'(x-u)$ is Lipschitz on $B(x_0,1)$ with constant $C\|r\|\langle x_0\rangle G(u)$, $G(u)=\mathbf{1}_{|u|\le1}|u|^{-p}+4\langle u\rangle^{-1-p+s}\in L^1$, and differentiable at x_0 whenever r' is differentiable at x_0-u; hence $\mathcal{R}_{2,\epsilon}r$ is differentiable everywhere with the stated derivative, and local Lipschitz continuity follows by the mean value theorem.

□

Lemma 8.22 ($\mathcal{R}_{2,\epsilon}$, near contributions)**.** *Let γ be admissible and fix $\vartheta\in C_c^\infty(\mathbb{R};[0,1])$ with $\vartheta=1$ on $[-\frac18,\frac18]$, $\operatorname{supp}\vartheta\subset[-\frac14,\frac14]$. There is $C=C(p,\gamma,\chi_+,\vartheta)$ such that for $0\le\epsilon\le\epsilon_c$, $r\in X_\gamma$ and $y\in\mathbb{R}$ (a.e. y in (2)):*

1. $\big|\int_{|y-z|\le\langle y\rangle/4}K_\epsilon(y,z)r'(z)dz\big|\le C|c_\epsilon|\|r\|_{X_\gamma}\langle y\rangle^{-\gamma}$;

2. $\left|\int \vartheta(\frac{y-z}{\langle y\rangle})\big[(\partial_y+\partial_z)K_\epsilon\, r'(z)+K_\epsilon\, r''(z)\big]dz\right| \le C|c_\epsilon|\|r\|_{X_\gamma}\langle y\rangle^{-\gamma+1/p}$;

and both integrals vanish for $y \le 0$.

Proof. Vanishing for $y \le 0$: Lemma 8.19(1). For $y > 0$: in the region $\langle z\rangle \approx \langle y\rangle$, so $|r'(z)| \le C\|r\|\langle y\rangle^{-\gamma+1/p}$, $|r''(z)| \le C\|r\|\langle y\rangle^{-\gamma+2/p}$; and $\int_{|y-z|\le\langle y\rangle/4}|y-z|^{-p}dz = \frac{2}{1-p}(\langle y\rangle/4)^{1-p}$. Then (1) is bounded by $C|c_\epsilon|\|r\|\langle y\rangle^{-p-2+1-p-\gamma+1/p} = C|c_\epsilon|\|r\|\langle y\rangle^{-\gamma+(1/p-2p-1)} \le C|c_\epsilon|\|r\|\langle y\rangle^{-\gamma}$ since $\frac{1}{p} < 2p+1$. In (2) (support of ϑ inside $|y-z| \le \langle y\rangle/4$), $|\dots| \le C|c_\epsilon|\langle y\rangle^{-p-2}\int|y-z|^{-p}(\langle y\rangle^{-1}|r'|+|r''|) \le C|c_\epsilon|\|r\|\langle y\rangle^{-\gamma+2/p-2p-1} \le C|c_\epsilon|\|r\|\langle y\rangle^{-\gamma+1/p}$.

□

Lemma 8.23 ($\mathcal{R}_{2,\epsilon}$, far contributions)**.** *Let γ be admissible and ϑ as in Lemma 8.22. There is C such that for $0 \le \epsilon \le \epsilon_c$, $r \in X_\gamma$ and (a.e.) $y \in \mathbb{R}$:*

1. $\left|\int_{|y-z|>\langle y\rangle/4} K_\epsilon(y,z)r'(z)dz\right| \le C|c_\epsilon|\|r\|_{X_\gamma}\langle y\rangle^{-p-2}$;
2. *the identity (boundary terms vanish)*

$$\int\Big[1-\vartheta\Big(\tfrac{y-z}{\langle y\rangle}\Big)\Big]\big[(\partial_y+\partial_z)K_\epsilon\, r'+K_\epsilon\, r''\big]dz = \int\Big[1-\vartheta\Big(\tfrac{y-z}{\langle y\rangle}\Big)\Big]\partial_y K_\epsilon\, r'\,dz - \frac{1}{\langle y\rangle}\int\vartheta'\Big(\tfrac{y-z}{\langle y\rangle}\Big)K_\epsilon\, r'\,dz \tag{8.10}$$

holds, and the left side is bounded by $C|c_\epsilon|\|r\|_{X_\gamma}\langle y\rangle^{-p-3}$.

Proof. (1) By Lemma 8.20 and $|y-z| > \langle y\rangle/4$, $\langle z\rangle^{-p} \le 1$: $|K_\epsilon| \le C|c_\epsilon|(\langle y\rangle^{-2p-2}+\langle y\rangle^{-p-2})$; multiply by $|r'|$ and integrate, using $\|r'\|_{L^1} \le C\|r\|_{X_\gamma}$ (Lemma 8.1, $\gamma - \frac{1}{p} > 1$). (2) On the support of $1-\vartheta(\frac{y-z}{\langle y\rangle})$, $|y-z| \ge \langle y\rangle/8 > 0$, so $z \mapsto K_\epsilon(y,z)r'(z)$ is locally absolutely continuous there with derivative $\partial_z K_\epsilon r' + K_\epsilon r''$. Split $(\partial_y+\partial_z)K_\epsilon = \partial_y K_\epsilon + \partial_z K_\epsilon$ and integrate by parts on $[-R,R]$; the boundary terms $(1-\vartheta)K_\epsilon r'$ at $\pm R$ tend to 0 ($|K_\epsilon| \le C|y-z|^{-1-p}$, r' bounded), and $\partial_z[1-\vartheta(\frac{y-z}{\langle y\rangle})] = \langle y\rangle^{-1}\vartheta'(\frac{y-z}{\langle y\rangle})$. All integrals converge absolutely (Lemma 8.21). First term on the right: by Lemma 8.20 and $|y-z| \ge \langle y\rangle/8$, it is $\le C|c_\epsilon|(\langle y\rangle^{-2p-3}+\langle y\rangle^{-p-3})\|r'\|_{L^1}$. Second term: ϑ' is supported in $\frac{1}{8}\langle y\rangle \le |y-z| \le \frac{1}{4}\langle y\rangle$, where Lemma 8.19(2) gives $|K_\epsilon| \le C|c_\epsilon|\langle y\rangle^{-2p-2}$; so it is $\le C|c_\epsilon|\langle y\rangle^{-2p-3}\|r'\|_{L^1}$.

□

Lemma 8.24 ($\mathcal{R}_{2,\epsilon}$ is small from X_γ to Y_γ)**.** *Let γ be admissible. There is $C = C(p,\gamma,\chi_+,C_0)$ such that for $0 \le \epsilon \le \epsilon_c$ and $r \in X_\gamma$: $\mathcal{R}_{2,\epsilon}r \in Y_\gamma$ and $\|\mathcal{R}_{2,\epsilon}r\|_{Y_\gamma} \le C\epsilon\|r\|_{X_\gamma}$.*

Proof. $\mathcal{R}_{2,\epsilon}r$ is locally Lipschitz (Lemma 8.21), so in $W^{1,\infty}_{\mathrm{loc}}$. Value: split at $|y-z| = \langle y\rangle/4$: $|\mathcal{R}_{2,\epsilon}r(y)| \le C|c_\epsilon|\|r\|(\langle y\rangle^{-\gamma}+\langle y\rangle^{-p-2}) \le C|c_\epsilon|\|r\|\langle y\rangle^{-\gamma}$ because $\gamma < 2p+1 < p+2$. Derivative: split the integrand of Lemma 8.21 with $\vartheta+(1-\vartheta)$: for $y \ge 0$, $|\partial_y\mathcal{R}_{2,\epsilon}r| \le C|c_\epsilon|\|r\|(\langle y\rangle^{-\gamma+1/p}+\langle y\rangle^{-p-3}) \le C|c_\epsilon|\|r\|\langle y\rangle^{-\gamma+1/p}$; for $y \le 0$ only the far part survives: $\le C|c_\epsilon|\|r\|\langle y\rangle^{-p-3} \le C|c_\epsilon|\|r\|\langle y\rangle^{-\gamma-1}$ since $\gamma+1 < p+3$. Finally $|c_\epsilon| \le C_0\epsilon$.

□

Lemma 8.25 (Conjugation of the linear operator)**.** *There are $\epsilon_c > 0$ (depending only on p,χ_+ and the constant C_0 of Lemma 3.12) and, for every admissible γ, a constant $C(\gamma)$ such that for all $0 \le \epsilon \le \epsilon_c$:*

1. *$\mathcal{U}_\epsilon, \mathcal{V}_\epsilon$ (Definitions 8.7 and 8.8) are $\mathbb{R}$-linear bijections of the space of all functions $\mathbb{R} \to \mathbb{R}$, independent of γ; they restrict to bijections $X_\gamma \to X_\gamma$, resp. $Y_\gamma \to Y_\gamma$, and $\|\mathcal{U}_\epsilon\|_{X_\gamma \to X_\gamma}, \|\mathcal{U}_\epsilon^{-1}\|_{X_\gamma \to X_\gamma}, \|\mathcal{V}_\epsilon\|_{Y_\gamma \to Y_\gamma}, \|\mathcal{V}_\epsilon^{-1}\|_{Y_\gamma \to Y_\gamma} \le C(\gamma)$;*

2. *$\mathcal{R}_\epsilon$ maps X_γ into Y_γ with $\|\mathcal{R}_\epsilon r\|_{Y_\gamma} \le C(\gamma)\epsilon \|r\|_{X_\gamma}$;*

3. *$\mathcal{V}_\epsilon\big(\mathcal{L}_0(\mathcal{U}_\epsilon r)\big) = \mathcal{L}_\epsilon r$ pointwise on $\mathbb{R}$, for all $r \in X_\gamma$;*

4. *$\|(\mathcal{V}_\epsilon^{-1} - I)G\|_{Y_\gamma} \le C(\gamma)\epsilon \sup_{y>1} \langle y\rangle^\gamma (|G| + |G'| + |G''|)$ for $G \in Y_\gamma \cap C^2((1,\infty))$;*

5. *for every $\delta > 0$: $\sup_{\mu \in \overline{B}(\mu_*, \delta)} \|(\mathcal{V}_\epsilon^{-1} - I)G_\mu\|_{Y_\gamma} \le C(\gamma)\, \epsilon\, \delta/|\mu_*|$.*

Proof. (1): Definitions 8.7 and 8.8, Lemmas 8.14 and 8.15. (2): Lemmas 8.18 and 8.24. (3): Lemma 8.13. (4): Lemma 8.16. (5): Lemma 8.17.

□

8.2 Inverse of the free operator

8.2.1 The free equation on the Fourier side

Lemma 8.26 (Shifted fractional kernel: domination and continuity)**.** *Let $0 < p < 1$, $A, M, N \ge 0$, and let $w : \mathbb{R} \to \mathbb{C}$ be locally Lipschitz with $|w(x)| \le M$ for every x and $|w'(x)| \le A\langle x\rangle^N$ for a.e. x. For every $B \ge 1$, $|x| \le B$ and $h \ne 0$,*

$$\frac{|w(x) - w(x-h)|}{|h|^{1+p}} \le A\langle B+1\rangle^N |h|^{-p} \mathbf{1}_{\{0<|h|\le 1\}} + 2M|h|^{-1-p}\mathbf{1}_{\{|h|>1\}}.$$

The right side is integrable in h. Consequently the integral defining $|D|^p w(x)$ is absolutely convergent for every x, $|D|^p w$ is continuous, and there is $C_{p,N} > 0$, independent of A, M, w, x, such that $||D|^p w(x)| \le C_{p,N}(A+M)\langle x\rangle^N$ for every x.

Proof. On the segment between $x - h$ and x, for $|h| \le 1$ and $|x| \le B$, the a.e. derivative bound is $A\langle B+1\rangle^N$. Local Lipschitz regularity implies absolute continuity there, so the fundamental theorem of calculus bounds the increment by $A\langle B+1\rangle^N|h|$. Outside $[-1,1]$ use $2M$. Integrate the powers separately; they are integrable because $p < 1$ near zero and $p > 0$ at infinity. For continuity, use this common majorant on a compact neighborhood of x in the shifted formula. For growth, apply the same argument with the local derivative bound $A\langle|x| + 1\rangle^N \le C_N A\langle x\rangle^N$, and use $\langle x\rangle^N \ge 1$ for the far part. □

Lemma 8.27 (Fractional integration by parts against Schwartz functions)**.** *Let $0 < p < 1$, let $w \in L^1(\mathbb{R};\mathbb{C})$ be bounded and locally Lipschitz, and suppose $|w'(x)| \le A\langle x\rangle^N$ a.e. for some $A, N \ge 0$. For every $\varphi \in \mathcal{S}(\mathbb{R};\mathbb{C})$,*

$$\int_{\mathbb{R}^2} \frac{|w(x) - w(x-h)|\,|\varphi(x)|}{|h|^{1+p}}\, dx\, dh < \infty,$$

and both pairings below converge absolutely and satisfy the bilinear identity

$$\int_{\mathbb{R}} (|D|^p w)(x)\varphi(x)\, dx = \int_{\mathbb{R}} w(x)(|D|^p\varphi)(x)\, dx.$$

No complex conjugation is used in these pairings.

Proof. Integrate the near/far majorants of Lemma 8.26 against $|\varphi(x)|$; polynomial weights times a Schwartz function are integrable. To exchange the differences, first restrict the h integral to $\delta < |h| < L$ with $0 < \delta < L$. Each separated term is then absolutely integrable, so translation in x and symmetry $h \mapsto -h$ transfer the difference from w to φ. Let $\delta \downarrow 0$ and $L \to \infty$ using the established domination. On the transferred side, the bound $|\varphi(x) - \varphi(x+h)| \le \min\{|h|\|\varphi'\|_\infty, 2\|\varphi\|_\infty\}$, together with $w \in L^1$, supplies the corresponding integrable majorant. Lean route: no truncation $\delta < |h| < L$; Fubini is applied directly to the two absolutely integrable double integrals, splitting the x-integral for each fixed h. □

Lemma 8.28 (The fractional multiplier in Schwartz-test form)**.** *Let $0 < p < 1$ and let $w \in L^1(\mathbb{R};\mathbb{C})$ be bounded and locally Lipschitz with $|w'(x)| \le A\langle x\rangle^N$ a.e. for some $A, N \ge 0$. For every $\varphi \in \mathcal{S}(\mathbb{R};\mathbb{C})$,*

$$\int_{\mathbb{R}} (|D|^p w)(x)\widehat{\varphi}(x)\,dx = \int_{\mathbb{R}} |\xi|^p \widehat{w}(\xi)\varphi(\xi)\,d\xi,$$

and both integrals are absolutely convergent. This is HasDistFourier$(|D|^p w, \xi \mapsto |\xi|^p\widehat{w}(\xi))$ *in the Fourier convention of Definition 1.2.*

Proof. Apply Lemma 8.27 with $\widehat{\varphi}$, which is Schwartz. The Schwartz multiplier identity in Lemma 1.8, applied to real and imaginary parts and converted to the present Fourier normalization, gives $|D|^p\widehat{\varphi} = \widehat{(|\cdot|^p\varphi)}$. Fubini then gives $\int w\,\widehat{(|\cdot|^p\varphi)} = \int \widehat{w}\,|\xi|^p\varphi$: its absolute majorant is $|w(x)|\,|\xi|^p|\varphi(\xi)| \in L^1(\mathbb{R}^2)$. The left pairing is integrable by the polynomial bound from Lemma 8.26. Lean route: the multiplier identity $|D|^p\hat{\varphi} = \widehat{|\xi|^p\varphi}$ is re-derived for complex Schwartz φ (the chapter-1 singular-integral lemma is stated only for real φ), from the integral $\int(1-\cos(az))|z|^{-1-p}dz = c\,|a|^p$. □

Lemma 8.29 (Fourier transform of the singular integral on L^1 functions)**.** *Let $w \in L^1(\mathbb{R};\mathbb{C})$ be bounded and locally Lipschitz with $|w'(x)| \le A\langle x\rangle^N$ a.e. for some $A, N \ge 0$. Then $|D|^p w$ (Definition 1.5) is continuous with $||D|^p w(x)| \le C\langle x\rangle^N$, and its distributional Fourier transform is the bounded-times-$|\xi|^p$ function $|\xi|^p\widehat{w}(\xi)$.*

Proof. Choose a nonnegative bound M for $|w|$. Continuity and polynomial growth follow from Lemma 8.26; Lemma 8.28 is precisely the required distributional Fourier identity, with absolute convergence of every test pairing. □

Lemma 8.30 (Fourier transform of an integrable weak derivative)**.** *Let $u : \mathbb{R} \to \mathbb{C}$ be locally Lipschitz, with $u, u' \in L^1(\mathbb{R})$ and $u(x) \to 0$ as $x \to \pm\infty$. Here u' is the a.e. derivative, represented by Lean's total derivative. For every $\xi \in \mathbb{R}$,*

$$\widehat{u'}(\xi) = i\xi\widehat{u}(\xi).$$

Both Fourier integrals are absolutely convergent; $\widehat{u}$ is continuous and $|\widehat{u}(\xi)| \le \|u\|_{L^1}$ for every ξ.

Proof. Integrate by parts on $[-R, R]$, using absolute continuity of u on compact intervals. The endpoint terms $u(\pm R)e^{\mp iR\xi}$ tend to zero. The two integral terms converge by the L^1 assumptions, proving the identity also at $\xi = 0$. Dominated convergence proves continuity of $\widehat{u}$, and the integral triangle inequality gives its bound. □

Lemma 8.31 (Fourier-test identity for the dilation term)**.** *Let $u : \mathbb{R} \to \mathbb{C}$ be locally Lipschitz with $u, u' \in L^1(\mathbb{R})$ and $u(x) \to 0$ as $x \to \pm\infty$. For every $\varphi \in \mathcal{S}(\mathbb{R}; \mathbb{C})$,*

$$\int_{\mathbb{R}} -xu'(x)\widehat{\varphi}(x)\,dx = -\int_{\mathbb{R}} \xi\widehat{u}(\xi)\varphi'(\xi)\,d\xi,$$

with both integrals absolutely convergent. In particular the distributional transform of $-xu'$ is $\partial_\xi(\xi\widehat{u})$, in this tested sense.

Proof. Since $\widehat{\varphi'}(x) = ix\widehat{\varphi}(x)$, the left integral equals $i\int u'(x)\widehat{\varphi'}(x)dx$. Fubini is justified by $u' \in L^1$ and $\varphi' \in L^1$, and gives $i\int \widehat{u'}\varphi'$. Substitute Lemma 8.30 and use $i^2 = -1$. The original left integrand is integrable because $x\widehat{\varphi}(x)$ is bounded; the right one is integrable because $\widehat{u}$ is bounded and $\xi\varphi'(\xi)$ is integrable. □

Lemma 8.32 ($\mathcal{L}_0$ on the Fourier side)**.** *Let $\gamma > 1 + \frac{1}{p}$ and $r \in X_\gamma$. Then $\widehat{r}$ is bounded and continuous, $\mathcal{L}_0 r$ is continuous and polynomially bounded, and in $\mathcal{S}'(\mathbb{R})$*

$$\widehat{\mathcal{L}_0 r}(\xi) = i\xi|\xi|^p\,\widehat{r}(\xi) + \widehat{r}(\xi) + \partial_\xi\big(\xi\,\widehat{r}(\xi)\big),$$

where ∂_ξ is the distributional derivative.

Proof. The weighted bounds and Lemma 8.1 imply $r, r' \in L^1$, boundedness of r', and $r \to 0$ at both infinities. Apply Lemma 8.30 to $u = r$. The function r' is locally Lipschitz and its a.e. derivative satisfies $|r''(x)| \le \|r\|_{X_\gamma}\langle x\rangle^{2/p}$, so Lemma 8.29 applies to r'. It gives continuity, polynomial growth and transform $i\xi|\xi|^p\widehat{r}$ for the fractional term. The local terms $r - xr'$ are continuous and polynomially bounded directly from the weighted bounds; their test identities are Fubini for $r \in L^1$ and Lemma 8.31. Adding the three absolutely convergent pairings gives the displayed identity. □

Definition 8.33 (The free inverse $\mathcal{T}_0$ and the subspace $\widetilde{Y}_\gamma$)**.** For $\gamma > 1$ and $G \in Y_\gamma$ (so $G \in L^1$ and $\widehat{G}$ is bounded and continuous) define $m_G : \mathbb{R} \to \mathbb{C}$,

$$m_G(\xi) := \int_0^1 t\,e^{-i(1-t^{p+1})\Phi_0(\xi)}\,\widehat{G}(t\xi)\,dt \;=\; \frac{e^{-i\Phi_0(\xi)}}{\xi^2}\int_0^\xi \eta\,e^{i\Phi_0(\eta)}\widehat{G}(\eta)\,d\eta \quad (\xi \neq 0), \tag{8.11}$$

(for $\xi < 0$, $\int_0^\xi$ is the oriented integral) and $\mathcal{T}_0 G := \check{m_G} \in \mathcal{S}'(\mathbb{R})$, i.e. the tempered distribution with $\widehat{\mathcal{T}_0 G} = m_G$. $\mathcal{T}_0$ is $\mathbb{R}$-linear and its definition does not involve γ. For admissible γ, $\widetilde{Y}_\gamma := \{G \in Y_\gamma : \ell(G) = 0\}$ (as in Definition 2.10). We say $\mathcal{T}_0 G \in X_\gamma$ if there is $r \in X_\gamma$ with $\widehat{r} = m_G$; this r is unique and is identified with $\mathcal{T}_0 G$.

Lemma 8.34 (Properties of the symbol m_G; the ODE in ξ)**.** *Let $\gamma > 1$ and $G \in Y_\gamma$. Then:*

1. *the two expressions in* (8.11) *agree for $\xi \neq 0$; m_G is continuous on $\mathbb{R}$, $|m_G| \le \frac{1}{2}\|G\|_{L^1}$, $m_G(0) = \frac{1}{2}\widehat{G}(0)$;*
2. *$m_G(-\xi) = \overline{m_G(\xi)}$; hence $\mathcal{T}_0 G$ is real (as a distribution);*
3. *$m_G \in C^1(\mathbb{R} \setminus \{0\})$ and $\xi\,m_G'(\xi) + \big(2 + i\xi|\xi|^p\big)m_G(\xi) = \widehat{G}(\xi)$ for $\xi \neq 0$; equivalently $\partial_\xi\big(\xi^2 e^{i\Phi_0(\xi)}m_G(\xi)\big) = \xi e^{i\Phi_0(\xi)}\widehat{G}(\xi)$;*

4. $\xi \mapsto \xi m_G(\xi)$ *is locally Lipschitz on* $\mathbb{R}$ *and* $\partial_\xi(\xi m_G) + m_G + i\xi|\xi|^p m_G = \widehat{G}$ *a.e. and in* $\mathcal{D}'(\mathbb{R})$.

Proof. (1) Substitute $\eta = t\xi$ and use $\Phi_0(t\xi) = t^{p+1}\Phi_0(\xi)$ ($t > 0$, both signs of ξ). Continuity: dominated convergence in the t-form ($|\widehat{G}| \le \|G\|_{L^1}$, $\int_0^1 t\,dt = \frac{1}{2}$). (2) $\widehat{G}(-\eta) = \overline{\widehat{G}(\eta)}$ (G real), Φ_0 odd. (3) FTC applied to the second expression ($\widehat{G}$ continuous), using $\Phi_0' = |\xi|^p$: $m_G' = -i|\xi|^p m_G - \frac{2}{\xi} m_G + \frac{\widehat{G}}{\xi}$. (4) For $\xi \neq 0$, $(\xi m_G)' = m_G + \xi m_G' = \widehat{G} - m_G - i\xi|\xi|^p m_G$, bounded on compacts; ξm_G is continuous at 0 (value 0); a continuous function, C^1 off 0 with locally bounded derivative, is locally Lipschitz, and its distributional derivative is the a.e. derivative. □

Remark 8.35 (Derivation of the formula). By Lemma 8.32, $\mathcal{L}_0 r = G$ becomes the first-order ODE in ξ $\xi\widehat{r}' + (2 + i\xi|\xi|^p)\widehat{r} = \widehat{G}$ on each half-line. The integrating factor is $\xi^2 e^{i\Phi_0(\xi)}$ (since $\Phi_0' = |\xi|^p$), so $\xi^2 e^{i\Phi_0}\widehat{r} = \int_0^\xi \eta e^{i\Phi_0}\widehat{G}\,d\eta + \kappa_\pm$ on $\pm\xi > 0$; the homogeneous solutions $\kappa_\pm e^{-i\Phi_0}\xi^{-2}$ are not bounded at $\xi = 0$, so continuity of $\widehat{r}$ (which holds for $r \in L^1$) forces $\kappa_\pm = 0$. This is Lemma 8.36.

Lemma 8.36 (Uniqueness: $\mathcal{L}_0$ is injective on X_γ)**.** *Let* $\gamma > 1 + \frac{1}{p}$, $r \in X_\gamma$, $G \in Y_\gamma$ *with* $\mathcal{L}_0 r = G$ *pointwise. Then* $\widehat{r} = m_G$, *i.e.* $r = \mathcal{T}_0 G$. *In particular* $\mathcal{L}_0 r = 0$, $r \in X_\gamma$ *implies* $r = 0$.

Proof. By Lemma 8.32, $\partial_\xi(\xi\widehat{r}) + \widehat{r} + i\xi|\xi|^p\widehat{r} = \widehat{G}$ in $\mathcal{D}'(\mathbb{R})$. On $(0,\infty)$ multiply by the smooth function $\xi e^{i\Phi_0}$ (Leibniz rule for distributions): $\partial_\xi(\xi^2 e^{i\Phi_0}\widehat{r}) = \xi e^{i\Phi_0}\widehat{G}$ in $\mathcal{D}'((0,\infty))$. Hence $\xi^2 e^{i\Phi_0}\widehat{r} - \int_0^\xi \eta e^{i\Phi_0}\widehat{G}\,d\eta$ has zero distributional derivative on $(0,\infty)$, so equals a constant κ_+ a.e., hence everywhere (both continuous). Let $\xi \to 0^+$: the left side tends to 0 since $\widehat{r}$ is bounded; so $\kappa_+ = 0$. Same on $(-\infty, 0)$. Thus $\widehat{r} = m_G$ off 0, and at 0 by continuity. □

Lemma 8.37 ($\mathcal{T}_0$ is a right inverse whenever $\mathcal{T}_0 G \in X_\gamma$)**.** *Let* $\gamma > 1 + \frac{1}{p}$ *and* $G \in Y_\gamma$, *and suppose* $r \in X_\gamma$ *satisfies* $\widehat{r} = m_G$. *Then* $\mathcal{L}_0 r = G$ *pointwise on* $\mathbb{R}$. *(No condition* $\ell(G) = 0$ *is assumed here; it is needed only to have* $\mathcal{T}_0 G \in X_\gamma$.*)*

Proof. By Lemma 8.32 and Lemma 8.34(4), $\widehat{\mathcal{L}_0 r} = \widehat{G}$ in $\mathcal{S}'$. By injectivity of the Fourier transform on $\mathcal{S}'$, $\mathcal{L}_0 r = G$ as distributions, hence a.e.; both are continuous, so everywhere. □

8.2.2 The functional ℓ

Lemma 8.38 (Truncated kernel of ℓ)**.** *For* $R > 0$, $y \in \mathbb{R}$ *put* $K_R(y) := -i\int_0^R e^{i(\Phi_0(\xi) - y\xi)}d\xi$. *Then for each* y *the limit* $K(y) := \lim_{R\to\infty} K_R(y)$ *exists, and there is* $C = C(p)$ *with*

$$\sup_{R>0} |K_R(y)| \le C\,\varpi(y), \qquad \varpi(y) = \langle y\rangle^{-1} \ (y \le 0), \quad \varpi(y) = \langle y\rangle^{\frac{1-p}{2p}} \ (y \ge 0).$$

Proof. Write $F_y(\xi) = e^{i(\Phi_0(\xi) - y\xi)}$ and $L_y = \int_0^1 F_y(\xi)d\xi + E_y(1)$, where E_y is the tail integral of Lemma 3.51. Splitting the proper integral at 1 shows $\int_0^R F_y \to L_y$, hence $K_R(y) \to -iL_y$. For $\eta > 0$, $E_y(\eta) = \int_\eta^1 F_y + E_y(1)$; continuity of the proper integral at $\eta = 0$ gives $E_y(\eta) \to L_y$, so the same tail estimate bounds $|L_y|$

by $C\varpi(y)$. Finally $\int_0^R F_y = L_y - E_y(R)$, hence $|K_R(y)| \le 2C\varpi(y)$ uniformly in $R > 0$. □

Proposition 8.39 (Continuity of ℓ)**.** *Let $\gamma > \beta + \frac{1}{p}$ (in particular every admissible γ). For every $G \in Y_\gamma$ the limit $\ell(G) = \lim_{R\to\infty} \int_0^R \xi e^{i\Phi_0(\xi)} \widehat{G}(\xi) d\xi$ of Definition 2.10 exists,*

$$\ell(G) = \int_{\mathbb{R}} G'(y) K(y)\, dy \qquad (K \text{ from Lemma 8.38, the integral absolutely convergent}),$$

and there is $C = C(p,\gamma)$ with $|\ell(G)| \le C\|G\|_{Y_\gamma}$. Thus $\ell : Y_\gamma \to \mathbb{C}$ is a continuous $\mathbb{R}$-linear functional and $\widetilde{Y}_\gamma$ is a closed subspace.

Proof. By Lemma 8.1, $\xi\widehat{G}(\xi) = -i\widehat{G'}(\xi)$, so by Fubini on $[0,R]\times\mathbb{R}$ ($G' \in L^1$), $\int_0^R \xi e^{i\Phi_0}\widehat{G}\, d\xi = \int_{\mathbb{R}} G'(y) K_R(y)\, dy$. Now $|G'(y)K_R(y)| \le C\|G\|_{Y_\gamma} v(y)$ with $v(y) = \langle y\rangle^{-\gamma-2}$ for $y \le 0$ and $v(y) = \langle y\rangle^{-\gamma+\frac{1}{p}+\frac{1-p}{2p}} = \langle y\rangle^{-\gamma+\frac{3-p}{2p}}$ for $y \ge 0$; $v \in L^1$ iff $\gamma > \frac{3-p}{2p} + 1 = \frac{p+3}{2p} = \beta + \frac{1}{p}$. Dominated convergence gives existence of the limit, the formula, and $|\ell(G)| \le C\|G\|_{Y_\gamma}\|v\|_{L^1}$. □

Lemma 8.40 (Tail form of m_G on $\widetilde{Y}_\gamma$)**.** *Let $\gamma > \beta + \frac{1}{p}$ and $G \in Y_\gamma$ with $\ell(G) = 0$. Then, with improper integrals $\int_\xi^\infty := \lim_{R\to\infty}\int_\xi^R$ and $\int_{-\infty}^\xi := \lim_{R\to\infty}\int_{-R}^\xi$ (which exist),*

$$m_G(\xi) = -\frac{e^{-i\Phi_0(\xi)}}{\xi^2}\int_\xi^\infty \eta e^{i\Phi_0(\eta)}\widehat{G}(\eta)\, d\eta \quad (\xi > 0), \qquad m_G(\xi) = \frac{e^{-i\Phi_0(\xi)}}{\xi^2}\int_{-\infty}^\xi \eta e^{i\Phi_0(\eta)}\widehat{G}(\eta)\, d\eta \quad (\xi < 0). \tag{8.12}$$

Proof. $\xi > 0$: $\int_0^\xi = \ell(G) - \lim_R \int_\xi^R = -\int_\xi^\infty$ (limit exists by Proposition 8.39). $\xi < 0$: substituting $\eta = -s$ and using $\widehat{G}(-s) = \overline{\widehat{G}(s)}$, Φ_0 odd: $\int_{-R}^0 \eta e^{i\Phi_0}\widehat{G}\, d\eta = -\overline{\int_0^R s e^{i\Phi_0}\widehat{G}\, ds} \to -\overline{\ell(G)} = 0$. Hence $\int_0^\xi = -\int_\xi^0 = \lim_R \int_{-R}^\xi$. □

Remark 8.41 (Why two real conditions). ℓ is $\mathbb{C}$-valued and $\mathbb{R}$-linear on the real space Y_γ; with $\ell(G_+) = 1$, $\ell(G_-) = i$ (Definition 2.15), $\ell : Y_\gamma \to \mathbb{C} \cong \mathbb{R}^2$ is onto, so $\widetilde{Y}_\gamma$ has real codimension two. The negative-frequency condition is automatic (proof of Lemma 8.40). Necessity (not needed for the main theorem, not formalized): if $\ell(G) = \lambda \ne 0$, then by (8.11), $m_G(\xi) = \lambda e^{-i\Phi_0(\xi)}\xi^{-2} + o(\xi^{-2})$ as $\xi \to +\infty$. For instance $m_{G_+}(\xi) = \chi_+(\xi)e^{-i\Phi_0(\xi)}\xi^{-2}$ ($\xi > 0$), so $\mathcal{T}_0 G_+ = -2\,\mathrm{Re}\, f_{0,+}$ with $f_{0,+}$ the seed of Lemma 4.33; its derivative is an oscillation of amplitude $\approx y^{-\beta}$ (stationary phase at $\xi = y^{1/p}$), whereas X_γ requires $|r'| \lesssim y^{-\gamma+1/p}$ with $\gamma - \frac{1}{p} > \beta$. By Lemma 8.36, $G_+ \notin \mathcal{L}_0(X_\gamma)$, and likewise G_-. So $\mathcal{L}_0(X_\gamma) \cap \mathrm{span}\{G_+, G_-\} = \{0\}$, and the projection Π onto $\widetilde{Y}_\gamma$ is unavoidable.

8.2.3 Frequency splitting

Definition 8.42 (Low/high frequency splitting of $\mathcal{T}_0$)**.** Fix $\chi \in C_c^\infty(\mathbb{R};[0,1])$ even, with $\chi = 1$ on $[-1,1]$ and $\mathrm{supp}\,\chi \subset [-2,2]$ (the same χ as in Proposition 3.39 and as in chapter 09). For $\gamma > 1$, $G \in Y_\gamma$:

$$\widehat{\mathcal{T}_{\mathrm{l}}G} := \chi\, m_G, \qquad \widehat{\mathcal{T}_{\mathrm{h}}G} := (1-\chi)\, m_G, \tag{8.13}$$

i.e. $\mathcal{T}_{\mathrm{l}}G(y) = \frac{1}{2\pi}\int \chi(\xi) m_G(\xi) e^{iy\xi} d\xi$ (a bounded C^∞ real function, as χm_G is bounded, compactly supported and Hermitian), and $\mathcal{T}_{\mathrm{h}}G = ((1-\check{\chi})m_G) \in \mathcal{S}'$. In Lean, the high-frequency operator is the continuous, polynomially bounded real representative with this transform when one exists, and is 0 otherwise, as for $\mathcal{T}_0$ in Definition 8.33. Membership in X_γ alone therefore does not establish its Fourier transform; the reconstruction statement and its norm bound are recorded separately in Lemma 8.59.

8.3 Low-frequency estimate

The only difficulty is that Φ_0 is not smooth at $\xi = 0$. We expand $e^{-i(1-t^\alpha)\Phi_0(\xi)}$ in powers of Φ_0; each power gives a compactly supported, non-smooth multiplier whose kernel decays like $\langle y\rangle^{-p-2}$ (Proposition 3.39), composed with a Hardy-type average.

Definition 8.43 (Intermediate space Z_γ)**.** For $\gamma \in \mathbb{R}$ let Z_γ be the space of continuous $H : \mathbb{R} \to \mathbb{C}$ with $H|_{(-\infty,-1)} \in W^{2,\infty}_{\mathrm{loc}}((-\infty,-1))$ and

$$\|H\|_{Z_\gamma} := \sup_{y\in\mathbb{R}} \langle y\rangle^\gamma |H(y)| + \operatorname*{ess\,sup}_{y\le -1} \langle y\rangle^\gamma \big(\langle y\rangle |H'(y)| + \langle y\rangle^2 |H''(y)|\big) < \infty.$$

Definition 8.44 (Averaging operators and low-frequency multipliers)**.** For $k \in \mathbb{N}_0$, $\gamma > 1$, $G \in Y_\gamma$:

$$\mathcal{A}_k G(y) := \int_0^1 (1-t^\alpha)^k G(y/t)\, dt. \tag{8.14}$$

Let

$$m_k(\xi) := \chi(\xi)\Phi_0(\xi)^k, \qquad H_k := \check{m}_k,\ \ H_k(y) = \frac{1}{2\pi}\int m_k(\xi) e^{iy\xi} d\xi, \qquad \mathcal{M}_k H := H_k * H. \tag{8.15}$$

Lemma 8.45 (Kernel bounds for H_k)**.** *There are C_1, C_2 (depending on p, χ) such that for all $k \ge 0$, $j \in \{0,1,2\}$, $y \in \mathbb{R}$:*

$$|\partial^j H_k(y)| \le C_1 C_2^k \langle y\rangle^{-p-2-j}, \tag{8.16}$$

hence $\|\partial^j H_k\|_{L^1} \le C_1' C_2^k$. Moreover $H_k \in C^\infty$, H_k is real for even k and purely imaginary for odd k, so $(-i)^k H_k$ is real.

Proof. The bound is Proposition 3.39 (same normalisation of $\check{}$ up to the factor $\frac{1}{2\pi}$). Smoothness: m_k is bounded with compact support. Parity: m_k is real with $m_k(-\xi) = (-1)^k m_k(\xi)$. □

Lemma 8.46 (Weighted convolution)**.** *Let $a \ge b > 1$. Then $\int_{\mathbb{R}} \langle y-z\rangle^{-a}\langle z\rangle^{-b} dz \le C_{a,b}\langle y\rangle^{-b}$ for all y.*

Proof. On $|z| \ge |y|/2$: $\langle z\rangle \ge \langle y\rangle/2$, so the contribution is $\le 2^b\langle y\rangle^{-b}\int\langle u\rangle^{-a}du$. On $|z| < |y|/2$: $|y-z| \ge |y|/2$, $\langle y-z\rangle^{-a} \le 2^a\langle y\rangle^{-a} \le 2^a\langle y\rangle^{-b}$, times $\int\langle z\rangle^{-b}dz$. □

Lemma 8.47 (Integrability and Fourier transform of the averaging operator)**.** *Let $\frac{3}{4} < p < 1$, $\gamma > 1$, and $G \in Y_\gamma$. For every integer $k \geq 0$, $\mathcal{A}_k G \in L^1(\mathbb{R})$ with $\|\mathcal{A}_k G\|_{L^1} \leq \frac{1}{2}\|G\|_{L^1}$, and for every $\xi \in \mathbb{R}$,*

$$\widehat{\mathcal{A}_k G}(\xi) = \int_0^1 t(1-t^{p+1})^k \widehat{G}(t\xi)\, dt.$$

The double integrals used to exchange the y and t integrations converge absolutely.

Proof. From the value part of the Y_γ norm and $\gamma > 1$, $G \in L^1$. For $t > 0$, substitution $y = tu$ gives $\int |G(y/t)|dy = t\|G\|_1$. Since $0 \leq (1-t^{p+1})^k \leq 1$ on $[0,1]$, Tonelli bounds the absolute double integral by $\|G\|_1 \int_0^1 t\, dt = \frac{1}{2}\|G\|_1$. Fubini is therefore valid, also after multiplication by $e^{-iy\xi}$; the same substitution gives the Fourier identity. The value assigned at $t = 0$ is irrelevant. □

Lemma 8.48 (Fourier transform of a low-frequency convolution)**.** *Let $\frac{3}{4} < p < 1$ and $k \geq 0$. Then $H_k \in L^1(\mathbb{R};\mathbb{C})$ and $\widehat{H}_k = m_k$ pointwise. For every $H \in L^1(\mathbb{R};\mathbb{C})$, the defining convolution $\mathcal{M}_k H(y)$ is absolutely convergent for every y, belongs to $L^1(\mathbb{R};\mathbb{C})$, and*

$$\|\mathcal{M}_k H\|_{L^1} \leq \|H_k\|_{L^1}\|H\|_{L^1}, \qquad \widehat{\mathcal{M}_k H}(\xi) = m_k(\xi)\widehat{H}(\xi) \quad (\xi \in \mathbb{R}).$$

Proof. The kernel lemma supplies $H_k \in L^1 \cap L^\infty$. The symbol m_k is continuous with compact support, so Fourier inversion with both m_k and its inverse transform integrable gives $\widehat{H}_k = m_k$ everywhere. Boundedness of H_k gives pointwise absolute convergence. Tonelli bounds the absolute integral of the convolution by the product of the L^1 norms. Fubini, followed by $u = y - z$, proves the Fourier product identity with the stated 2π convention.

□

Lemma 8.49 (Uniform Fourier-side Taylor series)**.** *Let $\frac{3}{4} < p < 1$, $\gamma > 1$, and $G \in Y_\gamma$. Put $M_p = 2^{p+1}/(p+1)$ and, for $n \in \mathbb{N}$,*

$$Q_n(\xi) = \sum_{k=0}^{n-1} \frac{(-i)^k}{k!} m_k(\xi)\widehat{\mathcal{A}_k G}(\xi), \qquad e_n = \sum_{k=n}^{\infty} \frac{M_p^k}{k!}.$$

The scalar series defining e_n converges and $e_n \to 0$. For every $n \in \mathbb{N}$ and $\xi \in \mathbb{R}$,

$$|\chi(\xi)m_G(\xi) - Q_n(\xi)| \leq \tfrac{1}{2}\|G\|_{L^1} e_n.$$

Thus $Q_n \to \chi m_G$ uniformly on $\mathbb{R}$; no upper restriction on γ is needed.

Proof. Substitute the Fourier formula for $\mathcal{A}_k G$ and move the finite sum into the t integral. On $\operatorname{supp}\chi$, $|(1-t^{p+1})\Phi_0(\xi)| \leq M_p$ for $0 \leq t \leq 1$. The exponential series remainder is bounded by e_n, uniformly in (t,ξ) there. Multiply by $t|\widehat{G}(t\xi)| \leq t\|G\|_1$, integrate t, and use $|\chi| \leq 1$. Outside the support both sides of the Fourier-series difference are zero. Summability of the scalar exponential series proves $e_n \to 0$ and the claimed uniform convergence.

□

Lemma 8.50 (Identification of a convergent series in the weighted space)**.** *Let $\frac{3}{4} < p < 1$, $\gamma > 1$, $G \in Y_\gamma$, and put $S_n = \sum_{k=0}^{n-1} (-i)^k \mathcal{M}_k \mathcal{A}_k G/k!$. Suppose $S_n \in X_\gamma^{\mathbb{C}}$ for every n, $S \in X_\gamma^{\mathbb{C}}$, and $\|S_n - S\|_{X_\gamma^{\mathbb{C}}} \to 0$ as $n \to \infty$. Then $\widehat{S} = \chi m_G$ pointwise and $S(y) = \mathcal{T}_{\mathrm{l}} G(y)$ for every $y \in \mathbb{R}$. This is a conditional assertion for every $\gamma > 1$; convergence in $X_\gamma^{\mathbb{C}}$ is not a conclusion.*

Proof. Since $\gamma > 1$, the weighted value bound gives $\|S_n - S\|_{L^1} \le C_\gamma \|S_n - S\|_{X_\gamma^{\mathbb{C}}}$, with all norms finite by the membership hypotheses. The Fourier transforms therefore converge uniformly to $\widehat{S}$. The first two lemmas identify these transforms with Q_n, and the third gives their uniform limit χm_G. This symbol is continuous, integrable and Hermitian by Lemma 8.34. Fourier inversion for the continuous L^1 function S gives $S = (\chi m_G)^\vee$ everywhere; Hermitian symmetry makes this inverse real, so it equals the real-part definition of $\mathcal{T}_{\mathrm{l}} G$.

□

Lemma 8.51 (Series representation of $\mathcal{T}_{\mathrm{l}}$)**.** *Let $\gamma > 1$, $G \in Y_\gamma$. Then $\widehat{\mathcal{A}_k G}(\xi) = \int_0^1 t(1-t^\alpha)^k \widehat{G}(t\xi)\,dt$, $\widehat{\mathcal{M}_k H} = m_k \widehat{H}$ for $H \in L^1$, and*

$$\chi m_G = \sum_{k \ge 0} \frac{(-i)^k}{k!}\, m_k\, \widehat{\mathcal{A}_k G} \quad \text{uniformly on } \mathbb{R}. \tag{8.17}$$

If moreover the series $S := \sum_k \frac{(-i)^k}{k!} \mathcal{M}_k \mathcal{A}_k G$ converges in $X_\gamma^{\mathbb{C}}$ (for $1 < \gamma < p+2$ this follows from Lemmas 8.53 and 8.57), then $\mathcal{T}_{\mathrm{l}} G = S$.

Proof. The two transform identities are Lemmas 8.47 and 8.48; Lemma 8.49 gives the uniform Fourier sum. Under the additional weighted-space convergence hypothesis, Lemma 8.50 identifies the limit with $\mathcal{T}_{\mathrm{l}} G$.

□

Lemma 8.52 (Second-derivative formula for $\mathcal{A}_k G$ on the negative axis)**.** *Let $\gamma > 1$, $G \in Y_\gamma$, $k \ge 0$. On $(-\infty, 0)$, $\mathcal{A}_k G \in C^1$ with $(\mathcal{A}_k G)'(y) = \int_0^1 (1 - t^\alpha)^k t^{-1} G'(y/t)\,dt$, $(\mathcal{A}_k G)'$ is locally Lipschitz, and for a.e. $y < 0$*

$$(\mathcal{A}_k G)''(y) = \begin{cases} -\dfrac{G'(y)}{y}, & k = 0, \\ -\dfrac{\alpha k}{y} \displaystyle\int_0^1 t^{\alpha-1}(1-t^\alpha)^{k-1} G'(y/t)\,dt, & k \ge 1. \end{cases}$$

Proof. For $y < 0$: $|t^{-1} G'(y/t)| \le \|G\| t^{-1} (t/|y|)^{\gamma+1}$ is integrable in t uniformly for y in compacts of $(-\infty, 0)$; integrating the difference quotient in y and using Fubini gives the first-derivative formula (continuity of the right side: substitute $u = y/t$ and use dominated convergence). Integration by parts in t, using $t^{-1} G'(y/t) = -\frac{t}{y} \partial_t [G(y/t)]$ and boundary terms $t(1-t^\alpha)^k G(y/t)\big|_{t=0}^{t=1} = \mathbf{1}_{k=0} G(y)$:

$$y(\mathcal{A}_k G)'(y) = B_k(y) - \mathbf{1}_{k=0} G(y), \qquad B_k(y) := \int_0^1 \big[(1-t^\alpha)^k - \alpha k t^\alpha (1-t^\alpha)^{k-1}\big] G(y/t)\,dt.$$

B_k is C^1 on $(-\infty, 0)$ with $B_k' = (\mathcal{A}_k G)' - \alpha k \int_0^1 t^{\alpha-1}(1-t^\alpha)^{k-1} G'(y/t)\,dt$ (as before). Hence $(\mathcal{A}_k G)' = (B_k - \mathbf{1}_{k=0} G)/y$ is locally Lipschitz, and differentiating

$y(\mathcal{A}_kG)' = B_k - \mathbf{1}_{k=0}G$ gives $(\mathcal{A}_kG)' + y(\mathcal{A}_kG)'' = B_k' - \mathbf{1}_{k=0}G'$ a.e., i.e. the claim. Lean route: fundamental theorem of calculus along $s \mapsto G(s/t)$, integration by parts in t with an ϵ-cutoff near $t = 0$, and Lebesgue differentiation for the a.e. second-derivative identity. □

Lemma 8.53 ($\mathcal{A}_k : Y_\gamma \to Z_\gamma$)**.** *Let $\gamma > 1$. There is $C = C(\gamma)$ with $\|\mathcal{A}_kG\|_{Z_\gamma} \le C\|G\|_{Y_\gamma}$ for all $k \ge 0$, $G \in Y_\gamma$ (in particular $\le C(1+k)\|G\|_{Y_\gamma}$ as in (8.19)).*

Proof. $\mathcal{A}_kG$ is continuous (dominated convergence). Since $\langle y/t\rangle \ge \langle y\rangle$ for $t \in (0,1]$ and $0 \le (1-t^\alpha)^k \le 1$: $|\mathcal{A}_kG(y)| \le \|G\|\langle y\rangle^{-\gamma}$. For $y \le -1$, by Lemma 8.52 and $\langle y/t\rangle \ge |y|/t$: $|(\mathcal{A}_kG)'(y)| \le \|G\| \int_0^1 t^{-1}(t/|y|)^{\gamma+1}dt \le \|G\||y|^{-\gamma-1} \le C\|G\|\langle y\rangle^{-\gamma-1}$; and, using $\alpha k \int_0^1 t^{\alpha-1}(1-t^\alpha)^{k-1}dt = 1$ and $|G'(y/t)| \le \|G\|\langle y\rangle^{-\gamma-1}$, $|(\mathcal{A}_kG)''(y)| \le \|G\|\langle y\rangle^{-\gamma-1}/|y| \le C\|G\|\langle y\rangle^{-\gamma-2}$. □

Lemma 8.54 (Differentiation of the low-frequency convolution)**.** *Let $\frac{3}{4} < p < 1$, $\gamma > 1$, $k \ge 0$, and $H \in Z_\gamma$. Then $\mathcal{M}_kH \in C^2(\mathbb{R};\mathbb{C})$, and for $j \in \{0,1,2\}$ and every $y \in \mathbb{R}$,*

$$\partial_y^j \mathcal{M}_kH(y) = \int_{\mathbb{R}} H_k^{(j)}(y-z)H(z)\,dz,$$

with absolute convergence of each integral. There is $C_\gamma > 0$, independent of H, with $\|H\|_{L^1} \le C_\gamma\|H\|_{Z_\gamma}$.

Proof. The value part of the Z_γ norm gives $|H(z)| \le \|H\|_Z\langle z\rangle^{-\gamma}$, which is integrable. Each derivative $H_k^{(j)}$, $j \le 2$, is continuous and bounded by the kernel lemma. Thus $\|H_k^{(j)}\|_\infty|H(z)|$ is a common majorant independent of y. Apply the dominated differentiation theorem twice, and dominated convergence to the second derivative; these statements establish differentiability rather than merely identities for a total derivative. □

Lemma 8.55 (Global weighted bounds for the low-frequency convolution)**.** *Let $\frac{3}{4} < p < 1$ and $1 < \gamma < p+2$. There exist $C, Q > 0$, depending only on p, γ and the fixed cutoff, such that for every $k \ge 0$, $H \in Z_\gamma$, $j \in \{0,1,2\}$ and $y \in \mathbb{R}$,*

$$|\partial_y^j \mathcal{M}_kH(y)| \le CQ^k\|H\|_{Z_\gamma}\langle y\rangle^{-\gamma}.$$

In particular this controls all positive-half-line X_γ weights and the negative weights on $[-2,0]$.

Proof. Insert the derivative representation and the bounds $|H_k^{(j)}(y-z)| \le C_1C_2^k\langle y-z\rangle^{-p-2-j}$ and $|H(z)| \le \|H\|_Z\langle z\rangle^{-\gamma}$. Apply weighted convolution with $a = p+2+j > \gamma = b > 1$ and take the maximum of the three constants. The extra positive-half-line weights $\langle y\rangle^{-j/p}$ are at most one; the negative weights $\langle y\rangle^j$ are bounded on $[-2,0]$. □

Lemma 8.56 (Derivative decay gain on the negative half-line)**.** *Let $\frac{3}{4} < p < 1$ and $1 < \gamma < p+2$. There exist $C, Q > 0$, depending only on p, γ and the fixed cutoff, such that for every $k \ge 0$, $H \in Z_\gamma$, $j \in \{1,2\}$ and $y \le -2$,*

$$|\partial_y^j \mathcal{M}_kH(y)| \le CQ^k\|H\|_{Z_\gamma}\langle y\rangle^{-\gamma-j}.$$

The near/far integrals in the proof and their integrations by parts are absolutely convergent.

Proof. In the derivative integral set $\nu_y(z) = \chi(8(y-z)/\langle y\rangle)$. On its support, $|y-z| \le \langle y\rangle/4$, $z < -1$ and $\langle z\rangle$ is comparable to $\langle y\rangle$. For the near part integrate by parts j times in z:

$$\int \nu_y(z) H_k^{(j)}(y-z) H(z)\,dz = \int H_k(y-z)\,\partial_z^j(\nu_y H)(z)\,dz.$$

Compact support inside $(-\infty,-1)$ removes boundary terms; $H \in C^1$ with locally Lipschitz derivative there justifies the second integration by parts using the a.e. second derivative. Leibniz and the Z_γ weights give $|\partial_z^j(\nu_y H)| \le C\|H\|_Z \langle y\rangle^{-\gamma-j}$ a.e.; use $\|H_k\|_1 \le C_1' C_2^k$. For the far part $|y-z| \ge \langle y\rangle/8$, so the kernel estimate bounds its absolute value by $CC_2^k\langle y\rangle^{-p-2-j}\|H\|_1$. The L^1 bound in Lemma 8.54 and $\gamma < p+2$ give the required power. Add the two contributions. Lean route: an indicator cutoff $|u| \le \langle y\rangle/8$ and the vanishing moments of H_k', H_k'' (with a second-order Taylor bound for H on the window) replace the integration by parts against a smooth cutoff. □

Lemma 8.57 ($\mathcal{M}_k : Z_\gamma \to X_\gamma$)**.** *Let $1 < \gamma < p+2$. There is $C = C(p,\gamma,\chi)$ such that for all $k \ge 0$ and $H \in Z_\gamma$: $\mathcal{M}_k H \in C^2$ and*

$$\|\mathcal{M}_k H\|_{X_\gamma^{\mathbb{C}}} \le C\, C_2^k\, \|H\|_{Z_\gamma}. \tag{8.18}$$

Proof. The convolution is C^2 by Lemma 8.54. Use Lemma 8.55 for $y \ge -2$ and for the value term on all of $\mathbb{R}$; Lemma 8.56 supplies the stronger derivative decay for $y \le -2$. Choose the maxima of the geometric bases and constants, sum the three weighted terms on each half-line, and take essential suprema. This proves finiteness of the $X_\gamma^{\mathbb{C}}$ norm as well as its quantitative bound; C^2 regularity gives the required local Lipschitz property of the derivative. □

Proposition 8.58 (Low-frequency estimate)**.** *Let $1 < \gamma < p+2$ (this contains every admissible γ). There is $C_{\mathrm{l}} = C_{\mathrm{l}}(p,\gamma,\chi)$ such that for every $G \in Y_\gamma$, $\mathcal{T}_{\mathrm{l}}G$ (Definition 8.42) is a real C^∞ function in X_γ with $\|\mathcal{T}_{\mathrm{l}}G\|_{X_\gamma} \le C_{\mathrm{l}}\|G\|_{Y_\gamma}$. No condition $\ell(G) = 0$ is needed.*

Proof. By Lemmas 8.53 and 8.57,

$$\|\mathcal{A}_k\|_{Y_\gamma \to Z_\gamma} \le C, \qquad \|\mathcal{M}_k\|_{Z_\gamma \to X_\gamma^{\mathbb{C}}} \le CC_2^k, \tag{8.19}$$

so $\sum_k \frac{1}{k!}\|\mathcal{M}_k\mathcal{A}_k G\|_{X_\gamma^{\mathbb{C}}} \le Ce^{C_2}\|G\|_{Y_\gamma}$. The partial sums are C^2 and converge with their first two derivatives locally uniformly, so the sum S is C^2, $\|S\|_{X_\gamma^{\mathbb{C}}} \le Ce^{C_2}\|G\|_{Y_\gamma}$ (norm is lower semicontinuous under pointwise convergence of r, r', r''), and $S = \mathcal{T}_{\mathrm{l}}G$ by Lemma 8.51. Each term $(-i)^k\mathcal{M}_k\mathcal{A}_k G = ((-i)^k H_k) * \mathcal{A}_k G$ is real (Lemma 8.45), so $S \in X_\gamma$.

□

Lemma 8.59 (Reconstruction from the two frequency pieces)**.** *Let $\frac{3}{4} < p < 1$, $\gamma > 1$, and $G \in Y_\gamma$. Suppose that $\mathcal{T}_{\mathrm{l}}G, \mathcal{T}_{\mathrm{h}}G \in X_\gamma$ and that the high-frequency representative satisfies $\widehat{\mathcal{T}_{\mathrm{h}}G} = (1-\chi)m_G$ in $\mathcal{S}'(\mathbb{R})$. Then*

$$\mathcal{T}_0 G = \mathcal{T}_{\mathrm{l}}G + \mathcal{T}_{\mathrm{h}}G, \qquad \widehat{\mathcal{T}_0 G} = m_G, \qquad \mathcal{T}_0 G \in X_\gamma,$$

and $\|\mathcal{T}_0 G\|_{X_\gamma} \le \|\mathcal{T}_{\mathrm{l}}G\|_{X_\gamma} + \|\mathcal{T}_{\mathrm{h}}G\|_{X_\gamma}$.

Proof. Apply Proposition 8.58 at $\min\{\gamma, p+1\} \in (1, p+2)$, using the monotone embedding of the weighted spaces, to obtain smoothness of $\mathcal{T}_{\mathrm{l}}G$ and its defining Fourier identity. By Lemma 8.34, m_G is bounded and continuous. The sum of the two representatives is continuous and polynomially bounded and has Fourier transform m_G, by linearity tested against Schwartz functions. Fourier uniqueness identifies it with $\mathcal{T}_0 G$. The X_γ regularity and triangle inequality give the remaining conclusions.

□

Remark 8.60. For the high-frequency part (chapter 09) one uses $\mathcal{T}_{\mathrm{h}}G = \mathcal{T}_0 G - \mathcal{T}_{\mathrm{l}}G$ with the same χ, together with Lemma 8.40; there the condition $\ell(G) = 0$ is essential. The proof of Theorem 2.22 (item 2) is then: for $G \in \widetilde{Y}_\gamma$, $r := \mathcal{T}_{\mathrm{l}}G + \mathcal{T}_{\mathrm{h}}G \in X_\gamma$ satisfies $\widehat{r} = m_G$ (Lemma 8.59), and $\mathcal{L}_0 r = G$ by Lemma 8.37.

Chapter 9

Linear inverse II: high frequency (paper §6.4)

9.1 Standing conventions

Throughout this chapter $p, \alpha = p+1, \beta = \frac{p+1}{2p}$ are as in Definition 2.1 ($\frac{3}{4} < p < 1$), and γ is called *admissible* if $\beta + \frac{1}{p} < \gamma < 2p+1$. Note $\beta + \frac{1}{p} = \frac{p+3}{2p}$, so admissible γ satisfy $\gamma - \frac{1}{p} > \beta > 1$. $\Phi_0(\xi) = \xi|\xi|^p/\alpha$ (so $\Phi_0'(\xi) = |\xi|^p$). The Fourier transform is $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-ix\xi}\,dx$ (Definition 1.2) and for an integrable symbol m we write $\check{m}(z) = \frac{1}{2\pi}\int_{\mathbb{R}} m(\xi)e^{iz\xi}\,d\xi$. m_G is the symbol of $\mathcal{T}_0$ (Definition 8.33), so $\widehat{\mathcal{T}_0 G} = m_G$ in $\mathcal{S}'$. $\chi \in C_c^\infty(\mathbb{R}; [0,1])$ is the fixed even cutoff of Definition 8.42 ($\chi = 1$ on $[-1,1]$, $\operatorname{supp}\chi \subset [-2,2]$), and $\mathcal{T}_{\mathrm{l}}, \mathcal{T}_{\mathrm{h}}$ are the low/high frequency parts of $\mathcal{T}_0$ defined there: $\widehat{\mathcal{T}_{\mathrm{l}} G} = \chi\, m_G$, $\widehat{\mathcal{T}_{\mathrm{h}} G} = (1-\chi) m_G$ ((8.13)). Unless said otherwise, "C" denotes a constant depending only on p, γ and χ (and on the further parameters M, L, k, ℓ when indicated by a subscript); it never depends on y, z, ξ, R or G.

9.2 Weights and elementary facts

Definition 9.1 (Weights)**.** For $\gamma \in \mathbb{R}$, $j \in \{0,1,2\}$ and $y \in \mathbb{R}$ put

$$w_{\gamma,j}(y) := \begin{cases} \langle y\rangle^{-\gamma+j/p}, & y \geq 0, \\ \langle y\rangle^{-\gamma-j}, & y < 0, \end{cases} \qquad w_E(z) := \begin{cases} \langle z\rangle^{-1}, & z \leq 0, \\ \langle z\rangle^{\frac{1-p}{2p}}, & z > 0. \end{cases}$$

Lemma 9.2 (Weights versus the norms of X_γ, Y_γ)**.** *Let γ be admissible.*

1. *If $r \in X_\gamma$ then $|r^{(j)}(y)| \leq \|r\|_{X_\gamma} w_{\gamma,j}(y)$ for $j = 0,1$ and all y, and for $j = 2$ and a.e. y. Conversely, if $r : \mathbb{R} \to \mathbb{R}$ is C^1 with r' locally Lipschitz and $|r^{(j)}| \leq A\, w_{\gamma,j}$ ($j = 0,1$ everywhere, $j = 2$ a.e.), then $r \in X_\gamma$ and $\|r\|_{X_\gamma} \leq 6A$.*
2. *If $G \in Y_\gamma$ then $|G(y)| \leq \|G\|_{Y_\gamma} w_{\gamma,0}(y)$ for all y and $|G'(y)| \leq \|G\|_{Y_\gamma} w_{\gamma,1}(y)$ for a.e. y; in particular $|G'(y)| \leq \|G\|_{Y_\gamma}\langle y\rangle^{-\gamma+1/p} \leq \|G\|_{Y_\gamma}$ a.e.*

3. *There is C with $\|G\|_{L^1} + \|G'\|_{L^1} + \int_{\mathbb{R}} |G'(z)|\, w_E(z)\, dz \le C\|G\|_{Y_\gamma}$ for all $G \in Y_\gamma$.*

4. *(Peetre) $\langle a+b\rangle \le \sqrt{2}\,\langle a\rangle\langle b\rangle$ for $a, b \in \mathbb{R}$; hence $\langle y-z\rangle^{-s} \le 2^{s/2}\langle y\rangle^{-s}\langle z\rangle^{s}$ for $s \ge 0$.*

5. *(Shifted derivative, paper eq. `eqn:linear_nonstationary_shifted_derivative`) There is C such that for $G \in Y_\gamma$, every $y \in \mathbb{R}$ and a.e. $z \in \mathbb{R}$: $|G'(y-z)| \le C\|G\|_{Y_\gamma}\langle z\rangle^{\gamma+1} w_{\gamma,1}(y)$.*

6. *For every y, z: $|G(y-z)| \le 2^{\gamma/2}\|G\|_{Y_\gamma}\langle y\rangle^{-\gamma}\langle z\rangle^{\gamma}$.*

Proof. (1),(2): read off Definitions 2.3 and 2.4; for the converse each of the two suprema in $\|r\|_{X_\gamma}$ is $\le 3A$. For (2), on $y \le 0$ we have $\langle y\rangle^{-\gamma-1} \le \langle y\rangle^{-\gamma+1/p}$, and $\gamma > 1/p$. (3): $\|G\|_{L^1} + \|G'\|_{L^1} \le C\|G\|$ is Lemma 8.1; directly: on $z \le 0$ the integrands are $\lesssim \langle z\rangle^{-\gamma}$, $\langle z\rangle^{-\gamma-1}$, $\langle z\rangle^{-\gamma-2}$; on $z \ge 0$ they are $\lesssim \langle z\rangle^{-\gamma}$, $\langle z\rangle^{-\gamma+1/p}$, $\langle z\rangle^{\frac{3-p}{2p}-\gamma}$, integrable since $\gamma > 1$, $\gamma - \frac{1}{p} > 1$ and $\gamma - \frac{3-p}{2p} > 1 \iff \gamma > \frac{p+3}{2p}$ (the admissibility threshold, used exactly here). (4): $1 + (a+b)^2 \le 1 + 2a^2 + 2b^2 \le 2(1+a^2)(1+b^2)$; apply with $a = y-z$, $b = z$. (5): if $y \ge 0$ use (2) and (4) with $s = \gamma - \frac{1}{p} \ge 0$, then $\langle z\rangle^{\gamma-1/p} \le \langle z\rangle^{\gamma+1}$. If $y < 0$ and $y - z \le 0$: $|G'(y-z)| \le \|G\|\langle y-z\rangle^{-\gamma-1}$ and (4) with $s = \gamma+1$. If $y < 0 < y - z$, then $z < y < 0$, so $|z| > |y|$, and $|G'(y-z)| \le \|G\| \le \|G\|\langle z\rangle^{\gamma+1}\langle y\rangle^{-\gamma-1}$. (6): (2) and (4) with $s = \gamma$. □

9.3 The positive high-frequency part and its kernel

Definition 9.3 (Positive high-frequency part; paper line 2904)**.** For admissible γ and $G \in \widetilde{Y}_\gamma$ define the function on $\mathbb{R}$

$$\widehat{\mathcal{T}_+ G}(\xi) := \mathbf{1}_{\{\xi>0\}}(1-\chi(\xi))\, m_G(\xi),$$

and (once Lemma 9.6 shows $\widehat{\mathcal{T}_+ G} \in L^1$) $\mathcal{T}_+ G(y) := \frac{1}{2\pi}\int_0^\infty \widehat{\mathcal{T}_+ G}(\xi) e^{iy\xi}\, d\xi$, a bounded continuous complex function.

Definition 9.4 (Tail integrals)**.** For $z \in \mathbb{R}$ and $\xi \ge 0$ put $\Psi_z(\eta) := \Phi_0(\eta) - z\eta$ and

$$E_z(\xi) := \lim_{N\to\infty} \int_\xi^N e^{i\Psi_z(\eta)}\, d\eta.$$

For admissible γ and $G \in Y_\gamma$ put $F_G(\xi) := \int_{\mathbb{R}} G'(z) E_z(\xi)\, dz$, $\xi \ge 0$.

Lemma 9.5 (Basic properties of the tail integrals)**.**

1. *The limit defining $E_z(\xi)$ exists for all $z \in \mathbb{R}$, $\xi \ge 0$; $(z,\xi) \mapsto E_z(\xi)$ is continuous on $\mathbb{R}\times[0,\infty)$; $E_z(\xi) \to 0$ as $\xi \to \infty$ for each z; and $\partial_\xi E_z(\xi) = -e^{i\Psi_z(\xi)}$.*

2. *There is $C_E = C_E(p)$ with $|E_z(\xi)| \le C_E\, w_E(z)$ for all $z \in \mathbb{R}$, $\xi \ge 0$.*

3. *For admissible γ there is C such that for $G \in Y_\gamma$: F_G is continuous on $[0,\infty)$, $|F_G(\xi)| \le C\|G\|_{Y_\gamma}$, and $F_G(\xi) \to 0$ as $\xi \to \infty$.*

Proof. (1) For $N' > N \geq N_z := \max\{\xi, (|z|+1)^{1/p}\}$ the phase derivative $\psi(\eta) = \eta^p - z \geq 1$ is increasing, and one integration by parts gives $\int_N^{N'} e^{i\Psi_z} = \big[\frac{e^{i\Psi_z}}{i\psi}\big]_N^{N'} + \int_N^{N'} \frac{\psi'}{i\psi^2} e^{i\Psi_z}$, of modulus $\leq 3/\psi(N) \to 0$: Cauchy criterion. The same estimate is locally uniform in (z,ξ), so the limit is continuous, and $|E_z(\xi)| \leq 3/\psi(\xi) \to 0$. The derivative formula is the fundamental theorem of calculus. (2) Lemma 3.51 (with its y renamed z) gives the bound for $\xi > 0$; let $\xi \downarrow 0$ using (1). (3) $|G'(z)E_z(\xi)| \leq C_E|G'(z)|w_E(z)$, integrable by Lemma 9.2(3); dominated convergence gives continuity and the limit 0.

□

Lemma 9.6 (Tail formula; the Remark at paper line 2909)**.** *Let γ be admissible and $G \in Y_\gamma$. (Items (2),(4) are Proposition 8.39 and Lemma 8.40 rewritten with the tail integrals E_z; we only need the form with F_G.)*

1. *For $0 \leq \xi < \Xi$: $\int_\xi^\Xi \eta e^{i\Phi_0(\eta)} \widehat{G}(\eta)\, d\eta = -i\big(F_G(\xi) - F_G(\Xi)\big)$.*
2. *The limit in Definition 2.10 exists and $\ell(G) = -iF_G(0) = -i\int_{\mathbb{R}} G'(z)E_z(0)\, dz$.*
3. *For $\xi > 0$: $m_G(\xi) = \xi^{-2} e^{-i\Phi_0(\xi)}\big(\ell(G) + iF_G(\xi)\big)$.*
4. *If $G \in \widetilde{Y}_\gamma$, then for $\xi > 0$*

$$\widehat{\mathcal{T}_+ G}(\xi) = i\,(1-\chi(\xi))\,\xi^{-2} e^{-i\Phi_0(\xi)} F_G(\xi) = -(1-\chi(\xi))\frac{e^{-i\Phi_0(\xi)}}{\xi^2}\int_\xi^\infty \eta e^{i\Phi_0(\eta)}\widehat{G}(\eta)\, d\eta$$

(the last integral as $\lim_{\Xi\to\infty}\int_\xi^\Xi$), and $|\widehat{\mathcal{T}_+ G}(\xi)| \leq C\|G\|_{Y_\gamma}\mathbf{1}_{\{\xi>1\}}\xi^{-2}$; in particular $\widehat{\mathcal{T}_+ G} \in L^1(\mathbb{R})$.

Proof. $G \in W^{1,1}(\mathbb{R})$ by Lemma 9.2(3) and $G(y) \to 0$ as $|y| \to \infty$, so $\widehat{G'}(\eta) = i\eta\widehat{G}(\eta)$, i.e. $\eta\widehat{G}(\eta) = -i\widehat{G'}(\eta)$. (1): by Fubini on $[\xi,\Xi]\times\mathbb{R}$ ($G' \in L^1$, bounded integrand), $\int_\xi^\Xi \eta e^{i\Phi_0}\widehat{G} = -i\int_{\mathbb{R}} G'(z)\int_\xi^\Xi e^{i\Psi_z(\eta)} d\eta\, dz = -i\int G'(z)(E_z(\xi) - E_z(\Xi))dz$. (2): let $\Xi \to \infty$ in (1) with $\xi = 0$ and use Lemma 9.5(3). (This is the representation of Proposition 8.39, since $K(y) = -iE_y(0)$ for the kernel of Lemma 8.38.) (3): Definition 8.33 gives $m_G(\xi) = \xi^{-2}e^{-i\Phi_0(\xi)}\int_0^\xi \eta e^{i\Phi_0}\widehat{G}$, and by (1),(2) $\int_0^\xi = -iF_G(0) + iF_G(\xi) = \ell(G) + iF_G(\xi)$. (4): put $\ell(G) = 0$ in (3); the second form is (1) with $\Xi \to \infty$. The bound follows from Lemma 9.5(3) and $1 - \chi = 0$ on $[0,1]$.

□

Lemma 9.7 (Reality; $\mathcal{T}_{\mathrm{h}} = 2\operatorname{Re}\mathcal{T}_+$)**.** *Let γ be admissible and $G \in Y_\gamma$ (real-valued). Then m_G is a bounded continuous function on $\mathbb{R}$ ($|m_G| \leq \frac{1}{2}\|G\|_{L^1}$) with $m_G(-\xi) = \overline{m_G(\xi)}$. If moreover $G \in \widetilde{Y}_\gamma$, then $(1-\chi)m_G \in L^1(\mathbb{R})$, $\mathcal{T}_{\mathrm{h}}G(y) = \frac{1}{2\pi}\int_{\mathbb{R}}(1-\chi(\xi))m_G(\xi)e^{iy\xi}d\xi$ is a bounded continuous real function (it represents the distribution $\mathcal{T}_{\mathrm{h}}G$ of Definition 8.42), and $\mathcal{T}_{\mathrm{h}}G = 2\operatorname{Re}\mathcal{T}_+G$. In particular $\mathcal{T}_+G$ is the inverse Fourier transform of $\mathbf{1}_{\{\xi>0\}}(1-\chi)m_G$.*

Proof. (Continuity, the bound and the symmetry are Lemma 8.34(1),(2).) The first formula in (8.11), $m_G(\xi) = \int_0^1 te^{-i(1-t^{p+1})\Phi_0(\xi)}\widehat{G}(t\xi)dt$ (substitution $\eta = t\xi$ in Definition 8.33, valid for $\xi \neq 0$ of either sign, and taken as the value at $\xi = 0$),

gives continuity and the bound $|m_G| \le \frac{1}{2}\sup|\widehat{G}| \le \frac{1}{2}\|G\|_{L^1}$. Since G is real, $\widehat{G}(-\eta) = \overline{\widehat{G}(\eta)}$, and Φ_0 is odd, so each integrand at $-\xi$ is the conjugate of the one at ξ. Integrability of $(1-\chi)m_G$ follows from Lemma 9.6(4) on $\xi > 0$ and the symmetry on $\xi < 0$. Since χ is even, $(1-\chi)m_G = \widehat{\mathcal{T}_+G}(\xi) + \overline{\widehat{\mathcal{T}_+G}(-\xi)}$ a.e., and the inverse Fourier transform of $\overline{\widehat{\mathcal{T}_+G}(-\cdot)}$ is $\overline{\mathcal{T}_+G}$. Hence $\mathcal{T}_{\mathrm{h}}G = \mathcal{T}_+G + \overline{\mathcal{T}_+G} = 2\,\mathrm{Re}\,\mathcal{T}_+G$.

□

Definition 9.8 (Frequency truncation)**.** For $R \ge 2$ put $\theta_R(\xi) := \chi(\xi/R)$ and, for admissible γ, $G \in \widetilde{Y}_\gamma$,

$$\mathcal{T}_+^R G(y) := \frac{1}{2\pi}\int_0^\infty \theta_R(\xi)\widehat{\mathcal{T}_+G}(\xi)e^{iy\xi}\,d\xi, \qquad \mathcal{T}_{\mathrm{h}}^R G := 2\,\mathrm{Re}\,\mathcal{T}_+^R G.$$

For $y \in \mathbb{R}$ put $R_0(y) := \max\{2, \langle y\rangle^{1/p}\}$.

Definition 9.9 (Kernels $K_{+,j}$; paper line 2920, paper eq. `eqn:linear_positive_kernel`)**.** For $j \in \{0,1,2\}$, $R \ge 2$ define the amplitude ($\xi > 0$; set $a_j^R(\xi) = 0$ for $\xi \le 0$)

$$a_j^R(\xi) := -\frac{i^{j+1}}{2\pi}(\chi(\xi) - 1)\,\theta_R(\xi)\,\xi^{j-2}$$

and the kernel

$$K_{+,j}^R(y,z) := \int_1^{2R} a_j^R(\xi)\,e^{i(y\xi - \Phi_0(\xi))}\,E_z(\xi)\,d\xi, \qquad y, z \in \mathbb{R}.$$

(Formally, with $\theta_R \equiv 1$ and the η-integral inside E_z written out, this is the paper's $K_{+,j}(y,z) = -\frac{i^{j+1}}{2\pi}\int_{1<\xi<\eta}(\chi(\xi)-1)\xi^{j-2}e^{i(\Phi_0(\eta)-\Phi_0(\xi)-z\eta+y\xi)}d\xi\,d\eta$.)

Lemma 9.10 (Smoothness after truncating the positive symbol)**.** *Let $\frac{3}{4} < p < 1$, let γ be admissible, let $G \in \widetilde{Y}_\gamma$, and let $R \ge 2$. For every integer $j \ge 0$ the function $\xi \mapsto \mathbf{1}_{\{\xi>0\}}(i\xi)^j\theta_R(\xi)\widehat{\mathcal{T}_+G}(\xi)$ is in $L^1(\mathbb{R})$ and vanishes outside $[1, 2R]$. The function $\mathcal{T}_+^R G$ is $C^\infty(\mathbb{R};\mathbb{C})$, and for every $j \ge 0$, $y \in \mathbb{R}$,*

$$\partial_y^j \mathcal{T}_+^R G(y) = \frac{1}{2\pi}\int_1^{2R}(i\xi)^j\theta_R(\xi)\widehat{\mathcal{T}_+G}(\xi)e^{iy\xi}\,d\xi.$$

Every integral here is absolutely convergent. For each j there is $C_{p,\gamma,R,j} > 0$, independent of G and y, such that $|\partial_y^j\mathcal{T}_+^R G(y)| \le C_{p,\gamma,R,j}\|G\|_{Y_\gamma}$.

Proof. The symbol bound in Lemma 9.6(4), together with $0 \le \theta_R \le 1$ and its support, gives an integrable majorant $C\|G\|_{Y_\gamma}\xi^{j-2}\mathbf{1}_{[1,2R]}$ for each order j. The same majorant for order $j+1$ controls the difference quotients of the jth derivative uniformly in y. Apply differentiation under the integral and dominated convergence, then induct on j. Integrating the majorant gives the final bound; no estimate uniform in R is asserted here.

□

Lemma 9.11 (An integrable majorant for the truncated kernel)**.** *Let $\frac{3}{4} < p < 1$ and $R \ge 2$. There is $C_{p,R} > 0$, independent of j, y, z, such that for $j \in \{0,1,2\}$*

and every $y, z \in \mathbb{R}$ the defining ξ-integral for $K^R_{+,j}(y,z)$ is absolutely convergent and

$$\int_1^{2R} |a_j^R(\xi) E_z(\xi)|\, d\xi \le C_{p,R} w_E(z), \qquad |K^R_{+,j}(y,z)| \le C_{p,R} w_E(z).$$

The kernel $(y,z) \mapsto K^R_{+,j}(y,z)$ is continuous on $\mathbb{R}^2$.

Proof. Use $|E_z(\xi)| \le C_E w_E(z)$ on $[1, 2R]$, where a_j^R is continuous and integrable. A valid common constant is $1 + C_E \sum_{j=0}^{2} \int_1^{2R} |a_j^R(\xi)| d\xi$. The exponential has modulus one. Joint continuity follows by dominated convergence on each compact set of (y,z), using continuity of $E_z(\xi)$ and boundedness of w_E on its z projection.

□

Lemma 9.12 (Fubini for the truncated kernel and the input derivative)**.** *Let $\frac{3}{4} < p < 1$, let γ be admissible, let $G \in Y_\gamma$, $R \ge 2$, and $j \in \{0,1,2\}$. For every $y \in \mathbb{R}$ the function*

$$(\xi, z) \longmapsto a_j^R(\xi) e^{i(y\xi - \Phi_0(\xi))} E_z(\xi) G'(z)$$

is integrable on $[1, 2R] \times \mathbb{R}$. Moreover $z \mapsto K^R_{+,j}(y,z) G'(z)$ is integrable and

$$\int_1^{2R} a_j^R(\xi) e^{i(y\xi - \Phi_0(\xi))} F_G(\xi)\, d\xi = \int_{\mathbb{R}} K^R_{+,j}(y,z) G'(z)\, dz.$$

There is $C_{p,\gamma,R} > 0$, independent of G, j, y, bounding the double integral of the absolute value by $C_{p,\gamma,R} \|G\|_{Y_\gamma}$.

Proof. The kernel factors are jointly continuous on $[1, 2R] \times \mathbb{R}$ and G' is measurable. Tonelli and Lemma 9.11 bound the double integral by $C_{p,R} \int |G'(z)| w_E(z) dz$, finite and bounded by Lemma 9.2(3). Fubini now applies to the absolutely integrable product. The inner integrals are precisely $F_G(\xi)$ and $K^R_{+,j}(y,z)$; the kernel majorant separately gives integrability of the latter pairing.

□

Lemma 9.13 (The differentiated truncated integral equals its kernel pairing)**.** *Let $\frac{3}{4} < p < 1$, let γ be admissible, $G \in \widetilde{Y}_\gamma$, and $R \ge 2$. For every $j \in \{0,1,2\}$ and $y \in \mathbb{R}$,*

$$\partial_y^j \mathcal{T}_+^R G(y) = \int_{\mathbb{R}} K^R_{+,j}(y,z) G'(z)\, dz,$$

with an absolutely convergent integral on the right.

Proof. Insert the projected tail formula $\widehat{\mathcal{T}_+ G} = i(1-\chi)\xi^{-2} e^{-i\Phi_0} F_G$ for $\xi > 0$ into Lemma 9.10. Check the normalization exactly: $(2\pi)^{-1} (i\xi)^j \theta_R\, i(1-\chi)\xi^{-2} = a_j^R$. Then apply Lemma 9.12. The integration by parts establishing $\widehat{G'} = i\xi \widehat{G}$ is already part of Lemma 9.6; it is not assumed anew.

□

Lemma 9.14 (Uniform error of frequency truncation)**.** *Let $\frac{3}{4} < p < 1$ and let γ be admissible. There is $C_{p,\gamma} > 0$, independent of G, R, y, such that for every $G \in \widetilde{Y}_\gamma$, $R \ge 2$ and $y \in \mathbb{R}$,*

$$|\mathcal{T}_+^R G(y) - \mathcal{T}_+ G(y)| \le C_{p,\gamma} R^{-1} \|G\|_{Y_\gamma}, \qquad |\mathcal{T}_{\mathrm{h}}^R G(y) - \mathcal{T}_{\mathrm{h}} G(y)| \le 2 C_{p,\gamma} R^{-1} \|G\|_{Y_\gamma}.$$

In particular both truncations converge uniformly on $\mathbb{R}$ as $R \to \infty$.

Proof. Since $\theta_R = 1$ on $[0, R]$ and $|1-\theta_R| \le 1$, the difference of the two positive-frequency integrals is bounded by $(2\pi)^{-1} C \|G\|_{Y_\gamma} \int_R^\infty \xi^{-2} d\xi$. Absolute convergence justifies subtracting the integrals. The integral is R^{-1} and the estimate is independent of y. Use Lemma 9.7, including its distributional Fourier identity and identification of the continuous representative, to replace $\mathcal{T}_{\mathrm{h}} G$ by $2 \operatorname{Re} \mathcal{T}_+ G$. Taking real parts gives the second inequality and the uniform limits.

□

Lemma 9.15 (Kernel representation of $\partial_y^j \mathcal{T}_+^R$)**.** *Let γ be admissible, $G \in \widetilde{Y}_\gamma$, $R \ge 2$. Then $\mathcal{T}_+^R G \in C^\infty(\mathbb{R})$, $|K_{+,j}^R(y,z)| \le C_R\, w_E(z)$, and for $j = 0, 1, 2$ and all y*

$$\partial_y^j \mathcal{T}_+^R G(y) = \int_{\mathbb{R}} K_{+,j}^R(y,z)\, G'(z)\, dz \qquad \textit{(absolutely convergent).}$$

Moreover $\sup_{y\in\mathbb{R}} |\mathcal{T}_+^R G(y) - \mathcal{T}_+ G(y)| \to 0$ *and* $\sup_y |\mathcal{T}_{\mathrm{h}}^R G - \mathcal{T}_{\mathrm{h}} G| \to 0$ *as* $R \to \infty$.

Proof. Smoothness is Lemma 9.10. The constant C_R and the kernel bound are Lemma 9.11; absolute convergence and the pointwise derivative identity are Lemmas 9.12 and 9.13. Finally Lemma 9.14 gives both uniform limits, with their correct representatives. □

9.4 Stationary and nonstationary splitting

Definition 9.16 (Stationary/nonstationary decomposition; paper line 2932)**.**

1. (paper eq. `eqn:linear_stationary_cutoffs`) For $y \in \mathbb{R}$, $\xi > 0$: $\chi_{\mathrm{s}}(y,\xi) := \chi\big(100(\xi - y^{1/p})/y^{1/p}\big)$ if $y > 0$, $\chi_{\mathrm{s}}(y,\xi) := 0$ if $y \le 0$; $\chi_{\mathrm{ns}} := 1 - \chi_{\mathrm{s}}$.
2. $a_j^{\mathrm{s},R}(y,\xi) := \chi_{\mathrm{s}}(y,\xi) a_j^R(\xi)$, $a_j^{\mathrm{ns},R}(y,\xi) := \chi_{\mathrm{ns}}(y,\xi) a_j^R(\xi)$.
3. (paper eq. `eqn:linear_stationary_kernel_split`) $K_{+,j}^{\mathrm{s},R}(y,z) := \int_1^{2R} a_j^{\mathrm{s},R}(y,\xi) e^{i(y\xi - \Phi_0(\xi))} E_z(\xi) d\xi$ and $K_{+,j}^{\mathrm{ns},R}$ likewise with $a_j^{\mathrm{ns},R}$; thus $K_{+,j}^R = K_{+,j}^{\mathrm{s},R} + K_{+,j}^{\mathrm{ns},R}$.
4. (paper eq. `eqn:linear_stationary_operators`) $\mathcal{T}_{+,j}^{\mathrm{s},R} G(y) := \int K_{+,j}^{\mathrm{s},R}(y,z) G'(z) dz$, $\mathcal{T}_{+,j}^{\mathrm{ns},R} G(y) := \int K_{+,j}^{\mathrm{ns},R}(y,z) G'(z) dz$, so that $\partial_y^j \mathcal{T}_+^R G = \mathcal{T}_{+,j}^{\mathrm{s},R} G + \mathcal{T}_{+,j}^{\mathrm{ns},R} G$ by Lemma 9.15.

Lemma 9.17 (Cutoff and amplitude bounds)**.** *There are constants $c_* \in (0,1)$ and C_ℓ ($\ell \ge 0$), depending only on p, χ, such that for all $R \ge 2$, $y \in \mathbb{R}$, $\xi > 0$, $j \in \{0,1,2\}$:*

1. *$\chi_{\mathrm{s}}(y,\xi) \ne 0 \Rightarrow y > 0$ and $|\xi - y^{1/p}| \le y^{1/p}/50$; $\chi_{\mathrm{s}}(y,\xi) = 1$ if $y > 0$ and $|\xi - y^{1/p}| \le y^{1/p}/100$.*
2. *$|\partial_\xi^\ell \chi_{\mathrm{s}}(y,\xi)| + |\partial_\xi^\ell \chi_{\mathrm{ns}}(y,\xi)| \le C_\ell \xi^{-\ell}$ for $\ell \ge 1$, and $0 \le \chi_{\mathrm{s}}, \chi_{\mathrm{ns}} \le 1$; moreover $\int_0^\infty |\partial_\xi \chi_{\mathrm{ns}}(y,\xi)| d\xi \le C_0$.*
3. *a_j^R is smooth on $(0,\infty)$, vanishes on $(0,1]$ with all derivatives, is supported in $[1, 2R]$, and $|\partial_\xi^\ell a_j^R(\xi)| \le C_\ell \xi^{j-2-\ell}$ (uniformly in R); $\int_0^\infty |\partial_\xi[(1-\chi)\theta_R](\xi)| d\xi \le C_0$.*

4. *If $\chi_{\mathrm{ns}}(y,\xi)\neq 0$ (or ξ is in its support) and $\xi\geq 1$ then $|y-\xi^p|\geq c_*(|y|+\xi^p)$; and $|y|+\xi^p\geq 2^{-1/2}\langle y\rangle$ for $\xi\geq 1$.*

5. *If $a_j^{\mathrm{s},R}(y,\xi)\neq 0$ then $y\geq y_*:=(50/51)^p$, $\frac{49}{50}y^{1/p}\leq\xi\leq\frac{51}{50}y^{1/p}$, and $\langle y\rangle\leq C_0\,y$.*

Proof. (1) $\operatorname{supp}\chi\subset[-2,2]$, $\chi=1$ on $[-1,1]$. (2) $\partial_\xi^\ell\chi_{\mathrm{s}}=(100y^{-1/p})^\ell\chi^{(\ell)}(\cdot)$, supported where $\xi\approx y^{1/p}$ by (1); the L^1 norm of $\partial_\xi\chi_{\mathrm{s}}$ is $\|\chi'\|_{L^1}$ (change of variables). (3) Product rule; $|\partial^\ell\theta_R|=R^{-\ell}|\chi^{(\ell)}(\xi/R)|\leq 2^\ell\|\chi^{(\ell)}\|_\infty\xi^{-\ell}$ on $\operatorname{supp}\theta_R\subset[0,2R]$; $1-\chi$ vanishes on $[-1,1]$; L^1 norms of derivatives of χ, θ_R are R-independent. (4) $y\leq 0$: $|y-\xi^p|=|y|+\xi^p$. $y>0$, $\xi\geq\frac{101}{100}y^{1/p}$: $\xi^p-y\geq(1-(\frac{100}{101})^p)\xi^p\geq\frac12(1-(\frac{100}{101})^p)(\xi^p+y)$. $y>0$, $\xi\leq\frac{99}{100}y^{1/p}$: $y-\xi^p\geq(1-0.99^p)y\geq\frac12(1-0.99^p)(y+\xi^p)$. Last claim: $|y|+\xi^p\geq\max(|y|,1)$. (5) $a_j^R(\xi)\neq 0$ needs $\xi>1$; combine with (1). □

9.4.1 Stationary part

Lemma 9.18 (Stationary kernel, near case)**.** *There is C (depending on p,χ) such that for all $R\geq 2$, $j\in\{0,1,2\}$, $y>0$ and $z\geq y/4$: $|K_{+,j}^{\mathrm{s},R}(y,z)|\leq C\langle y\rangle^{\frac{j}{p}-\frac{p+3}{2p}}\langle z\rangle^{\frac{1-p}{2p}}$.*

Proof. By Lemma 9.17(5) we may assume $y\geq y_*$ (else the kernel is 0), so $\langle y\rangle\approx y$. Apply Lemma 3.81 with $x_0=y^{1/p}$, $L=y^{1/p}/25$, $I=(x_0-L,x_0+L)$, $a=a_j^{\mathrm{s},R}(y,\cdot)$ (in $C_c^\infty(I)$ by Lemma 9.17(1)), $\Phi(\xi)=y\xi-\Phi_0(\xi)$, $\Psi=\Psi_z$, $\omega=y-z$ (so $\Phi+\Psi=\omega\xi$), $E=E_z$ (with $E'=-e^{i\Psi}$). Hypotheses: $\Phi'(x_0)=y-x_0^p=0$; on I, $\xi\in[\frac{24}{25}x_0,\frac{26}{25}x_0]$, so $|\Phi''|=p\xi^{p-1}\approx\kappa:=y^{(p-1)/p}$ and $|\Phi'''|=p(1-p)\xi^{p-2}\lesssim y^{(p-2)/p}\approx\kappa L^{-1}$; $|\partial^\ell a|\lesssim y^{(j-2)/p}y^{-\ell/p}\approx AL^{-\ell}$ $(\ell\leq 2)$ with $A=y^{(j-2)/p}$, by Lemma 9.17(2),(3); $|E_z|\leq M:=C_E\langle z\rangle^{\frac{1-p}{2p}}$ by Lemma 9.5(2) $(z>0)$. Conclusion: $|K_{+,j}^{\mathrm{s},R}|\lesssim y^{\frac{j-2}{p}}\big(\langle z\rangle^{\frac{1-p}{2p}}y^{\frac{1-p}{2p}}+y^{\frac{1-p}{p}}\big)$. As $z\geq y/4$, $y^{\frac{1-p}{2p}}\leq 4^{\frac{1-p}{2p}}\langle z\rangle^{\frac{1-p}{2p}}$, so both terms are $\lesssim y^{\frac{j-2}{p}+\frac{1-p}{2p}}\langle z\rangle^{\frac{1-p}{2p}}$ and $\frac{j-2}{p}+\frac{1-p}{2p}=\frac{j}{p}-\frac{p+3}{2p}$. □

Lemma 9.19 (Asymptotic expansion of the tail in the nonstationary regime)**.** *Let $z\in\mathbb{R}$, $\xi_0>0$, $c_1\in(0,1]$ with $\xi_0^p-z\geq c_1(\xi_0^p+|z|)$. Put $\psi(\eta)=\eta^p-z$, $g_0=1/(i\psi)$ and $Dh:=h'/(i\psi)$ on $[\xi_0,\infty)$. Then $\psi(\eta)\geq c_1(\eta^p+|z|)$ for $\eta\geq\xi_0$; for $k,\ell\geq 0$ there is $C_{k,\ell}=C_{k,\ell}(p,c_1)$ with*

$$|\partial_\eta^\ell D^k g_0(\eta)|\leq C_{k,\ell}\,\eta^{-k-\ell}(\eta^p+|z|)^{-k-1},\qquad \eta\geq\xi_0;$$

and for every $\xi\geq\xi_0$, $M\geq 0$,

$$E_z(\xi)=e^{i\Psi_z(\xi)}\sum_{k=0}^{M}(-1)^{k+1}D^kg_0(\xi)+(-1)^{M+1}\int_\xi^\infty e^{i\Psi_z(\eta)}\partial_\eta\big[D^Mg_0\big](\eta)\,d\eta,$$

the last integral being absolutely convergent.

Proof. $\psi(\eta)=(\eta^p-\xi_0^p)+(\xi_0^p-z)\geq(\eta^p-\xi_0^p)+c_1(\xi_0^p+|z|)\geq c_1(\eta^p+|z|)$. Derivative bounds: $\partial^\ell(1/\psi)$ is a sum of terms $\psi^{-1-r}\prod_{i=1}^r\psi^{(\ell_i)}$ with $\ell_i\geq 1$, $\sum\ell_i=\ell$, and $|\psi^{(\ell_i)}|\lesssim\eta^{p-\ell_i}\leq\eta^{-\ell_i}(\eta^p+|z|)$, so $|\partial^\ell(1/\psi)|\lesssim\eta^{-\ell}(\eta^p+|z|)^{-1}$; then induction on k with the Leibniz rule (D costs one derivative and

one factor $1/\psi$). Expansion: induction on M. For $M=0$, write $e^{i\Psi_z} = g_0\,\partial_\eta e^{i\Psi_z}$ and integrate by parts on $[\xi,N]$, $N\to\infty$ (boundary term at N is $O((N^p+|z|)^{-1})\to 0$). For the step, $\int_\xi^\infty e^{i\Psi_z}\partial(D^M g_0) = \int_\xi^\infty D^{M+1}g_0\,\partial e^{i\Psi_z} = -D^{M+1}g_0(\xi)e^{i\Psi_z(\xi)} - \int_\xi^\infty \partial(D^{M+1}g_0)e^{i\Psi_z}$. Absolute convergence: $|\partial D^M g_0| \lesssim \eta^{-M-1}(\eta^p+|z|)^{-M-1} \le c\,\eta^{-1-p}$ (as $M\ge 0$, $\eta\ge\xi_0>0$) is integrable at ∞. □

Lemma 9.20 (Stationary kernel, far case)**.** *For every $M\ge 0$ there is C_M (depending on p,χ,M) such that for all $R\ge 2$, $j\in\{0,1,2\}$, $y>0$, $z<y/4$: $|K^{\mathrm{s},R}_{+,j}(y,z)| \le C_M\langle y\rangle^{-M}(\langle y\rangle+|z|)^{-M}$.*

Proof. Assume $y\ge y_*$ (else $K=0$). Let $\xi_0=\frac{49}{50}y^{1/p}$; then $\xi_0^p\ge\frac{49}{50}y$, and $z<y/4$ gives $\xi_0^p - z \ge \frac12(\xi_0^p+|z|)$ (if $z\le 0$ trivially; if $0\le z<y/4$, $\xi_0^p - z\ge 0.73y$ and $\xi_0^p+|z|\le 1.25y$). Fix $N\ge 1$ and insert the expansion of Lemma 9.19 (order N, $c_1=\frac12$) for $\xi\in\operatorname{supp} a_j^{\mathrm{s},R}(y,\cdot)\subset[\xi_0,\frac{51}{50}y^{1/p}]$. Since $y\xi-\Phi_0(\xi)+\Psi_z(\xi)=\xi(y-z)$,

$$K^{\mathrm{s},R}_{+,j} = \sum_{k=0}^{N}(-1)^{k+1}\int a_j^{\mathrm{s},R}(y,\xi)e^{i\xi(y-z)}D^k g_0(\xi)d\xi + (-1)^{N+1}\int a_j^{\mathrm{s},R}(y,\xi)e^{i(y\xi-\Phi_0(\xi))}\int_\xi^\infty e^{i\Psi_z}\partial(D^N g_0)\,d\eta\,d\xi.$$

(paper eq. `eqn:linear_stationary_boundary_ibp`) $y-z\ge\frac35(y+|z|)\ge c(\langle y\rangle+|z|)$. Integrating by parts N times in ξ (compact support, no boundary terms) and using Lemma 9.17(2),(3),(5) and $\xi\approx y^{1/p}$, $\xi^p+|z|\approx y+|z|$ on the support (length $\lesssim y^{1/p}$): the k-th term is $\lesssim_N |y-z|^{-N}y^{1/p}y^{(j-2-N-k)/p}(y+|z|)^{-k-1} \lesssim \langle y\rangle^{(j-1-N-k)/p}(\langle y\rangle+|z|)^{-N-k-1}$. (paper eq. `eqn:linear_stationary_remainder_bound`) $\int_\xi^\infty \eta^{-N-1}(\eta^p+|z|)^{-N-1}d\eta \le N^{-1}\xi^{-N}(\xi^p+|z|)^{-N-1}$, so the last term is $\lesssim_N y^{1/p}y^{(j-2)/p}y^{-N/p}(y+|z|)^{-N-1}\approx\langle y\rangle^{(j-1-N)/p}(\langle y\rangle+|z|)^{-N-1}$. As $j\le 2$, choosing $N:=2M+1$ gives $(N+1-j)/p\ge M$ and $N+1\ge M$, whence the claim.

□

Proposition 9.21 (Stationary estimate)**.** *Let $c=\frac14$. There is C and, for every $M\ge 0$, C_M (depending on p,χ, M) such that for all $R\ge 2$, $j\in\{0,1,2\}$, $y,z\in\mathbb{R}$: $K^{\mathrm{s},R}_{+,j}(y,z)=0$ unless $y\ge y_*=(50/51)^p$, and*

$$|K^{\mathrm{s},R}_{+,j}(y,z)| \le \begin{cases} C\langle y\rangle^{\frac{j}{p}-\frac{p+3}{2p}}\langle z\rangle^{\frac{1-p}{2p}}, & z\ge cy,\\ C_M\langle y\rangle^{-M}(\langle y\rangle+|z|)^{-M}, & z<cy.\end{cases}$$

Proof. Vanishing: Lemma 9.17(5). The two cases are Lemmas 9.18 and 9.20.

□

9.4.2 Nonstationary part: symbol expansion

Lemma 9.22 (Nonstationary symbol expansion)**.** *Fix $R\ge 2$, $y\in\mathbb{R}$, $j\in\{0,1,2\}$ and put $\Theta_y(\xi):=y\xi-\Phi_0(\xi)$ (so $\Theta_y'=y-\xi^p$ for $\xi>0$). Define on $(0,\infty)$ (paper eq. `eqn:linear_nonstationary_symbols`)*

$$m^R_{j,0}(y,\xi) := \frac{a_j^{\mathrm{ns},R}(y,\xi)}{i(y-\xi^p)}, \qquad m^R_{j,k+1}(y,\xi) := -\frac{\partial_\xi m^R_{j,k}(y,\xi)}{i(y-\xi^p)},$$

set to 0 where $a_j^{\mathrm{ns},R}(y,\cdot)$ vanishes identically nearby (these are C^∞ functions of ξ supported in $[1,2R]$, since $a_j^{\mathrm{ns},R}(y,\cdot)$ vanishes on a neighbourhood of the zero $\xi=y^{1/p}$ of $y-\xi^p$). Then for $M\ge 1$:

1. *for every* $\eta \ge 1$, $\int_1^\eta a_j^{\mathrm{ns},R}(y,\xi)e^{i\Theta_y(\xi)}d\xi = e^{i\Theta_y(\eta)}\sum_{k=0}^{M-1} m_{j,k}^R(y,\eta) - \int_1^\eta e^{i\Theta_y(\xi)}\partial_\xi m_{j,M-1}^R(y,\xi)\,d\xi$;

2. *for every* $z\in\mathbb{R}$, *with* $K_{\mathcal{R}_{j,M}}^R(y,z) := -\int_1^{2R}\partial_\xi m_{j,M-1}^R(y,\xi)e^{i\Theta_y(\xi)}E_z(\xi)\,d\xi$,

$$K_{+,j}^{\mathrm{ns},R}(y,z) = 2\pi\sum_{k=0}^{M-1} m_{j,k}^{R\,\check{}}(y,\cdot)(y-z) + K_{\mathcal{R}_{j,M}}^R(y,z);$$

3. *(paper eq.* `eqn:linear_nonstationary_expansion`*) for admissible* γ *and* $G\in\widetilde{Y}_\gamma$, $\mathcal{T}_{+,j}^{\mathrm{ns},R}G(y) = 2\pi\sum_{k=0}^{M-1}\big[m_{j,k}^{R\,\check{}}(y,\cdot)*G'\big](y) + \mathcal{R}_{j,M}^R G(y)$ *with* $\mathcal{R}_{j,M}^R G(y) := \int K_{\mathcal{R}_{j,M}}^R(y,z)G'(z)dz$ *(paper eq.* `eqn:linear_nonstationary_remainder_definition`*); all integrals absolutely convergent.*

Proof. (1) Induction on M. On $\operatorname{supp} a_j^{\mathrm{ns},R}$, $a^{\mathrm{ns}}e^{i\Theta_y} = m_{j,0}^R\partial_\xi e^{i\Theta_y}$; integrate by parts on $[1,\eta]$; the boundary term at 1 vanishes since all $m_{j,k}^R(y,\cdot)$ vanish on $(0,1]$. Step: $-\int_1^\eta \partial m_{j,M-1}e^{i\Theta_y} = \int_1^\eta m_{j,M}\partial e^{i\Theta_y} = m_{j,M}(\eta)e^{i\Theta_y(\eta)} - \int_1^\eta \partial m_{j,M}e^{i\Theta_y}$. (2) For $N > 2R$, by Lemma 9.5(1), $E_z(\xi) = \lim_N\int_\xi^N e^{i\Psi_z}$ with $|\int_\xi^N e^{i\Psi_z}| \le 2C_E w_E(z)$, so by dominated convergence and Fubini on the bounded triangle $\{1\le\xi\le\eta\le N\}$, $K_{+,j}^{\mathrm{ns},R}(y,z) = \lim_N\int_1^N e^{i\Psi_z(\eta)}\int_1^\eta a_j^{\mathrm{ns},R}e^{i\Theta_y}\,d\xi\,d\eta$. Insert (1). Since $\Psi_z(\eta)+\Theta_y(\eta) = \eta(y-z)$, the first part gives $\sum_k\int_1^{2R} m_{j,k}^R(y,\eta)e^{i\eta(y-z)}d\eta = 2\pi\sum_k m_{j,k}^{R\,\check{}}(y,\cdot)(y-z)$; in the second part swap back (Fubini on the triangle) to get $-\int_1^{2R}\partial_\xi m_{j,M-1}^R e^{i\Theta_y}\int_\xi^N e^{i\Psi_z}d\eta\,d\xi \to K_{\mathcal{R}_{j,M}}^R(y,z)$. (3) Integrate (2) against $G'(z)$: $m_{j,k}^{R\,\check{}}(y,\cdot)$ is a Schwartz function (smooth compactly supported symbol) and $|K_{\mathcal{R}_{j,M}}^R(y,z)| \le C_R w_E(z)$, so every term is absolutely integrable (Lemma 9.2(3)). Lean route: (2) is proved by repeated integration by parts in ξ directly against E_z (using $\partial_\xi E_z = -e^{i\Psi_z}$ and $m_{j,k}^R(y,2R)=0$) instead of Fubini on the triangle; (3) by integrating (2) against G'.

□

Lemma 9.23 (Symbol-class bounds for $m_{j,k}$)**.** *For* $j\in\{0,1,2\}$, $k,\ell\ge 0$ *there is* $C_{k,\ell}$ *(depending on* p,χ*) such that for all* $R\ge 2$, $y\in\mathbb{R}$, $\xi > 0$*:*

$$|\partial_\xi^\ell m_{j,k}^R(y,\xi)| \le C_{k,\ell}\,\xi^{j-2-k-\ell}(|y|+\xi^p)^{-k-1}.$$

Moreover $i^{-(j+k)}m_{j,k}^R(y,\xi)\in\mathbb{R}$; $m_{j,0}^R = (i\xi)^j m_{0,0}^R$; *and, with* $b := (1-\chi)\chi_{\mathrm{ns}}(y,\cdot)\theta_R$,

$$m_{2,1}^R = -\frac{i}{2\pi}\Big[\frac{\partial_\xi b}{(y-\xi^p)^2} + \frac{p\,b\,\xi^{p-1}}{(y-\xi^p)^3}\Big], \qquad \xi m_{1,1}^R = -i\,m_{2,1}^R + \frac{b}{2\pi\,\xi\,(y-\xi^p)^2}.$$

Proof. On $\operatorname{supp}\chi_{\mathrm{ns}}(y,\cdot)\cap[1,\infty)$, $|y-\xi^p| \ge c_*(|y|+\xi^p)$ (Lemma 9.17(4)), and as in the proof of Lemma 9.19, $|\partial^\ell(y-\xi^p)^{-1}| \lesssim \xi^{-\ell}(|y|+\xi^p)^{-1}$ there. With $|\partial^\ell a_j^{\mathrm{ns},R}| \lesssim \xi^{j-2-\ell}$ (Lemma 9.17(2),(3)) the case $k=0$ follows from Leibniz; the recursion $m_{j,k+1} = i\,\partial_\xi m_{j,k}\cdot(y-\xi^p)^{-1}$ gains $\xi^{-1}(|y|+\xi^p)^{-1}$. Reality: $m_{j,0} = \frac{i^j}{2\pi}\,b\,\xi^{j-2}(y-\xi^p)^{-1}$ and each step multiplies by i times a real function. The explicit formulas: direct differentiation of $m_{2,0} = -\frac{1}{2\pi}b(y-\xi^p)^{-1}$ and $m_{1,0} = \frac{i}{2\pi}b\xi^{-1}(y-\xi^p)^{-1}$.

□

9.4.3 Nonstationary part: leading kernels

Lemma 9.24 (The function q)**.** *For $y \in \mathbb{R}$, $R \geq 2$, $\xi > 0$ put $q^R(y,\xi) := -\frac{1}{2\pi}\theta_R(\xi)\chi_{\rm ns}(y,\xi)(y-\xi^p)^{-1}$ (so $m^R_{2,0} = (1-\chi)q^R$; q^R is real, smooth on $(0,\infty)$, continuous on $[0,\infty)$ if $y \neq 0$, and $|q^R(0,\xi)| = \frac{1}{2\pi}\xi^{-p}$ near 0). There is C (depending on p,χ) such that*

1. *for all $R \geq 2$ and $\xi > 0$: $|q^R| \leq C(|y|+\xi^p)^{-1}$ and $|\partial_\xi^\ell q^R| \leq C_\ell \xi^{-\ell}(|y|+\xi^p)^{-1}$;*
2. *if $R \geq R_0(y)$: $|\partial_\xi q^R| \leq C\xi^{p-1}(|y|+\xi^p)^{-2}$ and $|\partial_\xi^2 q^R| \leq C\xi^{p-2}(|y|+\xi^p)^{-2}$ for all $\xi > 0$.*

Proof. (1) as in Lemma 9.23 (for $\xi < 1$ use $\chi_{\rm ns} = 1$ unless $\xi \approx y^{1/p}$, and Lemma 9.17(4) holds on the support of $\chi_{\rm ns}$ for all $\xi > 0$ by the same proof). (2) Derivatives falling on $(y-\xi^p)^{-1}$ give $p\xi^{p-1}(y-\xi^p)^{-2}$ and $p(p-1)\xi^{p-2}(y-\xi^p)^{-2} + 2p^2\xi^{2p-2}(y-\xi^p)^{-3}$, bounded as claimed. Derivatives falling on $\chi_{\rm ns}$ live where $\xi^p \approx y$, where $\xi^{-\ell}(|y|+\xi^p)^{-1} \approx \xi^{p-\ell}(|y|+\xi^p)^{-2}$. Derivatives falling on θ_R live on $[R,2R]$, where $R \geq R_0(y)$ gives $\xi^p \geq R^p \geq |y|$, so again $\xi^{-\ell}(|y|+\xi^p)^{-1} \leq 2\xi^{p-\ell}(|y|+\xi^p)^{-2}$.

□

Lemma 9.25 (Leading kernel $m_{2,0}$)**.** *For every $L \geq 0$ there is C_L such that for all $y \in \mathbb{R}$, $R \geq R_0(y)$, $z \neq 0$:*

$$\left|\operatorname{Re} \check{m}^R_{2,0}(y,\cdot)(z)\right| \leq C_L \begin{cases} |z|^{p-1}, & 0 < |z| \leq \langle y\rangle^{-1/p}, \\ \langle y\rangle^{-2}|z|^{-p-1} + \langle y\rangle^{-1}, & \langle y\rangle^{-1/p} < |z| \leq 1, \\ \langle y\rangle^{-1}|z|^{-L}, & |z| > 1. \end{cases}$$

Proof. Write $q = q^R(y,\cdot)$. Since q is real, $2\pi \operatorname{Re} \check{m}^R_{2,0}(z) = \int_0^\infty q\cos(z\xi)d\xi - \int_0^\infty \chi q\cos(z\xi)d\xi$ (both absolutely convergent: compact support, $|q| \lesssim \xi^{-p}$ near 0). Second term: $\leq \int_0^2 C(|y|+\xi^p)^{-1} \leq C\langle y\rangle^{-1}$. First bound, $0 < |z| \leq 1$: split at $1/|z|$; $\int_0^{1/|z|}|q| \leq C\int_0^{1/|z|}\xi^{-p} \lesssim |z|^{p-1}$; on $[1/|z|,\infty)$ integrate by parts once: $\leq |z|^{-1}(|q(1/|z|)| + \int_{1/|z|}^\infty |q'|) \lesssim |z|^{-1}(|z|^p + \int_{1/|z|}^\infty \xi^{-p-1}) \lesssim |z|^{p-1}$ (Lemma 9.24(1)). Second bound, $|y| \geq 1$, $0 < |z| \leq 1$: integrate by parts on $(0,\infty)$ (q is continuous at 0, $\sin 0 = 0$): $\int q\cos(z\xi) = -z^{-1}\int_0^\infty q'\sin(z\xi)$. On $(0,1/|z|)$ use $|\sin(z\xi)| \leq |z|\xi$ and Lemma 9.24(2): $\leq |z|^{-1}\cdot|z|\int_0^{1/|z|} C\xi^p|y|^{-2} \lesssim |y|^{-2}|z|^{-p-1}$. On $(1/|z|,\infty)$ integrate by parts again: $\leq |z|^{-2}(|q'(1/|z|)| + \int_{1/|z|}^\infty |q''|) \lesssim |z|^{-2}|y|^{-2}(|z|^{1-p}+|z|^{1-p})$. Total $\lesssim |y|^{-2}|z|^{-p-1}$. Combination: in the first region $\langle y\rangle^{-1} \leq 1 \leq |z|^{p-1}$; in the second region, if $|y| \geq 1$ use the second bound ($|y|^{-2} \leq 2\langle y\rangle^{-2}$), and if $|y| < 1$ then $|z| > 2^{-1/(2p)}$ so $|z|^{p-1} \lesssim 1 \lesssim \langle y\rangle^{-2}|z|^{-p-1}$. Third region: choose an integer $N \geq \max\{2, L\}$. $(1-\chi)q$ is smooth, supported in $[1,2R]$ and flat at 1; integrating by parts N times, $|\check{m}^R_{2,0}(z)| \leq \frac{|z|^{-N}}{2\pi}\int_1^\infty |\partial^N((1-\chi)q)| \lesssim |z|^{-N}\int_1^\infty \xi^{-N}(|y|+\xi^p)^{-1}d\xi \lesssim \langle y\rangle^{-1}|z|^{-N} \leq \langle y\rangle^{-1}|z|^{-L}$ (Lemma 9.24(1), Lemma 9.17(4)). Lean route: one works directly with $(1-\chi)q$ (supported in $[1,2R]$) instead of $q-\chi q$, via $2\pi\operatorname{Re}\check{m}^R_{2,0}(z) = \int_1^{2R+1}(1-\chi)q\cos(z\xi)\,d\xi$.

□

Lemma 9.26 (Leading kernels $m_{1,0}$, $m_{0,0}$)**.** *For all y, $R \geq 2$: $\partial_z \check{m}^R_{1,0}(y,\cdot) = m^R_{2,0}\check{}(y,\cdot)$ and $\partial_z m^R_{0,0}\check{}(y,\cdot) = m^R_{1,0}\check{}(y,\cdot)$. For every $L \geq 0$ there is C_L with, for all $y \in \mathbb{R}$, $R \geq R_0(y)$, $z \in \mathbb{R}$: $|\operatorname{Re}\partial_z m^R_{0,0}\check{}(y,\cdot)(z)| \leq C_L\langle y\rangle^{-1}\langle z\rangle^{-L}$ (paper eq. `eqn:linear_nonstationary_leading_kernel_j0`).*

Proof. $\partial_z \check{m} = i\xi\check{m}$ (Schwartz symbols) and $m_{j+1,0} = i\xi m_{j,0}$ (Lemma 9.23). Since $m_{1,0}$ is purely imaginary, $\operatorname{Re} m\check{}_{1,0}(0) = \frac{1}{2\pi}\operatorname{Re}\int m_{1,0} = 0$, so for $|z| \leq 1$, $|\operatorname{Re} m\check{}_{1,0}(z)| \leq \int_0^1 |\operatorname{Re} m\check{}_{2,0}(\pm s)|ds \lesssim \int_0^{\langle y\rangle^{-1/p}} s^{p-1}ds + \langle y\rangle^{-2}\int_{\langle y\rangle^{-1/p}}^1 s^{-p-1}ds + \langle y\rangle^{-1} \lesssim \langle y\rangle^{-1}$ by Lemma 9.25. For $|z| > 1$ choose an integer $N \geq \max\{L,1\}$ and integrate by parts N times in $m\check{}_{1,0}(z) = \frac{1}{2\pi}\int_1^\infty m_{1,0}e^{iz\xi}$: $\lesssim |z|^{-N}\int_1^\infty \xi^{-1-N}(|y|+\xi^p)^{-1} \lesssim \langle y\rangle^{-1}|z|^{-N}$. For $|z| \leq 1$, $\langle z\rangle^{-L} \geq 2^{-L/2}$; for $|z| > 1$, $|z|^{-N} \leq 2^{L/2}\langle z\rangle^{-L}$.

□

Lemma 9.27 (Uniform bound for the sine integral)**.** *There is an absolute constant C_{Si} such that $\left|\int_a^b \frac{\sin t}{t}dt\right| \leq C_{\mathrm{Si}}$ for all $0 < a < b$; consequently $\left|\int_1^\xi \frac{\sin(z\eta)}{\eta}d\eta\right| \leq C_{\mathrm{Si}}$ for all $z \in \mathbb{R}$, $\xi \geq 1$.*

Proof. For $b \leq 1$ the integrand is bounded by 1. For $a \geq 1$ integrate by parts: $|\int_a^b| \leq \frac{1}{a} + \frac{1}{b} + \int_a^b t^{-2} \leq 3$. Split otherwise. The second claim follows by substituting $t = z\eta$ ($z > 0$; odd in z; trivial for $z = 0$). □

Lemma 9.28 (Kernels with $k = 1$)**.** *For every $L \geq 0$ there is C_L such that for all $R \geq 2$, $y, z \in \mathbb{R}$:*

$$\sum_{j=0}^{1}\left|\operatorname{Re}\partial_z m^R_{j,1}\check{}(y,\cdot)(z)\right| + \left|\operatorname{Re} m^R_{2,1}\check{}(y,\cdot)(z)\right| \leq C_L\langle y\rangle^{-2}\langle z\rangle^{-L}.$$

Proof. Write $\partial_z m\check{}_{j,1} = i\xi\check{m}_{j,1}$. L^1 bounds: $\|\xi m_{0,1}\|_{L^1} \lesssim \int_1^\infty \xi^{-2}(|y|+\xi^p)^{-2} \lesssim \langle y\rangle^{-2}$; by the formula in Lemma 9.23, $\|m_{2,1}\|_{L^1} \lesssim \langle y\rangle^{-2}\int|\partial_\xi b| + \int_1^\infty \xi^{p-1}(|y|+\xi^p)^{-3}d\xi \lesssim \langle y\rangle^{-2}$, using Lemma 9.17(2),(3),(4) ($\|\partial_\xi b\|_{L^1} \leq C$ uniformly in y, R). Hence for $|z| \leq 1$ the terms with $\xi m_{0,1}$ and $m_{2,1}$ are $\lesssim \langle y\rangle^{-2}$. For $j = 1$: $m_{1,1}$ is real, so $\operatorname{Re}\partial_z m\check{}_{1,1}(z) = -\frac{1}{2\pi}\int \xi m_{1,1}\sin(z\xi)d\xi$. By Lemma 9.23, $\xi m_{1,1} = -im_{2,1} + B/(2\pi\xi)$ with $B := b(y-\xi^p)^{-2}$; the first part is bounded by $\|m_{2,1}\|_{L^1}$. For the second, with $S(\xi) := \int_1^\xi \frac{\sin(z\eta)}{\eta}d\eta$ ($S(1) = 0$, $|S| \leq C_{\mathrm{Si}}$ by Lemma 9.27, B compactly supported), $\int_1^\infty B\frac{\sin(z\xi)}{\xi}d\xi = -\int_1^\infty B'S\,d\xi$, bounded by $C_{\mathrm{Si}}\int|B'| \lesssim \langle y\rangle^{-2}\int|\partial b| + \int_1^\infty \xi^{p-1}(|y|+\xi^p)^{-3} \lesssim \langle y\rangle^{-2}$. For $|z| > 1$ choose an integer $N \geq \max\{L,1\}$, integrate by parts N times (smooth, supported in $[1,2R]$, flat at 1), and use $|\partial^N(\xi m_{0,1})| \lesssim \xi^{-2-N}\langle y\rangle^{-2}$, $|\partial^N(\xi m_{1,1})| + |\partial^N m_{2,1}| \lesssim \xi^{-1-N}\langle y\rangle^{-2}$ (Lemma 9.23, Lemma 9.17(4)); all integrable on $[1,\infty)$ as $N \geq 1$. Conclude as in Lemma 9.26.

□

Lemma 9.29 (Kernels with $k \geq 2$)**.** *For $k \geq 2$, $L \geq 0$ there is $C_{k,L}$ such that for all $R \geq 2$, $y, z \in \mathbb{R}$: $\sum_{j=0}^1 |\partial_z m^R_{j,k}\check{}(y,\cdot)(z)| + |m^R_{2,k}\check{}(y,\cdot)(z)| \leq C_{k,L}\langle y\rangle^{-k-1}\langle z\rangle^{-L}$.*

Proof. The symbols $i\xi m_{0,k}, i\xi m_{1,k}, m_{2,k}$ satisfy $|\partial^\ell(\cdot)| \lesssim_{k,\ell} \xi^{-k-\ell}(|y|+\xi^p)^{-k-1} \lesssim \xi^{-k-\ell}\langle y\rangle^{-k-1}$ on $[1,\infty)$ (Lemma 9.23, Lemma 9.17(4)), integrable for $k \geq 2$,

$\ell \geq 0$. For $|z| \leq 1$ use the L^1 norm ($\ell = 0$); for $|z| > 1$ integrate by parts an integer $N \geq \max\{L, 1\}$ times and use $|z|^{-N} \leq 2^{L/2}\langle z\rangle^{-L}$.

□

Lemma 9.30 (Symbol bounds)**.** *For every $L \geq 0$, $k \geq 0$ there are constants C_L, $C_{k,L}$ (depending on p, χ) such that for all $y \in \mathbb{R}$ and $R \geq R_0(y)$:*

1. *$k = 0$: paper eq. `eqn:linear_nonstationary_leading_kernel_j0`, i.e. $|\operatorname{Re}\partial_z \check{m}^R_{0,0}(y,\cdot)(z)| \leq C_L\langle y\rangle^{-1}\langle z\rangle^{-L}$, and (paper eq. `eqn:linear_nonstationary_leading_kernel_j12`) for $z \neq 0$, $|\operatorname{Re}\partial_z \check{m}^R_{1,0}(y,\cdot)(z)| + |\operatorname{Re}\check{m}^R_{2,0}(y,\cdot)(z)|$ is bounded by $2C_L$ times the three-region function of Lemma 9.25;*
2. *$k \geq 1$ (paper eq. `eqn:linear_nonstationary_higher_kernels`): $\sum_{j=0}^{1} |\operatorname{Re}\partial_z \check{m}^R_{j,k}(y,\cdot)(z)| + |\operatorname{Re}\check{m}^R_{2,k}(y,\cdot)(z)| \leq C_{k,L}\langle y\rangle^{-k-1}\langle z\rangle^{-L}$ (this part holds for all $R \geq 2$).*

Proof. Collect Lemmas 9.25 and 9.26 (using $\partial_z \check{m}_{1,0} = \check{m}_{2,0}$) and Lemmas 9.28 and 9.29. □

Lemma 9.31 (Weighted L^1 bounds for the kernels)**.** *Let γ be admissible and put $h^R_{j,k}(y,z) := \operatorname{Re}\partial_z \check{m}^R_{j,k}(y,\cdot)(z)$ for $j = 0, 1$ and $h^R_{2,k}(y,z) := \operatorname{Re}\check{m}^R_{2,k}(y,\cdot)(z)$. There is C such that for $0 \leq k \leq 4$, $j \in \{0,1,2\}$, $y \in \mathbb{R}$, $R \geq R_0(y)$ (paper eq. `eqn:linear_nonstationary_weighted_kernels`):*

$$\int_{\mathbb{R}} \langle z\rangle^{\gamma+1} |h^R_{j,k}(y,z)|\, dz \leq C\langle y\rangle^{-k-1}.$$

Proof. Take $L := \lceil\gamma\rceil + 3 > \gamma + 2$, so $\int\langle z\rangle^{\gamma+1}\langle z\rangle^{-L} < \infty$; this settles $k \geq 1$ and $j = 0$, $k = 0$. For $j = 1, 2$, $k = 0$: on $|z| > 1$ as before; on $|z| \leq 1$ ($\langle z\rangle^{\gamma+1} \leq 2^{(\gamma+1)/2}$) integrate the three-region bound: $2\int_0^{\langle y\rangle^{-1/p}} s^{p-1}ds + 2\langle y\rangle^{-2}\int_{\langle y\rangle^{-1/p}}^1 s^{-p-1}ds + 2\langle y\rangle^{-1} \leq \frac{2}{p}\langle y\rangle^{-1} + \frac{2}{p}\langle y\rangle^{-1} + 2\langle y\rangle^{-1}$. □

9.4.4 Nonstationary part: remainder

Lemma 9.32 (Remainder kernel)**.** *For $M \geq 2$ there is C_M (depending on p, χ, M) such that for all $R \geq 2$, $j \in \{0,1,2\}$, $y, z \in \mathbb{R}$: $|K^R_{\mathcal{R}_{j,M}}(y,z)| \leq C_M\langle y\rangle^{-M} w_E(z)$, i.e. $\lesssim_M \langle y\rangle^{-M}\langle z\rangle^{-1}$ for $z \leq 0$ and $\lesssim_M \langle y\rangle^{-M}\langle z\rangle^{\frac{1-p}{2p}}$ for $z > 0$.*

Proof. By Lemma 9.23, $|\partial_\xi m^R_{j,M-1}| \lesssim_M \xi^{j-2-M}(|y|+\xi^p)^{-M} \leq \xi^{-M}2^{M/2}\langle y\rangle^{-M}$ on $[1,\infty)$ (Lemma 9.17(4)), integrable since $M \geq 2$; and $|E_z(\xi)| \leq C_E w_E(z)$ (Lemma 9.5(2)).

□

9.5 The high-frequency estimate

Lemma 9.33 (Stationary contribution)**.** *Let γ be admissible. There is C such that for all $G \in \widetilde{Y}_\gamma$, $R \geq 2$, $j \in \{0,1,2\}$, $y \in \mathbb{R}$: $|\mathcal{T}^{\mathrm{s},R}_{+,j}G(y)| \leq C\|G\|_{Y_\gamma} w_{\gamma,j}(y)$ (paper eq. `eqn:linear_positive_stationary_contribution`).*

Proof. If $y < y_*$ the kernel vanishes (Proposition 9.21); in particular for $y \le 0$. Let $y \ge y_*$ (so $\langle y\rangle \approx y$, $w_{\gamma,j}(y) = \langle y\rangle^{-\gamma+j/p}$). On $z \ge y/4$, using $|G'(z)| \le \|G\|\langle z\rangle^{-\gamma+1/p}$: contribution $\lesssim \|G\|\langle y\rangle^{\frac{j}{p}-\frac{p+3}{2p}} \int_{y/4}^{\infty}\langle z\rangle^{-s}dz$ with $s := \gamma - \frac{3-p}{2p} > 1$ (admissibility), and $\int_{y/4}^{\infty}\langle z\rangle^{-s} \lesssim \langle y\rangle^{1-s}$, giving $\langle y\rangle^{\frac{j}{p}-\frac{p+3}{2p}+1-s} = \langle y\rangle^{-\gamma+\frac{j}{p}}$. On $z < y/4$: by Proposition 9.21 with $M := 3$, contribution $\le C_3\langle y\rangle^{-6}\|G'\|_{L^1} \lesssim \|G\|\langle y\rangle^{-6} \le \|G\|\langle y\rangle^{-\gamma+j/p}$ (as $\gamma < 3 < 6$). □

Lemma 9.34 (Nonstationary contribution)**.** *Let γ be admissible. There is C such that for all $G \in \widetilde{Y}_\gamma$, $j \in \{0,1,2\}$, $y \in \mathbb{R}$ and $R \ge R_0(y)$: $|\operatorname{Re}\mathcal{T}^{\mathrm{ns},R}_{+,j}G(y)| \le C\|G\|_{Y_\gamma}w_{\gamma,j}(y)$.*

Proof. Use Lemma 9.22(3) with $M = 5$. Remainder: by Lemma 9.32 and Lemma 9.2(3), $|\mathcal{R}^R_{j,5}G(y)| \lesssim \langle y\rangle^{-5}\int|G'|w_E \lesssim \|G\|\langle y\rangle^{-5} \le \|G\|w_{\gamma,j}(y)$ since $5 > \gamma + 2$. Convolution terms, $0 \le k \le 4$: since G' is real, $\operatorname{Re}[\check m * G'](y) = \int \operatorname{Re}\check m(z)G'(y-z)dz$. For $j = 0,1$ integrate by parts in z ($\check m$ Schwartz, G bounded, $G \in W^{1,1}_{\mathrm{loc}}$; no boundary terms): (paper eq. `eqn:linear_nonstationary_convolution_j01`) $= \int h^R_{j,k}(y,z)G(y-z)dz$, and by Lemma 9.2(6) and Lemma 9.31, $|\cdot| \le 2^{\gamma/2}\|G\|\langle y\rangle^{-\gamma}\int\langle z\rangle^{\gamma}|h^R_{j,k}| \lesssim \|G\|\langle y\rangle^{-\gamma-k-1} \le \|G\|w_{\gamma,j}(y)$ for $j \le 1$. For $j = 2$ (paper eq. `eqn:linear_nonstationary_convolution_j2`): by Lemma 9.2(5), $|\int h^R_{2,k}(y,z)G'(y-z)dz| \lesssim \|G\|w_{\gamma,1}(y)\int\langle z\rangle^{\gamma+1}|h^R_{2,k}| \lesssim \|G\|w_{\gamma,1}(y)\langle y\rangle^{-k-1} \le \|G\|w_{\gamma,2}(y)$ (for $y \ge 0$: $\langle y\rangle^{-\gamma+1/p-1} \le \langle y\rangle^{-\gamma+2/p}$; for $y < 0$: $\langle y\rangle^{-\gamma-2-k} \le \langle y\rangle^{-\gamma-2}$). Sum over $k \le 4$.

□

Lemma 9.35 (Truncated high-frequency bound)**.** *Let γ be admissible. There is C such that for all $G \in \widetilde{Y}_\gamma$, $j \in \{0,1,2\}$, $y \in \mathbb{R}$, $R \ge R_0(y)$: $|\partial_y^j\mathcal{T}^R_{\mathrm{h}}G(y)| \le C\|G\|_{Y_\gamma}w_{\gamma,j}(y)$.*

Proof. $\partial^j\mathcal{T}^R_{\mathrm{h}}G = 2\operatorname{Re}\partial^j\mathcal{T}^R_+G = 2\operatorname{Re}\mathcal{T}^{\mathrm{s},R}_{+,j}G + 2\operatorname{Re}\mathcal{T}^{\mathrm{ns},R}_{+,j}G$ (Definition 9.16(4)); bound $|\operatorname{Re}\mathcal{T}^{\mathrm{s}}| \le |\mathcal{T}^{\mathrm{s}}|$ and apply the two lemmas.

□

Lemma 9.36 (Passing to the limit $R \to \infty$)**.** *Let $w_0, w_1, w_2 : \mathbb{R} \to (0,\infty)$ be continuous, $A \ge 0$, and let $(r_R)_{R\ge 2}$ be real C^2 functions with $r_R \to r$ uniformly on $\mathbb{R}$ as $R \to \infty$. Suppose that for every $Y > 0$ there is $R_1(Y)$ such that $|r_R^{(j)}(y)| \le Aw_j(y)$ for $j = 0,1,2$, $|y| \le Y$, $R \ge R_1(Y)$. Then $r \in C^1(\mathbb{R})$, r' is locally Lipschitz (so $r \in W^{2,\infty}_{\mathrm{loc}}$), $|r| \le Aw_0$, $|r'| \le Aw_1$ everywhere and $|r''| \le Aw_2$ a.e.*

Proof. Fix Y and a sequence $R_n \to \infty$, $R_n \ge R_1(Y)$. On $[-Y,Y]$ the r'_{R_n} are uniformly bounded and uniformly Lipschitz ($|r''_{R_n}| \le A\max w_2$), so by Arzelà–Ascoli a subsequence converges uniformly to some g. Passing to the limit in $r_R(y) - r_R(0) = \int_0^y r'_R$ gives $r(y) - r(0) = \int_0^y g$, so $r \in C^1$, $r' = g$ (the limit is thus independent of the subsequence). Pointwise limits give $|r| \le Aw_0$, $|r'| \le Aw_1$, and $|r'(y) - r'(x)| \le A|\int_x^y w_2|$; hence r' is locally Lipschitz, differentiable a.e. (Rademacher/Lebesgue), and at every point of differentiability $|r''(x)| \le Aw_2(x)$ by continuity of w_2. □

Proposition 9.37 (High-frequency estimate)**.** *Let γ be admissible ($\beta + \frac{1}{p} < \gamma < 2p+1$). There is $C_{\mathrm{h}}(\gamma)$ such that for all $G \in \widetilde{Y}_\gamma$: $\mathcal{T}_{\mathrm{h}}G \in X_\gamma$ and $\|\mathcal{T}_{\mathrm{h}}G\|_{X_\gamma} \le C_{\mathrm{h}}(\gamma)\|G\|_{Y_\gamma}$.*

Proof. Apply Lemma 9.36 to $r_R := \mathcal{T}_{\mathrm{h}}^R G$ (real, C^∞), $r := \mathcal{T}_{\mathrm{h}} G$ (real, Lemma 9.7), $w_j := w_{\gamma,j}$, $A := C\|G\|_{Y_\gamma}$ and $R_1(Y) := \max\{2, \langle Y\rangle^{1/p}\} \geq R_0(y)$ for $|y| \leq Y$: uniform convergence is Lemma 9.15, the bounds are Lemma 9.35. Then Lemma 9.2(1) gives $\|\mathcal{T}_{\mathrm{h}} G\|_{X_\gamma} \leq 6C\|G\|_{Y_\gamma}$. $\mathbb{R}$-linearity of $\mathcal{T}_{\mathrm{h}}$ is clear.

□

9.6 Assembly: proof of the linear inverse theorem

Lemma 9.38 (Boundedness of $\mathcal{T}_0$)**.** *Let γ be admissible. For $G \in \widetilde{Y}_\gamma$, the real function $r := \mathcal{T}_{\mathrm{l}} G + \mathcal{T}_{\mathrm{h}} G$ lies in X_γ, satisfies $\widehat{r} = m_G$ (so r represents $\mathcal{T}_0 G$ in the sense of Definition 8.33) and $\|\mathcal{T}_0 G\|_{X_\gamma} \leq C_T(\gamma)\|G\|_{Y_\gamma}$ with $C_T(\gamma) := C_{\mathrm{l}}(\gamma) + C_{\mathrm{h}}(\gamma)$. The map $G \mapsto \mathcal{T}_0 G$ is $\mathbb{R}$-linear and given by a formula independent of γ: if $\gamma \leq \gamma'$ are admissible and $G \in \widetilde{Y}_{\gamma'}$ ($\subset \widetilde{Y}_\gamma$), $\mathcal{T}_0 G$ is the same function.*

Proof. Admissible γ satisfy $1 < \gamma < p + 2$ (since $2p + 1 < p + 2$), so Proposition 8.58 gives $\mathcal{T}_{\mathrm{l}} G \in X_\gamma$ with $\|\mathcal{T}_{\mathrm{l}} G\|_{X_\gamma} \leq C_{\mathrm{l}}(\gamma)\|G\|_{Y_\gamma}$, and Proposition 9.37 gives $\mathcal{T}_{\mathrm{h}} G \in X_\gamma$ with the bound $C_{\mathrm{h}}(\gamma)$. Both are L^1 functions (Lemma 8.1) whose Fourier transforms are χm_G and $(1-\chi)m_G$ (Definition 8.42, Lemma 9.7), so $\widehat{r} = m_G$; uniqueness of r is part of Definition 8.33 (injectivity of the Fourier transform). Lemma 8.59 gives the norm bound. γ-independence: m_G, χ, ℓ do not involve γ (Definitions 2.10 and 8.33) and $Y_{\gamma'} \subset Y_\gamma$, $X_{\gamma'} \subset X_\gamma$.

□

Lemma 9.39 (Assembly of the linear inverse theorem)**.** *Let $\epsilon_0, \delta, \overline{B}$ be as in Theorem 2.20, where $\epsilon_0 \leq \epsilon_c$ with ϵ_c the threshold of Lemma 8.25. Then:*

1. *For every admissible γ, $\ell : Y_\gamma \to \mathbb{C}$ is well defined and continuous: there is $C_\ell(\gamma)$ with $|\ell(G)| \leq C_\ell(\gamma)\|G\|_{Y_\gamma}$.*

2. *The $\mathbb{R}$-linear map $\mathcal{T}_0$ of Definition 8.33, defined on $\bigcup_{\gamma\ adm.} \widetilde{Y}_\gamma$ with values in functions $\mathbb{R} \to \mathbb{R}$ and given by a formula independent of γ, satisfies for every admissible γ: $\mathcal{T}_0(\widetilde{Y}_\gamma) \subset X_\gamma$, $\|\mathcal{T}_0 G\|_{X_\gamma} \leq C_T(\gamma)\|G\|_{Y_\gamma}$ for $G \in \widetilde{Y}_\gamma$, and $\mathcal{L}_0(\mathcal{T}_0 G) = G$ pointwise on $\mathbb{R}$ for $G \in \widetilde{Y}_\gamma$ ((2.2)).*

3. *For each $\epsilon \in [0, \epsilon_0]$ the maps $\mathcal{U}_\epsilon, \mathcal{V}_\epsilon$ of Definitions 8.7 and 8.8 ($\mathcal{U}_\epsilon r = r \circ \phi_\epsilon^{-1}$, $\mathcal{V}_\epsilon u = (\phi_\epsilon')^{p+1}\, u \circ \phi_\epsilon$; independent of γ) are $\mathbb{R}$-linear bijections of the space of all functions $\mathbb{R} \to \mathbb{R}$ such that for every admissible γ: $\mathcal{U}_\epsilon$ maps X_γ bijectively onto X_γ, $\mathcal{V}_\epsilon$ maps Y_γ bijectively onto Y_γ, there is $C_{UV}(\gamma)$ with $\|\mathcal{U}_\epsilon\|_{X_\gamma \to X_\gamma}, \|\mathcal{U}_\epsilon^{-1}\|_{X_\gamma \to X_\gamma}, \|\mathcal{V}_\epsilon\|_{Y_\gamma \to Y_\gamma}, \|\mathcal{V}_\epsilon^{-1}\|_{Y_\gamma \to Y_\gamma} \leq C_{UV}(\gamma)$ for $0 \leq \epsilon \leq \epsilon_0$, and $\mathcal{L}_\epsilon r = \mathcal{V}_\epsilon(\mathcal{L}_0(\mathcal{U}_\epsilon r))$ for all $r \in X_\gamma$.*

4. *For every admissible γ there is $C_G(\gamma)$ with $\sup_{\mu \in \overline{B}} \|(\mathcal{V}_\epsilon^{-1} - I)G_\mu\|_{Y_\gamma} \leq C_G(\gamma)\epsilon$ for $0 \leq \epsilon \leq \epsilon_0$ ((2.3)).*

Proof. This assembles the linear inverse theorem (paper lines 3295–3308) from the results above. *(1)* Proposition 8.39 (existence of the improper limit of Definition 2.10 and $|\ell(G)| \leq C\|G\|_{Y_\gamma}$), applicable as admissible $\gamma > \beta + \frac{1}{p}$. *(2)* Let $G \in \widetilde{Y}_\gamma$. By Lemma 9.38 (which combines (8.13), Proposition 8.58 and Proposition 9.37), $\mathcal{T}_0 G$ is represented by $r \in X_\gamma$ with $\widehat{r} = m_G$, the bound

holds, and $\mathcal{T}_0$ is γ-independent. Since admissible $\gamma > \beta + \frac{1}{p} > 1 + \frac{1}{p}$, Lemma 8.37 applies and gives $\mathcal{L}_0 r = G$ pointwise on $\mathbb{R}$. (This is the Fourier-side argument of Lemma 8.37.) *(3)* Lemma 8.25(1) (bijections of all functions, restriction to X_γ, Y_γ, uniform bounds with $C_{UV}(\gamma) := C(\gamma)$, γ-independence) and Lemma 8.25(3) (the identity $\mathcal{V}_\epsilon(\mathcal{L}_0(\mathcal{U}_\epsilon r)) = \mathcal{L}_\epsilon r$ pointwise for $r \in X_\gamma$), valid for $\epsilon \le \epsilon_0 \le \epsilon_c$. Here $\mathcal{L}_\epsilon$ is defined in Definition 2.17 with the $\mathcal{R}_\epsilon$ of Definition 8.9, which is precisely the remainder produced by the conjugation. *(4)* Lemma 8.25(5) with the δ of Theorem 2.20: $C_G(\gamma) := C(\gamma)\delta/|\mu_*|$, uniformly in $\mu \in \overline{B}$. (The uniformity in μ comes from $G_\mu = \operatorname{Re} z\, G_+ + \operatorname{Im} z\, G_-$, $|z| \le \delta/|\mu_*|$.) $\square$

Chapter 10

Nonlinear estimates (paper §7)

10.1 Standing notation and the decomposition of the nonlinearity

Throughout this chapter the parameters p, β are those of Definition 2.1: $\frac{3}{4} < p < 1$, $\alpha = p+1$, $\beta = \frac{p+1}{2p}$. We repeatedly use the elementary consequences

$$2\beta = 1+\tfrac{1}{p}, \qquad \beta > 1, \qquad -\beta+\tfrac{1}{p} = \tfrac{1-p}{2p} \in (0,1), \qquad \tfrac{1}{p} < 1+p, \qquad 2p+1 < p+2, \tag{10.1}$$

(the inequality $1/p < 1+p$ holds since $p > \frac{\sqrt{5}-1}{2}$), and, for $\beta + \frac{1}{p} < \gamma < 2p+1$, the facts $\gamma > 1$, $\gamma > p$ and $\gamma - \frac{1}{p} > \beta$.

We use the following notation from Proposition 3.10 and Lemma 3.12 (owned by Chapter 3; not redefined here):

- A_ϵ (Definition 1.10) and $B_\epsilon(u) = A_\epsilon(u)^{-1} = (1+\epsilon u^2)^{\alpha/2}$;
- the smooth kernel $\Gamma_\epsilon : \mathbb{R}^2 \to \mathbb{R}$ of (3.1), and the operator $\mathcal{I}_\epsilon(g)(y) = \int_{\mathbb{R}} \Gamma_\epsilon(g'(y), S_g(y,z))\,|y-z|^{-\alpha}\,dz$, defined for $g \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ with $g' \in L^\infty(\mathbb{R})$;
- the secant $S_g(y,z) = \frac{g(y)-g(z)}{y-z}$ for $z \neq y$;
- $W(y) = \max\{-y, 0\}$, the cutoff χ_+ ($\chi_+ = 1$ on $[2,\infty)$, $\chi_+ = 0$ on $(-\infty, 1]$), and $b_\epsilon(y) = c_\epsilon \chi_+(y) y^{-p}$, with c_ϵ and $|c_\epsilon| \le C\epsilon$ from Lemma 3.12;
- the kernel bounds (3.2), (3.3), (3.4) of Lemma 3.7: for $0 \le \epsilon \le 1$, $R > 0$, $j \ge 0$ there is $C = C(j,R,p)$ such that for $|u|, |v| \le R$

  $$|\nabla\Gamma_\epsilon| \le C\epsilon|u-v|(|u|+|v|), \quad |\nabla^2\Gamma_\epsilon| \le C\epsilon(|u|+|v|), \quad |\nabla^j\Gamma_\epsilon| \le C\epsilon,$$

  $$|(\partial_1+\partial_2)\Gamma_\epsilon(u,v)| \le C\epsilon|u-v|^2, \qquad |(\partial_1+\partial_2)\nabla\Gamma_\epsilon(u,v)| \le C\epsilon|u-v|, \qquad |\Gamma_\epsilon(u,v)| \le C\epsilon|u-v|^2(|u|+|v|).$$

  In particular $\nabla\Gamma_\epsilon(u,u) = 0$ and $\Gamma_\epsilon(u,u) = 0$.

The norms $\|\cdot\|_{X_\gamma}$, $\|\cdot\|_{Y_\gamma}$ are those of Definitions 2.3 and 2.4. Elements of X_γ are in $W^{2,\infty}_{\mathrm{loc}}$, hence C^1; all statements involving second derivatives hold for almost every y.

Definition 10.1 (The three nonlinear pieces)**.** Let $0 \le \epsilon \le 1$, let $\overline{f} \in C^2(\mathbb{R})$ with $\overline{f}' \in L^\infty(\mathbb{R})$, and let $r \in X_\gamma$ (for some $\gamma > 1/p$, so that $r' \in L^\infty$). Define the functions $\mathcal{N}_1(r) = \mathcal{N}_1^{\epsilon,\overline{f}}(r), \mathcal{N}_2(r), \mathcal{N}_3(r)$ on $\mathbb{R}$ by (paper `eq:nonlinear_decomposition`)

$$\begin{aligned}
\mathcal{N}_1(r) &= \big(B_\epsilon(\overline{f}' + r') - B_\epsilon(\overline{f}')\big)\big(\overline{f} - y\overline{f}'\big),\\
\mathcal{N}_2(r) &= \big(B_\epsilon(\overline{f}' + r') - 1\big)\big(r - yr'\big),\\
\mathcal{N}_3(r) &= \mathcal{I}_\epsilon(\overline{f} + r) - \mathcal{I}_\epsilon(\overline{f}) - b_\epsilon r'.
\end{aligned}$$

Note that $\mathcal{N}_j(0) = 0$ for $j = 1, 2, 3$.

Lemma 10.2 (Decomposition of $\mathcal{N}_{\epsilon,\overline{f}}$)**.** *Let $\beta + \frac{1}{p} < \gamma < 2p + 1$, let $0 \le \epsilon \le \min\{1, \epsilon_0\}$ where ϵ_0 is the constant of Lemma 8.25 (so that $\mathcal{R}_\epsilon$ of Definition 8.9 is defined), let $\overline{f} \in C^2(\mathbb{R})$ with $\overline{f}' \in L^\infty(\mathbb{R})$ and $\overline{f}'' \in L^\infty_{\mathrm{loc}}$, and such that $|D|^p\overline{f}'$ is defined in the sense of Definition 1.5; let $r \in X_\gamma$. Then $\mathcal{E}_\epsilon(\overline{f} + r)$ and $\mathcal{E}_\epsilon(\overline{f})$ are defined, and for every $y \in \mathbb{R}$*

$$\mathcal{N}_{\epsilon,\overline{f}}(r) = \mathcal{E}_\epsilon(\overline{f} + r) - \mathcal{E}_\epsilon(\overline{f}) - \mathcal{L}_\epsilon r = \mathcal{N}_1(r) + \mathcal{N}_2(r) + \mathcal{N}_3(r) - \mathcal{R}_\epsilon r.$$

In particular $\mathcal{N}_{\epsilon,\overline{f}}(0) = 0$.

Proof. Put $f = \overline{f} + r$; then $f \in W^{2,\infty}_{\mathrm{loc}}$ and $f' \in L^\infty$, so Proposition 3.10 applies to both f and $\overline{f}$:

$$\mathcal{E}_\epsilon(g) = |D|^p g' + B_\epsilon(g')(g - yg') + \mathcal{I}_\epsilon(g), \qquad g \in \{\overline{f}, \overline{f} + r\}.$$

Step 1 (fractional term). By linearity of $|D|^p$ on the class of Definition 1.5 (which contains $\overline{f}'$ and r'; note $r' \in L^1 \cap L^\infty$ since admissibility gives $\gamma > \beta + 1/p > 1 + 1/p$), $|D|^p(\overline{f}' + r') - |D|^p\overline{f}' = |D|^p r'$.

Step 2 (local term). Adding and subtracting $B_\epsilon(\overline{f}' + r')(\overline{f} - y\overline{f}')$,

$$B_\epsilon(\overline{f}'+r')(\overline{f}+r-y\overline{f}'-yr')-B_\epsilon(\overline{f}')(\overline{f}-y\overline{f}') = \mathcal{N}_1(r)+B_\epsilon(\overline{f}'+r')(r-yr') = \mathcal{N}_1(r)+\mathcal{N}_2(r)+(r-yr').$$

Step 3 (nonlocal term). By definition, $\mathcal{I}_\epsilon(\overline{f} + r) - \mathcal{I}_\epsilon(\overline{f}) = \mathcal{N}_3(r) + b_\epsilon r'$.

Step 4 (subtracting $\mathcal{L}_\epsilon$). By Definition 2.17, $\mathcal{L}_\epsilon r = |D|^p r' + r - yr' + c_\epsilon\chi_+ y^{-p} r' + \mathcal{R}_\epsilon r$, and by Lemma 3.12 $c_\epsilon\chi_+ y^{-p} = b_\epsilon$. Summing Steps 1–3 and subtracting: the terms $|D|^p r'$, $r - yr'$ and $b_\epsilon r'$ cancel exactly, leaving $\mathcal{N}_1(r) + \mathcal{N}_2(r) + \mathcal{N}_3(r) - \mathcal{R}_\epsilon r$.

Since $\mathcal{N}_j(0) = 0$ and $\mathcal{R}_\epsilon 0 = 0$, $\mathcal{N}_{\epsilon,\overline{f}}(0) = 0$. □

Remark 10.3. The free part $|D|^p r' + r - yr'$ of $\mathcal{L}_\epsilon$ is the linear part of the fractional and transport terms at $B_\epsilon \equiv 1$; the nonperturbative coefficient term $b_\epsilon r'$ is the "far-field model" part of $D\mathcal{I}_\epsilon(\overline{f})[r]$ (frozen at the tail W, see Lemma 10.30) and is subtracted inside $\mathcal{N}_3$ precisely because it was moved into $\mathcal{L}_\epsilon$; the conjugation remainder $\mathcal{R}_\epsilon$, which is *linear*, profile-independent, and $O(\epsilon)$ from X_γ to Y_γ by Lemma 8.25, appears with a minus sign. This explains the choice of the decomposition; the later far-field estimates are not used in the algebraic identity above.

10.2 Admissible profiles and elementary bounds

Definition 10.4 (Admissible profile)**.** Let $M > 0$. A function $\overline{f} \in C^2(\mathbb{R};\mathbb{R})$ is an *admissible profile with constant* M if for all $y \geq 0$

$$|\overline{f}(y)| \leq M\langle y\rangle^{-p}, \qquad |\overline{f}'(y)| \leq M\langle y\rangle^{-\beta}, \qquad |\overline{f}''(y)| \leq M\langle y\rangle^{-\beta+1/p},$$

and for all $y \leq 0$

$$|\overline{f}(y) + y| \leq M\langle y\rangle^{-p}, \qquad |\overline{f}'(y) + 1| \leq M\langle y\rangle^{-p-1}, \qquad |\overline{f}''(y)| \leq M\langle y\rangle^{-p-2}.$$

Lemma 10.5 (Admissibility is closed under C^2_{loc} limits)**.** *If $\overline{f}_k$ are admissible with constant M and $\overline{f}_k \to \overline{f}$ in $C^2_{\mathrm{loc}}(\mathbb{R})$, then $\overline{f}$ is admissible with constant M.*

Proof. Each of the six defining inequalities is a pointwise inequality between $\overline{f}_k^{(j)}(y)$ (or $\overline{f}_k(y)+y$, $\overline{f}_k'(y)+1$) and a fixed function of y; pass to the pointwise limit $k \to \infty$. □

Lemma 10.6 (Approximate profiles are admissible)**.** *There is $M_0 > 0$ such that for all $0 \leq \epsilon \leq \epsilon_0$ and $\mu \in \overline{B}(\mu_*, \delta)$ (the parameters of Theorem 2.20), $\overline{f}_{\epsilon,\mu}$ is an admissible profile with constant M_0.*

Proof. $\overline{f}_{\epsilon,\mu}$ is C^∞. Proposition 4.49 gives, uniformly in ϵ, μ, $|\overline{f}_{\epsilon,\mu}| \leq C\langle y\rangle^{-p}$ and $|\partial^j \overline{f}'_{\epsilon,\mu}| \leq C_j\langle y\rangle^{-\beta+j/p}$ $(j = 0, 1)$ for $y \geq 0$, and $|\partial^j(\overline{f}_{\epsilon,\mu} + y)| \leq C_j\langle y\rangle^{-p-j}$ $(j = 0, 1, 2)$ for $y \leq 0$. Take $M_0 = \max\{C, C_0, C_1, C_2\}$. □

Lemma 10.7 (Consequences of admissibility and of X_γ membership)**.** *Let $\beta + \frac{1}{p} < \gamma < 2p + 1$, $M, N > 0$, $\overline{f}$ admissible with constant M, $r \in X_\gamma$ with $\|r\|_{X_\gamma} \leq N$, and $f = \overline{f} + r$. There is $C = C(p, \gamma, M, N)$ such that:*

1. *$\|f'\|_{L^\infty(\mathbb{R})} \leq C$, $\|f\|_{L^\infty([-3,3])} + \|f''\|_{L^\infty([-3,3])} \leq C$, and $|f(y) - W(y)| \leq C\langle y\rangle^{-p}$ for all y; the same holds with f replaced by $\overline{f}$.*
2. *For $y \geq 0$: $|f'(y)| \leq C\langle y\rangle^{-\beta}$ and $|f''(y)| \leq C\langle y\rangle^{-\beta+1/p}$.*
3. *For $y \leq 0$: $|f'(y) + 1| \leq C\langle y\rangle^{-p-1}$ and $|f''(y)| \leq C\langle y\rangle^{-p-2}$.*
4. *For every $h \in X_\gamma$: $\|h\|_{L^\infty} \leq \|h\|_{X_\gamma}$, $\|h\|_{L^1} \leq C_\gamma\|h\|_{X_\gamma}$ with $C_\gamma = \int_{\mathbb{R}}\langle z\rangle^{-\gamma}dz < \infty$, and*

$$|h'(y)| \leq \|h\|_{X_\gamma}\begin{cases}\langle y\rangle^{-\gamma+1/p}, & y \geq 0\\ \langle y\rangle^{-\gamma-1}, & y \leq 0\end{cases}, \qquad |h''(y)| \leq \|h\|_{X_\gamma}\begin{cases}\langle y\rangle^{-\gamma+2/p}, & y \geq 0\\ \langle y\rangle^{-\gamma-2}, & y \leq 0.\end{cases}$$

In particular $\|h'\|_{L^\infty} \leq \|h\|_{X_\gamma}$ and $\|h''\|_{L^\infty([-3,3])} \leq 10^{2/p}\|h\|_{X_\gamma}$.

Proof. Item 4 is read off from the definition of $\|\cdot\|_{X_\gamma}$; $C_\gamma < \infty$ as $\gamma > 1$; and $\langle y\rangle^{-\gamma+1/p} \leq 1$ as $\gamma > 1/p$. For items 1–3 combine the admissible bounds with item 4 for $h = r$, using: for $y \geq 0$, $\langle y\rangle^{-\gamma} \leq \langle y\rangle^{-p}$ $(\gamma > p)$, $\langle y\rangle^{-\gamma+1/p} \leq \langle y\rangle^{-\beta}$ $(\gamma - 1/p > \beta)$, $\langle y\rangle^{-\gamma+2/p} \leq \langle y\rangle^{-\beta+1/p}$; for $y \leq 0$, $\langle y\rangle^{-\gamma-j} \leq \langle y\rangle^{-p-j}$. For $y \geq 0$, $f - W = f$; for $y \leq 0$, $f - W = f + y$ and $|f'| \leq 1 + |f' + 1|$. On $[-3, 3]$ all weights are bounded by $10^{2/p}$. □

Lemma 10.8 (Transport-factor bounds)**.** *Let $\beta+\frac{1}{p}<\gamma<2p+1$ and $\overline{f}$ admissible with constant M. There is $C=C(p,M)$ such that (paper* `eq:N1_profile_factor`*,* `eq:N1_profile_factor_derivative`*)*

$$|\overline{f}-y\overline{f}'|\le C\begin{cases}\langle y\rangle^{1-\beta}, & y\ge 0\\ \langle y\rangle^{-p}, & y\le 0\end{cases},\qquad |\partial_y(\overline{f}-y\overline{f}')|=|y\overline{f}''|\le C\begin{cases}\langle y\rangle^{1-\beta+1/p}, & y\ge 0\\ \langle y\rangle^{-p-1}, & y\le 0,\end{cases}$$

and for every $r\in X_\gamma$ and a.e. y (paper `eq:N2_transport_bound`*,* `eq:N2_transport_derivative_bound`*)*

$$|r-yr'|\le 2\|r\|_{X_\gamma}\begin{cases}\langle y\rangle^{1-\gamma+1/p}, & y\ge 0\\ \langle y\rangle^{-\gamma}, & y\le 0\end{cases},\qquad |\partial_y(r-yr')|=|yr''|\le \|r\|_{X_\gamma}\begin{cases}\langle y\rangle^{1-\gamma+2/p}, & y\ge 0\\ \langle y\rangle^{-\gamma-1}, & y\le 0.\end{cases}$$

Proof. Use $|y|\le\langle y\rangle$. For $y\ge 0$: $|\overline{f}|\le M\langle y\rangle^{-p}\le M\langle y\rangle^{1-\beta}$ because $1-\beta=\frac{p-1}{2p}>-p$; $|y\overline{f}'|\le M\langle y\rangle^{1-\beta}$; $|y\overline{f}''|\le M\langle y\rangle^{1-\beta+1/p}$. For $y\le 0$ write $\overline{f}-y\overline{f}'=(\overline{f}+y)-y(\overline{f}'+1)$, so $|\overline{f}-y\overline{f}'|\le 2M\langle y\rangle^{-p}$, and $|y\overline{f}''|\le M\langle y\rangle^{-p-1}$. For r: $|r|\le\|r\|\langle y\rangle^{-\gamma}\le\|r\|\langle y\rangle^{1-\gamma+1/p}$ and $|yr'|\le\|r\|\langle y\rangle^{1-\gamma+1/p}$ for $y\ge 0$; $|r|+|yr'|\le 2\|r\|\langle y\rangle^{-\gamma}$ for $y\le 0$; similarly for yr''. The identity $\partial_y(g-yg')=-yg''$ holds a.e. for $g\in W^{2,\infty}_{\mathrm{loc}}$. □

Lemma 10.9 (Elementary bounds for B_ϵ)**.** *Let $0\le\epsilon\le 1$, $R>0$, and $B_\epsilon(u)=(1+\epsilon u^2)^{\alpha/2}$ with $\alpha=p+1\in(1,2)$. Then for all $|u|\le R$:*

$$|B_\epsilon(u)-1|\le\tfrac{\alpha}{2}\epsilon u^2,\qquad |B_\epsilon'(u)|\le\alpha\epsilon|u|,\qquad |B_\epsilon''(u)|\le\alpha(1+R^2)\epsilon.$$

Proof. $B_\epsilon'(u)=\alpha\epsilon u(1+\epsilon u^2)^{\alpha/2-1}$ and, since $\alpha/2-1<0$, $(1+\epsilon u^2)^{\alpha/2-1}\le 1$, giving the second bound. Then $B_\epsilon(u)-1=\int_0^1 B_\epsilon'(tu)u\,dt$ gives the first. Finally $B_\epsilon''(u)=\alpha\epsilon(1+\epsilon u^2)^{\alpha/2-1}+\alpha(\alpha-2)\epsilon^2u^2(1+\epsilon u^2)^{\alpha/2-2}$, so $|B_\epsilon''|\le\alpha\epsilon+\alpha(2-\alpha)\epsilon^2R^2\le\alpha(1+R^2)\epsilon$. □

10.3 The local pieces $\mathcal{N}_1$ and $\mathcal{N}_2$

Lemma 10.10 (Difference of the B_ϵ coefficients)**.** *Let $0\le\epsilon\le 1$, $\beta+\frac{1}{p}<\gamma<2p+1$, $\overline{f}$ admissible with constant M, and $r_1,r_2\in X_\gamma$ with $\|r_j\|_{X_\gamma}\le N$. Let $\Delta B_\epsilon=B_\epsilon(\overline{f}'+r_1')-B_\epsilon(\overline{f}'+r_2')$ (paper* `eq:N1_delta_B`*). Then $\Delta B_\epsilon\in W^{1,\infty}_{\mathrm{loc}}$ and there is $C=C(p,\gamma,M,N)$ such that for a.e. y (paper* `eq:N1_delta_B_bound`*,* `eq:N1_delta_B_derivative_bound`*)*

$$|\Delta B_\epsilon(y)|\le C\epsilon\|r_1-r_2\|_{X_\gamma}\begin{cases}\langle y\rangle^{-\beta-\gamma+1/p}, & y\ge 0\\ \langle y\rangle^{-\gamma-1}, & y\le 0\end{cases},\quad |\partial_y\Delta B_\epsilon(y)|\le C\epsilon\|r_1-r_2\|_{X_\gamma}\begin{cases}\langle y\rangle^{-\beta-\gamma+2/p}, & y\ge 0\\ \langle y\rangle^{-\gamma-2}, & y\le 0.\end{cases}$$

Proof. Let $R=M+1+N$; by Lemma 10.7 $|\overline{f}'+r_j'|\le R$ and $|\overline{f}''+r_j''|$ obeys the bounds of items 2–3 there. Set $u_j=\overline{f}'+r_j'$ and $\delta=r_1-r_2$.

Value: by the mean value theorem, $|\Delta B_\epsilon|\le\sup_{t\in[0,1]}|B_\epsilon'(u_2+t(u_1-u_2))|\,|\delta'|\le\alpha\epsilon(|\overline{f}'|+|r_1'|+|r_2'|)|\delta'|$ by Lemma 10.9. For $y\ge 0$, $|\overline{f}'|+|r_1'|+|r_2'|\le C\langle y\rangle^{-\beta}$ and $|\delta'|\le\|\delta\|\langle y\rangle^{-\gamma+1/p}$; for $y\le 0$, $|\overline{f}'|+|r_j'|\le C$ and $|\delta'|\le\|\delta\|\langle y\rangle^{-\gamma-1}$.

Derivative: $B_\epsilon\circ u_j$ is the composition of a smooth function with a $W^{1,\infty}_{\mathrm{loc}}$ function, so it is in $W^{1,\infty}_{\mathrm{loc}}$ with derivative $B_\epsilon'(u_j)u_j'$ a.e.; hence

$$\partial_y\Delta B_\epsilon=B_\epsilon'(u_1)\delta''+\big[B_\epsilon'(u_1)-B_\epsilon'(u_2)\big](\overline{f}''+r_2'').$$

The first term is bounded by $\alpha\epsilon(|\overline{f}'|+|r_1'|)|\delta''|$; the second by $\alpha(1+R^2)\epsilon|\delta'|(|\overline{f}''|+|r_2''|)$. For $y \geq 0$: $\langle y\rangle^{-\beta}\langle y\rangle^{-\gamma+2/p}$ and $\langle y\rangle^{-\gamma+1/p}\langle y\rangle^{-\beta+1/p}$, both $= \langle y\rangle^{-\beta-\gamma+2/p}$. For $y \leq 0$: $\langle y\rangle^{-\gamma-2}$ and $\langle y\rangle^{-\gamma-1}\langle y\rangle^{-p-2} \leq \langle y\rangle^{-\gamma-2}$.

□

Lemma 10.11 (Lipschitz bound for $\mathcal{N}_1$)**.** *Let $0 \leq \epsilon \leq 1$, $\beta + \frac{1}{p} < \gamma < 2p+1$, $M, N > 0$, $\overline{f}$ admissible with constant M, and $r_1, r_2 \in X_\gamma$ with $\|r_1\|_{X_\gamma}, \|r_2\|_{X_\gamma} \leq N$. Then $\mathcal{N}_1(r_1) - \mathcal{N}_1(r_2) \in Y_\gamma$ and there is $C = C(p, \gamma, M, N)$ (independent of ϵ and $\overline{f}$) with*

$$\|\mathcal{N}_1(r_1) - \mathcal{N}_1(r_2)\|_{Y_\gamma} \leq C\epsilon\|r_1 - r_2\|_{X_\gamma}.$$

Proof. $\mathcal{N}_1(r_1) - \mathcal{N}_1(r_2) = \Delta B_\epsilon\,(\overline{f} - y\overline{f}')$, a product of $W^{1,\infty}_{\mathrm{loc}}$ functions. By the product rule and Lemmas 10.8 and 10.10, using $2\beta = 1 + 1/p$: for $y \geq 0$, $|\cdot| \leq C\epsilon\|\delta\|\langle y\rangle^{-\beta-\gamma+1/p+1-\beta} = C\epsilon\|\delta\|\langle y\rangle^{-\gamma}$ and $|\partial_y \cdot| \leq C\epsilon\|\delta\|(\langle y\rangle^{-\beta-\gamma+2/p+1-\beta} + \langle y\rangle^{-\beta-\gamma+1/p+1-\beta+1/p}) = 2C\epsilon\|\delta\|\langle y\rangle^{-\gamma+1/p}$; for $y \leq 0$, $|\cdot| \leq C\epsilon\|\delta\|\langle y\rangle^{-\gamma-1-p}$ and $|\partial_y \cdot| \leq 2C\epsilon\|\delta\|\langle y\rangle^{-\gamma-p-2}$. Multiplying by the weights of Definition 2.4 ($\langle y\rangle^{\gamma}$, resp. $\langle y\rangle^{\gamma-1/p}$ for the derivative on $y \geq 0$ and $\langle y\rangle^{\gamma+1}$ on $y \leq 0$) and taking suprema gives the claim. □

Lemma 10.12 (Coefficient bounds for $\mathcal{N}_2$)**.** *Under the hypotheses of Lemma 10.10, for $j = 1, 2$ and a.e. y (paper `eq:N2_coefficient_bound`, `eq:N2_coefficient_derivative_bound`)*

$$|B_\epsilon(\overline{f}'+r_j')-1| \leq C\epsilon \begin{cases} \langle y\rangle^{-2\beta}, & y \geq 0 \\ 1, & y \leq 0 \end{cases}, \qquad |\partial_y(B_\epsilon(\overline{f}'+r_j')-1)| \leq C\epsilon \begin{cases} \langle y\rangle^{-2\beta+1/p}, & y \geq 0 \\ \langle y\rangle^{-p-2}, & y \leq 0, \end{cases}$$

with $C = C(p, \gamma, M, N)$.

Proof. By Lemma 10.9, $|B_\epsilon(u_j) - 1| \leq \frac{\alpha}{2}\epsilon u_j^2$ and $|\partial_y B_\epsilon(u_j)| = |B_\epsilon'(u_j)u_j'| \leq \alpha\epsilon|u_j||\overline{f}'' + r_j''|$. Insert the bounds of Lemma 10.7: for $y \geq 0$, $|u_j| \leq C\langle y\rangle^{-\beta}$, $|u_j'| \leq C\langle y\rangle^{-\beta+1/p}$; for $y \leq 0$, $|u_j| \leq C$, $|u_j'| \leq C\langle y\rangle^{-p-2}$. □

Lemma 10.13 (Lipschitz bound for $\mathcal{N}_2$)**.** *Under the hypotheses of Lemma 10.11, $\mathcal{N}_2(r_1) - \mathcal{N}_2(r_2) \in Y_\gamma$ and there is $C = C(p, \gamma, M, N)$ with*

$$\|\mathcal{N}_2(r_1) - \mathcal{N}_2(r_2)\|_{Y_\gamma} \leq C\epsilon\|r_1 - r_2\|_{X_\gamma}.$$

Proof. With $\delta = r_1 - r_2$,

$$\mathcal{N}_2(r_1) - \mathcal{N}_2(r_2) = \big(B_\epsilon(\overline{f}' + r_1') - 1\big)(\delta - y\delta') + \Delta B_\epsilon\,(r_2 - yr_2').$$

Apply the product rule and Lemmas 10.8, 10.10 and 10.12 (the latter with $r = \delta$ and $r = r_2$, $\|r_2\| \leq N$).

$y \geq 0$, value: $\langle y\rangle^{-2\beta}\langle y\rangle^{1-\gamma+1/p} = \langle y\rangle^{-\gamma}$ (as $2\beta = 1+1/p$), and $\langle y\rangle^{-\beta-\gamma+1/p}\langle y\rangle^{1-\gamma+1/p} = \langle y\rangle^{-\gamma}\langle y\rangle^{1-\beta-\gamma+2/p} \leq \langle y\rangle^{-\gamma}$, since $1 - \beta + 2/p = \beta + 1/p < \gamma$ (again by $2\beta = 1 + 1/p$). $y \geq 0$, derivative: the four terms are $\langle y\rangle^{-2\beta+1/p}\langle y\rangle^{1-\gamma+1/p}$, $\langle y\rangle^{-2\beta}\langle y\rangle^{1-\gamma+2/p}$ (both $= \langle y\rangle^{-\gamma+1/p}$), and $\langle y\rangle^{-\beta-\gamma+2/p}\langle y\rangle^{1-\gamma+1/p}$, $\langle y\rangle^{-\beta-\gamma+1/p}\langle y\rangle^{1-\gamma+2/p}$ (both $= \langle y\rangle^{-\gamma+1/p}\langle y\rangle^{1-\beta-\gamma+2/p} \leq \langle y\rangle^{-\gamma+1/p}$).

$y \leq 0$, value: $1 \cdot \langle y\rangle^{-\gamma}$ and $\langle y\rangle^{-\gamma-1}\langle y\rangle^{-\gamma}$, both $\leq \langle y\rangle^{-\gamma}$. $y \leq 0$, derivative: $\langle y\rangle^{-p-2}\langle y\rangle^{-\gamma}$, $1 \cdot \langle y\rangle^{-\gamma-1}$, $\langle y\rangle^{-\gamma-2}\langle y\rangle^{-\gamma}$, $\langle y\rangle^{-\gamma-1}\langle y\rangle^{-\gamma-1}$, all $\leq \langle y\rangle^{-\gamma-1}$.

Each bound carries the factor $C\epsilon\|\delta\|_{X_\gamma}$. Multiplying by the Y_γ weights gives the claim. □

10.4 The nonlocal piece $\mathcal{N}_3$

10.4.1 Linearization and reduction to a derivative bound

Definition 10.14 (Linearization of $\mathcal{N}_3$)**.** Let $0 \le \epsilon \le 1$, $\overline{f}$ as in Definition 10.1, $r, h \in X_\gamma$ ($\gamma > 1/p$), and $f = \overline{f} + r$. For $y \neq z$ set (paper `eq:N3_integrand`)

$$K_{f,h}(y,z) = \partial_1\Gamma_\epsilon(f'(y), S_f(y,z))\, h'(y) + \partial_2\Gamma_\epsilon(f'(y), S_f(y,z))\, S_h(y,z),$$

and define (paper `eq:N3_variation`)

$$D\mathcal{N}_3(r)[h](y) = \int_{\mathbb{R}} \frac{K_{f,h}(y,z)}{|y-z|^\alpha}\, dz - b_\epsilon(y)h'(y).$$

The displayed formula for $K_{f,h}$ also applies to arbitrary $f, h \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ with bounded first derivatives; the weighted-space assumptions are only needed when estimating the resulting operator in Y_γ.

Lemma 10.15 (Secant identities)**.** *Let $g \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ and $y \neq z$. Then $S_g(y,z) = \int_0^1 g'(z+t(y-z))\,dt$; $y \mapsto S_g(y,z)$ is C^1 on $\mathbb{R}\setminus\{z\}$ with (paper `eq:N3_secant_derivatives`)*

$$\partial_y S_g(y,z) = \frac{g'(y) - S_g(y,z)}{y-z};$$

and, writing $I_{y,z}$ for the closed segment between y and z,

$$|g'(y) - S_g(y,z)| \le |y-z|\,\|g''\|_{L^\infty(I_{y,z})}, \quad |\partial_y S_g(y,z)| \le \|g''\|_{L^\infty(I_{y,z})}, \quad |\partial_y S_g(y,z)| \le \frac{2\|g'\|_{L^\infty}}{|y-z|},$$

$$|S_g(y,z)| \le \min\Big\{\|g'\|_{L^\infty},\ \frac{|g(y)|+|g(z)|}{|y-z|}\Big\}.$$

Proof. Fundamental theorem of calculus for the absolutely continuous g and g'. For the second bound: $g'(y) - S_g(y,z) = \int_0^1 (g'(y) - g'(z+t(y-z)))dt$ and $|g'(y) - g'(w)| \le |y-w|\|g''\|_{L^\infty(I_{y,z})}$, giving in fact $\frac{1}{2}|y-z|\|g''\|$. The quotient rule gives $\partial_y S_g$. □

10.4.2 Local singular-integral interfaces

Definition 10.16 (A dominated pair of value and derivative kernels)**.** Let $1 < \alpha < 2$, let $U \subset \mathbb{R}$ be a nonempty open interval, and let $A \ge 0$. For $s \neq 0$ put

$$\omega_0(s) = \frac{\min\{|s|^2, 1\}}{|s|^\alpha}, \qquad \omega_1(s) = \frac{\min\{|s|, 1\}}{|s|^\alpha},$$

and put $\omega_0(0) = \omega_1(0) = 0$. A pair of Borel measurable functions $H, J : \mathbb{R}^2 \to \mathbb{R}$ is a *dominated kernel pair on U with constant A* if:

1. $H(y,y) = J(y,y) = 0$ for every $y \in U$;
2. for each fixed z, $H(\cdot, z)$ is locally Lipschitz on $U \setminus \{z\}$ and $\partial_y H(y,z) = J(y,z)$ for a.e. $y \in U \setminus \{z\}$;
3. $|H(y,z)| \le A\omega_0(y-z)$ for all $y \in U$ and $z \neq y$;

4. there is a null set $E \subset U$ such that $|J(y,z)| \le A\omega_1(y-z)$ for every $y \in U \setminus E$ and every $z \neq y$.

Here local Lipschitz regularity is for the actual pointwise representative, and the null set in (4) is independent of z.

Lemma 10.17 (Integrability and exact integrals of the kernel envelopes)**.** *For every $1 < \alpha < 2$, the functions ω_0, ω_1 of Definition 10.16 are nonnegative, Borel measurable, and integrable on $\mathbb{R}$, with*

$$\int_{\mathbb{R}} \omega_0(s)ds = \frac{2}{3-\alpha} + \frac{2}{\alpha-1}, \qquad \int_{\mathbb{R}} \omega_1(s)ds = \frac{2}{2-\alpha} + \frac{2}{\alpha-1}.$$

For every $0 < r \le 1$ one also has $\int_{|s|<r} \omega_0(s)ds = 2r^{3-\alpha}/(3-\alpha)$ and $\int_{|s|<r} \omega_1(s)ds = 2r^{2-\alpha}/(2-\alpha)$.

Proof. Both functions are even. On $(0,1)$ they equal respectively $s^{2-\alpha}$ and $s^{1-\alpha}$, whose exponents exceed -1; on $(1,\infty)$ both equal $s^{-\alpha}$, whose exponent is less than -1. Split at 1, integrate the powers, and use symmetry. The value at 0 does not affect the integrals. The same computation on $(0,r)$ gives the truncated formulas; nonnegativity and Borel measurability follow directly from the definitions. □

Lemma 10.18 (Weak derivative of a fixed-z truncation)**.** *Let (H,J) be a dominated kernel pair on U with constant A and exponent $1 < \alpha < 2$. Fix $\vartheta \in C^\infty(\mathbb{R};[0,1])$ with $\vartheta = 0$ on $[-1,1]$ and $\vartheta = 1$ on $\mathbb{R} \setminus (-2,2)$, and put $M_\vartheta = \|\vartheta'\|_\infty$. For $0 < \eta < 1/2$ define*

$$F_\eta(y) = \int_{\mathbb{R}} \vartheta((y-z)/\eta)H(y,z)\,dz, \quad G_\eta(y) = \int_{\mathbb{R}} \left[\vartheta((y-z)/\eta)J(y,z) + \eta^{-1}\vartheta'((y-z)/\eta)H(y,z)\right] dz.$$

The value integral is absolutely convergent for every $y \in U$, and both summands defining G_η are absolutely integrable for a.e. $y \in U$. The actual function F_η is locally Lipschitz on U, $F_\eta' = G_\eta$ a.e. there, and

$$\|G_\eta\|_{L^\infty(U)} \le A\left(\frac{2}{2-\alpha} + \frac{2}{\alpha-1} + \frac{2^{4-\alpha}M_\vartheta}{3-\alpha}\eta^{2-\alpha}\right).$$

In particular, for every $a < b$ with $[a,b] \subset U$, $F_\eta(b) - F_\eta(a) = \int_a^b G_\eta(y)\,dy$.

Proof. For each z, multiply $H(\cdot,z)$ by the cutoff. This product vanishes near $y = z$, so it is locally Lipschitz on all of U. Its derivative is the displayed sum, with z held fixed. Apply the one-dimensional fundamental theorem on $[a,b]$ and then Fubini. Its hypotheses follow from Tonelli applied to the translated L^1 envelopes and the finite length of $[a,b]$. The cutoff derivative is supported where $\eta \le |y-z| \le 2\eta$ and has integral at most $2AM_\vartheta\eta^{-1}\int_0^{2\eta} s^{2-\alpha}ds$. The resulting increment identity and the displayed L^∞ bound prove local Lipschitz regularity and identify the a.e. derivative. □

Lemma 10.19 (Quantitative removal of the diagonal cutoff)**.** *With exactly the kernel and cutoff hypotheses of Lemma 10.18, let $F(y) = \int H(y,z)dz$ and*

$G(y) = \int J(y,z)dz$. *The first integral converges absolutely for every* $y \in U$ *and the second for a.e.* $y \in U$. *For every* $0 < \eta < 1/2$,

$$\sup_{y\in U} |F_\eta(y) - F(y)| \le \frac{2^{4-\alpha}}{3-\alpha} A\eta^{3-\alpha},$$

$$\|G_\eta - G\|_{L^\infty(U)} \le A\left(\frac{2^{3-\alpha}}{2-\alpha} + \frac{2^{4-\alpha}M_\vartheta}{3-\alpha}\right)\eta^{2-\alpha}.$$

The constants are independent of U, A, H, J, η *apart from the displayed factors.*

Proof. The omitted value and derivative kernels are supported on $|y-z| < 2\eta < 1$. Integrate respectively $A|y-z|^{2-\alpha}$ and $A|y-z|^{1-\alpha}$ there. Add the cutoff-derivative estimate from Lemma 10.18 for $G_\eta - G$. All estimates for J are taken outside its fixed exceptional null set. □

Lemma 10.20 (Local Lipschitz limit from convergence of derivatives)**.** *Let* $U \subset \mathbb{R}$ *be an open interval. Let* $F_n : U \to \mathbb{R}$ *be locally Lipschitz, let* $G_n, G \in L^\infty_{\mathrm{loc}}(U)$ *be measurable representatives, and suppose* $F_n' = G_n$ *a.e. on* U. *If* $F_n \to F$ *locally uniformly on* U *and* $\|G_n - G\|_{L^\infty(I)} \to 0$ *for every compact interval* $I \subset U$, *then the actual function* F *is locally Lipschitz,* $F' = G$ *a.e. on* U, *and for every* $a < b$ *with* $[a,b] \subset U$,

$$F(b) - F(a) = \int_a^b G(y)dy, \qquad |F(b) - F(a)| \le \|G\|_{L^\infty([a,b])}|b-a|.$$

Proof. Pass to the limit in $F_n(b) - F_n(a) = \int_a^b G_n$; the integral error is bounded by $|b-a|\|G_n - G\|_{L^\infty([a,b])}$. The increment estimate proves local Lipschitz regularity. On each compact subinterval, F differs by a constant from a primitive of G; the fundamental theorem for integrable functions gives the a.e. derivative. Exhaust U by countably many compact subintervals to obtain a single null set. □

Lemma 10.21 (Local domination for the linearized kernel)**.** *Let* $0 < p < 1$, $\alpha = 1 + p$, *and* $B, L \ge 1$. *There exists* $C = C(p, B, L) > 0$ *such that for every* $R \ge 1$, $0 \le \epsilon \le 1$, $H_0 \ge 0$, *and* $f, h \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ *satisfying*

$$\|f'\|_{L^\infty(\mathbb{R})} \le B, \quad \|h'\|_{L^\infty(\mathbb{R})} \le H_0, \quad \|f''\|_{L^\infty([-R-2,R+2])} \le L, \quad \|h''\|_{L^\infty([-R-2,R+2])} \le H_0,$$

the functions, set to zero on $y = z$,

$$H(y,z) = \frac{K_{f,h}(y,z)}{|y-z|^\alpha}, \qquad J(y,z) = \frac{\partial_y K_{f,h}(y,z) - \alpha K_{f,h}(y,z)/(y-z)}{|y-z|^\alpha},$$

form a dominated kernel pair on $U = (-R,R)$ *with constant* $C\epsilon H_0$. *Here* J *is defined by the Borel chain-rule expression using the classical* f'' *and* h'' *where they exist and zero otherwise; its values on the common exceptional null set of* y *may be set to zero for every* z. *More explicitly, for every* $y \in U, z \ne y$, $|K_{f,h}(y,z)| \le C\epsilon H_0 \min\{|y-z|^2, 1\}$, *and for a.e.* $y \in U$, *simultaneously for all* $z \ne y$,

$$|\partial_y K_{f,h}(y,z)| + \alpha|K_{f,h}(y,z)|/|y-z| \le C\epsilon H_0 \min\{|y-z|, 1\}.$$

Proof. Write $K = (\partial_1+\partial_2)\Gamma_\epsilon h' + \partial_2\Gamma_\epsilon(S_h - h')$. For $|y-z| \le 1$, use $|f'(y) - S_f| \le L|y-z|$ and $|h'(y) - S_h| \le H_0|y-z|$; the diagonal bounds give the quadratic vanishing of K. Differentiate this decomposition with z fixed. The diagonal bounds on $(\partial_1+\partial_2)\nabla\Gamma_\epsilon$ give one factor $|y-z|$ in each derivative term. For $|y-z| \ge 1$, use the global first-derivative bounds and the secant quotient rule. The common null set consists of the exceptional sets for f'' and h'' and their local bounds. Off $y = z$ the kernel is a composition of locally Lipschitz functions; its a.e. derivative has the stated Borel chain-rule expression. Increase C once to dominate both envelopes. □

Lemma 10.22 (Local domination for the profile kernel)**.** *Let $0 < p < 1$, $\alpha = 1+p$, and $B, L \ge 1$. There exists $C = C(p, B, L) > 0$ such that for every $R \ge 1$, $0 \le \epsilon \le 1$, and $g \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ with $\|g'\|_{L^\infty(\mathbb{R})} \le B$ and $\|g''\|_{L^\infty([-R-2,R+2])} \le L$, the pair*

$$H_g(y,z) = \frac{\Gamma_\epsilon(g'(y), S_g(y,z))}{|y-z|^\alpha},$$

$$J_g(y,z) = \frac{1}{|y-z|^\alpha}\left[\partial_1\Gamma_\epsilon(g'(y),S_g)g''(y) + \partial_2\Gamma_\epsilon(g'(y),S_g)\frac{g'(y)-S_g}{y-z} - \alpha\frac{\Gamma_\epsilon(g'(y),S_g)}{y-z}\right],$$

with diagonal values zero and g'' taken to be the classical derivative where it exists and zero otherwise, is a dominated kernel pair on $(-R,R)$ with constant $C\epsilon$. For the value numerator the bound is pointwise; for the derivative numerator it holds outside a null set of y independent of z. These envelope inequalities hold also for $|y| \le R$, pointwise for H_g and a.e. for J_g. The same conclusion on $(-R,R)$ holds with the local second-derivative hypothesis only on $[-2R,2R]$, provided $R \ge 1$, by using $|y-z| \le 1$ in the near region.

Proof. Use $|g'(y) - S_g| \le L|y-z|$ near the diagonal together with $|\Gamma_\epsilon(u,v)| \le C\epsilon|u-v|^2$ and $|\nabla\Gamma_\epsilon(u,v)| \le C\epsilon|u-v|$ on $[-B,B]^2$. The three terms of J_g are then bounded by $C\epsilon|y-z|^{1-\alpha}$. For $|y-z| \ge 1$, bounded first derivatives and the local bound for $g''(y)$ give $C\epsilon|y-z|^{-\alpha}$. Measurability and the off-diagonal weak chain rule follow as for the preceding linearized kernel. The segment from y to z stays in $[-2R,2R]$ whenever $|y| < R$, $|y-z| \le 1$, and $R \ge 1$. □

Lemma 10.23 (Differentiation in the profile parameter)**.** *Let $0 < p < 1$, $\alpha = 1+p$, $0 \le \epsilon \le 1$, and let $f, h \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ have bounded first derivatives. For every $y \in \mathbb{R}$, the function $t \mapsto \mathcal{I}_\epsilon(f+th)(y)$ is C^1 on $\mathbb{R}$, all its defining integrals are absolutely convergent, and*

$$\frac{d}{dt}\mathcal{I}_\epsilon(f+th)(y) = \int_{\mathbb{R}} K_{f+th,h}(y,z)|y-z|^{-\alpha}\,dz.$$

The derivative integral is absolutely convergent for every (t,y). Quantitatively, for every $B, L \ge 1$ and $T \ge 0$ there is $C = C(p,B,L,T) > 0$ such that if $R \ge 1$, $\|f'\|_\infty, \|h'\|_\infty \le B$ and $\|f''\|_{L^\infty([-R-2,R+2])}, \|h''\|_{L^\infty([-R-2,R+2])} \le L$, then for all $|y| < R$, $|t| \le T$, and $z \ne y$,

$$|\Gamma_\epsilon((f+th)'(y), S_{f+th}(y,z))| + |K_{f+th,h}(y,z)| \le C\epsilon\min\{|y-z|^2, 1\}.$$

Thus one integrable majorant in z works for all $t \in [-T,T]$ at each fixed y.

Proof. The bounds for $f + th$ are $B(1 + T)$ and $L(1 + T)$, so the two kernel lemmas give the displayed estimate. Linearity of secants and first derivatives gives the pointwise t derivative $K_{f+th,h}$. At fixed (y, z) it is continuous in t. Differentiate under the integral using the translated envelope $C\epsilon\omega_0(y - z)$, and apply dominated convergence once more for continuity of the derivative. Every t_0 belongs to the interior of some $[-T, T]$.

□

Lemma 10.24 (Variation formula for $\mathcal{N}_3$)**.** *Let $0 \le \epsilon \le 1$, $\beta + \frac{1}{p} < \gamma < 2p + 1$, $\overline{f}$ admissible with constant M, and $r, h \in X_\gamma$. Then:*

1. *for every y the integral defining $D\mathcal{N}_3(r)[h](y)$ converges absolutely;*
2. *for every y, the map $t \mapsto \mathcal{N}_3(r+th)(y)$ is C^1 on $\mathbb{R}$ with derivative $D\mathcal{N}_3(r+th)[h](y)$;*
3. *$D\mathcal{N}_3(r)[h] \in W^{1,\infty}_{\mathrm{loc}}$ and its weak derivative is (paper `eq:N3_variation_derivative`)*

$$\partial_y D\mathcal{N}_3(r)[h](y) = \int_{\mathbb{R}} \frac{\partial_y K_{f,h}(y, z) - \alpha K_{f,h}(y, z)/(y - z)}{|y - z|^\alpha}\, dz - \partial_y(b_\epsilon h')(y),$$

where $\partial_y K_{f,h}$ is computed by the chain rule using Lemma 10.15, and the integral converges absolutely for a.e. y.

Proof. Put $f = \overline{f} + r$. The admissible-profile and X_γ bounds give bounded f', h' and finite local bounds for f'', h''. For each $R \ge 1$, choose B, L, H_0 satisfying Lemma 10.21 on $[-R-2, R+2]$. Its value envelope proves (1). Apply Lemma 10.23 to $f + th$, then subtract the t-independent $\mathcal{I}_\epsilon(\overline{f})$ and differentiate $-b_\epsilon(r' + th')$ to obtain (2).

For (3), apply Lemmas 10.18 and 10.19 to the pair supplied by Lemma 10.21. Take $\eta_n = 1/(n+3)$. The truncated values converge uniformly on $(-R, R)$ and their weak derivatives converge in L^∞ there. By Lemma 10.20, the untruncated value integral is locally Lipschitz and its weak derivative is the absolutely convergent integral of the displayed fixed-z derivative kernel. Subtract $b_\epsilon h'$, which is locally Lipschitz because b_ϵ is smooth and h' is locally Lipschitz. Exhaust $\mathbb{R}$ by these intervals.

□

Lemma 10.25 (Mean value reduction)**.** *Let $0 \le \epsilon \le 1$, $\beta + \frac{1}{p} < \gamma < 2p + 1$, $\overline{f}$ admissible with constant M, $N > 0$ and $A \ge 0$. Suppose that for every $r \in X_\gamma$ with $\|r\|_{X_\gamma} \le N$ and every $h \in X_\gamma$, $\|D\mathcal{N}_3(r)[h]\|_{Y_\gamma} \le A\|h\|_{X_\gamma}$. Then for all $r_1, r_2 \in X_\gamma$ with $\|r_j\|_{X_\gamma} \le N$,*

$$\|\mathcal{N}_3(r_1) - \mathcal{N}_3(r_2)\|_{Y_\gamma} \le A\|r_1 - r_2\|_{X_\gamma}.$$

Proof. Let $h = r_1 - r_2$ and $\rho_t = r_2 + th$, $t \in [0, 1]$; $\|\rho_t\| \le N$ by convexity of the ball. By Lemma 10.24(2) and the fundamental theorem of calculus, for every y

$$G(y) := \mathcal{N}_3(r_1)(y) - \mathcal{N}_3(r_2)(y) = \int_0^1 D\mathcal{N}_3(\rho_t)[h](y)\, dt.$$

Hence $\langle y\rangle^\gamma |G(y)| \le \int_0^1 \langle y\rangle^\gamma |D\mathcal{N}_3(\rho_t)[h](y)| dt \le A\|h\|$. For the derivative: the function $(t, y) \mapsto D\mathcal{N}_3(\rho_t)[h](y)$ is jointly measurable (continuous in t, by (2)),

and its y-weak derivative $H_t(y)$ is bounded on $[0,1]\times[-R,R]$ (by the hypothesis, or by the local bounds in the proof of Lemma 10.24). For a test function $\varphi \in C_c^\infty$, Fubini gives $\int G\varphi'\,dy = \int_0^1 \int D\mathcal{N}_3(\rho_t)[h]\varphi'\,dy\,dt = -\int \big(\int_0^1 H_t\,dt\big)\varphi\,dy$, so $G' = \int_0^1 H_t\,dt$ a.e. For each half-line, apply the triangle inequality inside the t-integral to the weighted sum of the value and derivative, then use $\operatorname{ess\,sup}_y \int_0^1 q(t,y)dt \le \int_0^1 \operatorname{ess\,sup}_y q(t,y)dt$ for $q \ge 0$. Summing the two half-line estimates *inside the same t-integral* gives

$$\|G\|_{Y_\gamma} \le \int_0^1 \|D\mathcal{N}_3(\rho_t)[h]\|_{Y_\gamma}\,dt \le A\|h\|.$$

□

In the rest of this section we fix $0 < \epsilon \le 1$ (for $\epsilon = 0$, $A_0 \equiv 1$ so $\Gamma_0 \equiv 0$, $c_0 = 0$, $b_0 = 0$, and $\mathcal{N}_3 \equiv 0$), $\beta + \frac{1}{p} < \gamma < 2p+1$, $M, N > 0$, an admissible $\overline{f}$ with constant M, $r \in X_\gamma$ with $\|r\|_{X_\gamma} \le N$, $f = \overline{f} + r$, and $h \in X_\gamma$. "C" denotes a constant depending only on p, γ, M, N and the cutoff χ_+. By Lemma 10.7, $|f'| \le C$ and $|S_f| \le C$ everywhere, so all derivatives of Γ_ϵ below are evaluated in a fixed compact set $[-C_*, C_*]^2$ with $C_* = C_*(p,\gamma,M,N) \ge 1$, on which the kernel bounds listed at the start of the chapter hold with $R = C_*$.

10.4.3 Elementary integrals

Lemma 10.26 (Elementary integrals)**.** *Let $0 < p < 1$, $\alpha = 1 + p$. There is $C = C(p)$ such that for all $\lambda \ge 1$ and $a > 0$:*

$$\int_{\mathbb{R}} \frac{\min\{1, \lambda|\rho|\}^2}{|\rho|^{1+p}} d\rho = C_2\lambda^p, \quad \int_{\mathbb{R}} \frac{\min\{1,\lambda|\rho|\}}{|\rho|^{1+p}} d\rho = C_1\lambda^p, \quad \int_{|\rho|\le a} |\rho|^{1-p} d\rho = \tfrac{2a^{2-p}}{2-p}, \quad \int_{|\rho|\le a} |\rho|^{-p} d\rho = \tfrac{2a^{1-p}}{1-p}$$

$$\int_{|\rho|\ge a} |\rho|^{-\alpha} d\rho = \tfrac{2}{p}a^{-p}, \qquad \int_{|\rho|\ge a} |\rho|^{-\alpha-1} d\rho = \tfrac{2}{1+p}a^{-1-p}, \qquad \int_{|\rho|\ge a} |\rho|^{-\alpha-2} d\rho = \tfrac{2}{2+p}a^{-2-p},$$

with $C_1, C_2 \le C$. Moreover, for $\gamma > 1$ and $|y| \ge 2$, with $C_\gamma = \int \langle z\rangle^{-\gamma} dz$,

$$\int_{|y-z|\ge|y|/2} \frac{\langle z\rangle^{-\gamma}}{|y-z|^{\alpha+k}} dz \le 2^{\alpha+k} C_\gamma |y|^{-\alpha-k} \le 2^{2\alpha+2k} C_\gamma \langle y\rangle^{-\alpha-k}, \qquad k = 1, 2.$$

In applications $\lambda = \langle y\rangle^{1/p}$, so $\lambda^p = \langle y\rangle$; and $a = |y|/2$ with $|y| \ge 2$, so $a \ge \langle y\rangle/4$.

Proof. Substitute $\rho = s/\lambda$: $C_k = \int \min\{1,|s|\}^k |s|^{-1-p} ds$, finite because $k - 1 - p > -1$ near 0 and $-1 - p < -1$ at infinity. The power integrals are explicit. For the last bound, use $|y-z|^{-\alpha-k} \le (|y|/2)^{-\alpha-k}$ on the region and integrate $\langle z\rangle^{-\gamma}$; for $|y| \ge 2$, $\langle y\rangle \le \sqrt{2}|y| \le 2|y|$. □

10.4.4 Near field: $|y| \ge 2$, $|y - z| \le |y|/2$

Lemma 10.27 (Near-field increments)**.** *With the standing assumptions, let $|y| \ge 2$ and $|y - z| \le |y|/2$ (so z has the sign of y and $\frac{1}{2}\langle y\rangle \le \langle w\rangle \le 2\langle y\rangle$ for $w \in I_{y,z}$). Then, with $m(y,z) = \min\{1, \langle y\rangle^{1/p}|y-z|\}$, for $y \ge 2$ (paper `eq:N3_profile_increment`, `eq:N3_direction_increment`):*

$$|f'(y)| + |S_f(y,z)| \le C\langle y\rangle^{-\beta}, \qquad |f'(y) - S_f(y,z)| \le C\langle y\rangle^{-\beta} m(y,z), \qquad |f''(y)| + |\partial_y S_f(y,z)| \le C\langle y\rangle^{-\beta+1/p},$$

$|h'(y)| \le \|h\|\langle y\rangle^{-\gamma+1/p}$, $|h'(y)-S_h(y,z)| \le C\|h\|\langle y\rangle^{-\gamma+1/p} m(y,z)$, $|h''(y)|+|\partial_y S_h(y,z)| \le C\|h\|\langle y\rangle^{-\gamma+2/p}$

and for $y \le -2$:

$|f'(y)|+|S_f(y,z)| \le C$, $|f'(y)-S_f(y,z)| \le C\langle y\rangle^{-p-2}|y-z|$, $|f''(y)|+|\partial_y S_f(y,z)| \le C\langle y\rangle^{-p-2}$,

$|h'(y)| \le \|h\|\langle y\rangle^{-\gamma-1}$, $|h'(y)-S_h(y,z)| \le C\|h\|\langle y\rangle^{-\gamma-2}|y-z|$, $|h''(y)|+|\partial_y S_h(y,z)| \le C\|h\|\langle y\rangle^{-\gamma-2}$.

(Second-derivative statements hold for a.e. y.)

Proof. On $I_{y,z}$ all weights $\langle w\rangle^s$ are within a factor $2^{|s|}$ of $\langle y\rangle^s$. Then: $|S_g(y,z)| \le \sup_{I_{y,z}}|g'|$; $|g'(y) - S_g(y,z)| \le \min\{2\sup_I|g'|,\ |y-z|\sup_I|g''|\}$; $|\partial_y S_g| \le \sup_I|g''|$ (Lemma 10.15), applied to $g = f$ and $g = h$ with the bounds of Lemma 10.7. For $y \ge 2$ and $g = f$: $\min\{\langle y\rangle^{-\beta}, \langle y\rangle^{-\beta+1/p}|y-z|\} = \langle y\rangle^{-\beta} m$; for $g = h$ similarly with $-\gamma+1/p$ and $-\gamma+2/p$. For $y \le -2$ and $g = f$ use $\sup_I|f''| \le C\langle y\rangle^{-p-2}$; for $g = h$, $\sup_I|h''| \le C\|h\|\langle y\rangle^{-\gamma-2}$. □

Lemma 10.28 (Near-field integrand bounds)**.** *With the standing assumptions, $|y| \ge 2$ and $|y-z| \le |y|/2$, we have the decomposition (paper `eq:N3_near_decomposition`)*

$$K_{f,h} = (\partial_1+\partial_2)\Gamma_\epsilon\, h'(y) + \partial_2\Gamma_\epsilon\,(S_h(y,z) - h'(y)),$$

its y-derivative (paper `eq:N3_near_integrand_derivative`; all Γ_ϵ-derivatives at $(f'(y), S_f(y,z))$)

$$\begin{aligned}\partial_y K_{f,h} = {}& (\partial_{11}+\partial_{12})\Gamma_\epsilon\, f'' h' + (\partial_{12}+\partial_{22})\Gamma_\epsilon\,\partial_y S_f\, h' + (\partial_1+\partial_2)\Gamma_\epsilon\, h'' \\ & + \big(\partial_{12}\Gamma_\epsilon f'' + \partial_{22}\Gamma_\epsilon\partial_y S_f\big)(S_h - h') + \partial_2\Gamma_\epsilon(\partial_y S_h - h''),\end{aligned}$$

and the bounds (paper `eq:N3_near_integrand_bound`, `eq:N3_near_integrand_derivative_bound`)

$$|K_{f,h}| \le C\epsilon\|h\| \begin{cases} \langle y\rangle^{-2\beta-\gamma+1/p} m^2, & y \ge 2 \\ \langle y\rangle^{-\gamma-p-4}|y-z|^2, & y \le -2 \end{cases},$$

$$|\partial_y K_{f,h}| + \frac{|K_{f,h}|}{|y-z|} \le C\epsilon\|h\| \begin{cases} \langle y\rangle^{-2\beta-\gamma+2/p} m, & y \ge 2 \\ \langle y\rangle^{-\gamma-p-4}|y-z|, & y \le -2, \end{cases}$$

with $m = m(y,z)$ as in Lemma 10.27.

Proof. The decomposition is algebra. Its derivative: product and chain rule, with $\partial_y[f'(y)] = f''$, $\partial_y[S_f] = \partial_y S_f$, and the identity $\partial_1(\partial_1+\partial_2) + \partial_2(\partial_1+\partial_2)$ expanded; note that $\partial_y(S_h - h') = \partial_y S_h - h''$. Write $u = f'(y)$, $v = S_f(y,z)$, $|u-v| =: d$.

Value. $|(\partial_1+\partial_2)\Gamma_\epsilon| \le C\epsilon d^2$ and $|\partial_2\Gamma_\epsilon| \le C\epsilon d(|u|+|v|)$. For $y \ge 2$: $\epsilon\langle y\rangle^{-2\beta} m^2 \cdot \langle y\rangle^{-\gamma+1/p} + \epsilon\langle y\rangle^{-\beta} m\langle y\rangle^{-\beta}\cdot\langle y\rangle^{-\gamma+1/p} m$. For $y \le -2$: $\epsilon\langle y\rangle^{-2p-4}|y-z|^2\langle y\rangle^{-\gamma-1} + \epsilon\langle y\rangle^{-p-2}|y-z|\cdot\langle y\rangle^{-\gamma-2}|y-z|$.

Derivative, term by term, using $|(\partial_1+\partial_2)\nabla\Gamma_\epsilon| \le C\epsilon d$, $|\nabla^2\Gamma_\epsilon| \le C\epsilon(|u|+|v|)$ and Lemma 10.27. For $y \ge 2$ (all with factor $\epsilon\|h\|$): terms 1–2: $\langle y\rangle^{-\beta} m \cdot \langle y\rangle^{-\beta+1/p}\cdot\langle y\rangle^{-\gamma+1/p}$; term 3: $\langle y\rangle^{-2\beta} m^2\langle y\rangle^{-\gamma+2/p} \le \langle y\rangle^{-2\beta-\gamma+2/p} m$; term 4: $\langle y\rangle^{-\beta}\langle y\rangle^{-\beta+1/p}\langle y\rangle^{-\gamma+1/p} m$; term 5: $\langle y\rangle^{-2\beta} m\langle y\rangle^{-\gamma+2/p}$. For $y \le -2$: terms 1–2: $\langle y\rangle^{-p-2}|y-z|\langle y\rangle^{-p-2}\langle y\rangle^{-\gamma-1}$; term 3: $\langle y\rangle^{-2p-4}|y-z|^2\langle y\rangle^{-\gamma-2} \le \langle y\rangle^{-2p-\gamma-5}|y-z|$ (as $|y-z| \le \langle y\rangle$); term 4: $1\cdot\langle y\rangle^{-p-2}\langle y\rangle^{-\gamma-2}|y-z|$; term 5: $\langle y\rangle^{-p-2}|y-z|\langle y\rangle^{-\gamma-2}$. All are $\le \langle y\rangle^{-\gamma-p-4}|y-z|$.

Quotient $|K|/|y-z|$: for $y \ge 2$ use $m^2/|y-z| \le m\cdot m/|y-z| \le m\langle y\rangle^{1/p}$; for $y \le -2$ divide directly. □

Lemma 10.29 (Near-field estimates)**.** *With the standing assumptions and* $|y| \geq 2$ *(paper* `eq:N3_near_value`*,* `eq:N3_near_derivative`*):*

$$\int_{|y-z|\leq|y|/2} \frac{|K_{f,h}(y,z)|}{|y-z|^\alpha} dz \leq C\epsilon\|h\| \begin{cases} \langle y\rangle^{-\gamma}, & y \geq 2 \\ \langle y\rangle^{-\gamma-2p-2}, & y \leq -2, \end{cases}$$

$$\int_{|y-z|\leq|y|/2} \frac{|\partial_y K_{f,h} - \alpha K_{f,h}/(y-z)|}{|y-z|^\alpha} dz \leq C\epsilon\|h\| \begin{cases} \langle y\rangle^{-\gamma+1/p}, & y \geq 2 \\ \langle y\rangle^{-\gamma-2p-3}, & y \leq -2. \end{cases}$$

Proof. Integrate the bounds of Lemma 10.28 in $\rho = y - z$. For $y \geq 2$, extend to all $\rho \in \mathbb{R}$ and use Lemma 10.26 with $\lambda = \langle y\rangle^{1/p}$: value $\langle y\rangle^{-2\beta-\gamma+1/p} \cdot C\langle y\rangle = C\langle y\rangle^{-\gamma}$, derivative $\langle y\rangle^{-2\beta-\gamma+2/p} \cdot C\langle y\rangle = C\langle y\rangle^{-\gamma+1/p}$ (using $2\beta = 1+1/p$). For $y \leq -2$: $\langle y\rangle^{-\gamma-p-4} \int_{|\rho|\leq|y|/2} |\rho|^{1-p} \leq C\langle y\rangle^{-\gamma-2p-2}$ and $\langle y\rangle^{-\gamma-p-4} \int_{|\rho|\leq|y|/2} |\rho|^{-p} \leq C\langle y\rangle^{-\gamma-2p-3}$, using $|y| \leq \langle y\rangle$. The factor α is absorbed into C.

□

10.4.5 Far field for $y \geq 2$: subtraction of the model coefficient

Lemma 10.30 (Model coefficient subtraction)**.** *For* $y \geq 2$ *and every* $z < 0$, $S_W(y,z) = \frac{z}{y-z} \in (-1,0)$ *and* $\partial_y S_W(y,z) = -\frac{S_W(y,z)}{y-z}$*; for* $z \geq 0$, $z \neq y$, $S_W(y,z) = 0$*. Consequently, for* $y \geq 2$ *(paper* `eq:N3_model_coefficient`*)*

$$b_\epsilon(y) = \int_{z<0} \frac{\partial_1\Gamma_\epsilon(0, S_W(y,z))}{|y-z|^\alpha} dz = \int_{|y-z|\geq y/2} \frac{\partial_1\Gamma_\epsilon(0, S_W(y,z))}{|y-z|^\alpha} dz,$$

hence, writing $\widetilde{K}(y,z) = K_{f,h}(y,z) - \partial_1\Gamma_\epsilon(0, S_W(y,z))h'(y)$*, (paper* `eq:N3_far_subtraction`*,* `eq:N3_far_subtraction_derivative`*)*

$$\int_{|y-z|\geq y/2} \frac{K_{f,h}}{|y-z|^\alpha} dz - b_\epsilon(y)h'(y) = \int_{|y-z|\geq y/2} \frac{\widetilde{K}(y,z)}{|y-z|^\alpha} dz,$$

and for a.e. $y > 2$

$$\int_{|y-z|\geq y/2} \frac{\partial_y K_{f,h} - \alpha K_{f,h}/(y-z)}{|y-z|^\alpha} dz - \partial_y(b_\epsilon h')(y) = \int_{|y-z|\geq y/2} \frac{1}{|y-z|^\alpha} \Big(\partial_y - \frac{\alpha}{y-z}\Big) \widetilde{K}(y,z)\, dz.$$

(Here ∂_y *acts with* z *fixed; the integration region is not differentiated.)*

Proof. For $y > 0$: $W(y) = 0$, $W'(y) = 0$, and $W(z) = -z$ for $z < 0$, $W(z) = 0$ for $z \geq 0$, giving the formulas for S_W and $\partial_y S_W$. Since $\chi_+(y) = 1$ for $y \geq 2$, Lemma 3.12 gives $b_\epsilon(y) = \int_{\mathbb{R}} \partial_1\Gamma_\epsilon(0, S_W(y,z))|y-z|^{-\alpha} dz$; on $z \geq 0$ the integrand is $\partial_1\Gamma_\epsilon(0,0) = 0$ (as $\nabla\Gamma_\epsilon(u,u) = 0$). The set $\{z < 0\}$ is contained in $\{|y-z| \geq y/2\}$ (indeed $|y-z| > y$), and on $\{|y-z| \geq y/2\} \cap \{z \geq 0\}$ the integrand vanishes; this gives both forms of b_ϵ, and the first identity follows by linearity (both integrals converge absolutely: $|\partial_1\Gamma_\epsilon| \leq C\epsilon$, $\alpha > 1$).

For the derivative: on $y > 2$, $z < 0$, the integrand $\partial_1\Gamma_\epsilon(0, S_W(y,z))|y-z|^{-\alpha}$ is smooth in y, with y-derivative $\big(-\partial_{12}\Gamma_\epsilon(0, S_W)\frac{S_W}{y-z} - \alpha\frac{\partial_1\Gamma_\epsilon(0,S_W)}{y-z}\big)|y-z|^{-\alpha}$, dominated by $C\epsilon|y-z|^{-\alpha-1}$ uniformly for y in compact subsets of $(2,\infty)$; so differentiation under the integral is legitimate (paper `eq:N3_model_coefficient_derivative`).

Multiplying by $h'(y)$ and using the product rule: $\partial_y(b_\epsilon h') = \int_{z<0} |y-z|^{-\alpha}(\partial_y - \frac{\alpha}{y-z})\big(\partial_1\Gamma_\epsilon(0,S_W)h'(y)\big)dz$, and again the integrand vanishes on $z \geq 0$, so the region may be replaced by $|y-z| \geq y/2$. Subtract from the far part of Lemma 10.24(3), which converges absolutely.

□

Lemma 10.31 (Far-field integrand bounds, $y \geq 2$)**.** *With the standing assumptions, $y \geq 2$, $|y-z| \geq y/2$, and $\widetilde{K}$ as in Lemma 10.30:*

$$|\partial_1\Gamma_\epsilon(f'(y), S_f(y,z)) - \partial_1\Gamma_\epsilon(0, S_W(y,z))| \leq C\epsilon\langle y\rangle^{-1},$$

$$|\widetilde{K}(y,z)| \leq C\epsilon\|h\|\Big(\langle y\rangle^{-\gamma+1/p-1} + \frac{\langle y\rangle^{-\gamma} + \langle z\rangle^{-\gamma}}{|y-z|}\Big),$$

$$\Big|\partial_y\widetilde{K}(y,z)\Big| + \frac{|\widetilde{K}(y,z)|}{|y-z|} \leq C\epsilon\|h\|\Big(\langle y\rangle^{-\gamma+2/p-1} + \langle y\rangle^{1/p-1}\frac{\langle y\rangle^{-\gamma} + \langle z\rangle^{-\gamma}}{|y-z|}\Big)$$

(paper `eq:N3_far_coefficient_bound`*,* `eq:N3_far_integrand_bound`*,* `eq:N3_far_integrand_derivative_bo` *the last for a.e. y).*

Proof. Decomposition (paper `eq:N3_far_decomposition`): $\widetilde{K} = [\partial_1\Gamma_\epsilon - \partial_1\Gamma_\epsilon(0,S_W)]h'(y) + \partial_2\Gamma_\epsilon\, S_h(y,z)$, unmarked derivatives at $(f'(y), S_f(y,z))$.

Coefficient: the segment joining $(0,S_W)$ and $(f'(y),S_f)$ lies in $[-C_*,C_*]^2$, so by the mean value theorem and $|\nabla^2\Gamma_\epsilon| \leq C\epsilon$, the difference is $\leq C\epsilon(|f'(y)| + |S_f - S_W|)$. Now $|f'(y)| \leq C\langle y\rangle^{-\beta} \leq C\langle y\rangle^{-1}$ ($\beta > 1$) and $|S_f - S_W| = |S_{f-W}| \leq (|f-W|(y) + |f-W|(z))/|y-z| \leq 2C/|y-z| \leq 4C/y \leq C'\langle y\rangle^{-1}$ (Lemma 10.7(1)).

Value: $|\partial_2\Gamma_\epsilon| \leq C\epsilon$ and $|S_h| \leq (|h(y)| + |h(z)|)/|y-z|$, with $|h(w)| \leq \|h\|\langle w\rangle^{-\gamma}$ and $|h'(y)| \leq \|h\|\langle y\rangle^{-\gamma+1/p}$.

Derivative: with z fixed,

$$\partial_y\widetilde{K} = [\partial_1\Gamma_\epsilon - \partial_1\Gamma_\epsilon(0,S_W)]h'' + \big[\partial_{11}\Gamma_\epsilon f'' + \partial_{12}\Gamma_\epsilon\partial_y S_f - \partial_{12}\Gamma_\epsilon(0,S_W)\partial_y S_W\big]h'$$
$$+ \big[\partial_{12}\Gamma_\epsilon f'' + \partial_{22}\Gamma_\epsilon\partial_y S_f\big]S_h + \partial_2\Gamma_\epsilon\,\partial_y S_h.$$

By Lemma 10.15, $|\partial_y S_f| \leq 2\|f'\|_\infty/|y-z| \leq C\langle y\rangle^{-1}$, $|\partial_y S_W| = |S_W|/|y-z| \leq C\langle y\rangle^{-1}$, and $|\partial_y S_h| \leq |h'(y)|/|y-z| + (|h(y)| + |h(z)|)/|y-z|^2$. Also $|f''(y)| + \langle y\rangle^{-1} \leq C(\langle y\rangle^{-\beta+1/p} + \langle y\rangle^{-1}) \leq C\langle y\rangle^{1/p-1}$ (as $\beta > 1$, $1/p - 1 > -1$). With $|\nabla^2\Gamma_\epsilon| \leq C\epsilon$: the four terms are bounded by $\epsilon\langle y\rangle^{-1}|h''|$, $\epsilon\langle y\rangle^{1/p-1}|h'|$, $\epsilon\langle y\rangle^{1/p-1}|S_h|$, and $\epsilon(\langle y\rangle^{-1}|h'| + \langle y\rangle^{-1}|S_h|)$ (using $1/|y-z| \leq 2/y \leq C\langle y\rangle^{-1}$). Insert $|h''(y)| \leq \|h\|\langle y\rangle^{-\gamma+2/p}$. The bound on $|\widetilde{K}|/|y-z|$ follows from the value bound and $1/|y-z| \leq C\langle y\rangle^{-1} \leq C\langle y\rangle^{1/p-1}$. □

Lemma 10.32 (Far-field estimates, $y \geq 2$)**.** *With the standing assumptions and $y \geq 2$ (a.e. for the second bound) (paper* `eq:N3_far_plus_value`*,* `eq:N3_far_plus_derivative`*):*

$$\Big|\int_{|y-z|\geq y/2} \frac{K_{f,h}}{|y-z|^\alpha}dz - b_\epsilon h'(y)\Big| \leq C\epsilon\|h\|\langle y\rangle^{-\gamma},$$

$$\Big|\int_{|y-z|\geq y/2} \frac{\partial_y K_{f,h} - \alpha K_{f,h}/(y-z)}{|y-z|^\alpha}dz - \partial_y(b_\epsilon h')(y)\Big| \leq C\epsilon\|h\|\langle y\rangle^{-\gamma+1/p}.$$

Proof. By Lemma 10.30 it suffices to integrate the bounds of Lemma 10.31 against $|y-z|^{-\alpha}$ over $|y-z| \geq y/2$, using Lemma 10.26 with $a = y/2 \geq \langle y\rangle/4$ (all with factor $C\epsilon\|h\|$):

Value: $\langle y\rangle^{-\gamma+1/p-1} \cdot \langle y\rangle^{-p} + \langle y\rangle^{-\gamma}\langle y\rangle^{-1-p} + \langle y\rangle^{-\alpha-1} = \langle y\rangle^{-\gamma+1/p-1-p} + \langle y\rangle^{-\gamma-p-1} + \langle y\rangle^{-p-2}$. Each is $\leq \langle y\rangle^{-\gamma}$: the first since $1/p < 1+p$, the third since $\gamma < 2p+1 < p+2$.

Derivative: $\langle y\rangle^{-\gamma+2/p-1-p} + \langle y\rangle^{1/p-1}(\langle y\rangle^{-\gamma-1-p} + \langle y\rangle^{-p-2})$. Each is $\leq \langle y\rangle^{-\gamma+1/p}$: the first since $1/p < 1+p$, the second trivially, the third $\langle y\rangle^{1/p-p-3} \leq \langle y\rangle^{-\gamma+1/p}$ since $\gamma < p+3$.

□

10.4.6 Far field for $y \leq -2$ and the compact region $|y| < 2$

Lemma 10.33 (Far-field estimates, $y \leq -2$)**.** *With the standing assumptions and $y \leq -2$ (a.e. for the second bound) (paper* `eq:N3_far_minus_integrand_bound`*,* `eq:N3_far_minus_value`*,* `eq:N3_far_minus_derivative`*): for $|y-z| \geq |y|/2$,*

$$|K_{f,h}| \leq C\epsilon\|h\|\Big(\langle y\rangle^{-\gamma-1} + \frac{\langle y\rangle^{-\gamma} + \langle z\rangle^{-\gamma}}{|y-z|}\Big), \quad |\partial_y K_{f,h}| + \frac{|K_{f,h}|}{|y-z|} \leq C\epsilon\|h\|\Big(\langle y\rangle^{-\gamma-2} + \langle y\rangle^{-1}\frac{\langle y\rangle^{-\gamma} + \langle z\rangle^{-\gamma}}{|y-z|}\Big),$$

and

$$\int_{|y-z|\geq|y|/2} \frac{|K_{f,h}|}{|y-z|^{\alpha}}dz \leq C\epsilon\|h\|\langle y\rangle^{-\gamma}, \qquad \int_{|y-z|\geq|y|/2} \frac{|\partial_y K_{f,h} - \alpha K_{f,h}/(y-z)|}{|y-z|^{\alpha}}dz \leq C\epsilon\|h\|\langle y\rangle^{-\gamma-1}.$$

Moreover $b_\epsilon(y) = 0$ for $y \leq 1$.

Proof. $|\nabla\Gamma_\epsilon| + |\nabla^2\Gamma_\epsilon| \leq C\epsilon$ on $[-C_*, C_*]^2$. Value: $|K| \leq C\epsilon(|h'(y)| + |S_h|)$ with $|h'(y)| \leq \|h\|\langle y\rangle^{-\gamma-1}$, $|S_h| \leq (|h(y)| + |h(z)|)/|y-z|$. Derivative: $\partial_y K = \nabla\partial_1\Gamma_\epsilon \cdot (f'', \partial_y S_f)h' + \partial_1\Gamma_\epsilon h'' + \nabla\partial_2\Gamma_\epsilon \cdot (f'', \partial_y S_f)S_h + \partial_2\Gamma_\epsilon \partial_y S_h$, with $|f''(y)| + |\partial_y S_f| \leq C\langle y\rangle^{-p-2} + 2\|f'\|_\infty/|y-z| \leq C\langle y\rangle^{-1}$, $|\partial_y S_h| \leq |h'(y)|/|y-z| + (|h(y)| + |h(z)|)/|y-z|^2$, $|h''(y)| \leq \|h\|\langle y\rangle^{-\gamma-2}$ and $1/|y-z| \leq C\langle y\rangle^{-1}$. The quotient bound follows from the value bound and $1/|y-z| \leq C\langle y\rangle^{-1}$. Integration (Lemma 10.26, $a = |y|/2$): value $\langle y\rangle^{-\gamma-1-p} + \langle y\rangle^{-\gamma-1-p} + \langle y\rangle^{-p-2} \leq 3\langle y\rangle^{-\gamma}$ (as $\gamma < p+2$); derivative $\langle y\rangle^{-\gamma-2-p} + \langle y\rangle^{-1}(\langle y\rangle^{-\gamma-1-p} + \langle y\rangle^{-p-2}) \leq 3\langle y\rangle^{-\gamma-1}$ (as $\gamma < p+2$). The last claim is $\chi_+ = 0$ on $(-\infty, 1]$.

□

Lemma 10.34 (Compact region $|y| < 2$)**.** *With the standing assumptions, for $|y| < 2$ (a.e. for derivative statements):*

$$|K_{f,h}(y,z)| \leq C\epsilon\|h\|\min\{|y-z|^2, 1\}, \qquad |\partial_y K_{f,h}(y,z)| + \frac{|K_{f,h}(y,z)|}{|y-z|} \leq C\epsilon\|h\|\min\{|y-z|, 1\},$$

hence (paper `eq:N3_far_compact_value`*,* `eq:N3_far_compact_derivative`*)*

$$\int_{\mathbb{R}} \frac{|K_{f,h}|}{|y-z|^{\alpha}}dz + \int_{\mathbb{R}} \frac{|\partial_y K_{f,h} - \alpha K_{f,h}/(y-z)|}{|y-z|^{\alpha}}dz \leq C\epsilon\|h\|,$$

and $|b_\epsilon(y)h'(y)| + |\partial_y(b_\epsilon h')(y)| \leq C\epsilon\|h\|$. Consequently $|D\mathcal{N}_3(r)[h](y)| \leq C\epsilon\|h\|\langle y\rangle^{-\gamma}$ and $|\partial_y D\mathcal{N}_3(r)[h](y)| \leq C\epsilon\|h\|\min\{\langle y\rangle^{-\gamma+1/p}, \langle y\rangle^{-\gamma-1}\}$ for $|y| < 2$.

Proof. For $|y-z| \le 1$ we have $y, z \in [-3,3]$, where $\|f''\|_\infty + \|h''\|_\infty/\|h\| \le C$ (Lemma 10.7); so $|f'(y) - S_f| \le C|y-z|$, $|h'(y) - S_h| \le C\|h\||y-z|$, $|\partial_y S_f| \le C$, $|\partial_y S_h| \le C\|h\|$. The kernel-pair bounds of Lemma 10.21 on $(-2,2)$ (equivalently, the near decomposition of the integrand together with the kernel bounds) give $|K| \le C\epsilon\|h\||y-z|^2$; the five-term derivative formula gives $|\partial_y K| \le C\epsilon\|h\||y-z|$ (terms 1,2,4,5 have a factor $|y-z|$ via $d = |u-v|$ or $|S_h - h'|$; term 3 has d^2). For $|y-z| \ge 1$: $|K| \le C\epsilon(|h'(y)| + 2\|h\|_\infty) \le C\epsilon\|h\|$, and $|\partial_y K| \le C\epsilon(|h''| + |h'|(|f''| + |\partial_y S_f|) + (|f''| + |\partial_y S_f|)|S_h| + |\partial_y S_h|) \le C\epsilon\|h\|$ using $|\partial_y S_f| \le 2\|f'\|_\infty$, $|S_h| \le 2\|h\|_\infty$, $|\partial_y S_h| \le |h'(y)| + |S_h|$. Integrate with Lemma 10.26: $\int_{|\rho|\le 1} |\rho|^{2-\alpha} + \int_{|\rho|\ge 1} |\rho|^{-\alpha} < \infty$ and $\int_{|\rho|\le 1} |\rho|^{1-\alpha} < \infty$ (as $\alpha < 2$). For b_ϵ: on $[-2,2]$, $b_\epsilon = c_\epsilon \chi_+ y^{-p}$ is supported in $[1,2]$ where it is smooth with $|b_\epsilon| + |b_\epsilon'| \le C|c_\epsilon| \le C\epsilon$ (Lemma 3.12); and $|h'| + |h''| \le C\|h\|$ on $[-2,2]$. The consequence follows from Lemma 10.24(3), since on $|y| < 2$ all weights are comparable to 1. Lean route: the kernel pair on $(-2,2)$ is controlled by Lemma 10.21 and Lemma 10.17; the near-integrand lemma is not needed.

□

10.4.7 Conclusion for $\mathcal{N}_3$

Lemma 10.35 (Bound on the linearization)**.** *Let $0 \le \epsilon \le 1$, $\beta + \frac{1}{p} < \gamma < 2p+1$, $M, N > 0$, $\overline{f}$ admissible with constant M, $r \in X_\gamma$ with $\|r\|_{X_\gamma} \le N$, and $h \in X_\gamma$. Then $D\mathcal{N}_3(r)[h] \in Y_\gamma$ and*

$$\|D\mathcal{N}_3(r)[h]\|_{Y_\gamma} \le C\epsilon\|h\|_{X_\gamma}, \qquad C = C(p, \gamma, M, N).$$

Proof. If $\epsilon = 0$, $D\mathcal{N}_3 \equiv 0$. Otherwise, by Lemma 10.24 the value and the a.e. derivative of $D\mathcal{N}_3(r)[h]$ are given by absolutely convergent integrals, which for $|y| \ge 2$ we split into the near region $|y-z| \le |y|/2$ and the far region $|y-z| > |y|/2$ (the boundary $|y-z| = |y|/2$ has measure zero).

- $y \ge 2$: near part by Lemma 10.29, far part together with $-b_\epsilon h'$ by Lemma 10.32: $|D\mathcal{N}_3(r)[h]| \le C\epsilon\|h\|\langle y\rangle^{-\gamma}$, $|\partial_y D\mathcal{N}_3(r)[h]| \le C\epsilon\|h\|\langle y\rangle^{-\gamma+1/p}$.
- $y \le -2$: $b_\epsilon = 0$ near y; near part by Lemma 10.29 ($\langle y\rangle^{-\gamma-2p-2} \le \langle y\rangle^{-\gamma}$, $\langle y\rangle^{-\gamma-2p-3} \le \langle y\rangle^{-\gamma-1}$), far part by Lemma 10.33.
- $|y| < 2$: Lemma 10.34.

Multiplying by the weights of Definition 2.4 and taking suprema over $y \ge 0$ and $y \le 0$ yields the claim.

□

Lemma 10.36 (Lipschitz bound for $\mathcal{N}_3$)**.** *Under the hypotheses of Lemma 10.11, $\mathcal{N}_3(r_1) - \mathcal{N}_3(r_2) \in Y_\gamma$ and there is $C = C(p, \gamma, M, N)$ with*

$$\|\mathcal{N}_3(r_1) - \mathcal{N}_3(r_2)\|_{Y_\gamma} \le C\epsilon\|r_1 - r_2\|_{X_\gamma}.$$

Proof. Apply Lemma 10.25 with $A = C\epsilon$ from Lemma 10.35.

□

10.5 Continuity in the profile

Lemma 10.37 (Local $W^{1,\infty}$ structure of $\mathcal{I}_\epsilon$)**.** *Let $0 \le \epsilon \le 1$, $g \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ with $g' \in L^\infty(\mathbb{R})$, and set, for $z \neq y$,*

$$K_g^{(1)}(y,z) = \partial_1\Gamma_\epsilon(g'(y),S_g)g''(y)+\partial_2\Gamma_\epsilon(g'(y),S_g)\frac{g'(y)-S_g}{y-z}-(1+p)\frac{\Gamma_\epsilon(g'(y),S_g)}{y-z},$$

with $S_g = S_g(y,z)$. Then $\mathcal{I}_\epsilon(g) \in W^{1,\infty}_{\mathrm{loc}}$ with weak derivative (paper `eq:nonlinear_continuity_derivative`*) $\partial_y\mathcal{I}_\epsilon(g)(y) = \int_{\mathbb{R}} K_g^{(1)}(y,z)|y-z|^{-\alpha}dz$ (absolutely convergent for a.e. y). Moreover, for $R \ge 1$, $0 < \delta < 1$, with $L = \|g'\|_{L^\infty} + \|g''\|_{L^\infty([-2R,2R])}$, there is $C = C(p,R,L)$ such that (paper* `eq:nonlinear_continuity_near`*)*

$$\sup_{|y|\le R}\int_{|y-z|<\delta}\frac{|\Gamma_\epsilon(g'(y),S_g)|}{|y-z|^\alpha}dz \le C\epsilon\delta^{3-\alpha}, \qquad \operatorname*{ess\,sup}_{|y|\le R}\int_{|y-z|<\delta}\frac{|K_g^{(1)}(y,z)|}{|y-z|^\alpha}dz \le C\epsilon\delta^{2-\alpha},$$

and for $|y| \le R$, $|y-z| > 4R$: $|\Gamma_\epsilon(g'(y),S_g)| + |K_g^{(1)}(y,z)| \le C\epsilon$ and $|y-z| \ge \frac{2}{3}|z|$.

Proof. On each $(-R,R)$, Lemma 10.22 supplies a dominated kernel pair whose value integral is $\mathcal{I}_\epsilon(g)$ and whose derivative integral is the one stated here. Apply Lemmas 10.18 and 10.19 with $\eta_n = 1/(n+3)$ and then Lemma 10.20. The derivative convergence is in essential supremum norm; g'' need not be continuous. This proves the qualitative local Lipschitz assertion and formula. For the quantitative near bounds use the version of Lemma 10.22 with second derivatives controlled on $[-2R,2R]$, and integrate $C\epsilon|y-z|^{2-\alpha}$ and $C\epsilon|y-z|^{1-\alpha}$ over $|y-z| < \delta$. For $|y-z| > 4R$, the numerator bounds follow from its far-region estimates, and $|z| \ge |y-z| - |y| > 3R$ implies $|y-z| \ge (2/3)|z|$. □

Lemma 10.38 (Difference estimate away from the diagonal on a compact set)**.** *Let $0 < p < 1$, $\alpha = 1+p$, $R \ge 1$, $S \ge 5R$, $0 < \delta < 1$, and $B, L \ge 1$. There exists $C = C(p,R,S,\delta,B,L) > 0$ such that for every $0 \le \epsilon \le 1$ and $g_0, g_1 \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ satisfying $\|g_j'\|_{L^\infty(\mathbb{R})} \le B$ and $\|g_j''\|_{L^\infty([-S,S])} \le L$ $(j = 0,1)$, the following holds. Put*

$$e_S = \|g_1-g_0\|_{L^\infty([-S,S])}+\|g_1'-g_0'\|_{L^\infty([-S,S])}+\|g_1''-g_0''\|_{L^\infty([-S,S])}, \qquad E_y = \{z : |z| \le S,\ |y-z| \ge \delta\}.$$

The value-difference integrand below is absolutely integrable on E_y for every $|y| \le R$, and the derivative-difference integrand is absolutely integrable there for a.e. $|y| \le R$. Using $K_g^{(1)}$ from Lemma 10.37,

$$\sup_{|y|\le R}\int_{E_y}\frac{|\Gamma_\epsilon(g_1'(y),S_{g_1})-\Gamma_\epsilon(g_0'(y),S_{g_0})|}{|y-z|^\alpha}dz \le C\epsilon e_S,$$

$$\operatorname*{ess\,sup}_{|y|\le R}\int_{E_y}\frac{|K_{g_1}^{(1)}(y,z)-K_{g_0}^{(1)}(y,z)|}{|y-z|^\alpha}dz \le C\epsilon e_S.$$

The inequalities remain valid if E_y is replaced by any measurable subset.

Proof. On E_y, the segment between y and z stays in $[-S,S]$. Hence $|S_{g_1}-S_{g_0}| \le \|g_1'-g_0'\|_{L^\infty([-S,S])}$, and $|g_1'-g_0'|$, $|g_1''-g_0''|$ are bounded by e_S (the second a.e.). Both arguments of Γ_ϵ lie in $[-B,B]$. Apply the uniform Lipschitz bounds $C(p,B)\epsilon$ for Γ_ϵ and its first derivatives there, expand the products in $K_g^{(1)}$, and use the common bound L for g_j''. Every denominator is bounded by a power of δ^{-1}, giving a pointwise bound $C(p,\delta,B,L)\epsilon e_S$ for each absolute integrand, outside one null set in y. Integrate over a set of length at most $2S$. Restriction to a measurable subset only decreases the absolute integrals. □

Lemma 10.39 (Uniform tails for the value and derivative kernels)**.** *Let $0 < p < 1$, $\alpha = 1+p$, and $B, L \ge 1$. There exists $C = C(p,B,L) > 0$ such that for every $R \ge 1$, $S \ge 5R$, $0 \le \epsilon \le 1$, and $g \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ with $\|g'\|_{L^\infty(\mathbb{R})} \le B$ and $\|g''\|_{L^\infty([-2R,2R])} \le L$,*

$$\sup_{|y|\le R} \int_{|z|>S} \frac{|\Gamma_\epsilon(g'(y), S_g(y,z))|}{|y-z|^\alpha} dz + \operatorname*{ess\,sup}_{|y|\le R} \int_{|z|>S} \frac{|K_g^{(1)}(y,z)|}{|y-z|^\alpha} dz \le C\epsilon S^{-p}.$$

Both absolute integrals are finite (the derivative one for a.e. y), and the bound is independent of R, S, g, ϵ apart from the displayed factors and hypotheses.

Proof. For $|y| \le R$, $|z| > S \ge 5R$, one has $|y-z| \ge (4/5)|z| > 4R$. Bounded first derivatives control all secants, and $|\partial_y S_g| \le 2B/|y-z| \le 2B$. The formulas for the numerators therefore give the common bound $C(p,B,L)\epsilon$, with the derivative statement interpreted a.e. in y. Integrate $|z|^{-\alpha}$: $\int_{|z|>S} |z|^{-\alpha} dz = 2S^{-p}/p$. □

Lemma 10.40 (Local continuity of $\mathcal{I}_\epsilon$)**.** *Let $0 \le \epsilon \le 1$ and let $g_k, g \in W^{2,\infty}_{\mathrm{loc}}(\mathbb{R})$ satisfy (paper* `eq:nonlinear_continuity_profiles`*) $\sup_k \|g_k'\|_{L^\infty} + \|g'\|_{L^\infty} < \infty$ and $g_k \to g$ in $W^{2,\infty}_{\mathrm{loc}}$. Then for every $R > 0$, $\mathcal{I}_\epsilon(g_k) \to \mathcal{I}_\epsilon(g)$ in $W^{1,\infty}([-R,R])$ (paper* `eq:nonlinear_continuity_local`*).*

Proof. It suffices to consider $R \ge 1$. Choose a common bound B for all g_k', g' on $\mathbb{R}$ and a common bound L_R for all g_k'', g'' on $[-2R,2R]$; the latter exists by local $W^{2,\infty}$ convergence. Fix $0 < \delta < 1$ and $S \ge 5R$. Split both value and derivative integrals into

$$|y-z| < \delta, \qquad |y-z| \ge \delta \text{ and } |z| \le S, \qquad |z| > S.$$

For $|y| \le R$ this is a partition up to null boundaries: the last region has $|y-z| > 4R$. The sum of the two near contributions, for g_k and g, is bounded by $C_R\epsilon\delta^{2-\alpha}$ by Lemma 10.37. For each fixed S, local convergence supplies a common second-derivative bound L_S on $[-S,S]$ and $e_S(k) = \|g_k - g\|_{W^{2,\infty}([-S,S])} \to 0$. The compact-region integrals therefore tend to zero by Lemma 10.38. The tail integrals are bounded by $C_R\epsilon S^{-p}$, independently of k, S, by Lemma 10.39. Thus

$$\limsup_k \|\mathcal{I}_\epsilon(g_k) - \mathcal{I}_\epsilon(g)\|_{W^{1,\infty}([-R,R])} \le C_R\epsilon\big(\delta^{2-\alpha} + S^{-p}\big).$$

Let $S \to \infty$ and then $\delta \to 0$. All derivative estimates use essential suprema. In Lean the integrability of the value and derivative integrands on each region is obtained from the kernel-pair bounds of Lemma 10.22.

□

Lemma 10.41 (Local convergence of the nonlinear operator)**.** *Let $0 \le \epsilon \le 1$, $\beta + \frac{1}{p} < \gamma < 2p+1$, $r \in X_\gamma$, and let $\overline{f}_k$ be admissible with a common constant M with $\overline{f}_k \to \overline{f}$ in C^2_{loc}. Then for $j = 1, 2, 3$ and every $R > 0$, $\mathcal{N}_j^{\epsilon,\overline{f}_k}(r) \to \mathcal{N}_j^{\epsilon,\overline{f}}(r)$ in $W^{1,\infty}([-R, R])$.*

Proof. $\overline{f}$ is admissible with constant M (Lemma 10.5). $\mathcal{N}_1, \mathcal{N}_2$: on $[-R, R]$, $\overline{f}_k' + r' \to \overline{f}' + r'$ and $\overline{f}_k' \to \overline{f}'$ in $W^{1,\infty}$ with values in a fixed compact set; since B_ϵ is smooth, $B_\epsilon(\overline{f}_k' + r') \to B_\epsilon(\overline{f}' + r')$ and $B_\epsilon(\overline{f}_k') \to B_\epsilon(\overline{f}')$ in $W^{1,\infty}([-R, R])$ (the derivative is $B_\epsilon'(\cdot)(\overline{f}_k'' + r'')$). Also $\overline{f}_k - y\overline{f}_k' \to \overline{f} - y\overline{f}'$ in $W^{1,\infty}([-R, R])$ (its derivative is $-y\overline{f}_k''$) and $r - yr'$ is fixed. $W^{1,\infty}([-R, R])$ is a Banach algebra, so products converge. $\mathcal{N}_3$: apply Lemma 10.40 to $(g_k, g) = (\overline{f}_k + r, \overline{f} + r)$ and to $(g_k, g) = (\overline{f}_k, \overline{f})$; the hypotheses hold by Lemma 10.7(1) with constants uniform in k, and C^2_{loc} convergence implies $W^{2,\infty}_{\mathrm{loc}}$ convergence. The term $b_\epsilon r'$ does not depend on the profile.

□

Proposition 10.42 (Continuity in the profile)**.** *Let $0 \le \epsilon \le \min\{1, \epsilon_0\}$ (ϵ_0 of Lemma 8.25), $\beta + \frac{1}{p} < \gamma' < \gamma < 2p + 1$, $r \in X_\gamma$, $M > 0$, and let $\{\overline{f}_k\}$ be admissible profiles with common constant M such that $\overline{f}_k \to \overline{f}$ in $C^2_{\mathrm{loc}}(\mathbb{R})$ and such that Lemma 10.2 applies to each $\overline{f}_k$ and to $\overline{f}$. Then $\mathcal{N}_{\epsilon,\overline{f}_k}(r) \to \mathcal{N}_{\epsilon,\overline{f}}(r)$ in $Y_{\gamma'}$.*

Proof. By Lemma 10.5 $\overline{f}$ is admissible with constant M. By Lemma 10.2, $\mathcal{N}_{\epsilon,\overline{f}_k}(r) - \mathcal{N}_{\epsilon,\overline{f}}(r) = \sum_{j=1}^3 \big(\mathcal{N}_j^{\epsilon,\overline{f}_k}(r) - \mathcal{N}_j^{\epsilon,\overline{f}}(r)\big)$, the profile-independent term $\mathcal{R}_\epsilon r$ cancelling. By Lemmas 10.11, 10.13 and 10.36 with $r_1 = r$, $r_2 = 0$, $N = \|r\|_{X_\gamma}$ (and $\mathcal{N}_j(0) = 0$), each $\mathcal{N}_j^{\epsilon,\overline{f}_k}(r)$ and $\mathcal{N}_j^{\epsilon,\overline{f}}(r)$ lies in Y_γ with norm $\le C(p, \gamma, M, \|r\|)\epsilon\|r\|$; hence the differences are uniformly bounded in Y_γ (paper `eq:nonlinear_continuity_uniform_bound`). By Lemma 10.41 they converge to 0 in $W^{1,\infty}_{\mathrm{loc}}$. By Proposition 3.41 (second statement, with $G_k = \mathcal{N}_{\epsilon,\overline{f}_k}(r)$, $G = \mathcal{N}_{\epsilon,\overline{f}}(r)$) the convergence holds in $Y_{\gamma'}$.

□

10.6 Assembly: proof of the nonlinear estimates

Lemma 10.43 (Uniform estimates for admissible profiles)**.** *Let $\beta + \frac{1}{p} < \gamma < 2p + 1$, $M, N > 0$, $0 \le \epsilon \le \min\{1, \epsilon_0\}$ (ϵ_0 of Lemma 8.25), and $\overline{f}$ an admissible profile with constant M to which Lemma 10.2 applies. There is $C = C(p, \gamma, M, N)$, independent of ϵ and $\overline{f}$, such that for all $r, r_1, r_2 \in X_\gamma$ with norms $\le N$, $\mathcal{N}_{\epsilon,\overline{f}}(r) \in Y_\gamma$ and*

$$\|\mathcal{N}_{\epsilon,\overline{f}}(r_1) - \mathcal{N}_{\epsilon,\overline{f}}(r_2)\|_{Y_\gamma} \le C\epsilon\|r_1 - r_2\|_{X_\gamma}, \qquad \|\mathcal{N}_{\epsilon,\overline{f}}(r)\|_{Y_\gamma} \le C\epsilon\|r\|_{X_\gamma}.$$

Proof. By Lemma 10.2, $\mathcal{N}_{\epsilon,\overline{f}}(r_1) - \mathcal{N}_{\epsilon,\overline{f}}(r_2) = \sum_{j=1}^3 (\mathcal{N}_j(r_1) - \mathcal{N}_j(r_2)) - \mathcal{R}_\epsilon(r_1 - r_2)$ (linearity of $\mathcal{R}_\epsilon$). Lemmas 10.11, 10.13 and 10.36 bound the first three differences by $C\epsilon\|r_1 - r_2\|$; Lemma 8.25 gives $\|\mathcal{R}_\epsilon\|_{X_\gamma \to Y_\gamma} \le C_\gamma \epsilon$ uniformly in $0 \le \epsilon \le \epsilon_0$. The second estimate is the first with $r_2 = 0$, since $\mathcal{N}_{\epsilon,\overline{f}}(0) = 0$ (Lemma 10.2).

□

Lemma 10.44 (Nonlinear estimates: assembly)**.** *Let $\epsilon_0, \delta, \mu_*$ be as in Theorem 2.20, and assume (shrinking ϵ_0 if necessary) that $\epsilon_0 \le 1$ and ϵ_0 does not exceed the constant of Lemma 8.25. Let $\beta + \frac{1}{p} < \gamma < 2p+1$ and $N > 0$. Write $\overline{f} = \overline{f}_{\epsilon,\mu}$ for $0 \le \epsilon \le \epsilon_0$ and $\mu \in \overline{B} = \overline{B}(\mu_*, \delta)$. Then there is $C = C(p, \gamma, N)$, independent of $\epsilon \in [0, \epsilon_0]$ and $\mu \in \overline{B}$, such that for all $r, r_1, r_2 \in X_\gamma$ with $\|r\|_{X_\gamma}, \|r_1\|_{X_\gamma}, \|r_2\|_{X_\gamma} \le N$:*

1. *$\mathcal{N}_{\epsilon,\overline{f}}(r) \in Y_\gamma$ and $\|\mathcal{N}_{\epsilon,\overline{f}}(r)\|_{Y_\gamma} \le C\epsilon\|r\|_{X_\gamma}$ ((2.4));*
2. *$\|\mathcal{N}_{\epsilon,\overline{f}}(r_1) - \mathcal{N}_{\epsilon,\overline{f}}(r_2)\|_{Y_\gamma} \le C\epsilon\|r_1 - r_2\|_{X_\gamma}$ ((2.5));*
3. *for $\gamma_1 < \gamma_2$ as in Definition 2.1, each fixed $r \in X_{\gamma_2}$ and each fixed $\epsilon \in [0, \epsilon_0]$, the map $\overline{B} \ni \mu \mapsto \mathcal{N}_{\epsilon,\overline{f}_{\epsilon,\mu}}(r) \in Y_{\gamma_1}$ is continuous.*

This is exactly the conclusion of Theorem 2.24.

Proof. By Lemma 10.6, every $\overline{f}_{\epsilon,\mu}$ is admissible with the uniform constant M_0; it is C^∞ with $\overline{f}'_{\epsilon,\mu} \in L^\infty$, and $|D|^p \overline{f}'_{\epsilon,\mu}$ is defined (as required in Definition 1.26 for $\mathcal{E}_\epsilon(\overline{f}_{\epsilon,\mu})$ to make sense; Theorem 2.20(3)), so Lemma 10.2 applies.

Items 1–2: Lemma 10.43 with $M = M_0$; the constant $C(p, \gamma, M_0, N)$ is independent of ϵ, μ.

Item 3: $\overline{B} \subset \mathbb{C}$ is a metric space, so it suffices to prove sequential continuity. Let $\mu_k \to \mu$ in $\overline{B}$. By Theorem 2.20(4), $\overline{f}_{\epsilon,\mu_k} \to \overline{f}_{\epsilon,\mu}$ in C^2_{loc}; all are admissible with constant M_0. Apply Proposition 10.42 with $\gamma = \gamma_2$, $\gamma' = \gamma_1$ (admissible since $\beta + \frac{1}{p} < \gamma_1 < \gamma_2 < 2p+1$ by Definition 2.1). □

Chapter 11

Beyond the paper: the interior fractional Schauder estimate (cited in §4)

This chapter supplies a direct one-dimensional route to Lemma 3.26. The target agrees with Proposition 2.2 of the arXiv version of [ROS14], with $n = 1$ and $s = p/2$; Corollary 2.4 there is the related weighted-tail variant. The route below uses annular Fourier kernels and a cutoff commutator. It does not require formalizing the full boundary regularity theory.

Throughout, $0 < p < 1$, and $\beta > 0$ and $s = \beta + p$ are nonintegers unless stated otherwise. Write $N_a(u; I) = \|u\|_{C^a(I)}$ for the norm of Definition 3.1, $N_a(u) = N_a(u; \mathbb{R})$, and $k_a = \lfloor a \rfloor$, $\theta_a = a - k_a \in (0, 1)$. Global derivatives are ordinary derivatives; interval norms use derivatives within the interval. All integrals below are Lebesgue integrals. Finiteness must be established before using the real-valued Lean norm obtained with `ENNReal.toReal`. Constants may depend on the displayed exponents, derivative orders and the fixed cutoffs, but not on the input functions, dyadic index, or spatial variables.

11.1 Fixed data and global Holder transport

Definition 11.1 (Fixed cutoffs and dyadic operators)**.** Use the even cutoff $\chi = \rho$ of Definition 4.4. Set

$$\eta(x) = \mathrm{smoothTransition}((49-64x^2)/13), \quad \varphi(\xi) = \chi(\xi/2) - \chi(\xi), \quad q_p(\xi) = \begin{cases} \varphi(\xi)|\xi|^{-p}, & \xi \neq 0, \\ 0, & \xi = 0. \end{cases}$$

Let $K_p = \mathrm{Re}\,\check{q}_p$ and $K_{p,j}(x) = 2^{(1-p)j} K_p(2^j x)$ for $j \in \mathbb{N}$. For a real function v put

$$S_0 v = \mathrm{Re}\,\check{\chi}\widehat{v}, \qquad \Delta_j v = \mathrm{Re}\,\varphi(2^{-j}\,\cdot)\widehat{v}.$$

These use the project's $e^{-ix\xi}$ transform and $1/(2\pi)$ inverse normalization. Finally define

$$C_{\eta,p} w(x) = c_p \int_{\mathbb{R}} (\eta(x) - \eta(x-z)) w(x-z) |z|^{-1-p}\, dz,$$

with integrand zero at $z = 0$. These are total formulas; convergence and mapping properties are the separate lemmas below. Convolution means $(a * b)(x) = \int a(z)b(x - z)dz$.

Lemma 11.2 (Cutoff and annular-symbol properties)**.** *The cutoffs η, χ are smooth, real, even and valued in $[0, 1]$. The function η equals one on $[-3/4, 3/4]$ and has support in $[-7/8, 7/8]$; χ equals one on $[-1, 1]$ and has support in $[-2, 2]$. For every $p \in (0, 1)$, q_p and φ are smooth and supported in $\{\xi : 1 \leq |\xi| \leq 4\}$. All their derivatives vanish at zero.*

Proof. Apply the smooth-transition bounds to the explicit quadratic defining η. Use the properties of ρ for χ. Both terms in φ agree near zero; away from zero, $|\xi|^{-p}$ is smooth. Zero extension across that neighborhood is smooth. □

Lemma 11.3 (Global Holder translation and restriction)**.** *For every noninteger $a > 0$, real t, and $u \in C^a(\mathbb{R})$, the translate $\tau_t u(x) = u(x - t)$ belongs to $C^a(\mathbb{R})$ and $N_a(\tau_t u) = N_a(u)$. If $I = [b, c]$ with $b < c$, then $u|_I \in C^a(I)$ and $N_a(u; I) \leq N_a(u)$.*

Proof. Translate each global derivative and the two-point Holder quotient. Restriction uses equality of within derivatives and ordinary derivatives on a nondegenerate interval. Work first with the extended norms, then transfer the finite inequalities to reals. □

Lemma 11.4 (Global Holder product bound)**.** *For every noninteger $a > 0$ there is $C_a > 0$ such that for all real $u, v \in C^a(\mathbb{R})$, $uv \in C^a(\mathbb{R})$ and $N_a(uv) \leq C_a N_a(u) N_a(v)$.*

Proof. Apply the interval product bound on all intervals $[-R, R]$, $R \geq 1$, using a constant independent of R. The local derivatives agree with the global derivatives. Exhaust the line for the derivative suprema and the two-point Holder seminorm. □

Lemma 11.5 (Integration with a global Holder majorant)**.** *Let $a = k + \theta > 0$, $k \in \mathbb{N}$, $0 < \theta < 1$. Let $F : \mathbb{R}^2 \to \mathbb{R}$ and $B : \mathbb{R} \to [0, \infty)$ be such that $B \in L^1$, every section $F(\cdot, t)$ belongs to $C^a(\mathbb{R})$, and $N_a(F(\cdot, t)) \leq B(t)$ for every t. Suppose $(x, t) \mapsto \partial_x^r F(x, t)$ is measurable for $0 \leq r \leq k$. Then $U(x) = \int F(x, t)dt$ is absolutely defined for every x, lies in $C^a(\mathbb{R})$, and $N_a(U) \leq (k + 2) \int B(t)dt$. For every $r \leq k$, $\partial^r U(x) = \int \partial_x^r F(x, t)dt$ with absolute convergence.*

Proof. Restrict to compact intervals, applying the parameter-integral theorem with the real-to-complex bridge if needed. Its derivative-interchange proof gives the identities. Bounds independent of the interval give global C^k regularity and bound each of the $k + 1$ derivative suprema and the Holder seminorm by $\int B$. This suffices for the displayed constant. □

Lemma 11.6 (Translation differences of the fixed cutoff)**.** *For every noninteger $a > 0$ there is $C_a > 0$ such that for all $z \in \mathbb{R}$, $\eta - \tau_z \eta \in C^a(\mathbb{R})$ and $N_a(\eta - \tau_z \eta) \leq C_a \min\{|z|, 1\}$.*

Proof. For $|z| \leq 1$ use $\eta(x) - \eta(x - z) = z \int_0^1 \eta'(x - tz)dt$ and global translation bounds for η'. For $|z| > 1$ use the triangle inequality and $2N_a(\eta)$. Smooth compact support supplies finiteness of the cutoff norms in both cases. □

Lemma 11.7 (Quantitative cutoff extension from the interval)**.** *For every noninteger $a > 0$ there is $C_a > 0$ such that, for every $h \in C^a([-1,1])$, the function ηh, extended by zero outside $[-1,1]$, belongs to $C^a(\mathbb{R})$ and $N_a(\eta h) \le C_a N_a(h;[-1,1])$. Values of h outside $[-1,1]$ have no effect.*

Proof. Apply the interval product estimate. The support is separated from the endpoints by $1/8$, so extension by zero preserves the derivatives. For Holder pairs crossing outside the interval, either both derivative values vanish or their distance is at least $1/8$; use the sup bound. □

11.2 Local finiteness and the commutator

Lemma 11.8 (Near-part smoothness without bounds at infinity)**.** *Let $0 < p < 1$ and $w \in C^\infty(\mathbb{R})$, without global derivative bounds. The function $U(x) = c_p \int \chi(z)(w(x) - w(x-z))|z|^{-1-p}dz$ is smooth. For every $m \in \mathbb{N}$ its derivative is the same absolutely convergent integral with w replaced by $w^{(m)}$. On every compact x interval all differentiated integrands admit a common integrable majorant, using only the local bound for $w^{(m+1)}$ on the interval enlarged by two.*

Proof. Write $w^{(m)}(x) - w^{(m)}(x-z) = z\int_0^1 w^{(m+1)}(x - tz)dt$. The resulting majorant is a constant times $\mathbf{1}_{|z|\le 2}|z|^{-p}$. Apply dominated differentiation inductively; the point $z = 0$ is assigned value zero. □

Lemma 11.9 (Far-part smoothness by differentiating the kernel)**.** *Let $0 < p < 1$, and let w be smooth and bounded on $\mathbb{R}$. Put $\kappa_p(z) = (1 - \chi(z))|z|^{-1-p}$, with value zero near the origin. Every derivative of κ_p is integrable. The far part is*

$$c_p \left(w(x) \int \kappa_p - \int \kappa_p(x-z)w(z)dz \right).$$

It is smooth, and its convolution term can be differentiated any number of times by differentiating κ_p. Every such integral converges absolutely, locally uniformly in x.

Proof. For $|z| \ge 2$, the mth kernel derivative is bounded by $C_{p,m}|z|^{-1-p-m}$; the remaining region is compact and smooth. Use the smooth-convolution argument `c03_aux_conv_smooth`, or move that Schauder-free helper upstream. No derivatives of w at infinity enter the convolution bound. □

Lemma 11.10 (The local right-hand-side norm is finite)**.** *For $0 < p < 1$ and smooth bounded w, the singular integral defining $|D|^p w(x)$ converges absolutely for every x and is smooth on $\mathbb{R}$. In particular $|D|^p w \in C^a([-1,1])$ for every noninteger $a > 0$. This is a qualitative finiteness assertion; no bound by finitely many global derivatives of w is assumed or claimed.*

Proof. Split the integral with χ and add the two smooth functions. Smoothness on a neighborhood of a compact interval implies finiteness of each noninteger Holder norm there. □

Lemma 11.11 (Cutoff commutator identity and convergence)**.** *Let $0 < p < 1$ and w be smooth and bounded. Then all integrals in the identity*

$$|D|^p(\eta w) = \eta |D|^p w + C_{\eta,p} w$$

converge absolutely and the equality holds at every point.

Proof. Expand the difference of the two fractional kernels after putting them in the same shifted variable. The remaining numerator is $(\eta(x)-\eta(x-z))w(x-z)$, bounded by a constant times $\|w\|_\infty \min(|z|,1)$. Integrability permits splitting the integral. □

Lemma 11.12 (Global Holder bound for the commutator)**.** *For $0<p<1$ and noninteger $a>0$ there is $C_{p,a}>0$ such that every smooth $w\in C^a(\mathbb{R})$ satisfies $C_{\eta,p}w\in C^a(\mathbb{R})$ and $N_a(C_{\eta,p}w)\le C_{p,a}N_a(w)$.*

Proof. The C^a norm of $(\eta-\tau_z\eta)\tau_z w$ is bounded by $C_a\min(|z|,1)N_a(w)$. All required mixed derivatives are measurable. Integrate this majorant against $|z|^{-1-p}$; its scalar integral is $2/(1-p)+2/p$. This is precisely where $p<1$ is used. □

Lemma 11.13 (Quantitative localized right-hand side)**.** *For $0<p<1$ and noninteger $\beta>0$ there is $C_{p,\beta}>0$ such that, for smooth $w\in C^\beta(\mathbb{R})$, $v=\eta w$ is smooth and compactly supported and $h=|D|^p v$ belongs to $C^\beta(\mathbb{R})$ with*

$$N_\beta(h)\le C_{p,\beta}\big(N_\beta(w)+N_\beta(|D|^p w;[-1,1])\big),\qquad \|v\|_\infty\le N_\beta(w).$$

Proof. The local norm is finite by the first lemma. Apply quantitative cutoff extension to $|D|^p w$ and add the commutator estimate. The fixed cutoff satisfies $0\le\eta\le 1$. □

11.3 Annular kernels, cancellation and scaling

Lemma 11.14 (Finite dyadic telescoping)**.** *For every $n\in\mathbb{N}$ and $\xi\in\mathbb{R}$,*

$$\chi(\xi)+\sum_{j=0}^{n-1}\varphi(2^{-j}\xi)=\chi(2^{-n}\xi).$$

The right side tends to one as $n\to\infty$ and is bounded in absolute value by one.

Proof. Telescope the defining difference. For fixed ξ, eventually $|2^{-n}\xi|\le 1$. □

Lemma 11.15 (Real Schwartz inverse kernels)**.** *For $0<p<1$, the inverse transforms of q_p and χ are real Schwartz functions. Their Fourier transforms are respectively q_p and χ, in the project's convention. In particular, K_p and every derivative are integrable; for every $m\in\mathbb{N}$ and $a\ge 0$, $\int |z|^a|K_p^{(m)}(z)|dz<\infty$.*

Proof. Bundle smooth compactly supported symbols using `HasCompactSupport.toSchwartzMap`. Apply Schwartz Fourier inversion with the 2π normalization bridge. Reality follows from even real symbols. Schwartz decay with an integer weight greater than $a+1$ proves the weighted integrability. □

Lemma 11.16 (Vanishing polynomial moments)**.** *For $0<p<1$ and all integers $r,m\ge 0$, the moment integral converges absolutely and*

$$\int_{\mathbb{R}} z^r K_p^{(m)}(z)dz=0.$$

Proof. The transform of $K_p^{(m)}$ is $(i\xi)^m q_p(\xi)$, whose derivatives all vanish at zero. Apply the Fourier differentiation formula for Schwartz functions and evaluate there. Record the project's normalization explicitly; the zero conclusion is independent of its nonzero factor. □

Lemma 11.17 (Exact dyadic scaling and weighted norms)**.** *For $0 < p < 1$, $j, m \in \mathbb{N}$, and $a \geq 0$,*

$$K_{p,j}^{(m)}(x) = 2^{(m+1-p)j} K_p^{(m)}(2^j x), \qquad \widehat{K}_{p,j}(\xi) = \varphi(2^{-j}\xi)|\xi|^{-p},$$

where the symbol at zero is defined to be zero. Moreover,

$$\int |z|^a |K_{p,j}^{(m)}(z)|dz = 2^{(m-p-a)j} \int |z|^a |K_p^{(m)}(z)|dz, \qquad \int z^r K_{p,j}^{(m)}(z)dz = 0 \quad (r \in \mathbb{N}).$$

Proof. Differentiate the scaled kernel and substitute $u = 2^j z$ in the integrals. For the Fourier transform, $2^{-pj} q_p(2^{-j}\xi) = \varphi(2^{-j}\xi)|\xi|^{-p}$. This scaling factor is essential to the gain of p derivatives. □

11.4 Factorization and block estimates

Lemma 11.18 (Compact-input fractional Fourier bridge)**.** *Let $0 < p < 1$ and $v \in C_c^\infty(\mathbb{R};\mathbb{R})$. The function $m(\xi) = |\xi|^p \widehat{v}(\xi)$ is integrable and $|D|^p v(x) = \check{m}(x)$ pointwise, with absolute convergence on both sides.*

Proof. Apply the Schwartz multiplier identity of Lemma 3.25; it is valid for all $0 < p < 1$. The Fourier transform of v is Schwartz, so multiplication by $|\xi|^p$ preserves integrability. Use the real-to-complex bridge for the displayed identity. □

Lemma 11.19 (Dyadic block factorization)**.** *For $0 < p < 1$, $v \in C_c^\infty(\mathbb{R};\mathbb{R})$, $h = |D|^p v$, and every $j \in \mathbb{N}$,*

$$\Delta_j v(x) = (K_{p,j} * h)(x) \qquad (x \in \mathbb{R}).$$

The convolution and the double integral used in this identity are absolutely convergent.

Proof. Substitute the inverse-transform expression for h. Fubini follows from $\|K_{p,j}\|_1 \||\xi|^p \widehat{v}\|_1 < \infty$. The two symbols multiply to $\varphi(2^{-j}\xi)$, also at zero. This proof does not require a general distributional transform theorem for h. □

Lemma 11.20 (Global Holder Taylor remainder)**.** *Let $\beta = k + \theta > 0$, $k \in \mathbb{N}$, $0 < \theta < 1$. There is $C_\beta > 0$ such that for every $h \in C^\beta(\mathbb{R})$ and $x, z \in \mathbb{R}$,*

$$\left| h(x - z) - \sum_{r=0}^{k} \frac{h^{(r)}(x)(-z)^r}{r!} \right| \leq C_\beta N_\beta(h)|z|^\beta.$$

Proof. For $k = 0$ use the Holder inequality. For $k \geq 1$, use Taylor's integral formula through order $k - 1$, subtract the constant kth derivative inside its remainder, and use its Holder bound. The estimate is global and works for both signs of z. □

Lemma 11.21 (Cancellation estimate for differentiated blocks)**.** *For $0 < p < 1$, noninteger $\beta > 0$, and every $m \in \mathbb{N}$, there is $C_{p,\beta,m} > 0$ such that for every $h \in C^\beta(\mathbb{R})$ and $j \in \mathbb{N}$, $K_{p,j} * h$ is smooth, its mth derivative is the absolutely convergent convolution $K_{p,j}^{(m)} * h$, and*

$$\|\partial^m (K_{p,j} * h)\|_\infty \le C_{p,\beta,m} 2^{(m-\beta-p)j} N_\beta(h).$$

Proof. Differentiate the Schwartz kernel, using boundedness of h and local translation majorants. Subtract the Taylor polynomial of $h(x-z)$ under the integral; every polynomial term vanishes by the kernel moments. Bound the remainder and use the exact weighted scaling. No additional derivatives of h are required. □

Lemma 11.22 (Low-frequency bounds from the value norm)**.** *For every $m \in \mathbb{N}$ there is $C_m > 0$ such that, for every real $v \in C_c^\infty(\mathbb{R})$, $S_0 v = \chi^\vee * v$ is smooth and $\|\partial^m S_0 v\|_\infty \le C_m \|v\|_\infty$. Every differentiated convolution is absolutely convergent.*

Proof. Fubini identifies the inverse-transform definition with convolution. Differentiate the Schwartz kernel and use its derivative L^1 norm.

□

11.5 Reconstruction of the higher Holder norm

Lemma 11.23 (Uniform reconstruction, including differentiated partial sums)**.** *Let $v \in C_c^\infty(\mathbb{R};\mathbb{R})$ and $P_n v = S_0 v + \sum_{j=0}^{n-1} \Delta_j v$. For every integer $m \ge 0$, $\partial^m P_n v \to \partial^m v$ uniformly on $\mathbb{R}$ as $n \to \infty$.*

Proof. The differentiated partial sum is the real inverse transform of $(i\xi)^m \chi(2^{-n}\xi)\widehat{v}(\xi)$. Dominated convergence in L^1_ξ follows from the Schwartz decay of $\widehat{v}$. The inverse-transform $L^1 \to L^\infty$ bound and Fourier inversion give the conclusion. This compact-input argument avoids building a general theorem on differentiating infinite series. □

Lemma 11.24 (The scalar dyadic increment estimate)**.** *For every $0 < \theta < 1$ there is $C_\theta > 0$ such that for every $d \ge 0$ and $n \in \mathbb{N}$,*

$$\sum_{j=0}^{n-1} \min\{2^{-\theta j}, d\, 2^{(1-\theta)j}\} \le C_\theta d^\theta.$$

For every $a > 0$, $\sum_{j \ge 0} 2^{-aj}$ converges, with sum $(1 - 2^{-a})^{-1}$.

Proof. If $d = 0$ the first sum is zero; if $d \ge 1$ use the decreasing geometric series. For $0 < d < 1$, choose $J \in \mathbb{N}$ with $2^J d \le 1 < 2^{J+1} d$. Sum the increasing geometric expression for $j \le J$ and the decreasing one for $j > J$. Both are bounded by a constant times d^θ, uniformly in the partial-sum cutoff. □

Lemma 11.25 (Compact-input global Schauder estimate)**.** *Let $0 < p < 1$, $\beta > 0$, and assume β and $s = \beta + p$ are nonintegers. There is $C_{p,\beta} > 0$ such that, for every real $v \in C_c^\infty(\mathbb{R})$ with $h = |D|^p v \in C^\beta(\mathbb{R})$,*

$$v \in C^{\beta+p}(\mathbb{R}), \qquad N_{\beta+p}(v) \le C_{p,\beta}\big(\|v\|_\infty + N_\beta(h)\big).$$

Proof. Put $k = \lfloor s \rfloor$, $\theta = s - k$. For $0 \le m \le k$, sum the block bounds $2^{(m-s)j}$ and pass to the uniformly reconstructed derivative. For the kth derivative increment, each block is bounded by a constant times $N_\beta(h) \min(2^{-\theta j}, |x-y| 2^{(1-\theta)j})$. Use the scalar increment lemma and pass from finite sums to $v^{(k)}$ by uniform convergence. The low block is controlled by its kth and $(k+1)$st derivative bounds, splitting $|x-y| \le 1$ from $|x-y| \ge 1$. Assemble the finite Holder norm with the norm constructors. The $(k+1)$st derivative block estimate is used only for increments: its infinite series is not claimed to be summable. □

11.6 Assembly of the estimate

For Lemma 3.26, take $v = \eta w$. Lemma 11.13 supplies the global C^β norm of $|D|^p v$ and the value bound for v. Apply Lemma 11.25, then restrict to $[-1/2, 1/2]$ using Lemma 11.3. Since $\eta = 1$ on a neighborhood of this interval, the restriction agrees with w, including its within derivatives. Absorb the two fixed constants.

Remark 11.26 (Lean modules). The declarations of this chapter are formalized in `GSQG/SchauderEstimate/*.lean`. The Schauder estimate Lemma 3.26 and the smoothness bootstrap are in `GSQG/Sec4_Analysis/Smoothness.lean`.

Chapter 12

Beyond the paper: derivation from the 2D patch equation

This chapter derives the graph equation from a two-dimensional weak patch system and proves its converse for the smooth unbounded graphs used in Theorem 1.31. All declarations of this chapter are formalized in Lean. The scope is weak patch solutions with smooth interfaces before the corner time T: no classical vector velocity on the interface and no continuation beyond T is asserted. For background on contour dynamics and the divergent tangential trace see [HS18, Section 3.1]; their parameter equals $2-\alpha$ in the convention used here. The rotation, density orientation and time normalization are fixed explicitly below.

We work with $1<\alpha<2$ and with a positive kernel coefficient c until the normalization in Section 12.1. This includes $\alpha = p+1 \in (7/4, 2)$ in the application. The space $E=\mathbb{R}^2$ has its Euclidean norm and Lebesgue measure. In Lean, use `EuclideanSpace R (Fin 2)` and explicit volume-preserving coordinate maps; the default norm of a product of two real normed spaces is not the Euclidean radius. Write $J(x_1,x_2)=(-x_2,x_1)$, so $\nabla^{\perp}=J\nabla$. Spatial and spacetime test functions are smooth and compactly supported in their indicated open domains. Constants in quantitative statements are chosen before the variables in which uniformity is asserted.

12.1 The two-dimensional operator and its normalization

Definition 12.1 (Kernel, adjoint cancellation and renormalized potential)**.** Let $E=\mathbb{R}^2$ with the Euclidean norm, and write $(x,y)\in E$ for the point with coordinates x,y. For a vector field $Q: E\to E$ put $\operatorname{div} Q=\partial_1 Q_1+\partial_2 Q_2$. For $z\neq 0$ set

$$P_{\alpha,c}(z)=c|z|^{-\alpha}, \qquad K_{\alpha,c}(z)=\alpha c\, Jz\,|z|^{-\alpha-2}=-J\nabla P_{\alpha,c}(z),$$

and set both kernels to zero at the single point zero. For a vector test function $\Phi : E \to E$ define

$$B^*_{\alpha,c}\Phi(Y) = \int_E K_{\alpha,c}(z) \cdot (\Phi(Y+z) - \Phi(Y))\, dz.$$

For a bounded measurable real density θ define

$$V_{\alpha,c}[\theta](X) = \int_E [P_{\alpha,c}(X-Y) - \mathbf{1}_{|Y|>1} P_{\alpha,c}(-Y)]\theta(Y)\, dY.$$

These are total formulas; the next lemmas establish their analytic meanings. The unsubtracted integral $P * \theta$ is not asserted to converge for an infinite patch.

Definition 12.2 (Canonical and front normalizations)**.** Put

$$A_\alpha = \frac{2^{2-\alpha}\pi\Gamma(1-\alpha/2)}{\Gamma(\alpha/2)}, \qquad c_{\mathrm{std}} = A_\alpha^{-1}, \qquad c_{\mathrm{front}} = \mathrm{calpha}(\alpha{-}1), \qquad \kappa_\alpha = \frac{c_{\mathrm{front}}}{c_{\mathrm{std}}}.$$

The two-dimensional Fourier convention is $\widehat{f}(\xi) = \int_E f(X) e^{-iX\cdot\xi} dX$, with inverse factor $(2\pi)^{-2}$. The canonical operator $-\nabla^\perp(-\Delta)^{-1+\alpha/2}$ has multiplier $-iJ\xi|\xi|^{\alpha-2}$, given value zero at $\xi = 0$. Identification of the real-space operator with this multiplier is a theorem below. The two coefficients are kept distinct; the final application records the corresponding time change.

Lemma 12.3 (Euclidean coordinates and radial majorants)**.** *Coordinate identification $E \leftrightarrow \mathbb{R} \times \mathbb{R}$ preserves Lebesgue measure, and $|z|^2 = z_1^2 + z_2^2$. For $r > 0$,*

$$\int_{|z|<r} |z|^{-\alpha} dz = \frac{2\pi}{2-\alpha} r^{2-\alpha}, \qquad \int_{|z|>r} |z|^{-\alpha-1} dz = \frac{2\pi}{\alpha-1} r^{1-\alpha}.$$

The displayed nonnegative integrands are integrable on their respective domains. Consequently there is $C_{\alpha,c} > 0$ such that for every $\xi \in E$,

$$\int_E |K_{\alpha,c}(z)|\, |e^{-iz\cdot\xi} - 1| dz \le C_{\alpha,c} |\xi|^{\alpha-1},$$

with the integral zero when $\xi = 0$.

Proof. Use the volume-preserving Euclidean-to-coordinate map and radial integration in dimension two. For the last bound split at $r = |\xi|^{-1}$ and use $|e^{-is} - 1| \le \min(|s|, 2)$. The cases $\alpha > 1$ and $\alpha < 2$ control different ends of the integral. □

Lemma 12.4 (L^1 translation estimate for test fields)**.** *For every vector test function $\Phi : E \to E$ and every $z \in E$,*

$$\int_E |\Phi(Y+z) - \Phi(Y)| dY \le \min\left(|z| \int_E \|D\Phi(Y)\| dY,\ 2 \int_E |\Phi(Y)| dY \right).$$

All the displayed integrals are finite.

Proof. Apply segment FTC to $\Phi(Y + tz)$, integrate its norm, and use Fubini and translation invariance. The second bound is the triangle inequality. This is a finite-dimensional test-function lemma. □

Lemma 12.5 (Absolute adjoint pairing)**.** *For every vector test Φ, the function $(Y,z) \mapsto K_{\alpha,c}(z) \cdot (\Phi(Y+z) - \Phi(Y))$ is absolutely integrable on $E \times E$. The inner integral defining $B^*\Phi(Y)$ converges absolutely for every Y, and*

$$\|B^*\Phi\|_{L^1(E)} \le C_{\alpha,c}(\|\Phi\|_{L^1(E)} + \|D\Phi\|_{L^1(E)}).$$

Thus $\int \theta B^*\Phi$ is absolutely convergent for every bounded measurable θ.

Proof. Use Tonelli for the norm majorant and split $|z| \le 1$ from its complement. Pointwise convergence uses the bounded derivative of the test field near zero and its bounded values at infinity. □

Lemma 12.6 (Test potential and adjoint divergence identity)**.** *For a vector test Φ, let $A_\Phi(Y) = \int_E P_{\alpha,c}(X-Y)\Phi(X)dX$. This integral is absolutely convergent, A_Φ is C^1, and*

$$B^*\Phi = -\operatorname{div}(JA_\Phi).$$

If $\operatorname{supp}\Phi \subset \{|X| \le L\}$, then for $|Y| > 2L+1$, $|A_\Phi(Y)| \le C_{\alpha,c}|Y|^{-\alpha}\|\Phi\|_1$. In particular $B^*(\nabla\zeta) = 0$ for every scalar test ζ.

Proof. Differentiate the compact test factor after shifting variables. Integration by parts in the locally integrable potential kernel gives the adjoint identity; retain the cancellation at zero. For a gradient test, the scalar curl is zero by equality of mixed derivatives. Lean route: A_Φ is the convolution $P * \Phi$ (P even), differentiated on the compact test factor; the adjoint identity $B^*\Phi(Y) = \int P(z)\operatorname{curl}\Phi(Y+z)dz$ is proved for the regularized kernel $P_\delta = c(|z|^2+\delta^2)^{-\alpha/2}$ by whole-space integration by parts and oddness of K_δ, then $\delta \to 0$ by dominated convergence, keeping the cancellation $\Phi(Y+z)-\Phi(Y)$ until the limit. □

Lemma 12.7 (Absolute convergence of the renormalized potential)**.** *For every $R \ge 1$ there is $C_{\alpha,c,R} > 0$ such that for every bounded measurable θ with $|\theta| \le M$, $M \ge 0$, and every $|X| \le R$,*

$$\int_E |P(X-Y) - \mathbf{1}_{|Y|>1}P(-Y)|\,|\theta(Y)|dY \le C_{\alpha,c,R}M.$$

The integral defining $V[\theta]$ is therefore absolutely convergent at every X, measurable and locally bounded. Changing the subtraction radius to another positive radius changes it by a finite constant independent of X.

Proof. On $|Y| > 2R+2$, use the mean-value estimate $|P(X-Y)-P(-Y)| \le C_{\alpha,c}R|Y|^{-\alpha-1}$. In the remaining bounded region integrate the translated singular kernel using its radial integral. Measurability follows from parameter integration. The difference of two subtraction choices is an integral over a bounded annulus and contains no X. □

Lemma 12.8 (Renormalized potential and its distributional gradient)**.** *For every bounded measurable θ, the defining integral for $V_{\alpha,c}[\theta](X)$ converges absolutely for every X and defines a locally bounded measurable function. For every vector test Φ,*

$$\int_E \theta B^*\Phi = -\int_E V_{\alpha,c}[\theta]\,\operatorname{div}(J\Phi).$$

Both sides converge absolutely. This is $B_{\alpha,c}\theta = -J\nabla V_{\alpha,c}[\theta]$ in distributions. Replacing the subtraction radius 1 by another positive radius changes V by a finite constant independent of X and leaves the velocity unchanged.

Proof. For $|Y| > 2|X|+2$, the subtracted kernel is bounded by $C_{\alpha,c}|X||Y|^{-\alpha-1}$. Near $Y = X$ use local integrability of $|X - Y|^{-\alpha}$. Establish local bounds by integrating the singular kernel, not by taking a pointwise supremum over its moving pole. Integrate by parts in the kernel with a small-radius cutoff; its error is $O(r^{2-\alpha})$. Keep the subtraction until Fubini is justified. Its contribution against $\operatorname{div}(J\Phi)$ then vanishes because the integral of a compactly supported divergence is zero. □

Lemma 12.9 (Gamma layer representation of the singular kernel)**.** *For $z \neq 0$,*

$$|z|^{-\alpha} = \frac{1}{\Gamma(\alpha/2)} \int_0^\infty s^{\alpha/2-1} e^{-s|z|^2} ds, \qquad K_{\alpha,c}(z) = \frac{2c}{\Gamma(\alpha/2)} Jz \int_0^\infty s^{\alpha/2} e^{-s|z|^2} ds.$$

Both scalar parameter integrals converge absolutely. For each $\xi \in E$, multiplying the second representation by $e^{-iz\cdot\xi} - 1$ yields an absolutely integrable double integrand in $(s, z) \in (0, \infty) \times E$.

Proof. Substitute $u = s|z|^2$ in the Gamma integral and use $\Gamma(1 + \alpha/2) = (\alpha/2)\Gamma(\alpha/2)$. Tonelli on the norm reduces double-integrability to the radial cancellation majorant. □

Lemma 12.10 (Two-dimensional Gaussian transform and first moment)**.** *For every $s > 0$ and $\xi \in E$, the following integrals converge absolutely:*

$$\int_E e^{-s|z|^2 - iz\cdot\xi} dz = \frac{\pi}{s} e^{-|\xi|^2/(4s)}, \qquad \int_E z e^{-s|z|^2 - iz\cdot\xi} dz = -\frac{i\pi}{2s^2} \xi e^{-|\xi|^2/(4s)}, \qquad \int_E z e^{-s|z|^2} dz = 0.$$

The vector formulas are interpreted componentwise after complexification.

Proof. Apply the inner-product-space Gaussian Fourier integral in dimension two with phase coefficient $-i$, or coordinate Fubini and the one-dimensional formula. Differentiate in ξ under the integral, with a Gaussian moment majorant, for the first moment. Oddness gives the final identity. Keep the $e^{-iz\cdot\xi}$ convention explicit. □

Lemma 12.11 (Inverse-parameter Gamma integral)**.** *For every $a, b > 0$, the integrand is integrable on $(0, \infty)$ and*

$$\int_0^\infty s^{-1-a} e^{-b/s} ds = b^{-a}\Gamma(a).$$

Proof. Use the decreasing substitution $u = b/s$, including the absolute Jacobian. Gamma integrability proves convergence at both endpoints before the change of variables. □

Lemma 12.12 (Gaussian evaluation of the velocity symbol)**.** *For every $\xi \in E$, the vector integral is absolutely convergent and*

$$\int_E K_{\alpha,c}(z)(e^{-iz\cdot\xi} - 1) dz = -icA_\alpha J\xi|\xi|^{\alpha-2},$$

where the right side is zero at $\xi = 0$. In particular $A_\alpha, c_{\mathrm{std}}, \kappa_\alpha > 0$.

Proof. Use the Gamma representation $K(z) = \frac{2c}{\Gamma(\alpha/2)} Jz \int_0^\infty s^{\alpha/2} e^{-s|z|^2} ds$. Absolute Fubini follows from the radial cancellation majorant before evaluating the Gaussian. The two-dimensional Gaussian first moment is $\int z e^{-s|z|^2 - iz\cdot\xi} dz = -i\pi\xi(2s^2)^{-1} e^{-|\xi|^2/(4s)}$. Substitution $u = |\xi|^2/(4s)$ gives A_α. Handle $\xi = 0$ separately. Gamma positivity gives $c_{\mathrm{std}} > 0$; use the already proved positivity of `calpha` for the ratio. □

Lemma 12.13 (Identification with the canonical two-dimensional operator)**.** *For every real Schwartz function f on E, the integral $B_{\alpha,c} f(X) = \int K(z)(f(X - z) - f(X))dz$ converges absolutely, belongs to $L^1(E; E)$, and has Fourier multiplier $-icA_\alpha J\xi|\xi|^{\alpha-2}$, componentwise. Equivalently, $B_{\alpha,c} f = -cA_\alpha \nabla^\perp (-\Delta)^{-1+\alpha/2} f$. For bounded densities the adjoint pairing and renormalized potential give the same distributional extension.*

Proof. Fourier-transform the absolutely integrable difference kernel using Fubini, or insert Schwartz inversion and the symbol integral. The bound $C|\xi|^{\alpha-1}$ justifies the latter exchange. Differentiation of the inverse transform of $|\xi|^{\alpha-2}\widehat{f}$ is legitimate by absolute integrability. Duality is meaningful for bounded densities because $B^*\Phi \in L^1$; no divergent unsubtracted potential or undefined product of tempered distributions is used. □

12.2 Graph geometry and finite-domain calculus

Definition 12.14 (Admissible graphs, patches and canonical representatives)**.** Let I be a nonempty open time interval. A graph $F : \mathbb{R} \times I \to \mathbb{R}$ is admissible if it is jointly smooth on $\mathbb{R} \times I$ (no smoothness is required at the endpoints of I or outside $\mathbb{R} \times I$) and for every compact interval $J_0 \subset I$ there is $M_{J_0} \geq 0$ with $|F_x(x,t)| \leq M_{J_0}$ for all $x \in \mathbb{R}, t \in J_0$. No global bound on F, F_{xx} or time derivatives at spatial infinity is assumed. Put $\gamma_t(s) = (s, F(s,t))$, $\tau_t(s) = (1, F_x(s,t))$, $\Omega_t = \{(x,y) : y < F(x,t)\}$, and $\theta_F = \mathbf{1}_{\Omega_t}$. Define, for $X \notin \Gamma_t = \gamma_t(\mathbb{R})$,

$$U_{\alpha,c}[F](X,t) = -c \int_{\mathbb{R}} \tau_t(s)|X - \gamma_t(s)|^{-\alpha} ds,$$

and give U value zero on Γ_t. This candidate boundary representation is not assumed to satisfy the two-dimensional velocity law; that is proved in Section 12.4. Define separately

$$W_{\alpha,c}[F](x,t) = c \int_{\mathbb{R}} [F_x(x,t) - F_x(s,t)]|\gamma_t(x) - \gamma_t(s)|^{-\alpha} ds.$$

The graph normal is $(-F_x, 1)/\sqrt{1 + F_x^2}$; W is its normal velocity multiplied by $\sqrt{1 + F_x^2}$. It is not the value of the arbitrarily extended vector field on the graph.

Lemma 12.15 (Measurability, null graphs and a distance bound)**.** *Each patch is measurable, each graph has two-dimensional measure zero, and the spacetime graph has three-dimensional measure zero. For a fixed-time slope bound $|F_x| \leq M$, let $h = s - x$ and $\delta = y - F(x,t)$. Then*

$$h^2 + \delta^2 \leq 2(1 + M^2)\big(h^2 + (y - F(s,t))^2\big).$$

The same constants work on a compact time interval with its common slope bound.

Proof. Openness of the subgraph follows from continuity; Fubini makes each vertical graph section a null singleton. Use $|F(s,t) - F(x,t)| \le M|h|$ and the squared triangle inequality. □

Lemma 12.16 (Finite graph-domain flux formula)**.** *Let $R > 0$, $F \in C^1(\mathbb{R})$, $H > 0$ with $-H < F(s)$ for $|s| \le R$, and $Q : E \to E$ be C^1 on a neighborhood of the closed domain $D = \{(s,y) : |s| < R, -H < y < F(s)\}$. Then*

$$\int_D \operatorname{div} Q = \int_{-R}^{R} (Q_2 - F'Q_1)(s, F(s))ds - \int_{-R}^{R} Q_2(s,-H)ds + \int_{-H}^{F(R)} Q_1(R,y)dy - \int_{-H}^{F(-R)} Q_1(-R,y)dy.$$

Every integral is absolutely convergent.

Proof. Coordinate Fubini and vertical FTC give the Q_2 terms. Differentiate $s \mapsto \int_{-H}^{F(s)} Q_1(s,y)dy$, then use one-dimensional FTC. This is the graph-specific substitute for a general Gauss theorem. Lean route: for the x-derivative term substitute $y = (F(s) + H)\tau - H$, which maps the domain to the rectangle $[-R,R] \times [0,1]$, then use Fubini there and FTC in s and in τ. □

Lemma 12.17 (Flux formula for an unbounded subgraph and a compact test field)**.** *For a C^1 graph with bounded slope and a compactly supported C^1 vector field Q,*

$$\int_{y<F(x)} \operatorname{div} Q(X)dX = \int_{\mathbb{R}} (Q_2 - F'Q_1)(x, F(x))dx,$$

with absolute convergence.

Proof. Use $H_R = |F(0)| + (M+1)R + 1$. Once R is large enough, the artificial sides lie outside the support of Q. Pass to the limit in the compactly supported volume integral. □

12.3 The candidate velocity as an actual function

Lemma 12.18 (Off-graph velocity convergence and size)**.** *For $|F_x| \le M$ the contour integral for $U(X,t)$ converges absolutely at every $X \notin \Gamma_t$. There is $C_{\alpha,c,M} > 0$, independent of F, X, such that*

$$|U((x,y),t)| \le C_{\alpha,c,M}|y - F(x,t)|^{1-\alpha} \quad (y \ne F(x,t)).$$

Proof. Bound $|\tau|$ by $\sqrt{1+M^2}$ and compare the denominator with $h^2 + \delta^2$, using $|F(s,t) - F(x,t)| \le M|s-x|$ directly. The one-dimensional integral of $(h^2+\delta^2)^{-\alpha/2}$ scales as $|\delta|^{1-\alpha}$ and converges because $\alpha > 1$. □

Lemma 12.19 (Local integrability and joint contour-test Fubini)**.** *For an admissible graph, U is jointly measurable and locally integrable on $E \times I$. For a compact spatial set Q and compact time interval $J_0 \subset I$, $\int_{J_0}\int_Q |U| < \infty$. For every compactly supported spatial test Φ, the integrand $|\tau_t(s)||X - \gamma_t(s)|^{-\alpha}|\Phi(X)|$ is integrable in (s, X); its bound is uniform in $t \in J_0$ for tests supported in a fixed compact set with bounded sup norm.*

Proof. Integrate $|y - F(x,t)|^{1-\alpha}$ over bounded vertical intervals; the bound is uniform in the center $F(x,t)$ since $1 - \alpha > -1$. For joint contour-test integrability, use Tonelli: bounded s uses local two-dimensional integrability of $|X - \gamma(s)|^{-\alpha}$; large $|s|$ uses horizontal separation and the integrable tail $|s|^{-\alpha}$. Lean route: the vertical estimate and the graph nullness are proved directly in coordinates (Tonelli in `lintegral`); joint measurability on $E \times I$ uses a piecewise modification, since F is not controlled for $t \notin I$. Do not replace this Tonelli argument by a pointwise integral assigned zero on the singular graph. □

Lemma 12.20 (Spatial smoothness and divergence-free velocity)**.** *For each fixed time, U is spatially C^1 on $E \setminus \Gamma_t$ and $\operatorname{div}_X U = 0$ there. Its first spatial derivatives are jointly continuous in (X, t) off the spacetime graph. No time differentiability of U is claimed from the present hypotheses.*

Proof. On off-graph compact subsets the vertical gap is bounded away from zero and the global slope is bounded uniformly in time. The kernel and its first spatial derivative have tails bounded by $C|s|^{-\alpha}$ and $C|s|^{-\alpha-1}$, respectively. For divergence, use $\operatorname{div} U = c\int \partial_s |X - \gamma_t(s)|^{-\alpha} ds$. FTC on $[-R, R]$ and horizontal separation make both endpoint terms vanish. □

12.4 Identification of the candidate with the two-dimensional velocity law

Lemma 12.21 (Artificial-boundary errors vanish)**.** *Fix a time and $|F'| \le M$. Put $H_R = |F(0)| + (M+1)R + 1$ and $D_R = \{|x| < R, -H_R < y < F(x)\}$. For every vector test Φ, the sum of the three artificial-boundary contributions in the finite graph flux formula for $Q = JA_\Phi$ has modulus at most $C_{\alpha,c,M,F(0),\Phi} R^{1-\alpha}$ for all sufficiently large R. The constant and lower threshold for R are chosen independently of R.*

Proof. Each artificial side is at distance at least $R/2$ from the test support once R is large. Its length is at most $C_{M,F(0)}R$. Multiply this length by the $R^{-\alpha}$ potential bound. Lean route: that far-field bound, $|A_\Phi(Y)| \le |c|2^\alpha |Y|^{-\alpha}\|\Phi\|_1$ for $|Y| \ge 2L$, is proved directly from $|X - Y| \ge |Y|/2$ on the test support, without citing the earlier lemmas. The exponent $1 - \alpha < 0$ is essential here. □

Lemma 12.22 (Bulk-to-contour velocity identity)**.** *For a fixed smooth bounded-slope graph, every vector test Φ satisfies*

$$\int_E \theta_F(Y) B^*\Phi(Y) dY = -\int_{\mathbb{R}} A_\Phi(\gamma(s)) \cdot \tau(s) ds = \int_E U[F](X) \cdot \Phi(X) dX.$$

All integrals are absolutely convergent. Hence the candidate contour velocity is a locally integrable representative of the independently defined two-dimensional operator $B_{\alpha,c}\theta_F$.

Proof. Apply the finite flux formula to JA_Φ on D_R and use $B^*\Phi = -\operatorname{div}(JA_\Phi)$. On the graph, $n\,ds = (-F', 1)ds = J\tau\,ds$, so $(JA_\Phi)\cdot n\,ds = A_\Phi\cdot\tau\,ds$. The outer minus sign gives the negative contour velocity. The artificial sides vanish by the preceding lemma; dominated convergence applies to $\theta_{D_R}B^*\Phi$, which is dominated by $|B^*\Phi| \in L^1$. The graph is null, so the value of U on it is irrelevant. Joint contour-test Fubini gives the second equality. □

Lemma 12.23 (Bulk principal values agree with the weak velocity)**.** *Set $u_\varepsilon(X) = \int_{|z|>\varepsilon} K(z)\theta_F(X-z)dz$ for $\varepsilon > 0$. Each integral is absolutely convergent. Off the graph, the limit as $\varepsilon \downarrow 0$ exists and equals the absolutely convergent desingularized integral $\int K(z)(\theta_F(X-z) - \theta_F(X))dz$. The functions u_ε converge in $L^1_{\mathrm{loc}}(E)$ to $U[F]$; the desingularized expression and $U[F]$ therefore agree a.e. No boundary vector principal value is asserted.*

Proof. Oddness gives $\int_{|z|>\varepsilon} K(z)dz = 0$. Off the graph the density is constant in a ball of radius comparable to the vertical gap, so the desingularized integrand vanishes there. Uniformly in ε, its norm is bounded by $C|y - F(x)|^{1-\alpha}$, an L^1_{loc} majorant: it is bounded by a multiple of the contour majorant $\int |\tau||X - \gamma(s)|^{-\alpha}ds$, which is locally integrable by joint contour-test Fubini. The limit represents $B\theta_F$ by the adjoint pairing and thus equals U a.e. Use the locally-integrable test-function uniqueness theorem, not an unproved pointwise trace claim. □

12.5 The finite normal trace

Lemma 12.24 (Normal-speed convergence and continuity)**.** *The integral defining $W[F](x,t)$ converges absolutely and W is continuous on $\mathbb{R}\times I$. For $|x| \le R$, $t \in J_0 \Subset I$, a bound for the integrand in the variable $h = s - x$ is*

$$cL_{R,J_0}|h|^{1-\alpha}\mathbf{1}_{0<|h|\le 1} + 2cM_{J_0}|h|^{-\alpha}\mathbf{1}_{|h|>1},$$

where L_{R,J_0} bounds $|F_{xx}|$ on $[-R-1, R+1]\times J_0$.

Proof. Use the mean-value bound for the numerator near $s = x$ and the global slope bound in the tail. The denominator is at least $|s-x|^\alpha$. Shift to $h = s - x$ before dominated convergence. □

Lemma 12.25 (Two-sided vertical normal trace with an error estimate)**.** *For $d \neq 0$ put $W_d(x,t) = U_2(x, F(x,t)+d, t) - F_x(x,t)U_1(x, F(x,t)+d, t)$. For every $R > 0$, $J_0 \Subset I$ there is $C = C(\alpha, c, R, J_0, F) > 0$ such that*

$$\sup_{|x|\le R,\, t\in J_0} |W_d(x,t) - W(x,t)| \le C|d|^{2-\alpha} \quad (0 < |d| \le 1).$$

Thus both vertical normal traces exist and agree with W.

Proof. Combine the two velocity components inside their absolutely convergent off-graph integrals. The resulting numerator is $c(F_x(x,t) - F_x(x+h,t))$. Split $|h|$ into $(0,|d|)$, $(|d|,1)$ and $(1,\infty)$. Use the near majorant in the first part and the kernel derivative bound $|\partial_d(h^2 + (F(x,t)+d-F(x+h,t))^2)^{-\alpha/2}| \le C_\alpha|h|^{-\alpha-1}$ in the other two. The bounds involve only the same compact-time slope and local curvature constants. Lean route: the convergence of the limit integral W is re-derived from the same pointwise majorant, after the substitution $h = |d|r$, so the separate continuity lemma for W is not needed here. □

Lemma 12.26 (Flat-graph sign and trace check)**.** *For $F = 0$ and $d \neq 0$,*

$$U(x,d) = -c\,C_\alpha |d|^{1-\alpha}(1,0), \qquad C_\alpha = \int_{\mathbb{R}} (1+s^2)^{-\alpha/2} ds \in (0,\infty), \qquad W = 0.$$

In particular the tangential velocity diverges in magnitude at the boundary while its normal component is zero. Assigning any values to U on the null graph leaves its weak equations unchanged.

Proof. Substitute $s = x + |d|r$ in the velocity integral. Positivity and tail integrability give $C_\alpha > 0$. Use nullness of the graph to justify representative independence. □

12.6 Transport and equivalence with the front equation

Definition 12.27 (Weak patch system with coefficient c)**.** A scalar (respectively vector) spacetime test on $E \times I$ is a smooth compactly supported function $E \times \mathbb{R} \to \mathbb{R}$ (respectively $E \times \mathbb{R} \to E$) whose support lies in $E \times I$. A bounded measurable density $\theta : E \times I \to \mathbb{R}$ and a measurable $u \in L^1_{\mathrm{loc}}(E \times I; E)$ obey the coefficient-c velocity law if, for every vector spacetime test Φ,

$$\int_I \int_E u(X,t) \cdot \Phi(X,t) dX dt = \int_I \int_E \theta(Y,t) B^*_{\alpha,c}(\Phi(\cdot,t))(Y) dY dt.$$

They solve the weak patch system if also, for every scalar spacetime test ζ,

$$\int_I \int_E \theta(X,t)[\partial_t \zeta(X,t) + u(X,t) \cdot \nabla_X \zeta(X,t)] dX dt = 0.$$

All pairings must be absolutely integrable. For $\theta = \theta_F$ boundedness is automatic. With $c = c_{\mathrm{std}}$ this is the canonical gSQG operator, as identified in Section 12.1. The product θu is formed only after obtaining the locally integrable velocity representative.

Lemma 12.28 (Spacetime velocity law and uniqueness of the representative)**.** *For every admissible F, the representative $U[F]$ satisfies the coefficient-c velocity law on $E \times I$. Every other locally integrable velocity satisfying that law for θ_F equals $U[F]$ a.e. in spacetime. Such velocities are divergence-free distributionally in space.*

Proof. Integrate the spatial identity in time; compact-time slope bounds and test-function seminorms justify Fubini. Apply locally-integrable test-function uniqueness componentwise to the difference. For divergence use gradient tests and $B^*(\nabla\zeta) = 0$. □

Lemma 12.29 (Spatial weak flux across the actual graph)**.** *For every fixed $t \in I$ and scalar spatial test ζ,*

$$\int_{\Omega_t} U(X,t) \cdot \nabla\zeta(X) dX = \int_{\mathbb{R}} W(x,t)\zeta(x, F(x,t)) dx.$$

Both integrals are absolutely convergent, with estimates uniform on compact time intervals for tests with a common compact support and bounded first derivatives.

Proof. First integrate over $y < F(x,t) - d$, $d > 0$. To apply the compact-field formula, multiply $U\zeta$ by a smooth function of the vertical gap that equals one for gap at least d and zero for gap at most $d/2$. This field is C^1 across the excluded original graph. On the truncated domain its divergence is $U \cdot \nabla\zeta$. Pass $d \downarrow 0$: local integrability controls the volume term, and the cancelled normal trace controls the boundary term, with continuity of W for the dominated boundary limit. Never pass to separate vector boundary traces. □

Lemma 12.30 (Time derivative of the moving-patch pairing)**.** *For every scalar spacetime test ζ,*

$$\int_I \int_E \theta_F \partial_t \zeta = -\int_I \int_{\mathbb{R}} F_t(x,t)\zeta(x,F(x,t),t)dxdt,$$

with absolute convergence.

Proof. Write the spatial pairing as $\int dx \int_{-H}^{F(x,t)} \zeta(x,y,t)dy$, with H below the vertical test support and below F on its compact horizontal/time projection. Differentiate by endpoint FTC and parameter differentiation. Integrate the resulting total time derivative over the compact test support; its endpoint terms vanish. Lean route: translate $y = F(x,t) + s$ and set $Z(x,s,t) = \zeta(x,F(x,t) + s,t)$, so that $\partial_t\zeta(x,F+s,t) = \partial_t Z - F_t\partial_s Z$; then Fubini and FTC on $s \le 0$ and in t, with no differentiation under the integral sign. □

Lemma 12.31 (Lifting tests from the graph)**.** *For every scalar test $\psi \in C_c^\infty(\mathbb{R}\times I)$ there is a scalar spacetime test $\zeta \in C_c^\infty(E \times I)$ satisfying $\zeta(x,F(x,t),t) = \psi(x,t)$. Consequently a continuous $r : \mathbb{R} \times I \to \mathbb{R}$ whose integral against all these traces is zero must vanish pointwise.*

Proof. Choose a fixed smooth compactly supported η with $\eta(0) = 1$ and set $\zeta(x,y,t) = \psi(x,t)\eta(y - F(x,t))$. The function F is bounded on the compact support of ψ, making the lifted support compact. Its time support stays inside I, so extension by zero outside I is smooth. Finish by test-function uniqueness and continuity on the open domain. □

Theorem 12.32 (Two-dimensional patch equation is equivalent to the graph equation)**.** *Let F be admissible. Then $(\theta_F, U_{\alpha,c}[F])$ solves the coefficient-c weak patch system if and only if*

$$F_t(x,t) = c\int_{\mathbb{R}} \frac{F_x(x,t) - F_x(s,t)}{((x-s)^2 + (F(x,t) - F(s,t))^2)^{\alpha/2}} ds \quad (x \in \mathbb{R}, t \in I).$$

The same equivalence holds with any locally integrable u satisfying the velocity law, interpreted up to a.e. equality. Every front integral is absolutely convergent.

Proof. The transport pairing equals $\int (W - F_t)\zeta(x,F,t)dxdt$. One direction follows by substitution; in the other use lifted tests and continuity of $W - F_t$. A.e. equality of velocities preserves all transport pairings because θ_F is bounded. □

Lemma 12.33 (Local time continuity and the initial-value formulation)**.** *For $R > 0$ and $s, t \in I$,*

$$\int_{[-R,R]^2} |\theta_F(X,t) - \theta_F(X,s)|dX \le \int_{-R}^{R} |F(x,t) - F(x,s)|dx.$$

Thus $t \mapsto \theta_F(t)$ is continuous into $L^1_{\mathrm{loc}}(E)$. If the equivalent equations above hold and $t_0 < t_1$ both lie in I, let ζ be smooth on a neighborhood of $E \times [t_0, t_1]$, have a common compact spatial support there, and vanish for t sufficiently close to t_1. It may be nonzero at t_0. Then

$$\int_{t_0}^{t_1} \int_E \theta_F(\partial_t \zeta + U \cdot \nabla \zeta) + \int_E \theta_F(X, t_0)\zeta(X, t_0) dX = 0.$$

For an initial endpoint outside I, this last assertion requires additionally a locally uniform graph limit and a common global slope bound on a one-sided time neighborhood; it is not implied by interior compact-time bounds alone.

Proof. Integrate the difference of two half-line indicators on vertical slices. Uniform continuity of F on compact horizontal/time sets gives the local time trace. For the initial-value identity use only the weak transport equation, tested with $\chi_h(t)\zeta$ for a time cutoff χ_h rising from 0 at t_0 to 1 at $t_0 + h$, together with the L^1_{loc} continuity just proved. Retain the lower endpoint term in the time pairing identity; its sign is positive in the weak formula. Initial-value tests are restrictions of smooth ambient functions, not globally smooth functions required to vanish before t_0; that latter restriction would force the initial term to be zero. For the stated endpoint extension, start at $t_0 + h$ and use the uniform-slope integrability estimate and the L^1_{loc} density trace as $h \downarrow 0$. Lean route: for that endpoint case, the velocity bound and the common slope bound give integrability of U on $K \times (t_0, t_1]$; the limit F_0 is continuous, and the L^1_{loc} density limit comes from the slice estimate above applied to the glued graph equal to F_0 at t_0 and to F for $t > t_0$. □

12.7 Transfer of the front theorem to the canonical patch system

Lemma 12.34 (Uniform slope for the self-similar family)**.** *For every profile supplied by Theorem 1.31, the graph $F_{\epsilon,T}$ is admissible on $(-\infty, T)$ and satisfies $|F_x(x,t)| \le \sqrt{\epsilon}(C+1)$ for all $x, t < T$, with the profile-bound constant C. No global curvature bound is required.*

Proof. The two profile estimates imply $|f'| \le C + 1$. Differentiate the ansatz to obtain $F_x = \sqrt{\epsilon} f'(x/L(t))$. Joint smoothness is already part of the main theorem. Local curvature bounds follow on spacetime compacts lying strictly below T. □

Lemma 12.35 (Exact time normalization)**.** *Let $p = \alpha - 1$, $\kappa = \kappa_\alpha > 0$ and let an admissible F solve the normalized front equation on I. Then $G(x,t) = F(x, t/\kappa)$, on $\{t : t/\kappa \in I\}$, is admissible and satisfies the coefficient-c_{std} graph equation. If $I = (-\infty, T/\kappa)$, the new interval is $(-\infty, T)$. For the self-similar ansatz its physical scale is*

$$L_{\mathrm{std}}(t) = (\alpha(T-t)/\kappa)^{1/\alpha}.$$

Proof. The literal normalized graph equation has coefficient c_{front}. Time differentiation contributes $1/\kappa = c_{\mathrm{std}}/c_{\mathrm{front}}$. The ansatz with normalized blow-up

time T/κ gives the displayed scale. Positivity of κ follows directly from positivity of the Gamma factors in A_α and of `calpha`. Preserve the density jump equal to one; do not absorb the coefficient by silently changing the patch strength. □

Theorem 12.36 (Canonical two-dimensional patches forming a corner)**.** *Let $\alpha \in (7/4, 2)$ and $p = \alpha - 1$. There exist $\epsilon_0, C > 0$, depending only on α, such that for every $0 < \epsilon < \epsilon_0$ there is a profile f_ϵ satisfying all profile estimates of Theorem 1.31 with constant C and the following property. For every $T > 0$, define*

$$G(x,t) = \sqrt{\epsilon} L_{\mathrm{std}}(t) f_\epsilon(x/L_{\mathrm{std}}(t)), \qquad \theta(X,t) = \mathbf{1}_{X_2 < G(X_1,t)}, \qquad u = U_{\alpha, c_{\mathrm{std}}}[G].$$

Then on $E \times (-\infty, T)$ the pair is a weak solution of

$$\partial_t \theta + \operatorname{div}(\theta u) = 0, \qquad u = -\nabla^\perp (-\Delta)^{-1+\alpha/2} \theta,$$

with the canonical normalization defined above and with the initial trace at $t = 0$. The interface is smooth for $t < T$ and $\|G(\cdot,t) - \sqrt{\epsilon} x_-\|_\infty \le C\sqrt{\epsilon}(T-t)^{1/\alpha}$. For each such ϵ there is $b_\epsilon > 0$, independent of T and δ, such that for every $T, \delta > 0$ there is $t_\delta < T$ with $\sup_{|x|<\delta} |\partial_x^2 G(x,t)| \ge b_\epsilon (T-t)^{-1/\alpha}$ whenever $t_\delta < t < T$. Moreover, with $\theta_*(X) = \mathbf{1}_{X_2 < \sqrt{\epsilon}(X_1)_-}$, $\theta(t) \to \theta_*$ in $L^1_{\mathrm{loc}}(E)$ as $t \uparrow T$. This is an infinite-area patch result; no global L^1 assumption on θ is made.

Proof. Apply the completed front theorem at normalized time T/κ and use the equivalence theorem with coefficient c_{std}. The initial time zero is interior to the time interval. Corner convergence acquires the fixed factor $\kappa^{-1/\alpha}$ and the curvature lower bound the fixed factor $\kappa^{1/\alpha}$; positivity preserves both claims. On $[-R, R]^2$, the density error is at most $2R$ times the uniform graph-height error. No classical vector velocity at the interface or continuation beyond the corner time is claimed. □

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